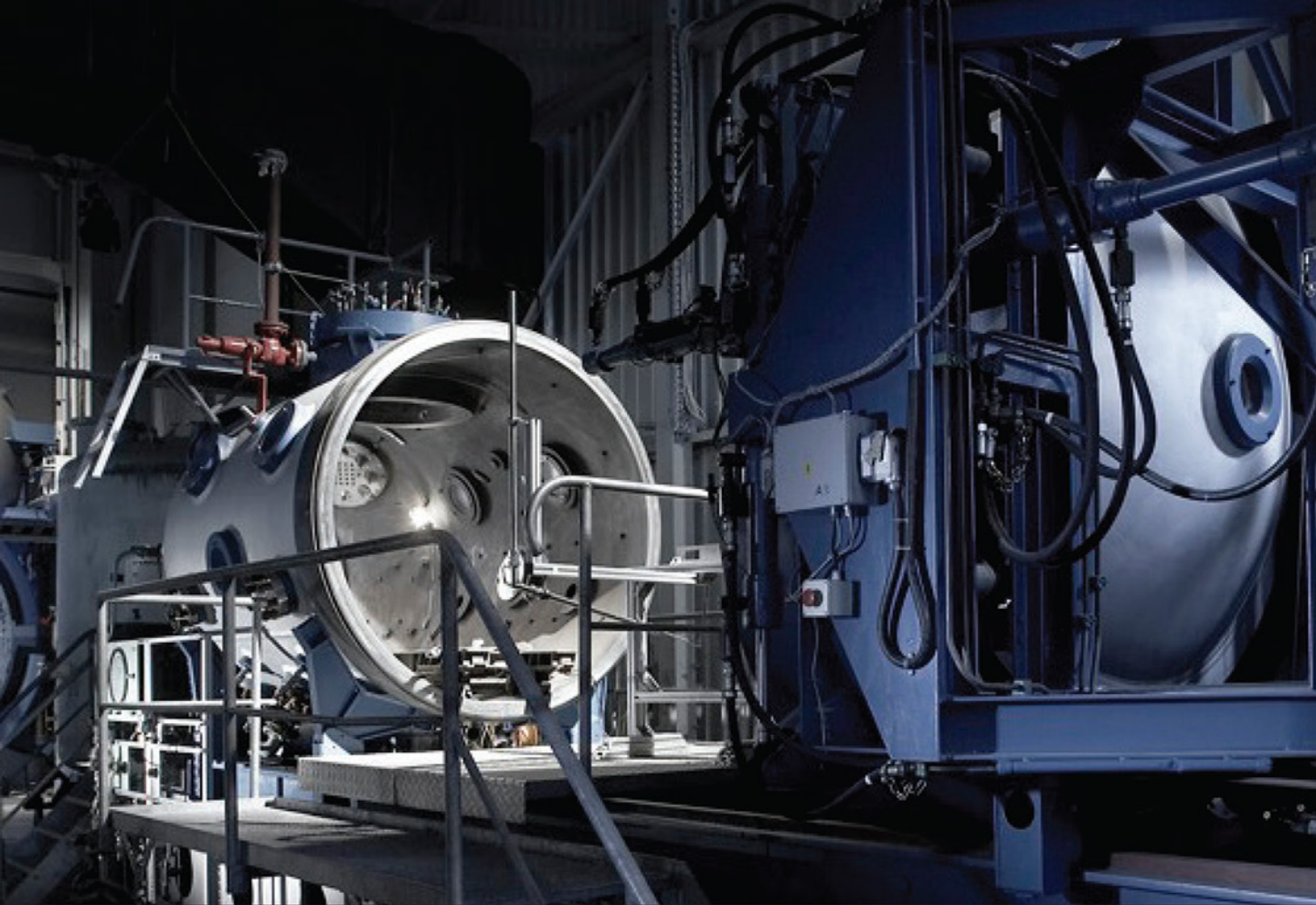




HYDROGEN SAFETY FOR ENERGY APPLICATIONS

Engineering Design, Risk Assessment,
and Codes and Standards

Edited by
Alexei Kotchourko
Thomas Jordan



Hydrogen Safety for Energy Applications

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and Codes and Standards

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Preface

Hydrogen safety and the associated research have a long history, in fact mainly rooted in nuclear safety. Not only Fukushima but also other previous major nuclear accidents were accompanied by explosions of accidentally produced hydrogen – air mixtures. Although the corresponding safety research studies received considerable funding and provided a broad scientific basis, the nowadays envisaged applications for the intentional use of hydrogen as an energy carrier imply quite different conditions for the accidental phenomena and scenarios.

Also, the experience gained from the large-scale use of hydrogen in the petrochemical industry, in the fertilizer production, and from the use as a rocket fuel in space applications only partially addresses the safety issues associated with the new applications of hydrogen developed for the defossilization of our future energy system.

Since the first energy crisis in the early 1970s, hydrogen was investigated in several waves as an energy carrier and as a potential replacement of fossil fuels. Safety was always considered crucial especially for the wide-spread decentralized use in the public domain. Untrained customers needed fail-safe technologies, easy to handle with obvious and convincing safety features. However, the hydrogen safety knowledge remained geographically and thematically fragmented and only partially applicable. Harmonized methodology and standardization were almost impossible on this basis.

These deficits initiated the public support for founding the Network of Excellence HySafe, with the mission to integrate and institutionally steady hydrogen safety research. With an initial geographical focus on Europe, the International Association IA HySafe today has more than 50 members from all over the world.

The main activities and products of HySafe are the organization of the biennial International Conference on Hydrogen Safety, the organization of Research Priorities Workshops, development and conducting educational and training programs, and the documentation of the thereby continuously further developed the state of the art via the Biennial Report on Hydrogen Safety (BRHS). All elements have been broadly endorsed by European, North American, and Asian public funding bodies or safety-related institutions, such as FCH JU, US DoE, NEDO, IEA, IPHE, and ISO, and by key industries.

The latest version of the BRHS was published more than 10 years ago as an online Wiki version and lived a quite hidden life. However, the current interest and huge investments in hydrogen technologies inspired reviving this activity and compiling a new book with a slightly different format, containing the current knowledge and state of the art promoted by the HySafe community. Obviously the released BRHS issues served as one of the main resources and many former contributors acted also as authors for this book.

Therefore the intended purpose of the book is to serve as a desktop reference for students, researchers, and engineers. In seven chapters it summarizes all relevant knowledge and describes necessary models and tools for safe design and operations of hydrogen technologies. In the introductory chapters it provides the fundamental background on hydrogen and the associated technologies. On this background the relevant accidental phenomena and consequences are described in detail. Risk assessment methodologies and safety measures are addressed in the subsequent chapters. Here it is important to note that strictly speaking “safety” is acceptable risk and therefore

involves probabilistic and societal concepts. In particular, for emerging technologies like hydrogen technologies, robust probabilistic data are missing and society is lacking hands-on experience. Therefore in many places, and in particular, in these chapters about risk assessment and safety measures, we are rather dealing with hazards, physical effects, or accidental consequences on a deterministic basis. These labels become important for discriminating hazard and safety distances for instance. Despite the induced inconsistency, the term safety is quite frequently used in a more general meaning, also in this book. The last chapter of the book deals with relevant regulations, standards, and other codes, which actually represent the main societal aspects. More psychological issues such as risk perception, acceptance, and trust in technology are not covered in this book.

In particular, the technologies and standards will further develop quickly because of the huge investments currently made for new developments and market activation. This implies that the concerned chapters might need updates in a few years. However, we hope that they will remain a valuable resource in the long term.

Finally, before letting the reader dive into the more technical details, the editors would like to thank all contributors and particularly acknowledge the generous support of the IA HySafe for producing this book.

Hydrogen fundamentals

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Chapter Outline

1.1 Physical properties of hydrogen	2
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Hydrogen is the first element in the periodic table with the atomic number 1. It is the lightest and most abundant element in the universe representing 75mass% or 90vol% of all matter. On Earth, it is mostly found in compounds with almost every other element. Considering just the compound water, the hydrogen contents bound in the total water supply of the world is in the order of 10^{14} t. Hydrogen exists also as a free element in the atmosphere, but only to the extent of less than 1 ppm by volume. Free ionic hydrogen is more reactive than molecular hydrogen, the nonpolar covalent compound of two hydrogen atoms.

Paracelsus (1493–1541) observed a “wind” emanating when iron was dissolved in sulfuric acid, but still considered this as “air.” Robert Boyle (1627–91) collected the gas in a vessel, but also described that as a different kind of air. It was in 1776, when Henry Cavendish (1731–1810) produced and analyzed hydrogen as a distinct species, “inflammable gas,” different from “ordinary air” and “fixed air,” the only other gases known at that time. It was given the name *hydrogenium* or “water maker” in 1783 by Antoine Lavoisier, who proved by sending an electric spark into a mixture of air and “inflammable air” and precisely measuring the combustibles and their products that the result was water composed of hydrogen and oxygen. At this early stage already, in 1783, hydrogen gas was found by the Frenchman Jacques Charles as an alternative to hot air and applied to fill a balloon and start his first ascension.

At standard temperature and pressure conditions, hydrogen is a colorless, odorless, tasteless, nontoxic, noncorrosive, nonmetallic diatomic gas, which is, in principle, physiologically not dangerous. There are three isotopes of hydrogen that can be found in nature. The most common

hydrogen isotope is protium ^1H , H-1, H, atomic mass 1.007825 u; here u is the “unified atomic mass unit” or Dalton (Da), defined as 1/12-th of the mass of a C-12 nucleus and equivalent to $1.660539040(20) \times 10^{-27}$ kg. The second isotope is the stable deuterium (^2H , H-2, D, atomic mass 2.014102 u) or heavy hydrogen discovered in 1932 by Urey and coworkers. Deuterium has a natural fraction of 0.014% with physical and chemical properties slightly different from H-1. Nearly all D in natural hydrogen is in combination with hydrogen atoms, the diatomic HD with a fraction of 0.032% in natural hydrogen; the existence of molecular deuterium, D_2 , is highly improbable. The third hydrogen isotope is the radioactive tritium (^3H , H-3, T, atomic mass 3.016049 u) discovered in 1934 by E. Rutherford. It undergoes β -decay at a half-life of 12.3 years. Tritium occurs naturally in traces only due to the steadily on-going nuclear reactions in the atmosphere, but also as a remnant from above-ground nuclear weapon testing in the past. Other isotopes of hydrogen are the highly unstable isotopes H-4, H-5, H-6, and H-7 that have been synthesized in the meantime.

1.1 Physical properties of hydrogen

Hydrogen can be considered an ideal gas over a wide range of (not too low) temperatures and (not too high, up to 10 MPa) pressures. At some point, however, like any other substance that is sufficiently cooled or compressed, hydrogen will act like a real gas. This behavior when departing from an ideal gas (Fig. 1.1) can be described, for example, by the Redlich – Kwong – Soave equation of state (Soave, 1972):

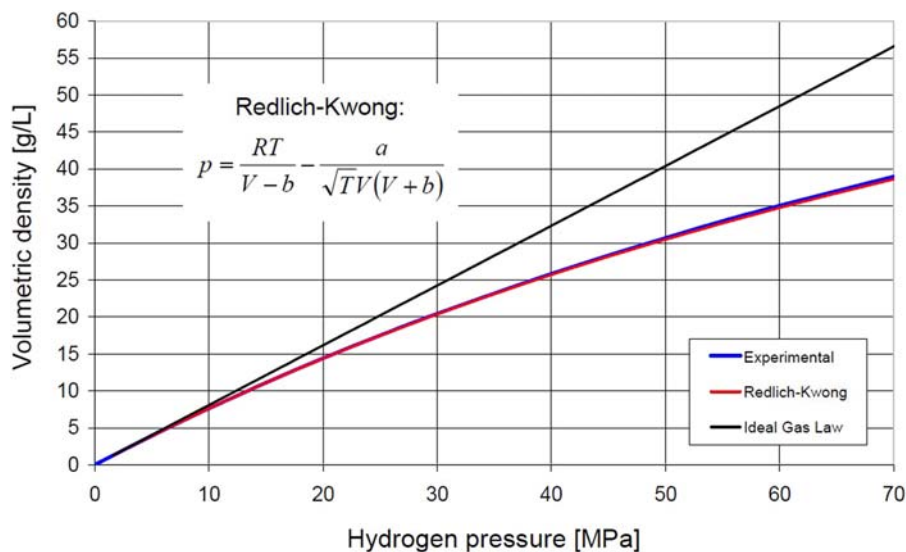


FIGURE 1.1

Deviation of the Redlich – Kwong equation of state for hydrogen from the ideal gas law.

$$p = \frac{RT}{V_m - b} - \frac{a\alpha}{V_m(V_m + b)}$$

with $a = 0.42747 \frac{R^2 T_c^2}{p_c}$ and $b = 0.08664 \frac{RT_c}{p_c}$

and for hydrogen: $\alpha = \alpha(T_r) = 1.202 \exp\{-0.30288 T_r\}$, with $T_r = \frac{T}{T_c}$ where p is the pressure (Pa); T is the temperature (K); V_m is the molar volume (m^3/mol); R is the universal gas constant = $8.3145 \text{ J}/(\text{mol} \cdot \text{K})$; T_r is the reduced temperature; T_c is the critical temperature = 33.25 K ; p_c is the critical pressure = 1.297 MPa .

One of hydrogen's most important characteristics is its low volume-related density of 84 g/m^3 @ normal temperature and pressure (NTP), which makes it necessary for any practical applications to either compress the hydrogen or liquefy it. Liquid hydrogen (LH_2) has a high density of 70.8 kg/m^3 compared to the gaseous form, which allows for a substantial volume reduction by a factor of 845 versus the gaseous state at ambient conditions. Therefore hydrogen in the form of a liquid is typically used for storage/transportation purposes if volume is an issue. This density, however, is low compared to water. Its specific gravity is 0.07 meaning that, in case of LH_2 spills onto water, more than 90% would be moving above the water surface. As the figure indicates, there is also a cryogenic range, which yields even higher densities than the liquid state if at sufficiently high pressures and sufficiently low temperatures.

Hydrogen gas is highly diffusive and highly buoyant: it is positively buoyant above a temperature of 22K , that is over (almost) the whole temperature range of its gaseous state, and rapidly mixes with the ambient air upon release. The diffusion velocity is proportional to the diffusion coefficient and varies with temperature according to T^n with n in the range between 1.72 and 1.8. Diffusion in multicomponent mixtures is usually described by the Stefan – Maxwell equation (Maxwell, 1965), which is different from the Fickian law of diffusion. Corresponding diffusion rates of hydrogen in air are larger by about a factor of 4 compared to those of air in air. The rising velocity under the influence of (positively) buoyant forces cannot be determined directly, as they are dependent on the density difference between hydrogen and air as well as on drag and friction forces. Also shape and size of the rising gas volume as well as atmospheric turbulence have an influence on the final velocity of the rising gas. The positive buoyancy of hydrogen is a favorable safety effect in unconfined areas, but it may cause a hazardous situation in (partially) confined spaces where the hydrogen can accumulate, for example, underneath a roof. Both diffusion and buoyancy determine the rate at which the gas mixes with the ambient air. The rapid mixing of hydrogen with the air is a safety concern, as it leads very soon to flammable mixtures, which on the other hand—for the same reason—also will quickly dilute to the nonflammable range.

Hydrogen coexists in two isomeric forms, ortho and para hydrogen, a quantum mechanics feature that was discovered in 1926 by the German physicist Werner Heisenberg. A small energetic difference is given, if the spins of the two protons of a hydrogen molecule are either aligned parallel (ortho) or antiparallel (para). The existence of the two forms was proven experimentally in 1929 by Bonhoeffer and Harteck using charcoal as catalyst for the separation. The partition is dependent on the temperature. Normal hydrogen at room temperature is a mixture of 75% ortho and 25% para hydrogen. In the lower temperature range $<80\text{K}$, para hydrogen is the more stable form. At 20K , in thermal equilibrium, the concentrations are 99.8% para and 0.2% ortho. The rate of conversion between ortho and para states is with $0.0114/\text{h}$ slow in the gas phase. The noncatalyzed transition takes place over a longer period (about 3–4 days), until a new equilibrium state is reached.

However, magnetic impurities and also small oxygen concentrations are able to accelerate ortho – para conversions raising the rate by several orders of magnitude [very good: $\text{Fe}(\text{OH})_3$] to the order of hours. The conversion from ortho to para is an exothermal reaction with a conversion energy of 270 kJ/kg at room temperature, which increases with decreasing temperature. At temperatures below 77K, it is almost constant at 523 kJ/kg.

Boiling point of hydrogen at atmospheric pressure is 20.3K, below which the hydrogen gas turns into a liquid. The liquefaction of hydrogen, first realized by James Dewar in 1898, is a highly energy-intensive process usually conducted in several separate steps of multistage compression and expansion (cooling). During the cooling phase, additional energy is needed to remove the heat generated from both the condensation process (446 kJ/kg) and from the conversion of normal H_2 (75% ortho, 25% para) to para- H_2 (523 kJ/kg).

As is shown in Fig. 1.2, different pathways are possible to cool the H_2 gas down to its boiling point, but all require the same minimum specific exergy. The type of process selected is depending on different parameters such as the real energy demand determined by the efficiencies of the single compressing and cooling steps, the amounts of irreversible losses, and last but not least, the availability of respective systems and their costs.

There is a minimum specific work (exergy), $A_{\min}$, required for the liquefaction of hydrogen at ambient conditions. It is composed of (Vander Arend et al., 1964)

1. the work for cooling the gas to the boiling point determined by the Carnot cycle

$$A = (T_o - T_s) \times \Delta s = \frac{(T_o - T_s)}{T_s} \times \Delta Q$$

with $\Delta Q = T_s \times \Delta s$ the heat withdrawn at T_s ,

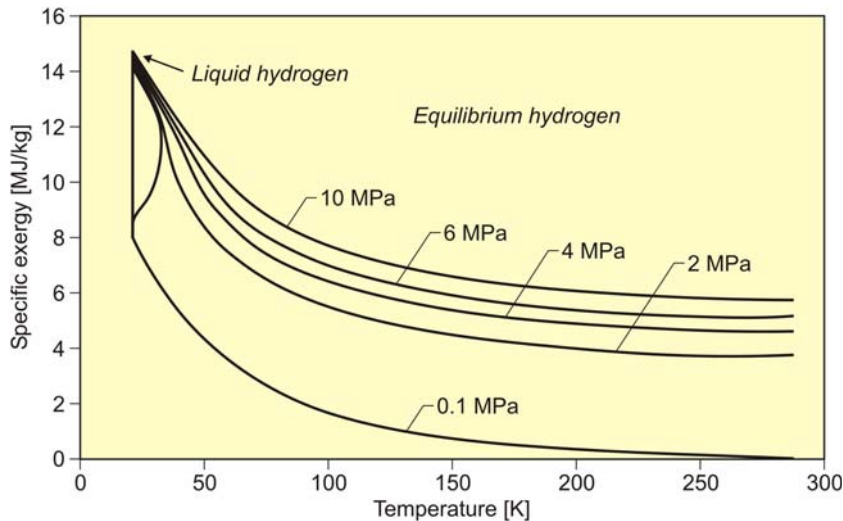


FIGURE 1.2

Specific exergy required for the liquefaction of hydrogen (Quack, 2002).

2. the work for condensation of the gas, and
3. the work for the temperature-dependent ortho – para conversion.

resulting in

$$A_{\min} = \int_{T_o}^{T_s} \frac{T_o - T}{T} c_p dT + \frac{T_o - T_s}{T_s} \Delta H_v + \int_{C_1}^{C_2} \frac{T_o - T_s}{T_s} C_{o \rightarrow p} dC$$

$$= 1.58 + 1.70 + 0.62 = 3.90 \frac{\text{kWh}}{\text{kg}}$$

where T_o is the ambient temperature (K); T_s is the boiling point (K); c_p is the specific heat capacity (temperature dependent) [J/(kg•K)]; ΔH_v is the heat of vaporization (J/kg); C is the concentration of ortho-hydrogen; $C_{o \rightarrow p}$ is the conversion energy from ortho to para hydrogen (J/kg).

Table 1.1 shows the minimum liquefaction work and its single contributions for hydrogen and a comparison with respective data for some other gases. The total minimum energy required for liquefying hydrogen quoted in Peschka (1992) is 3.92 kWh/kg or 14.1 MJ/kg. The comparison of hydrogen with the other fluids shows that for helium, due to the lower boiling point, the contribution from cooling to the liquefaction work is dominant, whereas condensation work is small. In contrast, for methane and nitrogen with much higher boiling points, the cooling work is small compared to the condensation work. But for both, total liquefaction energy required is smaller than that for hydrogen and helium.

Another advantage of LH₂ is its extreme cleanliness, which makes it beneficial to a variety of industries such as the glass, metals, or electronics industries. A major drawback of LH₂ is its significant energy consumption of about one-third of its heat of combustion in the production process. Another one is the enormous effort needed for the storage vessel design with its typical features of a double-walled container and high performance insulation. Still, there is the unavoidable loss by boil-off, which is to maintain the cold temperature in the tank and which increases with decreasing tank size. A safety issue arises also if LH₂ is inadvertently released in a confined area as its violent boiling changes the local atmosphere and may cause harm to people present. In an enclosed space, final pressure may even rise to 172 MPa, which certainly overpressurizes systems to bursting.

The vaporization rate is even enhanced when ortho-hydrogen is stored. The heat liberated during the ortho – para conversion at 20K (670 kJ/kg) is larger than the latent heat of vaporization at the same temperature (446 kJ/kg), which means that normal LH₂ is able to vaporize completely even in a perfectly insulated vessel. This represents a safety issue requiring a design of the

Table 1.1 Minimum liquefaction energy for different gases (Peschka, 1992).

Gas	Boiling point (K)	Heat of vaporization (kJ/kg)	Minimum work (kWh/kg)		
			CoolingNTP → BP	Condensation	Total
Hydrogen	2.28	445.59	2.242 ^a	1.666	3.91
Methane	111.63	509.88	0.077	0.230	0.31
Nitrogen	77.34	201.	0.055	0.156	0.21
Helium	4.22	20.9	1.917	0.398	2.32

^aIncludes ortho – para conversion.

hydrogen loop, which is able to remove the heat of conversion in a safe manner. Ortho – para conversion can be done adiabatically, simply using a catalyst bed, however, on the expense of a temperature increase of the flowing hydrogen. There is also an isothermal method, putting the catalyst into LN₂ or LH₂ at boiling point with temperature remaining constant (Stang, Neksa, & Brendeng, 2006). In today's liquefaction plants, this process step is performed already at the site of liquefaction by applying a catalyst resulting in LH₂ with almost a 100% of ortho-hydrogen.

The release of hydrogen as a liquefied gas usually results in the accumulation and formation of a liquid pool on the ground, which expands, depending on the volume spilled and the release rate, in radial direction away from the release point, and which also immediately starts to vaporize (Verfondern & Dienhart, 2007). The most dominant heat source is heat transport from the ground. This contribution is based on the relationship between heat flux density and the temperature difference between cryogen and surface ("Nukiyama curve") (Brentari, Giarratano, & Smith, 1965). Fig. 1.3 shows the comparison of experimentally derived heat flux densities with correlations recommended for the film boiling regime ("Kutateladze") and the nucleate boiling regime ("Breen & Westwater") (Dienhart, 1995). Upon contact with the ground, the cryogen will in a short initial phase slide on a vapor cushion (film boiling) due to the large temperature difference between liquid and ground. The vaporization rate is comparatively low and if the ground is initially water, no ice will be formed. With increasing coverage of the surface, the difference in temperatures decreases until—at the Leidenfrost point—the vapor film collapses resulting in enhanced heat transfer via direct contact (nucleate boiling). On water, there is the chance of ice formation, which, however, depending on the amount of mass released, will be hindered due to the violent boiling of the cryogen.

A further temperature decrease below the boiling point eventually results in the generation of solid hydrogen and a two-phase fluid of coexisting liquid and solid hydrogen or slush hydrogen

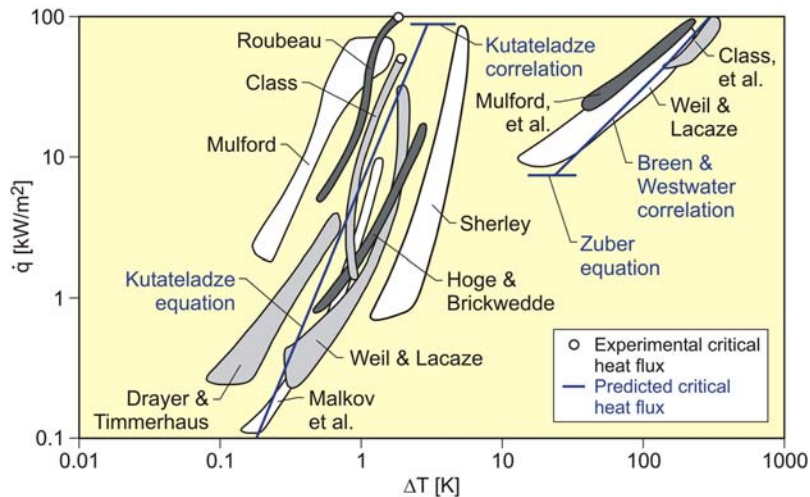


FIGURE 1.3

Correlations of the Nukiyama curve for hydrogen based on experimentally covered ranges for the heat flux density (Dienhart, 1995).

(SLH₂), respectively. A mixture of 50% solid and 50% liquid offers the advantages of a higher density by up to 16%, a higher heat capacity by up to 18%, and a prolongation of the storage time of the cryogen as the solid melts and absorbs heat (Ohira, 2015). Applications of SLH₂ are seen as a rocket fuel, but also as a refrigerant for superconducting power transmission. Due to the fact that the hydrogen vapor pressure is strongly reduced with decreasing temperatures, from 98 kPa (which is about atmospheric pressure) at 20K down to 13 kPa at 13K, SLH₂ systems must be designed to safely operate far below atmospheric pressure. A safety risk is arising from such a pressure below that of the local atmosphere demanding protection against air entrainment into the system, which would solidify the air and warm up the system. Another safety concern is electric charge accumulation in SLH₂ flows (as it involves solid particles), which may become a source of ignition at the presence of air; this effect is less severe for flowing LH₂. NASA has studied SLH₂ characteristics under conditions of production, storage, and transfer (NASA, 1997; Sonntag, Squires, & Park, 1988).

The triple point is the temperature (13.8K) and pressure (7.2 kPa), at which all three phases of solid, liquid, and gaseous hydrogen can exist in equilibrium (Fig. 1.4). The boiling point increases with pressure to the critical point, which is given by $T_c = 33.15\text{K}$, $p_c = 1.296\text{ MPa}$ with a critical density of $\gamma_c = 31.4\text{ kg/m}^3$. A pressure increase beyond the critical point has no further influence.

There is no liquid phase of hydrogen existing above its critical temperature. If hydrogen (or any other fluid) is maintained above its critical temperature and pressure is applied, a single phase “supercritical fluid” is formed, which is characterized by T_c and p_c . It is gas-like in that it is compressible, it is liquid-like in that it has a comparable density, and there is some transitory state in between characterized by strong structural fluctuations causing the unusual behavior of fluid properties near their critical point. It also exhibits higher flow rates as compared with liquids. There is a strong dependence of the thermophysical properties of cryogenic hydrogen on temperature and

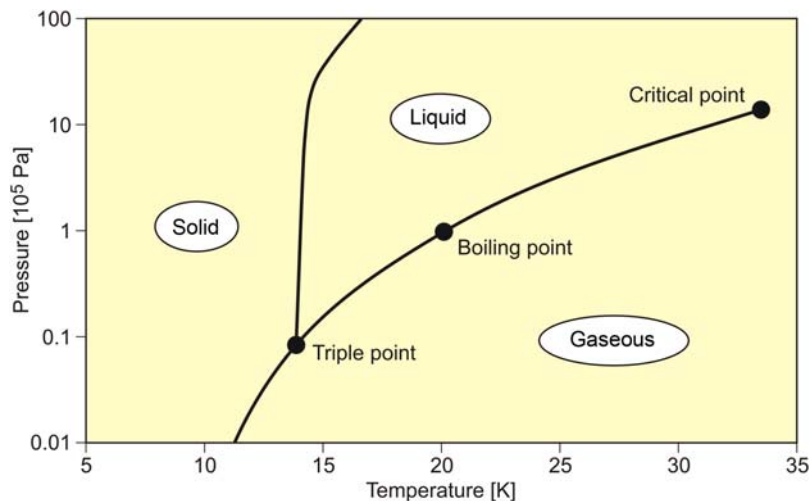


FIGURE 1.4

Phase diagram of hydrogen.

pressure in the supercritical state. They vary especially in the near-critical region. C_p has a maximum at the so-called pseudocritical temperature (“thermal spike phenomenon”). Supercritical hydrogen might undergo a turbulent-to-laminar transition due to the dependence of viscosity on temperature. Heat transfer coefficients are unpredictable in the transition regime, and are much lower in the laminar regime.

For any given pressure, a real gas has an inversion temperature T_{inv} . The so-called Thomson – Joule effect arising from the forces between the gas molecules means that the temperature of a real gas decreases ($T < T_{inv}$) or increases ($T > T_{inv}$) upon expansion (depressurization) at constant enthalpy. It is “internal work” against the van der Waals forces acting among the molecules, which is used to cool or heat the gas. All locations where there is no temperature change are forming the so-called inversion curve (Fig. 1.5).

Unlike most other gases, the inversion temperature of H_2 gas is with 193K at atmospheric pressure much lower than ambient temperature. A safety concern is that the sudden depressurization of a GH_2 storage vessel may lead to an ignition. For example, the temperature increase is 6K, if a sudden pressure drop from 20 MPa to ambient pressure takes place. The chance of a spontaneous ignition just by that effect, however, is small. An explosion is more likely to occur because of electrostatic charging of dust particles during the depressurization or autoignition at high temperatures or other mechanisms such as spark discharges from isolated conductors, brush discharges, corona discharges (Astbury & Hawksworth, 2005).

Because of its small size, its small molecular weight, and its low viscosity, hydrogen may cause a safety issue with respect to the propensity of the gas to leak at a larger molecular flow rate than other gases. Diffusion in small amounts is even possible through intact materials, in particular organic materials, which may lead to gas accumulation in confined spaces. This behavior also holds for the liquid state. Leakage rates are by a factor of 50 higher than for water and by a factor of 10

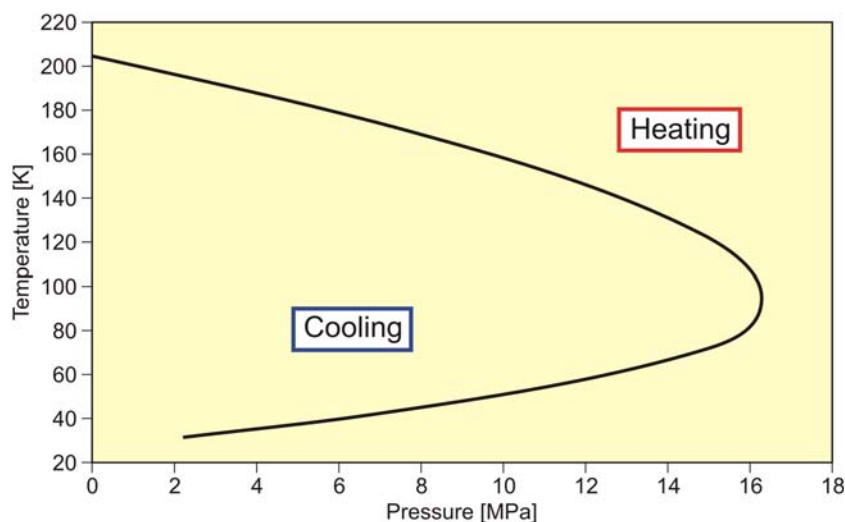


FIGURE 1.5

Inversion temperature of hydrogen.

compared to nitrogen. Unlike invisible “normal” gaseous leakages, the release of LH_2 or gaseous hydrogen at cryogenic temperature will usually produce a visible water vapor fog. The addition of an odorant or colorant would ease the detection of small leaks; however, this is not practicable in most situations and not feasible for LH_2 .

Hydrogen gas dissolved in liquids will permeate into adjoining vessel materials. At elevated temperatures and pressures, hydrogen attacks mild steels severely, causing decarburization and embrittlement. This is a serious concern in any situation involving storage or transfer of hydrogen gas under pressure. Proper material selection, for example, special alloy steels, and technology are required to prevent embrittlement.

Hydrogen is both in the gaseous and liquid phase essentially an insulator. Above some critical “breakdown” voltage of ~ 300 V, where ionization turns the hydrogen into a plasma, it becomes an electrical conductor. At extreme, but accessible pressures (~ 200 – 300 GPa) and temperatures (~ 4400 K), hydrogen will make a phase transition to liquid metallic hydrogen and at even higher pressures (~ 450 GPa) to solid metallic hydrogen, which may be superconducting at room temperature (McMinis, Clay, Lee, & Morales, 2015). The existence of the metallic phase of hydrogen predicted in 1935 was eventually proven in a shock compression test in 1996 (Weir, Mitchell, & Nellis, 1996). Metallic hydrogen is accepted to exist in the interior of Saturn and Jupiter, but has no practical application on Earth so far.

1.2 Chemical properties of hydrogen

Hydrogen is able to react chemically with most other elements. Hydrogen reacts both with nonmetals (high electronegativity) and with metals (low electro-negativity) to form either ionic or covalent hydrides (e.g., HCl , H_2O). The electronegativity is a measure for the attraction of electrons to the nucleus and its difference to the partner’s electronegativity defines the character of the bond: nonpolar covalent (difference = 0), polar covalent, or ionic (difference high). The electronegativity of hydrogen is 2.20 (Pauling scale).

As a fuel hydrogen represents a clean, environmentally benign energy carrier. In connection with oxygen, hydrogen is highly flammable over a wide range of concentrations. A stoichiometric hydrogen – air mixture, where all fuel is consumed upon reaction, that is, where maximum combustion energy is released, contains 29.5 vol% of hydrogen. The combustion product of hydrogen is water vapor. It burns in a nonluminous, almost invisible pale blue, hot flame to water vapor liberating the chemically bound energy as heat (gross heat of combustion). The flame temperature of a burning (premixed stoichiometric) hydrogen – air mixture is max. 2403 K.

The mass-related energy density of hydrogen is very high; 1 kg of hydrogen contains 132.5 MJ, which is approximately 2.5 times more energy than is contained in 1 kg of natural gas. The energy content of hydrogen is given as either lower heating value (LHV) of 242 kJ/mol or as higher heating value (HHV) of 286 kJ/mol. The difference of 15.6%, which is large compared to other gases, is due to the heat liberated upon condensation of the water vapor.

There is a wide flammability range of hydrogen (at room temperature) between 4 and 75 vol% of concentration in air and up to 95 vol% in oxygen. This means, on the other hand, that a minimum oxygen concentration of 5% is necessary to keep a flame propagating. The lower flammability limit (LFL)

as the minimum amount of fuel that supports combustion, is usually the “more important” limit, as it will be reached first in a continuous leakage. The flammability range widens with higher temperatures.

The change of flammability limits of H_2 – air mixtures in the higher temperature range above room temperature up to 400°C can be expressed with the following empirical linear equations (Schroeder & Holtappels, 2005), for the LFL

$$c_{LFL} = c_{LFL}(T_0) \times [1 - K_L(T - T_0)](\text{vol}\%)$$

where T is the temperature given in (K), $T_0 = 273.15\text{K}$, $c_{LFL}(T_0) = 4.1\%$, and the factor $K_L = 0.00157/\text{K}$.

and for the upper flammability limit (UFL)

$$c_{UFL} = c_{UFL}(T_0) \times [1 - K_U(T - T_0)](\text{vol}\%)$$

where $c_{UFL}(T_0) = 74.5\%$, and the factor $K_U = 0.00044/\text{K}$.

In the range below room temperature, the influence of the temperature on the flammability limits is expressed in the modified Burgess – Wheeler equation for the LFL, which is for hydrogen at ambient pressure (Zabetakis, 1967):

$$c_{LFL} = c_{LFL}(300\text{K}) - \frac{3.14}{\Delta H_c}(T - 300) = 4.0 - 0.013(T - 300)(\text{vol}\%)$$

where ΔH_c is the net heat of combustion $= 242 \text{ kJ/mol}$.

The respective equation for the UFL valid for the temperature range $150\text{--}300 \text{ K}$ is (Eichert, Deimel, Hoffmann, & Pültz, 1992):

$$c_{UFL} = 74.0 + 0.026(T - 300)(\text{vol}\%)$$

A more recent experimental study [Kuznetsov 2013] concludes a slightly modified linear relationship between flammability limits and temperature recommending the following equations:

$$c_{LFL} = 4.64 - 0.0067 T(\text{vol}\%)$$

in the temperature range $-150^\circ\text{C} \leq T \leq 400$, with T in $^\circ\text{C}$, and

$$c_{UFL} = 73.8 + 0.033 T(\text{vol}\%)$$

in the temperature range $-60^\circ\text{C} \leq T \leq 400$, with T in $^\circ\text{C}$.

All above LFL and UFL correlations are depicted in Fig. 1.6.

Also a pressure effect on the flammability range of H_2 – air and H_2 – O_2 mixtures was identified (Schroeder & Holtappels, 2005). Respective experimental data indicate an increase of the LFL with increasing initial pressure, different from most other gaseous hydrocarbons. Correspondingly, the UFL decreases with increasing initial pressure, however, exhibiting an abnormal behavior from $\sim 2 \text{ MPa}$, where UFL values increase until $\sim 5 \text{ MPa}$, before they slightly drop again.

For the determination of LFL and UFL of mixtures of fuels, the Le Chatelier rule is the most commonly applied method:

$$\frac{1}{L_m} = \sum \frac{y_i}{L_i}$$

where y_i are the volume fraction and L_i the flammability limit of fuel i . Le Chatelier's rule was found to be in accurate agreement with experimental data for the system H_2 – CO , but shows

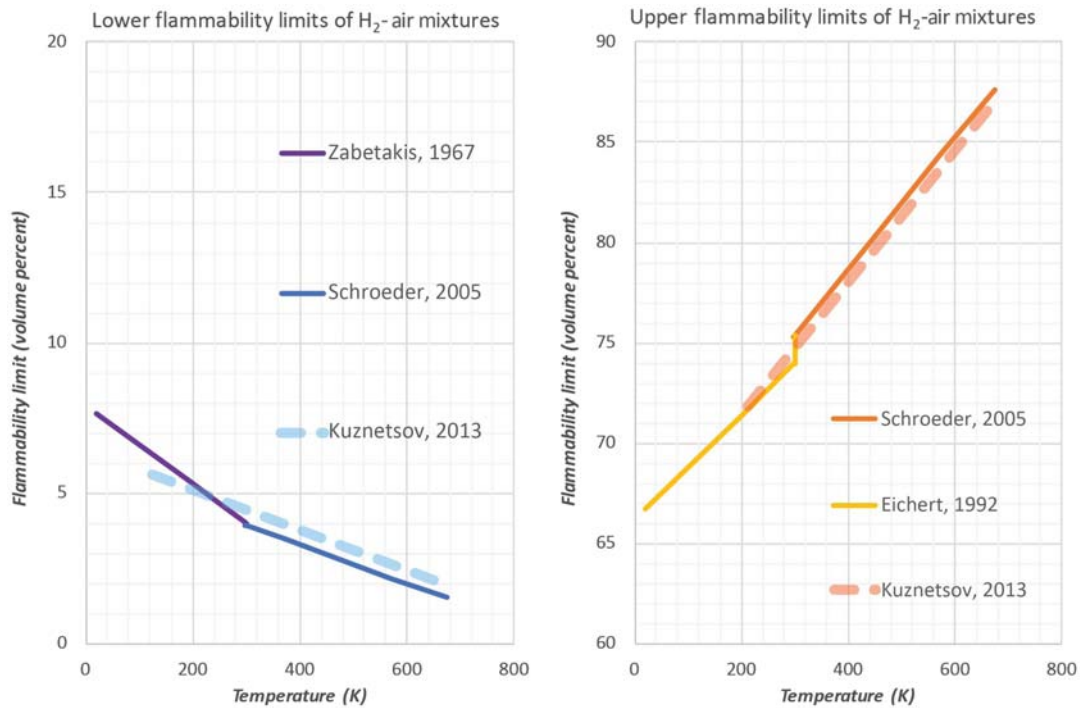


FIGURE 1.6

Flammability limits in hydrogen-air mixtures, LFL (left) and UFL (right) (Kuznetsov, 2013).

deviations up to 35% for the systems methane – propane or methane – ethane (Choudhuri, 2005). The GEXCON handbook (Bjerketvedt, Bakke, & Wingerden, 1997) states that “the formula does not work properly for H_2 .”

The potential for an explosion of a flammable hydrogen – air mixture is very high. The auto-ignition temperature, which is the minimum temperature of a hot surface that can ignite a flammable mixture, is with 858K relatively high, but can be lowered by catalytic surfaces. Hydrogen gas does not have a flash point as it is already a gas at ambient conditions. It means that cryogenic hydrogen will flash at all temperatures above its boiling point of 20K.

The minimum ignition energy, that is, the spark energy required to ignite the “most easily ignitable hydrogen concentration in air” is with 0.02 mJ very low, much lower than for hydrocarbon – air mixtures. A weak spark or the electrostatic discharge, for example, by a human body that is about 10 mJ, would suffice for an ignition; this is, however, no different from other burnable gases. The minimum ignition energy is further decreasing with increasing temperature, pressure, or oxygen contents.

It is known from the experience that a hydrogen – air gas cloud evolving from the inadvertent release of hydrogen upon the failure of a storage tank or pipeline liberates only a small portion of

its thermal energy contents in a typical unconfined hydrogen explosion, which is in the range of 0.1%–10%, in most cases <1% (Lind, 1975).

The thermal radiation emitted from hydrogen combustion essentially originates from the water vapor within the hydrogen flame. The emissivity factor of water vapor, however, is substantially lower ($\varepsilon < 0.1$) than that of hydrocarbon flames ($\varepsilon \sim 1$) (Arthur, 1960). Therefore despite its high flame temperature, the burning hazard is small compared to other fuels. It also results in the nonvisibility of hydrogen flames, which makes a hydrogen fire difficult to recognize and localize. In contrast, other gases containing carbon also generate—in addition to water vapor and carbon dioxide—soot within the flame. Soot is an efficient infrared emitter, leading to a strong thermal radiation and visible flames. An advantage of hydrogen-air fires is the fact that there is no smoke generation (assuming no other material be ignited), which is important for confined areas.

The laminar burning velocity in a flammable gas mixture, defined as the speed with which a smooth plane combustion wave advances into a stationary flammable mixture, is a pertinent property of the gas depending on temperature, pressure, and concentration. The burning velocity of hydrogen in air at stoichiometric ambient conditions is 2.37 m/s. It is higher than that of other hydrocarbon fuel–air mixtures because of its fast chemical kinetics and high diffusivity. This comparatively high burning velocity results in a greater chance for a transition of the combustion mode from deflagration to detonation. The detonability range is usually given to be 18–59 vol % of hydrogen concentration. In pure oxygen, the detonation range is extended to 15%–90% (Zabetakis, 1967), but will typically decrease at the presence of a diluting substance (Breitung, 2005). The detonation velocity may reach values in the order of 2000 m/s.

The rapid combustion of a hydrogen–air mixture cloud results in the formation of a pressure wave, which strongly depends on the combustion mode. In the deflagration of a free gas cloud, the expanding reaction products are pushing the fresh gas forward while overpressure is steadily decreasing. In contrast, a detonation creates a compression wave characterized by an initial pressure spike followed by exponential decay. In a closed vessel, explosion pressures increase by a factor of up to ~ 8 (stoichiometric mixture) for a deflagration and up to ~ 16 for a detonation (Schroeder & Holtappels, 2005).

In the case of cold hydrogen gas cloud evolving from an open LH₂ pool, it needs to be considered that cold hydrogen gas is less volatile compared to ambient gas and thus more prone to the formation of a flammable mixture with air. Furthermore, LH₂ in direct contact with the ambient air quickly contaminates itself due to condensation and solidification of air constituents. This effect carries the danger of potentially plugging valves, vents, or other equipment. In addition, due to the different boiling points of nitrogen (77.3K) and oxygen (90.2K), the oxygen condenses first upon cooling down or vaporizes last upon warming up, both situations always connected with an oxygen-enriched condensate forming shock-explosive mixtures. Also liquid or solid oxygen in combination with another combustible material, even if solid, may form highly explosive mixtures. Examples are LH₂ plus solid air having an O₂ fraction of >40%, or liquid oxygen spilled onto asphalt (Zabetakis, 1967). The ignition of premixed LH₂ with condensed air significantly enhances burning rates with a large variation. The flame will become wider and taller, and the radiated energy increases (Urano, Horiguchi, Iwasaka, Tokuhashi, & Hashiguchi, 1986).

Measurements of the ranges of flammable hydrogen–air mixtures following the vaporization of LH₂ compared with the measured height of the visible cloud resulting from the condensation of water vapor have shown that the visible cloud does not necessarily coincide with the flammable

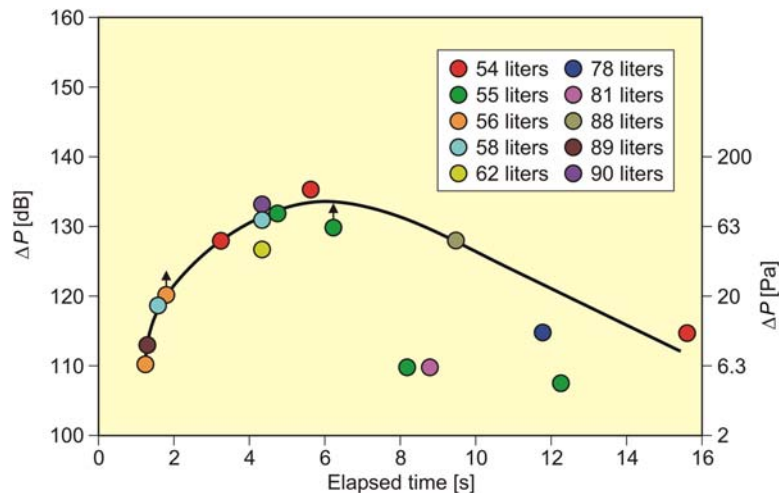


FIGURE 1.7

Overpressures measured at 49 m distance after ignition of vapor cloud above LH₂ pools versus time of ignition after spillage (Zabetakis & Burgess, 1960).

hydrogen – air mixture) meaning there might be flammable portions outside the visible cloud and also nonflammable portions inside the visible cloud (Edeskuty & Stewart, 1996).

A series of LH₂ spill and ignition tests in the open atmosphere, with quantities in the range of 54–90 ℓ, were conducted with the purpose of measuring the overpressures. The results at a distance of ~ 50 m shown in Fig. 1.7 as a function of the ignition time after spillage demonstrate that the blast pressures produced were relatively small and were depending on the time delay for ignition (Zabetakis & Burgess, 1960). They were found to increase with delay time, until after more than 5–6 s of delay, they were decreasing again as soon as the H₂ concentration in the rising and diffusing vapor clouds became smaller.

1.3 Physiological hazards associated with hydrogen

Hydrogen is classified as nontoxic, nonacid, and noncarcinogenic, being a simple asphyxiant with no threshold limit value or LD50 (lethal dose 50%) value established (NASA, 1997).

Vaporization of released LH₂ affects the composition of the atmosphere, particularly in (partially) confined areas, carrying the risk of asphyxiation. The enormous liquid/ambient expansion ratio combined with condensation of O₂ from the ambient air and burning of flammable H₂ – air mixtures leads to a significant dilution of the local atmosphere. An oxygen volume fraction of less than 19.5% is considered by NASA to be dangerous to humans; less than 8% will be lethal within minutes (Table 1.2). Alarm levels are generally set at 19% of oxygen.

Direct contact with LH₂ or with surfaces at very low temperature causes cryogenic “burns” similar to thermal burns. Living tissue will freeze except for very brief contact periods where

Table 1.2 Impact on humans by an atmosphere with decreasing oxygen contents (Edeskuty & Stewart, 1996).

Oxygen contents in air (%) @ 1atm total pressure	Symptoms
~ 21–19	None
~ 19–15	Reduced reaction times, no visible effects
~ 15–12	Heavy breathing, rapid heart beat, impaired attention or coordination
~ 12–10	Dizziness, faulty judgment, poor muscular coordination, rapid fatigue, lips slightly bluish
~ 10–8	Nausea, vomiting, inability to move, loss of consciousness followed by death
~ 8–6	Brain damage after 4–8 min, death within 8 min
< 6	Coma after 40 s, respiratory failure, death

Table 1.3 Physiological effects of blast overpressures (Edeskuty & Stewart, 1996).

Maximum overpressure (kPa)	Effect on personnel
7	Knock personnel down
35	Eardrum damage
100	Lung damage
240	Threshold for fatalities
345	50% fatalities
450	99% fatalities

the temperature difference between cryogen and skin is still high (film boiling regime) and heat transfer small. The freezing of skin onto a cold surface can lead to severe damage upon removal. Prolonged skin exposure to cold hydrogen may result in frostbite. A symptom is short-lived local pain. Frozen tissues are painless and appear waxy, with a pale whitish or yellowish color. Thawing of the frozen tissue can cause intense pain. Shock may also occur. Prolonged inhalation of cold vapor or gas may cause serious lung damage. Particularly eyes are sensitive to cold. A longer exposure to cold temperatures after a large spill lowers the body temperature resulting in hypothermia, organ dysfunction, and respiratory depression (Edeskuty & Stewart, 1996).

Physiological effects as anticipated upon the explosion of a flammable gas mixture are listed in Table 8–2; the effects arising from the accidental release and combustion of hydrogen are treated in Section 8.4. In the deflagration of a free hydrogen – air gas cloud, the maximum overpressure is in the order of 10 kPa. An overpressure of 7 kPa is still deemed not dangerous; at 7 kPa, people would fall down to the ground; at 35 kPa, damage of ear drums is expected; 240 kPa is considered a threshold value above which fatalities must be taken into account (Table 1.3).

1.4 Influence of hydrogen on materials

The stress that a structural material is able to withstand is determined by its ductility (Fig. 1.8). A material is elastic if, after being elongated under stress, it returns to its original shape and volume as soon as the stress is removed. At a certain strain, it departs from linearity, that is the material will retain a permanent elongation. The applied stress is the so-called “yield stress.” A further increase of the strain eventually reaches the “ultimate or tensile stress” beyond which the stress steadily decreases until rupture. In contrast, a brittle material does not exhibit a permanent elongation phase and rather breaks abruptly without any warning as soon as it is exposed to its tensile stress (Edeskuty & Stewart, 1996).

Hydrogen has long been recognized to have a deleterious effect on some metals by changing their physical properties. It is basically due to the presence of hydrogen atoms dissolved in the metal grid and accumulating in disturbed lattice regions, with the result of potentially premature cracking. Research on this embrittlement effect has led to the definition of three different categories (Dutton, 1984; Edeskuty & Stewart, 1996):

1. **Hydrogen reaction embrittlement.** Hydrogen reaction embrittlement is a phenomenon in which the hydrogen chemically reacts with a constituent of the metal to form a new microstructural element or phase such as a hydride or gas bubbles (“blistering”), for example, methane gas if combined with carbon, or steam if combined with oxygen. These reactions usually occur at higher temperatures. They result in the formation of blisters or expansions from which cracks may start to weaken the metal.
2. **Internal hydrogen embrittlement.** Internal hydrogen embrittlement means that hydrogen is introduced into the metal during its processing, for example, chemical reactions with water to form metal oxide and liberate hydrogen. It is a phenomenon that may lead to the structural

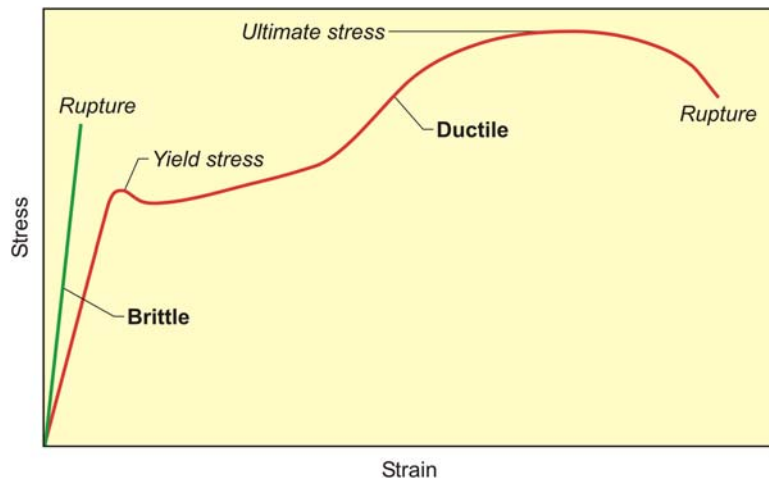


FIGURE 1.8

Ductile and brittle behavior of materials (Edeskuty & Stewart, 1996).

failure of material that never has been exposed to hydrogen before. Internal cracks are initiated showing a discontinuous growth. Not more than 0.1–10 ppm hydrogen on the average are involved. The effect is observed in the temperature range between -100°C and $+100^{\circ}\text{C}$ and is most severe near room temperature.

3. *Environmental hydrogen embrittlement.* Environmental hydrogen embrittlement means that the material was subjected to a hydrogen atmosphere, for example, storage tanks. Absorbed and/or adsorbed hydrogen modifies the mechanical response of the material without necessarily forming a second phase. The effect occurs when the amount of hydrogen that is present, is more than the amount that is dissolved in the metal. The effect strongly depends on the stress imposed on the metal. It also maximizes at around room temperature.

To which extent metal is degraded by hydrogen is influenced by strain (tension, frequency, stretch velocity), specimen geometry (notches, bad welds, geometric inhomogeneities, surface defects), the medium (pressure variation, temperature, time of exposure, H_2 purity), and material (chemical composition, microstructure, fabrication, heat treatment, welding quality) (Kusssmaul & Deimel, 1995).

Several mechanisms have been proposed, which might explain at least partially the degradation of metal by hydrogen embrittlement and might act simultaneously:

1. The formation of hydrides can lead to new hydrogen-related phases, which may be brittle and also may have a lower density than the pure metal leading to internal stress.
2. The hydrogen distribution in metal under stress is highly nonuniform, which can lead to locally increased hydrogen-enhanced plasticity causing local microscopic deformation and eventually failure.
3. The lattice decohesion effect is presumed to cause embrittlement by a decrease in the atomic bonding strength in the presence of hydrogen. A fracture occurs when the stress exceeds the cohesive stress.
4. Molecular hydrogen precipitation forming high pressures and compound formation are other mechanisms identified.

The above ideas help understand the observations that whether or not a metal is susceptible to embrittlement by hydrogen or a hydrogen compound, depends on the metal and also its metallurgical history, which affect the migration behavior of hydrogen within the metal. The embrittlement is strongly connected with locally high hydrogen concentrations that can be caused by stress-enhanced diffusion rates to lattice defects and reaction sites to initiate cracks (Kesten, 1985). Cracks grow when hydrogen concentrations reach a critical level; crack growth stops when the crack has grown through the H_2 -enriched region. The ability to explain the observations, however, was not accompanied by the development of nondestructive methods to detect early stages of degradation and the ability to predict with certainty the susceptibility of new alloys to hydrogen (Louthan & Morgan, 1996).

For the polymeric liner material used in Type IV storage pressure vessels, H_2 permeation has to be considered, an issue inherent to a polymer matrix in contact with a gas. Different from the other (metallic) types of H_2 containers, for which permeation is negligible, measurement of the permeation rate is here generally required to prove its value below a specified upper limit. For H_2 storage tanks in vehicles it is currently defined in the EU regulation as $1 \text{ Nm}^3/\text{h}/\ell$ -water capacity, while a

corresponding HYSAFE proposal is 61 Nm²/h/ℓ-water capacity (Adams, Bengaouer, Cariteau, Molkov, & Venetsanos, 2009; Barthelemy, 2007).

With today's existing sets of regulations and guidelines, there is a well-established list of materials acceptable to contain hydrogen under precisely defined operating conditions. This holds for both gaseous and LH₂.

Cryogenic temperatures can affect structural materials. With decreasing temperature, ultimate stress and yield stress increase for most metals, generally connected with a corresponding drop in fracture toughness, which is a measure of the material's ability to resist crack propagation. The lower the toughness, the smaller is the tolerable crack length. A material can change from ductile to brittle behavior as soon as the temperature falls below its "nil-ductility temperature," which is sometimes considerably higher than the temperature of the cryogen. It is a particular problem in cryogenic equipment exposed to periodic temperature changes. Several accidents with failure of a cryogen storage tank have been traced to originate from cold embrittlement, for example the severe accident with the rupture of a 4250 m³ LNG tank in Cleveland, United States, in 1944 (Zabetakis, 1967).

Low temperatures can also affect materials by thermal contraction causing large thermal stresses, if the system cannot accommodate the differential thermal contraction of the materials. The thermal expansion coefficient is a function of temperature. For many materials cooled down from room to cryogenic temperature, more than 90% of the total contraction experienced will have already taken place at 77K. Ballpark figures of thermal contraction are 0.3% in iron-based alloys, 0.4% in aluminum, or over 1% in many plastics (Edeskuty & Stewart, 1996). Cryogenic vessels or piping systems must account for this contraction to avoid large thermal stresses.

The components of a cryogenic system usually undergo a thermal gradient, some only during cool-down or warm-up phases, others even at steady-state operation. Strong gradients, particularly if nonlinear, result in stresses that, under certain circumstances, may lead to rupture. Thermal gradients are of particular significance in systems with stratified two-phase flows of cryogenics.

An additional requirement for materials containing LH₂ is a sufficient fracture viscosity. Adequate materials are aluminum alloys, austenitic steels, and steels with a high Ni fraction. The need for weight reduction in mobile applications has led to the development of composite materials, a matrix of plastic or metal in which glass or carbon fibers are embedded. Qualities of the composites are a higher strength compared with steel, a lower heat conductivity, and almost no thermal expansion in fiber direction. With decreasing temperature, however, composites are prone to enhanced embrittlement and lower strength as well as a higher thermal expansion perpendicular to the fiber direction.

Materials for hydrogen containing components have to meet the requirements of high strength and high ductility to enable a high system pressure and have to be appropriate for welding. Respective materials are ultrafine grain steels and thermomechanically treated steels. Usually applied dense coatings on the surface of the ferritic steels containing gaseous hydrogen prevent the adsorption of hydrogen unless they are damaged by plastic deformation. Other sensitive areas are welding seams and surface quality. Also a high degree of purity of the hydrogen favors embrittlement, which could be influenced by adding gaseous promoters (e.g., H₂S) or inhibitors (e.g., O₂ and CO) or inert gases (N₂ and noble gases) depending on the specific operating conditions (Kussmaul & Deimel, 1995).

Materials with sufficiently high strength and high ductility, working successfully at low temperatures, include aluminum and most of its alloys, copper and its alloys, nickel and some of its

alloys, as well as austenitic stainless steels and should be used in cryogen containing systems. EN (European Standards) and ISO (International Organization for Standardization) standards cover the requirements for selecting appropriate materials resisting to cold embrittlement.

For many materials, specific heat exhibits strong temperature dependence below 200K showing that at cryogenic temperatures, much less heat is required to raise the temperature of a body than at ambient temperatures. As an example, the heat capacity of aluminum is reduced from about 950 J/(kg•K) at ambient temperature to less than 10 J/(kg•K) at 20K.

1.5 Environmental impact of hydrogen in the atmosphere

There are no significant environmental hazards associated with the accidental discharge of hydrogen due to its nontoxic character.

Presently, the major sources for hydrogen emissions into the atmosphere are the photochemical oxidation of atmospheric methane and volatile organic compounds with estimated 50% and the combustion of fossil fuels and biomass, for example, forest fires or traffic emissions (~ 40%). Stratospheric hydrogen is controlled by the production from methane oxidation and the destruction due to the OH reaction. In addition, hydrogen is exchanged with the troposphere. The main removal mechanism from the atmosphere is by dry deposition and uptake in soils (~ 75%) or through oxidation by hydroxyl free radicals OH (~ 25%) (Schultz, Diehl, Brasseur, & Zittel, 2003). Hydrogen concentrations in the troposphere and stratosphere are constant at a level of ~ 0.5 ppm, which winds up to a total H₂ mass of 175 million t in the atmosphere.

Hydrogen is not a direct greenhouse gas, as it does not absorb electromagnetic radiation within the infrared spectrum. An adverse effect is identified with hydrogen acting as a significant sink for OH, whose concentration further decreases with increased atmospheric concentrations of hydrogen. This in turn could increase the atmospheric lifetime of greenhouse gases and other pollutants in the stratosphere and may reduce stratospheric ozone concentrations. On the other hand, hydrogen constitutes only about 5% of the average OH sink, and OH reacts much stronger to changes in reactive NO_x emissions, which are believed to significantly perturb the amount of ozone in the upper troposphere and lower stratosphere region. The anticipated changes in OH levels due to changes in the atmospheric hydrogen concentration are marginal when hydrogen is atmospherically degraded in the lower stratosphere where the persistent ice (from the water) may have a cooling effect that again influences the temperature-dependent ozone depletion mechanism (Groß, Feck, Vogel, & Riese, 2010).

In a global hydrogen infrastructure, emissions of molecular H₂ can occur along the whole hydrogen process chain. A realistic upper limit for the global average leak rate is 3% for the whole hydrogen energy chain (Schultz et al., 2003). Any diffusive, targeted, or accidental release will emit predominantly into the atmosphere. Its oxidation results in the formation of water vapor (clouds), but it is not expected to raise considerably the overall content of water vapor in the troposphere, as also the present fossil fuel economy emits significant amounts of water vapor from combustion and cooling.

There is still a large uncertainty around the atmospheric budget of molecular hydrogen. Therefore except for the expected strong reduction of CO and CO₂ emissions, no final conclusions

Table 1.4 Physical and chemical properties of hydrogen.

Parameter	Hydrogen	Unit
Atomic weight		u (unified atomic mass unit)
Standard atomic mass	1.0078225	
Protium	1.00782503207	
Deuterium	2.0141017778	
Tritium	3.0160492777	
Molecular weight	2.01594	g/mol
Stoichiometric fraction in air	29.53	vol%
Boiling point (BP)	20.28	K
Melting point (MP)	14.01	K
Triple point (TP)		
Temperature	13.8	K
Pressure	7.2	kPa
Critical point:		
Temperature	33.145	K
Pressure	1.2964	Mpa
Density	31.263	kg/m ³
Electronegativity	2.20	Pauling scale
Density		
Gas @ STP ⁽¹⁾	0.08988	kg/m ³
Gas @ BP	1.338	
(Para) liquid @ BP	70.78	
Slush with 50 mass% solid @ TP	81.48	
Solid @ 4K	88.0	
Expansion ratio liquid/ambient	845	
Diffusion coefficient @ NTP ⁽²⁾	0.61×10^{-4}	m ² /s
Diffusion velocity @ NTP	< 0.02	m/s
Buoyant velocity	1.2–9.1	m/s
Gas constant	4124.3	J/(kg•K)
Specific heat (constant p)		
Gas @ STP	14.30	kJ/(kg•K)
Gas @ BP	12.15	
Liquid @ BP	9.69	
Liquid @ TP	6.51	
Slush with 50 mass% solid at TP	11.4	
Thermal conductivity		
Gas @ NTP	0.187	W/(m•K)
Gas @ BP	0.01694	
Liquid @ BP	0.09892	
Viscosity		
Gas @ NTP	8.948	μPa s
Gas @ BP	1.079	
Gas @ TP	0.73	
Liquid @ BP	13.32	
Surface tension @ BP	1.92×10^{-3}	N/m

(Continued)

Table 1.4 Physical and chemical properties of hydrogen. *Continued*

Parameter	Hydrogen	Unit
Vapor pressure		
Normal ⁽³⁾ @ TP	7200	Pa
Para @ TP	7040	
Solid @ 10K	257	
Para/ortho ratio		
@ STP	25 / 75	
@ BP	99.8 / 0.2	
Inversion temperature @ 0.1 MPa	193	K
Heat of conversion		
From ortho to para	703	kJ/kg
From normal to para	527–670	
Heat of melting (fusion) @ MP	58.8	kJ/kg
Heat of vaporization @ BP	445.6	kJ/kg
Vaporization index ⁽⁴⁾	8.9	K•cm ³ /J
Vaporization rate of LH ₂ pool	4.2–8.3	mm/s
Heat of sublimation	379.6	kJ/kg
Speed of sound		
Gas @ NTP	1294	m/s
Gas @ BP	355	
Liquid @ BP	1093	
Inversion (Joule – Thomson) temperature	193	K
Flammability limits in air	4.0–75.0 ⁽⁵⁾	vol%
Detonability limits in air	13–70 ⁽⁶⁾	vol%
Minimum ignition energy	1.9×10^{-5}	J
For detonation	~ 10,000	
Autoignition temperature in air	793–1023 (858)	K
Hot air jet ignition temperature	943	K
Gross heat of combustion (Higher Heating Value, HHV) @ 15°C	286.1	kJ/mol
	12.75	MJ/m ³
	141.86	MJ/kg
Net heat of combustion (Lower Heating Value, LHV) @ 15°C	241.7	kJ/mol
	10.8	MJ/m ³
	119.93	MJ/kg
Wobbe Index ⁽⁷⁾		
Upper	48.34	MJ/m ³
Lower	40.90	
Flame temperature (theoretical)	2318	K
Burning rate of LH ₂ pool	0.5–1.1	mm/s
Laminar burning velocity in air	2.65–3.25	m/s
Visible laminar flame front velocity	18.6	m/s
Deflagration pressure ratio	8.15	
Quenching distance @ NTP	0.64	mm
Maximum experimental safe gap @ NTP	0.08	mm
Adiabatic flame temperature	2318	K
Emission coefficient	< 0.1	
Detonation velocity	1480–2150	m/s

Table 1.4 Physical and chemical properties of hydrogen. *Continued*

Parameter	Hydrogen	Unit
Chapman – Jouguet velocity	1968	m/s
Chapman – Jouguet detonation pressure ratio p_C/p_0	15.6	
Energy release	2.82	MJ/kg mixture
Detonation cell size	15	mm
Critical tube diameter	0.2	m
Detonation initiation energy	0.88	g TNT ⁽⁸⁾
Detonation induction distance @ NTP	Length/diameter ~ 100	
TNT ⁽⁸⁾ equivalent (based on LHV)	28.7	kg TNT/kg-H ₂

can be made at present about the potential environmental benefits and risks of the move from a fossil fuel-based energy industry to a hydrogen-based industry (Grooß, Feck, Vogel, & Riese, 2010).

1.6 Hydrogen properties

Main characteristic properties of hydrogen are summarized in Table 1.4 taken from various sources.

1. STP (Standard temperature and pressure): 273K, 101325 Pa.
2. NTP (Normal temperature and pressure): 293K, 101325 Pa.
3. “normal” hydrogen = 75% ortho + 25% para.
4. Indicates the relative ease of a substance to vaporize.
5. LFL is valid for upward propagation of flame. For downward propagation, LFL values is 5.3vol %.
6. Valid for weak ignition. For a high-energy igniter, a range of 11.6–74.9 was quoted.
7. The upper/lower Wobbe index is defined as the ratio of the HHV/LHV and the square root of the relative density, which again is the ratio of the fuel gas density to the density of dry air under the same pressure and temperature conditions.
8. Formula: C₇H₅N₃O₆; name: trinitrotoluene or 2-methyl-1,3,5-trinitrobenzene 1 kg TNT corresponds to an energy released at detonation of 4.184 MJ.

References

- Adams, P., Bengaouer, A., Cariteau, B., Molkov, V., Venetsanos, A.G. (2009). Allowable hydrogen permeation rate from road vehicle compressed gaseous storage systems in garages—Part 1: Introduction, scenarios, and estimation of allowable permeation rate. In *Proceedings of the third international conference on hydrogen safety ICHS-3*, Ajaccio.

- Arthur, D. Little (1960). Final report on an investigation of hazards associated with the storage and handling of liquid hydrogen, *Report C-61092*. Arthur D. Little Inc., Cambridge, MA.
- Astbury, G.R. & Hawksworth, S.J. (2005). Spontaneous ignition of hydrogen leaks: A review of postulated mechanisms. In *Proceedings of the first international conference on hydrogen safety ICHS-I*, Pisa.
- Barthelemy, H. (2007). Safety of hydrogen cylinders and pressure vessels, In *Second summer school on hydrogen safety*, Belfast.
- Bjerketvedt, D., Bakke, J. R., & Wingerden, K. Van (1997). Gas explosion handbook. *Journal of Hazardous Materials*, 52, 1–150.
- Breitung, W. (2005). Sicherheitsaspekte zukünftiger Wasserstoff-Technologien. *FZK Nachrichten*, 37(3), 140–145.
- Brentari, E.G., Giarratano, P.J., Smith, R.V. (1965). Boiling heat transfer for oxygen, nitrogen, hydrogen and helium, US-DOC, National Bureau of Standards, Technical note no. 317, Washington, DC.
- Choudhuri, A.R. (2005). *Investigation on the flame extinction limit of fuel blends*. Combustion and Propulsion Research Laboratory at the University of Texas at El Paso, TX.
- Dienhart, B. (1995). *Ausbreitung und Verdampfung von flüssigem Wasserstoff auf Wasser und festem Untergrund*, (Dissertation), Report Jül-3155, Research Center Jülich.
- Dutton, R. (1984). Materials degradation problems in hydrogen energy systems. *International Journal of Hydrogen Energy*, 9, 147–155.
- Edeskuty, F. J., & Stewart, W. F. (1996). *Safety in the handling of cryogenic fluids. The International Cryogenics Monograph Series*. New York: Plenum Press.
- Eichert, H., Deimel, P., Hoffmann, M., & Pültz, G. (1992). Gefährdungspotential bei einem verstärkten Wasserstoffeinsatz. Stuttgart: Deutsches Zentrum für Luft- und Raumfahrt.
- Groß, J.-U., Feck, T., Vogel, B., Riese, M. (2010). Impact of H₂ emissions of a global hydrogen economy on the stratosphere. In *Proceedings of the eighteenth world hydrogen energy conference (WHEC'18)*, Essen.
- Kesten, M. (1985). Physikalischer Wasserstoffangriff. In D. Uebing, & D. Schlegel (Eds.), *Einflußgrößen der Zeitsicherheit bei technischen Anlagen*. Rheinland: F. Vieweg & Sohn, Verlag TÜV.
- Kussmaul, K., & Deimel, P. (1995). *Materialverhalten in H₂-Hochdrucksystemen*. VDI Berichte No. 1201 (pp. 87–101). Düsseldorf: Düsseldorf: VDI-Verlag.
- Kuznetsov, M., Czerniak, M., Grune, J., Jordan, T. (2013). Effect of temperature on laminar flame velocity for hydrogen-air mixtures at reduced pressures. *Proc. 5th Int Conf Hydrogen Safety (ICHS-5)*, Brussels (2013), paper 231.
- Lind, C. D. (1975). What causes unconfined vapor cloud explosions. *Loss Prevention*, 9, 101–105.
- Louthan, M. R., & Morgan, M. J. (1996). Some technology gaps in the detection and prediction of hydrogen-induced degradation of metals and alloys. *Journal of Nondestructive Evaluation*, 15, 113–120.
- Maxwell, J. C. (1965). On the dynamical theory of gases. The Scientific Papers of J. C. Maxwell (pp. 26–78).
- McMinis, J., Clay, R. C. III, Lee, D., & Morales, M. A. (2015). Molecular to atomic phase transition in hydrogen under high pressure. *Physical Review Letters*, 114(10), 1–6, 105305.
- NASA (1997). *Report of the Presidential Commission on the Space Shuttle Challenger Accident (1986)*. <<http://science.ksc.nasa.gov/shuttle/missions/51-l/docs/rogers-commission/table-of-contents.html>>.
- Ohira, K. (2015). Slush hydrogen production, storage, and transportation. In Basile, A. (Ed.), *Compendium of hydrogen energy. Volume 2: Hydrogen storage, transportation and infrastructure*. Woodhead Publishing Series in Energy, pp. 53–90.
- Peschka, W. (1992). *Liquid hydrogen: Fuel of the future*. Wien New York: Springer-Verlag.
- Quack, H. (2002) Die Schlüsselrolle der Kryotechnik in der Wasserstoff-Energiewirtschaft. *Klima Luft- und Kältetechnik* 3 (2002) 157–161. Available from <https://www.docplayer.org/30408407-Die-schlüsselrolle-der-kryotechnik-in-der-wasserstoff-energiewirtschaft.html>

- Schroeder, V. & Holtappels, K. (2005). Explosion characteristics of hydrogen-air and hydrogen-oxygen mixtures at elevated temperatures. In *Proceedings of the first International conference on hydrogen safety ICHS-1*, Pisa.
- Schultz, M. G., Diehl, T., Brasseur, G. P., & Zittel, W. (2003). Air pollution and climate-forcing impacts of a global hydrogen economy. *Science (New York)*, 302, 624–627.
- Soave, G. (1972). Equilibrium constants from a modified Redlich – Kwong equation of state. *Chemical Engineering Science*, 27(6), p1197–p1203.
- Sonntag, M., Squires, K.B., Park, Y. (1988). Slush hydrogen, gelled hydrogen, and gelled-slush hydrogen. NASP Contractor Report 1027, NASA Langley Research Center.
- Stang, J.H., Neksa, P., Brendeng, E. (2006). On the design of an efficient hydrogen liquefaction plant. In *Proceedings of the sixteenth world hydrogen energy conference WHEC'16*, Lyon, paper 324.
- Urano, Y., Horiguchi, S., Iwasaka, M., Tokuhashi, K., Hashiguchi, Y. (1986). *Hazards of burning liquefied hydrogen—Part 1: Flame of stable burning—Part 2: Flame of abnormal burning*. National Chemical Laboratory for Industry, 81, pp. 143 – 157 (In Japanese).
- Vander Arend, P.C. et al. (1964). The liquefaction of hydrogen' In Scott, R.B. et al. (Eds.) *Technology and uses of liquid hydrogen*. Oxford: Pergamon Press Ltd.
- Verfondern, K., & Dienhart, B. (2007). Pool spreading and vaporization of liquid hydrogen. *International Journal of Hydrogen Energy*, 32, 2106–2117.
- Weir, S. T., Mitchell, A. C., & Nellis, W. J. (1996). Metallization of fluid molecular hydrogen at 140 GPa (1.4 Mbar). *Physical Review Letters*, 76(11), 1860–1863.
- Zabetakis, M. G., & Burgess, D. S. (1960). Research on the hazards associated with the production and handling of liquid hydrogen. Report No. In *WADD TR. (60-141)*. OH: Wright Air Development Division.
- Zabetakis, M. G. (1967). *Safety with cryogenic fluids*. New York: Plenum Press.

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Hydrogen technologies

2

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Chapter Outline

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2.1 History and potential future of hydrogen technologies

Hydrogen technologies have matured since the detection of hydrogen and its initial use as a lifting gas about 250 years ago. In 1804 Swiss inventor Francois Isaac de Rivaz patented the first hydrogen car. The four-wheel prototype (see Fig. 2.1) was using an internal combustion engine (ICE) running on hydrogen and oxygen gas. The hydrogen gas was contained in a balloon and the ignition was provided via an electrical Volta starter. The combustion lifted the piston in the cylinder and by falling back it was driving a wheel via a ratchet. This might be considered a prototypical system consisting of a hydrogen storage, pipes, armatures, and a convertor.

About 40 years after the first electrolyser developed by the military scientist Ritter, the scientists Schönbein and Sir Grove detected and demonstrated the reverse technology, the fuel cell almost simultaneously in 1839 (see Fig. 2.2).

However, all this early hydrogen technology was not developed further. Instead alternating current electricity and ICEs running on gasoline became the dominating technologies for energy transfer and mobility. Only in the 1960s hydrogen technologies, in particular fuel cells, experienced a boost with the race for the space and for military applications.

Today hydrogen is mainly used at large scale in different chemical industries. Synthesis of ammonia and methanol and increasingly desulphurization and refining of crude oil in the petrochemical industry induce the major demand for hydrogen as a chemical commodity. For those applications hydrogen is typically produced onsite by reforming fossil hydrocarbons. For economic reasons, about 60% of the world hydrogen production is based on steam reforming (SR) of natural gas [steam methane reforming (SMR)] and coal gasification (CG) today with a considerable CO₂ footprint—this hydrogen is sometimes called "gray" hydrogen (IRENA, 2020). Besides there is a broad dispersed use of hydrogen in different processes, for instance in the food, electronics, and metallurgical industry. So far, the only significant use as an energy carrier, as a fuel respectively, has been realized in the intensified space programs, where rocket propulsion benefits from the high gravimetric energy density of hydrogen.

Hydrogen is, however, emerging as a major element for a future sustainable and decarbonized energy economy where it provides unique opportunities for large scale energy storage and flexible coupling of the different energy sectors. The main driving forces to introduce hydrogen as an energy carrier are the limited unequally distributed fossil fuel resources and more and more dominantly environmental concerns on urban pollution and climate change.

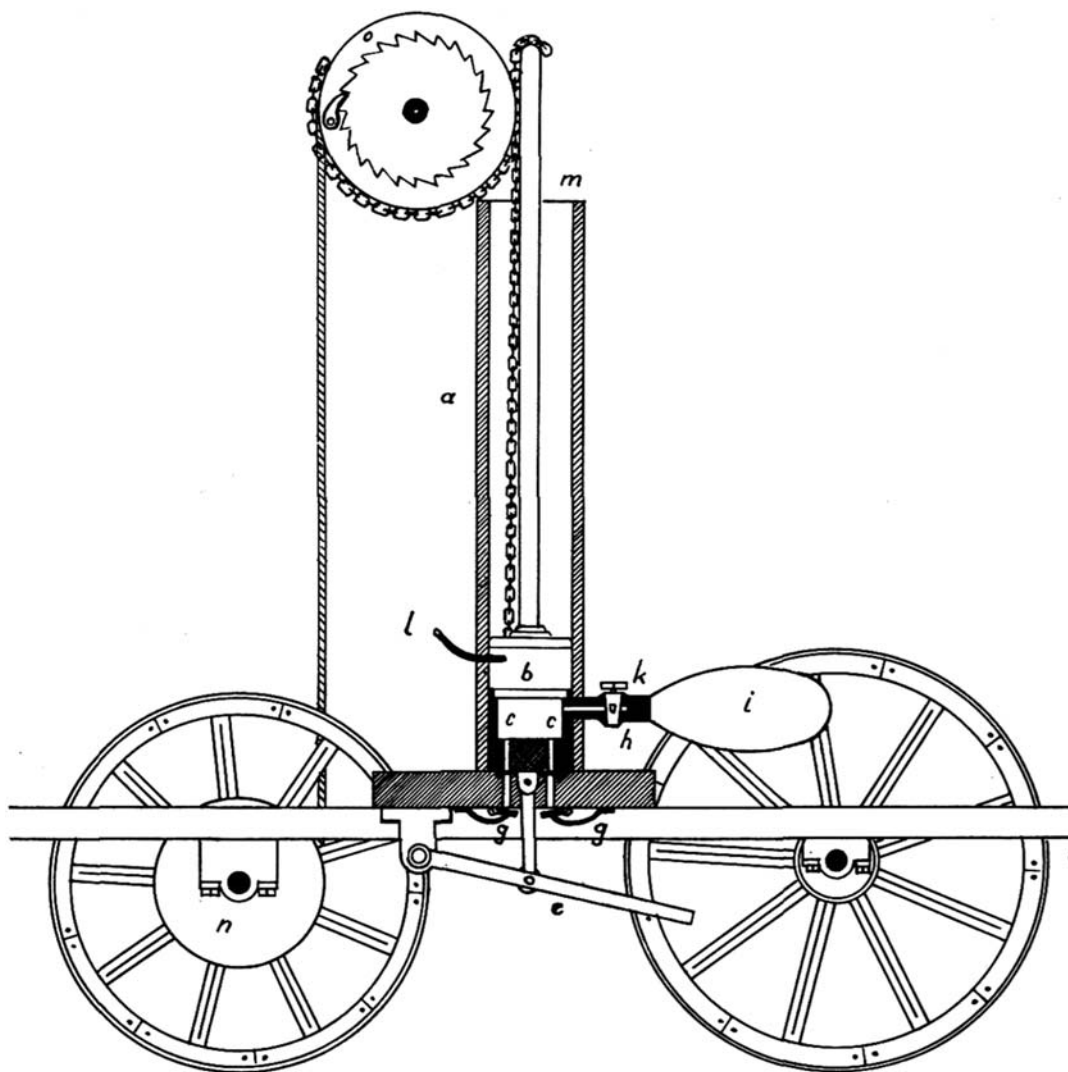
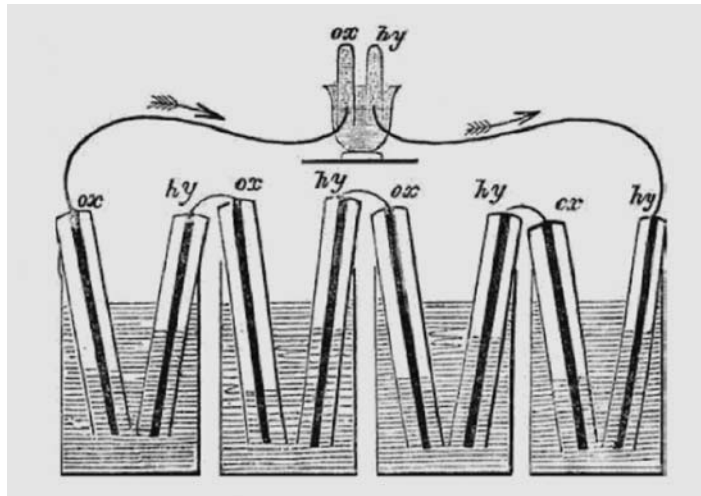


FIGURE 2.1

First hydrogen driven car patented and built by de Rivaz ([Eckermann, 2001](#)).

Hydrogen as an energy carrier is thought to take a key role in sector coupling, to combine and to apply different renewable and sustainable energy sources. In particular, the flexible coupling of electricity and the gas grid via hydrogen will allow decarbonization of applications, whose carbon footprint otherwise would be difficult to be reduced. The worldwide transition toward renewable energies, the implied needs for energy storage, and sector coupling will increase the demand for green hydrogen. However, as long as there is not sufficient wind and solar-based electricity and

**FIGURE 2.2**

First fuel cell set-up by Sir Grove in 1839.

electrolyzer capacity for producing green hydrogen, intermediate alternative pathways for CO₂ lean hydrogen production might be established, which are for instance based on fossil fuels with C or CO₂ separation and use, or on hydrogen production with the help of nuclear energy (those inventories are frequently called “turquoise” or “blue” hydrogen).

Although currently the carbon reduction policies put a strong pressure on the implementation of hydrogen as an energy carrier it is still far away from having a significant market share. Whereas in general the relatively high maturity of hydrogen production, storage, and conversion technologies may be considered supportive, it is rather the lacking scales and the closely related relatively high specific costs, which are still blocking a broader rolling out of these associated technologies. In a few niches relevant scales could be reached. The most prominent application where hydrogen was implemented broadly is material handling. Here safe indoor refueling with short fueling times made the technology more economically attractive than battery-based solutions.

The scaling up of the use of hydrogen as an energy carrier will require innovation, new developments, and different approaches compared to the established industry uses. Especially for the quite innovative applications in mobility and in other distributed applications where untrained consumers will get in contact with hydrogen technologies, the operational domain will be likely expanded into extreme conditions. The high pressure levels required for automotive applications or the cryogenic temperatures associated with liquid hydrogen impose new challenges in the proper design for achieving acceptable risk levels. The change of the energy system toward a system with a more distributed production and conversion topology implied with more renewable energy sources, brings the technology closer to the consumers, what requires more robust, fail-safe designs. Achieving similar or even better safety performance compared to established energy carriers and fuels, shall not impose additional costs.

In general, hydrogen-based mobility, in particular high mileage cars, buses, and heavy-duty transport, are seen as important and attractive applications for reducing emissions in city centers

and to comply with CO₂ reduction requirements. At least in Japan, Germany, and California refueling stations for hydrogen-fueled vehicles are slowly becoming part of the general supply infrastructure, like conventional gas stations. Hydrogen and its associated technologies therefore have to reach safety acceptance by the public and approval by the relevant authorities. National authorities, industry, and research institutes must work closely together to facilitate the required supply infrastructure and the development of a reliable framework of standards and regulations allowing safe introduction of commercial hydrogen vehicles and refueling stations. An improved understanding and a broader knowledge related to safety aspects of applied hydrogen technologies are providing the vital basis for these developments and will ensure a safe transition and the high investments needed, while keeping the safety-specific costs below acceptable limits. The same aspects are valid for other applications, for example, stationary use of hydrogen fuel cells.

In the long term, the vision is the transition to a “hydrogen society” where hydrogen, derived without pollution from renewable energy sources, will become a clean and widely used energy carrier, managed closely coupled to renewable electricity, with the leading role in applications requiring larger amounts of energy at low weight, or where electricity might not be used directly. Hydrogen production paths will start from solar, wind, water (tidal energy, currents), biological, and other renewable energy sources, in theory leading to nearly inexhaustible and economic supplies of hydrogen. Coupling renewable energy resources with hydrogen storage will reduce the impact of low/variable capacity factors and enable energy to be supplied when and where needed. Hydrogen can then be utilized in combined heat/power generation, in industry, and in every form of land, water, or airborne transport. It is expected that fuel cells will be the preferred choice for converting hydrogen to electricity, as this is the cleanest and most efficient means to put this energy vector to use in the respective application. However, for heat application and for the drivetrains of special machines, direct combustion of hydrogen and robust ICEs running on hydrogen will have their share. Generally, the safe supply infrastructure, consisting of efficient and cost-effective hydrogen storage, transport and distribution, is key to the provision of renewable power on demand in all the different energy sectors.

Finally, all these new means of using hydrogen raises the important question of how to ensure the safe introduction of this new energy carrier for use by the general public. The next chapters will provide the necessary basis for answering this question and highlight the most relevant safety issues specific for the respective technology.

2.2 Hydrogen production processes

2.2.1 Introduction

Almost all hydrogen on earth is found chemically bound in two material classes, mainly in combination with oxygen as water or in combination with carbon as organic substances. For water as well as for organic substances it takes energy to separate hydrogen. This energy may be largely recovered if the reverse processes are initiated. This is why hydrogen is not an energy source but a chemical energy storage or carrier.

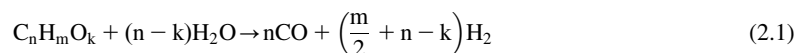
Currently hydrogen is primarily produced from fossil resources, in the first place by reforming of natural gas, currently about 43% of the world's annual production of about 115 Mt (IEA, 2019),

but also with processes such as partial oxidation (POX) of oil or the gasification of coal, accounting for 16%, see Fig. 2.3. Biomass gasification, which is still in the demonstration phase, is used on a minor scale to produce a hydrogen-rich fuel gas. The alternative carbon free pathways for producing green hydrogen will mainly rely on splitting water, either by electricity in water electrolysis, heat, or chemical processes. The use of electrolysis dedicated to hydrogen production has been limited to approximately 1% world wide mainly because of the typically high costs of electricity. However, most of the by-product hydrogen, accounting for about 41% of the world's annual hydrogen production, is actually produced in chlorine-alkaline electrolysis.

2.2.2 Hydrocarbon splitting processes

2.2.2.1 Steam reforming

Endothermal SR of light hydrocarbons, in particular the SMR, is technically mature and commercially the widest adopted technology for hydrogen production on industrial scale. Reforming technology is mainly used in the petrochemical and fertilizer industries for the production of so-called captive hydrogen, what means on-purpose and onsite produced. The process, as shown in Fig. 2.4, is characterized by a relative high yield, which is rooted in the fact that both input materials, the organic feed and the water itself, are split and contribute to the hydrogen production. The hydrocarbons are oxidized and the steam is reduced in this redox reaction. The general SR reaction is provided in Eq. (2.1).



The SR reaction of methane, the main constituent of natural gas, is given in Eq. (2.2).

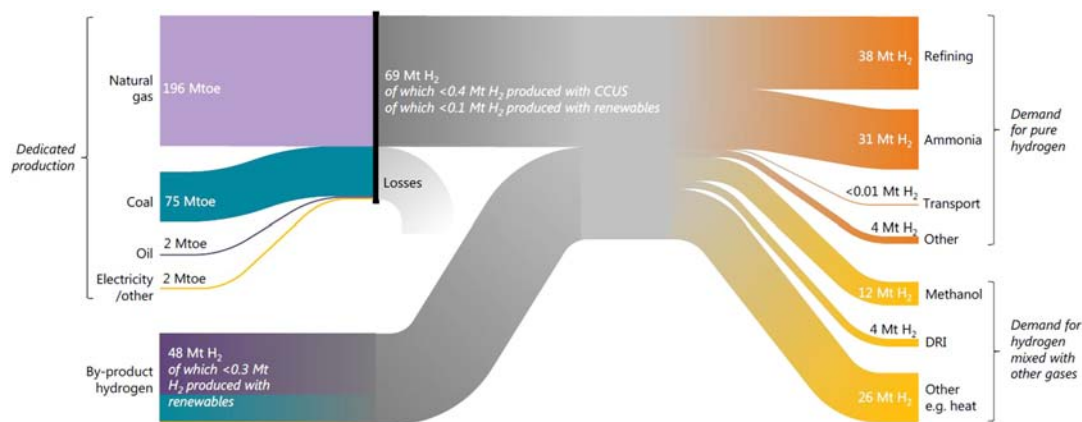
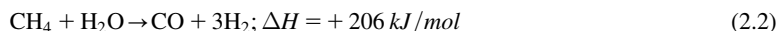


FIGURE 2.3

Today's hydrogen value chains (IEA, 2019).

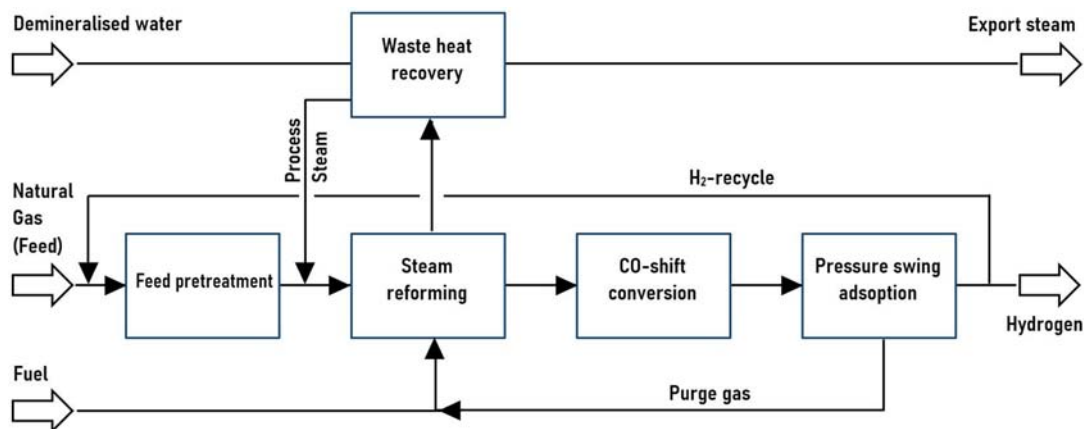


FIGURE 2.4

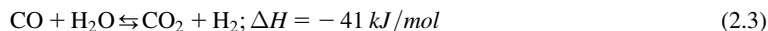
Block scheme of steam methane reforming.

High reforming temperatures in the order of 700°C–950°C, low pressures and high steam to methane ratio (S/C) favor a high methane conversion. Optimum pressure range is 2.5–3 MPa resulting in a hydrogen yield of 86%–90% (Uhde, 2003). If excess steam is injected the equilibrium is shifted toward the products and the undesired production of carbon, which might deposit on the catalyst, is prevented.

Typically vertical tubes are used as open reactors. A critical factor in the reactor design is keeping the tube wall temperature uniform and hot enough to promote the reforming reaction. Two types of heater designs have been employed for this purpose (Elshout, 2010): the side-firing reforming furnace and the roof-fired heater (see Fig. 2.5).

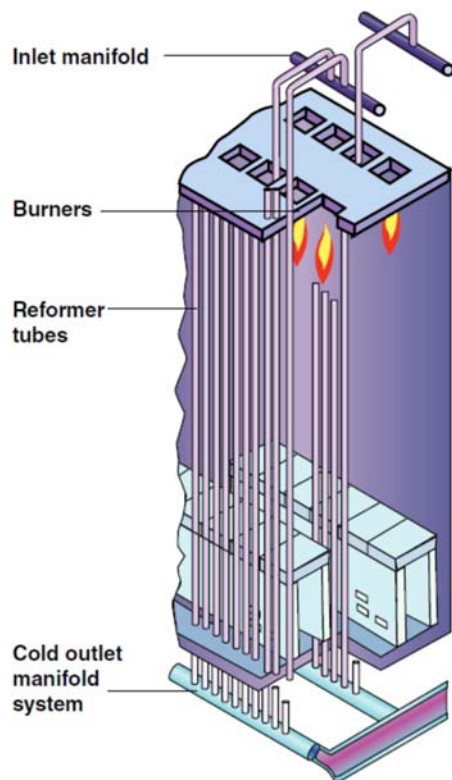
Presently, large steam reformer units with up to about 1000 reactor tubes have a production capacity of around 150,000 Nm³/h. Future reformer plants are designed with about double the capacity. Those plants typically use more than one catalyst at different temperatures to optimize the efficiency and the H₂ output. Smaller SMR units for local H₂ production have a capacity of around 150 Nm³/h. However, the first experience with small SMR for H₂ production at refueling stations, in the CUTE project for instance, was indicating that the technology is considerably less mature at these scales (Giron, 2007).

The whole SMR process (see Fig. 2.4), as established in the refining industries (Ruthardt, 2008) for instance, requires initial desulfurization of the feed and demineralization of the feed water, a high and low temperature CO shift conversion [see (Eq. 2.3)], and a final purification by pressure swing adsorption (PSA).



Industrial SMR processes, as the one shown in Fig. 2.6, may be designed to deliver CO, synthesis gas and heat, process steam, respectively, besides pure hydrogen.

Overall, the different process steps need considerable amount of energy. The total balance for such a plant is that 1 Nm³ of methane allows the production of 2.5 Nm³ of hydrogen, which corresponds to an overall efficiency of the process of around 65%–75%. It is rather difficult to get much

**FIGURE 2.5**

Primary reformer tubes with roof-fired heaters.

higher efficiencies in practice. One way to reduce the CO₂ footprint of this hydrogen production pathway is to supply the high temperature heat either by nuclear power or by concentrated solar.

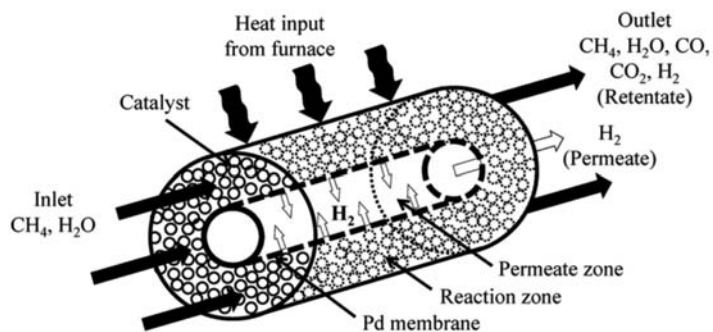
An advanced reforming technique is the membrane reforming (Shirasaki et al., 2009). This process uses microporous ceramic membranes made of Pa-based alloy and a Ni-based catalyst, which can perform SR reaction, shift reaction, and H₂ separation simultaneously, that is, without shift converter and PSA stages. The simultaneous processes allow to lower the reaction temperature down to around 550°C posing less stringent requirements to the materials. Fig. 2.7 shows a typical geometrical arrangement of the membrane as a tube-like set-up with the heated reactants in the outer volume and the products generated on the inside.

Such systems are compact and may provide higher efficiencies. Technical feasibility of the membrane reforming system was demonstrated by the Tokyo Gas Co., Japan, with test runs up to 1500 hours achieving a hydrogen production rate of 15 Nm³/h and a 76% conversion of the natural gas (Hori, 2004). The recent development of protonic membrane reactors promise even higher efficiencies and allow for simultaneous pressurization of the hydrogen by applying an electric voltage over the membrane (Harald Malerød-Fjeld, 2017).

**FIGURE 2.6**

Industrial steam methane reformer.

Courtesy of Linde.

**FIGURE 2.7**

Principle of membrane reformer (Shirasaki et al., 2009).

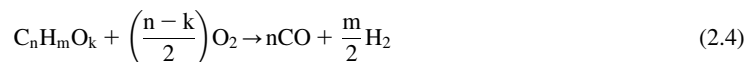
Another advanced reforming technique operates at supercritical water conditions (Boukis, Galla, Müller, & Dinjus, 2007). As the process is conducted at pressure considerably above the critical pressure of 22 MPa, the product, a gas mixture with a typically high hydrogen fraction (depending on the liquid feed), does not have to be pressurized. On one hand this process improves the efficiency and on the other hand introduces new hazards in combination with the quite corrosive atmosphere associated with supercritical water.

Furthermore, in high temperature fuel cells like solid oxide fuel cells (SOFC), the heat generated during the electrochemical reaction inside the fuel cell is sufficient to maintain the endothermic reforming of natural gas described above. For this internal reforming, a distinction is made in principle between two variants, direct internal reforming and indirect internal reforming, also known as integrated reforming.

Because of the low costs currently mainly natural gas is used for the above reforming processes; the processes might be also used to convert biogas, in the case of supercritical water reforming even wet biomass, into hydrogen.

2.2.2.2 Partial oxidation

POX is the substoichiometric oxidation of preferentially heavy hydrocarbons. The process is exothermic and needs careful control of oxygen supply. Eq. (2.4) gives the generic equation of reaction for POX.



The large scale implementations require an air separation unit. The high temperatures and suitable preprocessing allow to use feeds, heavy oils, or even solid organic materials, which otherwise would be difficult to process.

2.2.2.3 Autothermal reformation

The autothermal reforming (ATR) is a combination of SR and POX. The ATR process combines the endothermic reaction of the hydrocarbon feed with steam and the exothermic reaction of the hydrocarbon feed with oxygen, such that no major external heat exchange, for instance for the preheating of the feed, is required. This is depicted in Fig. 2.8.

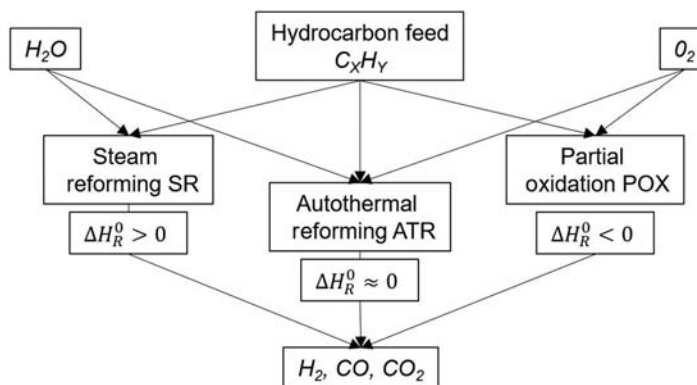
Accordingly, the educts and products of both reactions are combined. Eq. (2.5) summarizes the ATR reaction of methane.



The ATR not only combines the energetic properties and materials but also other characteristics like efficiency and hydrogen yield, which is typically 40%. ATR is the preferred reforming technology for small to medium scale, where instead of oxygen air may be used. This reduces the system complexity seriously.

2.2.2.4 Pyrolysis

Hydrocarbons may be also thermally decomposed at elevated temperatures in an inert atmosphere. The interesting fact is, that by this pyrolysis reaction generally no CO_2 is generated, but the carbon will emanate as black carbon and graphite, respectively. The latter may be considered rather as a

**FIGURE 2.8**

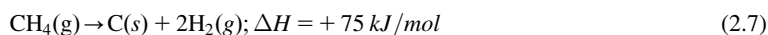
Autothermal reforming as a combination of steam reforming and partial oxidation.

product. Depending on the quality of this product, the overall economy of the process might become attractive.

The general endothermic reaction equation of pyrolysis is shown in Eq. (2.6).



For methane Eq. (2.6) translates into Eq. (2.7).

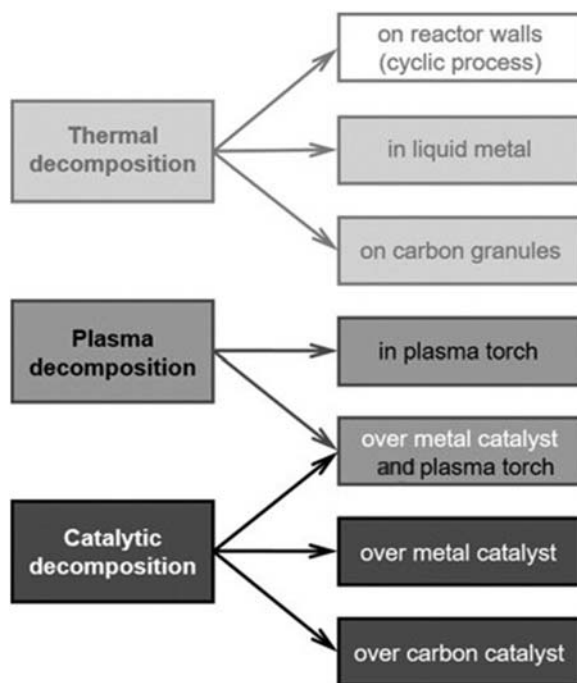


Therefore the decomposition of methane requires only 1/8 of the energy to decompose water (see following Section 2.2.3).

According to Schneider (2020), the pyrolysis processes may be divided into three main categories: thermal, plasma, and catalytic decomposition, see Fig. 2.9.

For the thermal decomposition, temperatures well above 1000°C are required. In the catalytic conversion the required temperature may be reduced. A typical example for catalytic decomposition of methane is the pyrolysis in a fluidized bed of catalytic particles, which carry the fine carbon particles after the reaction (Pohlenz, 1962). The similar Hazer process, using a fluidized bed of iron ore as catalyst, has recently gained a lot of attention (HazerGroup, 2020). Another promising approach is developed at KIT, Germany (Plevan et al., 2015). It uses a liquid metal reactor for supplying the heat and the catalytic function. Methane is bubbled from below into the liquid tin column and the product gases are extracted from above the liquid surface. The relatively light carbon particulates swim on the tin surface and may be easily removed.

In a plasma reactor the required energy is supplied by electricity. Based on research work at SINTEF Norway in the 1990s, the Arker Kvaerner process has been developed. A plasma reactor using direct current and hollow cylindrical graphite electrodes was used and demonstrated for plasma reforming of methane in a 1-MW pilot in Hofors, Sweden. In 1997 KARBOMONT, a partnership between Kvaerner and Canada's Société Générale du Financement de Québec, started an up to 20,000 ton/year carbon black and up to 50 million Nm^3 H_2 plant at Montreal East, Canada.

**FIGURE 2.9**

Categories of methane pyrolysis processes.

However, as it failed to produce commercially viable grades of carbon black, the plant operations were stopped one year later.

Generally, the processes have different technological maturity and are not yet used commercially.

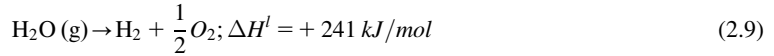
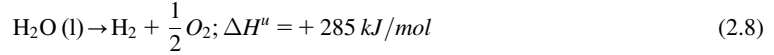
2.2.2.5 Safety challenges with hydrocarbon-based hydrogen production

All processes for using hydrocarbons for hydrogen production presented here are characterized by their relative high operational temperature and the therefore required industrial scales. Industries applying those processes have good long-term experience and a good safety track with few accidents. However, scaling down these technology and bringing them closer to the untrained public users generated problems, in particular with components facing these extreme temperature and cycling operations. This is not a principal problem, it is rather associated with lacking experience of the component suppliers with those reduced scales.

2.2.3 Water splitting processes

Water may be decomposed into its constituents oxygen and hydrogen in an endothermic reaction, which is the reverse of the fundamental combustion reaction of hydrogen. Obviously, the required

energy depends on whether the educt water enters the reaction as a liquid or as a gas, that is, as steam. The standard reaction enthalpies are then the upper heat and the lower heat of reaction, respectively, as shown in Eqs. (2.8) and (2.9). The difference is the standard heat of evaporation.



The required energy may be supplied by electricity, as in the electrolysis, or by heat, that is, a pyrolytic decomposition of water usually supported by catalysts. The dependency of the reaction enthalpy ΔH , representing the minimum total energy required, the free enthalpy ΔG , and entropic losses $T\Delta S$ on the temperature, to produce hydrogen is shown in Fig. 2.10 (Gallandat, 2017).

The jump-like reduction of the total energy required at 100°C is rooted in the latent heat of water, which has to be provided below this temperature at standard pressure and may be omitted above.

2.2.3.1 Electrolysis

In electrolysis water is split into hydrogen and oxygen using electricity. The typical set-up of an electrolyzer cell consists of two electrodes, an anode and a cathode, separated by an electrolyte and a diaphragma, separator or a membrane for separating the produced gases (see Fig. 2.11).

The actual voltage required in a real electrolysis process is composed of the ideal cell voltages, as given in Fig. 2.10, the operational losses, like ohmic losses, parasitic losses, and electrode overpotentials. Therefore a direct current voltage of typically 2 V must be applied to the electrodes of

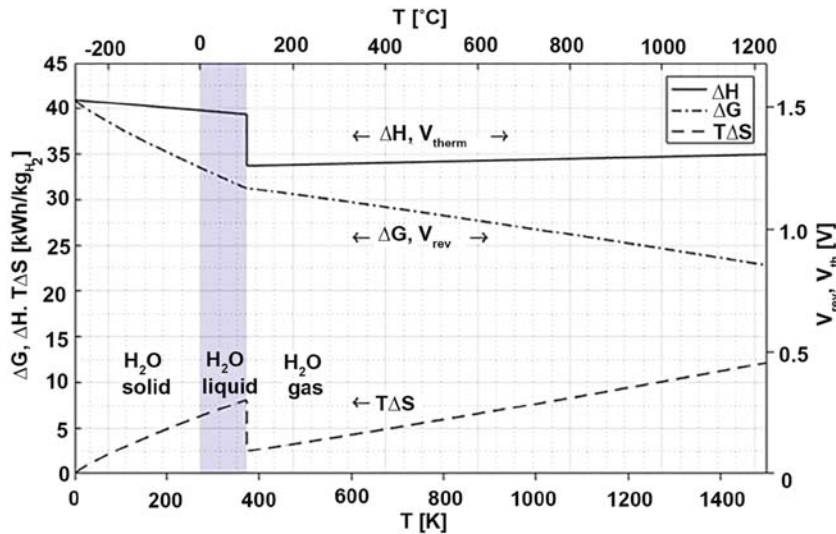
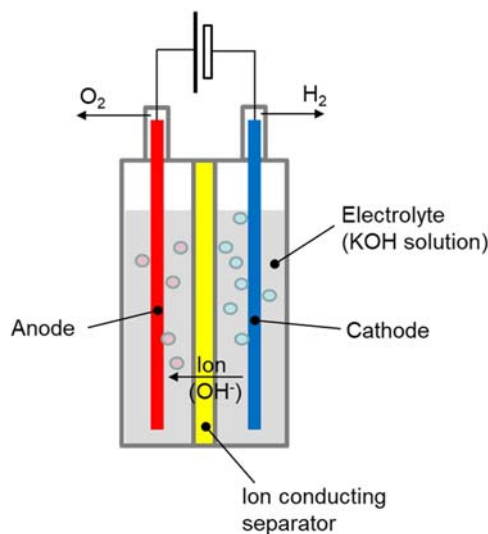


FIGURE 2.10

Thermodynamics and corresponding voltages of water dissociation as a function of temperature (Gallandat, 2017).

**FIGURE 2.11**

Main components of an electrolysis cell (e.g., alkaline electrolysis).

the cell to generate hydrogen on the cathode and oxygen on the anode. For an efficient technical implementation, several cells are installed in series forming a stack. Thus higher voltages, in the order of 400 V, are applied that reduce electric losses. In such a stack the internal electrodes are realized as bipolar plates. One side of a bipolar plate acts as anode and the other as cathode. Bipolar plates have to be electrically well-conducting, withstand the strong chemically corrosive forces, and prevent gas cross-over despite their very small thickness. Typically, they are made from graphite, titanium or stainless steel.

The actual electrolyzer technology is depending on the electrolyte used. The electrolyte is conducting the ions but isolates against electrons. It is determining the operational regime, in particular temperature and must be combined with suitable electrode and catalytic material. Table 2.1 gives an overview on the current electrolyzer technology.

As for all electrolyzer systems, purification of water and products, rectifier and product gas cleaning is required. For alkaline systems additional reprocessing of alkaline solution is necessary. Plant operation is simple, highly flexible, and appropriate for off-peak electricity use.

2.2.3.2 Alkaline electrolysis

Alkaline electrolysis is established for about 100 years in chemical industry for producing relatively pure hydrogen and other gases, in particular chlorine. Approximately 48 Mt of H₂ are being generated actually as a by-product of the chlorine production and not all of this inventory is used in hydrogen applications.

The liquid electrolyte is typically 30% KOH solution in water, which is quite corrosive. It contains the products H₂ and O₂ as bubbles in the gaseous phase and to a small degree solved in the liquid electrolyte itself. The electrolyte is pumped into phase separators. The electrodes are

Table 2.1 Overview on electrolyzer technology.

	Alkaline electrolysis (AEL)	Proton exchange membrane electrolysis (PEMEL)	Solid oxide electrolysis (SOEL)
Operation temperature	60°C–80°C	50°C–80°C	700°C–1000°C
Electrolyte	Potassium hydroxide	Acidic proton conducting membrane (e.g., NAFION)	Solid oxides, ceramics like yttrium stabilized ZrO
Ions transported	OH^-	H^+	O^{2-}
Partial reaction on anode side	$2\text{OH}^- \rightarrow \frac{1}{2}\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^-$	$\text{H}_2\text{O} \rightarrow \frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^-$	$\text{O}^{2-} \rightarrow \frac{1}{2}\text{O}_2 + 2\text{e}^-$
Partial reaction on cathode side	$2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	$\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + \text{O}^{2-}$
Technical maturity	Commercial	Commercial	Experimental
Typical production capacity per stack	< 1000 Nm ³ /h	< 100 Nm ³ /h	< 10 Nm ³ /h

specially coated Raney nickel and silver. A thin OH-conducting diaphragma separates the electrolyte compartments associated with the cathode and anode.

First alkaline electrolyzers for hydrogen production were developed by Norsk Hydro (today NEL) in Norway, where cheap electricity from hydropower is available. Alkaline electrolysis has become a mature technology at both small (1 kW) and large (100 MW) scale, see Fig. 2.12. Today's units are available in sizes up to about 2 MW(e) corresponding to $\sim 470 \text{ Nm}^3/\text{h}$ of H_2 production with multiple units being modularly combined to larger capacities. One of the largest integrated installation has been installed in Assuan, Egypt, with a production capacity of $40,000 \text{ Nm}^3/\text{h}$ with an electric input of 200 MW. They typically have an availability of $>98\%$ and an energy consumption of 4.1 kWh/Nm^3 (45 kWh per kg of H_2) operating at about atmospheric pressure and $60^\circ\text{C} - 80^\circ\text{C}$ (Gallandat, 2017). Relating these specific energy requirements to the upper heat of reaction of 3.55 kWh/Nm^3 , see Eq. (2.8), yields an efficiency of up to 87%.

Prototypes operated at a pressure up to 30 MPa help to save compression energy. In addition, the high pressure may reduce electric losses by the reducing size of the gas bubbles in the electrolyte.

2.2.3.3 Polymer membrane electrolysis

The more advanced method is solid polymer electrolyte membrane (PEM) water electrolysis, which can be operated at higher current densities due to volume reduction compared to cells with a liquid electrolyte. Typical operation temperatures are $60^\circ\text{C} - 80^\circ\text{C}$. This membrane electrolysis is simpler in its design and promises a longer lifetime, less maintenance costs, and a higher efficiency. The requirement of electricity will be reduced to values below 4 kWh/Nm^3 of H_2 . However, small-scale units ($8\text{--}260 \text{ Nm}^3/\text{h}$) exhibit lower efficiencies. High-pressure systems are established in the smaller power range with pressures of 3 MPa achieved. The bipolar plates are made from titanium. The catalysts on the anode side is typically iridium. Main problems are the still high cost of membrane manufacturing and the mass production of large bipolar plates, the electrodes, which have to withstand the quite challenging corrosive conditions at operations. However, the technology benefits a lot from the development of

**FIGURE 2.12**

Alkaline electrolyzer at Akzo Nobel.

Courtesy of NEL.

almost identical components and materials used for the PEM fuel cell, which is in the focus for mobility applications and gains ground also for stationary electricity production.

2.2.3.4 High-temperature electrolysis

Another promising variant of electrolysis is the high temperature electrolysis (HTEL) of steam. An operation at temperatures above 600°C offers the advantage of a smaller specific electricity requirement compared to conventional electrolysis (see Fig. 2.10). The savings in electricity have to be compensated with high temperature heat supply. Some high temperature electrolyzers may be operated as fuel cells, what is also known as reversible solid oxide cell.

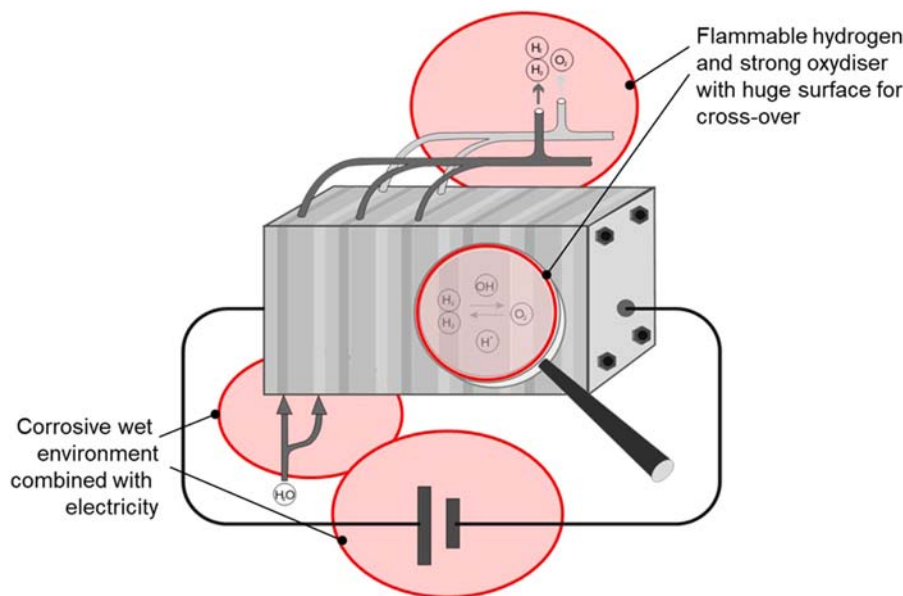
High temperature electrolyzers are using molten carbonates (MCEL) or solid oxides (SOEL) as electrolytes, similar to the corresponding fuel cell technology.

The main challenges are material compatibilities with high temperatures in general and the limitations regarding operational transients. Certain alloys of high temperature steels, in particular nickel, may be depleted and disposed on other functional elements like the electrolyte, leading to loss in performance, function, or even to safety critical states.

The maturity of HTEL is still low compared to classical alkaline electrolysis for instance. Still a lot of research is done in the field of suitable materials for electrodes, electrolytes, and packages, see [Oliver Posdziech \(2019\)](#).

2.2.3.5 Safety challenges with electrolysis

As depicted in Fig. 2.13 electrolyzers combine several hazards, in particular flammable gases in close vicinity with their reaction partners, like oxygen or chlorine, chemically aggressive media with the acidic or caustic electrolytes and electricity.

**FIGURE 2.13**

Intrinsic hazards of water electrolysis.

The separation of the gases, realised by the membrane or by the diaphragma has to be well-designed to avoid—in particular for the PEMEL and HTEL—diffusion across these very thin sheets (so called “cross-over”) under varying operational conditions.

The pumped electrolyte and the process water have to be cleaned carefully and in the phase separators impurities of H_2 in the O_2 branch and vice versa have to be controlled and removed before the product gases are further processed.

Furthermore, electric hot spots might develop generating a run-away reaction within the cell.

2.2.3.6 Thermochemical (hybrid) cycles

Thermochemical cycles can be used to split water through a series of thermally driven chemical reactions where the net result is the production of hydrogen and oxygen at much lower temperatures than direct thermal decomposition, which requires about 2000°C . All supporting chemical substances are regenerated and recycled, and remain—ideally—completely in the system. Thermochemical cycles are being investigated mainly for using low cost high temperature heat input from solar or nuclear power.

Numerous instances of such cycles have been proposed in the past and checked against features such as reaction kinetics, thermodynamics, number and separation of substances, material stability, processing scheme, and cost analysis (Beghi, 1985). Some of the most promising cycles include those based on the sulphur family, which all have in common the thermal decomposition of sulphuric acid at high temperatures (typically $\sim 900^\circ\text{C}$). One cycle considered with a high priority is the iodine sulphur process, which was originally developed by the US company General

Atomics and later modified and successfully demonstrated by JAEA in Japan. The facility consisted of more than 10 process units primarily made of glass and quartz with a hydrogen production rate achieved of 30 NI/h. The next step that started in 2005 is the design and construction of a pilot plant with a production rate of 30 Nm³/h of H₂. The theoretical limit of efficiency for the total process is assessed to be about 50%, assuming ideal reversible chemical reactions. A best estimate was found to be around 33%–36% (Goldstein, Borgard, & Vitart, 2005), but it is hoped that 40%–50% be achievable. The economic constraints, however, are caused by the availability and price of iodine.

Other processes combine chemical cycles and electrolysis, for instance the Westinghouse Sulfur Cycle (Brecher, Spewock, & Warde, 1977). Although this reduces the number of involved materials and reactions and may improve efficiencies, the process relies again—at least partially—on the use of traditionally expensive electricity.

2.2.3.7 Safety challenges with chemical cycles

The decomposition of SO₄ and HI were found to cause severe corrosion problems. In general, the handling of sulphuric acid at high temperature is hazardous, in particular if the solution has to be heated in poorly controlled locations like in a solar heat receiver. If the high temperature heat, representing a hazard in itself, is provided by nuclear power, a careful determination of safety distances has to balance economy and safety aspects. In addition, a safe and efficient transfer of high temperature heat, by a helium heat pipe for instance, must be integrated in the system design.

2.2.4 Hydrogen cleaning processes

After the actual production process the product hydrogen usually must be processed, what includes pressurization and purification. The purification techniques broadly fall into three categories:

1. Chemical (catalytic purification).
2. Physical (PSA, cryogenic separation, metal hydride separation).
3. Selective diffusion (noble metal membrane diffusion, polymer membrane diffusion, solid polymer electrolyte cells).

Obviously, each technique has limitations as well as advantages. The actual purification process depends on the type of impurities that are specific for the different production pathways. For instance, for reforming processes the remaining carbon monoxides may be removed by preferential oxidation, selective methanization, or cleaned by physical processes, like diffusion via membranes or absorption on large surfaces of special absorbing materials. Electrolysis provides typically higher purities. The main impurities in electrolysed hydrogen are oxygen, water, and electrolyte aerosols or particulates. Oxygen is typically removed by catalytic cleaning, aerosols and water are removed in a spurger or filter.

The qualities of hydrogen traded are typically labeled as 3.0, 4.5, or 5.0 corresponding to 99.9%, 99.995%, or 99.999% pure hydrogen, respectively. As low temperature fuel cells might suffer from certain impurities (especially halogens) even at ppm level, the remaining impurities are further qualified in the ISO standards for refueling stations, for instance.

2.2.4.1 Pressure swing absorption

PSA is a physical process using the preferred absorption of impurities at higher pressure on special absorbing materials providing huge specific surfaces (e.g., zeolites or activated carbon).

PSA is usually a discontinuous process conducted in five steps:

- 1. Step: Absorption:** Under high pressure the feed is directed in a bed of absorber material. Impurities are absorbed and the clean hydrogen leaves the absorber bed.
- 2. Step: Saturation:** When the contamination in the cleaned gas exceeds a critical value the absorber is saturated.
- 3. Step: Depressurization:** The pressure in the absorber is reduced and the absorbed impurities are released into the gas stream now switched in the off-gas branch, also named tail gas branch.
- 4. Step: Purge:** The absorber is purged with the feed until the impurities become smaller than a certain minimum value.
- 5. Step: Pressurization:** The exit of the absorber is switched to the clean product gas branch and the absorber bed is pressurized again and the process restarts with Step 1.

To achieve a continuous flow of products several absorbers, typically 3–6, are combined.

PSA is the most mature and widely used purification process in industry. This is why even a specific safety standard has been developed for this technology, ISO/TS 19883:2017.

2.2.4.2 Membrane separation

There are different membrane materials that can be considered for hydrogen separation and purification. The currently widest used membrane type is metallic. Metallic membranes are typically dense sheets that H_2 permeates through as its components protons and electrons. So, it requires catalytic dissociation of the molecular hydrogen on the surface of the membrane, conduction of electrons and protons, and recombination on the “clean” side of the membrane. Typically, amorphous Pd – Ag alloys, with 23% Ag (Grashoff, 1983), are used for such membranes.

Because of costs, pressure losses, and fabrication issues alternative membrane materials are under development. Promising technologies with similar functionality are silica and proton exchange membranes. The latter obviously benefits from the corresponding fuel cell and electrolyzer developments.

Another new approach is hydrogen rejection and contaminate permeation by the use of carbon-based membranes. They have distinct economic advantage and minimize the need for H_2 recompression steps (Ockwig, 2007).

2.2.4.3 Cryogenic separation

The cryogenic separation is based on the fact that hydrogen has, with the exception of helium, the lowest boiling temperature of all materials. This means all contaminants, but helium, will condense and freeze when approaching 20K. Contaminants in the liquid or solid phase may be easily separated by physical means, like filters or density-based methods. Cryogenic distillation is also used for hydrogen isotope separation (Bhattacharyya, 2016), representing an alternative approach to the use of electrolysis. Here the slightly different boiling points of isotopic hydrogen species are used. Technologies for associated hydrogen liquefaction are provided in Section 2.2.5

2.2.5 Hydrogen liquefaction

With 70 g/L the density of liquid hydrogen is about five times larger than compressed hydrogen at 20 MPa and still considerably higher than compressed hydrogen at 70 MPa (see Fig. 2.14), what makes liquid hydrogen an attractive option for weight critical applications like aerospace, for scaling up supply infrastructures, and mid to long distance transport of hydrogen.

However, liquefaction of hydrogen is an energy intense process as hydrogen has to be cooled down to the boiling temperature of about 20 K at standard pressure. At least 14.2 MJ (11.8% of the lower heat of reaction) are required to liquefy 1 kg normal hydrogen. This value is larger than for other gases and includes removal of tangible heat (5.8 MJ/kg), latent heat (6.2 MJ/kg), and the heat released by ortho-para shift (2.2 MJ/kg), see Fig. 2.15. Currently established technology requires in the order of 50 MJ. Feasibility studies showed a potential to reduce this to 22 MJ.

The principle available processes for cooling are grouped in external cooling by a colder medium in a heat exchanger and internal cooling, where the medium itself undergoes thermodynamic changes associated with a temperature drop. The latter are split in processes where the media provides mechanical work in expansion turbines “expanders” or in piston machines and in isenthalpic throttling in Joule – Thomson valves.

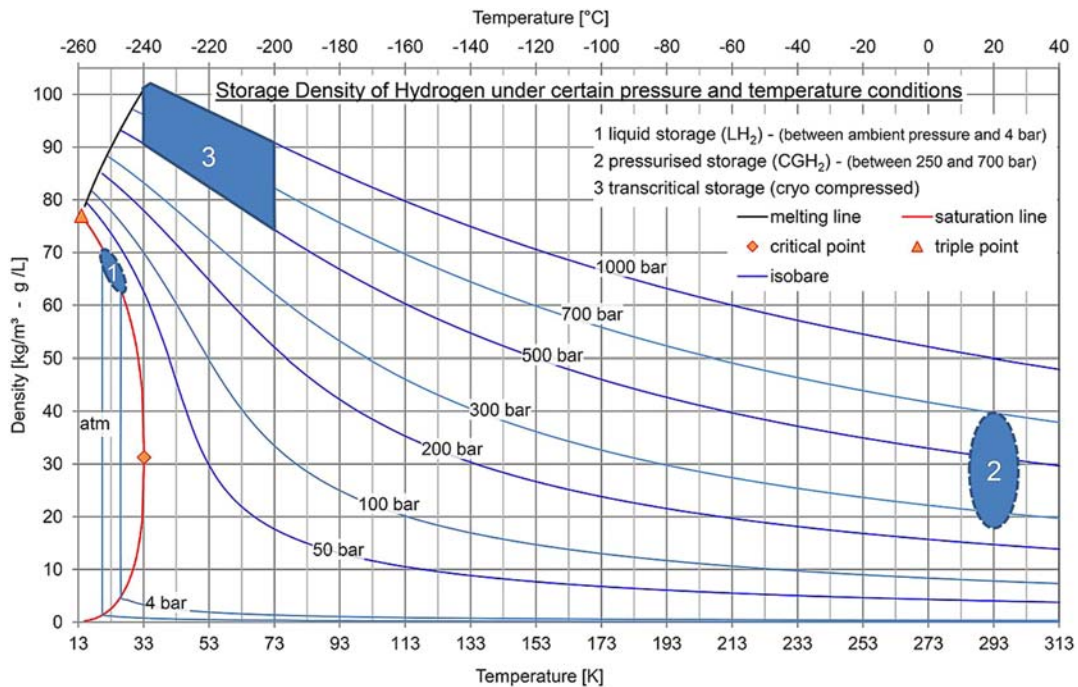


FIGURE 2.14

Density of normal hydrogen dependent on temperature at different pressure levels (Kuhn, 2015).

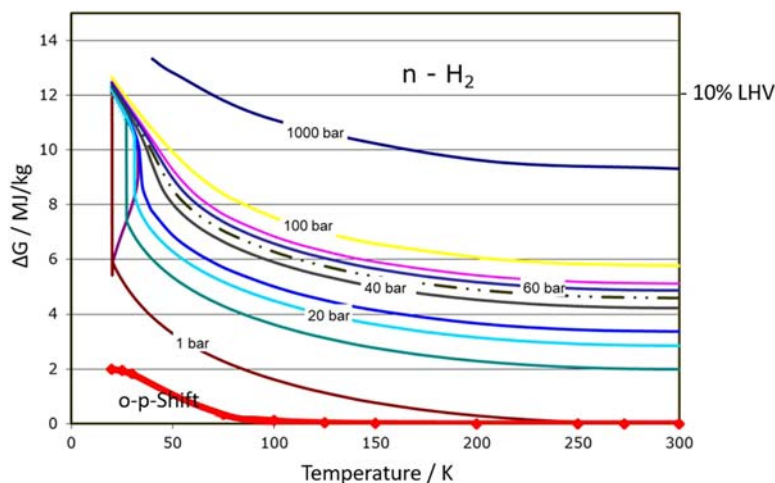


FIGURE 2.15

Free enthalpy differences quantifying the minimum energy required for pressurizing and cooling normal hydrogen with a reference point at standards starting from standard conditions (Haberstroh, 2015).

2.2.5.1 Linde – Hampson process

The Linde – Hampson method is a thermodynamic process, where isothermal compression and subsequent isobaric cooling with the nonliquefied cold gas is done in a heat exchanger. Joule – Thomson expansion connected with an irreversible change in entropy is used as the refrigeration process. Because of the relative low inversion temperature of hydrogen, a precooling, for instance with liquid nitrogen, is applied. Despite its simplicity and reliability, this method has become less attractive compared to modern ones, where cooling is carried out in almost reversible expansion machines, what improves energy efficiency.

2.2.5.2 Claude process

The commonly applied method in large-scale liquefaction plants is the Claude process combined with liquid nitrogen (LN_2) precooling, where the necessary refrigeration is provided in four main steps.

1. Multistage compression of hydrogen gas, approaching the ideal iso-thermal process by removing the compression heat.
2. Precooling with liquid nitrogen ($> 80\text{K}$).
3. Multistage cooling by expansion of fractions of the hydrogen flow in expanders, combined with heat exchangers and feeding back the expanded hydrogen into the compression step 1 ($> 30\text{K}$).
4. Expanding of the residual hydrogen in a Joule – Thomson valve (20K).

Joule – Thomson expansion is applied for the final step to avoid two-phase flow in the expander.

Fig. 2.16 shows a simplified flowsheet of the Claude process and the thermodynamics of the process in a T-s diagram.

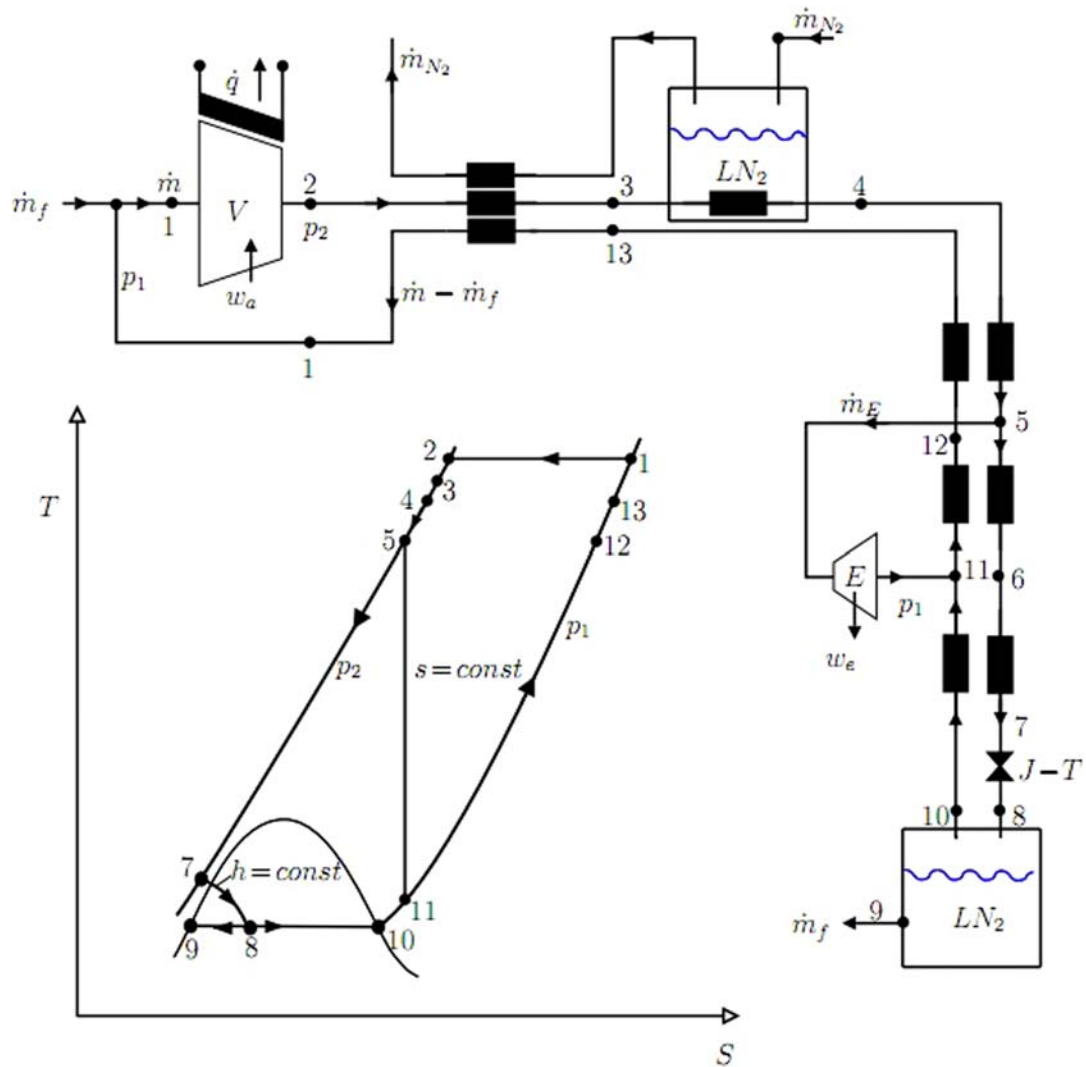


FIGURE 2.16

Flowsheet of a simplified Claude process with LN₂ precooling (Peschka, 1992).

2.2.5.3 Magnetic refrigeration process

The magnetic refrigeration process takes advantage of the entropy difference and the associated adiabatic temperature change upon application or removal of magnetic fields in the special materials, which should have either para- or ferromagnetic behavior. For efficiency reasons ferromagnetic material with appropriate Currie temperature should be applied. The actual cooling effect happens during the isentropic demagnetization of the ferromagnetic material.

Currently, this method is only relevant for temperatures below 5K and its use for hydrogen liquefaction rather on a research level. It is expected that at least 15 separate cooling stages are necessary to cool down hydrogen to the boiling point, provided strong magnetic fields with a magnetic induction of about 10 T may be applied. A typical temperature decrease in the order of 1K/T is expected, see Peschka (1992). Suitable ferromagnetic materials with Curie point temperatures in the range of 20 – 300K have to be identified or developed.

However, the process appears promising because of its potentially compact design, long lifetime, low capital investment, and higher efficiency with an estimated liquefaction work of about 26 MJ/kg.

2.2.5.4 Real world implementation

Large scales reduce the specific heat ingress at typical temperature differences in the order of 250K from the ambient to the liquid hydrogen. Most of the large liquefiers have been installed worldwide for providing rocket fuel for space applications. For efficiency reasons typically the Claude process with LN₂ precooling is further equipped with external Brayton coolers, which are implementing a reverse closed Brayton cycle with turbo compression and expansion using helium or mixed refrigerants as operating medium.

The feed hydrogen has to be purified carefully to avoid plugging by condensed and frozen impurities. The expansion is split in up to six stages. Several ortho-para converters using either Fe₃O₄ or active carbon as catalytic material are integrated below 100K.

All cold parts are mounted in a cold box, which is thermally insulated for instance with perlite.

Worldwide there are about 29 large scale liquefiers in operation with production capacity from 1 ton to more than 60 tons of LH₂ per day. The largest capacities are installed in the United States. In addition, there are several laboratory-scaled liquefiers in operation with a capacity of a few kg/day (Aasadnia, 2018).

Further improvement in efficiency is expected with the development of new materials and new compression/expansion technology.

2.2.5.5 Safety issues

A major problem when producing and handling liquid hydrogen is the potential contamination of the hydrogen with air or other impurities, which will, with the exception of helium, freeze and might block pipes, filters, or armatures. Countermeasures are careful purity control of the feed hydrogen and purging the cold boxes with helium.

On the exterior of poorly insulated containers or pipes the cryogenic temperatures may condense air with hazardous enrichment of oxygen. Liquified or frozen solid oxygen promotes ignition and oxidizes easily materials that are usually nonflammable.

The extreme low temperatures require careful selection of materials. Conventional carbon steels will suffer from a transition to nil ductility (NDTT). Aluminum or stainless steels are typically suitable structural materials for cryogenic hydrogen and welded connections are preferred to screwed connections.

Because of the heat uptake the contained liquid hydrogen warms up and evaporates, what increases the pressure. Therefore boil-off management and proper safety valve design are required.

If cryogenic hydrogen is leaking, it might lead also to condensation of air and to hazardous oxygen enrichment. Initially the leaking cryogenic hydrogen is as heavy or even heavier as ambient air. This suppresses buoyancy effects and promotes dispersion of flammable mixtures on ground

level. Leaks may be detected by the associated temperature drop and visually identified via the fog formed from condensating ambient humidity.

2.2.6 Large scale versus small scale and centralized versus decentralized production

At present, most hydrogen is produced on-site in commercial, large-scale SMR units dedicated to the needs of the chemical and petrochemical industries. On-site production means flexible, on-purpose production with no need for transportation.

In the future centralized hydrogen production will refer to large-scale systems connected to a hydrogen delivery/distribution network transporting the product hydrogen to the point of use in gaseous or liquid state via pipeline or truck. Centralized large facilities are motivated by the efforts to decrease specific production cost by increasing the unit size (economy of scale) and allow for high capacity factors. Also, the use of large hydroelectric power or nuclear primary energy only makes sense for centralized H₂ production on a large scale.

Renewable energy sources with their low power densities and typically intermittent availability will motivate a rather distributed hydrogen generation pattern. The distributed hydrogen production from renewables has to be tuned to the potentially alternative use of renewable energy and direct injection (DI) in existing electricity and gas grids. This applies to hydrogen production via electrolysis with wind or PV electricity and to hydrogen production from biomass, which will be limited in size simply because of the difficulty to transport biomass. Both widely available electricity and gas grids can help transferring the point of use of distributed renewable energy to more centralized locations and optimizing the patterns for more centralized or decentralized hydrogen production.

For future applications of hydrogen as part of the energy economy, the installation of a network of small-scale H₂ production units as integral parts of energy hubs appears to be a good short-term approach for the introduction phase (Maroufmashat, et al., 2016). Market prospects for stationary and mobile fuel cell applications have already led to the development of small-scale H₂ units on the prototype level to be either part of the required infrastructure for fuel cell vehicles (FCV) or for feeding local grids for residential stationary fuel cell systems. Small SMR or electrolyser units, which are competitors at this scale, are attractive for early low-demand stages. They require less absolute capital investment and no transport and delivery infrastructure. On the other hand, there are drawbacks in terms of limited efficiency and high H₂ cost, because they are lacking the advantages of the economy of scale factor and of the improved storage efficiency of large plants. Furthermore, operation and control of a network of many small H₂ units require a cost-effective smart grid control and more efforts for establishing and maintaining consistent high quality and safety standards.

With respect to the projected networks of H₂ refueling stations, a comparative cost analysis study has shown that for consumptions lower than 600 Nm³/h, the delivery of LH₂ by tank truck represents the most economic option compared to onsite production, gaseous batch transport, or dedicated pipelines. However, at larger scales dedicated hydrogen pipelines turn out to be most economic (Krieg, 2012).

In any case the actual distribution and use cases determine the required purity and pressure levels to be provided by the hydrogen production. If the distributed hydrogen production is connected to a pipeline grid, a problem may also be seen in the mixing of H₂ streams from different

sources unless minimum quality requirements are set for each source. On the other hand, if the distributed production of hydrogen is not connected to a larger grid, but rather combined with distributed usage, the required purity and pressure levels might be adjusted to the actual requirements, which give freedom for economic optimization.

2.3 Hydrogen storage and transport

2.3.1 Introduction

Today hydrogen is used worldwide mainly for industrial applications. It is produced onsite and on demand, what minimizes transport and storage needs. The future use of hydrogen as a chemical energy storage or carrier, respectively fuel, will change this and effective and safe storage and transport of hydrogen will be required.

As a chemical energy carrier it provides highest gravimetric energy density (see Fig. 2.17). This makes hydrogen the preferred rocket fuel for space crafts for instance.

However, with its low density hydrogen needs a relatively large volume to be stored at standard conditions, where 1 kg of hydrogen takes a volume of approximately 11 m³ (corresponding to a density of 0.089 kg/m³). Therefore, storing hydrogen is a challenge if space is limited, as it is with

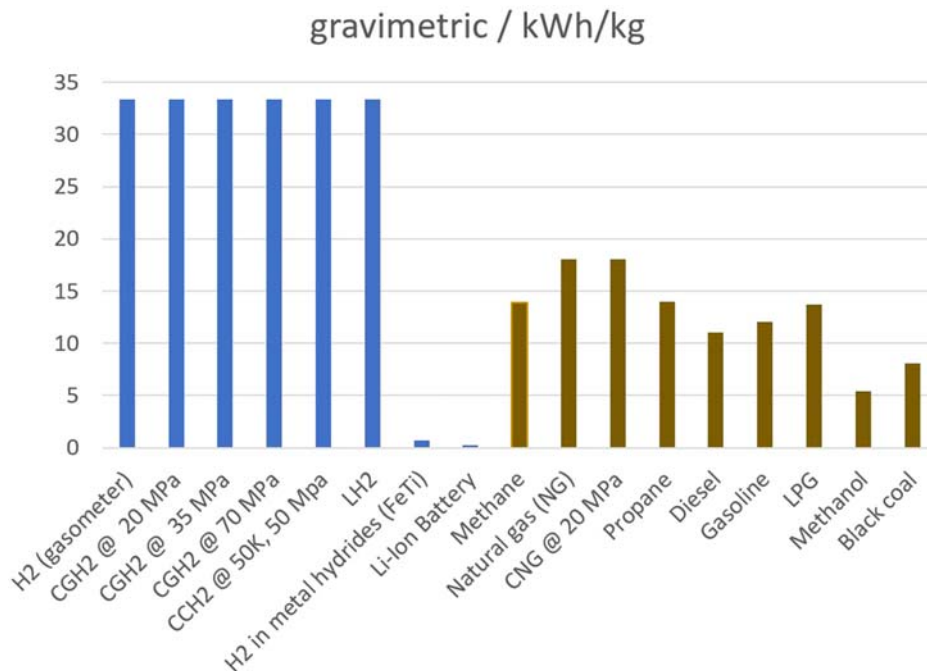


FIGURE 2.17

Comparison of gravimetric energy densities (lower heating value) of different energy carriers.

mobile applications in particular. Fig. 2.18 compares the volumetric storage densities of energy carriers, which for gaseous energy carriers obviously depend on pressure and temperature.

To reduce these space requirements there are practically three different approaches:

1. Compression.
2. Liquefaction.
3. Physical or chemical binding to liquid or solid storage materials.

All three approaches introduce new problems, either with regard to efficiency or to hazards implied with high pressure or cryogenic temperatures. In particular the classical solid storage materials like iron and titanium add huge masses and thereby cancel the advantage of the pure hydrogen's weight performance.

To overcome inherent problems related to the three fundamental approaches also combined, so-called hybrid storage technologies are investigated (see Fig. 2.19). The most prominent example is the cryocompressed storage, combining the density gains provided by cryogenic temperature and high pressure.

As all approaches require specific containers and further infrastructure, for technical solutions rather the weight and volume of the storage system than those parameters of the contained hydrogen have to be considered.

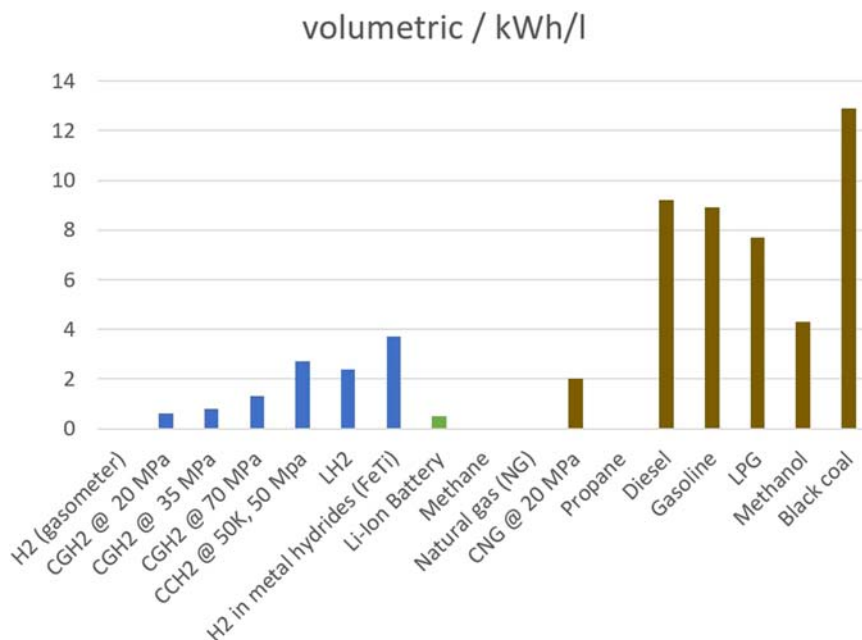
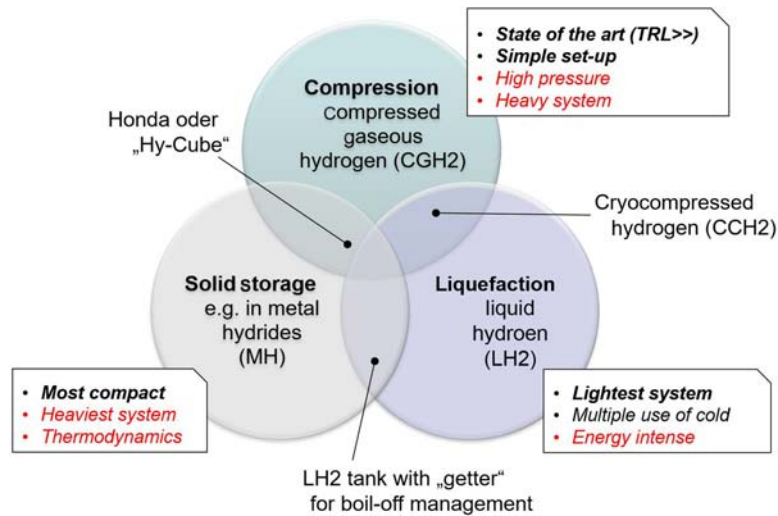


FIGURE 2.18

Comparison of volumetric energy densities of different energy carriers (hydrogen at NTP ~“gasometer” conditions 0.003 kWh/L).

**FIGURE 2.19**

Hydrogen storage technologies.

The following sections present some details related to compressed gaseous and solid state storage of hydrogen. The liquefaction has been treated in detail in [Section 2.2.5](#). The corresponding system aspects are covered in [Section 2.6](#).

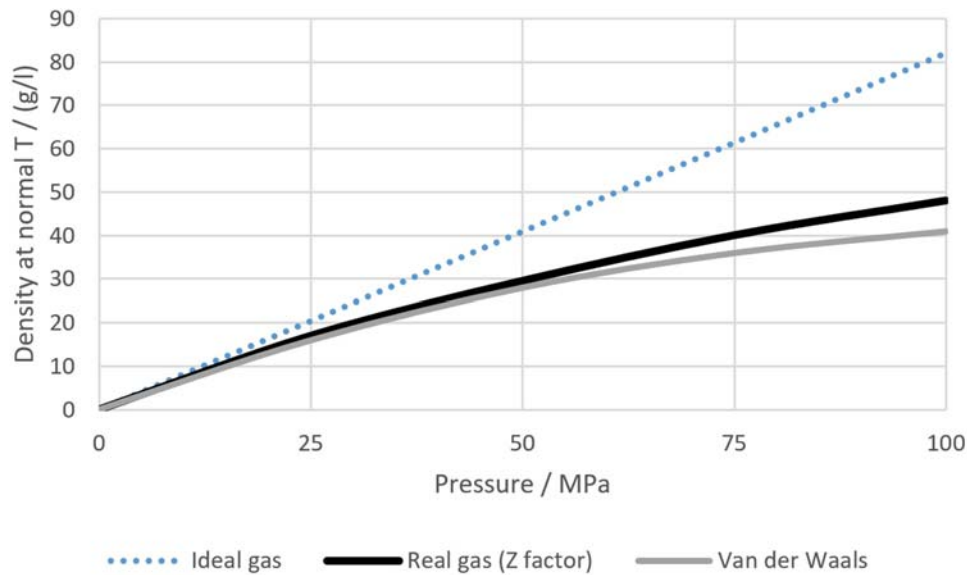
2.3.2 Compressed gaseous storage

To reduce space requirements hydrogen may be compressed to pressures of 4 MPa up to 100 MPa depending on the actual application. Limitations for storing hydrogen at even higher pressures are rooted in the real gas effects, reducing the density gains at higher pressure and the at least linearly increasing mass of the container and energy required for compression. Above 15 MPa the deviations from ideal gas behavior become evident (more than 10% in density, see [Fig. 2.20](#) and [Table 2.2](#)). In this domain and above the gas behavior is not well-modeled with the ideal equation of state (EOS), but a real gas EOS, such as van-der-Waal EOS or other advanced EOS, should be applied. For technical applications like compression, the approximation with real gas factor Z , also known as compressibility factor, is sufficiently exact. [Eq. \(2.10\)](#) defines the dimensionless Z .

$$Z = \frac{pv}{RT} \quad (2.10)$$

[Table 2.2](#) provides Z values for some frequently encountered pressure levels, determined with the reference EOS of NIST at a temperature of 20°C.

[Fig. 2.20](#) compares the densities of normal hydrogen as a function of pressure. The curves are determined with the ideal gas EOS, the van-der-Waals EOS, and the real gas factors for a constant temperature of 20°C. The standard van-der-Waals EOS does not provide a good approximation as

**FIGURE 2.20**

Density of normal hydrogen as a function of pressure.

Table 2.2 Density and real gas factor Z for compressed normal hydrogen at 20°C.										
p/MPa	0.1013	5	10	20	30	35	40	50	70	100
$\rho/(\text{kg}/\text{m}^3)$	0.083	4	7.8	14.7	20.8	23.7	26.3	31.2	39.7	49.9
Z	1	1.033	1.06	1.125	1.192	1.217	1.251	1.318	1.45	1.648

it is derived from the gas behavior close to the critical point, that means moderate pressure and cryogenic temperature.

2.3.2.1 Compression work

To determine the required compression work an EOS of hydrogen applicable to the respective pressure range has to be used. In addition, a caloric equation is needed. To simplify the calculations an ideal adiabatic isentropic process is considered.

In general the technical compression work is the integral of specific volume (reverse density) over pressure, see Eq. (2.11).

$$W = \int_{p_1}^{p_2} v dp = \int_{p_1}^{p_2} \frac{dp}{\rho} \quad (2.11)$$

With the above idealization and with the ideal gas EOS Eq. (2.11) turns into Eq. (2.12)

$$W_{\Delta S=0,\text{ideal}} = \frac{\gamma}{\gamma-1} RT_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right] \quad (2.12)$$

with initial pressure p_1 and the ratio of specific heats $\gamma = \frac{c_p}{c_v} = 1.4$.

The real compression process is not isentropic. Therefore the work for a real compression process may be approximated via the efficiency related to the ideal adiabatic compression process according to Eq. (2.13):

$$W_{\text{compression}} = \frac{W_{\Delta S=0,\text{ideal}}}{\eta_{\text{adiabatic}}} \quad (2.13)$$

$\eta_{\text{adiabatic}}$ is around 0.75–0.85. In addition, multiplying the efficiency of the electric drive of the compressor with about 0.9 one might estimate the really required energy for compression. For a compression to 70 MPa the ideal isentropic process requires about 23.3 MJ/kg and together with the above efficiencies about 25% of the lower heating value would be required.

However, the real process might be rather approximated by an interpolation of the above ideal process and an isothermal process. Eq. (2.14) defines the ideal isothermal compression work for an ideal gas:

$$W_{\Delta T=0,\text{ideal}} = RT \ln \frac{p_2}{p_1} \quad (2.14)$$

The ideal isothermal work required to compress hydrogen from standard conditions again to 70 MPa amounts to 7.9 MJ/kg, what is considerably less than for the isenthalpic process. The use of a real gas EOS yields more realistic values; see Eq. (2.15) for van-der-Waals and Eq. (2.16) for Z factors.

$$W_{\Delta T=0,\text{real}} = RT \ln \frac{p_2 - b}{p_1 - b} + \frac{a}{V_2} - \frac{a}{V_1} \quad (2.15)$$

$$W_{\Delta T=0,Z} = ZRT \ln \frac{p_2}{p_1} \quad (2.16)$$

The real compression is best approximated by a polytropic change of state. The required work, which is obviously larger than the ideal isentropic, is calculated with Eq. (2.17).

$$W_{\text{polytropic}} = \frac{n}{n-1} RT_1 \left[\left(\frac{p_2}{p_1} \right)^{\frac{n-1}{n}} - 1 \right] \quad (2.17)$$

with polytropic coefficient n . Again, the actual work of compression is larger than the calculated work and alternatively maybe determined via the polytropic efficiency η_p as shown in Eq. (2.18).

$$W_{\text{compression}} = \frac{W_{\Delta S=0,\text{ideal}}}{\eta_p} \quad (2.18)$$

The polytropic efficiency is often quoted by manufacturers. From this value the polytropic coefficient may be derived according to Eq. (2.19).

$$n = \frac{\eta_p \gamma}{1 + \eta_p \gamma - \gamma} \quad (2.19)$$

The conclusion is that the isothermal process is most efficient for compression. However, there is no straightforward technical realization of this process. Instead the ideal process is approximated by a multistage polytropic compression with intermediate cooling. For this a cost benefit consideration will give an optimum for 2–5 pressure steps for pressurization to 70 MPa, depending on the compressor type and the required mass flow.

2.3.3 Solid state storage

Hydrogen reacts at elevated temperatures with many transition metals and their alloys to form hydrides. Hydrogen can be stored in this form at higher densities than by simple compression or even in liquefied or cryocompressed state. On the other side, the weight of the metallic hosting material is quite high compared to the mass of hydrogen actually stored.

Metal hydrides begin as intermetallic compounds, produced in a similar way as any other metal alloy. They exhibit one important difference. When metal hydrides are exposed to hydrogen at certain pressures and temperatures, they absorb large quantities of the gas and form metal hydride compounds. When heat is applied or/and the pressure is reduced, the stored gas is released.

2.3.3.1 Absorption process

The absorption process is actually a sequence of a physisorption and a chemisorption subprocess. When molecular hydrogen from the hydrogen gas comes into contact with the surface of a hydrogen storage metal hydride material, it dissociates catalytically into atomic hydrogen, what enters and diffuses into the bulk material. This is the main part of the physisorption. Then, in the bulk material hydrogen is either stored on interstitial locations (interstitial hydrides) or is integrated chemically into the lattice (metallic hydrides or intermetallic compounds). This chemical binding process, the chemisorption, releases energy and changes the bulk material density and other properties. Metal hydrides literally trap hydrogen within the alloy, much like a sponge absorbs water.

2.3.3.2 Desorption process

When heat is applied and/or pressure is reduced below the equilibrium pressure, hydrogen atoms migrate to the surface of the metal hydride and recombine there into hydrogen molecules. From there it may be taken as hydrogen gas. Ideally the metal structure reestablishes and the original metal crystal structure is formed again.

Therefore the storage reaction, as given in Eq. (2.20) in a strongly simplified way, is reversible. More precisely, like all chemical reactions, it is an equilibrium reaction, where the balance of educts and products depends on temperature and pressure, heat of formation, and densities of all involved materials.



In fact, the key to practical use of metal hydrides is their ability to both absorb and release the same quantity of hydrogen many times without deterioration. This is defining the reversibility.

The hydrogen absorbing behavior of metal hydride alloys is characterized using equilibrium pressure-temperature-composition data. These data are determined by keeping an alloy sample at constant temperature while precisely measuring the quantity of hydrogen absorbed and the pressure at which sorption occurs. The quantity of hydrogen absorbed is expressed in terms of alloy composition, either as an atomic ratio of hydrogen atoms to the number of atoms in the base metal alloy, or as the

capacity of hydrogen in the alloy on a weight percent basis. Hydride alloys can be engineered to operate at different temperatures and pressures by modifying alloy composition and production techniques.

On a system level, it is important to provide huge surfaces for the exchange of heat and material, which means hydrogen. The rate at which a hydride alloy can absorb or release hydrogen is dependent upon the rate at which heat can be transferred into or out of the alloy. Increasing the heat transfer rate allows the processing of higher flow rates. Therefore to improve the performance of the storage systems based on metal hydrides, researchers must find ways to increase the proportion of hydrogen in the hydrides, while maintaining the reversibility of the reaction within a reasonable temperature and pressure range.

Today the class of metal hydride material used in practical applications is the conventional low temperature hydrides. Low temperature hydrides release hydrogen at near ambient temperature and pressure (different groups of materials exist, based on Ti, Zr, V, rare earth, or other compounds). However, their reversible hydrogen storage density is usually between 1.5wt% and 1.8wt%.

The high temperature metal hydrides are mainly based on Mg. These materials need operating temperatures above 230°C (260°C–280°C) to release the hydrogen. They are capable of theoretically storing about 7wt% with about 5–6wt% being reached at lab scale today.

A new class of metal hydrides, the so-called medium temperature materials and particular the alanates (e.g., NaAlH_4 or LiAlH_4) are currently being investigated with high expectations. Based on the use of light metals such as Mg and Al, storage densities of 4.5–5.0wt% at 130°C have been shown with the theoretical maximum being 5.5wt% or 4.5 system wt%.

Fig. 2.21 provides an overview on the volumetric and gravimetric storage performance of some selected metal hydrides (Züttel, 2003).

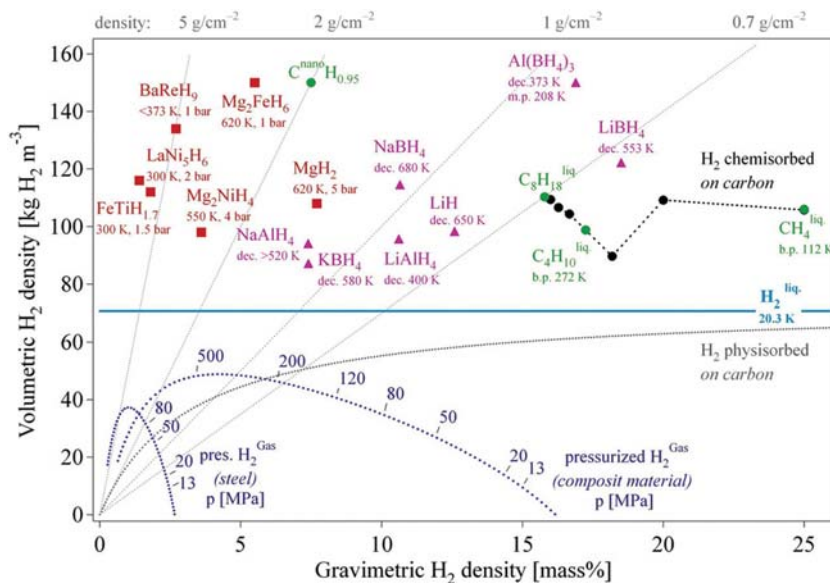


FIGURE 2.21

Volumetric and gravimetric hydrogen density of some selected hydrides.

Züttel, A. (2003). *Materials for hydrogen storage*. (Elsevier, Hrsg.) *Materials Today*, S. 24–33.

Therefore metal hydrides promise very compact storage solutions, but are too heavy for many transport applications. The lightest hydrides are approaching the 7% weight performance requirement of the automotive industry, but require a quite complex thermal management. The interplay of heat and chemical energy provides opportunities for sector coupling solutions mainly for stationary energy storage. However, reactivity of this medium to air, water, or other fluids is a safety concern that remains to be addressed.

Alternative developments are addressing adsorbing materials like zeolites, metal organic frameworks, and carbon nanotubes. Most of those materials are better performing at low temperature and might be considered in combination with cryostorage, as boil-off adsorber for instance.

Besides the reversible storage materials there are other solid materials, either containing or releasing hydrogen in special reactions. A prominent example is lithium activated alumina that releases considerable amounts of hydrogen when wetted with water (Valery Rosenband, 2010). Therefore this is for single use and the materials have to be reprocessed separately. However, because of the safety advantages these materials have in common with the reversible hydrides, they may be an attractive option for hydrogen supply of systems that are relatively light and easy in handling, finally. Therefore application for drones for instance is considered.

2.3.4 Batch transport of hydrogen

Depending on the scale and the desired use, hydrogen batch transport is based either on compressed or liquefied hydrogen. Despite the higher efficiencies rooted in the higher densities of LH₂, in most cases high-pressure gaseous hydrogen is preferred over liquid hydrogen. This is because of the small numbers of liquefiers available worldwide and the higher energies required for liquefaction. For long distance transport, however, liquid hydrogen is the preferred option.

Road transport of gaseous hydrogen is presently carried out using trucks with steel cylinders of up to 90 L at 20–30 MPa pressure or large seamless cylinders called “tubes” of up to 3000 L at 20–25 MPa, see Fig. 2.22. For transport in larger scale pressure of up to 50–60 MPa or even higher may be employed. A 40-ton truck delivers about 26 tons gasoline to a conventional gasoline filling station. A 40-ton truck carrying compressed hydrogen can deliver only 400 kg, because of the weight of the 20 MPa pressure vessels. On an energy scale this corresponds to about only 5% of the energy contained in a gasoline or diesel truck transport.

The LH₂ trailers, as shown in Fig. 2.23, are rather limited by the maximum dimensions of the transport vehicles. Therefore typically they come with a total weight of 25 tons including 2–4 tons LH₂.

Therefore one LH₂ trailer suffices to transport the same hydrogen inventory as at least 5 trailers and up to 15 trailers with compressed gaseous hydrogen (CGH₂). In addition the mass-specific investment costs for LH₂ trailers are considerably lower than for CGH₂ trailers, see Fig. 2.24.

Currently, batch transport of hydrogen is focussed on road transport. Neither rail nor waterborne transport are established in a broader way.

2.3.5 Transport using pipelines

Even though hydrogen distributed in pipelines demands better/more tightness for the pipe material, for seals and fittings, and rises specific materials compatibility issues, the procedure is well-known and has been safely in use for many years in industrial areas for local distribution (DoE, 2021).

**FIGURE 2.22**

Tube trailer for compressed gaseous hydrogen transport.

Courtesy of Air Liquide.

**FIGURE 2.23**

Liquid hydrogen trailer.

Courtesy of Linde.

The total length of installed hydrogen pipelines amounts to more than 2000 km worldwide. However, this is still modest as compared to a complete national or even international gas transport network, delivering energy mainly for heating purposes in private homes or industry.

Hydrogen's growing importance for decarbonizing the energy systems and its function as an intermediate storage of green excess electricity will lead to a hydrogen network of pipelines in order to connect new large scale production sites with end users and applications with high

**FIGURE 2.24**

Capacities and investment costs for compressed gaseous hydrogen and LH₂ trailer transport.

demand. Therefore in the long run, hydrogen will be directly delivered via pipelines to industry customers, filling centers, train, or ship fueling stations, to more distributed use cases, for example, middle-sized combined heat and power generation, and even to home heating devices. Prior to this situation, decentralized hydrogen production will take advantage of the existing natural gas infrastructure, which will be adapted to hydrogen stepwise. The first step of mixing hydrogen into the natural gas grid up to 20% volumetrically is now demonstrated in several projects, see HyDeploy (HyDeploy, 2021) and THyGA (THyGA, 2020) project websites, for instance. No modifications are deemed necessary up to those concentrations as the hazard classification of the gas mixture does not change compared to natural gas. However, the metering has to be adjusted to account for the different caloric value. Later dedicated backbones, new transfer pipelines for 100% hydrogen, will be built, see H21 project website (h2_2020). In the long term the 100% conversion to hydrogen will account for the new gas production and use patterns.

In a pipe of a given size and pressure, hydrogen flows about three times faster, but because it also contains about three times less energy per volume, a comparable amount of energy gets through the pipe. Fig. 2.25 compares the pressure losses in such a configuration of pipeline with a diameter of 100 cm along a length of 100 km (Siemens Energy, Gascade Gastransport GmbH, & Nowega, 2020).

Therefore more energy is required for pumping hydrogen instead of natural gas. The low volumetric energy density of hydrogen demands higher flow rates, which in turn leads to greater flow resistance. In most of the natural gas infrastructure one or two stages turbo compressors are used. These compressors require a power of up to 30 MW. For an identical pipeline set-up the compression power has to be raised by a factor 3–5, if hydrogen instead of natural gas is transferred at comparable heat content. In summary, the energy transfer capacity of a given pipeline configuration is with 80%–90% slightly lower when it carries hydrogen instead of natural gas.

Compared to the huge investment for new pipelines [> 0.5 Mio €/km for a 20-inch pipeline (Leighty, 2006)] refurbishing existing natural gas pipelines only takes 15%–20% of those costs. Depending on the hydrogen content, which should be used in the pipeline, this infrastructure can be maintained or adapted according to the following criteria:

1. Up to approx. 10% H₂, the compressor can generally continue to be used without major adaptation.

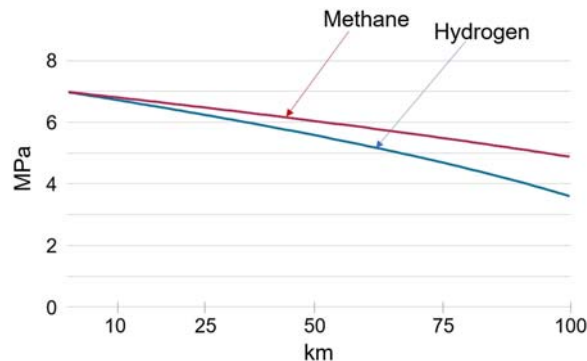


FIGURE 2.25

Comparison of pressure losses for methane and hydrogen transferred in a 100-km pipeline with 100 cm inner diameter (Siemens Energy, Gascade Gastransport GmbH, & Nowega, 2020).

2. Up to approx. 40% H₂, the compressor housing can be retained; impellers and feedback stages as well as gear boxes have to be adapted.
3. From approx. 40% H₂ the whole compressor must be replaced.

Besides the specific energy costs and suitable pressure levels for the transport, the material compatibility is a major issue for hydrogen pipelines and the reuse of natural gas pipelines. This concerns embrittlement of the bulk material, leak tightness of seals, and the incompatibility of gas compressor lubrication.

In summary, conversion of the existing steel pipelines from natural gas to hydrogen operation is economically feasible to the scale needed for the ramp-up of a hydrogen economy. A significant reduction in the lifetime of high-pressure pipelines due to the influence of hydrogen does not seem likely (Wezel, 2019). Further considerations with respect to pipe materials and technical implementation and a high pressure pipeline operated at 100 MPa are presented in Section 2.6.6.

Transport of cryogenic liquid hydrogen in pipelines is actually limited to short distances. Typical applications of liquid hydrogen pipelines are only encountered at space flight and research centers for filling rocket tanks or testing rocket motors, respectively. As pipes geometrically provide large specific surfaces they must be insulated quite carefully. Similar as cryostats they are made from an inner tube containing the LH₂ at a temperature in the order of 20K, which is contained in an outer tube. The evacuated space in between those concentric tubes is insulated with multilayer super-insulating material. The tube material is typically made from stainless steel, sometimes from alumina.

2.3.6 Future pathways and safety challenges

Scaling-up implies quantitative and qualitative changes. Qualitative changes are implied in particular with the high pressures and cryogenic temperatures required for efficient storage and transport. High pressure pipelines, operated at a pressure of 100 MPa, and LH₂ storage and pipelines built

from plastic material are under development. Storage and transport are also key for sector coupling. Therefore hybrid storage systems for storing electric energy, thermal energy, and chemical energy are under development. For example, the hybrid energy storage investigated in the LIQHYSMES project combines superconducting magnetic energy storage with the storage of chemical energy as LH_2 . Those developments, however, also introduce new hazards.

In any hydrogen system storage and transport are associated with largest localized accumulation or flow of the energy carrier, implying highest hazards and vulnerabilities. In particular, for mobile or transport solutions, a less controlled environment and interference with the public have to be considered.

More detailed treatment of the specific safety challenges is given in the respective components-related sections below.

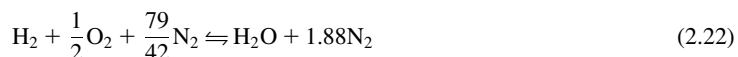
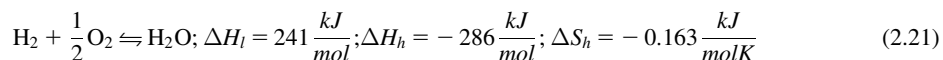
2.4 Hydrogen energy conversion

2.4.1 Introduction

The energy carrier hydrogen is not only characterized by its flexibility in production and its unique properties as chemical energy storage, but also by its flexible ways to efficiently supply carbon-free energy in various forms to different applications. For supplying heat, hydrogen may be used as a flammable gas and a fuel, respectively, where the heat of combustion may be released at relative high temperature. If, however, electricity is required, the free enthalpy of hydrogen may be also directly converted to electricity in fuel cells. Both pathways are generally characterized by relatively high conversion efficiencies and no CO_2 emissions at the point of use.

2.4.2 Combustion

Like other fuels or flammable gases, hydrogen may be used in combination with oxygen or air in combustion processes. The main product of this reaction is water. For special applications like rocket motors, chlorine and fluoride may also be considered as a reaction partner for hydrogen, providing even higher temperatures in the combustion process. However, for most applications this is not an option, because of the availability and costs of those media, the extreme reactivity of the mixtures and the hazards, in particular toxicity, associated with the reaction products. The combustion reactions with oxygen and air are given in Eqs. (2.21) and (2.22).



The heat released in the combustion process with oxygen or air may be used for heating purposes in industry or for domestic heating. The released heat may also be used to drive heat power cycles as in conventional ICEs for vehicles or as in gas turbines for power stations for electricity generation.

Although the actual reaction produces only water and is therefore a potentially very clean process, the combustion with air might produce nitrogen oxides, in particular if the combustion is close

to stoichiometric conditions and the flame temperature reaches relatively high values. However, there are some strategies to reduce this effect far below the level of NO_x production in conventional fuel combustion. The properties of hydrogen allow stable combustion in the very lean region with high air ratios λ , which reduce the flame temperature accordingly. Another strategy is the exhaust gas recirculation or the use of catalytic burners. In all cases the reaction occurs at considerably lower temperatures.

Besides being carbon-free hydrogen differs from other gaseous fuels like methane, propane, or gasoline vapor by providing highest diffusivity, widest flammability range, lowest ignition energy, smallest quenching gap size, highest autoignition temperature, and flame velocity. This makes hydrogen a very stable and efficient fuel. However, the very low density reduces the volumetric power density drastically, if no special remedies are applied.

2.4.2.1 Internal combustion engines

Hydrogen is a well-defined fuel with many advantageous properties for ICE. The simplified equation Eq. (2.23) for the principal thermodynamic efficiency of the combustion process promises higher efficiencies for hydrogen compared to gasoline.

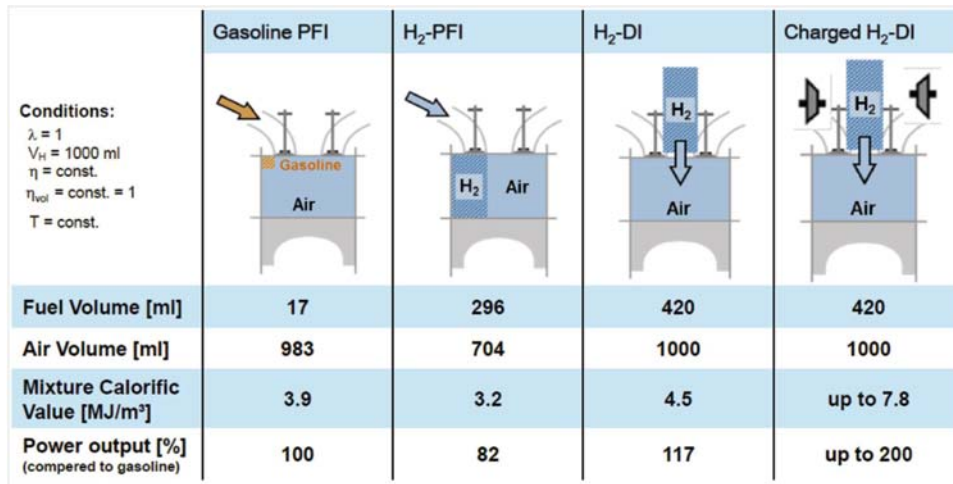
$$\eta_{th} = 13 \frac{1}{\left(\frac{V_1}{V_2}\right)^{\gamma-1}} \quad (2.23)$$

Assuming a fixed compression ratio $V_1/V_2 = 10$ and with the ratios of specific heats γ of the fuel air mixtures ($\gamma_{H_2/air} = 1.4$; $\gamma_{gasoline/air} = 1.3$), Eq. (2.23) gives 0.6 for H₂ and 0.5 for gasoline. With the higher autoignition temperature higher compression ratio could be applied for hydrogen potentially.

The wide flammability range allows using a wide range of mixtures for the combustion process. Lean mixtures reduce the maximum temperatures in the flames and thereby the formation of NO_x. On the other hand lean mixtures reduce further the low volumetric energy and power density associated with hydrogen. However, the wide flammability range reduces the unburnt fraction, what is supported further by the good mixing properties of the highly diffusive gas. Therefore losses are typically reduced implying a potentially high efficiency. This was shown in the HyICE project with BMW as a leading automotive project partner (Fickel, 2006). Efficiencies up to 50% could be demonstrated with the BMW 750 h, which uses a modified standard 12-cylinder ICE in a bifuel mode.

The mass ratio air/fuel with 137.3 g/4 g = 34.3 is for hydrogen more than double compared to conventional fuels. This ratio can be further extended into the lean region up to 180, what corresponds to an equivalence ratio of 0.2. However, there are obvious limits regarding the air supply at these extreme lean regions, in particular with regard to the decreased power rating. Turbo-charging and DI, instead port fuel injection, are measures to compensate the low volume-specific power density for a hydrogen ICE. Fig. 2.26 compares the volumetric power density of those different strategies (Korn, 2019).

The ignition properties imply that an Otto process with dedicated ignition system is needed for hydrogen. The small ignition energy and the wide flammability range on the other hand guarantee reliable ignition even for lean mixtures. Despite the relatively high autoignition temperature transitions into detonations, the so-called knocking, are promoted by the high flame velocities and the flame acceleration driven by instabilities characteristic for lean mixtures. The tendency for knocking with gaseous fuels is

**FIGURE 2.26**

Estimation of the potential of hydrogen combustion (combustion processes with port fuel injection and direct injection) (Korn, 2019).

described with the phenomenological methane number (MN), similar as the octane number for liquid fuels. On this empirical scale methane takes an $MN = 100$ and hydrogen takes an $MN = 0$ (Scharlipp, 2017). It should be noted that the knocking tendency might be controlled with appropriate ignition timing.

Because of the small flame quenching gap (0.6 mm compared to 2 mm with methane), hydrogen burns very close to structural surfaces. Again, this helps minimizing unburnt quantities. On the other hand this allows the flame to escape the control volume even through very small gaps at the valves for instance, and thus may lead to a flashback of the flame into the premixing chamber, if no DI is used.

Hydrogen ICE for the drivetrain of buses was successfully demonstrated in several projects, like in the H2ICE subproject of HyFLEET:CUTE (Stolzenburg & Kentzler, 2010). Fig. 2.27 shows the turbocharged hydrogen engine built by MAN for those buses.

Conversion of diesel engines is commercially offered by the company Keyou for instance. Rotary engines usually provide more space for rearrangements for fuel and air supply, for the exhaust extraction, and ignition system modification. Mazda was demonstrating the feasibility of a hydrogen-driven rotary engine with the RX8.

Generally, the hydrogen-fueled ICE offers at least competitive efficiencies at higher power ratings. It is easy to cool with conventional means and tolerant with regards to fuel impurities. As a robust, mature, and cost-efficient solution applications are envisaged for heavy-duty, long-distance, and off-road applications.

Hydrogen may also fuel ICEs as stationary gas motors, used for Combined Heat and Power (CHP). Also here the combustion technology is mature and robust and allows for fuel flexible solutions, where mixtures of natural gas and hydrogen might be used without major changes. The conversion of Stirling engines, frequently used for CHP, is even less challenging due to the different way these machines are provided with heat from exterior (for this reason Stirling engines are actually not ICEs).

**FIGURE 2.27**

Picture of prototype turbo-charged hydrogen engine MAN H 2876 LUH01.

Courtesy of MAN Nutzfahrzeuge.

2.4.2.2 Gas turbines

Conversion of gas turbines or even new designs of gas turbines to run on hydrogen is driven by companies like Siemens, Rolls Royce, and Kawasaki. Gas turbines provide high power densities at relatively low specific costs. However, the relative high temperatures involved and the working principle make them the preferred solution for middle to high power stationary installations or for aircrafts.

The replacement of hydrocarbons in gas turbines for power stations is driven by the increasing costs for CO₂ treatment or release. Gas turbines operated with hydrogen and oxygen avoid the need for removal of NO_x in the exhaust gases, but require a quite energy-intense air separation instead, if the oxygen is not produced together with hydrogen by electrolysis.

The conversion of an existing design to hydrogen fuel has to review and adapt the following components to the hydrogen-specific properties:

1. the fuel pump;
2. the fuel injection;
3. the fuel control unit;
4. the heat exchanger;
5. and in case the combustion chamber geometry.

For aircrafts the whole fuel systems, including piping and tanks etc., have to be replaced. For aviation, cryogenic storage of hydrogen might be the preferred option. If LH₂ is used as the stored fuel, however, thermal treatment of the fuel and pressurization for injection have to be considered.

Historically, there has been several projects for aircraft turbines running on hydrogen. Positive experience has been already made in the 1950s by US and Russian developers. In 1956/57 a B-57

Canberra was operated with liquid hydrogen. During the flight one of the two J65 turbines were switched to the hydrogen operation. Mainly the fuel control unit was changed.

In the former UdSSR it was the group of engineers around Kusnezow, which converted a NK-8–2U Turbofan-engine to LH₂ operation. With this installation the first time a full flight including take-off and landing was demonstrated. Similar activities were restarted in the 1980s, where Tupolev presented the first passenger aircraft TU154 running on hydrogen. About 10 years later Tupolev and Daimler – Chrysler Aerospace cooperated in the Cryoplane project.

Key characteristics of hydrogen-fueled gas turbines are the wide control range close to the lean limits, reducing the flame temperature and the specific NO_x emissions (Faaß, 2001). These advantageous properties are shown in Fig. 2.28.

2.4.3 Fuel cells

Fuel cells convert the chemical energy of the fuel directly into electric energy, which is the inverse function of an electrolyser, see Fig. 2.29.

Whereas in combustion the oxidation reaction takes place with the reactants, fuel, and oxidizer, mixed in the same volume, in the fuel cell the reaction is split into two partial reactions: the hydrogen (or other fuel) oxidation on the anode and the oxygen reduction on the cathode. Both reaction zones have to be separated to avoid mixing of the reactants and direct exchange of electrons like in the combustion reaction. This is provided by an electrolyte, which allows ions to flow and thereby close the electric circuit, but inhibits electron transport.

Therefore the functional elements are similar as in electrolyzers. Because of this proximity the technological challenges related to fuel cells are similar to those electrolyzers and new achievements in technology development are beneficial for both technologies.

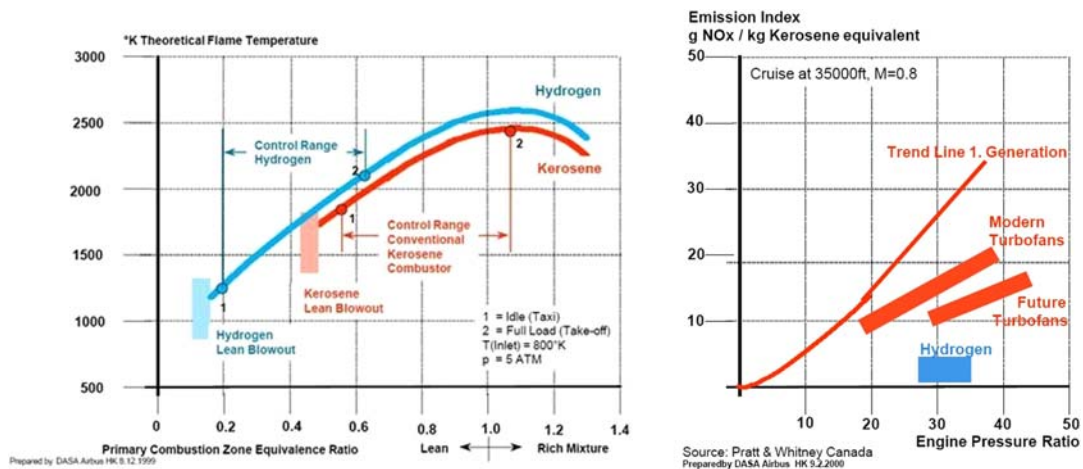
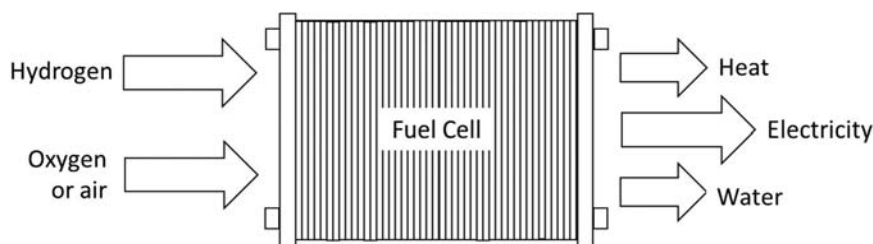


FIGURE 2.28

Flame temperature versus Equivalence Ratio (left) and NO_x Emission Index versus Engine Pressure Ratio (right) (Faaß, 2001).

**FIGURE 2.29**

Principal function of a fuel cell.

Fuel cells differ from other galvanic elements, in particular batteries, as they separate the functions of energy storage and conversion. The chemical energy, that means the educts are supplied from outside, whereas batteries include the chemical storage inside. Although fuel cells have still a relatively low power density, they are easily configured in modular systems.

The applications of fuel cells reach from

1. very small microsystems energy supply,
2. replacement of batteries in portable systems like laptops, digital cameras (see Fig. 2.30 left),
3. energy supply for electric mobility on roads, on water or even airborne,
4. stationary or mobile auxiliary power units, and
5. large blocks for industry power supply (see Fig. 2.30 right).

Although fuel cells have a quite long history their development was blocked by the dominance of alternating current technologies and of combustion engines, favored because of very cheap fossil fuels. Only in the 1960s the fuel cell development was pushed by the “space race” and a little later by the first “energy crisis” initiating a search for alternative and independent energy supplies. The very sensitive alkaline fuel cell (AFC) was then replaced by the more stable systems using acidic electrolytes. In the 1980s high temperature fuel cells, like the molten carbonate and the SOFC, were further developed. High temperature fuel cells provide higher power densities and are less sensitive to fuel impurities as they need less catalytic support. With the high temperatures they may integrate reforming of hydrocarbon feeds, what makes them suitable even for operations with natural gas as fuel, and the high temperature off-gas, water steam, might be used in further conversion processes, like a steam turbine, for increasing overall system efficiency. Therefore they have been applied successfully in CHP industry projects.

2.4.3.1 Thermodynamics and efficiencies of a fuel cell

As described in the introduction, the total reaction of fuel oxidation is split into two partial reactions on the two electrodes, the anode reaction and the cathode reaction. The actual partial reactions are determined by the ions transported via the electrolyte. Generally, electrons are delivered to the anode (generator convention) and extracted from the cathode.

The driving forces for these reactions are the different chemical potentials of the partial reactions. Without the ion conduction in the electrolyte or without the current in the external circuit a thermodynamic equilibrium (voltage) is generated and the reaction stops. Similar as for the

**FIGURE 2.30**

Scales of fuel cells.

Courtesy of Toshiba (left), Courtesy of Fuel Cell Energy (right).

electrolysers, the theoretically maximum voltages may be calculated based on fundamental energy terms involved. Due to the constraints imposed by the second law of thermodynamics only a fraction of the enthalpy, the Gibbs potential G , may be used for the voltage generation and the irreversible part $T\Delta S$ is lost as heat. Eqs. (2.24) and (2.25) define the ideal reversible cell voltages with this potential, the number of mols of electrons involved in the reaction $n = 2$, and the Faraday constant F , defining the electrical charge of 1 mol of electrons. The values with superscript l (“lower”) corresponds to the product water in gaseous phase, those with superscript u (“upper”) correspond to the reaction with liquid water as product.

$$U_{\text{rev}}^l = \frac{\Delta G_0^l}{nF} = 1.19\text{V} \quad (2.24)$$

$$U_{\text{rev}}^u = \frac{\Delta G_0^u}{nF} = 1.23\text{V} \quad (2.25)$$

Relating the maximum energy that could be converted to electricity, the Gibbs potential, to the total enthalpy of the reaction allows defining the maximum, ideal efficiency of a fuel cell.

$$\eta_{\text{max}} = \frac{\Delta G}{\Delta H} = 1 - T \frac{\Delta S}{\Delta H}; \eta_{\text{max}}^l = 83.3\%; \eta_{\text{max}}^u = 94.5\% \quad (2.26)$$

Eq. (2.26) exhibits inverse temperature dependency than ideal heat power cycles, in particular Carnot. Therefore efficiency decreases with temperature, making fuel cells especially attractive for small scale, distributed applications.

Real operations will introduce additional losses on membrane electrode assemblies (MEAs), stack, and system level, which will decrease the generated voltages and delivered electrical energy. Therefore real efficiencies are always considerably lower than those provided in Eq. (2.26).

The losses on MEA level are caused by the passage and transport of reacting gases at the electrodes, recombination losses on the catalyst, and resistive losses of the ion and electron currents. Typically, the losses at the anode are far smaller than on the cathode. The voltage efficiency describes the ratio of the actual voltage provided in operation and the reversible cell voltage:

$$\eta_U = \frac{\Delta U}{U_{rev}} = 1 - \frac{\Delta U_D + \Delta U_R + \Delta U_{Diff}}{U_{rev}} \quad (2.27)$$

with voltage drop due to a mixed potential (catalyst and oxidation of impurities on the cathode) ΔU_D , the so-called activation losses, resistive Ohmic losses ΔU_R , and the diffusion limitations of the gas supply/water removal ΔU_{Diff} . Typical values for ΔU are 0.6 – 0.9V associated with total efficiencies up to 75%. The losses are reflected in characteristic performance diagram provided in Fig. 2.31.

On stack level additional ohmic and parasitic losses, and on system level energy consumption of auxiliary system components like blowers, humidifiers, coolant pumps etc. have to be accounted for. Including those balance of plant efforts system efficiencies of up to 60% are realistic, even for small systems.

2.4.3.2 Fuel cell stack

As technical applications require higher voltages than those delivered by a single cell, several cells, consisting of MEAs, are connected in series, forming a stack. The MEAs are electrically connected by the bipolar plates (see Fig. 2.32), which serve as anode for one cell on one side and as cathode for the next cell on the other side. The bipolar plates provide a sophisticated channel structure for gas distribution to make sure that the gases—hydrogen on the anode side and air or oxygen on the cathode side—are equally distributed on the whole rectangular or circular cross-section of the stack and that the produced water is collected and removed properly. They contribute to the stack cooling and support the neighboring membranes or gas diffusion layer mechanically.

Therefore bipolar plates must be good electrical conductors, must withstand the corrosive forces introduced by the electric potentials and the electrolyte, and inhibit gas, in particular hydrogen

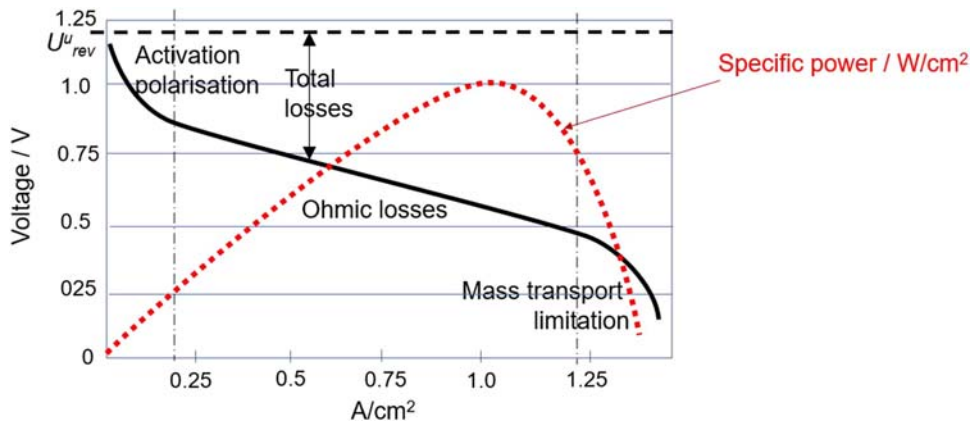


FIGURE 2.31

Voltage-current characteristic of a typical fuel cell.

**FIGURE 2.32**

Bipolar plates.

Courtesy of FuelCellStore.

permeation. High temperature fuel cells add high temperature and thermal stresses to these challenges. Either graphite or stainless steel with special coatings, depending on the fuel cell type, is used as materials for bipolar plates. The stack of MEAs is then packed in between the two end electrodes, which have a slightly different design than the bipolar plates.

2.4.3.3 Fuel cell types

Similar as electrolyzers, fuel cells are named by the electrolyte and categorized by the operating temperature.

2.4.3.3.1 Low temperature fuel cells

Low temperature fuel cells work at temperatures below 100°C. The product water is in liquid phase. At these temperatures catalysts are necessary to provide high enough reaction rates. The catalysts as well as the electrolyte require a high purity feed.

Alkaline fuel cell. These fuel cells use a liquid caustic potassium hydroxide solution (30–45wt%) as electrolyte. They can be operated in the wide temperature range of 20°C – 90°C, typically at 60°C.

The partial reactions on the electrodes are:



The AFCs are highly efficient as the kinetics of the limiting oxygen reduction reaction prefers the caustic electrolyte to acidic media. On the other hand, the serious disadvantage of this electrolyte is its sensitivity against carbon dioxide CO₂. Even a few ppm CO₂ (> 10 ppm), like the concentration in ambient air, will lead to the formation of carbonates, which are accumulated in the electrolyte and lead to a degradation of the fuel cell. Therefore the typical AFC needs high purity hydrogen and oxygen feed gases.

The low operation temperatures and the relatively small corrosivity of the potassium hydroxide solution do not cause serious material problems. All materials that favor the hydrogen dissociation, like the noble metals platinum, palladium, ruthenium, and their nickel or silver alloys, maybe used as catalysts on the electrodes. Therefore the anode material is usually Raney nickel. For the cathode Raney-silver is used, alternatively active carbon dotted with noble metal particles. Raney-nickel is a pyrophore nickel-based material, which is produced by alloying pure nickel with silver, silicium, or zink. After milling the nonactive metal parts are dissolved and a sponge-like metal structure remains, with a very high specific surface. The actual current conducting layer is made from pure nickel and connected to the electrode. The frames of the cells are made of plastics.

Other important components of an AFC are the electrolyte circulation pumps, phase separators, and electrolyte reprocessing (water extraction from anode side).

AFC systems have been the earliest developments and matured already in the 1950s, in particular in the United States and Europe. It has proven its function and applicability in military and space applications, in the NASA Apollo (see Fig. 2.33) and Space Shuttle spaceships for instance. The sensitivities to CO₂ and the quite complex liquid electrolyte management, however, have limited their market share. **Proton exchange membrane fuel cell (also polymer electrolyte fuel cell).** The electrolyte of the proton exchange membrane fuel cell (PEMFC) is a thin gas tight, proton conducting solid membrane. Nafion is the brand name for the sulfonated tetrafluoroethylene-based fluoropolymer-copolymer material for the membrane, which was discovered in the late 1960s by Walther Grot of DuPont.

The following partial reactions occur on the electrodes:



In the PEMFC the membrane has three functions:

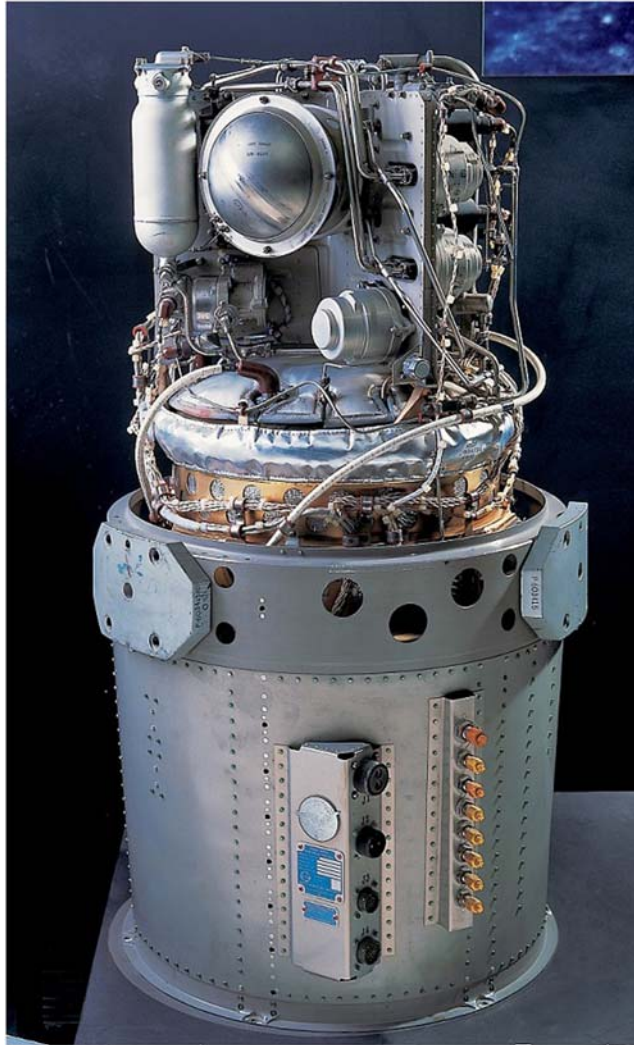
1. Ion conductor as the electrolyte.
2. Carrying the catalyst.
3. Separating the gas feeds.

Between bipolar plate and the membrane, the PEMFC usually has a gas diffusion layer. The principle elements of a PEMFC MEA are depicted in Fig. 2.34.

To maintain its ion conduction function the membrane has to be humidified. This limits the operational temperature range to 0°C – 100°C, as otherwise the membrane would be damaged either by freezing of the water contained or by drying out. The typical operational temperature range is 20°C–70°C. At those temperatures, in particular, the cathode has to be loaded with catalytic Pt particles, what makes the PEMFC quite sensitive to catalyst poisoning impurities and expensive.

Sufficient surface for catalysis has to be provided. The small solid Pt particles have to be accessible for the ions in the humidified solid electrolyte and for the educt gases. Liquid product water has to be removed from these three phases contact zones efficiently and the catalytic particles have to be electrically connected to the electrode.

Because of the thin electrolyte foil (50 – 150 μm) and its high conductivity, PEMFCs have a high current (5 A/cm² and higher) and power density and provide similar efficiencies like AFCs. PEMFCs generally show good start-up behavior and are well-suited for dynamic load following.

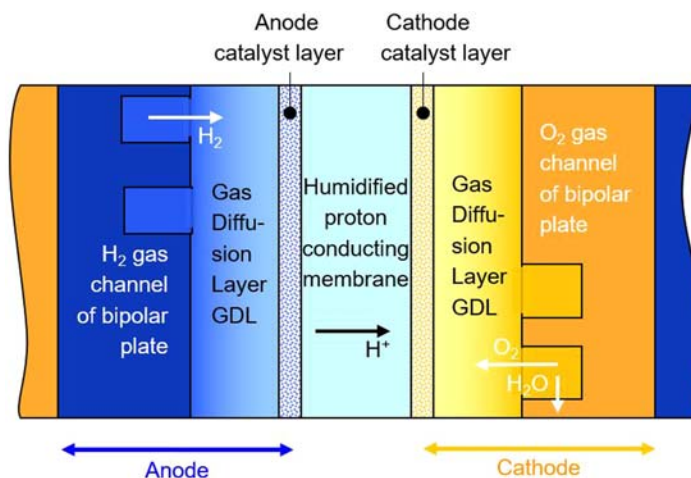
**FIGURE 2.33**

Alkaline fuel cell for Apollo mission.

Courtesy of NASA.

Because of the high power densities and general simplicity introduced with a solid electrolyte, PEMFCs are the preferred choice for mobility (see Fig. 2.35), portable and small stationary applications.

Direct methanol fuel cells are using similar set-up for the MEA. However, the efficiencies are far lower and the stability of the membranes in the presence of methanol is still a problem.

**FIGURE 2.34**

MEA of PEMFC.

Current research tries to shift the operational temperature to higher values ($\sim 140^{\circ}\text{C}$) by a pressurized operation or by new membrane materials. Those high temperature PEMFCs are promising with regard to higher power densities, better coolability, less expensive catalytic material, and lower sensitivity to impurities.

2.4.3.3.2 Middle temperature fuel cells

Phosphoric acid fuel cell. The phosphoric acid fuel cell (PAFC) is operated at 200°C with a highly concentrated phosphoric acid (H_3PO_4) solution as catalyst. The application of an acidic electrolyte allows CO_2 in the hydrogen and air feed, as the CO_2 does not react in the acidic medium. The partial reactions are identical to the PEMFC, what explains the broader communalities of both technologies.

The electrolyte is bound in a fabric-like matrix made from Teflon-covered SiC. Similar as for PEMFCs, the bipolar plates are made from graphite and the electrodes are designed as gas diffusion electrodes. The anode base material is usually a tungsten carbide alloy; the cathode base material is a graphite-platinum compound. Both porous media provide again large specific surfaces for the reactions.

Because of the higher temperatures the requirements for the catalyst and the purity of the feeds are lower compared to the PEMFC. It tolerates higher CO levels up to 1%mol. This makes the PAFC a good candidate for an operation coupled to natural gas reforming. In addition, the higher temperature makes PAFCs more flexible for multiple use of heat, steam, and electricity as frequently required in industrial-scaled (electric power 100–400 kW) stationary applications (see Fig. 2.36).

In some installations, like energy supply for compute centers, the fuel cell operation may reduce the oxygen level in connected closed rooms, what enhances fire safety and protection of valuable equipment including data storage (labeled “quadruple use”).

**FIGURE 2.35**

PEMFC stack NM12 with 135 kW for automotive application.

Courtesy of ElringKlinger.

2.4.3.3.3 High temperature fuel cells

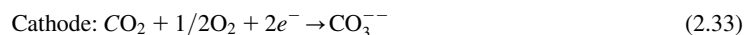
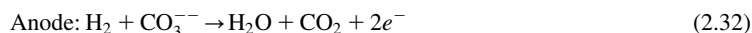
Molten carbonate fuel cell. The molten carbonate fuel cell (MCFC) is operated at approximately 650°C. It has carbon dioxide as a reactant and is therefore perfectly suited to cope with feeds characterized by high carbon impurities.

**FIGURE 2.36**

PAFC stationary installation with 100 kW.

Courtesy of Fuji Electric.

The reactions of the electrodes of an MCFC are:



The electrolyte is molten alkali carbonate (Li_2CO_3 , K_2CO_3) fixed by a ceramic matrix made from LiAlO_2 . The material for both electrodes is nickel. The nickel on the cathode is oxidized during start-up.

Noble metals for catalysts are not needed because of the high temperatures. A special feature of the MCFC is the gas management, which mixes the off-gas from the anode with its high CO_2 content in the cathode feed, usually air. At the cathode CO_2 and oxygen, air respectively, form carbonate ions, which are the ions transported in the electrolyte. Opposite to the other fuel cell types an increased level of CO_2 even improves its performance. Special designs allow integrating the reformer stage in the fuel cell system. This is why MCFC systems typically use natural gas as fuel.

One technical difficulty is the right material selection for the fuel cell frame's bipolar plates, as the molten carbonates are very corrosive. Another problem is the solution of the NiO from the cathode into the electrolyte. The solved nickel diffuses to the anode and forms there precipitates, which might even cause electrical shorts.

Internationally the MCFC technology has demonstrated its availability and performance in many demonstration projects. In Europe, for instance, CFC Solutions/MTU marketed the MCFC-based "Hot Module" (245 kW electric and 180 kW thermal power output), which was successfully demonstrated in industry and hospitals for CHP and backup power solutions. In 2011 the company stopped the further development and marketing for economic reasons. In the United States the company Fuel Cell Energy (before Energy Research Corporation) is participating in several demonstration projects, in Japan Hitachi, IHI und Mitsubishi Electric develop further the MCFC technology.

Solid oxide fuel cell. The SOFC is operated at highest temperatures, 800°C–1000°C. The reactions on the electrodes are:



Ceramics made from yttrium stabilized zirconium oxide functions as electrolyte. It conducts electrical current via oxide ions O_2^{2-} , therefore representing a caustic medium. At temperatures above 750°C this electrolyte provides a sufficiently good ion conductivity combined with gas tightness and negligible electron conductivity. The high temperature allows including the reforming stage in the fuel cell, similar as for the MCFC.

The anode is made from nickel zirconium oxide and the cathode material is based on mixed oxides like lanthanum-strontium manganese. Because of the small resistivity of the electrolyte and the even higher operating temperatures, the SOFC allows for higher current densities than the MCFC.

Worldwide there are several companies researching and developing SOFC technology, because of the high power densities and low sensitivities with respect to fuel impurities. Currently there are only classical planar types developed. In the past a tubular design was promoted by Siemens, see Fig. 2.37.

The high temperatures cause several material problems, for instance compatibility of the ceramic and metallic structures or chromium diffusion and relocation, causing electrolyte degradation. Fast start-up and cycling operation is widely limited because of thermal inertia and induced stresses. On the other side the high operation temperatures make coupling with a steam cycle process highly efficient. The SOFC produces electrical current and steam as offgas with at 850°C,

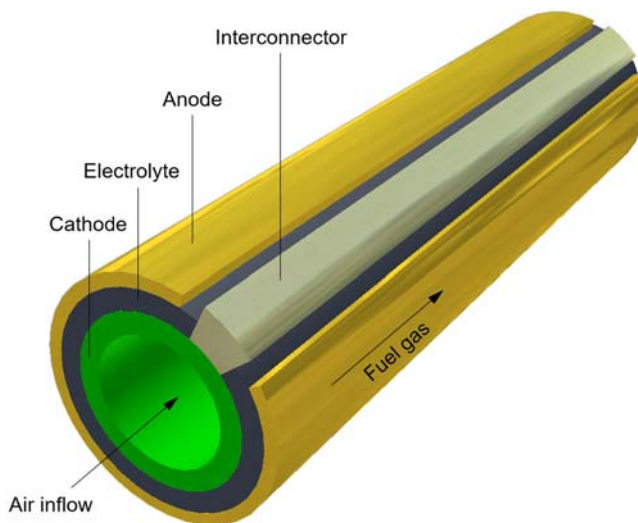


FIGURE 2.37

Tubular solid oxide fuel cells.

Courtesy of Siemens.

which is expanded in a steam turbine. The total chemical to electrical energy conversion efficiency of such a coupled process might reach 70%. Another attractive feature of SOFC is the potential for a reversible operation, which would allow to use the same installation as electrolyser.

Fig. 2.38 summarizes the presented fuel cell types with increasing operational temperature and ions transported via the electrolyte.

2.4.3.4 Fuel cell systems

As indicated in the efficiency considerations, the fuel cells need additional infrastructure, the balance of plant, to provide electrical energy. This infrastructure comprises the gas cleaning components, gas/air compression, off gas/water management, heat management like cooling cycles and an

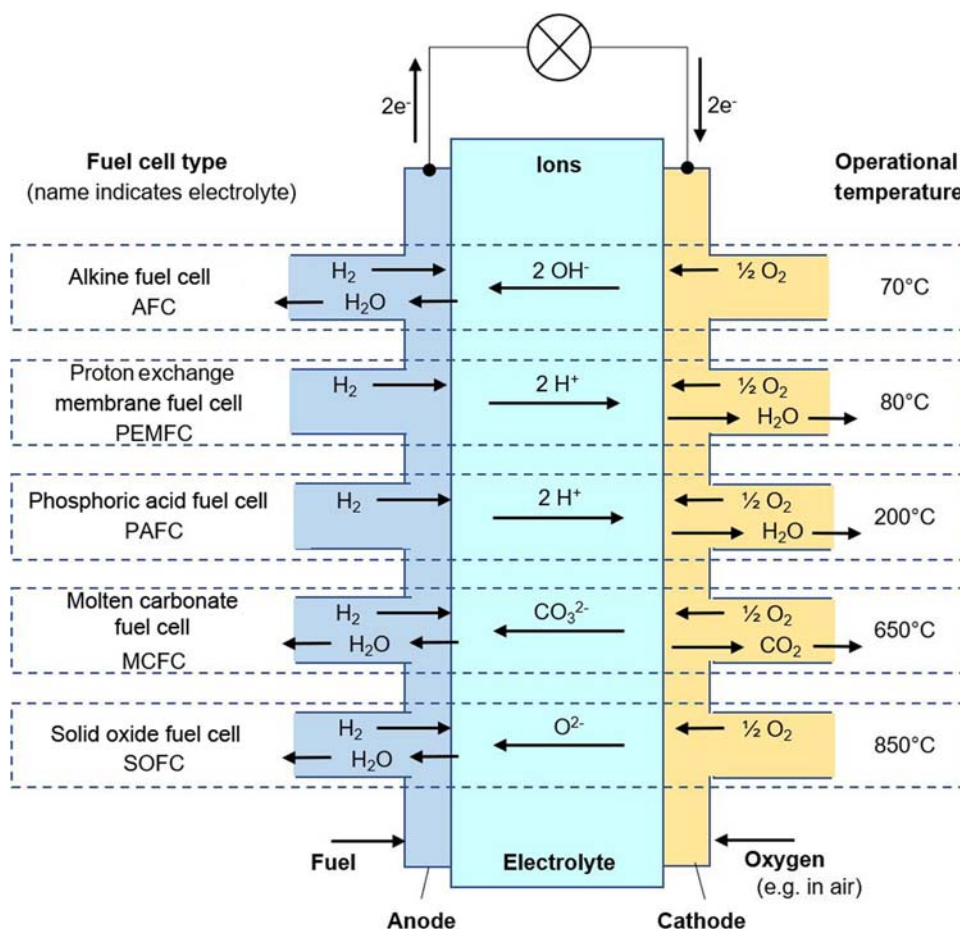


FIGURE 2.38

Fuel cell types with ions transported and typical operational temperatures.

electrical subsystem like inverter etc. Fig. 2.39 shows a light-duty vehicle fuel cell system with the stack and the balance of plant as used for a cost analysis of the US DoE (Papageorgopoulos, 2018).

Depending on the fuel cell type and the actual application there are different requirements concerning the purity of the fuel gas and the oxidiser. The purity requirements for the AFC, PEMFC, and partially also for the PAFC are quite high, which might increase the number of cleaning stages, which increases the system complexity and costs.

There is the general tendency that with increasing operation temperature

1. the purity requirements decrease,
2. the efficiencies decrease slightly,
3. material problems increase.

The current research efforts are targeting new membranes and catalysts for high temperature polymer electrolyte fuel cells with higher power densities, less sensitivity to impurities in the gases supply, and a better low temperature performance with less costly catalytic material. These systems are the most promising candidates for the currently designed fuel cell vehicles (FCVs). Other developments are addressing reversible solid oxide technology and alkaline membrane fuel cells, combining the advantages of AFCs and PEMFCs.

Two key problems are still to be resolved for a broader market entry: the price per kW has to be reduced further (factor 2 from the current large scale extrapolated 50 US\$/kW_{System} to about 30 US\$/kW_{System}), in particular for the automotive applications, and the lifetime of the MEAs has

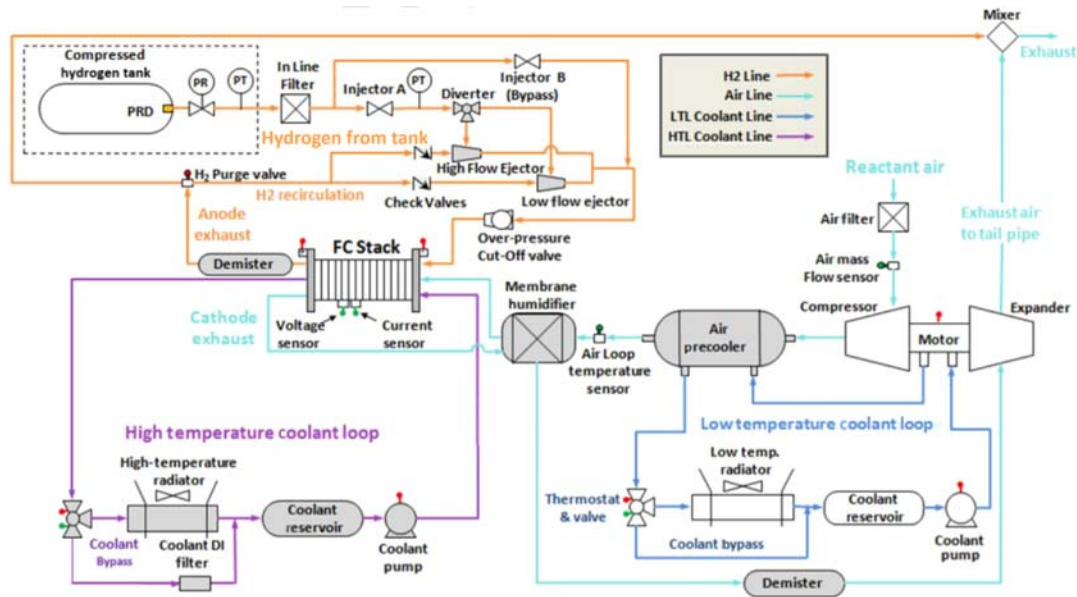


FIGURE 2.39

Automotive fuel cell system: fuel cell stack and balance of plant, including air loop, fuel loop, and high and low-temperature liquid coolant loops (Papageorgopoulos, 2018).

to be improved for stationary applications (also by a factor of approximately 2) particularly for natural gas fueled systems.

2.4.3.5 Specific safety issues with fuel cells

In fuel cells, usually only small inventories of hydrogen and oxygen are present. However, the huge surfaces of the very thin bipolar plates and membranes increase the probability for cross-over or membrane fracture with subsequent mixing of the reactants. This is furthered by the relative corrosive environment provided by the electrolytes. Catalytic elements might then act as ignition source.

The gas rooms on the electrodes have to be sealed to prevent leakage via the huge cross-section of the stack. Therefore small leakage of hydrogen from the stack has to be assumed and for many applications a vented housing has to be provided for the stack.

The off-gases, in particular from the anode side, may contain hydrogen. Some stacks still need regular purging with pure hydrogen and then considerable inventories of hydrogen will be released with the purged gas via the exhaust pipe. Hydrogen might also enter the cooling fluid and then accumulate in the reservoir container. Therefore typically catalytic hydrogen recombination and hydrogen sensors should be considered at the exhaust pipe exit and in the cooling fluid containers.

2.5 Hydrogen applications and their supply infrastructure

2.5.1 Introduction

Hydrogen applications are combining hydrogen storage, transfer and energy conversion, and in some cases even the production. The applications define systems with a minimum number of interfaces for exchanging hydrogen. Although this chapter generally focusses on new applications of hydrogen as an energy carrier, the first subchapter addresses established industry use cases. The industry applications provide particular opportunities for a fast decarbonization at large scale. This concerns applications like refining and ammonia production, where traditionally hydrogen is used, and also new innovative processes where additional hydrogen demand will be generated, like direct reduction for steel production.

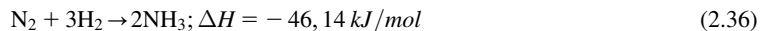
2.5.2 Industrial applications

In all concerned industrial processes green hydrogen may substitute the gray hydrogen produced. The strong pressure for decarbonization, exerted via an appropriate CO₂ pricing, accompanied by a continuous reduction of the price for green energy promotes new pathways for large scale green hydrogen supply or production, in particular electrolysis. Even in industry processes where hydrogen played a minor role so far, like steel production, green hydrogen will become an important carrier for green energy and will allow efficient and deep CO₂ reductions in applications otherwise difficult to address.

2.5.2.1 Green ammonia production

The annual worldwide production of ammonia by Haber–Bosch process is about 180–200 Mt, what accounts for a yearly demand of hydrogen of about 35 Mt, see [Dolci \(2018\)](#) and [IEA \(2019\)](#). The ammonia itself is mainly needed for fertilizer production.

The exothermal synthesis reaction of the Haber–Bosch process is provided in [Eq. \(2.36\)](#).



The reaction requires high pressures, 25 – 35 MPa, and medium temperatures, 450°C–550°C. Ferrite is used as catalyst.

China is the world's biggest producer of ammonia with about one-third of the world production. European ammonia production is of the order of 20 Mt. In a European context, a ton of ammonia requires about 8–10 MWh (feedstock and energy), if hydrogen is produced using SMR. The total energy consumption grows up to 10–12 MWh if hydrogen is produced through electrolysis. When SMR hydrogen production is involved, an average of about 1.6–1.9 tonnes of CO₂ are emitted per tonne of synthesized NH₃. The amount of CO₂ emitted rises to more than 4 tonnes of CO₂ per tonne of synthesized NH₃ if hydrogen is produced through coal gasification. A standard SMR hydrogen production route consumes even about 15% less energy than that required for producing the same amount of hydrogen through electrolysis. However, if electrolytic hydrogen is used, CO₂ emissions can be close to zero if the electricity source does not carry any significant carbon footprint, that is, if the electricity is produced by renewable energy.

Greening the hydrogen needed for ammonia synthesis through electrolysis will require additional significant and reliable sources of renewable electricity. Considering a so-called “global scale” plant producing around 800 kt of ammonia per year, the energy consumption of the plant using an electrolyser for hydrogen production will be around 8 TWh/year. An estimate provided by OCI nitrogen is that completely greening their annual hydrogen production (more than 1 Mt) would require an installed renewable capacity of around 3 GW (considering a capacity factor around 50%, like with off-shore wind). This corresponds to the entire currently installed renewable capacity in the Netherlands.

The substitution of fossil fuel-derived hydrogen with green hydrogen in the ammonia synthesis process is technically straightforward and does not present any challenge. There have been examples in the past (such as the Norsk Hydro Glomfjord plant), and there are currently some ammonia plants running on by-product hydrogen from other industrial processes such as the chlor-alkali process. The thermal balance of the plant does not seem to be critically impacted by the substitution of the SMR step by electrolysis. In particular, steam currently generated as a by-product of SMR can be easily substituted. Water consumption for electrolysis is not deemed to be a limiting factor.

2.5.2.2 Green refinery

With about 38 Mt per year, crude oil refining causes worldwide largest demand for hydrogen. Currently hydrogen is used in the refining process to remove the sulphur from the final products, or to maximize the conversion of heavier fractions into more valuable products (hydrocracking). Because of the increasing demand for refined oil products, mainly in Asia, and the decreasing qualities of the crude oil, the worldwide demand for hydrogen for refining is continuously increasing. However, in some economies, for instance in Europe, oil consumption is rather stagnating. The European refining capacity amounts to ~14 million barrels per day, equal to about 2 Mt of crude oil ([BP, 2018](#)).

Typically, the majority of the hydrogen used for these conversion processes is derived from cracking of the crude hydrocarbons, and it covers between 50% and 100% of the hydrogen required. The current need for additional hydrogen is usually covered by on-site SMR, which produces about 10 tons of CO₂ for each ton of hydrogen produced. A medium conversion refinery with a capacity of 10 Mt oil/year consumes approximately 50 kt of hydrogen per year. In this case, the ad hoc hydrogen production using SMR is around 30% of the total hydrogen produced by the refinery and accounts for about 15% of the current emissions associated with the refining industry (Dolci, 2018).

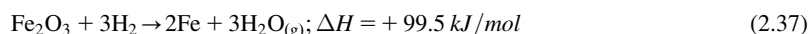
Completely replacing the 50 kt/y of hydrogen for an average refinery would require an installed electrolyser capacity of about 340 MW, operated with an estimated yearly energy consumption of about 2.5–2.75 TWh. As demand in the oil refining process is not really flexible, hydrogen production must be fairly constant, green electricity is not. Low capacity factors of the electrolyser and storage of hydrogen would generate additional costs. So, an intermediate option would be the introduction of a hydrogen production system where SMR and electrolysis are coupled and used flexibly or the import of green hydrogen. Introduction of significant electrolyser capacity in a refinery environment will also have to be carefully assessed from a safety point of view.

2.5.2.3 Green steel

Current steel production causes about 7% of the worldwide CO₂ emissions (Manfred Fischedick, 2014). Therefore considerable research and development activities have been addressing technologies with the highest long-term potential for CO₂ reductions in the iron and steel industry. The most promising technologies are carbon capture and storage in combination with a top gas recycling for the blast furnace, direct reduction with electric arc furnace (EAF), and a rather immature technology with high future potential, the iron ore electrolysis also called electrowinning (Stefan Lechtenböhmer, 2018).

Hydrogen direct reduction (H-Dr) refers to the use of green hydrogen in the direct reduction process (Valentin Vogl, 2018). Since the 1970s direct reduction processes are using natural gas or coal (Wörtler, 2013), whereas H-Dr uses hydrogen produced from renewable sources to reduce the carbon footprint.

Iron ore pellets are reduced to direct reduced iron (DRI), also known as sponge iron, along the endothermal reduction reactions, see Eqs. (2.37) and (2.38).



These reactions take place in a fluidized bed reactor, which minimizes sticking of the iron ore particles during reduction and allows the direct use of fine ore instead of pellets (Chunbao Charles, 2010). However, as the direct reduction does not allow the separation from gangue, only ores with high iron and low gangue content can be used.

Carbon is added to the process either in the pelletizing or in the direct reduction process. DRI has to be stored in special vessels, as it tends to reoxidize quickly under atmospheric conditions. However, DRI can be compacted to hot-briquetted iron (HBI), which allows for longer storage but reduces the porosity of the iron product. DRI or HBI are then transformed to steel in EAF, where also scrap can be added to the process.

Due to the necessary big scale of green hydrogen production, electrolysis is expected to be the main process for hydrogen production. Thus H-Dr runs mainly on electricity and can be almost CO₂-free. The process shown in Fig. 2.40 offers various options for flexible production and could be used to balance renewable power loads.

Recent analyses (Jernkontoret, 2018) suggested production costs slightly higher than those of current integrated steel plants. However, low electricity prices and a high carbon price, as well as the increased use of scrap in the process, can improve the competitiveness of H-Dr significantly. Import of green hydrogen via seaports from abroad or via pipelines from larger production hubs provide further options for large scale hydrogen supply and cost reductions in green steel production.

2.5.3 Material handling

Hydrogen energy and fuel cells have become competitive solutions and are widely applied in material handling in the warehouses of key companies. Major truck OEMs (Linde MH, Still GmbH, Jungheinrich, Crown Europe, Toyota Europe, Yale/Hyster, etc.) have their systems in commercial operation at customers mainly in North America and Europe (for instance at Colruyt, Belgium;

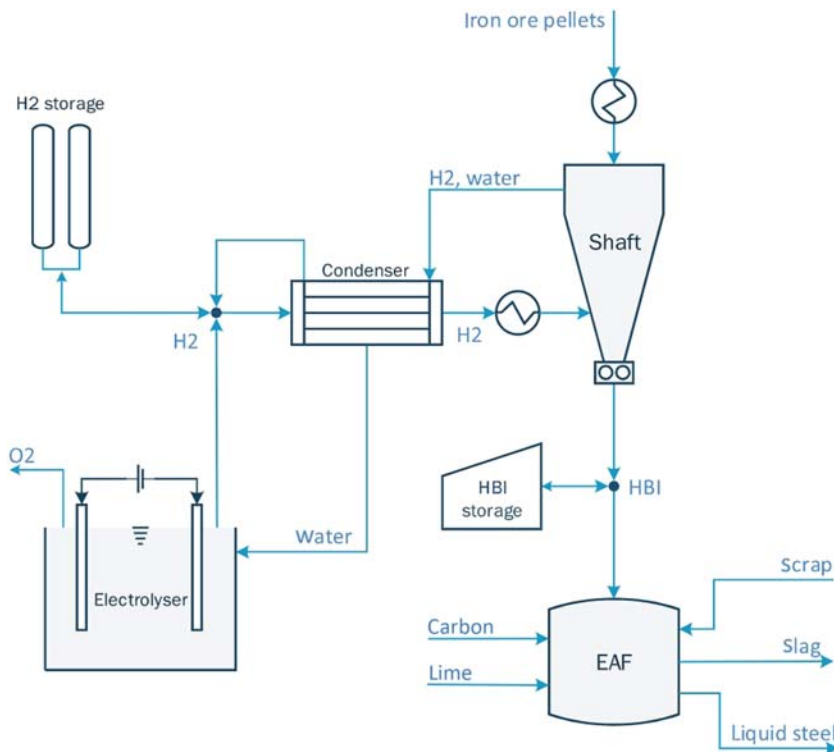


FIGURE 2.40

Hydrogen direct reduction process (Valentin Vogl, 2018).

Carrefour, FM Logistics, Prelocenter in France; BMW, Daimler in Germany, Honda in the United Kingdom; Walmart, Amazon BMW, Daimler in the United States). Fig. 2.41 shows representatives of the different truck classes.

Unlike batteries, fuel cells can be rapidly refueled, boosting productivity by eliminating the time and cost associated with battery change-outs and charging. Fuel cell-powered lift trucks can reduce the labor cost of refueling/recharging by up to 80% and require 75% less space as compared with battery recharging infrastructure. With their proven performance capabilities, short refueling times, and the potential for increasing productivity, fuel cells can be cost-competitive with batteries on a lifecycle basis (DoE, 2016).

In a three-shift operation and with indoor refueling, hydrogen-driven industrial trucks are reliable, safe, and cost-efficient alternative to conventional battery-driven systems. Forklifts may operate at consistent full speed and power for a whole shift. Refilling takes less than 3 minutes compared to 10–20 minutes battery charging times. The lifetime of a PEM fuel cell system is up to two times longer than for conventional batteries. Fig. 2.42 shows a prototypical hydrogen system for a material handling truck.

Indoor refueling (see Fig. 2.43) is state-of-the-art with robust standards, see SAE J 2601/3 Fueling Protocols for Gaseous Hydrogen Powered Industrial Trucks (forklifts) (SAE, 2014).

Following basic safety principles, however, the mass storage for those indoor refueling points are located outside the warehouses, typically in the free space with appropriate distance to the building and to doors or windows in particular.

2.5.4 Light duty and cars

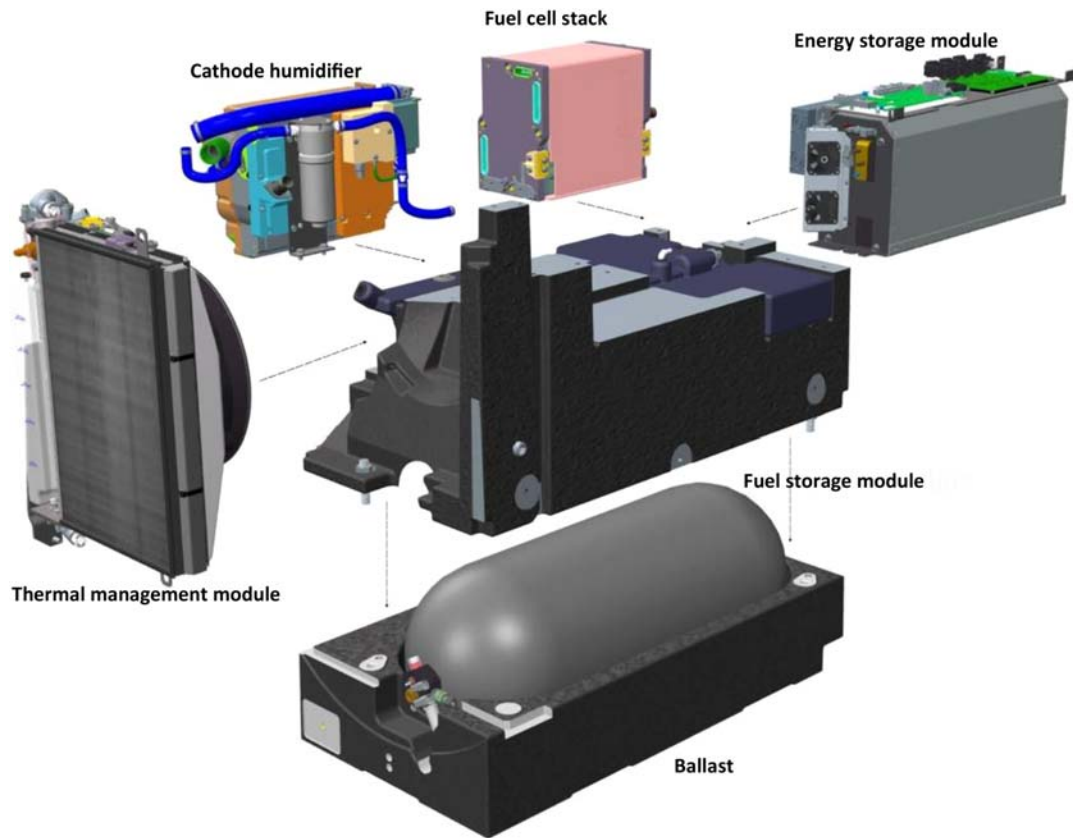
Despite the recent successful developments of mainly Toyota with Mirai and Mirai 2, Honda with the FCV and Hyundai Nexo (see Fig. 2.44), the light duty and cars technology are experiencing strong competition with battery electric vehicles. This competition is also impacting the development of corresponding fueling, charging, and infrastructure, respectively.



FIGURE 2.41

Classes 1, 2, and 3.

Courtesy of Plugpower.

**FIGURE 2.42**

Typical system for a material handling truck drivetrain.

Courtesy of Plugpower.

However, for long-distance or multiple shift light-duty transport, hydrogen-based range extenders for battery electric drivetrains or even dedicated hydrogen drivetrains may turn out to be economically more attractive than pure battery electric solutions.

A typical automotive system, like the one shown in Fig. 2.45, consists of a hydrogen storage with a refilling interface, and an onboard converter, which transforms the chemical energy into mechanical energy for propulsion. These basic elements have to be connected with suitable piping and equipped with further components, like valves and measurement and safety devices. Typically, the drivetrains are hybrid systems as the hydrogen subsystem is electrically combined with a battery. The battery is able to store and reinject electrical energy at high power, but is limited in the actual energy content. The less dynamic hydrogen system provides the main energy for achieving long ranges and short refueling times, similar as those of conventional fossil fuel-based mobility.

**FIGURE 2.43**

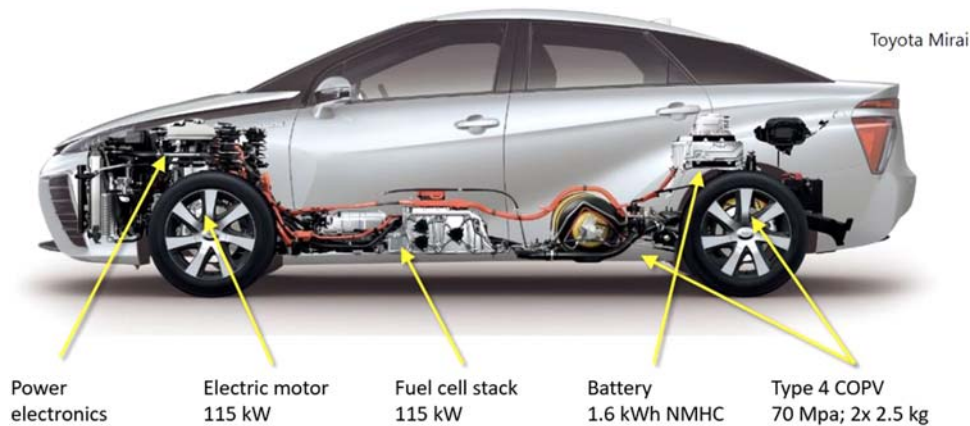
Indoor refueling station.

Courtesy of Plugpower.

**FIGURE 2.44**

Hyundai NEXOs refueled at hydrogen filling station in Saclay, Paris.

For mobile applications, besides the volumetric density the gravimetric aspect is also important. Therefore for a hydrogen-driven car, the tank system has to store more than 7% weight hydrogen to be acceptable for automotive industry. This explains the necessity for light-weighted structural material like glass and carbon fiber composites etc. For onboard storage of typically 3–7 kg of

**FIGURE 2.45**

Hydrogen drivetrain of the MIRAI.

Courtesy of Toyota.

hydrogen, currently compressed gaseous storage of hydrogen in Type 4 composite overwrapped pressure vessels operated at 35 MPa or 70 MPa nominally is the most mature solution.

The refueling of these vessels requires a relatively fast pressurization from a minimum pressure in the order of a few MPa to the nominal pressures, associated with a 100% state-of-charge. This might even require excess pressure of about 15%. As this fast pressurization heats up the gases in the pressure vessels considerably and the compound material used for Types 2 – 5 are quite sensitive to elevated temperature (glass transition temperature for carbon fiber composite $\sim 120^{\circ}\text{C}$), the hydrogen has to be precooled, at least for the 70 MPa refueling. The SAE J2601 standard (SAE, 2014) prescribes different precooling temperatures, either in the order of -20°C or even -40°C to keep the temperature of the vessel materials below 85°C , which has been defined as a limit. Therefore the pressure vessel itself, and its integration, has to account for the thermal and mechanical cycling induced by the refueling process.

The armature of the pressure vessels usually integrates a pressure reduction/limiter function and a thermally initiated pressure relief device (TPRD) for protection against external heat attack, fire loads respectively. TPRDs are usually realized by glass break—historically by eutectic metal fuses. The latter showed deficiencies, and in a few accidents the eutectic melt was prematurely frozen. So the fuse did not open fully and the vessel ruptured. Usually there is no safety valve foreseen to protect the vessel against overpressurization by refueling. The connection(s) to the energy converter is usually realized with a pressure reducer and a small diameter piping operated at moderate pressure of less than 1 MPa.

There had been liquid hydrogen storage solutions demonstrated by BMW with a small series. Although offering certain advantages with the low pressure and high densities the boil-off of cryogenic liquid hydrogen represents a considerable problem. Fig. 2.46 shows a BMW Hydrogen 7 at a refueling at the CEP hydrogen filling station in Berlin.

In principle, the problem related to the boil-off might be mitigated with a cryocompressed storage, typically operated at 50K and 35 MPa nominally. Such a system might be filled with cryocompressed hydrogen, LH_2 , or with compressed 35 MPa and represents a quite versatile solution.

**FIGURE 2.46**

BMW Hydrogen 7 liquid filling at the CEP hydrogen filling station in Berlin.

Courtesy of BMW

Solid storage solutions, in particular conventional metal hydrides (Fe and Ti-based storage material), are considered to be too heavy for light duty and cars. Although the light metal hydrides come close to the 7% weight performance (mass hydrogen/mass storage system), they require complex thermodynamic management for heat generated during refilling and required for extracting hydrogen.

The conversion of the chemical energy might be realized in a fuel cell or in a conventional ICE. As indicated above, both concepts may be hybridized by combining the converter with a battery and electric motors. The fuel cell, however, offers full electrification with potentially higher conversion efficiencies compared to conventional ICE-based system. The hydrogen ICE provides stronger fuel flexible solutions, more experience, and typically higher robustness at lower costs. For energy-intensive transport, like off-road and heavy-duty long-distance applications, the hydrogen ICE might provide even higher efficiencies than a fuel cell-based drivetrain.

2.5.4.1 Light-duty vehicle refueling

At the end of 2019 more than 430 hydrogen fueling stations were in operation worldwide. Three hundred and thirty stations were accessible as public stations ([Ludwig-Bölkow-Systemtechnik, 2021](#)). More impressive is the dynamics, as there was an increase of the public stations by a factor of 4 compared to the number of public stations in 2015. There are fueling stations integrated in conventional fueling stations and dedicated hydrogen fueling stations. The ISO 19880 provides the internationally agreed set of standards for the light-duty, high pressure gaseous fueling.

A typical fueling station, as shown in Fig. 2.47, consists of a mass storage, typically gaseous or liquid if liquid hydrogen is delivered, a compressor, several parallel high pressure storage buffers for cascade fueling, a cooling block for the 70 MPa option, a dispenser including metering, and a standardized fueling connector. For safe and efficient fueling ambient temperature, information on hydrogen tank temperature and pressure must be determined with appropriate measurement devices. In addition, a break-away coupling and electrical grounding must be provided.

The hydrogen might be produced on-site by a reformer (fuel processing technology) or by on-site electrolysis. Alternatively, the hydrogen may be delivered by batch transport, liquid or compressed, or by pipeline. For the case with liquid mass storage, either a vaporizer is needed before the compressor or a cryogenic pump might be used. For the latter a thermal conditioning might be needed. The functional relations of the main components are shown in the flow diagram in Fig. 2.48.

2.5.4.2 Safety issues with light duty and cars including the respective infrastructure

The requirements for ensuring safe design and operation of the hydrogen vehicles are set by the UNECE Global Technical Regulation on Hydrogen and FCV GTR 13. These requirements are legally binding for automotive industry.

The standards and protocols for the fueling are defined in the above-introduced SAE and ISO standards. They typically become legally binding, only when referred in respective regulation.

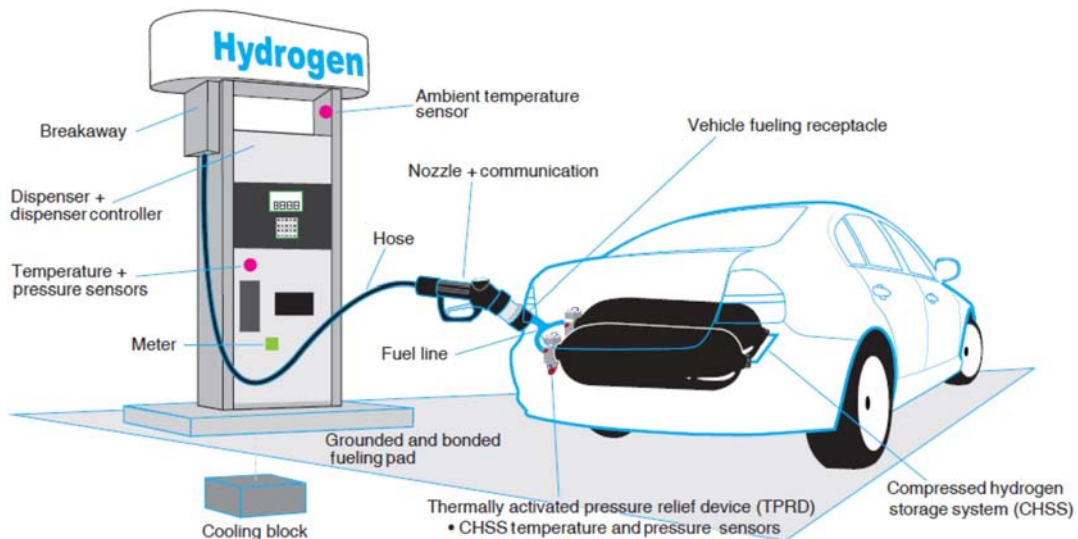
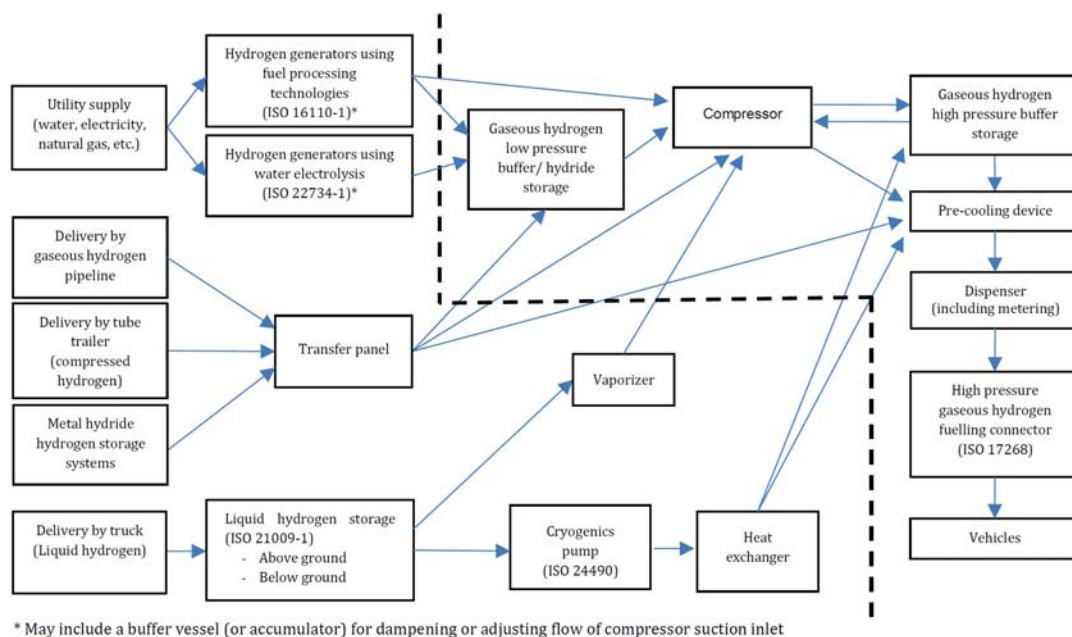


FIGURE 2.47

Example of the fueling station dispensing system key components including the vehicle high pressure hydrogen system.

Courtesy of ISO.

**FIGURE 2.48**

Functional relations of the main components of a light-duty fueling station.

Courtesy of ISO.

Only a few accidents with light-duty vehicles have been recorded. None of the published accidents was linked with catastrophic failure of the high pressure storage system. However, it should be noted that failure of onboard pressure vessels was observed with compressed natural gas vehicles. Some of those accidents were associated with different technologies (Type 1), but others were using quite similar pressure vessels (Type 4) as used for hydrogen vehicles. The root causes were undetected mechanical predamage, mechanical impact, and thermal or chemical attack.

A systematic assessment of fueling station accidents (Junji Sakamoto, 2016) showed that most of the cases are rooted in defective sealings or in installation errors, mainly inadequate torque of screwed connections and fittings, respectively. The vibrations introduced by mechanical compression are generating additional cyclic loads, which have to be accounted for in the design of systems and the high pressure connections in particular. Besides, noncertified components and welding defects were causing incidents or accidents. Three major accidents with considerable financial losses and human injuries have been reported so far. The station in the City of White Plains, NY, United States, August 21, 2008, involved a fire and smaller explosions. It was caused by a pressure switch, failing because of inappropriate welding material. The accident in Rochester, NY, USA, August 26, 2010, destroyed the whole facility and injured two persons. The root cause is unknown. The recent accident at the Sandvika (Kjorbo) station in Norway, June 10, 2019, was caused by inappropriately mounted, screwed connection to one of the high pressure storage containers (see Fig. 2.49). Lessons learnt from these accidents have to be implemented in the next generation of

**FIGURE 2.49**

Hydrogen explosion at hydrogen station at Sandvika, Norway.

Courtesy of nrk.no.

fueling stations, while reducing overconservatism, which is causing extra costs and raising unjustified concerns in the public view.

2.5.5 Buses, heavy duty, and railways

2.5.5.1 Buses

Bus and heavy-duty applications have the advantage that whole fleets might be operated with a small number of dedicated fueling stations. Buses provide visibility and heavy duty provide scale. Moreover, for long distance heavy-duty transport, hydrogen FC and ICE-based drivetrains are superior to battery electric drivetrains, as the latter are too heavy, have too long charging times, and finally are too expensive.

There are different types of buses. Standard or standard city buses have a typical length of 12 m. Articulated buses with a typical length of 18 m provide more space for passengers. Fuel cell and H₂ ICE-based buses were built for both categories.

Several bus projects, like the European CUTE, HyFLEET-CUTE, CHIC, HYTRANSIT, HIGHVLOCITY, JIVE, JIVE2, H2BUS Europe, or the US transit bus projects, have been demonstrating the feasibility of zero emission public transport with green hydrogen and helped maturing the involved technology. Most of the FC buses built by VanHool, EvoBus, Solaris, Wrightbus, Skoda, VDL, Hino, etc. so far were using similar platforms as for diesel hybrid or battery buses. The small series implied huge costs and prices, which only allowed for publically subsidized operations. MAN was developing and demonstrating a few H₂ ICE-based buses for Munich airport services and Volvo was refurbishing their natural gas ICE for mixtures of natural gas and H₂.

Similar as for the light-duty vehicles also the H₂ FC buses are hybridized, which means that they have a battery or a supercapacitor for recuperation and for drive modes requiring higher than average power, in particular acceleration. Supercapacitors are less frequently used. They typically store less energy but they are less sensitive to low temperatures than batteries. Either serial or parallel hybridization with small or large batteries are used. In the extreme, the latter case is rather a battery electric bus with an FC-based range extender. The typical power rating for an FC drivetrain

with a small battery is about 120 kW. This is surprisingly low, but the FC only has to supply the average power required, peak demand is satisfied with the battery. The electric coupling of battery and FC requires direct current converters, which had to be developed specifically for these applications. All in all, the recuperation increases efficiency in particular for inner city routes, with frequent stops and low average velocity.

A prototypical set-up of a city bus is shown in Fig. 2.50. The hydrogen storage vessels, the stacks, and other subsystems, like the batteries and the cooler fans for the separate FC cooling media, are mounted on the top of the bus, as city buses usually require a low floor technology. Together with the air conditioning system also mounted on the roof, this shifts the center of gravity upwards and the vehicle thus tilts considerably more when cornering, if the chassis is not modified accordingly. The electric motor equipment varies from one central electric motor, saving some costs, to wheel hub drives, increasing the moving mass. All electric motors still come with gear boxes.

The typical efficiencies are in the order of 7–10 kgH₂/100 km depending on the topology of the route, the driver, and weather conditions. This includes all comfort functions like air conditioning, pneumatic drives for doors etc. Only half of the energy of such a modern bus is actually required for driving. The current standards for hydrogen storage are Type 4 vessels with a nominal working pressure of 35 MPa. As the operations of the vessels require always a minimum pressure of about

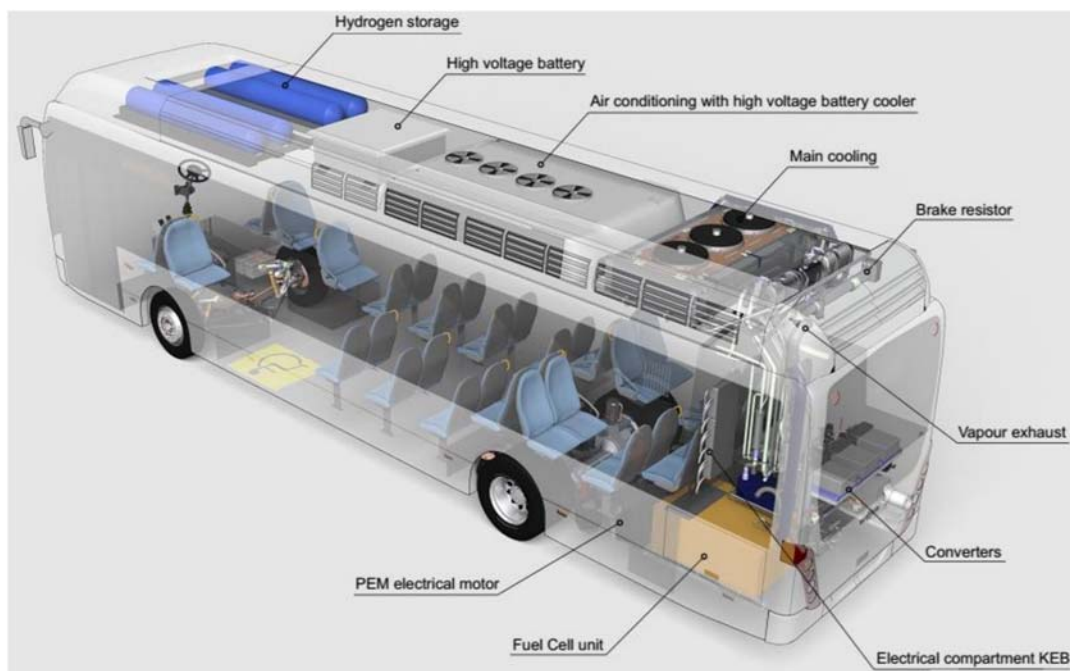


FIGURE 2.50

A prototypical set-up of a city bus.

Courtesy of EvoBus.

5 MPa, not all the total nominal stored may be extracted. Thus with typically available 30 kg a range of up to 450 km is practically possible, even at low temperature conditions.

2.5.5.2 Heavy-duty trucks

Quite demanding CO₂ fleet emission requirements for heavy-duty applications make hydrogen as a clean fuel more and more relevant. Tax incentives further promote this emission-free commercial transport. For 40 tons total weight and required ranges in the order of 1000 km only hydrogen provides an economic solution for deep decarbonization in road transport.

For storage of up to 100 kg of hydrogen currently CGH₂ at 35 – 70 MPa, cryocompressed hydrogen, and liquid hydrogen are investigated. The currently most mature storage solution is inherited from the bus systems and consists of a set of 35 MPa Type 4 vessels, which are typically integrated behind the driver cabin or above the rear axis. The LH₂ cryostats might be positioned at the same locations where the conventional fuel diesel is stored. Two cryostats, each with about 500 L empty volume have to be installed for the envisaged ranges. For energy conversion, either PEM fuel cells or a H₂ ICE might be chosen depending on the actual application and further criteria. Technologies may be also derived from the bus application, where more experience is available.

In June 2017 the Zurich engineering company ESORO received the road approval for the world's first FC heavy-duty vehicle. It built and developed the fuel cell truck in the 35-tonne category with the necessary load capacity able to fulfill Coop's logistical requirements. The truck was powered by a PowerCell S3 PEM FC stack.

For a field test of commercial haulage firms Hyundai shipped the first 10 fuel cell XCIENT trucks (see Fig. 2.51) to Switzerland in 2020. The aim of the Swiss project driven by Coop is to have 1600 trucks in service by 2025, meaning that there is scope for production on the order of hundreds of units a year.

XCIENT is powered by a 190-kW hydrogen FC system with two 95-kW FC stacks. Seven large hydrogen tanks offer a combined storage capacity of around 32 kg of hydrogen. The driving range



FIGURE 2.51

XCIENT fuel cell truck.

Courtesy of Hyundai.

per charge for XCIENT Fuel Cell is about 400 km, which was developed with a balance between the specific requirements from the potential commercial fleet customers and the charging infrastructure in Switzerland. Refueling for each truck takes approximately 8 to 20 minutes. The trucks provide sufficient range for the specified international distribution services. Hyundai's next goal is to reach 1000 km between refueling stops, which should come close to the average trucker's shift length.

Similar developments are on the way by other truck builders, like Toyota, Nikola, Iveco, MAN, and Daimler Trucks. The latter presented the GenH₂ prototype with an LH₂ tank system in 2020.

2.5.5.3 Railway transport

Similar as for heavy duty, railway transport offers relative huge potential for CO₂ savings. Especially for services in remote areas and tracks without electrification, the replacement of diesel locomotives is economically attractive, accounting for the railway-specific lifetimes and amortization cycles. "Hydrail" is the generally used term describing all forms of rail vehicles, large or small, which use on-board hydrogen fuel as a source of energy to power the drivetrain with an FC or an ICE as the central energy converter.

After several design studies and prototypes, which were also stimulated by the idea of introducing fuel cell-driven locomotives for mining services in Canada and South Africa, the first commercial hydrogen fuel cell-driven passenger train Cordia iLint was presented by Alstom in 2014. The first train was delivered as part of a €81 million contract signed in November 2017 to supply 14 Alstom iLints to Lower Saxony, Germany by 2021. Letters of intent with three more northern German states for 44 trains and deliveries to Austria demonstrate the attractiveness of the technical solution.

The Cordia iLint drivetrain consists of a 390-kW underframe-mounted traction motor. The serial hybrid unit has a 200-kW fuel cell that charges a 225-kW battery to give a peak power output of 425 kW per coach. The maximum speed of the 107-ton heavy train is 140 km/h. On routes with frequent stops, energy savings from regenerative braking of around 30% is feasible. Roof tanks on each coach hold 89 kg hydrogen at 35 MPa giving a range of between 600 km and 800 km. Refueling is possible within 15 minutes (Shirres, 2018).

2.5.5.4 Safety issues with buses, heavy duty, and railways

Public street operations of commercial vehicles are rough and demanding. The stored inventories are huge compared to light-duty vehicles. The buses involve many passengers and the storage vessels mounted on the roof are quite vulnerable to fire attack and mechanical impact (e.g., too low tunnels).

The fueling of the captive fleets is usually realized at dedicated stations with professionally trained staff. The steady operations and protected environment evidently improve the safety performance. Therefore only a few minor incidences are known from the CUTE bus projects for instance.

Neither for the buses nor for the heavy duty there are standards for fueling, no standards for crash tests and no commonly accepted scenarios for risk assessment in general. However, work for refueling protocols in international projects has started.

2.5.6 Maritime

Transport by ships is the most cost-effective means of moving long-distance freight, representing about 90% of the global physical trade by volume. The maritime sector alone consumes approximately 5% of the global oil demand, and is responsible for about 2.5% of the global energy-related emissions of CO₂ (IEA, 2019). The extensive use of diesel and heavy fuel oils in the maritime sector results in significant emissions of sulphur and particulate matter, which contribute to the deterioration of the air quality in ports and harbors. The International Maritime Organization (IMO), until 1982 the Inter-governmental Maritime Consultative Organization, is a specialized agency of the UN dedicated to the safety and security of shipping, as well as the prevention of marine and atmospheric pollution from ships. In 1973, IMO adopted the International Convention for the Prevention of Pollution from Ships (MARPOL), with an objective to minimize pollution of the oceans and seas. In 1997, the Marine Environment Protection Committee in IMO adopted MARPOL Annex VI on air pollution.

Because of the relative huge CO₂ footprint and the increasingly demanding regulations, hydrogen is gaining a lot of attention for maritime applications. However, the spectrum of potential applications is wide, from fuel cell-driven military submarines (HDW Thyssen-Krupp), via sports and tourist boats, inland waterway and international transport to large passenger ships. For larger ships and larger ranges liquid storage is the preferred storage option. For safety critical applications, like the submarines, solid hydrides are used.

Recently, there has been a focus on hydrogen for ferry services. Besides design studies like the US Seabreeze, the US Water-Go-Round and the Norwegian ferry built by NORLED (see Fig. 2.52) will be first vessels that go for commercial services.

Most of the current designs foresee on-deck installation of all hydrogen technology for better ventilation performance. Although hydrogen is very light this design shifts the gravicenter up and would make a mono-hull design quite instable. This and the better use of space favors a multihull or catamaran design.



FIGURE 2.52

On-deck arrangement of LH₂ storage, fuel cell, and vent mast on the NORLED ferry.

Courtesy of NORLED.

2.5.6.1 Bunkering

In maritime applications fueling is labeled bunkering. In a feasibility study on bunkering of gaseous fuels (Finn Vogler, 2012), three different bunkering methods were identified: shore-to-ship, ship-to-ship, and truck-to-ship, see Fig. 2.53. The prevalent method of bunkering is the bunker boat method. The arrangements between the bunker vessels and the ships can be of very different kinds.

The liquid hydrogen bunkering process includes inert gas purging, pipe cooling, and the refueling itself. To bunker a large container with a capacity of 3500 kg LH₂, 4 hours of purging and cooling and 1 hour of refueling are required.

Alternatively to bunkering hydrogen in gaseous or liquid state via a transfer line, a batch based bunkering with swapping mobile tank units, containers respectively, could be an attractive solution. The full containers are loaded onto the ship and the empty ones removed later. While the ship is in motion, the hydrogen is pumped from the mobile add-on tank into the tank on the ship allowing to decrease the times of stops for refueling significantly.

2.5.6.2 Safety issues related to maritime applications

Ships are often operated under very demanding conditions, with strong impacts caused by the weather and the sea in far-off and unprotected locations. Evacuation of human beings and intervention from outside are difficult. This is why there is a long history in special precautions and live-

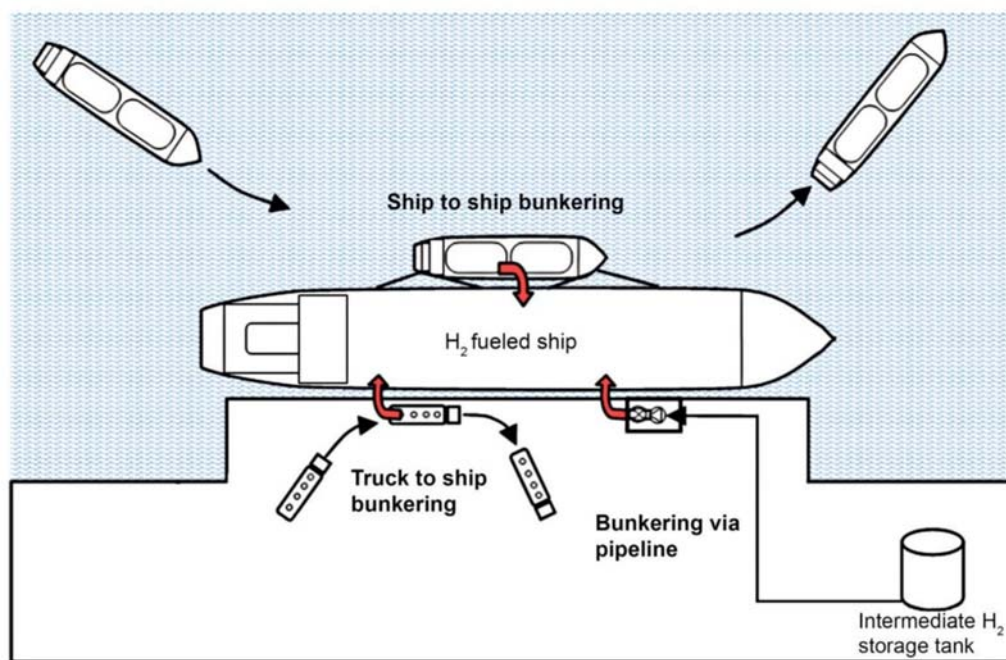


FIGURE 2.53

Three possible ways of bunkering hydrogen (Finn Vogler, 2012).

saving regulations for maritime applications, which are typically more demanding than for other technologies. On top comes hydrogen as a flammable gas with its special hazard profile.

Special topics not yet resolved are:

1. Optimal large scale venting strategies—radiation/blast loads from ignited events.
2. Tolerable blast and impulse loads (how high pressures are tolerable for structures and people when duration is only a few milliseconds).
3. Ventilation strategies for underdeck installation of hydrogen storage and connection to the power converter and acceptability of significant releases (5 g/s, 10 g/s, 20 g/s) into confinement.
4. LH₂ spills, material compliance, and ventilation of cryogenic gas mixtures.

2.5.7 Aircraft and space

As aircraft and space applications are highly weight critical, hydrogen is actually the ideal fuel. Especially for space applications cryogenic liquid hydrogen and liquid oxygen have been the preferred binary fuel for rocket motors since the first developments. In addition, for rocket motors the simple molecule and the associated high conversion of heat into translational velocities yield a relative high thrust.

Supported by the experience gained in the space missions in 1960 to 1980 several studies of liquid hydrogen-fueled aircrafts have been initiated in the United States and in the former UdSSR. In the late 1950s a B-57 Canberra was operated with liquid hydrogen. During the flight one of the two J65 turbines were switched to the hydrogen operation. Mainly the fuel control unit was changed and an evaporator was added. A little later in the former UdSSR a group of engineers around Kusnezow converted an NK-8—2U Turbofan-engine to liquid hydrogen. With this aircraft the first time a full hydrogen-fueled flight including take-off and landing was demonstrated. Similar activities were restarted in the 1980s, where Tupolev presented the first passenger aircraft TU154 running on hydrogen. In the 1990s Tupolev and Daimler – Chrysler Aerospace cooperated in the Cryoplane project. All these developments with refurbished gas turbines culminated in the 2020 announcement of Airbus to introduce its first liquid hydrogen-fueled mid-range aircrafts by 2025.

Only recently electric flying and the demand for drones with long mission times made hydrogen fuel cells an increasingly interesting option.

The Antares DLR-H₂ research aircraft is the world's first manned aircraft powered exclusively by hydrogen fuel cells. The official first flight of the Antares DLR-H₂ took place in Hamburg on July 7, 2009. In November 2009, the fuel cell's ability to function under low pressure conditions at an altitude of more than 2000 m was demonstrated at a flight altitude of 2558 m. The aircraft uses a single Type 4 composite vessel for compressed gaseous hydrogen (CGH₂) storage. The vessel was clamped externally beneath the right wing. The fuel cell system was mounted beneath the other wing in a special container.

In 2019 a 800-W fuel cell power module by Intelligent Energy was integrated in an electric drone to achieve a new Guinness World Record for the longest multirotor flight time. The flight was conducted by South Korean company MetaVista. The drone, shown in Fig. 2.54, flew for 12 hours, 7 minutes and 5 seconds using liquid hydrogen to fuel the module. The previous record stood at 2 hours, 6 minutes 7 seconds.

**FIGURE 2.54**

World record drone fueled with liquid hydrogen.

Courtesy of MetaVista.

2.5.8 Portables

Several prototypes for portable applications have been developed, mainly as a light replacement of batteries for laptop computers or professional TV video cameras. A few grams of hydrogen are stored in small metallic pressure vessels or in metal hydride cartridges. This would require the establishment of a special supply chain for providing centrally filled and returning empty storage containers. This infrastructural requirement and the steady improvement of battery performance hindered a broader success of portable hydrogen applications.

2.5.9 Heat applications

Decarbonization of heat production via hydrogen is addressed in several national strategies and tested in several projects, mainly in Germany and the United Kingdom. Generally, the main pathway is characterized by a continuous increase of injection of green hydrogen into the existing natural gas grid, up to 30% by volume, and a later construction of pure hydrogen transfer lines, as a sort of green backbones for the gas grid. A more mild transition is offered by using the green hydrogen and captured CO₂ for producing synthetic natural gas, which can be injected into the natural gas grid without limitation.

The characteristics of hydrogen as a gas for heating have been introduced in the chapter on ICEs. Compared to methane it has a lower oxygen demand, reduces emissions, comes with lower density and viscosity, higher diffusivity, a wider flammability range, and higher laminar flame speed. [Table 2.3](#) lists the most relevant properties of methane and hydrogen and mixtures of both.

The relatively high autoignition temperature and small minimum experimental safe gap width render hydrogen as a gas of the group IIC according to [European Committee for Electrotechnical Standardisation CENELEC \(2018\)](#). [Table 2.4](#) provides an overview on the classification groups and some examples for the different classes.

Table 2.3 Comparison gas properties of methane, hydrogen, and mixtures.

Gas	Flammability limits x_i in vol. %		Limiting oxygen concentration Mol%	Lower heating value $H_{f,n}$ kWh/m ³	Wobbe index $W_{s,n}$	Relative air demand 1
	LFL	UFL				
100% CH ₄	4.2	16.6	10.1	11.06	14.85	100%
10% _{vol} H ₂ in CH ₄	4.2	18.2	9.6	10.12	14.40	91%
25% _{vol} H ₂ in CH ₄	4.2	21.2	9.1	9.18	13.95	81%
50% _{vol} H ₂ in CH ₄	4.0	29.0	7.9	7.30	13.07	63%
100% H ₂	4.1	77.0	4.3	3.54	13.43	25%

Table 2.4 Flammable gas classification groups.

Gas group	Temperature class—AIT (°C)					
	T1 450 >	T2 300–450	T3 200–300	T4 135–200	T5 100–135	T6 85–100
IIA MESG > 0.9 mm	Methane Propane Ammonia	Methanol Ethanol	Diesel Fuel oils Petrol	—	—	—
IIB 0.9 ≥ MESG > 0.5	Coal gas	Ethylene	—	Diethyl ether	—	—
IIC 0.5 mm > MESG	Hydrogen	Acetylene	—	—	—	Carbon disulphide

The safety implications of mixing hydrogen into the natural gas grid were studied in several projects. From the European project NATURALHY (Lowesmith, Mumby, Hankinson, & Puttock, 2011), dedicated safety research (Schröder, 2016) and field tests, like HyDeploy (HyDeploy, 2021) at Keele University UK, in Klanxbüll-Neukirchen (klanx_2020) and Fläming (Wasserstoff im Gasnetz, 2020) in Germany, there are well-proven conclusions, that mixing up to 20% into the natural gas does not change the general combustion characteristics and hazard qualifications of the hydrogen/natural gas mixture compared to natural gas. Most of the parameters are less affected by admixing hydrogen to these levels than the variation, which is associated with the different natural gas qualities, for example, within the ranges of the definition of H-gas and L-gas. This is shown in Fig. 2.55, mapping caloric values and the Wobbe index of different H and L-gas qualities and extreme values with addition of up to 25% vol hydrogen.

Only a few projects address the build-up of a dedicated hydrogen gas grid, transporting pure hydrogen for heating purposes. The most prominent example is the H21 Leeds City Gate project (h21_2021) where a transition of the Leeds, United Kingdom, gas grid to 100% hydrogen is planned. To this end even an intermediate large scale conventional production of hydrogen based on natural gas reforming is planned.

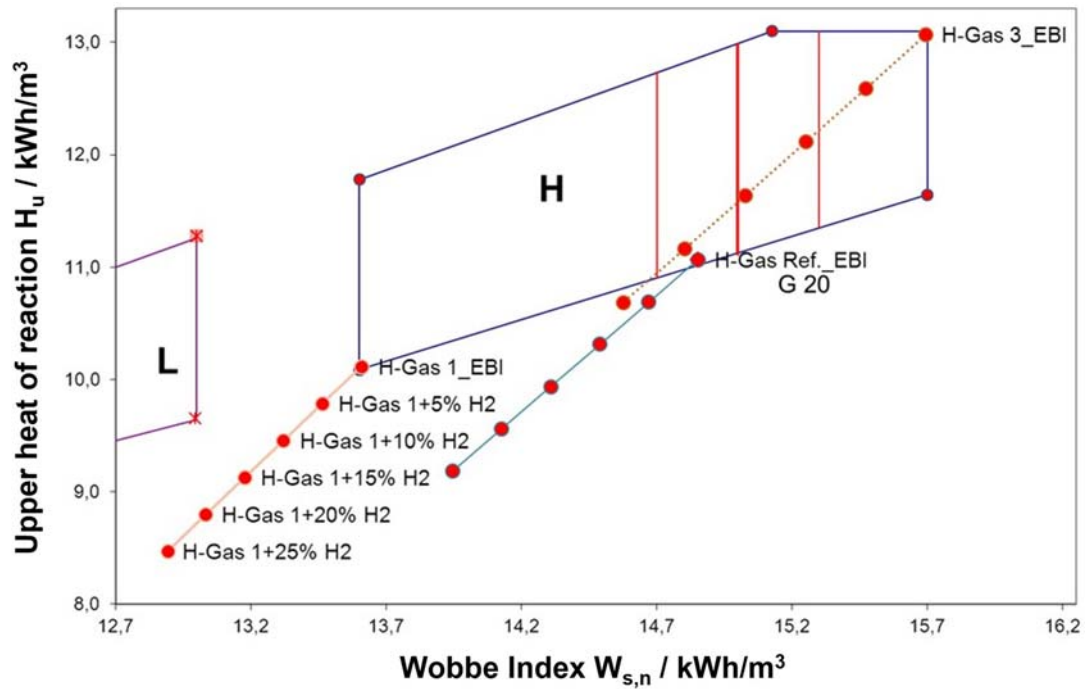


FIGURE 2.55

Upper heating value versus Wobbe index of H and L-gas and mixtures with hydrogen (Dörr, 2020).

2.5.9.1 Safety issues related to heat applications

The main open issues according to the recent HySafe Research Priorities Workshop (HySafe Research Priorities Workshops, 2020) outcome are as follows:

1. Materials issues (steel, PE etc. including effects of long exposure and jointing).
2. Hydrogen/natural gas mixtures (detection, mixing local and downstream in large grids with multipoint injection, appliance testing).
3. Pure hydrogen systems (production, purity, odorants, colorant).
4. Leaks in buildings and buried pipework including tracking effects etc.

2.6 Hydrogen systems major components

2.6.1 Compressors

Because of the small density, high diffusivity, and purity requirements special compressor technology is required for hydrogen. The principal classification of compressors distinguishes oscillating, reciprocating, rotating piston, and centrifugal compressors. The piston compressors typically

provide smaller mass flux at higher pressure ratios, whereas the rotating piston compressors provide larger volume flow at lower pressure ratios and pressure heads, respectively. Fig. 2.56 maps the compressor technologies into the coordinates volume flow and pressure rise ratio.

For the large process industry and pipeline operations multistage radial turbo compressors with volume flow rates typically above 50000 Nm³/h are applied. Slightly smaller volume flows are compressed with screw compressors. This technology is applied in small liquefaction processes, for instance.

For small volume flows, but high pressure ratios piston compressors are used. Depending on the hydrogen purity requirements three different subtypes are applied:

1. Oil lubricated piston compressors only for applications where hydrogen is used for saturation of hydrocarbons or combustion; not suitable for fuel cell usage or later liquefaction because of potential contamination.
2. Water lubricated piston compressors, when small hydrocarbon remainders are not acceptable.
3. Nonlubricated piston compressors for highest purity requirement. Piston rings are made from polytetrafluoroethylene and leaked gases may be recycled.

The piston usually is driven by an electric motor and a crankshaft drive. In intensifiers this electric drivetrain is replaced with a hydraulically or pneumatically driven piston, installed in the so-called

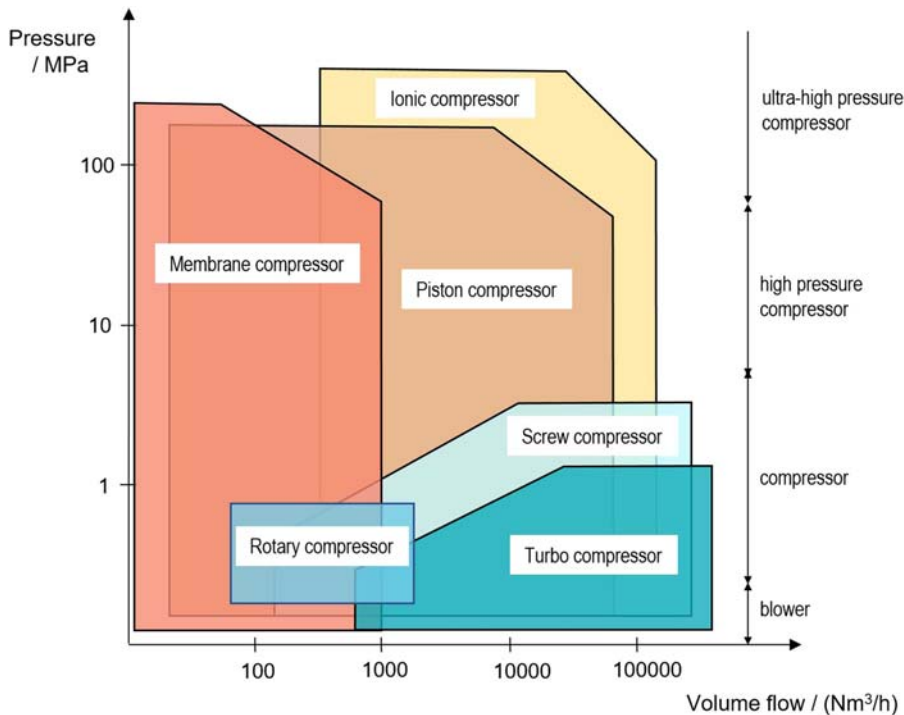


FIGURE 2.56

Hydrogen compressor types.

motive cylinder. The most common arrangement is a central motive cylinder combined with two gas compressing cylinders, one on each side of the motive cylinder (see Fig. 2.57). The gas compressing cylinders are connected in series via check valves to realize a two-stage compressing unit.

In operation, the force of the motive piston is balanced by gas pressure acting on the gas piston. As the hydraulic cylinder strokes, gas is compressed and displaced from one gas cylinder while simultaneously filling the other gas cylinder.

As the flow of the hydraulic or pneumatic piston may be driven by automatic valves no electric energy is required to drive or control those compressors, what removes a potential source for ignition from those machines working on flammable gases.

In diaphragma compressors and membrane compressors, the piston is replaced by a hydraulically driven membrane. Membrane compressors remove the sealing problems, material issues, and losses of the piston compressors. However, this comes with a reduced volume flow capacity and the requirement to maintain the steadily degrading membrane.

In the ionic compressors the solid piston itself or the interface of the solid piston and the hydrogen is realized with an ionic liquid material. Ionic liquids are molten salts, which have a very low vapor pressure. As a liquid they seal the gas volume perfectly and allow for high efficiencies and high pressure ratios by minimizing the dead volume. The power rating is also higher than for conventional piston compressors.

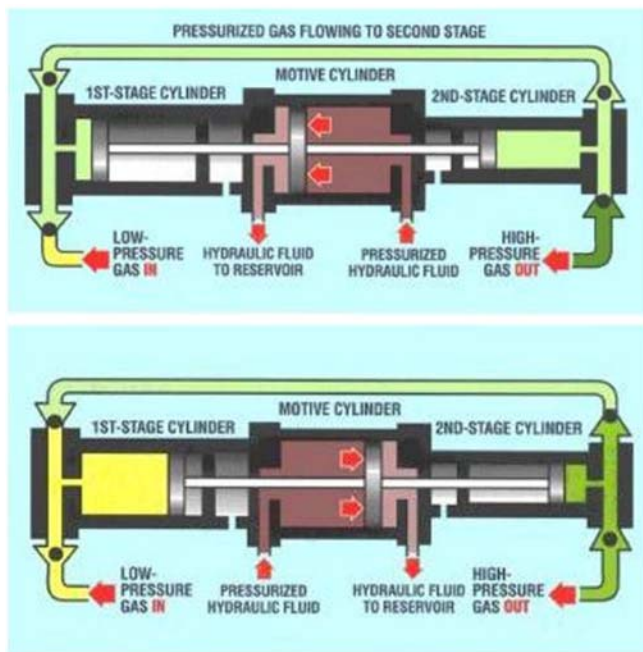


FIGURE 2.57

Principal operation and realization of a two-stage intensifier.

Hydro-Pac.

2.6.1.1 Future pathways and safety issues of compressors

Compressors, even those applicable for hydrogen compression, are mature products. The power-specific pricing scales well with size and efficiencies are relatively high, provided the compressor is operated close to the design point.

Better efficiencies are achieved by compressing the nearly noncompressible liquid hydrogen with a liquid hydrogen pump combined with a subsequent “evaporation” process. This process is particularly promising for 70 MPa fueling stations based on liquid hydrogen mass storage, as the required energy intense precooling might be omitted and the overall footprint of the installation is much smaller compared to a station with gaseous storage and compression. However, the maturity of liquid hydrogen pumps is relatively low and comes at high investment costs. Thermal compression might be combined with cryogenic hydrogen. Alternatively, hydrogen adsorbed and desorption with a temperature swing process might be used for compression without moving parts.

Another promising development are electrochemical compressors, which have a similar set-up as PEM electrolyzers. Instead of reducing water, hydrogen is catalytically dissociated and oxidized at the low pressure side; then the protons are “pumped” by the electric field to the cathode where hydrogen recombines at high pressure.

Generally, compressors represent a cost-intensive and safety critical component in hydrogen systems as they inject energy into the hydrogen and raise pressure and temperature of the compressed media to high levels. The oscillating or rotating piston in conventional compressors induces vibrations, which stresses connection bearings and sealings with cyclic loading. Therefore careful installation and maintenance are required, inducing considerable operational costs.

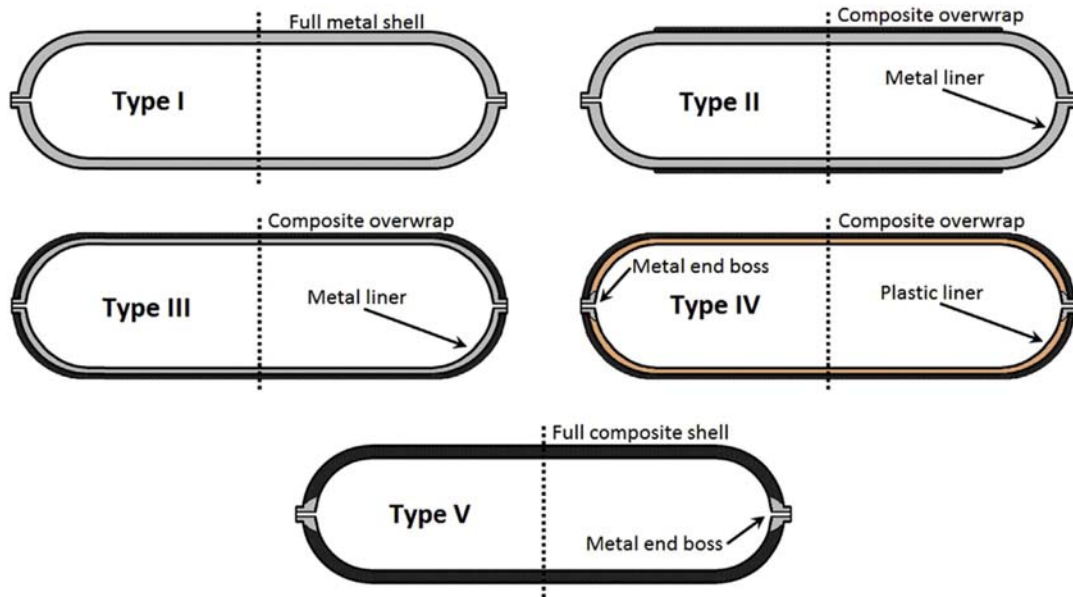
2.6.2 Pressure vessels

CGH₂ storage is state of the art for stationary, mobile, and transport applications. It represents the most mature, market ready solution (Barthelemy, 2012). The typical shape of a pressure container resembles a more or less short tube with half spherical closures at both ends. Mainly driven by the need of the automotive industry for light-weighted gaseous storage at highest pressure, new vessel technologies have been developed. Depending on the materials applied, five pressure vessel types are distinguished (see Fig. 2.58).

Type 1 is an all-metal cylinder, actually too heavy for automotive solutions, but heavily used for stationary applications with moderate storage pressures (see Fig. 2.59).

Type 2 is reducing the specific weight by applying a glass or carbon fiber layer, hoop wrapped on the cylindrical section. Type 3 extends the fiber wrapping on the spherical end caps and reduces further the wall thickness of the inner metal container, which is also called liner. The liner, typically made from alumina, provides still a load carrying function of about 30%. For Type 4 vessels the metal liner is replaced by a plastic liner, which is able to suppress permeation losses. Type 5 vessels, which are currently under development, do not have a dedicated liner but integrate the permeation barrier function in the compound material itself.

Depending on the inventory required, available space, costs, and the integration options several vessels are combined as bundles. In particular at fueling stations several bundles are connected to form a cascade, which allows maximum availability of highest pressure.

**FIGURE 2.58**

Types of pressure vessels (Fowler, Orifici, & Wang, 2016).

Components of a typical compressed hydrogen storage system include:

1. the container itself;
2. the check valve;
3. the shut-off valve; and
4. the TPRD.

The hydrogen tanks used in vehicles are predominantly Type 4 today. Fig. 2.60 shows a prototypical Type 4 vessel with a few further functional and protective elements.

For ranges of about 500–600 km the onboard stored inventory for a passenger vehicle is about 5 kg at nominally 70 MPa. For the integration this inventory is distributed in 2–3 vessels, which are usually connected via high pressure lines.

The SAE standard 2701 provides suitable protocols for filling those inventories safely at a time of about 3 minutes (see Section 2.5.4.1). To prevent too high temperature in the compound material and in the plastic liner the hydrogen has to be precooled to -20°C or -40°C . For the larger inventories needed for buses and heavy-duty trucks (30–100 kg), there are no standards available currently. However, as those applications usually provide sufficient space for the pressure vessels, the nominal working pressure is typically set to 35 MPa. This relaxes the precooling requirements. The typical fueling duration of heavy-duty trucks and buses is in the order of 20 minutes, where a maximum mass flux of about 120 g/s may be realized per fueling point. For larger inventories, like for trains or ships, parallel fueling of subsets of the vessels should be considered.

**FIGURE 2.59**

Stationary storage system.

Shell.

2.6.2.1 Specific safety aspects of automotive storage

From a safety perspective vehicle tanks are exposed to more demanding environment with higher probability for accidental loads by crash and fire. Type 3 and in particular Type 4 (and tentatively Type 5) are vulnerable in a thermal attack caused by external fire. Without protection a Type 4 vessel might fail in less than 10 minutes. Therefore the vessels are protected with the TPRD. A TPRD opens via a glass bulb fuse and releases the stored hydrogen via a critical cross-section with a typically dimension of 2–4 mm in diameter. The actual orifice sizing represents a compromise with regard to fast depressurisation on one side and avoiding pressure peaking phenomena in closed spaces like garages on the other side. Thermal shielding might mitigate the setup additionally. The TPRD is often integrated with a pressure reducing and other control elements.

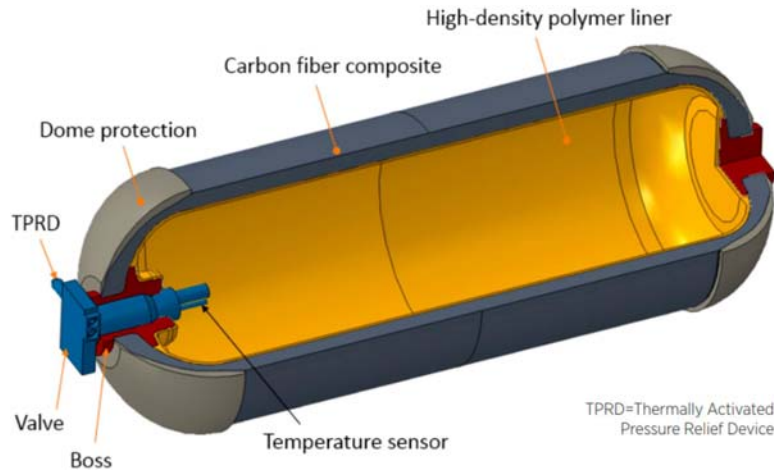
The relative sensitive carbon fibers are sometimes protected with a thin outer layer of glass fibers, which are mechanically more robust.

The required test series for certification of an automotive pressure vessel design are defined in the GTR 13 regulations (ECE, 2021), which are adopted by national regulations.

For nondestructive, in-service examination of the state-of-health of a vessel acoustic emissions—during loading or deloading—and eddy current inspections are under development. However, for the time being there is no reliable methodology available.

2.6.3 Large scale stationary storage containments

For seasonal energy storage underground and underwater storage facilities are considered to be of strategic importance to match hydrogen production and demand and to ensure energy reliability. In research and demonstration projects salt caverns turned out to be suitable to store hydrogen similarly like natural gas safely. Such caverns have typical volume of up to 1 Mio m³ and may be operated at pressures up to 20 MPa, provided there is a solid geological covering layer. They may be constructed in a depth up to 2000 m by solving the salt with water. Besides testing the conversion of similar existing natural gas caverns also construction of such huge dedicated hydrogen storage

**FIGURE 2.60**

Type 4 composite overwrapped hydrogen pressure vessel (Etienne Rivard, 2019).

facilities is realized for commercial purposes in the United Kingdom (e.g., Teesside) and the United States (e.g., Clemens Dome, Spindletop, Moss Bluff).

2.6.3.1 Specific safety aspects

For the geo-mechanical safety of caverns, required minimum thicknesses of salt layers above (hanging wall) and below (foot wall) the cavern are defined as functions of the cavern diameter. The minimum thickness of the hanging wall is suggested as 75% of the cavern diameter, whereas the minimum foot wall thickness is 20% of the cavern diameter (Wang, 2015). Moreover, a minimum limit for the height-to-diameter minimum ratio for salt caverns is defined as 0.5. Depleted oil and gas fields or aquifers are considered not suitable because of the contamination of the stored hydrogen.

2.6.4 Cryostats

Similar to compressed storage, there are two main classes of LH_2 storage vessels. There are cryostats for stationary applications and for transport. Representatives of both classes are shown in Figs. 2.61 and 2.62.

Cryogenic vessels have been commonly used for more than 40 years for the storage and transportation of liquefied gases including liquid hydrogen. In order to manage storage at -253°C , for large inventories ($> 100 \text{ m}^3$ volume) double-walled vacuum-insulated tanks are used. Such tanks, cryostats respectively, consist of an inner pressure vessel and an external protective jacket. The volume between inner vessel and jacket is filled with compressed perlite under vacuum. Perlite is an inorganic amorphous volcanic glass that represents a good trade-off between cost and insulation properties. For smaller cryostats ($< 100 \text{ m}^3$) a multilayer insulation is used instead. This insulation is composed of several layers of a metallic-coated plastic sheet, separated by a grid-like spacer. The heat absorption for the small automotive LH_2 storage tanks with an internal volume of about



FIGURE 2.61

3800 m³ LH₂ storage at Kennedy Space Flight Center in Florida.

NASA.



P0044080 BMW Group PressClub: www.press.bmwgroup.com
 © BMW AG Nur für Presse Zwecke / For press purposes only 06/2008
 LH₂ Leichtbau-Formtank mit integrierten Nebensystemen (Schnittmodell)
 LH₂ lightweight formtank with integrated auxiliary systems (cross-sectional model)

FIGURE 2.62

Cross-sectional model of an LH₂ lightweight form tank with integrated auxiliary system for automotive application.

BMW AG.

100 L, like the one shown in Fig. 2.62, is thus reduced to about 1 W. This heat input leads to evaporation and via a pressure limiting valve to the boil-off of the evaporated cold gases. The boil-off corresponds to a loss of 1.5% of the stored energy per day for small cryostats. The typically stored mass of about 7 kg will be lost in 2 months if the car was not used in this phase.

The following boil-off management may reduce these losses or at least reduce the associated risk with released hydrogen by:

1. cold combustion with air in catalytic recombiners,
2. storing the boil-off gases in metal hydrides or other adsorbers,
3. recycling in a reliquefaction process,
4. direct energetic use in a fuel cell, for instance.

Obviously, the involved temperatures are demanding not only regarding the design of the actual storage but also regarding the compatibility of all connected technologies, like measurement techniques, armatures, valves, and piping. Also fuel cells actually require higher temperature of the feed hydrogen. However, the cooling power of the extracted gases might be used also for cooling purposes.

Therefore the typical liquid hydrogen storage system (LHSS), as shown in Fig. 2.63, consists of the following components:

1. insulated liquefied hydrogen storage container(s);
2. shut off devices(s);
3. a boil-off system;
4. pressure relief devices;
5. a level gage for measuring the filling level;

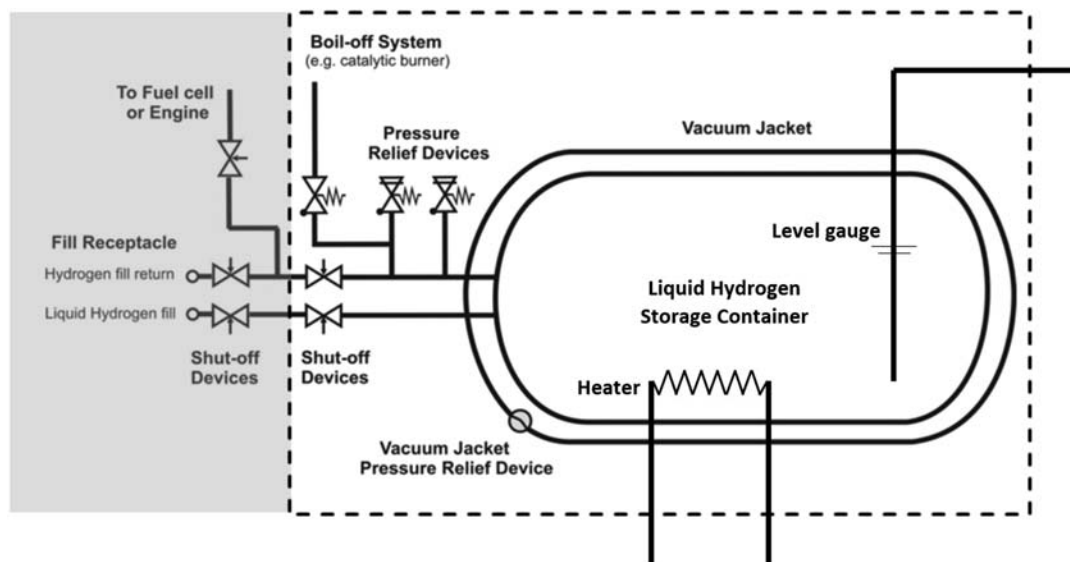


FIGURE 2.63

Liquid Hydrogen Storage System (based on: UN ECE GTR13).

6. a heater, either electric or heat exchanger air, for higher extraction rates; and
7. interconnecting piping (if any) and fittings between the above components.

Prototypical cryocompressed CCH_2 storage vessels are based on similar design. However, the inner insulated pressure vessel has been realized with a Type III vessel (Kircher, Greim, Burtcher, & Brunner, 2011). This allows for a nominal working pressure of 30 – 35 MPa at a storage temperature down to 50K. The single phase supercritical state makes obsolete the level gage and the classical boil-off gas management.

2.6.4.1 Safety challenges with cryostats

Liquid hydrogen storage should be always operated with a certain distance from the critical point, that means the nominal working pressure should be well below 1.3 MPa and the temperature below 33K. Boil-off, safety valves, and rupture disk should be designed properly. Design criteria for safety valves might be derived from Zoller (2018).

Cryostats should be positioned in the free and provisions for a potential spill should be taken. Cooling down and purging with hydrogen or helium should be conducted carefully to avoid accumulation of condensed air in the system.

Knowledge gaps are related to the potential of boiling liquid expanding vapour explosion (BLEVE) and crash worthiness of cryostats. This relates to conventional designs with austenitic steels and in particular to innovative designs using plastic material.

2.6.5 Solid state storage systems

Conventional metal hydride storage systems are operated at ambient temperature and moderate pressures. This reduces the intrinsic hazards. The relatively high specific weight, however, limits the application to stationary or safety critical, weight tolerant mobile applications like submarines.

Metal hydride storage systems must provide huge specific surfaces for the exchange of heat via a cooling liquid and for adsorption and desorption of the hydrogen. Besides the actual material performance this surface determines the rate at which a hydride alloy can absorb or release hydrogen. Therefore actual systems consist of tubular containers for the metal hydride itself, penetrated with pipework for introducing and extracting the hydrogen and cooling or heating the bulk material with a liquid coolant, either water or oil.

2.6.5.1 Safety challenges with solid state storage systems

Hydrogen can be stored in the form of hydrides at higher volumetric densities than by simple compression or even in liquefied or cryocompressed state. As storage pressure is quite low and heat is required to release the hydrogen, this method reduces safety concerns, in particular, about leakage that can be a problem with compressed hydrogen and LH_2 . However, as metal hydride material may react spontaneously, when exposed to air, water, or oil (Fichtner, 2017), other specific safety issues need to be addressed. Another issue might be toxicity of the dust-like metallic material.

2.6.6 Pipes and fittings

The issue of pipes as structural elements transferring hydrogen among subsystems and for the actual technology for connecting pipe elements finally breaks down to the classical structural mechanics design rules and to the issue of appropriate material selection. For the pipes, the fittings, and for the sealings, materials have to be chosen to fulfill primarily the requirements defined by the structural loads and operational conditions implied with the actual application. Simultaneously, the material has to cope with the potential of hydrogen to degrade the material properties. The degradation processes are usually affecting the ductility and therefore are summarized as hydrogen embrittlement (HE). Besides the technical requirements economic aspects also have to be considered.

For transporting gaseous hydrogen in particular the implied high pressures and the high purity of hydrogen represent the main challenges. From general experience low-strength carbon steels, aluminum and copper alloys, and austenic steels are generally good candidates, see Table C.1 of ISO TR 15916 (ISO, 2015), for instance. The group of metals that are highly susceptible to HE includes high-strength, martensitic-aging, and ferritic steels. Where chrome typically inhibits diffusion of hydrogen, nickel might have adverse effects and furthers HE (Balitskii, 2010).

Although having a different scope, namely manufacturing of steel cylinders, the standard ISO 11114-4 (iso_p4_2017, ISO 11114-4:2017 Transportable gas cylinders—Compatibility of cylinder and valve materials with gas contents—Part 4: Test methods for selecting steels resistant to hydrogen embrittlement, 2017) provides useful information about suitable tests for selecting metallic materials resistant to HE. The standard ISO 15649:2001 (ISO, 2001, ISO 15649:2001 Petroleum and natural gas industries—Piping) on piping for petroleum and natural gas industries is used as a guideline also for hydrogen technologies. This standard is applicable to piping within facilities and for packaged equipment, with exclusion of transportation pipelines and associated plant. The standard EN 13480 referred by the European pressure equipment directive 2014/68/EU (Commission, 2014) is divided in seven parts specifying the requirements for industrial piping systems and supports made of metallic materials. The tube elements are standardized in European standards EN 10216 and EN 10217.

In any case either the pipe elements have been certified, tested, labeled, and listed for the intended hydrogen use or the designer has to do the expensive qualification process supported by a certification body accepted by respective authorities.

Depending on the pipe dimensions and operation conditions for the connection of pipe segments either screwed connections, fittings, or welded connections are possible. Screwed connections come with the need for proper sealings, the need for control of installation, and repeated control of leak tightness in operation. Fittings are standardized in the European norm EN 10253 for instance and flanges in the EN 1092 and the ASME code 16.5/16.47 (ASME, 2020). The welded connections may be considered as leak tight; however, the influence of the welding process on the base materials and the hydrogen compliance of the weld material might introduce new issues.

It remains to be investigated whether for certain types of steel, welds, and sealings an adjustment of the operating parameters is required. Depending on the steel grade used for the pipelines and the fittings and the operating conditions of the pipeline, a reduction in toughness and ductility can lead to the growth of already existing crack-like defects. In these cases, the service life of the pipeline will be shortened. Initially existing crack-like defects, especially on the inner side of the

pipeline, existence of hydrogen in atomic form, and strong dynamic changes in pipeline pressure are the most critical factors. Therefore a comprehensive and continuous integrity management of the system is recommended in order to avoid additional risks. The observation and analysis of the material conditions, as central part of this integrity management, is carried out by physical, internal, and external inspection equipment and monitoring systems as well as pipeline tests. Special technologies and inspection devices exist for pipelines that detect various changes in the pipeline in operation. An essential means of today's condition assessment and maintenance of natural gas pipelines is the so-called "pigging technology." Depending on the inspection technology used, these "pigs" allow a recurring inspection of the pipe wall for existing and possibly developing anomalies. The existing maintenance concepts and tools can be adapted to the requirements of the hydrogen transport. In 2017 and 2019, a hydrogen transport pipeline built in 1996 was inspected for a study in the United States with appropriately designed "pigs." The required tool components were adapted to ensure resistance to uneven wear. At a pressure of 2 MPa and a flow rate of 13,000 Nm³/h, the tool was able to safely assess the pipeline. The inspection was completed with 100% sensor coverage (ROSEN Group, 2020).

2.6.6.1 Special pipes

At the Infraser industrial park in Frankfurt, Germany, a 100-MPa pipeline is connecting the central by-product hydrogen production and compressing site with the fueling station at a distance of 1.7 km (Heinrich Lienkamp, 2012). The high pressure pipeline is characterized by a quite small inner and outer diameter (in the order of 25 mm), what reduces the required wall thickness and the specific costs per length. In addition, the small dimension is advantageous with regard to safety, as it implicitly limits the fracture surface and the released mass fluxes in case of a worst case full bore rupture.

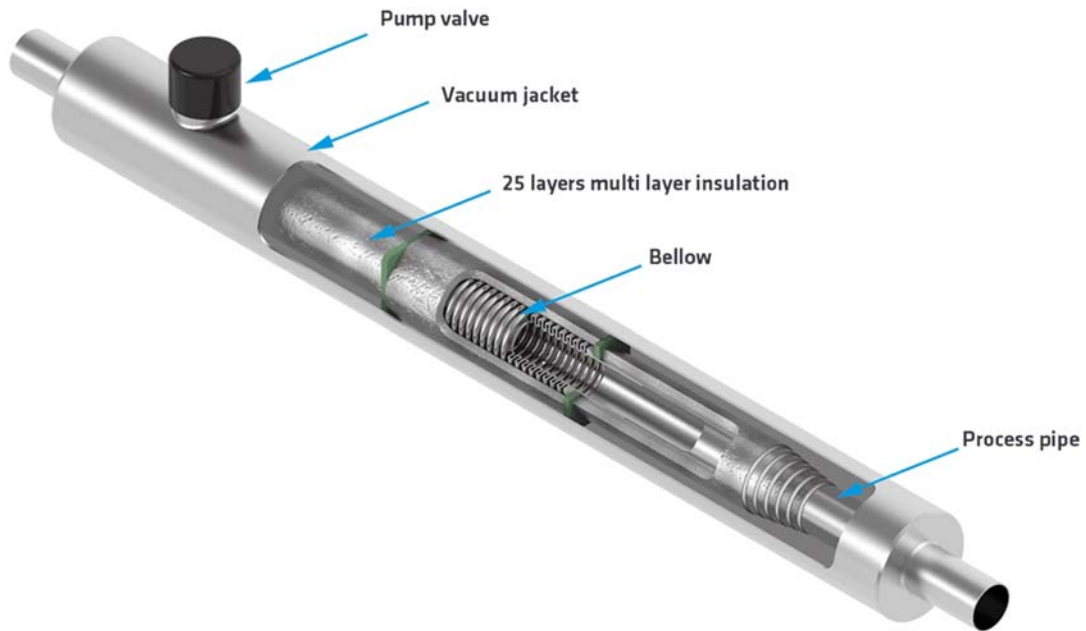
Pipes for transferring cryogenic liquid hydrogen must comply with the extreme low temperature of LH₂ (about 20K) and the associated insulation requirements. Similar concepts as for the cryostats (see Section 2.6.4) are used. A prototypical transfer pipe for LH₂ therefore consists of at least two concentric tubes combined with superinsulation material in the vacuum-insulated gap. There are rigid or flexible variants of these Vacuum Superinsulated Transfer Lines (VSTL). Fig. 2.64 shows a rigid VSTL and Fig. 2.65 shows a flexible transfer line with a Johnston coupling, which are used for flexible detachable connections.

The Johnston coupling consists of a male and female connector, which can be easily connected and secured. No welding or special knowledge is required for the operators, but a slightly higher heat intake compared to welded couplings has to be taken into account.

In the research and development project icefuel, research groups from KIT together with industry-led partners Evonik and LEONI tested a flexible plastic pipe for combined transport of LH₂ and electricity via high temperature superconductor (see Fig. 2.66). The insulation of the thin cable was realized with superinsulating material and a liquid nitrogen shield. The flexible pipeline could be used to supply residential areas with chemical and electrical power and information with an outer diameter of 40 mm only (Georg Markowz, 2010).

2.6.7 Valves

Valves are devices that work to control, regulate, or direct flow within a system or process.

**FIGURE 2.64**

Vacuum superinsulated transfer line for LH_2 .

DEMACO.

In most cases, valves fall into one of three categories:

1. **Manual valves:** Typically adjusted by hand, these valves use handwheels, hand levels, gear wheels, or chains to actuate.
2. **Actuated valves:** For hydrogen pneumatic systems, hydraulic systems, or solenoids, these valves allow remote control and automation for high-precision or large-scale applications.
3. **Automatic valves:** Some valves activate when a specific flow condition is met. Examples include check valves closing during backflow or pressure release valves activating when an overpressure condition is detected.

Common types of valves are ball, butterfly and gate valves, needle valves, check valves, and pressure relief valves. The function varies from isolation (ball, butterfly, gate), regulation (ball, needle) to safety (check, pressure relief). A special safety valve with no-return characteristics is the TPRD used to protect hydrogen carrying pressure vessels in a fire.

Extremely low internal and external leakage is essential for hydrogen-carrying valves to ensure stable system operation. Low hydrogen consumption, even when not in operation, is always a priority. The valves used must function precisely and reliably—even in stressful installation situations such as extreme temperature environments or vibrations. At the same time, they must meet the highest safety requirements throughout their entire service life.

**FIGURE 2.65**

At NASA's Kennedy Space Center in Florida, test and operations support contract engineers and technicians inspecting a liquid hydrogen supply hose at Launch Pad 39B. The hose is terminated with a Johnston coupling.

Photo credit: NASA/Frankie Martin.

2.6.7.1 Safety issues with valves

All the components of such a complex device like a valve must cope with the challenges introduced with hydrogen. The sealings (see [Section 2.6.8](#)) have to limit permeative losses to the level of technical leak tightness and the structural materials must avoid hydrogen-induced embrittlement. Therefore the bulk material of the valves is typically stainless steel. Particular attention has to be paid to the springs, used in check valves or regulating valves, as the material used for these springs (high strength steels) is extremely susceptible for HE. By the way, this applies also for pressure gauges.

The former design of TPRDs was based on eutectic melting lot, which melted when critical temperature was reached. However, in a few accidents with natural gas pressure vessels the out-flowing cold gases prevented the full opening of the release cross-section, what resulted in a rupture of the affected cylinder. Therefore the melting lot was replaced by a glass bulb as activating device, similar as in water sprinklers.

For premixed systems, like hydrogen – air or hydrogen – oxygen mixtures in particular, the valve opening dynamics should be slow enough not to induce shocks, which might lead to ignition and subsequent explosive loads.

Valve suppliers typically certify their products for use with hydrogen. They might be self-certifying or contract a certifying or listing body, like UL. If this is not provided, the costs for a dedicated certification might easily become prohibitive.

2.6.8 Seals

Seals are safety critical elements for screwed connections, fittings, valves couplings, and other sub-systems like electrolyser and fuel cell stacks. The main problems associated with hydrogen sealing

**FIGURE 2.66**

Icefuel cable sample (outer diameter ca. 40 mm; max transport capacity: 100–200 kW lower heat of reaction).

LEONI, Nürnberg (Georg Markow, AD (2010). Icefuel® – An infrastructure system for cryogenic hydrogen storage, distribution and decentral use. Proceedings of the WHEC. 78-1. Essen, Germany: Schriften des Forschungszentrums Jülich / Energy & Environment).

are again rooted in the special properties of hydrogen, in particular with its high diffusivity, the permeation capability, and its potential for deteriorating material properties. These issues turn into real challenges, when considering the extreme operational conditions encountered in the new applications.

A distinction is made between static and dynamic seals. Static seals are usually used when there is no relative motion between mating surfaces. Dynamic seals are used when there is motion between surfaces. Dynamic seals should also show good static sealing functions when motion does not take place, but should not prevent movement when motion is required.

In addition to permeation, the solubility of hydrogen in the sealing material potentially plays a crucial role. The absorbed hydrogen possibly leads to swelling of the sealing material, what may affect the strength of the seal essentially. Current research investigates the swelling of elastomers in hydrogen. In the case of a hydrogen intercalation in the sealing material, a sudden expansion of the volume of the stored hydrogen occurs during a very rapid pressure decrease. This can lead to the destruction of the material if the gas cannot escape fast enough from the material or the tensile strength of the material is too low. The suitability of materials to resist these so-called explosive decompressions has to be considered in the selection of the sealing material.

In principle there are two suitable material groups for hydrogen sealing: metallic materials, in particular copper alloys, and nonmetallic polymer materials, like polychlorotrifluoroethylene or polytetrafluoroethylene. Teflon, Neoprene, Dacron, Mylar, Viton, Buna-N and Kel-F are examples of suitable products available commercially. Although the latter polymer materials are suitable for low temperature services (down to -40°C), they are too brittle at cryogenic temperature, with exemption of Kel-F and Teflon (ISO, ISO/TR 15916:2015, 2015).

A specifically challenging static sealing task is encountered during refueling of an automotive hydrogen storage tank (Albrecht, 10AD). The critical loading conditions are induced by the relatively high pressure (70–80 MPa) and the relatively wide temperature range (down to -40°C). With increasing hydrostatic pressure the so-called free volume of the polymer material is reduced,

what brings the material closer to its glass transition, that is, closer to the glassy state, which makes it less suitable for sealing purposes. The temperature above which the polymer properties convert from a rigid to the flexible, elastic state is defined as the glass transition temperature T_g .

The hydrostatic pressure in an O-ring, for example, depends on the system pressure and the stress conditions in the cross-section. The rate at which T_g increases seems to be material-independent and amounts to about 0.2K/MPa as an average value for overall elastomers. Consequently, when pressure increases from atmospheric to 100 MPa the glass transition temperature T_g shifts by about +20K. Below T_g the elastomer experiences not only the transition from a flexible to a rigid, glassy state, but also other properties change adversely. For instance, the “free volume,” which characterizes the gaps between molecular chains, decreases by a factor of 2–3. The latter effect yields a specific volume reduction well beyond the pure thermal shrinking. This means that the resilience of the squeezed O-ring is reduced substantially when temperature falls below the T_g value.

In electrolyzers and fuel cells seals must also guarantee certain electrical properties, besides the primary mechanic function of sealing. Typically, they should be good electrical isolators.

Therefore designing a safe sealing function is a complex engineering task. The ISO TC 197 has recognized this and initiated recently a working group defining suitable tests and qualification procedures for O-ring sealing for hydrogen applications.

References

- Aasadnia, M. M. (2018). Large-scale liquid hydrogen production methods and approaches: A review. *Applied Energy*, 212, 57–83. Available from <https://doi.org/10.1016/j.apenergy.2017.12.033>.
- ASME, B16.5. (2020). <https://www.asme.org/codes-standards/find-codes-standards/b16-5-pipe-flanges-flanged-fittings-nps-1-2-nps-24-metric-inch-standard>.
- Balitskiia, A. V. (2010). The low-cycle fatigue of corrosion-resistant steels in high pressure hydrogen. *Procedia Engineering*, 2, 2367–2371. Available from <https://doi.org/10.1016/j.proeng.2010.03.253>.
- Barthelemy, H. (2012). Hydrogen storage—Industrial perspectives. *International Journal of Hydrogen Energy*, 37, 17364–17372. Available from <https://doi.org/10.1016/j.ijhydene.2012.04.121>.
- Beghi, G. (1985). Development of thermochemical and hybrid processes for hydrogen production. *International Journal of Hydrogen Energy*, 10(7–8), 431–438. Available from [https://doi.org/10.1016/0360-3199\(85\)90071-0](https://doi.org/10.1016/0360-3199(85)90071-0).
- Bhattacharyya, R. B. (2016). Simulation studies of the characteristics of a cryogenic distillation column for hydrogen isotope separation. *International Journal of Hydrogen Energy*, 41(9), 5003–5018. Available from <https://doi.org/10.1016/j.ijhydene.2016.01.106>.
- Boukis, N., Galla, U., Müller, H., & Dinjus, E. (2007). Biomass gasification in supercritical water experimental progress achieved with the VERENA pilot plant. *1st COE-INES Int. Symp. on Innovative Nuclear Energy Systems for Sustainable Development of the World*.
- BP. (2018). *BP statistical review of world energy - 67th edition*.
- Brecher, L. E., Spewock, S., & Warde, C. J. (1977). The Westinghouse sulfur cycle for the thermochemical decomposition of water. *International Journal of Hydrogen Energy*, 2(1), 7–15, doi:10.1016/0360-3199(77)90061-1.
- Chunbao Charles, X. U. (2010). A brief overview of low CO₂ emission technologies for iron and steel making. *Journal of Iron and Steel Research, International*, 3, 1–7.

- Commission, E. (2014). *Pressure equipment directive 2014/68/EU*. https://ec.europa.eu/growth/sectors/pressure-gas/pressure-equipment/directive_en.
- Daniel Albrecht, D.M. (10. March 2014). *KGK Rubberpoint*. <https://www.kgk-rubberpoint.de/>>; <<https://www.kgk-rubberpoint.de/9244/seals-for-high-pressure-tank-systems-in-fuel-cell-cars/>>
- Dolci, F. (2018). *Green hydrogen opportunities in selected industrial processes*. JRC Technical Reports.
- DoE, U. (2016). https://www.energy.gov/sites/prod/files/2016/12/f34/fcto_early_markets_mhe_fact_sheet.pdf.
- DoE, U. (2021). <https://www.energy.gov/eere/fuelcells/hydrogen-pipelines>.
- Dörr, H. (2020). 4. 11. *Wasserstoff und Wasserstoffzumischung zu Erdgas*. Karlsruhe, Germany: DVGW Forschungsstelle / EBI KIT.
- ECE, U. (2021). *Global technical regulation on hydrogen and fuel cell vehicles*. ECE/TRANS/180/Add.13: <https://www.unece.org/fileadmin/DAM/trans/main/wp29/wp29wgs/wp29gen/wp29registry/ECE-TRANS-180a13e.pdf>.
- Eckermann, E. (2001). *World History of the Automobile*. Society of Automotive Engineers.
- Elshout, R. (2010). *Hydrogen production by steam reforming*. *Chemical Engineering*. Available from <https://www.chemengonline.com/hydrogen-production-by-steam-reforming/?printmode=1>.
- Etienne Rivard, M. T. (2019). Hydrogen storage for mobility: A review. *Materials*, 12(1973). Available from <https://doi.org/10.3390/ma12121973>.
- European Committee for Electrotechnical Standardisation CENELEC. (2018). *EN IEC 60079-0 explosive atmospheres—Part 0: Equipment—General requirements*. Brussels: European Committee for Electrotechnical Standardisation CENELEC.
- Faaß, R. (2001). *Cryoplane*. Hamburg, Germany. https://www.fzt.haw-hamburg.de/pers/Scholz/dglr/hh/text_2001_12_06_Cryoplane.pdf, Cryoplane.
- Fichtner, M. (2017). *INT-Research-Materials for Hydrogen Storage*. Karlsruhe: KIT.
- Fickel, H.-C.E. (2006). *EU Funded H2ICE Project*. Long Beach, USA: HyICE. NHA.
- Finn Vogler, B.S. (2012). *Machbarkeitsstudie zum Bunkern von Flüssiggasen in deutschen Häfen*. GL.
- Fowler, C. P., Orifici, A. C., & Wang, C. H. (2016). A review of toroidal composite pressure vessel optimisation and damage tolerant design for high pressure gaseous fuel storage. *International Journal of Hydrogen Energy*, 41(47), 22067–22089. Available from <https://doi.org/10.1016/j.ijhydene.2016.10.039>, S.
- Gallandat, N. R. (2017). An analytical model for the electrolyser performance derived from materials parameters. *Journal of Power and Energy Eng*, 5, 34–49. Available from <https://doi.org/10.4236/jpee.2017.510003>, S.
- Georg Markowz, AD (2010). *Icefuel® – An infrastructure system for cryogenic hydrogen storage, distribution and decentral use*. *Proceedings of the WHEC. 78-1*. Essen, Germany: Schriften des Forschungszentrums Jülich / Energy & Environment.
- Giron, E. (2007). *The hydrogen refueling plant in Madrid*. *International Journal of Hydrogen Energy* (32). Elsevier. Available from <http://doi.org/10.1016/j.ijhydene.2006.10.040>.
- Goldstein, S., Borgard, J.-M., & Vitart, X. (2005). Upperbound and best estimate of the efficiency of the IS cycle. *International Journal of Hydrogen Energy*, 30, 619–626.
- Grashoff, G. J. (1983). The purification of hydrogen. *Platinum Metals Review*, 27, 157–169.
- H21—Pioneering a UK hydrogen network. (2021). <https://www.h21.green>.
- H21—Pioneering a UK hydrogen network. (2020). <https://www.h21.green>.
- HazerGroup. (2020). Hazer Group. HG. <https://www.hazergroup.com.au>.
- Haberstroh, C. (6. March 2015). *Kryogene Wasserstofftechnologie*. Karlsruhe, Germany.
- Harald Malerød-Fjeld, D. C.-T.-M. (2017). Thermo-electrochemical production of compressed hydrogen from methane with near-zero energy loss. *Nature Energy*. Available from <https://doi.org/10.1038/s41560-017-0029-4>.
- Heinrich Lienkamp, A. R. (2012). Achievements and lessons learnt in the EU project Zero Regio. *Procedia—Social and Behavioral Sciences*, 48, 2201–2210. Available from <https://doi.org/10.1016/j.sbspro.2012.06.1193>, S.

- Hori, M. (2004). Synergistic hydrogen production by nuclear-heated steam reforming of fossil fuels. In *Proceedings of the first COE-INES international symposium on innovative nuclear energy systems for sustainable development of the world*.
- HyDeploy, (2021). Available from <https://hydeploy.co.uk>
- HySafe Research Priorities Workshops. (2020). <https://hysafe.info/activities/research-priorities-workshops/>. IA HySafe.
- IEA. (2019). The future of hydrogen. Paris: IEA. <https://www.iea.org/reports/the-future-of-hydrogen>.
- ISO. (2001). *ISO 15649:2001 Petroleum and natural gas industries—Piping*. <https://www.iso.org/standard/28195.html>. ISO.
- ISO, TR. (2015). *ISO/TR 15916:2015*. <https://www.iso.org/standard/56546.html>.
- ISO, Part 4. (2017). *ISO 11114-4:2017 Transportable gas cylinders—Compatibility of cylinder and valve materials with gas contents—Part 4: Test methods for selecting steels resistant to hydrogen embrittlement*. <https://www.iso.org/standard/64587.html>.
- IRENA. (2020). *Green hydrogen: A guide to policy making*. <http://www.irena.org>. IRENA.
- Jernkontoret. (2018). <https://www.jernkontoret.se/en/vision-2050/carbon-dioxide-free-steel-production/>
- Junji Sakamoto, R. S. (2016). Leakage-type-based analysis of accidents involving hydrogen fueling stations in Japan and USA. *International Journal of Hydrogen Energy*, 41(46), 21564–21570. Available from <https://doi.org/10.1016/j.ijhydene.2016.08.060>, S.
- Korn, T. (2019). The new highly efficient hydrogen internal combustion engine as ideal powertrain for the heavy-duty sector. In *Proceedings Internationaler Motorenkongress 2019. Springer Fachmedien*. https://doi.org/10.1007/978-3-658-26528-1_23.
- Krieg, D. (2012). *Konzept und Kosten eines Pipelinesystems zur Versorgung des deutschen Straßenverkehrs mit Wasserstoff*. <http://www.fz-juelich.de/zb/juwel>. Schriften des Forschungszentrums Jülich.
- Kuhn. (2015). *Hydrogen and methane testing field at the ILK.*, <https://www.ilkdresden.de/en/service/research-and-development/project/hydrogen-test-area-at-ilk-dresden/>, ILK.
- Kircher, O., Greim, G., Burtscher, J., & Brunner, T. (2011). Validation of cryo-compressed hydrogen storage (CCH₂)—A probabilistic approach. *Proc, ICHS2011*. <http://conference.ing.unipi.it/ichs2011/papers/258.pdf>.
- Leighty, W.H. (2006). Compressorless hydrogen transmission pipelines deliver large-scale. *WHEC 16 / 13–16 June 2006*. Lyon, France.
- Lowesmith, B., Mumby, C., Hankinson, G., & Puttock, J. S. (2011). Vented confined explosions involving methane/hydrogen mixtures. (Elsevier, Hrsg.) *International Journal of Hydrogen Energy*, 36(3), 2337–2343, S.
- Ludwig-Bölkow-Systemtechnik GmbH. (2021). *H2Stations.org*.
- Manfred Fischedick, J. M. (2014). Techno-economic evaluation of innovative steel production. *Journal of Cleaner Production*, 84, 563–580, S.
- Maroufmashat, A., Fowler, M., Khavas, S. S., Elkamel, A., Roshandel, R., & Hajimiragha, A. (2016). Mixed integer linear programming based approach for optimal planning and operation of a smart urban energy network to support the hydrogen economy. *International Journal of Hydrogen Energy*, 7700–7716. Available from <https://doi.org/10.1016/j.ijhydene.2015.08.038>, S.
- Ockwig, N. W. (2007). Membranes for hydrogen separation. *Chemical Reviews*, 4078–4110, S.
- Oliver Posdziech, T. G. (2019). System development and demonstration of large-scale high-temperature electrolysis. *ECS Transactions*, 91(1), 2537. Available from <https://doi.org/10.1149/09101.2537ecst>, S.
- Papageorgopoulos, D. (2018). Direct hydrogen fuel cell electric vehicle cost analysis: System and high-volume manufacturing description, validation and outlook. *Journal of Power Sources*, 399, 304–313. Available from <https://doi.org/10.1016/j.jpowsour.2018.07.100>.
- Peschka, W. (1992). *Liquid hydrogen, fuel of the future*. Springer. doi:10.1007/978-3-7091-9126-2

- Plevan, M., Geißler, T., Abánades, A., Mehravaran, K., Rathnam, R. K., Rubbia, C., ... Wetzel, Th (2015). Thermal cracking of methane in a liquid metal bubble column reactor: Experiments and kinetic analysis. *International Journal of Hydrogen Energy*, 40(25), S. 8020–8033. <https://doi.org/10.1016/j.ijhydene.2015.04.062S>.
- Pohlenz, J. B. (1962). New process promises low-cost hydrogen. *Oil Gas Journal*, 60(17).
- ROSEN Group. (2020). Hydrogen getting into focus. In *Proceedings of the fifteenth pipeline technology conference*. Berlin, Germany.
- Ruthardt (Dr K.M.S.) (2008) *Reliability and availability—Integrated hydrogen production schemes for refineries*. 2nd Iran Oil Refining Forum. https://ucpcdn.thyssenkrupp.com/_legacy/UCPhyssenkruppBAIS/assets/files/download_1/hydrogen_1/uhde_publications_pdf_en_15000014.pdf.
- SAE. Protocol. (2014). Fueling protocols for light duty gaseous hydrogen surface vehicles: https://www.sae.org/standards/content/j2601_201407, SAE.
- Scharlipp, S. (2017). *Untersuchung des Klopffverhaltens methanbasierter Kraftstoffe*. Springer Vieweg. <<https://doi.org/10.1007/978-3-658-20205-7>>
- Schneider, S. B. (2020). State of the art of hydrogen production via pyrolysis of natural gas. *ChemBioEngReviews*, 7, 150–158. Available from <https://doi.org/10.1002/cben.202000014>, S.
- Schröder, V. (2016). *Abschlussbericht zum Forschungsvorhaben 2539, Sicherheitstechnische Eigenschaften von Erdgas-Wasserstoff-Gemischen*. Berlin, Germany: BAM.
- Shirasaki, Y., Tsuneki, T., Ota, Y., Yasuda, I., Tachibana, S., Nakajima, H., & Kobayashi, K. (2009). Development of membrane reformer system for highly efficient hydrogen production from natural gas. *International Journal of Hydrogen Energy*, 34(10), 4482–4487. Available from <https://doi.org/10.1016/j.ijhydene.2008.08.056>, Elsevier.
- Shirres, (2018). <https://www.icheme.org/media/9621/david-shirres-is-hydrogen-the-answer-070319-v2.pdf>.
- Stefan Lechtenböhrer, C.S. (2018). *Climate innovations in the steel industry—Deliverable 2.2 REINVENT Project Nr. 730053*.
- Siemens Energy, Gascade Gastransport GmbH., & Nowega, GmbH. (2020). <https://assets.siemens-energy.com/siemens/assets/api/uuid:3d4339dc-434e-4692-81a0-a55adbcaa92e/200915-whitepaper-h2-infrastructure-en.pdf>.
- Stolzenburg, K., & Kentzler, M. (2010). Lessons learned from hydrogen infrastructure operation in the HyFLEET:CUTE Project. In *Proceedings of the (WHEC2010)*, pp. 78259–6263). Essen, Germany: Schriften des Forschungszentrums Jülich / Energy & Environment. <https://core.ac.uk/download/pdf/34994735.pdf>.
- Uhde. (2003). *Hydrogen*. <http://www.uhde.biz/informationen/broschueren.en.html>.
- THyGA. (2020). Testing hydrogen admixture for gas applications. European Commission and Fuel Cells and Hydrogen Joint Undertaking (FCH 2 JU). <https://thyga-project.eu>.
- Valentin Vogl, M. Å. (2018). Assessment of hydrogen direct reduction for fossil-free steelmaking. *Journal of Cleaner Production*, 203, 736–745.
- Valery Rosenband, A. G. (2010). Application of activated aluminum powder for generation of hydrogen from water. *International Journal of Hydrogen Energy*, 10898–10904, S.
- Wasserstoff im Gasnetz. (2020). Von Wasserstoff im Gasnetz: <<https://www.avacon-netz.de/de/avacon-netz/forschungsprojekte/wasserstoff-im-gasnetz.html>> abgerufen.
- Wasserstoff Einspeisung Klanxbüll. (2020). <https://www.sh-netz.com/de/schleswig-holstein-netz/innovation/wasserstoff-einspeisung-klanxbuell.html>.
- Wang, T. Y. (2015). Safety evaluation of gas storage caverns located close to a tectonic fault. *Journal of Natural Gas Science and Engineering*, 281–293.
- Wezel, B. (2019). Europe with “Renewable Gas Ready” turbines. *ASME Global Gas Turbine News*, S. 56–57.
- Wörtler, M.S.-T. (2013). *Steel’s contribution to a low carbon Europe 2050*. Boston.
- Zoller, C. (2018). *Experimental investigation and modelling of incidents in liquid helium cryostats*. Karlsruhe: KIT.
- Züttel, A. (2003). Materials for hydrogen storage. *Materials Today*, 6, 24–33.

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Phenomena relevant to accidents

3

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Industrial accidents connected with hydrogen usually result from explosions of the hydrogen. The explosion is typically the last event in the following sequence: initial release of hydrogen in the atmosphere, then dispersion in the surrounding when hydrogen is mixed with air creating burnable mixture, and, after that, ignition of burnable mixture.

In this chapter, main features of the phenomena before and during such accidents are presented. The chapter content arrangement follows the designated event sequence. First, a concise survey of the hydrogen release, dispersion, and ignition is given. Then, the principles of combustion, including deflagration, detonation, and deflagration to detonation transition (DDT), are presented. Flame acceleration as a basic feature for the safety is presented in more details with accentuation on the factors affecting flame acceleration, such as confinement and obstruction of the surroundings.

This description is aimed to provide basic picture of the accident-related problems in the context of industrial background. The exposition has only introductory character and in some parts introduces necessary simplifications for the sake of clarity. For the in-depth look, a comprehensive list of the presented references can be used.

3.1 Hydrogen release

In most accidental situations, the complications start with the discharge of hydrogen in the amounts, which later can bring the system to the unwanted consequences. The variety of releases is quite wide, and is defined by many factors, including the amount and discharge rate of the escaping gas, the type of break in the equipment from which the discharge occurs, the surrounding spatial conditions, etc. As soon as liquid hydrogen is also expected to be commonly used simultaneously with compressed hydrogen, the hydrogen releases as a gas or as a liquid and/or a two-phase flow can take place in the possible accidents.

Due to resemblances of the release properties, few major classes can be distinguished:

- permeation, typically leak through equipment material;
- jet release, can be purely gaseous or two-phase, typically through breach in the equipment under pressure;
- pool vaporization, typically after liquid hydrogen spills from equipment on the surrounding surface;
- explosive release, typically during the equipment vessel complete disintegration due to the excessive pressure inside system.

Following this classification, this section describes the main phenomena characterizing these major classes.

3.1.1 Gaseous releases

3.1.1.1 Permeation leaks

Use of the term permeation implies penetration of the hydrogen through the walls of some solid enclosure, in which hydrogen is stored. The process comprises the molecular diffusion of the permeant through the membrane bounding the gas. The permeation is controlled by the properties of the membrane and by the concentration gradient across the membrane. Permeation can have place for most materials, such as plastic, natural materials, and metals. The permeability of metals is typically much lower than that of plastic. This fact explains why permeation from the hydrogen storage tanks, which do not include metallic containment layer, is significantly higher. For the storage tanks, the permeation intensity can be affected also by the internal tank pressure and temperature, area of contact surface with permeant, and duration of the permeant store.

Permeation of hydrogen through metal (Schefer et al., 2006) starts with adsorption of hydrogen molecules at the internal membrane surface, followed by their dissociation to the atomic hydrogen, which diffuses through the membrane, continues with the recombination to the molecular hydrogen at the external surface, and finally by their desorption.

At steady state, expression for the rate of permeation of hydrogen by diffusion reads as

$$J = DS\sqrt{P}/l,$$

here J is the permeation rate ($\text{mol H}_2 \text{ s}^{-1} \text{ m}^{-2}$), D is the diffusivity of hydrogen in the membrane material, and S is the solubility of hydrogen in the membrane material. The permeability is a product of the solubility and the diffusivity, which has exponential dependence on temperature T .

$$\Phi = DS = \Phi_0 \exp(E_\Phi/RT).$$

Hydrogen permeability for several metals are provided in Table 3.1. These data are taken from literature; they are based on experimental data at elevated temperatures and extrapolated to near ambient temperatures. This fact, probably, explains high scatter in the cited data.

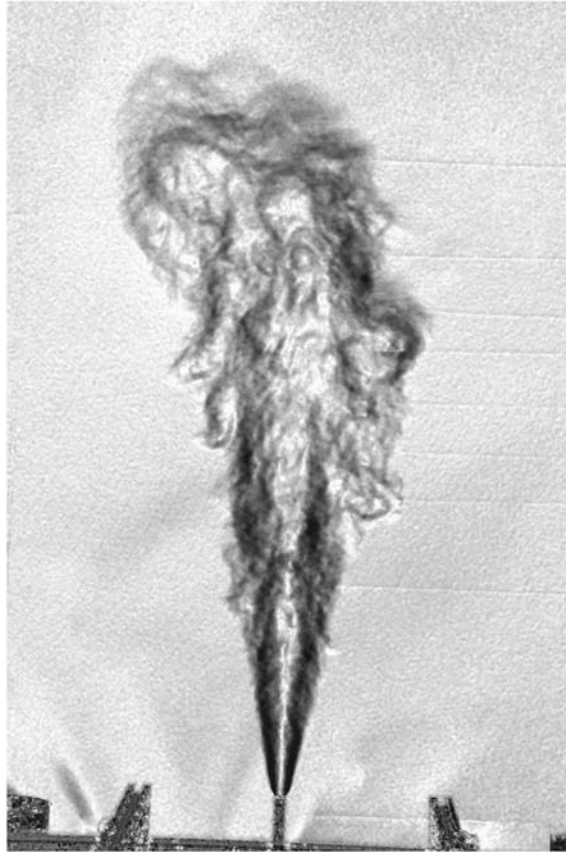
In Table 3.1, permeability Φ_0 and Φ are in $\text{mol H}_2 \text{ s}^{-1} \text{ m}^{-1} \text{ MPa}^{-1/2}$, and activation energy E_Φ is in kJ/mol .

3.1.1.2 Jets and plumes

Release through breach in the vessel depending on the conditions of the discharge can produce jet or plume. When the flow is controlled in major by buoyancy, the term plume is used. For the flows

Table 3.1 Literature data for hydrogen permeability.

Material	Φ	Φ_0	E_Φ at 298 K	Reference
Pure Fe	$6.9 \cdot 10^{-11}$	$5.35 \cdot 10^{-5}$	33.6	Yamanishi, Tanabe, and Imoto (1983)
Stainless Steel 403	$1.9 \cdot 10^{-12}$	$5.90 \cdot 10^{-5}$	42.7	Oriani, Hirth, and Śmiałowski (1985)
Stainless Steel 316 L	$2.3 \cdot 10^{-15}$	$3.90 \cdot 10^{-4}$	64.1	Xiukui, Jian, and Yiyi (1989)
Aluminum	$6.3 \cdot 10^{-16}$	$1.00 \cdot 10^{-6}$	52.5	Song, Du, Xu, and Long (1997)

**FIGURE 3.1**

Hydrogen jet in choked regime. Visualization made using background-oriented schlieren (BOS) technique.

controlled mainly by momentum, the term jet is preferred. Characteristic image of jet is presented in Fig. 3.1. Often for the weak jets with low momentum, the term buoyant jet is used.

In case of hydrogen leak, due to the low density the buoyant force is always directed upwards, even in the case of cryogenic leak. The buoyant force is initially directed downwards with time after heating of cold hydrogen and mixing with ambient air changes direction.

Usually, during ascending of plume from the source, it expands due to entrainment of the surrounding gas at its peripheries. The plume shapes can be affected also by flow in the ambient, for example, by the wind.

To characterize relations between the flow momentum and external field (often buoyancy) the dimensionless number Fr named after William Froude was introduced.

$$Fr = \frac{U_0}{\sqrt{g_0 l_0}},$$

here U_0 is a characteristic flow velocity, g_0 is in general a characteristic external field, and l_0 is a characteristic length. For the analysis of the dynamic for jets and plumes it is convenient to use densimetric Froude number.

$$Fr = \sqrt{\frac{\rho_0 U_0^2}{(\rho_\infty - \rho_0) g d_0}},$$

where U_0 , ρ_0 , and d_0 are the fluid velocity, the fluid density, and the fluid diameter at the break point, respectively; g is the gravity acceleration and ρ_∞ is the bulk density.

The Froude number can characterize the flow depending on the controlling regime and separate momentum jets from plumes. Gebhart, Jaluria, Mahajan, and Sammakia (1988) introduced criteria for the separation of the regions of the vertical upward propagating flow. In Table 3.2, the expressions describing parameters of jets and plumes for three regions, including inertial, intermedium, and buoyant, are listed.

The x-direction is along the centerline of the flow, u_c , ρ_c , and c_c are the fluid velocity, the fluid density, and concentration at the centerline, respectively; r is distance from centerline.

The first similarity solutions for plumes and buoyant jets were proposed by Batchelor (1954) and the classical bulk model for isolated jets and plumes was presented in Morton, Taylor, and Turner (1956). Later it was generalized to take into account time-dependence in the various fluxes driving the flow in Scase, Dalziel, and Hunt (2006).

For the plume modeling in most cases several assumptions are introduced, which match conditions for fully developed turbulent plume and substantially simplified formulation:

- the Boussinesq approximation is valid;
- the radial distribution is either Gaussian or flat;
- the rate of entrainment into the plume is proportional to the local velocity; the entrainment coefficient is about 0.1 for vertical jets and plumes;
- set of conservation equations gives a complete description of the flow in most cases, for example, for a simple rising plume the solution predicts the expanding with the height plume with constant half-angle of about 6 – 15 degrees.

Table 3.2 Recommended expressions for the three regions of a vertical upward structure (Gebhart et al., 1988).

Regions	Inertia	Intermedium	Buoyancy
Structure	Jet	Forced plume (or buoyant jet)	Buoyant plume
Bounds	$Fr^{-\frac{1}{4}} \left(\frac{\rho_0}{\rho_\infty} \right)^{-\frac{1}{4}} \frac{x}{d_0} < 0.5$	$0.5 \leq Fr^{-\frac{1}{4}} \left(\frac{\rho_0}{\rho_\infty} \right)^{-\frac{1}{4}} \frac{x}{d_0} \leq 5$	$5 < Fr^{-\frac{1}{4}} \left(\frac{\rho_0}{\rho_\infty} \right)^{-\frac{1}{4}} \frac{x}{d_0}$
$\frac{u_c}{U_0}$	$6.2 \left(\frac{\rho_0}{\rho_\infty} \right)^{-\frac{1}{2}} \left(\frac{x}{d_0} \right)^{-1}$	$7.26 Fr^{-0.05} \left(\frac{\rho_0}{\rho_\infty} \right)^{\frac{9}{20}} \left(\frac{x}{d_0} \right)^{-1}$	$3.5 Fr^{-\frac{1}{3}} \left(\frac{\rho_0}{\rho_\infty} \right)^{-\frac{1}{3}} \left(\frac{x}{d_0} \right)^{-\frac{1}{5}}$
$\frac{\rho_c - \rho_\infty}{\rho_\infty - \rho_0}$	$5 \left(\frac{\rho_0}{\rho_\infty} \right)^{-\frac{1}{2}} \left(\frac{x}{d_0} \right)^{-1}$	$0.44 Fr^{\frac{1}{16}} \left(\frac{\rho_0}{\rho_\infty} \right)^{-\frac{7}{16}} \left(\frac{x}{d_0} \right)^{-\frac{5}{4}}$	$9.35 Fr^{\frac{1}{6}} \left(\frac{\rho_0}{\rho_\infty} \right)^{-\frac{1}{3}} \left(\frac{x}{d_0} \right)^{-\frac{4}{5}}$
$\frac{c_c}{C_0}$	$6.34 \left(\frac{x}{d_0} \right)^{-1}$		$12.16 Fr^{\frac{1}{3}} \left(\frac{\rho_0}{\rho_\infty} \right)^{-\frac{1}{3}} \left(\frac{x}{d_0} \right)^{-\frac{6}{5}}$
$\frac{u}{u_c}$	$\exp \left(-92 \left(\frac{r}{x} \right)^2 \right)$		$\exp \left(-90 \left(\frac{r}{x} \right)^2 \right)$
$\frac{c}{c_c}$	$\exp \left(-52 \left(\frac{r}{x} \right)^2 \right)$		$\exp \left(-80 \left(\frac{r}{x} \right)^2 \right)$

3.1.1.3 Subsonic and sonic jets

Hydrogen accidental discharges through a breach or a hole in the pressurized vessels for the case of jet regime, depending on the degree of compression in the vessel, can be either subsonic or choked (also called sonic). The regime is altered when pressure difference is so that the flow in the orifice reaches conditions when Ma is equal to 1.

If the flow is subsonic, no shock waves appear by the exit from the orifice and the jet is called expanded. In contrast to this, the choked regime is characterized by the pressure and temperature jumps at different locations after gas exit. The flow pattern can be overexpanded or underexpanded, depending on the relations between ambient pressure and the pressure in the vessel. For hydrogen, expanded jet exists if ratio of the pressure inside vessel to ambient is below 1.899.

Steady-state choked flow occurs when the ambient pressure P_0 is below critical value p^* . For the ideal gas, critical value can be calculated using (Landau & Lifshitz, 1987)

$$\frac{p^*}{P_0} = \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}},$$

where γ is specific heat C_p/C_v .

For the hydrogen with a heat capacity ratio γ equal to 1.41, critical pressure is $p^* = 0.526 \cdot P_0$. For critical pressure 1 bar, the pressure in the vessel, which makes the flow choked, is 1.899 bar.

When the flow is choked, the mass flow rate is (Landau & Lifshitz, 1987)

$$\dot{m} = S_{\min} \sqrt{\gamma P_0 \rho_0} \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma + 1}{\gamma - 1}},$$

here P_0 and ρ_0 is pressure and density of the gas in vessel, $S_{\min}$ is effective cross-sectional area of the orifice. Note that, despite that gas velocity in the orifice is constant, the mass flow rate is not constant and depends on the pressure and density of the gas in the vessel and does not depend on the gas parameters outside.

The flow rate is also influenced by the shape of the opening, friction, and the length of the duct between the vessel and the ambient. To characterize such influence, a discharge coefficient C_d is often used; $S_{\min} = C_d \cdot A$, where A is geometrical cross-sectional area.

For the correct evaluation of the discharge rate, accurate values for the hydrogen density should be used. The discrepancy in the density estimation for the ideal gas and for real values could reach considerable values.

In Table 3.3, hydrogen densities at temperature 288 K and pressures 200 bar and 700 bar for real and for ideal gas are listed (Marshall, Liquide, & Scientifique, 1976).

In the choked flow, the region near the orifice exit exhibits complex shock wave pattern involving oblique waves, Mach disk, diamond structures [see, e.g., Franquet, Perrier, Gibout, & Bruel (2015)].

Table 3.3 Comparison between real and ideal gas properties.

Pressure (bar)	Real density (kg/m ³)	Ideal density (kg/m ³)	Relative error (%)
200	14.96	16.85	12.6
700	40.18	58.90	46.6

The characteristic details of checked flow soon after injection begin are good reproduced in the numerical simulations, as for example, Fig. 3.2. The region downstream the shocks, where the flow is expanded, can be described using correlations for the subsonic jet, introducing concept of notional nozzle. In these correlations the jet discharge diameter d_0 is replaced by the virtual notional nozzle diameter d_{not} .

Several authors (Birch, Brown, Dodson, & Swaffield, 1984; Birch, Hughes, & Swaffield, 1987; Ewan & Moodie, 1986) have proposed different approaches to evaluate notional nozzle diameter d_{not} . For the mean axial concentration, the dependence on the downstream distance z and virtual origin displacement a reads as

$$\eta = \frac{kd}{z+a} \left(\frac{\rho_a}{\rho_g} \right)^{1/2},$$

where k is axial decay constant, d is orifice diameter, ρ_a and ρ_g are ambient and gas densities, respectively.

For the axial decay constant k , different values have been proposed for better fit of the reality. In Birch et al. (1984) an average value of 4.9 is given, while in Birch et al. (1987), value of 5.4 is provided.

Notional nozzle approach has been validated by the authors and despite known shortcomings has demonstrated reasonable performance. In Houf and Schefer (2005), authors used the Birch approach to determine the concentration decay of underexpanded hydrogen jets taking, for the first time, into account non-ideal behavior of gas at high pressures.

The inventive theory aimed to calculate concentration change in underexpanded hydrogen jets without using notional nozzle diameter but density of hydrogen at real nozzle exit instead was developed and successfully validated by Molkov and Saffers (2011). This novel approach demonstrated that the similarity law in the original form suggested by Chen and Rodi (1980) can be applied to both expanded and underexpanded jets without modification.

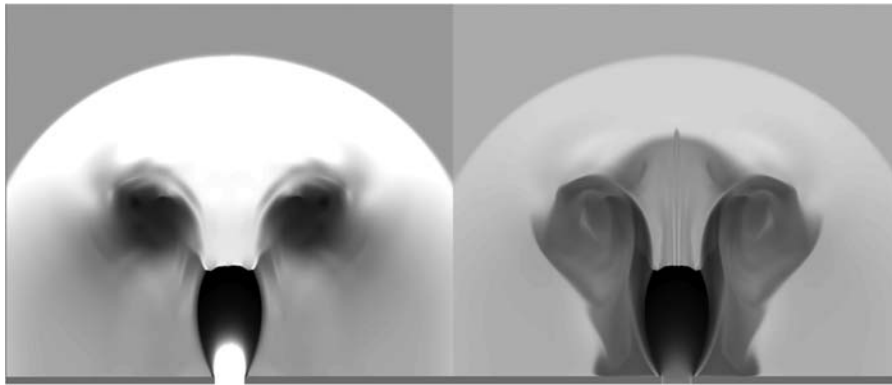


FIGURE 3.2

Numerical simulation (Kotchourko, Lelyakin, Yanez, & Halmer, 2017) of the 3D structure of underexpanded hydrogen jet. Left image shows distribution of pressure and the right image shows distribution of temperature.

3.1.2 Multiphase releases

Understanding of the cryogenic liquid discharge, which runs in conjunction with gaseous release, requires quantification of the phenomena, which significantly differ from phenomena in the pure gaseous hydrogen release. Specific for the two-phase release studies is accounting for the diverse gas behavior at low temperatures, phase change, heat transfer from the surrounding environment, peculiar effect of weather conditions, for example, temperature, humidity, wind, and dissimilar effect of obstacles and confinement.

If release occurs from a pressure vessel, the rapid pressure reduction results in spontaneous vaporization or boiling also called flash of the liquid. Depending on the leak location and size, the two-phase flow can transform into two-phase jet, and in case of intense flow it reaches the ground and forms a pool of liquid hydrogen. Both jet and pool vaporize producing a burnable plume with the parameters, such as composition, height, dimension, defined for the most part by the configuration of the release source. This section describes the main phenomena that lead to the cloud formation in two-phase releases.

3.1.2.1 Two-phase jets

The efforts to describe two-phase jet were started essentially in last decades of the 20th century. In [Bricard and Friedel \(1998\)](#) a survey of hazardous dispersion with special attention to two-phase jet conditions was presented. In this review, two-phase jet specifics together with the few integral and multidimensional models are discussed.

When the mixture passes the orifice, parameters of the mixture changes severely from the state inside the vessel to the expanded conditions. The key physical phenomena, which take place during this transformation, include

- flashing if the liquid is sufficiently superheated;
- gas expansion when the flow is choked;
- liquid fragmentation.

The essential parameters related to these phenomena, and which are required in the modeling for the description of these phenomena, include the flash fraction, the jet mean temperature, velocity and diameter, and the droplet size.

The amount of vapor after flashing is usually evaluated by assuming isenthalpic depressurization of the mixture between the outlet and the position outside where thermodynamic equilibrium at ambient pressure is reached. In case of the choked flow, gas part expands to the ambient pressure within one or two orifice diameters. Consequently, the flow accelerates and expands, so the jet enlarges its diameter. Similarly to the model for pure gas jet, the momentum and mass balance can provide values of the expanded state velocity and jet diameter. In such consideration, it is assumed that no air is entrained in the jet region. Remarkably, the flow speed can be increased by a factor as high as 10 in this region, which has important consequences on the downstream dispersion. According to [Fauske and Epstein \(1988\)](#), the parameters of the expanded state (index 2) are controlled by the state in the orifice plane (index 1).

$$d_2 = d_1 \left(\frac{\rho_1 u_1}{\rho_2 u_2} \right)^{\frac{1}{2}},$$

$$u_2 = \frac{u_1(P_1 - P_2)}{\rho_1 u_1}.$$

The adequate model accounting for the droplets in two-phase flow expects substantial understanding of the liquid phase behavior, including formation of the droplets and their following dynamics. In the two-phase jet, the droplets can appear due to flashing or due to liquid break-up by gas-dynamic force impact. In the first case, the fragmentation results from the intense boiling and exploding of bubbles in the superheated liquid, while in the second case, it is caused by the excitation of the liquid surface instabilities. For the flashing mechanism, there are still lack in the experimental data and the reliable models. For the gas-dynamic fragmentation, it is accepted that the maximum stable drop size is usually given by a critical Weber number, which represents the ratio of inertia over surface tension forces

$$\text{We}_{\max} = \frac{U^2 \rho_g d_{\max}}{\sigma},$$

where σ is the surface tension of the liquid, ρ_g is the gas density, $d_{\max}$ is maximum stable droplet diameter, and U the mean relative velocity between particles and gas. An influence of the maximum Weber number for the multiphase release is reviewed in [Bricard and Friedel \(1998\)](#), and for the droplet breakup in [Gelfand \(1996\)](#).

The existing integral models are divided in two categories, one deals with both near-field and far-field and the other considers the near-field only. The models are all based on the balance equations for mass, momentum, energy, and species, integrated over the jet cross-section; they mainly differ by the adopted closure relations. Several reviews of different methodologies for mass flow rate evaluation in the liquid and two-phase flow releases is made in, for example, [Hanna and Strimaitis \(1989\)](#); [Lees \(1996\)](#).

In [Houf and Winters \(2013\)](#), the authors developed a set of integral models targeted to describe the release process of a small and slow leak (at very low Mach number) from an LH₂ storage system. The description recognizes four phases in the release process: a leak followed by three turbulent entrainment periods. To predict the hydrogen – air mixture state, an equilibrium thermodynamic model based on the NIST REFPROP subroutines ([Lemmon, Huber, & McLinden, 2013](#)) was used. In another paper ([Winters & Houf, 2011](#)), the same authors presented a similar multizones model for high-pressure liquid release, adding a model for the zone of underexpanded flow (Mach number > 1). For gaseous hydrogen leak model, they performed validation calculation of the concentration on the centerline of the leak, demonstrating an approving agreement between the model results and the experimental data. [Friedrich, Breitung, Stern, & Veser \(2012\)](#) performed experiments on release and combustion of cryogenics hydrogen jets, providing an estimate of safety distances and an extrapolation model for other jet conditions. A topical review ([Ekoto et al., 2014](#)) on the liquid hydrogen release and behavior provides valuable analysis on the present achievement in the modeling and in the available experimental data.

3.1.2.2 Liquid H₂ pool spreading and vaporization

A pool of liquid hydrogen appearing due to the spill from the storage, depending on the heat balance with surroundings can exhibit varied behavior. The heat balance for the spill is controlled by the competition between heat contribution from the ground and from the atmosphere, and heat sink

due to evaporation. In case of liquid hydrogen spill, the dominant part in most cases belongs to the heat flow from the ground, which, for certain conditions, could reach 80%–90% of the total input. Radiation, for example, for the burning pool, can also provide significant contribution.

With a high release rate, the pool on the ground steadily increase its size and thickness with time until it reaches an equilibrium state. When the release ends, amount of the liquid in the pool starts to decrease from the surface and when the thickness of the liquid becomes smaller, 1–2 mm, the surface breaks into separate spots and gaps in the surface appear.

Shortly after pool formation, a vapor pad is settled due to the film boiling. With the enlarging of the pool, the vapor pad degrades, so the liquid contacts directly with the ground surface, and the film boiling is changed to nucleate boiling. The character of the vaporization in case of the liquid and of the solid below the spill differs essentially. The principal difference is in the dynamics of the ground temperature. For the underlying liquid, due to the natural convection the temperature of the ground and correspondingly the heat flux remain almost constant. For the solid ground, due to the gradual cooling proportional to $\sim 1/\sqrt{t}$ the vaporization rate decreases. In case if moisture is present in the solid ground, the ice formation can provide additional heat source, promoting vaporization significantly.

When liquid hydrogen is released at the water surface, a phenomenon called rapid phase transition (RPT) could be observed. RPT is an event when liquefied gas vaporizes violently upon coming in contact with water. This is also called a physical explosion, as despite the energy liberation is small in comparison with chemical energy of the explosion involving combustion, the shocks waves can be observed.

For the creation of the reliable models, better understanding of this initial phase together with more experimental data on the dispersion phase are required. The fact that still limited number of experiments are available in the literature was indicated in [Kotchourko et al. \(2014a\)](#).

The first experiments of LH₂ spill are dated 80s of the last century (e.g., [Chirivella & Witcofsky, 1986](#); [Dienhart, 1995](#); [Verfondern & Dienhart, 1997, 2007](#); [Witcofski & Chirivella, 1984](#)). In the last decade, intensive study of the liquid hydrogen spill was performed at Health and Safety Laboratory from Health and Safety Executive, UK (HSL/HSE) ([Hooker, Willoughby, & Royle, 2011](#); [Royle & Willoughby, 2011](#)). In these experiments, it was confirmed that if release is made on the sufficiently cool ground, a pool of liquid hydrogen could be formed, and, furthermore, it was observed that oxygen and nitrogen undergo solidification, forming a solid layer on the ground. Recently, interesting results were obtained in the frames of the experimental campaign of EC Project PRESLHY (e.g., [Jordan, Jallais, & Bernard, 2019](#)).

Contemporary CFD models in general demonstrate ability to predict values observed in experiment with the accuracy within a factor 2. To illustrate the CFD modeling status, few examples of the integral model are discussed. In [Venetsanos et al. \(2009\)](#) CFD model was used to examine the model potential to reproduce the NASA experiments ([Chirivella & Witcofsky, 1986](#); [Witcofski & Chirivella, 1984](#)) with a release rate of 9.5 kg/s. By modeling the source as a two-phase jet and including the heat transfer from the pool to the ground, the good agreement between experiments and simulations was demonstrated. However, some measurement sensors revealed noticeable deviations, which authors attributed to the incomplete accounting of the environmental conditions. Among others, the following CFD modeling achievements should be mentioned: [Baraldi et al. \(2009\)](#); [Ichard, Hansen, Middha, and Willoughby \(2012\)](#); [Molkov, Makarov, and Prost \(2005\)](#); [Statharas et al. \(2000\)](#); [Winters and Houf \(2011\)](#).

Beside CFD studies, analytical mathematical models were also developed to describe specific stages of the liquid hydrogen release. Kim and coworkers (Kim, Do, Choi, & Han, 2012) positively applied perturbation techniques to solve a simple physical model that describe the LH₂ pool spreading.

3.2 Hydrogen dispersion

3.2.1 Hydrogen transport and mixing with air

If the released hydrogen is ignited earlier than it has created premixed cloud, the ignition normally is not followed by pressure waves producing substantial loads, and combustion of the leaking hydrogen transforms to the standing flame. On the contrary, the delay in the ignition can and usually leads to the explosion with the significant pressure loads due to the intensive combustion, which commonly develops in the premixed cloud.

In the course of hydrogen release process, three clearly recognized periods could be differentiated. Early stage, when released hydrogen is still poorly mixed with air, completely unmixed, or mixed only in peripheral zones. Middle stage, when hydrogen due to buoyant forces rises upward allowing surrounding air to entrain into the inner regions, thus reducing concentration and making mixture burnable. In the final stage, due to the continuous mixing concentration of hydrogen drops below flammability limit, making mixture unburnable.

3.2.1.1 Molecular and turbulent mixing

The process of hydrogen mixing with air plays key role in the potential of the hydrogen release to be dangerous. Depending on the release characteristics and the surrounding conditions, the mixing process can proceed either in molecular or in turbulent regime.

To characterize relative significance of the advection and the diffusion, a nondimensional Péclet number Pe was introduced. The Péclet number is a class of dimensionless numbers relevant in the analysis of transport phenomena

$$Pe = \frac{uL}{D} = Re Sc,$$

it is ratio of flow convection (advection) rate of a characterized physical quantity to the rate of diffusion of the same quantity driven by an appropriate gradient. Here D is a diffusion coefficient, u is the flow velocity, and L is a characteristic length scale. Péclet number can be interpreted as a product of another two nondimensional numbers important for the transport of the viscous media, namely Reynolds number Re and Schmidt number Sc .

The Reynolds number Re is the ratio of inertial forces to viscous forces

$$Re = \frac{uL}{\nu} = \frac{\rho uL}{\mu},$$

here ν is kinematic viscosity, u is the flow velocity, L is characteristic length scale, μ is dynamic viscosity, and ρ is density. The Reynolds number is exceptionally important dimensionless quantity in fluid mechanics. It allows characterizing flow parameters in different flow conditions. Reynolds

number serves as exclusive discriminator between laminar regime by low Reynolds numbers, and turbulent regime at high Reynolds numbers.

Schmidt number is defined as the ratio of momentum diffusivity (kinematic viscosity) and mass diffusivity. It characterizes the relative thickness of the hydrodynamic layer and mass-transfer boundary layer

$$Sc = \frac{\nu}{D} = \frac{\mu}{\rho D},$$

here ν is kinematic viscosity, D is molecular mass diffusion, μ is dynamic viscosity, and ρ is density.

If $Pe \ll 1$, the diffusion is the dominant mechanism in the transport of the fluid, while for $Pe \gg 1$, the convection (advection) gets control.

In engineering appliances, the Péclet number is often very large. It could be connected with the fact that for the large times the characteristic lengths $L \sim ut$ defining fluid movement becomes much larger than diffusion scale and convection (advection) dominates. Thus for the industrial applications in the laminar regime, the molecular diffusion usually plays only minor role.

In the turbulent flows in contrast to the case of laminar movements despite that the mechanism of the mixing in the final stage is still defined by molecular diffusion, the effectiveness of the turbulent mixing due to development of large interface surface, generated by motions of the different size eddies existing in the high Reynolds number turbulent streams, appears tremendously high.

Turbulent mixing in accordance to the classical interpretation (Eckart, 1948) can be considered as a three-stage process. In the first stage, called entrainment, rather large volumes of the pure components and sharp gradients at their interfaces, which are distinctly observable, exist. In the second stage, referred as stirring or dispersion, the component volumes are distorted and a rapid increase of the extent of the interfacial regions with high gradients is observed. In the last stage, the extensive diffusion leads to the smoothing of the gradients with homogenization of the mixture (Batchelor, 1959).

The turbulent mixing phenomena depending on the flow characteristics in its turn can be separated into three different classes (Dimotakis, 2005). The events involving passive mixing, as occurs between passive scalars, belong to the first class. As example, mixing of gases with close densities, or with small temperature difference can be mentioned. The dispersion process is driven by turbulence, but an influence on the fluid dynamic is weak.

The mixing of the different density or different temperature gases in the gravitational field belongs to the second class. For this class, the mixing is coupled with the fluid dynamics, as in Rayleigh – Taylor (Rayleigh, 1882; Taylor, 1950) and Richtmyer – Meshkov (Meshkov, 1969; Richtmyer, 1960) instabilities.

In third class, the mixing is also coupled with the gas dynamics and produces changes of the fluid, for example, producing nonuniformities in composition, density, pressure. Most combustion phenomena, deflagrations, detonations, belongs to this class.

As it was indicated in Dimotakis (2005), the progress in the study of turbulent mixing has mostly been confined to first class, for example, for grid/isotropic turbulence, channel and pipe flows, and free shear layers and jets. The theory, modeling, and even empirical knowledge for second and third classes are less developed and can be characterized as mostly open research topics.

Turbulent flow is characterized by the continuous change of the characteristic lengths of the local fluid movements. Such hierarchy of the scales with the corresponding change of the local energy content is called turbulent energy cascade.

For the characterization of the energy cascade specific length scales were introduced. The mostly evident and natural scale is integral length scale based on the characteristic size of large eddies in turbulent motion. Large eddies appear due to the perturbation of the mean flow and acquire energy from it and from the mutual interaction with other eddies. The principal part of the turbulent energy participating in the energy cascade is in large eddies. The final dissipation of the kinetic energy takes place at microscopic scales where molecular processes dominate and the turbulent kinetic energy due to viscosity dissipates into heat.

In [Kolmogorov \(1941\)](#), the author introduced hypothesis on the statistics properties of the turbulent motion: for very high Reynolds numbers the characteristics are determined only by the kinematic viscosity ν and the rate of energy dissipation ε . He has proposed the microscale

$$\eta = \left(\frac{\nu^3}{\varepsilon} \right)^{1/4},$$

which is the smallest scale in turbulent motion and is called Kolmogorov scale.

3.2.2 Dispersion in open environment

3.2.2.1 Atmospheric turbulence

The mixing of most hydrogen leaks could be characterized as belonging to the second class, as hydrogen at the ambient temperatures has considerably lower density in comparison with the air.

The strong buoyancy definitely affects the dispersion process, which depending on the source of the release will vary the total picture. However, for the most types of the sources, as liquid spill evaporation, high momentum jet release for both single-phase and two-phase releases, catastrophic vessel rupture, and a plume from a weak leak from equipment, the atmospheric turbulence plays significant role in the subsequent mixing in the open environment.

The turbulence is a motion, which is characterized by irregular and random changes in pressure and instant flow velocity. One of the basic properties of the turbulence is existence of the hierarchy of different random fluctuating velocity scales, or eddies. Atmospheric turbulence scales can range from global patterns defined by large-scale wind field down to local and near-instantaneous fluctuations in average wind direction and speed.

When the wind moves the air over the ground, the ground surface friction and internal wind shear slow down the air movement in the lower air positions. At greater heights above the ground, the air speed is mostly independent on the ground level conditions.

The atmospheric turbulence is affected by the obstacles on the ground surface, sheer force between the ground and the upper atmosphere, and between wind field and realized gas cloud. It generally increases with the stronger wind and with greater number, height, and spacing of ground surface obstacles. The ground characteristics affect atmospheric turbulence and consequently changes of the wind speed with height.

For the characterization of the atmospheric boundary layer, in [Pasquill and Smith \(1983\)](#) the authors proposed categorization of the wind profiles. They have introduced the categories A, B, C for unstable, D for neutral, E, F for stable conditions for average vertical density and temperature stratification.

Neutral conditions occur when heat flow between the ground and the atmosphere is small, as for example, by the windy cloudy weather. Unstable conditions occur in daytime when the warm

ground leads to an unstable density stratification with warm lower air layer underneath cooler upper air layer.

Transient conditions appear when the warm air moves up and expands during the movement, resulting in the high mixing rate in the vertical direction and rapid dilution of the released gas. For hydrogen cloud dispersion at normal conditions such situation is most commonly encountered.

The stable conditions are usually realized at night when the atmosphere is cooled from below, which results in the stable vertical temperature and density profile. Such stable stratification can restrict vertical mixing rate, leading to the vertically elongated gas cloud.

In contrast to the strong dependence of the vertical mixing rate on the stability category, the horizontal dispersion perpendicular to the wind is weakly dependent on it.

The considered categorization of the boundary layer is convenient for the simple cases of the flat terrain with near constant weather conditions.

In the case when the ground has multilevel complicated relief, as in the urban areas, the mixing flow can become quite complex with large-scale vortices and channeled streams between buildings (Perdikaris, 1993). When wind blows across buildings or industrial structures, behind the structures a low-pressure zone can appear. The plume drifting over such area can be pulled down into this region, increasing mixing close to the ground and creating hazardous cloud in the obstructed/confined area (Randerson, 1984).

3.2.2.2 Small releases

Small releases in the open atmosphere, even cryogenic, usually rapidly obtain ambient temperature, and due to the high buoyancy, quickly rises and mixes with the air. The rate of the dispersion and dilution is predominantly defined by the atmospheric turbulence.

3.2.2.3 Jet releases

Jet releases are intensively studied for more than 50 years, and the scaling properties were in detail described starting with the classical work of Abramovich (1963) and later of Chen and Rodi (1980), who proposed the correlation describing axial concentration

$$\frac{C(x)}{C_0} = K \frac{d_0}{x - x_0} \left(\frac{\rho_\alpha}{\rho_g} \right)^{\frac{1}{2}},$$

here x is distance from the exit along centerline, $C(x)$ is the volumetric concentration at location x , C_0 is the concentration at the outlet nozzle, d_0 is the jet discharge diameter, ρ_α is the density of the ambient air, ρ_g the density of the gas at ambient conditions, x_0 is the position of the hyperbola asymptote, and K is a constant equal to 5. The concentration of hydrogen has Gaussian distribution along jet radius is inversely proportional to the distance to the nozzle along the axis of the jet.

Expansion of the classical similarity law for underexpanded jets and its validation for both expanded and underexpanded jets is presented in Molkov (2012). For turbulent jets, concentration on the centerline decreases inversely proportional to the distance from orifice for round jets and to the square root of distance for planar jets. Few detailed studies of the orifice shape influence on the hydrogen concentration in the expanded jet area were performed in Scase et al. (2006) and Makarov and Molkov (2013).

If jet is released when wind is sufficiently strong, as the jet loses momentum and dilutes, the jet tends to be bended in the direction of the wind. This leads to the change of the momentum direction close to the wind orientation, and creates a small dynamic pressure difference between upwind and downwind sides of the plume. This initiates internal gas circulation so that the jet splits up into two oppositely rotating vortices (Zhang & Ghoniem, 1993).

During dilution, the velocity and density of a plume approaches the ambient values and at this later stage the plume mixes passively.

Small jet releases, which are buoyancy controlled, do not follow scaling law for the momentum-controlled jets and usually mixed faster. In Molkov and Saffers (2011), authors have proposed general engineering correlation for the jet flame length for both buoyant and momentum jets, thus providing uniform tool for the prediction of the jet hazardous potential.

Proximity of jet to the surface can significantly alter parameters of the hydrogen distribution. Some pilot works showed the increase of the flammable length and diverse concentration distribution properties. In Bénard et al. (2007), authors found a transient behavior short after initiation of the release created by the recirculation zone between nozzle and the surface.

In case when liquefied hydrogen forms the jet, two-phase mixture can create highly inhomogeneous object, where fast evaporation of the liquid particles can result locally in raised hydrogen concentration. In Kneebone and Prew (1974) it was reported that for the two-phase jet with the increase of the liquid fraction, the flammable cloud also increases.

3.2.2.4 Liquid pool

In the case of the liquid hydrogen spills, the hydrogen dispersion in the atmosphere occurs during and after vaporization of the cryogenic liquid. Just vaporized hydrogen has density higher than air until the temperature reaches 22 K, when density becomes equal to the air density. This period of negative buoyancy is changed shortly due to the rapid heating of the mixing flow and creation of the rising plume above the pool. For the large spills, the vaporized gas can affect environmental conditions changing wind profile in the area of the cloud. Under such conditions known as “vapor blanket,” the atmospheric wind is lifted up by the vapor, making the wind inside cloud decreasing nearly to zero.

In case of long-lasting leaks or high release rates of liquid hydrogen, the mixing is controlled by the intense turbulence generated by the source. For the gradual pool vaporization, the stable stratification, which typically appears, reduces boundary layer turbulence, stabilizing the buffer layer.

In case if the heat exchange between the mixture and environment is negligible, the adiabatic mixing approach can be applied. In such case the hydrogen concentration in the mixture can be evaluated, if the mixture temperature and state of the air (pressure, temperature, and humidity) are known. Comparison with the experimental observations has confirmed reasonable accuracy of this approach.

As atmospheric moisture always exists, depending on the relative humidity, the condensation and solidification of the water vapor can affect mixing process. The heat balance in this case is distorted by the phase transition, and the buoyancy is changed with the change of cloud parts' density. The condensation of the atmospheric water vapor can make visible an outline of the cold hydrogen cloud. Interesting that at near normal temperatures and humidity $\sim 50\%–60\%$, the contour of the visible image coincides with the border of the flammable mixture.

3.2.2.5 Catastrophic releases

In the case of large opening or complete pressurized vessel failure, strong pressure waves will propagate from the opening outward. The hydrogen from the vessel will expand very rapidly until the pressure inside vessel becomes atmospheric. In case of liquid hydrogen, such rapid spreading will force flashing for the part of the liquid, and spill for the rest. The cloud will grow initially due to near isentropic expansion, and then the expanded cloud fed additionally from the spillage (if present) will form a rising cloud where dispersion is controlled by the buoyancy and atmospheric turbulence.

3.2.3 Dispersion in confined environment

When hydrogen leak happens within enclosure, depending on the geometry and construction properties of the enclosure and the leak characteristics, the following relevant situations can be important for the safety:

- localized accumulation in the closed areas;
- creation of the extended layered burnable mixtures with stratified concentration distribution;
- enforced mixing and turbulization of the leaked hydrogen due to natural and imposed convective motion, which can be intensified by presence of ventilation systems.

These phenomena, which increase a risk of the ignition event leading to an explosion inside enclosures, present particular issues with respect to safety. Localized accumulations can appear near poorly vented dead-ends of the equipment or compartment sections or below ceiling in rooms. Turbulent well-mixed combustible mixture can be formed near leak in the vicinity of ventilation openings.

Competition between local accumulation and convective motion can result either in a stable stratified distribution of concentration or in the formation of a homogeneous layer if the convective flows at the top of the enclosure are high enough (Cariteau et al., 2011a,b; Studer et al., 2012). The factor that mostly controls the resulting distribution of the hydrogen in enclosure is presence of ventilation.

In a closed unventilated case, depending on the strength of the source and volume of the enclosure, one of three regimes can be observed:

- homogeneous mixture distributed in the whole volume;
- homogeneous layer in upper part of the volume;
- stratified (inhomogeneous) layer in upper part of the volume.

The governing number is modified Richardson number

$$Ri = \frac{V^{1/3}g}{U^2},$$

here V is case volume, U is characteristic velocity of leaking gas, and g is gravitational constant.

The listed regimes are observed for the following ranges of Richardson number $Ri < 2.5 \cdot 10^{-3}$, $2.5 \cdot 10^{-3} < Ri < 3$, and $Ri > 3$ correspondingly.

For the first two regimes an engineering “zonal” model was proposed in Cleaver, Marshal, and Linden (1994). The model demonstrated good accuracy in comparison with experimental data

(Cariteau & Tkatschenko, 2012; Gupta, Brinster, Studer, & Tkatschenko, 2009). For the third regime an analytical model was proposed earlier in Worster and Huppert (1983) and demonstrated good accuracy for the larger orifice diameters more than 20 mm in experiments (Cariteau et al., 2011b) and in simulations (Venetsanos et al., 2009). For the smaller diameter as 5 mm and less, only linear empirical correlations exist, which require further analysis.

In Benteboula, Bengaouer, and Cariteau (2009), by analysis of simplified mathematical models, the effects of release velocities, diameters, durations, mass flow rates, and flow regimes on the vertical distribution of the gas concentration were studied. It was found that for relatively longer time gas discharge for jet-like or plume-like flows the model provides reasonable estimates. Note that such models can deliver only rough ideas on the mixture distribution, as the practically important details, such as aspect ratio of enclosure, leak source position and orientation, volume congestion, and other similar particulars, are not taken into account.

3.2.3.1 Natural and forced ventilation

Presence of ventilation in the enclosure brings another degree of complexity into the system. The models predicting distribution dynamics are more sophisticated while in simple situation are still able to produce reasonably accurate results.

For naturally ventilated enclosure, few models were proposed. For the enclosure with one vent, “mixed regime” model was presented by Linden (1999). The model considers properties of the mixture distribution during the transient and steady state phases assuming that the blend is in fully mixed state and there is no external influence such as wind blowing-in through the vent.

The author also proposed a model for steady-state parameters accounting for the second vent located in the lower part of the volume, when the first vent positioned as in the one-opening model in the upper part of the enclosure. The model describes parameters of the well-mixed layer giving concentration of the blend and position of the interface. Both models were intensively validated against experiment of different scales and demonstrated good accuracy (Barley & Gawlik, 2009; Cariteau et al., 2011a; Merilo, Groethe, Colton, & Chiba, 2011; Swain, Filoso, Grilliot, & Swain, 2003).

In Lowesmith, Hatkinson, Spataru, and Stobbart (2007) a model for two openings, which accounts for the external wind with various directions, was proposed. The “zonal” model provides depiction of the transient phase, extending the earlier model of Linden.

For the setting with possible hydrogen leak, ventilation system as mitigation means has a goal to maintain concentration of hydrogen below the lower flammability limit. The conventional ventilation systems include exhaust openings, chimneys, ducts, and can contain additional equipment such as heat exchangers, fans, grids, or other components.

The performance of ventilation system is defined by the leak rate and the geometrical arrangement of the areas, which should be ventilated. In case of forced ventilation, the fans supplying air have to be able to provide about 25 times more air than hydrogen to maintain lower hydrogen concentration.

Under certain conditions in the enclosure with one vent a “pressure peaking” phenomenon was observed (Shentsov, Molkov, & Kuznetsov, 2015). The phenomenon is characterized by the existence of a peak on the overpressure transient in an enclosure with vent(s) at some conditions, and the expected overpressure can significantly exceed the steady-state pressure.

3.2.3.2 Nonuniformities in mixed distribution

By small leaking gas speed or in the areas where the leaking hydrogen has lost the momentum and where convective flow motion is slow, the accumulated hydrogen can form a nonuniform stratified structure.

Typically, for the inhomogeneous layer of hydrogen in air, the concentration of hydrogen is at maximum directly above the leak location and decreases in the vertical direction downward and on the periphery of the cloud outward if the cloud did not reach the bounding walls.

In large enclosures with buoyant plume or jet leak, the formed mixed layer is usually stratified. The conditions describing transition from well-mixed to stratified nonuniform cloud were introduced by Peterson (1994). For the case with buoyant jet source, the condition of stable stratification reads as

$$\frac{H_{sf}}{d_{bjo}} Ri_{bjo}^{\frac{1}{3}} \left(1 + \frac{d_{bjo}}{4\sqrt{2} \alpha_T H_{sf}} \right) > 1,$$

here H_{sf} is the height of the enclosure, d_{bjo} is jet orifice diameter, $\alpha_T = 0.05$, and Ri_{bjo} is jet Richardson number

$$Ri_{bjo} = \frac{(\rho_a - \rho_0) g d_{bjo}}{\rho_a U_0^2},$$

ρ_a is density of the atmosphere, ρ_0 is density of the leaked gas.

In Schatzmann (1979), a differential equation model for the prediction of spreading and rising of submerged round buoyant jets discharged into a stratified, flowing ambient fluid is presented. The model is able to operate with flows ranging from the simple momentum jet to buoyant plumes in stratified cross-winds with temperature inversion.

Nonuniform cloud can appear to be a phenomenon that brings not fully understood and therefore underestimated risk into systems with hydrogen. In the stratified nonuniform distributions, propagating flames have increased likelihood to meet areas where high-speed regimes of combustion are very probable. In addition, the presence of the concentration gradients can promote strong flame acceleration potentially leading to detonation or quasi-detonation regimes.

As release of hydrogen is one of the typical consequence of the postulated accident in nuclear power station, the dispersion of hydrogen and particularly stratified dispersion were intensively and thoughtfully studied in nuclear safety (e.g. Allelein et al., 2007; Lowesmith, Hankinson, Spataru, & Stobbart, 2009; Marangon & Carcassi, 2014).

3.3 Hydrogen ignition

Occurrence of an accident situation assumes ignition of the combustible mixture. There is a number of conceivable ignition sources, among which the most essentials include electrical sparks, fused wires, incendiaries, hot surfaces, rapid adiabatic compression, shock waves, and catalytic materials. The principal mechanism of ignition consists in heating of a mixture portion to high enough temperatures when the chemical reaction starts and heats up the neighboring portions of the mixture allowing this process to propagate further in a self-maintaining mode.

For hydrogen – air mixture to ignite, the mixture has to be in the certain concentration limits and the energy of the ignition source has to be higher of the specific level. The concentration limits depend also on the temperature and pressure in the mixture. For characterization of the energy of ignition a “minimum ignition energy” term is used, it corresponds to the minimal energy that is necessary to obtain ignition of the quiescent mixture by normal pressure of 1013.25 hPa and 20°C.

Numerous studies to determine these parameters were performed since the mid of 20th century (e.g., Coward & Jones, 1952; Kumar, 1985; Lovachev et al., 1973), and their values are determined with good accuracy and consistency. In technical report (ISO, 2015), the basic parameters defining ignition properties are summarized. As an example, a table demonstrating dependence of the flammability limits on the temperature is shown in Table 3.4.

For hydrogen, the minimum ignition energy for hydrogen – air stoichiometric mixture is extremely low having only 0.017 mJ (ISO, 2015). Such low values can be easily achieved by almost any conceivable source.

At the normal conditions, hydrogen is a gas, therefore it does not have flash point, which is the lowest temperature at which vapors of a volatile liquid ignites. However, cryogenic liquefied hydrogen can flash at all temperatures above its boiling point of 20 K.

Often a term spontaneous ignition is used. Under this term a process of self-heating due to internal exothermic reaction followed by the rapid increase of temperature and finally establishing of self-sustaining combustion, is assumed. Spontaneous ignition can be initiated by the above-mentioned ignition sources, or as in case of charcoal can be an intrinsic property of the substance.

The physical phenomenon that results in spontaneous ignition is usually referred as autoignition. For hydrogen – air mixtures, the chain-branching mechanism provides temperature rise with consecutive reaction rate acceleration up to establishing of stable reaction regime. Depending on the mixture composition, pressure and temperature, and mixture vessel characteristics, ignition can happen or not. In Fig. 3.3, a diagram for the explosion limits of hydrogen – oxygen mixture is presented. Typically, when the heat release in the reaction exceeds heat losses due to recombination or to collisions with walls, the autoignition can take place. In the figure, this situation corresponds to the lower peninsula and the situation with the quenching of the reaction relates to the upper peninsula without possibility of autoignition.

The lowest temperature at which the autoignition event occurs is referred as autoignition temperature. This temperature typically decreases as the pressure or oxygen concentration increases. In

Table 3.4 Dependence of hydrogen flammability limits on mixture temperature (DIN_EN_1839, 2004).

Temperature (°C)	LFL (% vol.)	UFL (% vol.)	LAC (% vol.)	LOC (% vol.)
20	4.1	75.6	20.4	4.3
100	3.4	77.6	19.1	4.0
200	2.9	81.3	15.0	3.2
300	2.0	83.9	10.9	2.3
400	1.4	87.6	6.2	1.3

LFL, Lower flammability limit for H₂ in air; *UFL*, Upper flammability limit for H₂ in air; *LAC*, limiting air concentration; *LOC*, limiting oxygen concentration.

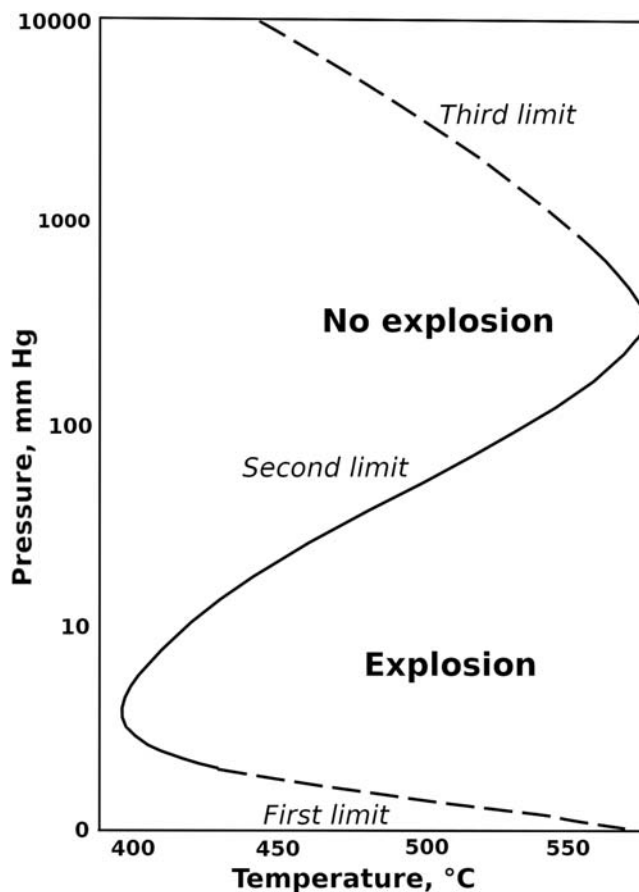


FIGURE 3.3

Ignition limits for a stoichiometric mixture of hydrogen with oxygen (Lewis & von Elbe, 1987).

the normal conditions for pressure about 1 bar, in mostly open conditions when the influence of walls can be neglected, the temperature of autoignition as it is referred in literature equals to 585°C.

3.3.1 Static electricity and electric spark

An electric spark as an ignition source is a rapid electrical discharge appearing when a sufficiently high electric field creates ionized channel through hydrogen – air mixture. The spark is created when produced electric field exceeds certain breakdown value, for air it is about 30 kV/cm. The value of the breakdown voltage depends on the characteristics of the mixture (pressure, humidity, impurities, or additives, etc.) while the characteristics of the discharge current depend largely on the properties of the electrical circuit participating in the discharge. The detailed model with the

experimental justification of hydrogen – air ignition by electric spark was presented in Thiele et al. (2002).

In case when the discharge is caused by the accumulated static electricity, it is called electrostatic discharge. It can happen in the form of the normal electric spark, and also as corona discharge from sharp electrodes or brush discharge from blunt electrodes. Such a discharge cannot be characterized by a certain energy, but the ignition potential can still be equaled to the normal spark. In von Pidoll, Brzostek, and Fröchtenigt (2004), a guidance was recommended allowing to correlate the minimum ignition energy and the charge transferred during release. Further elaboration of the conditions was performed in CENELEC (2019), with the maximum tolerable charge transferred for hydrogen is given as 10 nC.

In Astbury and Hawksworth (2007), authors have performed a critical review of several incidents and have concluded that there is a distinct possibility that observed spontaneous ignitions may be of an electrostatic origin.

3.3.2 Hot surface

Ignition by a hot surface can happen when the combustible mixture locally is heated to the temperatures usually well above the autoignition temperature. The preheated volume should be large enough to provide self-sustained further flame propagation. In addition, the conditions of the ignition can be influenced by the shape of the surface, its material, and parameters of the gas flow as its velocity, turbulence, etc. (Buckel & Chandra, 1996; Lewis & von Elbe, 1987; Proust, Carleton, Bothe, & Hawksworth, 2000).

The same authors have found that temperatures igniting H₂ mixtures with air and oxygen are between 640°C and 930°C, remarkably, that easy ignition is achieved mostly for the leaner H₂-air mixtures and in the tests with very small hot surfaces in Proust et al. (2000) and Hawksworth et al. (2005) it was suggested that concentrations in the range of 10% – 15% are the most easily ignited.

When hot surface is catalytic, the effective reduction of the autoignition temperature can be radically downgraded, for example, in Cho and Law (1986), it was found that ignition temperature on platinum surface was reduced to 70°C.

3.3.3 Mechanical friction and impact

The nature of the ignition by mechanical friction is most likely directly similar to the ignition by the hot surface. In early work (Brearley & Tolson, 1995) where power levels and contact loads required to ignite combustible mixture were studied, it was noted that ignition took place mostly by hot spot close to the point of contact. In the later works (Hawksworth et al., 2004), the ignition was observed at temperature 530°C in the hot areas and with few sparks produced at low speed movements.

In case of metal or other stiff materials impact, formation of small particles is highly probable. Such particles can have extremely high temperature and serve as an ignition source for combustible mixtures. Powell (1984) reported that light impact with energy below 1 J of aluminum, magnesium, and few metals from Ti-subgroup produces burning particles with temperature exceeding 2000°C easily igniting flammable mixtures.

3.3.4 Miscellaneous

Among several accidental hydrogen releases followed by the spontaneous ignition, there were several instances that have been investigated, but no satisfactory explanation has been produced. The energy of the ignition in case of hydrogen is tremendously small in contrast to the customary flammable gases as hydrocarbon-air mixtures, so the source of the ignition can be fairly unusual.

In case of propagating shock wave when it enters a dead end by the reflection or focusing in oblique reflections, conditions allowing a portion of the gas to be compressed and heated to the state close to autoignition, an ignition with the following flame formation can occur.

Largely similar situation is realized in the area surrounding initiation of explosive. Strong shock propagating from the explosion source brings the gas to the preconditioned state to make ignition possible. In contact with the open flame the transition of the combustion process through the contact surface interface is very likely if contact surface and duration of the contact is sufficiently large.

From a highly specialized point of view, the hydrogen – air mixture is chemically stable. It can be stored for long periods without any transformation of composition or heat release. However, in presence of other oxidizers, such as fluorine and chlorine or catalytic materials, the hydrogen reacts very actively and becomes unstable with spontaneous ignition.

In Proust et al. (2000), it was demonstrated that optical radiation has a potential to cause ignition in flammable gases. Four main effects were found important for the possibility of ignition:

- radiation absorbed by particle surface heating them up to autoignition conditions;
- heat up of gas volume, where the optical wavelength matches an absorption band of the gas;
- photochemical ignition by radiation in the ultraviolet wavelength range;
- direct laser induced breakdown of the gas at the focus of a strong beam.

3.4 Combustion of hydrogen

The hydrogen burns in almost invisible pale blue hot flame producing water vapor and releasing during chemical reaction energy almost completely as heat.

In combustion, two base categories are distinguished. One is called premixed combustion, and for the second one a term nonpremixed combustion is used. In hydrogen – air mixtures both regimes occur. Premixed combustion takes place if hydrogen and air or any other oxidizing gaseous species are mixed prior to the ignition. Nonpremixed combustion refers to the conditions when hydrogen and oxidizer are separated before and during the process. The governing phenomena and potential consequences are quite different for both regimes and will be discussed in different sections in this chapter.

Premixed combustion of hydrogen can occur in quite different regimes. The visible combustion speed for laminar regimes can proceed with velocities up to approximately 20 m/s, while for turbulent regimes it can range from tens to hundreds meter per second, and for detonation of stoichiometric hydrogen – air mixture it is about 2000 m/s.

3.4.1 Laminar premixed flames

The fundamental phenomena in combustion is laminar flame in the premixed burnable gas blend. The study of the laminar flames is exceptionally complicated as the flame is a dynamic object that

is hot, thin, fast, and having complex internal structure. The knowledge of the laminar flames provides information, which is not only helpful for the simple flat flames but also for the corrugated and wrinkled flames due to the flow turbulence.

The burning velocity of flammable hydrogen – air mixtures increases with concentration for lean mixtures reaching 2.55 m/s at stoichiometric conditions, grows up to 3.2 m/s at 40.1% and decreases with further enrichment. For hydrogen – oxygen mixtures the burning velocity reaches its maximum of 11.75 m/s.

3.4.1.1 Structure of the reaction zone and flame temperature

As the first attempts to understand the nature of the flame propagation mechanism, it was hypothesized that heat conduction is the primary process controlling the propagation. After the works of Zel'dovich of the 1940s, it is common to accept few statements.

- The flame has a unique propagation speed.
- The flame transmits heat upstream by thermal conductivity.
- The flame has thin high-temperature zone where reaction occurs, the zone upstream before the mixture ignites is preheat zone.

In such “thermal” flame concept, the internal structure of the laminar flame is defined by the collective effects of the chemical reaction thermodynamics and kinetics and transport properties of the mixture. The basic structure of the laminar flame is shown in Fig. 3.4.

The thermo-chemical and transport properties of the species are studied since the pioneering works of A. Fick, who introduced the now well-known laws governing the transport of mass through diffusive processes. Today the solid array of data on the transport properties of the different materials are available in the databases, as for example, *NIST Reference Fluid Thermodynamic and Transport Properties Database (REFPROP)* (Lemmon et al., 2013).

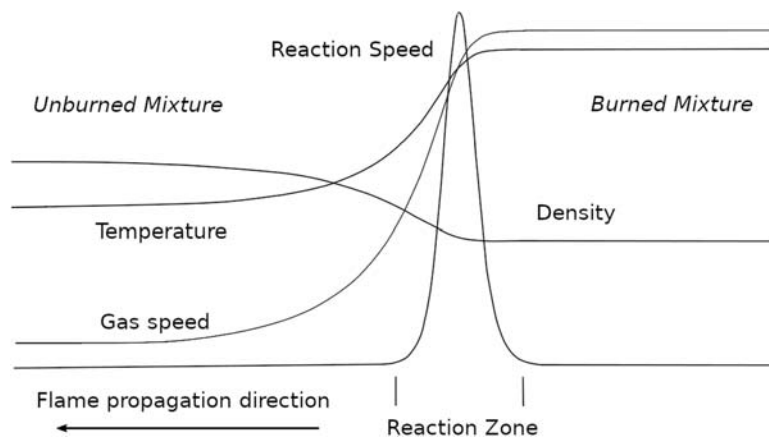
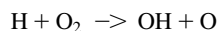


FIGURE 3.4

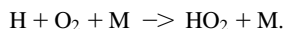
Laminar flame structure.

The reaction mechanisms, which are used for the calculations of flame properties, are intensively studied since mid of the 20th century. First achievements in the development of the experimental methods for measurements of the fast chemical kinetics were obtained by British chemists R.G.W. Norrish and George Porter, and German chemist Manfred Eigen, who were awarded the Nobel Prize for Chemistry in 1967. Since the initial studies, the detailed descriptions of the kinetics mechanisms were intensively revised and extended. Currently, the implementation of the modern transport database (Jasper et al., 2014) together with last revisions of the hydrogen kinetic scheme (Konnov, 2019) provide kinetic model of high quality and performance.

For the laminar hydrogen flames, it has been definitely proven that diffusion of hydrogen atoms into the preheat zone is the primary mechanism that provides flame propagation through chain-branching reactions. Hydrogen atoms participate in two competing reactions, one is chain-branching reaction



and another one is recombination reaction.



The competition between chain-branching fast reaction by high temperature and slow but temperature-independent recombination leads to the final equilibrium. In the preheat zone where temperature is relatively low, the recombination reaction dominates, and lowering of the unburned mixture temperature leads to the decrease of the burning velocity. In contrast, for the pressure change a negative effect is observed.

The chemical conversion from reactants to products in the course of the combustion process is followed by release of energy. In the case of lossless isobaric process, the combustion products obtain the temperature called adiabatic flame temperature, which is controlled by initial pressure, temperature, and the mixture composition. The characteristic dependence of the adiabatic temperature on fuel-oxidizer ratio is given in Fig. 3.5. The similar dependence of the laminar flame speed demonstrate intrinsic relation of the chemical energy supply and flame propagation rate.

For moderate conditions of the hydrogen combustion as well as for most hydrocarbon – air mixture combustions, the brutto reaction can be used for the evaluation of the adiabatic flame temperature. To exemplify a use of simple correlations, the expression of Chang and Rhee (1983) can be referred

$$T_{ad} = a(1 + b \cdot \ln \phi + c \cdot \ln^2 \phi),$$

where

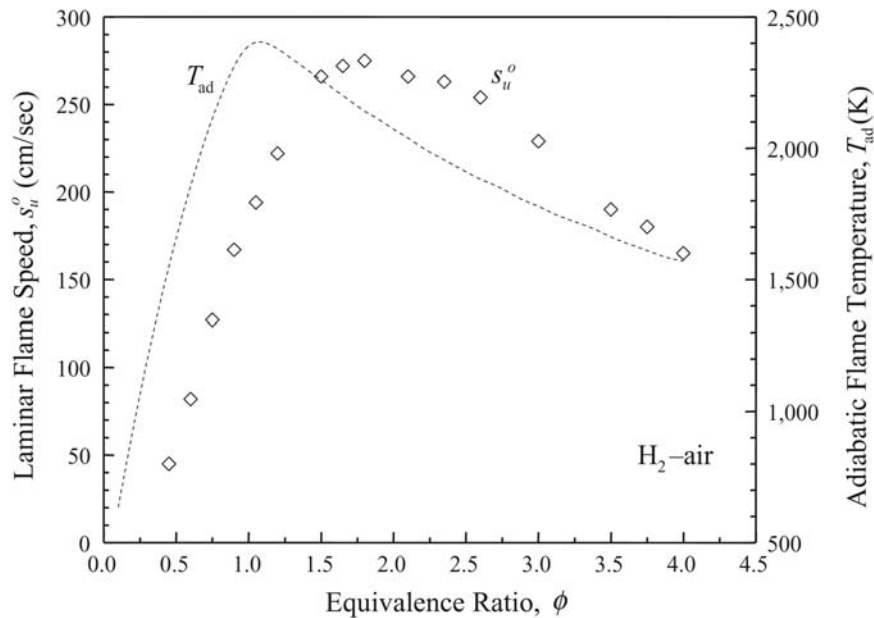
$$a = a_1 + a_2 \ln (P/P_0),$$

$$b = b_1 + b_2 \ln (P/P_0),$$

$$c = c_1 + c_2 \ln (P/P_0),$$

$$\phi = \frac{Y_{fuel}}{(Y_{fuel})_{St}}$$

Here T_{ad} is adiabatic flame temperature in K, P is pressure, $P_0 = 1$ atm, Y is mole fraction. The coefficients for a , b , and c for hydrogen – air mixture are given in Table 3.5.

**FIGURE 3.5**

Dependence of adiabatic flame temperature and laminar flame speed on the fuel equivalence ratio ϕ , for hydrogen – air mixtures at initial conditions of 1 bar and 298.15 K (Law, 2006).

Table 3.5 Coefficients for calculation of adiabatic flame temperature.

	a, K	b, K	c, K
Coefficients indexed by 1	2412.2	0.473	0.0217
Coefficients indexed by 2	17.9	0.021	0.0191

3.4.1.2 Dependence of laminar burning velocity on mixture composition, pressure, and temperature

Laminar flame speed is a fundamental property of the premixed combustion. When the flame propagates in the motionless and uniform mixture of burnable reactants and does not undergo stretching or wrinkling, the velocity of such flame is the laminar flame speed. According to the pioneering work of Mallard and Le Chatelier, the laminar speed depends only on the properties of the mixture.

$$S_L = \sqrt{\chi \dot{f} \frac{T_b - T_i}{T_i - T_u}},$$

where χ is thermal conductivity, $\dot{f}$ is chemical reaction rate, T_b , T_i , and T_u are temperatures of the burnt, ignition, and unburned mixture correspondingly.

Later the theory was revised and improved with the quite significant input made by [Zeldovich \(1951\)](#) and Frank-Kamenetskii. In the presentation of Semenov, who generalized their theory, similarly to the initial Mallard–Le Chatelier theory, the author used an ignition temperature, which actually does not exist in the flame, but by approximation he has eliminated it from the equation to obtain finally more practical results. Such approach is associated to one named as *activation energy asymptotic* currently.

The further theory improvements were achieved when accounting of the radicals' diffusion was proposed. The introduction of the particle diffusion idea was first made by [Lewis and Von Elbe \(1934\)](#) and substantial input in the concept of radical diffusion were made later in works of [Tanford \(1947\)](#), [Tanford and Pease \(1947\)](#).

The followed advanced theory development showed that even most complex approaches still did not provide required quality of the theoretically predicted flame characteristics. It is finally recognized that the exact flame parameters can be obtained from computation of the fluid dynamics together with detailed resolution of the chemical reactions with their heat effects.

Uncertainties in the predictions of the theoretical models initiated experimental researches aimed to determine flame speed characteristics and especially the effect of temperature and pressure variations. Due to this fact, currently the experimental study remains the unique actual source on the laminar flame characteristics.

In many applications, and for practical reasons, it is often convenient to assume that chemical processes are described by a single one-step global reaction. In such cases, instead of looking for explicit experimental data or performing detailed simulations, it is usually feasible to rely on the approximation of the existent verified data. The laminar burning velocity in this case will depend on several variables characterizing the mixture, for example, heat conductivity, reaction order, activation energy, temperature of the mixture ([Zeldovich, Barenblatt, Librovich, & Makhviladze, 1985](#)).

The laminar burning velocity dependence on pressure, temperature, and fuel composition in hydrogen – air – steam mixtures is mostly characterized by the power function

$$S_L(\phi, T, p) = S_{L,0}(\phi) \left(\frac{T}{T_0} \right)^\alpha \left(\frac{p}{p_0} \right)^\beta.$$

Dependence of the flame velocity on the mixture composition were measured by a number of authors (e.g., [Dahoe, 2005](#); [Dowdy, Smith, Taylor, & Williams, 1991](#); [Iijima & Takeno, 1986](#); [Koroll, Kumar, & Bowles, 1993](#); [Takahashi, Mizomoto, & Ikai, 1983](#); [Tse, Zhu, & Law, 2000](#)), and currently data for H₂ – air mixtures with equivalence ratio ranging from ≈ 0.2 –5.5 are known. In [Fig. 3.6](#), data on the laminar burning velocity from several authors are shown. Note that relatively high dispersion of the experimental results exist especially for the lean mixtures with low equivalence ratio. The considerable deviations in the experimental data initiated intensive surveys to calculate numerically laminar speed on the basis of the detailed kinetics instead of creating of the analytical approximation of the experimental data.

Since 80s of the last century, several detailed mechanisms were proposed ([Maas & Warnatz, 1988](#); [Lutz, Kee, & Miller, 1988](#); [Li, Zhao, Kazakov, & Dryer, 2004](#); [Mueller, Kim, Yetter, & Dryer, 1999](#); [Konnov, 2019](#)) accounting on the different aspects of the elementary reactions and molecular transport. If some of these schemes demonstrate excellent predicative capabilities for the certain range of parameters, they can perform unsatisfactorily beyond this range. Alternative approach to direct simulation of the laminar flame speed was demonstrated in [Szabó et al. \(2012\)](#),

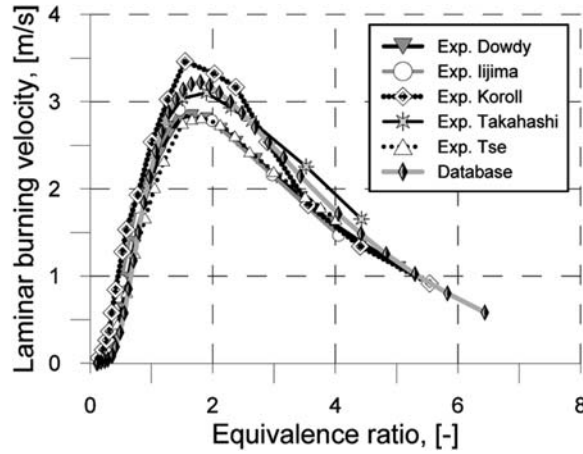


FIGURE 3.6

Laminar burning velocity dependence on equivalence ratio at 300 K and 1 bar. Comparison of calculated data (Szabó et al., 2012) with experimental results by Iijima and Takeno (1986), Koroll et al. (1993), Dowdy et al. (1991), Takahashi et al. (1983), Tse et al. (2000).

where the authors proposed a new correlation describing the laminar burning velocity of hydrogen/air mixtures as a function of temperature, pressure, and mixture composition is proposed (see Fig. 3.6). The correlation was created in two steps: first, the laminar burning velocity was calculated with the detailed reaction mechanism of Maas and Warnatz (1988); second, the results obtained were validated against experimental data and a new heuristic approximation based on them was formulated. The correlation is a set of simple mathematical expressions and provides a good approximation of the laminar burning velocity in the range between 200 K and 600 K for temperature, between 0.1 bar and 10 bar for pressure, and up to 20 mol% steam dilution.

A rise of the temperature of the reactants independently on the mixture composition typically increases the reaction rate. In Iijima and Takeno (1986), authors found that the exponent depends on the equivalence ratio and have proposed formula for temperature interval of 291 – 500 K. Hu, Huang, He, and Miao (2009) observed, both in measurements and in calculations with the CHEMKIN code (Kee, Rupley, & Miller, 1989), that the exponent depends only on temperature in a wide range. The extension of the general dependence was proposed by Koroll et al. (1993), including the variation of the thermal conductivity λ

$$S_L(\phi, T) = S_{L,0}(\phi) \sqrt{\frac{\lambda(T)}{\lambda(T_0)}} \left(\frac{T}{T_0} \right)^\alpha,$$

with $T_0 = 298$ K, and utilized the expression for the exponent (Liu & MacFarlane, 1983), which depends only on the hydrogen concentration

$$\alpha = 1.571 + 0.3839 \cdot (0.42 - X_{H_2}); \text{ for } X_{H_2} < 0.42,$$

$$\alpha = 1.571 - 0.2476 \cdot (0.42 - X_{H_2}); \text{ for } X_{H_2} > 0.42.$$

Variation of the pressure has a more complex influence on the laminar burning velocity. In the study of Behrendt and Warnatz (1985), it was found that the pressure exponent is variable with positive values for increasing pressure ($\beta > 0$) up to one bar and negative for higher pressures ($\beta < 0$). In addition, they have discovered that for the stoichiometric mixtures, the pressure exponent also depends on the nitrogen dilution. Shebeko et al. (1995) realized that an increase of temperature will alter the pressure at which a maximum of laminar flame speed is achieved at higher values. In addition, they concluded that the pressure exponent is a function of temperature also. Iijima and Takeno (1986) proposed an expression for the exponent that depends linearly on the equivalence ratio. Based on their measurements, Verhelst and Sierens (2007) tried to improve the results of Iijima and Takeno and suggested a dependence that leads to a change of sign at $\varphi = 0.4$. Using results obtained with CHEMKIN, Hu et al. (2009) created a correlation valid at ambient temperature for pressures between 1 bar and 80 bar. Consequently, the discussed formulas consider the pressure exponent either as a function of equivalence ratio or of pressure, disregarding the complex dependencies of β on equivalence ratio, temperature, pressure, and diluting concentration. Mixtures diluted by nitrogen or steam have lower flame temperatures in comparison with the undiluted. These diluents in general slow down diffusion processes and steam in particular affects the reaction kinetics reducing the laminar burning velocity, see Le Cong and Dagaut (2008) and Shebeko et al. (1995). This influence was quantified by Koroll et al. (1993) by measuring the laminar burning velocity of hydrogen – air mixtures with H_2 concentrations from 9 mol% to 70 mol% and steam dilution up to 42 mol% and ultimately producing correlation that takes into account the change of thermal conductivity λ due to dilution

$$S_L(\phi, T, x_{dil}) = S_L(\phi, T) \cdot \sqrt{\frac{\lambda_{dil}(T)}{\lambda_{pure}(T)}} \cdot \left(1 - \frac{x_{dil}}{x}\right).$$

The exhaustive summary of the available data on the laminar burning velocity for hydrogen – air mixtures is given in Dahoe (2005) and is presented in Fig. 3.7.

3.4.2 Turbulent premixed flames

In most practical situations, the flow conditions affect the combustion process in a way that very soon after ignition the flow becomes turbulent. Turbulence leads to more effective mixing and consequently to more intensive chemical reaction, which can finally lead to the resulting flame velocities exceeding laminar flame speed two order of magnitude.

Particular interest for the practical purposes is focused on the studying of the turbulent flame speed, which is assigned to the average propagating velocity of the zone with intensive chemical reaction through premixed turbulent flow of reacting components.

In the laminar flow, it is implicitly assumed that the flow does not affect chemical mechanism. On the contrary, in case of turbulent combustion such influence can and actually usually occurs.

To understand the influence level from the flow turbulence on the burning process, the comparison of the characteristic times of the individual phenomena from both turbulence and chemical interaction is required. If the chemical reaction proceeds fast enough, that is, the chemical characteristic time is much shorter than the characteristic time of turbulent fluctuations, the course of the reaction is generally not disturbed. In the other limit, for intensive turbulence, the characteristic

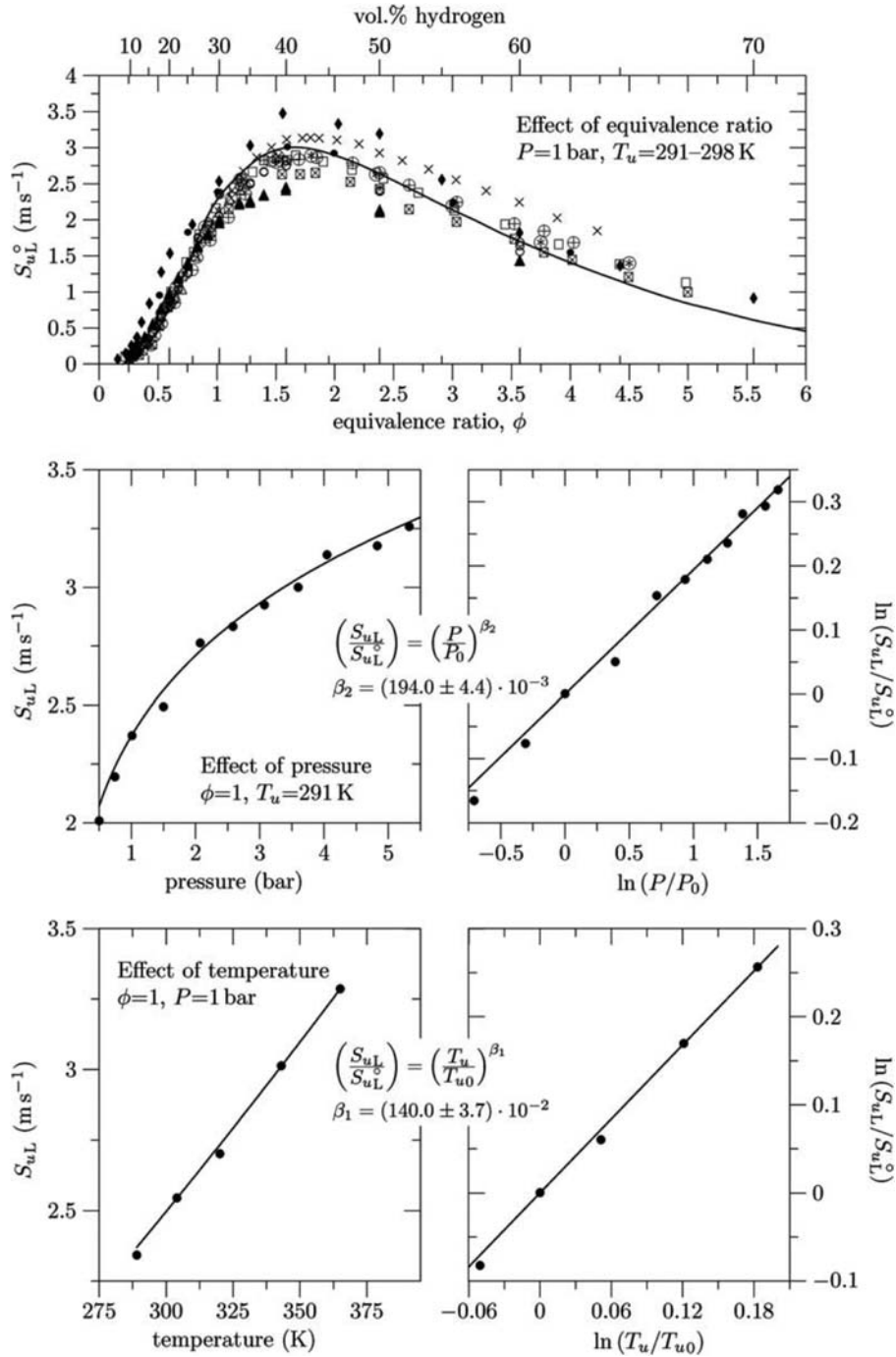


FIGURE 3.7

Effect of equivalence ratio, pressure, and temperature on the laminar burning velocity of hydrogen–air mixtures (Dahoe, 2005).

time of turbulence is much shorter than characteristic chemical time, the flame structure, including energy release rate, is significantly changed.

3.4.2.1 Turbulence scales

To understand behavior of turbulent combustion systems, an appropriate starting point is to consider the fluid parameters, which control the system properties. Turbulence is exceptionally a complex phenomenon, involving processes of rather different time and length scales.

Usual approach to consider turbulence is to separate averaged macroscopic motion of a fluid and fluctuating part of the motion as a representative of the turbulence. The flow typically has at least one characteristic velocity U and one characteristic length scale L of the geometric environment where the flow occurs. Thus the Reynolds number Re for such system reads as

$$Re = \frac{U \cdot L}{\nu},$$

where ν is a characteristic kinematic viscosity. Note that ν is approximately proportional to T^2 , and consequently, during combustion when temperature can easily be changed by a factor of three, the Re drops by an order of magnitude. Thus combustion possibly can dump the turbulence.

Such defined macroscopic Reynolds number can be used for global characterization of the turbulence. However, the parameters that are directly connected with the turbulence fluctuations can provide more straightforward understanding of the turbulent combustion.

First, the turbulent kinetic energy $q = U'^2/2$, U' is a characteristic of root-mean-square velocity fluctuations, is a measure of the turbulence intensity. Weak turbulence corresponds to $U'/U < 1$ and intense turbulence has U'/U of the order unity.

In addition, it is useful to consider length scales characterizing the turbulence rather than geometry, with one length scale l_t related to the large eddy sizes, often referred to as the integral scale. It is of the order of magnitude of the apparatus characteristic size but typically smaller. This length is used for the definition of a turbulent Reynolds number

$$Re_t = \frac{U' \cdot l_t}{\nu},$$

which emphasizes turbulence influence clearer than global Re .

Another length scale characterizing turbulence is one at which molecular effects become significant. From the energy-cascade theory established in 40s of the 20th century by [Kolmogorov \(1941\)](#), it is accepted that the turbulence is fed by the energy of the largest scales as it is produced by the flow motion. The energy then drifts from the largest scales to the consequently smaller scales until it reaches the scale when it dissipates by the viscous effects.

This scale can be defined in terms of dissipation rate ([Batchelor, 1993](#))

$$\varepsilon = \frac{dq}{dt} \sim d \frac{U'^2}{dt} \sim \frac{U'^3}{l_t},$$

here ε is energy dissipation rate reflecting transfer of turbulent energy from macro-scale to micro-scale. This relationship is named after its author as Kolmogorov law. The length scale

l_K , which relates the dissipation rate for the macroscale l_t and the microscale where molecular viscous effect is dominant,

$$l_K = \left(\frac{\nu^3}{\varepsilon} \right)^{\frac{1}{4}},$$

is called Kolmogorov scale. Using this definition, the microstructure can be additionally characterized by Kolmogorov velocity and time scale

$$U_K = (\nu \varepsilon)^{\frac{1}{4}} \quad t_K = (\nu \varepsilon)^{\frac{1}{2}},$$

Note that

$$l_K = \left(\frac{\nu^3 l_t}{U^3} \right)^{\frac{1}{4}} = \left(\frac{\nu^3 l_t^4}{U^3 l_t^3} \right)^{\frac{1}{4}} = \left(\frac{l_t^4}{Re_t^3} \right)^{\frac{1}{4}} = \frac{l_t}{Re_t^{3/4}}.$$

Thus the Kolmogorov length represents the distance at which dissipation stops the transfer of the energy over the scales and defines a cut-off of the turbulence spectrum.

Another supplementary length, with the size between l_t and l_K , which naturally appears in the equations for turbulent flows, is the Taylor length l_T , which characterizes appearance of strain in viscous medium.

$$l_T = \left(\frac{\nu}{U' l_t} \right)^{\frac{1}{2}} = \frac{l_t}{Re_t^{1/2}}.$$

For the reacting flow, laminar burning velocity S_L , flame length δ_L , and representative chemical time τ_c characterize combustion process (Peters, 1991)

$$\delta_L \approx \frac{\nu}{S_L} \quad \tau_c \approx \frac{\delta_L}{S_L}.$$

To distinguish between fast and slow chemical reaction, the Damköhler number Da , which relates characteristic mixing time τ_m and characteristic chemical reaction time τ_c

$$Da = \frac{\tau_m}{\tau_c} = \frac{l_t/U'}{\tau_c} = \frac{l_t S_L}{U' \delta_L}.$$

For large Damköhler numbers, when the reaction time is short (fast chemistry), the reaction zone is thin and often flame sheet becomes unstable and wrinkling can disturb initially flat flame surface. For small Da numbers, the chemistry is slower than mixing, which leads to the thickening of the flame zone up to the well-stirred regime of combustion.

A derivative from Da number, which is applicable for characterization of a combustion with accounting of microscale turbulence parameters, is Karlovitz number Ka , obtained by substitution of the integral length scale by Kolmogorov scale

$$Ka = \frac{\tau_c}{\tau_K} = \frac{U_K \delta_L}{l_K S_L}.$$

If $Ka < 1$, the premixed turbulent flame falls into the category of corrugated flamelets and wrinkled flamelets, otherwise into the thin reaction zone or broken reaction zone flames.

3.4.2.2 Interaction between turbulence and flames: turbulent burning velocity

The complexity of the combustion processes makes their investigation one of the extremely hard scientific task. When the turbulence arrives on the scene, the difficulty of combustion examination increases manifold. The turbulence itself is one of the most challenging scientific tasks due to the dynamically changing structure of the turbulent flows, which includes broad spectrum of time and length scales with energy transfer between them; and the complex interaction between flame and turbulence brings additional level of complexity.

In contrast to the laminar flames, a turbulent flame speed is not simply defined by the mixture properties; it does also depend on the turbulence parameters of the flow where the flame propagates.

The turbulence can greatly influence the observable propagation flame speed. If the characteristic turbulence length is relatively large in comparison with flame thickness, the flame front sheet is wrinkling increasing its surface area and consequently the effective fuel consumption rate. If the flame thickness is greater than the characteristic turbulence length, then the turbulence is able to affect the internal structure of the flame. Depending on the conditions, this could lead to the two opposite scenarios. In some cases, turbulence enhances heat transfer through the flame toward fresh mixture and, therefore boosts the flame speed. In other cases, it could lead to the effective loss of energy due to an intensive mixing inside the flame sheet, and consequently to the local or even global quench of the flame.

In his early paper [Damköhler \(1947\)](#), for the large scale lower intensity turbulence, assumed that turbulent flame speed should be proportional to the area of the surface of the corrugated laminar flame

$$S_t = S_L \cdot \frac{A_L}{A_{average}},$$

here S_t is turbulent flame speed, S_L is laminar flame speed, A_L is total surface area of the laminar flame, and $A_{average}$ is averaged over time area of the turbulent flame. In addition, he has introduced an idea that the area ratio is controlled by the turbulence intensity

$$\frac{A_L}{A_{average}} = 1 + \frac{U'}{S_L},$$

where U' is velocity fluctuations in the unburned mixture upstream the flame and finally

$$S_t = S_L \left(1 + \frac{U'}{S_L} \right).$$

Since this first proposal, scientists are often trying to obtain correlations between laminar and turbulent flame speed in similar form. As an example, [Gülder \(1991\)](#) using fractal geometry ideas proposed the expression

$$S_t = S_L \left(1 + 0.7 \left(\frac{U'}{S_L} \right)^{\frac{1}{2}} \left(\frac{L}{\delta} \right)^{\frac{1}{4}} \right),$$

where L is integral turbulence scale and δ is turbulent flame thickness. Later, in [Bradley, Lau, Lawes, and Smith \(1992\)](#) another formula was proposed

$$S_t = 1.53 (Ka Le)^{-0.3},$$

based on the correlation between turbulent flame speed and flame stretch rate. In modern correlations, authors are successfully applying accounting of the dependence on the increased flame surface area by introduction of wrinkling factor, which is used as an additional multiplication factor for the correlations considering only other factors.

3.4.2.3 Borghi diagram and interpretation of combustion regimes

Depending on the interrelations between turbulence and flame parameters, the flame can propagate in rather different regimes. From the comparison of the chemical and flow scales a flame type can be easily identified. Following [Glassman, Yetter, and Glumac \(2014\)](#) the subsequent regimes can be recognized:

- wrinkled flame for $\delta_L < l_K$;
- severely wrinkled flame for $l_K < \delta_L < l_t$;
- flamelets in eddies $l_t < \delta_L < L$;
- distributed reaction front for $L < \delta_L$.

To characterize different flame propagation regimes, several specific terms assigned to the key property of the regime were introduced. For a flame with containing small scale folds, wrinkled flame term is used; for a surface containing large scale folds, the term corrugated flame is used.

In case when the flame zone is fragmented into a set of pockets containing reactants only or combustion products only, the term flamelet is usually applied. If such pocket has a mixture of both reactant and product or when unbroken flame sheet becomes thicker, it is usual to speak about thickened flames or thickened wrinkled flames. If with the increased level of turbulence, the flame becomes to be smeared out in space, the term well-stirred reactor regime is used. Each of these combustion regimes exhibits specific properties and has characteristic burning velocity.

To facilitate the interpretation of the dependences, an idea of a schematic representation of the regimes of premixed turbulent combustion in terms of velocity and length scale ratios have been proposed by [Borghi \(1988\)](#), [Peters \(1988\)](#), [Abdel-Gayed, Bradley, and Lung \(1989\)](#), [Poinsot, Veynante, and Candel \(1991\)](#), and many others (see [Fig. 3.8](#)).

Using definitions of the turbulence scales, the lines $Re = 1$, $Ka = 1$, and $Da = 1$ represent boundaries between different regimes of premixed turbulent combustion. Other borders of interest are the line that separates the wrinkled flamelets from the wrinkled flamelets with pockets (corrugated flamelets), and the line denoted by $L/\delta_L = 1$, which separates thickened flames from the well-stirred reaction zone.

The lower left corner of the diagram represents laminar flames. The boundary is approximately assigned to the value of $Re = 1$, which, however, for the real configurations, could differ significantly. In this area only stretch of the flame can affect the flame speed, as turbulence level is too low.

In the turbulent area with fast chemistry and weak flame stretch ($Re > 1$, $Da > 1$, $Ka < 1$), wrinkled flames and flamelets can exist. In the region with strong flame stretch $Ka > 1$ regime with the thickened (distributed) reaction zone is located. For region between $Da > 1$ and $Ka > 1$, thickened wrinkled flames appear. When intensity of turbulence exceeds level when the flame thickness increases such that $Da < 1$, the thickened flames emerge.

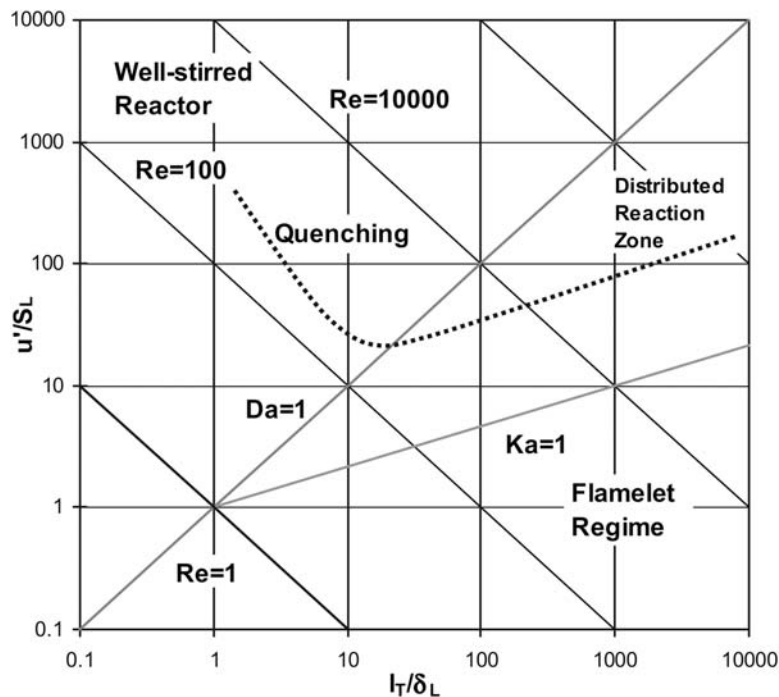
**FIGURE 3.8**

Diagram of the combustion regimes depending on the turbulence scale and turbulence intensity (Borghi diagram).

Source: Slightly modified from Vesper, A. et al. (2011). *The structure and flame propagation regimes in turbulent hydrogen jets*. International Journal of Hydrogen Energy, 36, 2351–2359.

The condition $Da < 1$ implies that characteristic mixing time is shorter than reaction time. Under such conditions, the flow fluctuations affect the course of the chemical reaction, and the laminar flame structure is distorted by these motions. The burning zone in such regime is rather a stripe where reaction takes place over the whole width of it. For the even larger values of Da number when $L/\delta > 1$, the regime of the well-stirred reactor could be observed.

Note that despite that the diagram has more illustrative than definitive character, it could be exceptionally useful for analysis of the turbulent combustion phenomena.

3.4.3 Laminar diffusion flames

In contrast to the premixed combustion when fuel and oxidizer are mixed before the burning starts, the situation when fuel and oxidizer are separated and coming into reaction by separate flows is referred as nonpremixed or diffusion combustion. For the diffusion flame, the process is controlled by diffusion as well as by convection and therefore the burning rate is delimited by the rate of component mixing rather than by the chemical kinetics. Diffusion flames can appear when pure

hydrogen release is ignited before it mixes with the surrounding air, as for example, at the break of pipeline or storage tank or by evaporation of liquid hydrogen.

The name diffusion flame was first used by [Burke and Schumann \(1928\)](#), who presented mathematical description of the flame established at the outlet of the two concentric ducts supplying fuel and oxidizer separately. They put forward a concept of infinitely fast chemistry and correspondingly infinitely thin flame front, which allowed them to successfully predict flame shape and height. Their hypothesis made it possible to understand the observed different shapes of diffusion flame. Depending on the mutual relation of the fuel and oxidizer, two regimes can be distinguished: underventilated when oxidizer is not enough for full oxidation of the fuel, and overventilated in opposite case. The first who made detailed measurements of species' distributions in a concentric laminar H_2 – air diffusion flame were [Hottel and Hawthorne \(1949\)](#).

The simplified approach of thin flame has demonstrated that it can provide reliable data of the global parameters, while the internal structure of the diffusion flame has a wider region over which the composition changes and chemical reactions can take place. It is usually accepted that the flame front is located at the stoichiometric iso-surface; however, the actual picture is more complex. In numerical simulation (see [Fig. 3.9](#)) of Burke – Schumann diffusion flame with detailed chemical mechanism it was found that in reality reaction zone can be of several millimeters thick ([Fukutani, Kunioshi, & Jinno, 1991](#)).

Depending on the speed of the flow, the flame shape and height are changed and demonstrate different behavior for both momentum- and buoyancy-controlled jets ([Roper, 1977](#); [Roper, Smith, & Cunningham, 1977](#)). In their paper the authors demonstrated that for slot burner with length L

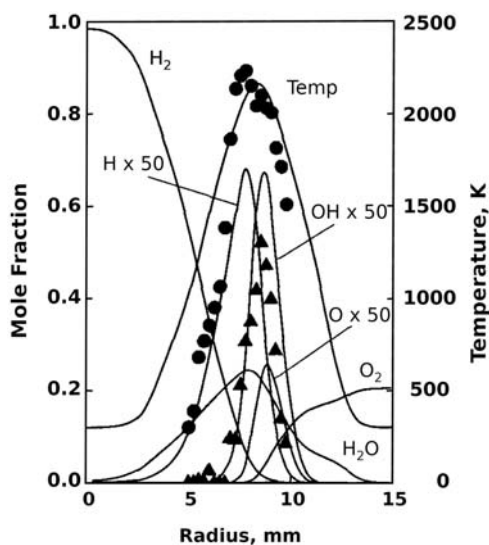


FIGURE 3.9

Comparison of the simulated temperature and few species profiles (curves) and experimental profiles of OH concentration (triangles) and temperature (circles) ([Fukutani et al., 1991](#)).

and width x for momentum-controlled jets, the flame height is regulated mostly by the flow rate of the fuel Q and diffusivity D

$$z_f \sim \frac{Qd}{DL},$$

while for a buoyancy-controlled jet, the correlation has another characteristic dependence

$$z_f \sim \left(\frac{Q^2}{DL^2(2g)^{1/2}} \right)^{2/3} \sim \left(\frac{Q^4}{D^2L^22g} \right)^{1/3}$$

and does not depend on the slot width.

3.4.4 Turbulent diffusion flames

In practical situations for hydrogen leaks, the speed of releases is typically high enough for jets to be turbulent. The Froude number is accordingly increased, that is, the regime of flow is momentum-controlled.

The molecular diffusion is disturbed in the turbulent mixing layer and the local effective diffusion time scale can be of the same magnitude as local chemical time scale. Under such conditions, an assumption of the infinitely fast chemistry is not valid anymore and specific phenomena of the turbulent diffusion combustion, which cannot exist in laminar mode, namely, lift-off and blow-off, appears. For highly intensive turbulence, due to rapid stretching of the flame surface or/and by the intensive mixing, local quenching can occur.

In Hawthorne, Weddell, and Hottel (1949) experimentally, and in Liñán and Williams (1993) theoretically the evolution of the diffusion flame was studied. In the laminar regime the height of flame increases with the increase of flow speed (see Fig. 3.10), until the transition into the turbulent regime when the flame height remains constant independently on the flow velocity. With increase

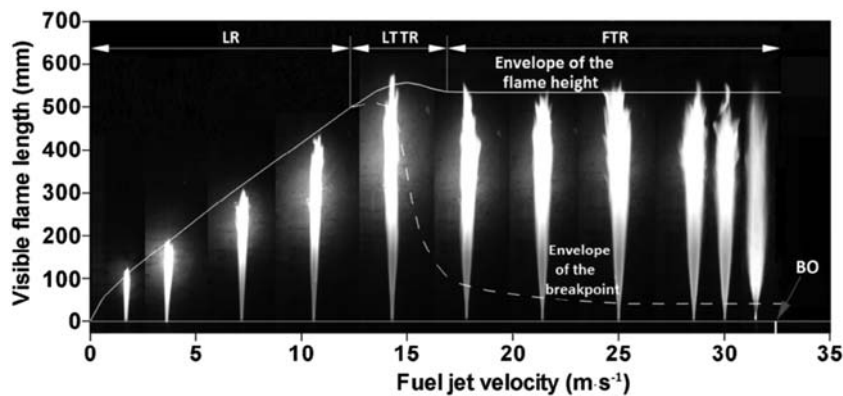


FIGURE 3.10

Evolution of diffusion flame regime with jet speed on the example from experimental flame images (Kang et al., 2016). LR, laminar; LTTR, laminar-to-turbulent; TR, turbulent regimes; BO, blow-off.

of flow speed, the height of the flame appearance (lift-off) also increases, until the complete flame depletion (blow-off).

The flame height in turbulent regime in contrast to the laminar regime is defined by only one parameter, which is flow exit diameter d

$$z_f \sim d,$$

this fact was first reported by [Hottel and Hawthorne \(1949\)](#). The transition to the fully turbulent regime can be characterized by Re number and for hydrogen occurs by Re equal to approximately 2000.

Jet fires can be expected on openings of storage tanks with compressed or liquid hydrogen, or leakages from fuel cell stack. Such jets might impose materials or humans at the distances up to 15 m, depending on the release conditions, mainly vessel pressure, mass flow rate, and momentum.

The prediction of the turbulent nonpremixed hydrogen jet flame parameters such as length, width, and radiative properties is, among others, one of the challenging safety research tasks. Intensive experimental and theoretical studies have been performed for understanding of the influence of initial pressure, temperature, break size, and form ([Grune, Sempert, Kuznetsov, & Breitung, 2011](#); [Houf & Schefer, 2005](#); [Veser et al., 2011](#)); optical nonintrusive techniques and especially background-oriented schlieren method (see, e.g., [Fig. 3.11](#)) are successfully adopted for jet flames studies ([Deimling et al., 2011](#); [Kotchourko et al., 2014b](#)).

A comprehensive critical review of jet flame research is presented in [Molkov and Saffers \(2013\)](#), where also a universal correlation for underexpanded jet is proposed. In contrast to the traditional approach based on the Froude number only, the new one accounts additionally for Mach and Reynolds numbers. The correlation is validated for pressures from 0.1 MPa to 90.0 MPa, temperatures from 80 K to 300 K, and leak diameters from 0.4 mm to 51.7 mm. The authors have shown that three distinct flame regimes exist, including conventional buoyancy-controlled, momentum-dominated “plateau” for expanded jets, and momentum-dominated “slope” for

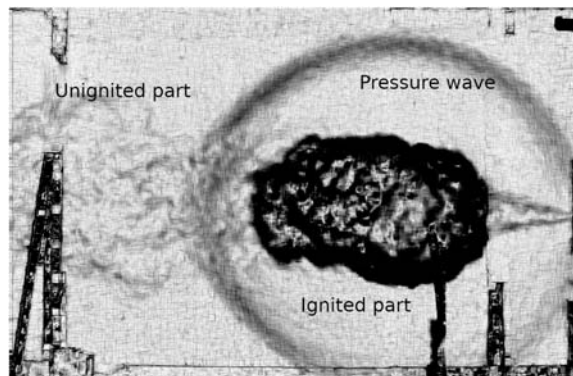


FIGURE 3.11

Visualization of the ignition of hydrogen jet using background-oriented schlieren technique. The ignited part of the jet and pressure wave resulting from the ignition clearly stand out against the jet part without flame.

underexpanded jets. In Fig. 3.12 the dimensionless flame length dependence on the jet parameters is presented.

$$\frac{\rho_N}{\rho_S} \left(\frac{U_N}{C_N} \right)^3 = \frac{g \cdot \mu_N}{\rho_S \cdot C_N^3} \cdot Re \cdot Fr,$$

where the viscosity μ_N was calculated as

$$\mu_N = \mu_{293} \cdot \frac{293 + K_{Suth}}{T_N + K_{Suth}} \left(\frac{T_N}{293} \right)^{\frac{3}{2}},$$

Here Sutherland constant for hydrogen was taken equal to 72 K and the dynamic viscosity ($\mu_{293} = 8.76 \cdot 10^{-6}$ Pa s), and Re and Fr are determined by parameters of hydrogen flow in the actual nozzle exit $Re = (\rho_N U_N D) / \mu_N$ and $Fr = U_N^2 / (g D)$.

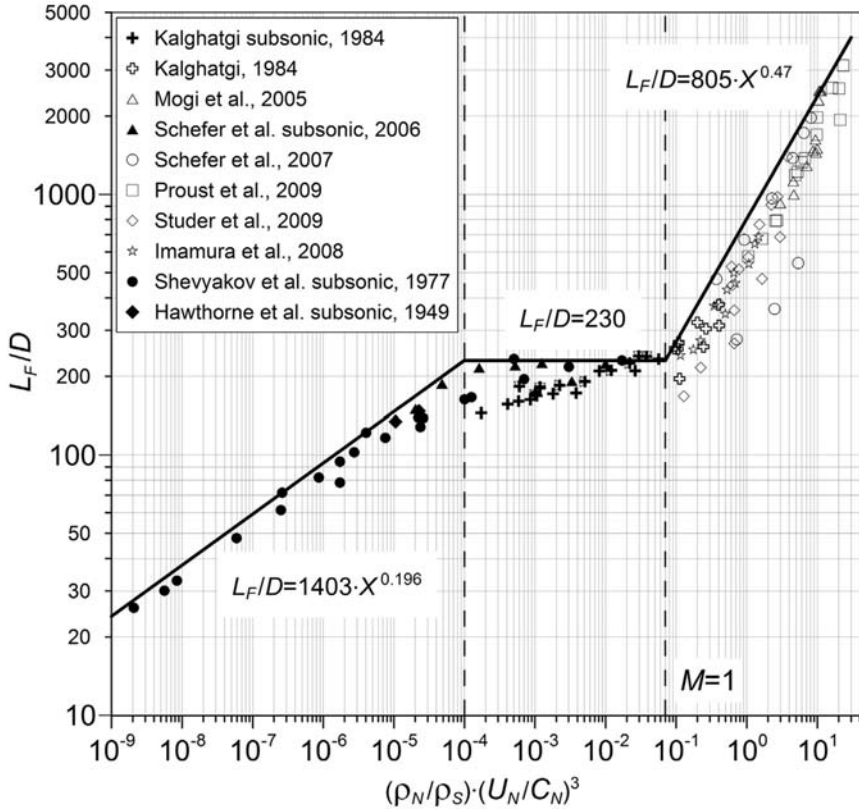


FIGURE 3.12

The dimensionless correlation for hydrogen jet flames (in formulas “X” denotes the similarity group $(rN/rS) (U_N/C_N)^3$).

It may be noted that while the jet fires are well-studied at laboratory scales, including complex phenomena as air entrainment, at larger scale further progress is still demanded. The waste field of research is in the field of jet fires in congested environment or for jets impinging process equipment, structural barriers, etc. The potential of the fast flame acceleration due to intensive turbulization or even explosion due to the delay in ignition when the hydrogen can accumulate and partially mix with air bring additional risk for severe damages by jet fires.

3.5 Deflagration

Deflagration is a specialized term for flames propagating in subsonic regime. In contrast to the detonation when the initiation of the chemical reaction in the next layer of the burning material is performed by the traveling ahead shock wave, in deflagration the onset of the reaction in the neighboring gas is provided by heat transfer mechanism, either molecular or turbulent.

3.5.1 Deflagration basics

3.5.1.1 Turbulence and deflagration speed

As the laminar flame is intrinsically hydrodynamically unstable, shortly after ignition flame undergoes an influence of instabilities. This type flame instabilities are caused by mutual peculiarities in diffusion, thermal expansion, buoyancy, heat losses processes. Flame instabilities have different time and space scales and act in different forms. The characteristic lengths can range from ~ 1 cm (Markstein, 1964) for cellular structures appearing in uniform flow, up to 50 cm (Lind & Whitson, 1974) for wavy perturbations on large scale expanding flames. The parametric acoustic instability can have characteristic frequency from few to hundreds of Hertz.

As the mostly studied and often noticeably affecting combustion in everyday conditions are the hydrodynamic Darrieus–Landau instability, the Rayleigh–Taylor instability, and thermal-diffusive instabilities. The subject is intensively examined and reviewed in a number of articles and surveys (e.g., Clavin, 1994; Sivashinsky, 1983). For premixed flames, a primary role is played by Darrieus–Landau instability appearing from thermal expansion at the flame front. For nonpremixed flames, main role is played by thermo-diffusive instabilities while the Darrieus–Landau instability does not affect flame significantly (Matalon, 2007).

The intrinsic instabilities can affect flame propagation to an substantial extent, when the flame speed grows maximally to several times. A further speed-up of the process takes place only if the flame interacts with surroundings. When moving flow resulted from the propagating flame meets congestion on the way of propagation, the flow in front of the flame becomes turbulent. When the flame enters such region, the effective combustion rate increases manifold. First, the flame is stretched and wrinkled by flow nonuniformities, increasing its area and consequently combustion rate. Furthermore, the intensive gas motion in turbulent vortices increases effective heat and species transfer between layers also accelerating flame propagation. The increase of combustion rate intensifies the flow in front of flame, that is, amplifies turbulence production with further increase of combustion rate. Thus the process is developed with strong positive feedback.

3.5.1.2 Accelerated and fast deflagrations

Deflagration is a combustion process with velocities ranging from centimeters per second to hundreds of meters per second and can produce pressure waves up to several bar in case of fast deflagration in hydrogen – air mixtures. For the slow deflagration the generated pressure wave typically has gradually rising front located far before the burning zone. The overpressure of such wave is in the range of hundreds of millibar and duration is in the range of seconds. For the fast deflagration, despite that the leading pressure wave and chemical reaction complex are still decoupled, the distance between them is considerably shorter and the overpressure can exceed several bars with duration of several hundreds of milliseconds (see Fig. 3.13).

One of the most remarkable features of the propagating hydrogen flame is their propensity to accelerate. It is well-known that hydrogen flames easily start to accelerate in the congested environment. However, in the combustion experiments in the smooth tubes it was observed that flames also accelerate on a quite short distance after ignition.

From engineering practical point of view the most important is how fast the flame moves, as this parameter predominantly defines the possible pressure loads resulting from accidental combustion. Numerous experimental, numerical, and analytical studies addressed the turbulent combustion properties (Bradley, 1992; Ciccarelli & Dorofeev, 2008; Clavin, 1994; Libby & Williams, 1994; Peters, 1999). For evaluating the effective combustion speed, a set of correlation dependencies was proposed by different authors. These correlations have different level of detailization and take into consideration dependencies on the mixture composition, pressure and temperature, turbulence characteristics as intensity and scale. Often the generalized formula reads as

$$S_t = S_L \cdot \phi \left(\frac{u'}{S_L}, \frac{l_t}{\delta}, Le, Ka, Da, \dots \right).$$

Highly turbulent flow under certain conditions instead of acceleration can also decrease combustion rate, which could even result in local or global quenching. Such decrease in combustion

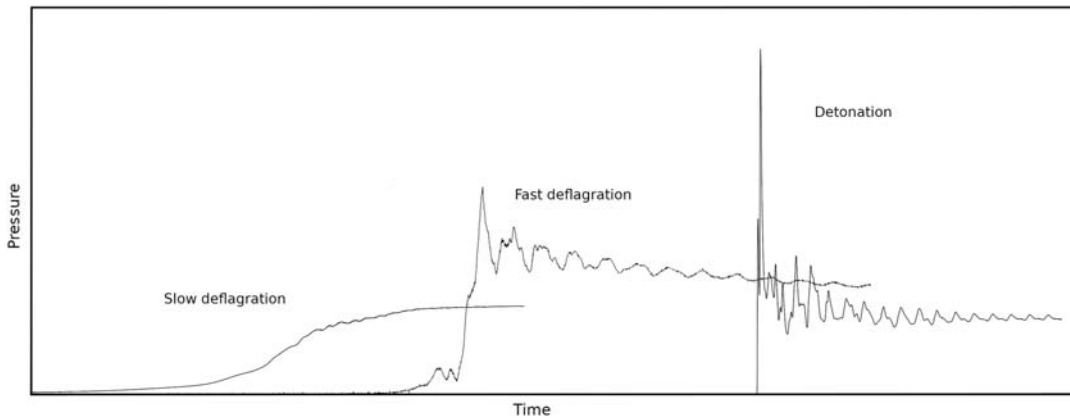


FIGURE 3.13

Typical profile of the pressure wave generated by slow and fast deflagration and detonation.

rate can originate from the excessive mixing of reactants in the fresh mixture and reaction products reducing temperature of the reaction zone, or due to the drastic stretching of the flame resulting in the gaps in flame surface.

Strong flame acceleration, which is referred to the processes in highly congested and confined environment, is a complex process of interaction between pressure waves and reacting substances. Advancing flame generates pressure waves, which after reflections from the obstacles and surrounding walls enter flame reaction zone. Interaction of the pressure waves with reacting media initiates development of flame instabilities, which cause flame distortion and consequently waves' amplification. In such conditions, several rapidly advancing instabilities can occur. From the appearing shear the Kelvin – Helmholtz instability and from the density difference the Rayleigh – Taylor instability typically turn up. In the compressible media, when acceleration of the flow takes place, the Kelvin – Helmholtz and the Richtmyer – Meshkov (analog of Rayleigh – Taylor) instabilities are initiated.

In practice, strong flame acceleration often leads to the ultimate stage of the acceleration and change in the flame propagation regime, namely to the transition to detonation.

In some experiments (Teodorczyk, Lee, & Knystautas, 1991), a regime was observed when the amplification of the reflected waves led not to a detonation transition, but to stable combustion with flame speeds from 400 m/s to 1200 m/s. For such particular fast deflagration regime, the authors observed decoupled shock and reaction zone complex and have shown that its mechanism is controlled through turbulization by (1) shock flame interaction; (2) Rayleigh – Taylor instability; (3) auto-ignition by rapid entrainment and mixing in the large recirculating eddies in the obstacles.

3.5.2 Deflagration in open atmosphere

For the conditions of open areas where no enclosing walls or barriers exist, the combustion regime is controlled by the possibility for turbulence to develop on the obstacles' presence and boundary layer conditions in the burning space. When pressure waves generated by the flame leave the combustion zone, they start to decay rapidly when the distance from the source grows, as shown in Fig. 3.14.

In the far zone, regardless of the combustion regime, the decaying pressure wave is transformed into nearly acoustic wave, the amplitude of which is determined only by the amount of hydrogen participating in the reaction. In contrast, in the near zone the amplitude strongly depends on the combustion speed in the reacting zone.

Following Dorofeev (2007), maximum overpressure $P^* = \min(P_1^*, P_2^*)$ and impulse $I^* = \min(I_1^*, I_2^*)$ depend on dimensionless distance and flame speed U_f , with

$$P_1^* = 0.34/(R^*)^{\frac{3}{2}} + 0.062/(R^*)^2 + 0.0033/(R^*)^3,$$

$$P_2^* = (U_f/C)^2(\sigma - 1)/\sigma \cdot (0.083/R^* + 0.14/(R^*)^2),$$

$$I_1^* = 0.0353/(R^*)^{0.968},$$

$$I_2^* = (U_f/C)(\sigma - 1)/\sigma(1 - 0.4(U_f/C)(\sigma - 1)/\sigma)(0.06/R^* + 0.04/(R^*)^2 - 0.0025/(R^*)^3),$$

here for the distance, pressure, and impulse, the Sach's presentation is used $R^* = RP_0^{\frac{1}{3}}/E^{\frac{1}{3}}$, $P^* = P/P_0$, and $I^* = IC/E^{\frac{1}{3}}P_0^{\frac{2}{3}}$, P_0 and C are atmospheric pressure and sound speed, and E is released energy.

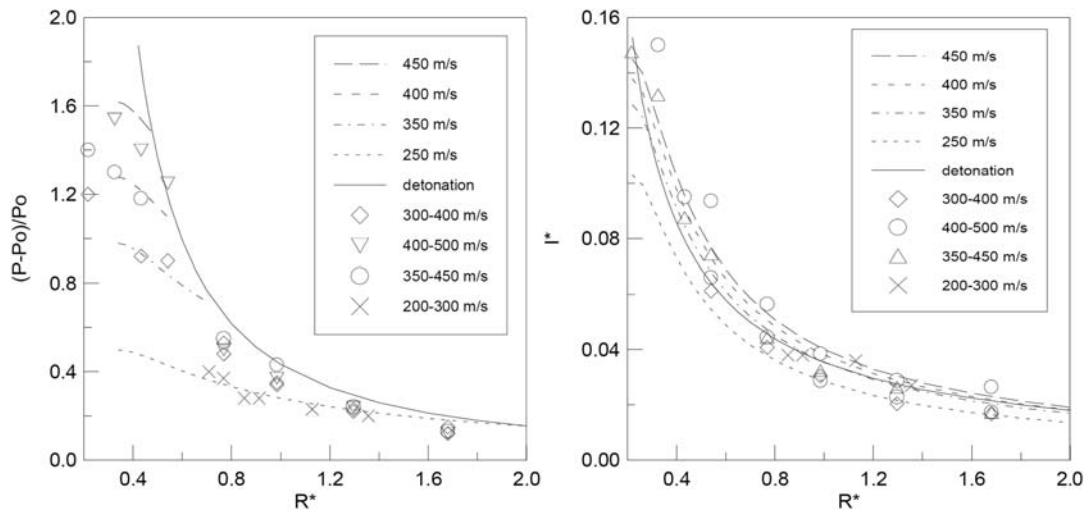


FIGURE 3.14

Positive overpressure and impulse in air blast wave versus distance for deflagrations with different flame speeds (Dorofeev, 1995).

3.5.3 Confined deflagrations

Combustion in the confined volumes differs from combustion in open space by noticeable pressure increase. In the open space, gas has possibility to expand and therefore it loses some energy on the expansion against surrounding pressure. In the confined systems, the whole energy released in the chemical reaction remains in the system and increases the internal energy of the gas. Thus in general in confined systems combustion occurs at higher temperatures and pressures than for the open conditions.

For stoichiometric hydrogen – air mixture, theoretical temperature value for thermodynamically equilibrated products will achieve 2748.3 K in case of constant volume conditions, and 2379.9 K for constant pressure. The obtained pressure for the first case is 8.11 bar.

3.5.3.1 Deflagrations in tubes and in a system of connected vessels

Flame acceleration, which is routinely observed in the experiments in the combustion tubes, is known to be a random process and is rarely reproduced in detail. However, the main stages are consistently repeated in most such tests.

When burnable mixture is ignited by relatively weak ignition source, the flame initially propagates in laminar regime or in slow deflagration regime with velocities much below sound speed. The gas ahead of the flame front is moving relatively slow with characteristic gradual growth on the pressure wave front.

As the flame propagates, different effects as development of flame instabilities, formation of boundary layer at the tube walls, flame interaction with possible obstructions in the tube, lead to the flame acceleration. The flame acceleration initiates formation of strong pressure wave moving

ahead of the flame front and overtaking the precursor perturbation from slow combustion. The newly formed wave can propagate with velocities up to 1000 m/s.

In such conditions in the mixture between slow precursor and strong pressure wave, due to small perturbation a local point explosion can happen. The local explosion is characterized by significant transient overpressures in some cases and by the ability to transform to the stable detonation with typical velocities about 2000 m/s for near stoichiometric mixtures.

In Fig. 3.15 (Teodorczyk & Thomas, 2003), typical representation of the pressure dynamics observed in the flame acceleration test is given. Pressure record in diagram (A) refers to slow deflagration with pressure amplitude about 0.5 bar; diagram (B) can be referred to the state demonstrating formation of a precursor shock in case of fast deflagration; diagram (C) belongs to the state

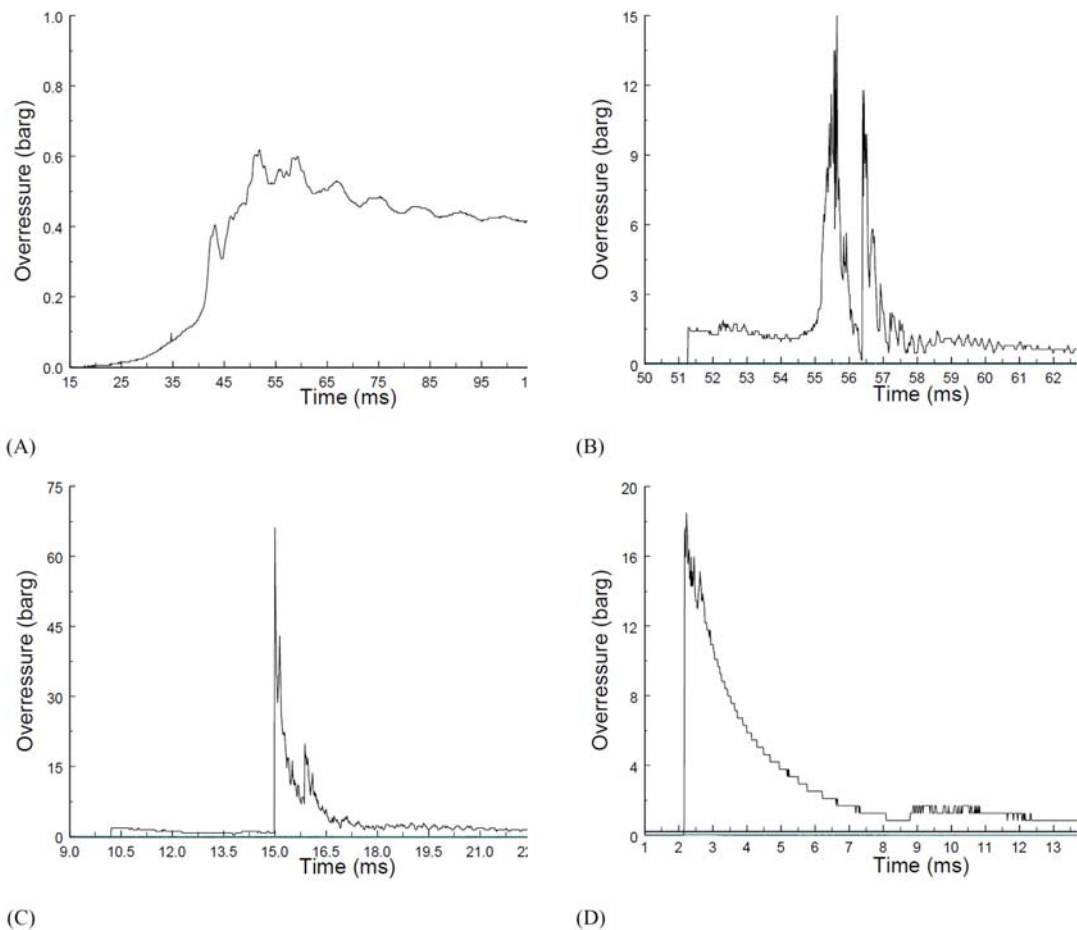


FIGURE 3.15

Pressure wave generated by propagating combustion in different regimes (Teodorczyk & Thomas, 2003).

soon after DDT event with overdriven detonation, characterized by pressure peak amplitude exceeding steady state detonation value several times. For comparison, diagram (D) presents typical pressure record of a regular detonation wave.

3.5.3.2 Amplification of pressure effect due to precompression (Mache effect)

In most practical situations, the speed of combustion is less than speed of sound; therefore during flame propagation, the pressure in the closed vessel is uniform across the volume where combustion occurs. However, the temperature distribution after such process is quite different. As there are no strong media movements connected with combustion, the flame propagates involving every neighboring portion consecutively one after another. As a result, the temperature distribution after combustion is nonuniform. The final temperature of gas depends not only on the pressure in the vessel, but also on the pressure at which the combustion took place. Every gas portion first is compressed before chemical reaction, then it is heated by the reaction, and after that it is further compressed due to combustion of the residual gas. Therefore as a result, after combustion in a closed vessel, the different locations have different temperatures. The regions that were burned at the beginning and then adiabatically compressed have higher temperature as the regions that were first compressed and then burned out. This effect was described in [Flamm and Mache \(1917\)](#), and is named after Mache. This effect is clearly exhibited for combustion in a closed spherical vessel with a central ignition, where the gas does not mix but move due to expansion of reaction products ([Zeldovich et al., 1985](#)).

3.5.4 Vented deflagrations

Vented deflagration and vented explosion are combustion processes in which the pressure wave generated during burning of a mixture enclosed in a vessel or a cavity comes out through a vent of the enclosure. The vent can be either preexisting as a construction peculiarity or created by the pressure wave destroying the chamber wall (see [Fig. 3.16](#)).

After the burnable mixture releases through the vent, the formed cloud can be ignited and depending on the release conditions burns out or explodes producing expanded pressure waves.

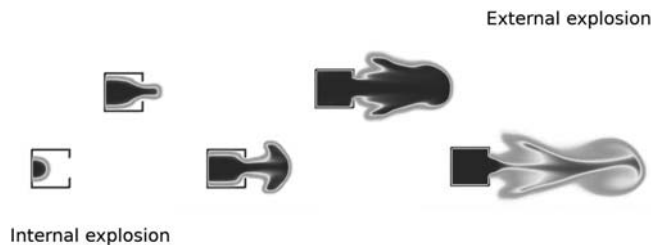


FIGURE 3.16

Representation of vented explosion development.

3.5.4.1 Multipeaks structure of pressure transients and underlying physical phenomena

Pressure waves generated during vented deflagration are characterized by multipeak structure. Depending on the deflagration conditions, the number of peaks and their parameters can considerably vary. In Fig. 3.17 (Cooper, Fairweather, & Tite, 1986), detailed structure of the pressure wave resulted from deflagration of 10% natural gas in air mixture in enclosure with 0.76 m^3 volume and venting area $V^{2/3}/9.2 \text{ m}^2$ with initially closed vent cover with 3.3 kg/m^2 , is presented.

Following the authors' explanation (Cooper et al., 1986; Gardner & Hulme, 1995), the four pressure peaks structure is discussed here. The first peak P1 is connected with the formation of pressure wave after vent opening at moment (a).

After moment (b) when the volume of exhausted gas is equal to the volume of the generated gas, and moment (c) when the burnt gas starts to flow out from the vessel, a peak P2 arrives, which possibly corresponds to the external explosion of the previously vented unburnt gas (at moment d) or to the intensive outflow of the hot combustion products followed by abrupt pressure drop (Harris, 1983).

Peak P3 is characterized by the relatively low magnitude but long duration and refers to the maximum of the internal combustion inside the vessel when flame has maximum surface, usually when flame approaches vessel walls. Before this, the flame surface was increasing constantly, then the flame continues to burn in isolated pockets of unburnt mixture while its area is decreasing. The pressure decreases also accordingly to the overall combustion rate reduction.

Peak P4 has usually strong oscillations attributed to acoustic resonance inside the vessel, resulting in high reaction rate leading to the significant peak values.

During internal and external explosion, the flame instabilities can affect the dynamics of the process. After the external explosion, the Helmholtz instability can be developed with wave length larger than characteristic size of the vessel and which in its turn can induce appearance of Rayleigh – Taylor instability. The pressure oscillations can provoke also revealing of a cellular structure in the flame, giving rise to very high combustion rates. Therefore the damping of the instabilities can be an effective mean for the risk reduction during vented explosions. Also, in case when the relieve pressure exceeds 7.5 kPa , the peaks P3 and P4 are usually not present.

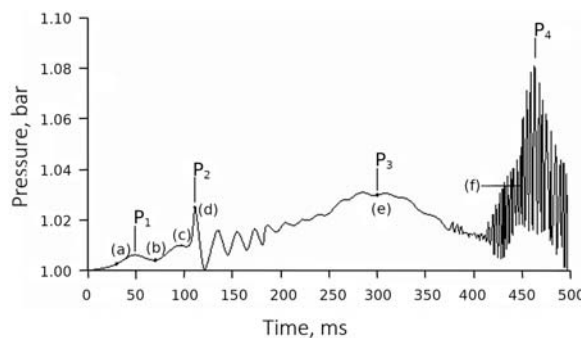


FIGURE 3.17

Multipeak pressure wave generated by vented explosion (Cooper et al., 1986).

Since the beginning of the vented explosion studies, it was found that the strong turbulence is developed during such processes promoting significant flame acceleration.

The difficulty in the formulation of the consistent correlation is connected with the fact that effective discharge coefficient μ strongly depends on the deflagration conditions. In a series of studies, it was found that the ratio of the turbulence factor χ to the discharge coefficient μ is important for the explosion pressure, rather than the turbulence level itself.

The generalizing correlations for the vent size effect on the generated turbulence was consistently developed and presented by Molkov et al., including impact of the mixture composition and elevated initial pressures (Molkov et al., 2004). The following correlation for venting generated turbulence has been obtained by processing a wide range of experimental data on vented gaseous deflagrations.

$$\frac{\chi}{\mu} = \alpha \left[\frac{(1 + 10 \cdot V^{\frac{1}{3}})(1 + 0.5 \cdot Br^{\beta})}{1 + \pi_v} \right]^{0.4} \pi^{0.6},$$

here the empirical coefficients α and β are equal for hydrogen – air mixtures to 1.0 and 0.8, respectively, and for hydrocarbon – air mixtures to 1.75 and 0.5; V is dimensionless volume (numerically equal to enclosure volume in m^3), dimensionless initial pressure $\pi = P_i/(1 \text{ bar})$, and P_i = initial pressure (bar abs = 10^5 Pa), dimensionless vent closure release pressure $\pi_v = P_v/P_i = (P_{\text{stat}}/P_i + 1)$ with P_v is vent closure release pressure, bar abs., P_{stat} is vent closure release pressure used in the NFPA 68, bar gauge, and P_i is initial pressure, bar abs.; and the Bradley number is

$$Br = \frac{F}{V^{2/3}} \cdot \frac{c_{ui}}{S_{ui}(E_i - 1)},$$

here F denotes vent area, V is enclosure volume, c_{ui} is speed of sound; S_{ui} is burning velocity at initial conditions, E_i is combustion products expansion coefficient.

With the growth of Bradley number, for example, by enlargement of the vent area, the turbulence factor will also increase. However, by any intensification of the combustion due to growth of the burning velocity, the turbulence factor will decrease.

In several vented explosion tests, it was found that the external explosion can be highly dangerous. Solberg, Pappas, and Skramstad (1980) have found that the flame from 4-m vessel can propagate up to 30 m outside the enclosure and flame speed can reach 100 m/s. Concurrently, Harrison and Eyre (1987) and Swift and Epstein (1987) have found that the external explosion can affect the internal process and suggested that major factor in this process is contained in decrease of the mass flow rate through the vent. Study of the external explosion scaling (Catlin, 1991) demonstrated that flame begins at the vent and pressure grows proportionally to the vent flow speed. Numerical confirmation of the models and theoretical analysis were performed in a series of papers by Molkov (Molkov, Makarov, & Puttock, 2006).

3.6 Transition from deflagration to detonation

Detonation in the burnable mixture can occur due to a number of reasons. Usually, the onset of detonation is separated into two wide groups distinguished by the way of the detonation initiation, namely: direct initiation (initiation by high explosive, by spark, by strong shock, etc.) and transition

from deflagration to detonation. The last process is usually observed in the accelerated flames propagating in the closed volumes.

3.6.1 Phenomenology of flame acceleration and deflagration to detonation transition

One of the best-understood events observed during DDT is a process referred as an-explosion-in-an-explosion mechanism. Realization of this mechanism usually takes place in the area of strongly turbulent flow between primary shock and front of the flame, which generates pressure waves propagating toward primary shock and overtaking it with subsequent formation of a detonation wave (see Fig. 3.18).

Despite the fact that, most likely, “an-explosion-in-an-explosion” mechanism is one of the most often occurring event, DDT can also happen as a result of an interaction between primary shock with local ignition forming a secondary explosion. Emergence of such secondary explosions was observed on the flame front, shortly behind primary shock, and on the contact surfaces formed by secondary shocks (Oppenheim, Manson, & Wagner, 1963).

The development of detonation in tubes was studied by many authors for at least since the mid of the last century (Oppenheim et al., 1963), and can be presented as the following sequence of events:

- Due to the accelerating movement of a laminar flame, a shock wave is formed ahead of the flame front, which at this stage of the process can have wrinkled combustion front.
- Well-developed turbulent flow is formed due to the accelerated gas movement.
- Stochastic emergence of a point explosion, which could appear due to the implosion of a pocket of unburned medium formed within the reaction zone.
- This point explosion results in the appearance of two strong waves propagating from the origin in opposite directions with multiple transverse waves in between them. The first wave propagates toward unburned mixture and forms later detonation in the fresh unburned mixture. The second one is a wave passing back through burned or still burning mixture toward the origin and is often referred as retonation wave.
- Most probably, the obstacles in the flow as well as wall roughness can play a key role in the appearing of such secondary explosion, as the center of the explosion waves is found most frequently near wall or cluttering barrier.
- The pressure, which is attained at the location of the local explosion, is typically quite high and can considerably exceed stationary values. The downstream propagating detonation wave moves initially in unstable “overdriven” regime, with pressure and velocity higher than in stable detonation.
- The self-sustained or Chapman – Jouguet (CJ) detonation is achieved at the final stage after stabilization of the “overdriven” regime.

3.6.2 Effect of mixture and environmental properties

Transition from deflagration to detonation, in contrast to direct initiation of detonation, which is characterized by instantaneous creation of detonation wave from powerful source, does not require any additional (external) energy source.

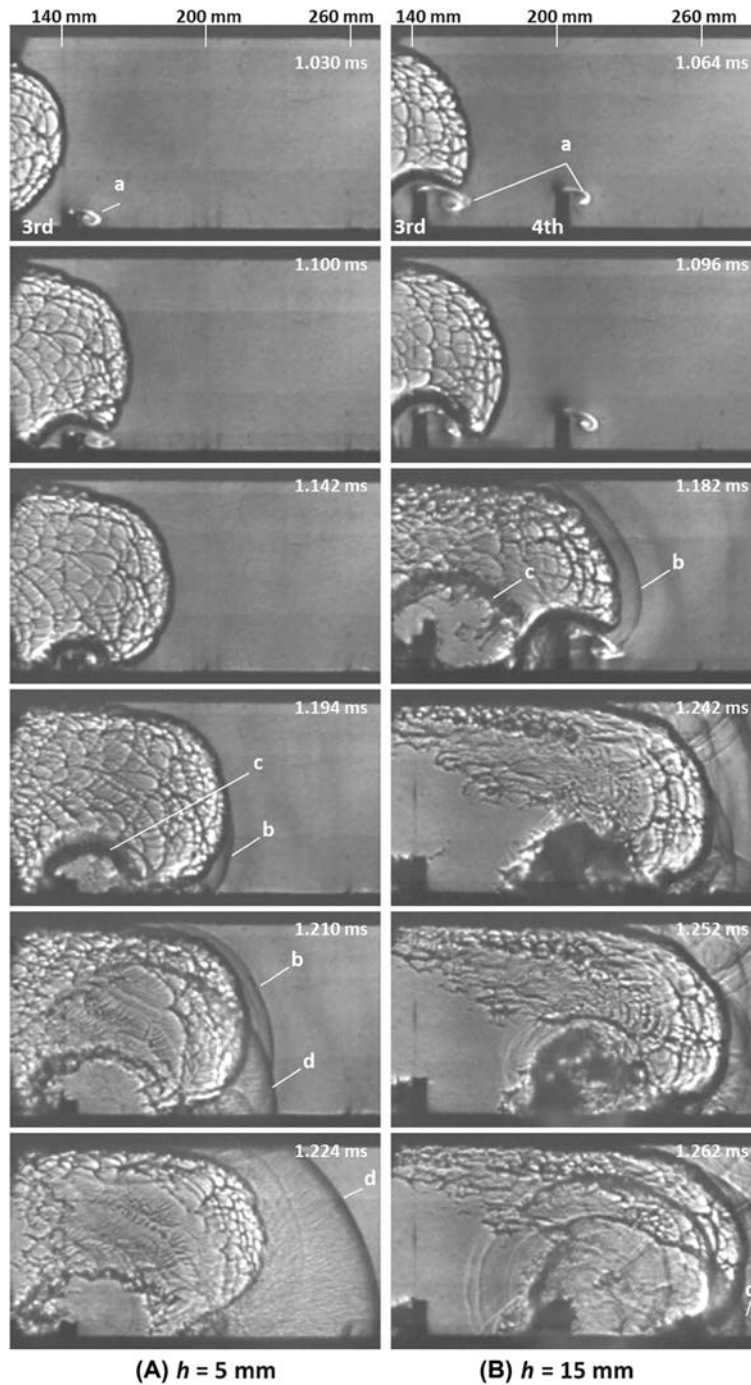


FIGURE 3.18

Sequence of schlieren photographs illustrating the transition from deflagration to detonation phenomenon for two cases of obstacle heights, h (Maeda, Minami, Okamoto, & Obara, 2016). Here a points to vortex behind the obstacle, b to shock wave, c to volumetric explosion behind the obstacle, and d to detonation wave.

However, the energy, which is released during local explosion and then together with expanding “overdriven” detonation, should suffice to produce self-sustained detonation later. In case if the energy release is inadequate, for example, it can be too slow or too low, the local explosion will lead to the decaying detonation when flame and shock lose their linkage and the flame-shock complex breaks down.

Several theoretical models describing DDT exists. First, an idea of the role of the nonuniformity was proposed by [Zeldovich, Librovitch, Makhviladze, and Sivashinsky \(1970\)](#) and later was developed by [Lee and Moen \(1980\)](#), who assumed that successful onset of detonation can be achieved if a spatial and temporal coherence of the energy release exists. This mechanism called SWACER (Shock Amplification by Coherent Energy Release) was initially proposed for explanation of the photochemical detonation initiation in $H_2 - Cl_2$ mixtures.

The concept of the mechanism assumes that the necessary coherence is achieved by preconditioning of the mixture. In the preconditioned conditions, the ignition delay time τ increases in a predefined way so that the flame front propagates at a velocity $V_0 = \partial\tau/\partial x - 1$. The spatial gradients in the ignition delay time can be achieved due to the gradient of temperature or free radical concentration. Turbulent mixing of the hot combustion products and unburned gas also can provide necessary conditions for onset of detonation when hot spot (or local explosion) arises together with shock arrival to its location. DDT was observed by [Knystautas, Lee, Moen, and Wagner \(1979\)](#) when burning turbulent jet was injected into quiescent burnable gas.

The mathematical aspects of the mechanism were analyzed in detail by [Kapila, Schwendeman, Quirk, and Hawa \(2002\)](#), also conditions and phase-wave mechanism realization are discussed by [Khokhlov, Oran, and Wheeler \(1997\)](#) and by [Oran and Gamezo \(2007\)](#). The role of turbulent mixing together with local quenching to provide required spatial gradient in the chemical ignition delay time in the onset of detonation was considered.

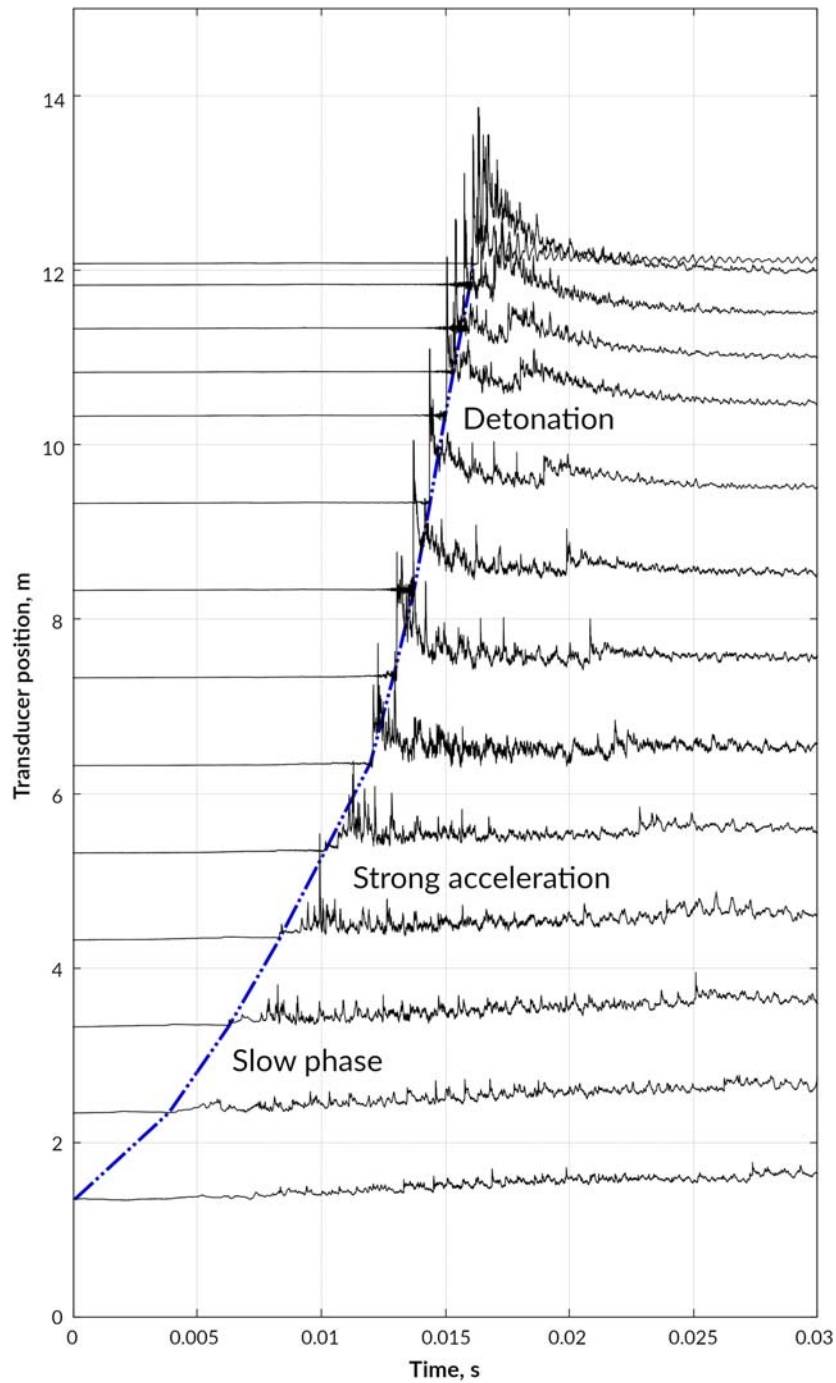
Note that coherent release of energy by itself does not guarantee the onset of detonation, the amount of the released energy should be large enough.

Another detonation onset mechanism is transfer to detonation as a result of a shock reflection. It was observed by many authors, for example, by [Gilbert and Strehlow \(1966\)](#). A decisive feature of this “strong” mechanism of initiation is the fact that the ignition delay time before the onset of self-sustained reaction in the unburned mixtures is very short. The strong limit in the coherence theory ([Meyer & Oppenheim, 1971](#)) appeared to be consistent with the observations.

3.6.3 Criteria for spontaneous flame acceleration to supersonic flame speed

Typically, before the actual onset of detonation a phenomenon referred as flame acceleration is observed. The initially slow flame propagation is transformed to the constantly speeding up process with first gradual and then rapid increase of the flame velocity, formation of precursor shock and considerable rise of pressure in the propagation wave. Such sequence of events, which leads subsequently to the detonation, is illustrated in [Fig. 3.19](#).

In almost all cases when transition to detonation is observed, excluding direct and “strong” initiation, the final transition usually is associated with local explosion. However, without flame acceleration in the initial phase, when necessary gradient in thermal auto-ignition delay time in the mixture is formed, the detonation onset looks very different. Thus determination of conditions when considerable flame acceleration is possible is of primary importance.

**FIGURE 3.19**

Flame acceleration and deflagration-to-detonation transition in combustion tube with 15% H_2 in air mixture.

In Dorofeev et al. (2001), authors have demonstrated that a possibility for strong flame acceleration is in great extent defined by the mixture properties, namely by the mixture reactivity. The experimental data on the combustions in confined volumes from a wide variety of test facilities with the different characteristic lengths and the geometric configurations were analyzed. The analysis showed that strong flame acceleration was observed only for the sufficiently sensitive mixtures (Fig. 3.20).

The parameter, which controls the possibility of flame acceleration, was identified as an expansion ratio $\sigma = \rho_u / \rho_b$, ratio between density of unburnt and burnt mixture at constant pressure. An expression of critical expansion ratio (for values below σ^* the flame acceleration is not observed) reads as

$$\sigma^* = 0.9 \cdot 10^{-5} x^3 - 0.19 \cdot 10^{-2} x^2 + 0.1807x + 0.2314,$$

where x is dimensionless activation energy $x = E_a / RT_u$. For hydrogen – air mixtures, this parameter is approximately equal to 25 and, therefore for critical value σ^* gives approximately value of 3.7. The last expression was obtained for the fully confined systems and therefore should be applied carefully keeping in mind its limitations.

In Alekseev, Kuznetsov, Yankin, and Dorofeev (2001), authors have analyzed systems with partial transverse venting and have found the critical value σ^* is modified proportionally to vent ratio $\alpha = S_v / S_e$ (ratio between surface of vent S_v and surface of enclosure S_e). The larger the vent area,

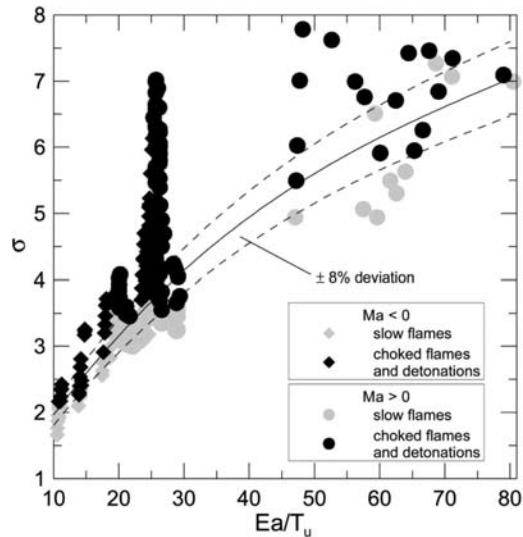


FIGURE 3.20

Observed combustion regime as a function of expansion ratio σ and dimensionless activation energy E_a / RT_u for hydrogen – air and hydrocarbon – air mixtures for closed systems.

Picture reproduced with permission from Dorofeev, S.B. et al. (2001). Evaluation of limits for effective flame acceleration in hydrogen mixtures. Journal of Loss Prevention in the Processes Industries, 14, 583–589.

the more reactive mixtures are necessary for development of fast flames (Fig. 3.21). The resulting expression for critical expansion ratio reads as

$$\sigma_{cr} = \sigma^*(1 + 2.24\alpha),$$

where σ^* is critical expansion ratio for closed systems.

Using findings of Dorofeev et al. (2001) and Alekseev et al. (2001), a criterion for strong flame acceleration in the closed vessels and vessels with partial venting with necessary conditions was formulated: the mixture can exhibit strong flame acceleration only if its expansion ratio is larger than critical expansion ratio σ_{cr} . To be more practical, this statement is to be complemented with additional precisising condition: strong flame acceleration can take place only if there is enough room for actual development of supersonic flame. This geometric limitation is connected with necessity of run-up distance for flame acceleration process, which can be evaluated for tube-like configurations as having value of 20–40 diameters (see, e.g., Kuznetsov, Alekseev, Matsukov, & Dorofeev, 2005).

Note that application of the proposed criteria in the actual industrial conditions can be hindered as they were obtained using data in confined and confined with partial venting systems.

In Grune et al. (2017) and Kuznetsov and Grune (2019), authors have analyzed the systems that can have more practical significance. They have studied flat layers, which can be formed near the

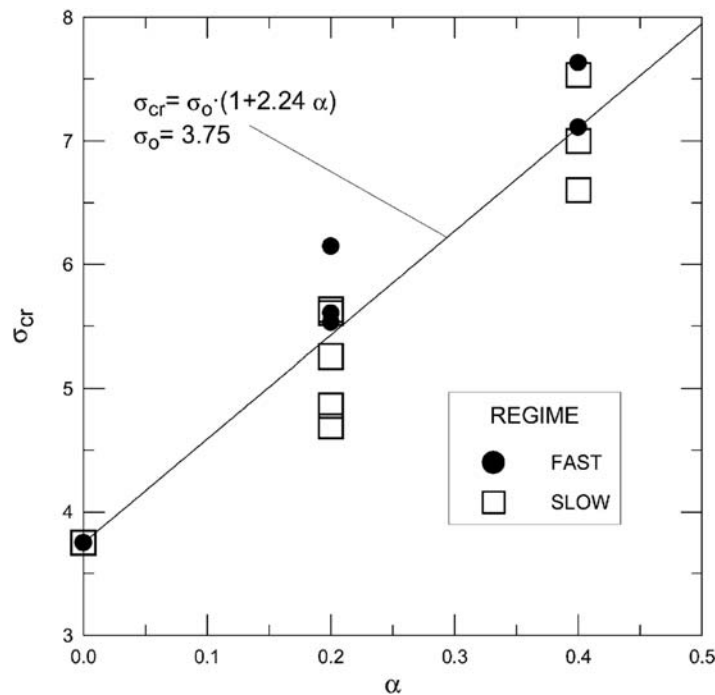


FIGURE 3.21

Observed combustion regime as a function of expansion ratio σ and vent ratio α . Dependence of critical expansion ratio σ_{cr} on α (Alekseev et al., 2001).

ceilings or horizontal obstacles after hydrogen leaks. They have analyzed critical condition for different regimes of flame propagation as function of layer thickness and roughness of the channel. It was found that thinner layer suppresses the detonation onset and even with a roughness, the flame may quench or, in thicker layer, is available to accelerate to speed of sound. The detonation may occur only in a channel thicker than 4 mm (Fig. 3.22).

In Grune et al. (2013), authors have found that critical expansion ratio for effective flame acceleration to sonic speed linearly depends on the reciprocal layer thickness, thus introducing a criterion for the flame acceleration in layered configuration (Fig. 3.23).

In Yáñez et al. (2011) flame propagation regimes for the stoichiometric hydrogen – air mixtures in an obstructed semiconfined flat layer have been numerically investigated. The authors concluded that the following parameters as thickness of the layer of H_2 – air mixture, blockage ratio, and distance between obstacles and obstacle size are key factor defining combustion regime. As a result of parametric study, a criterion that separates experiments resulting in *slow* subsonic from *fast* sonic propagations regimes was proposed.

$$\sigma > \sigma^* \left(0.575 \frac{h}{H} \frac{1}{BR} + 0.136 \frac{D}{H \cdot BR} \right),$$

here h is size of obstacles, H is channel height, D is distance between obstacles, and BR is blockage ratio (Fig. 3.24).

Note that the generalization of this criterion to lower degrees of confinement up to open systems should provide highly valuable information for evaluation of flame acceleration potential in safety analysis.

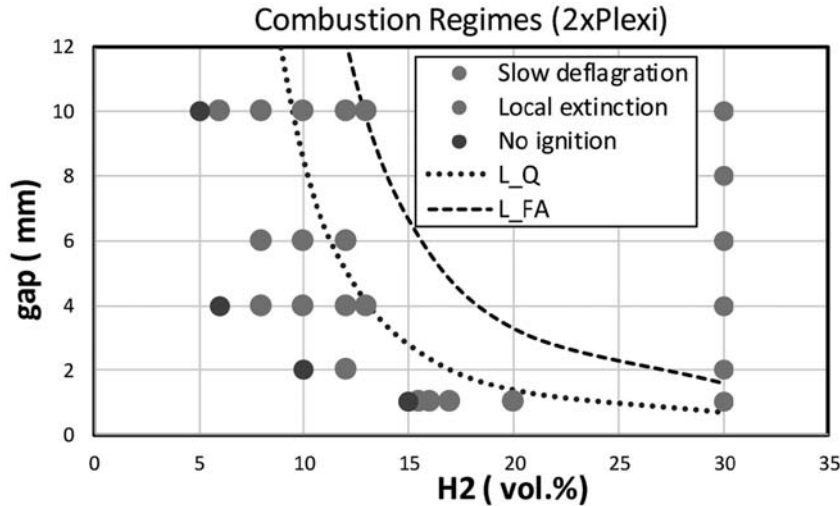
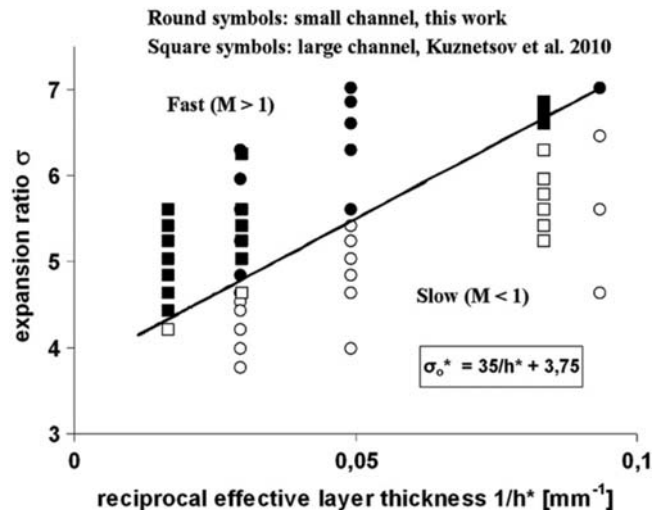
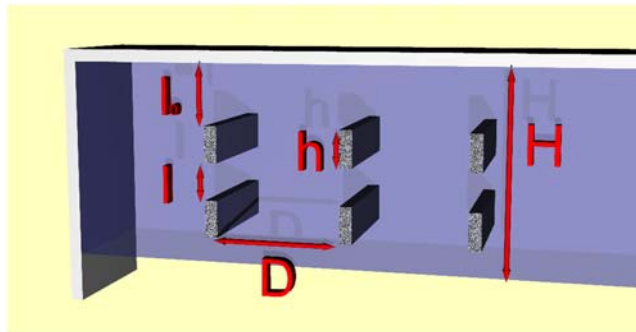


FIGURE 3.22

Diagram of state of combustion regimes for different hydrogen – air mixtures in a layer geometry of different thicknesses (100% of channel length congested) (Kuznetsov & Grune, 2019).

**FIGURE 3.23**

Critical conditions for an effective flame acceleration as function of expansion ratio versus reciprocal effective layer thickness h_{eff} : sonic flame and detonations (solid points); subsonic flame (open points).

**FIGURE 3.24**

Main geometrical variables of the parametric study of Yáñez et al. (2011).

3.6.4 Criteria for establishment of stable detonation

In both industrial and research scales, the DDT process is well-understood only qualitatively. However, an accurate quantitative theory that can reliably predict onset and self-sustained stable propagation of detonation is still not available.

Due to this fact and taking into account that DDT has principally stochastic nature, the DDT predictions, which can be obtained from CFD simulations, always have considerable degree of uncertainty. To avoid ambiguity in the forecast of DDT and provide possibility to make evaluation

of the worst-case scenario, a criterion for possibility of detonation to exist would significantly facilitate evaluation of the hazard risk for safety experts in practice.

In Dorofeev et al. (2000) an attempt to systematize conditions in which self-sustained detonation can take place was done. The massive array of literature data together with the data from experiments carried out by authors were used. It was shown that depending on geometry of the facility, the mixture of the same composition could exhibit different behavior. In the facilities with characteristic length larger than certain value (mostly defined by mixture reactivity) the mixture does detonate, and in smaller facilities the detonation does not occur. The results of analysis are summarized in Fig. 3.25. Solid line $L = 7\lambda$ in the figure separates analyzed cases with observed detonation from cases without detonation. The accuracy of this separation was estimated by authors as remaining within 30% of λ value.

Parameter λ , which is used as controlling parameter for characterization of mixture sensitivity, is detonation cell size. Detonation cells are imprint of complex three-dimensional structure of detonation wave and their size is defined by mixture properties only (e.g., H_2 concentration, initial temperature) and does not depend on geometric configuration of detonating mixture. The values of detonation cells are often known only with inaccuracy of 100%, therefore utilization of 7λ rule has to be made with clear understanding of the correlation uncertainty.

A criterion built on basis of 7λ correlation says that for detonation existence as self-sustained process, it is necessary that characteristic size of combustible mixture (cloud or room size) should

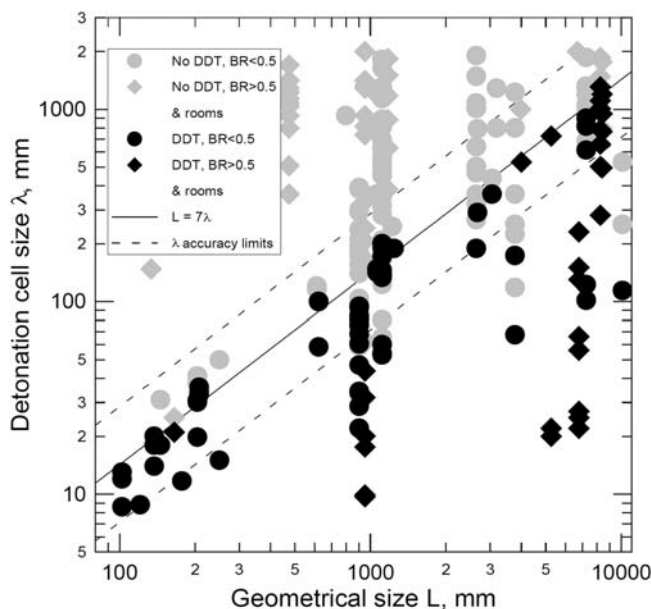


FIGURE 3.25

Critical conditions for onset of detonations in rooms and channels with obstacles. Deviations from line $L = 7\lambda$ are within 30% (Dorofeev, Sidorov, Kuznetsov, Matsukov, & Alekseev, 1999).

exceed 7λ (necessary condition). If detonation develops as a transition from fast deflagration, the corresponding criterion for flame acceleration has to be fulfilled as well.

Note that this criterion together with criterion for flame acceleration has limited area of applicability, as for nonuniform mixtures with spatially dependent properties (e.g., concentration) assessments of detonation cell size and expansion ratio can have considerable variations in the values.

3.7 Detonation

The term detonation refers to the high-speed (supersonic) regime of combustion. In this regime, combustion wave propagates together with leading shock in self-sustained regime, which is characterized by exceptionally raised velocity and consecutively high amplitude of the pressure shock. For stoichiometric hydrogen – air mixture in normal conditions, CJ detonation reaches ≈ 1970 m/s in the speed and ≈ 19 bar in the amplitude of the pressure wave.

Detonation can be initiated by an explosion either of condensed explosives or another gas mixture, by electrical wire explosion, high-voltage, or laser sparks, etc. Such detonation start, which is characterized by immediate self-sustained regime, is usually referred as a direct initiation.

3.7.1 Detonation basics

3.7.1.1 Detonation direct initiation

The rapid energy release during the process of the direct initiation results in the creation of the decaying strong shock propagating outward from the energy source. The high temperatures produced by the shock initiate chemical reaction, which lags behind and fades out when the shock decays. Depending on the intensity of the initiating source, the concurrency between the energy loss in the decaying shock, and energy supply into the shock due to exothermic chemical reaction, result in either initiation or failure of detonation. If the shock obtains enough energy during initiation to allow self-sustaining structure within detonating complex to form, the wave will propagate further without failure.

In the first attempt to estimate critical conditions made in Zel'dovich, Kogarko, and Simonov (1956), the idea was that time before the shock will decay to stationary CJ values has to exceed the characteristic chemical time. Later this idea was further developed combining it with the results of similarity theory for strong shocks by Lee, Lee, and Knystautas (1966) where it was formulated in shock-pressure–shock-radius terms for cylindrical case. Using formulation for spherical geometry, critical energy of initiation E^* (Lee & Higgins, 1999; Lee, 1977) reads as

$$E^* \sim \rho D_{CJ} \Delta^3,$$

here ρ is gas density, D_{CJ} is CJ detonation velocity, and Δ is the induction length.

The experiments of Higgins and Bruckner (1996), where spheres of different sizes were fired at the CJ velocity into a hydrogen – oxygen – argon mixture at various initial pressures, demonstrated good agreement with this theory.

In Guirao, Knystautas, and Lee (1989), it was reported that minimal energy of initiation by high explosive has the mixture with 30.61% of H_2 in air. In this case, detonation ignition was achieved

by using of 1.09 g of tetryl with equivalent energy release equal to 4.65 kJ. At the same time for the leanest mixture, which was possible to initiate in the referred experiments with 17.33% H₂ in air, it requires 36.4 g of tetryl with equivalent energy release of 155.4 kJ.

3.7.1.2 Rankine – Hugoniot curve

In contrast to the general theory of detonation, which still has many not fully understood aspects, a wide spectrum of experiences is available for one-dimensional models of detonation. Common for all one-dimensional models is that detonation wave is presented as a complex containing leading shock and a chemical reaction joined to it. For the clarity, it is assumed that shock velocity remains constant for the mixtures with constant parameters.

If the details of the chemical reaction can be neglected, the integral laws for conservation on the jump can provide comprehensive insight into the gas dynamic of the shock wave relations before and after shock.

In the coordinate system connected with shock

$$\begin{aligned}\rho_1 v_1 &= \rho_2 v_2 \equiv j, \\ p_1 + \rho_1 v_1^2 &= p_2 + \rho_2 v_2^2, \\ w_1 + \frac{v_1^2}{2} &= w_2 + \frac{v_2^2}{2}\end{aligned}$$

here index 1 refers to the gas toward which the jump is moving and index 2 refers to gas remaining behind the jump. Notation for gas parameters: ρ – density, v – gas velocity, p – pressure, w – enthalpy. Introducing specific volume $V = 1/\rho$ and using internal energy ϵ instead of enthalpy can obtain the relation of the thermodynamic properties on both sides of the jump

$$\epsilon_1 - \epsilon_2 = -\frac{1}{2}(V_1 - V_2)(p_1 + p_2)$$

and for the density of the mass flux through the jump

$$j^2 = \frac{p_2 - p_1}{V_1 - V_2}$$

The first relation is called Rankine – Hugoniot curve, and the second is known as Rayleigh line.

In case if energy release q takes place on the jump in ideal gas with constant specific heat γ , the Rankine – Hugoniot relation can be presented as

$$\frac{\gamma}{\gamma - 1}(p_2 V_2 - p_1 V_1) - q = \frac{1}{2}(V_1 + V_2)(p_2 - p_1)$$

In Fig. 3.26, a schematic of the possible states for the system with energy release is shown. The initial state of the system is denoted as position S. The final state of the system is presented as RH curve localized above the initial point, as in the case of energy release (combustion), pressure, and volume increase during the reaction. The final state curve can be divided into five parts, which correspond to the different regimes of combustion.

Region III, which is separated by lines $p = \text{const}$ and $V = \text{const}$ going through initial state P_1, V_1 , is impossible for the real physical situation. In this region $V_2 > V_1$ and $p_2 > p_1$, that is, from the Rayleigh line equation mass flux becomes imaginary.

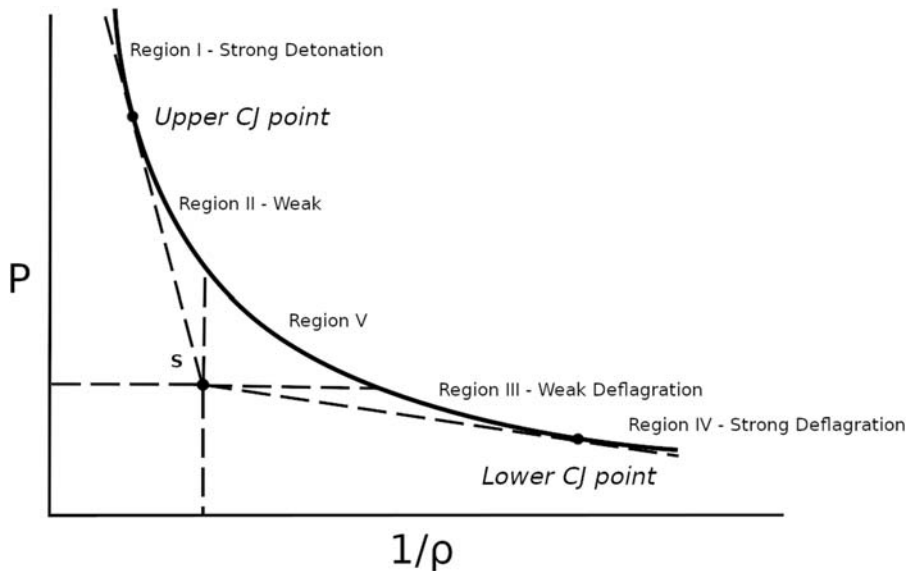


FIGURE 3.26

Rankine – Hugoniot curve describing solutions of combustion waves.

The Regions I and II represent detonation branches of the RH curve. In this branch $p_2 > p_1$ and $V_2 < V_1$, that is, conditions correspond to the compression wave when pressure and density become higher than in the initial state. Region I corresponds to regime of strong detonation and Region II relates to the weak regime of detonation. In Region I, flow velocity relative to the shock speed is subsonic, that is, any perturbation appearing in the flow will weaken the shock moving the final state position over the curve down. In contrast to it, in Region II, the flow velocity is supersonic, that is, any perturbation appearing in this region will overtake and strengthen shock, moving the final state position over the curve up. Thus the tangent to the curve of the final states is the only position, which provides stable propagation of the chemical reaction—shock complex.

The Regions III and IV, where $p_2 < p_1$ and $V_2 > V_1$, represent deflagration wave in which rarefaction wave is moving behind the burning wave. Similarly, as for the detonation branch, the deflagration branch has only one stable position for the final state, which is defined by the tangent to the RH curve in the deflagration region.

3.7.1.3 Chapman – Jouguet detonation

Exactly in the form originated from the study of Rankine – Hugoniot, based on the uniqueness of the solution for the final state, first [Chapman \(1899\)](#) and then [Jouguet \(1906\)](#) proposed CJ condition, which relates sound speed c and products flow velocity v_2 with detonation speed

$$v_2 + c_2 = D_{CJ}.$$

The CJ theory assumes that chemical reaction is infinitely fast and is completed directly at the shock. Taking into account this condition, an expression for the maximum energy that can be released at the front depending on the shock Mach number M reads as

$$q_{CJ} = c_1^2 \frac{(M_{CJ}^2 - 1)^2}{2(\gamma_2^2 - 1)M_{CJ}^2},$$

here C_S is sound speed in the reactants, γ is specific heat for the products, and q is heat of combustion. Note that the simplest CJ detonation theory, which neglects all three-dimensional effects as well as complexity of the reaction details assuming that reactants from its initial state instantaneously transformed into the products, excellently predicts parameters of detonation wave.

Despite the real success of the CJ theory, further development of the theory of detonation allowed to bring features that are more realistic in the description of the detonation front. Three authors (Neumann, 1942; Zel'dovich, 1940; Doering, 1943) independently proposed a model, which takes into account finite speed of the chemical reaction. In the simplest form of the ZND theory in the contrast to CJ theory, it is assumed that the chemical reaction does not start immediately on the shock. The shock amplitude is defined by the relation for the jump in compressed media and reaches the value referred as von Neumann peak. The chemical reaction starts only after certain time, which is often named as induction period, and lasts over some distance from the start, which is named as reaction zone. In the frames of ZND model the CJ condition remains valid for the plane where chemical reaction ends, and therefore detonation speed in the ZND theory remains the same.

If initial state of the explosive mixture and thermodynamic data of the chemical reaction are known, the Rankine – Hugoniot relation with CJ condition allows to obtain uniquely the final state of the gas behind the detonation wave using some mathematics, for example, CANTERA code (Goodwin, Speth, Moffat, & Weber, 2021). Example of such calculation for CJ and ZND theories are given in Table 3.6. Detonation parameters were calculated for different diluted hydrogen–oxygen mixtures with initial temperature 298.15 K and pressure $1.013 \cdot 10^5$ Pa. In the column headings, index “e” means experimental data, index “a” refers to the experimental data averaged over interval 20–80 mks, and index “c” means calculated data. Experimental data are taken from Edwards, Williams, and Breeze (1959).

The examples shown in Table 3.6 demonstrate that accuracy of the CJ theory results is quite high and can be used for the practical engineering calculations with high degree of credibility.

In cases when detonation speed is high enough, so that M_{CJ} is noticeably higher than 1, many of the relations can be considerably simplified to produce convenient but still acceptably accurate formulas

Table 3.6 Comparison of calculated and experimental detonation properties for $H_2 - O_2$ – dilutant mixtures.

Mixture	P_{CJ} , bar	T_{CJ} , K	P_{vN} , bar	T_{vN} , K	P_e , bar	P_a , bar	D_s , m/s	D_e , m/s
$2H_2 + O_2 + 3.76N_2$	15.25	2932.2	28.03	1531.7	-	-	1970	-
$2H_2 + O_2$	18.33	3668.1	33.48	1766.1	27.1	18.6	2839	2819
$H_2 + O_2$	17.37	3454.9	31.78	1670.6	26.1	18.3	2322	2320
$2H_2 + O_2 + 3Ar$	17.36	3353.3	32.00	2031.3	-	-	1929	1800

$$D \approx (2q(\gamma_2^2 - 1))^{\frac{1}{2}},$$

$$p_2 \approx \frac{2q(\gamma_2 - 1)}{V_1},$$

$$T_2 \approx \frac{2q\gamma_2}{(\gamma_2 + 1)c_2},$$

with additionally exact relations

$$D = c_2 \frac{(\gamma_2 + 1)}{\gamma_2},$$

$$v_2 = \frac{c_2}{\gamma_2},$$

$$\frac{V_2}{V_1} = \frac{\rho_1}{\rho_2} = \frac{\gamma_1}{\gamma_1 + 1},$$

$$v_2 = \left(2q \frac{\gamma_2 - 1}{\gamma_2 + 1}\right)^{1/2}.$$

Under certain conditions, as for example, during DDT event, the detonation wave can move faster than is predicted by the theory. This regime is called an overdriven detonation. The amplitude of the pressure wave in such cases appears to be significantly higher even for relatively moderate increase of the detonation speed (Teodorczyk, 1992) as it is shown in Table 3.7.

3.7.1.4 Detonation limits

The research practice in the studies of detonations in gases unambiguously demonstrates that depending on mixture composition and ambient conditions, there are cases when detonation could not be initiated even with strong ignitions source. Clear influence on the possibility to ignite detonation is brought by the mixture composition, particularly by mutual concentrations of hydrogen and, in case of atmospheric conditions, oxygen or another oxidizer, ρ .

From the tube experiments, it is known that together with mixture content, the geometry (tube diameter) and roughness of the internal surfaces are playing critical role in the detonation onset and/or propagation likelihood.

In the mixtures near stoichiometry, detonation front usually has clearly formed cellular structure. For the mixtures far from the stoichiometry in both lean and reach limits, detonation exhibits unstable behavior and often first turns into multihead regime and then detonation front starts to move helically having a spiral path inside the tube. This regime is referred as spinning detonation. In Fig. 3.27, relative dependence of detonation regime on the fuel concentration and tube diameter is shown. In the figure,

Table 3.7 Theoretical pressure (bar) for overdriven detonation in stoichiometric hydrogen mixtures.

M/M _{CJ}	1.0	1.05	1.1	1.2	1.3	1.4
H ₂ /Air	16.10	22.75	27.01	35.14	43.52	52.31
H ₂ /O ₂	19.44	27.83	32.94	42.76	52.81	

the tube diameter is plotted against fuel concentration around the lean limit. In the region above the curve, known as the self-sustained detonation region, multihead detonations propagate.

In Table 3.8, the concentration limits for the possibility to detonate are shown. For comparison, the combustion concentration limits are given with the view to illustrate the fact that detonation limits are always more narrow. Note that detonation limits can vary considerably depending on the geometrical sizes of the vessels, and for the volumes with larger characteristic sizes, the detonation limits become wide.

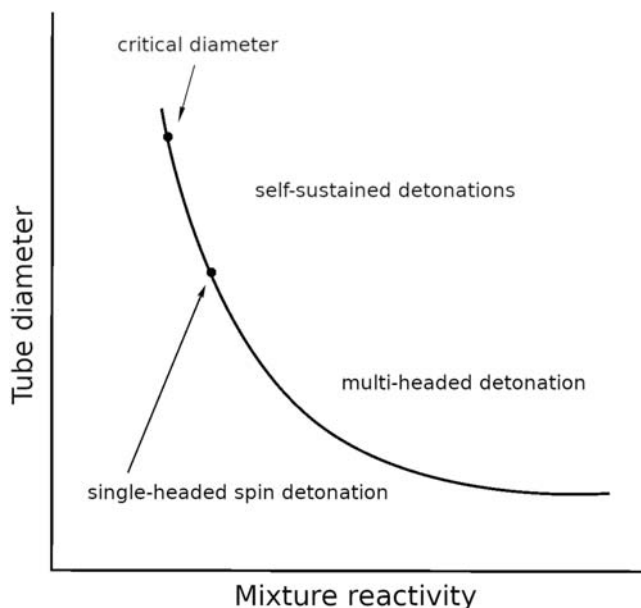


FIGURE 3.27

Effects of fuel concentration and tube diameter on the onset of single-head spin detonation.

Table 3.8 Experimental deflagration and detonation limits for hydrogen.					
Mixture	Deflagration	Detonation	Detonation	Deflagration	Comment
	Lean limit [% H ₂ vol.]		Rich limit [% H ₂ vol.]		
H ₂ –air	4	18.3	59	74	Kuo (2005)
H ₂ –air		17.36			Guirao et al. (1989)
H ₂ –air		12.5			Dorofeev et al. (2000)
H ₂ –O ₂		15	90	93.9	Kuo (2005)
H ₂ –O ₂		15	63.5		Kogarko and Zeldovich (1948)
H ₂ –O ₂	4.6	13.6	70		Tieszen et al. (1986)

Often in safety literature one can find the references to detonability limits regularly ranging between 18% and 59% vol. (Lewis & von Elbe, 1987). This concept appeared in analogy to the flammability limits and values of limits were obtained as a result of the corresponded experiments. The state-of-the-art understanding does not assume actual limits for the concentration of the mixture, rather connects the critical concentration with the scale of the facility or exploding mixture. As it is demonstrated in Table 3.8, the experimentally observed lower limit for hydrogen – air mixture is at least as low as $\sim 13\%$ vol.

Experimental studies for the lean diluted hydrogen – oxygen mixtures, as, for example, by Edwards and Morgan (1977), Gao, Ng, and Lee (2015), have identified another regime of detonation propagation, when regular pulsation of the leading front speed takes place. The frequency of such pulsations depends on the mixture reactivity and diluent nature. The peak values of the front speed can lead to the exceeding values in the pressure local maximums. The detonation in such regime is called “galloping” detonation. Note that possibility for galloping detonation can occur in mixtures, which appears in industrial conditions, and can realize during accident events.

Self-sustained detonation is typically observed in the geometries with constant geometries, as for example, in the tubes with constant diameter. If detonation transfers into areas with larger diameter, it can decay. It was found that the detonation decay and possible further reinitiation depends on the tube diameter. The critical geometrical parameters are commonly characterized using representative parameter λ , which is average dimension of the detonation cellular structure elements (detonation cell) and is defined by the chemical properties of the mixture only.

For the detonation to survive during transfer from a tube d to the semiconfined space outside the tube it is necessary the tube diameter exceeds 13λ for circular tubes (relevant example in Figs. 3.28), 10λ for planar tubes, and for sufficiently wide rectangular channels the critical height approaches a value of 3λ . For the onset of single-head spinning detonation in the smooth circular tubes, the critical tube diameter should be at least λ/π . First the criteria was proposed by Kogarko and Zeldovich (1948), later it was confirmed by Lee (1985) and Dupre, Knystautas, and Lee (1985). In Fig. 3.29, the dependence of critical tube diameter 13λ for hydrogen and some

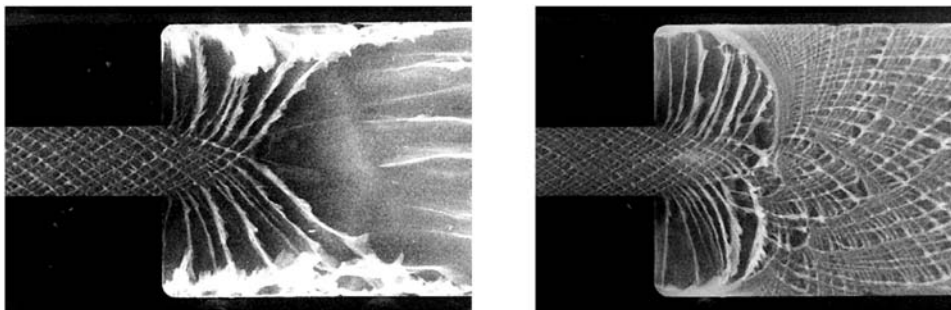


FIGURE 3.28

Example of the transition of the detonation to larger volume (Bauwens & Dorofeev, 2019).

hydrocarbons on equivalence ratio are shown, and in Fig. 3.30, the critical tube diameter for the onset of spinning detonation λ/π for the same mixtures are shown.

In Kuznetsov et al. (1997), it was found that mechanism in the spontaneous initiation of a detonation can be closely connected with the problem of the detonation transition to less sensitive mixture. Possibility to reinitiate detonation after decay in the inert layer and to transit to the less sensitive mixture were started by Bull and Elsworth (1981), Thomas, Sutton, and Edwards (1991), Teodorczyk and Benoan (1996). They showed that possibility of reinitiation increases with the increase of detonation strength in the original mixture and sensitivity of the acceptor and decrease with the length of inert layer. It was also found that length of original mixture driver can be a critical parameter.

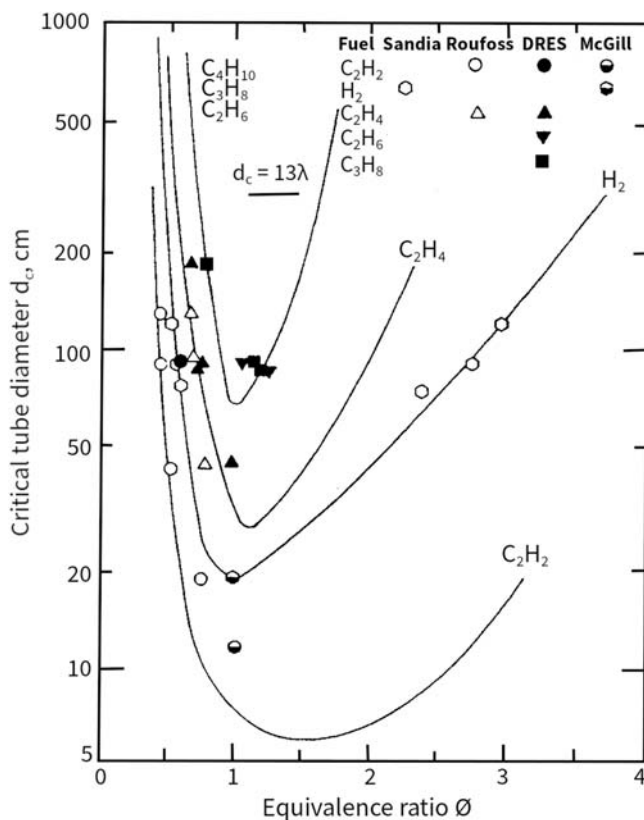
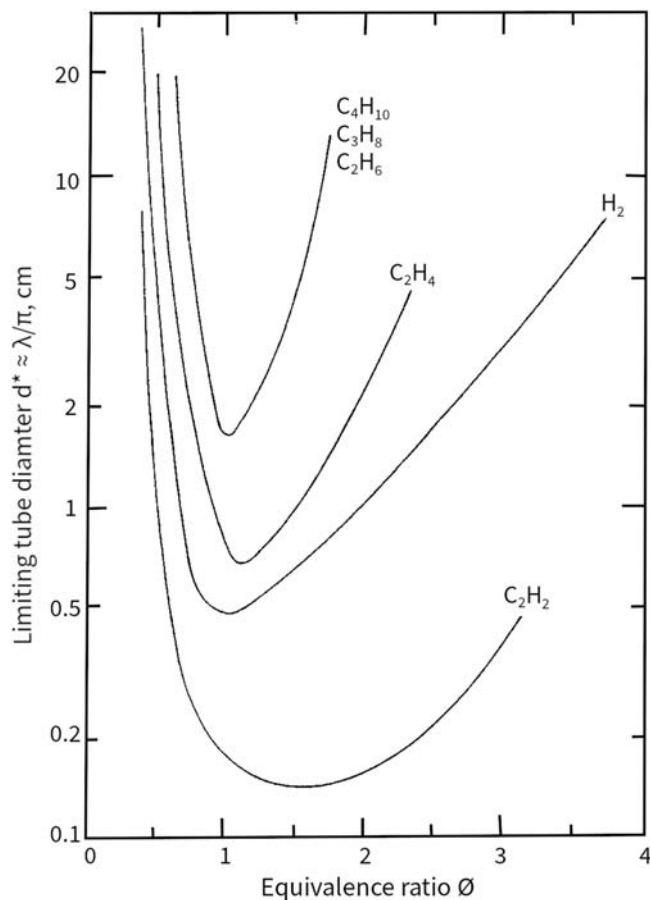


FIGURE 3.29

Correlation of a critical tube diameter with established criteria 13λ for different fuel – air mixtures (Kuo, 2005).

**FIGURE 3.30**

Dependence of critical tube diameter λ/π on mixture composition (Kuo, 2005).

3.7.1.5 Detonation front structure

Standard practice of evaluating possible hazards, which can come from detonation process, is usually based on the theory of steady state one-dimensional detonation. The predictions of such theory are well-known and usually can have sufficient accuracy to provide necessary information for industrial environment.

However, there are numerous examples demonstrating nonstationary and multidimensional internal structure of the detonation wave. This fact actually affects expectations on the maximum of local pressures developing inside the detonation wave front during detonation propagation.

Detonation waves in the mixtures, which are far enough from detonation limits, have complex internal multidimensional structures. Already the classical works on the structure of detonation

waves (e.g. Strehlow, 1969, 1970; Strehlow, Maurer, & Rajan, 1969; Voitsekhovskii, Mitrofanov, & Topchiyan, 1969) made it possible to detect the following types of detonation waves:

- front structures, which have highly constant values of main characteristics averaged in time and regularly repeated structure;
- front structures, which have constant averaged characteristics but no regular structure;
- front structures, which exhibit really nonstationary behavior.

In case of lean or reach mixtures, which are close to the detonation limits, the detonations occur exhibiting characteristics of the second or third types. Unfortunately, most of the experimental works were performed for the mixtures far from detonation limits, and detonations of lean/reach mixtures are not studied in satisfactory volume. Note that the role of front structure, as a rule, appears to be more significant in case of lean mixtures detonations, which are typical for industrial accidents.

Multidimensional structure of the detonation front (see Fig. 3.31) includes leading shock wave and a number of transverse waves, which propagates perpendicular to the leading shock, reflecting from each other and from bounding walls. The surface of the leading shock consists of the sequence of convex parts, which start chemical reaction, and concave parts, which are fast decaying waves. Additional reaction zones are located behind the transverse waves where the reaction completes.

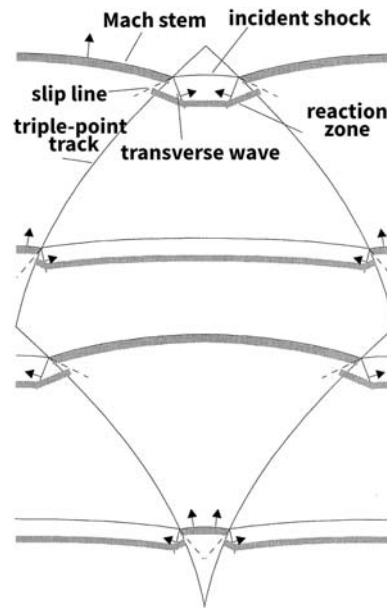


FIGURE 3.31

Schematic presentation of multidimensional detonation front structure (Gavrikov, Efimenko, & Dorofeev, 2000).

3.7.1.6 Detonation cell size

In Fig. 3.31 of the previous section, a diamond-shaped form demonstrates a typical trajectory of triple points from intersecting shock waves. Such trajectory is called detonation cell and can be easily obtained experimentally on soot surfaces located in detonating gases. These cells form a cellular structure and are characterized by their two lengths: longitudinal size L and transverse size S . These two lengths are varying depending on type of burnable gas and initial conditions; however, the relation $S \approx 0.6 L$ are kept usually constant.

In Fig. 3.32, an example of detonation cell structure in stoichiometric hydrogen – air mixture is shown. In the figure a soot track of irregular detonation cellular structure in stoichiometric H_2 – air mixture is shown (Kuznetsov, Alekseev, & Dorofeev, 2000; Kuznetsov et al., 2002). Average transverse cell size of the test is shown in the figure $S \approx 1$ cm. The detonation cell size is kept constant for the given components for the same initial condition and depends only on the mixture composition, and therefore is often used as a measure of mixture reactivity.

Usually the detonation cell size reaches its minimum at the stoichiometry composition and grows for leaner and richer mixtures. In Fig. 3.33 dependence of the detonation cell size for H_2 – air mixtures on initial H_2 concentration and mixture temperature and comparison with the engineering model are shown. The figure is presented as it is in Breitung et al. (2000) with the data originated from Ciccarelli et al. (1994); Kaneshige and Shepherd (1997).

3.7.1.7 Detonation stability. Steady and unsteady detonations

All gaseous detonations propagating averagely with parameters close to CJ regime have strong unsteadiness on the propagating front resulting in the footprint of the triple points trajectories as detonation cell. Commonly, using the regularity of the cellular structure the overall detonation process can be characterized qualitatively.

Alternatively, more quantitative method was proposed to use the value of effective activation energy of equivalent one-step reaction. For hydrogen mixtures, such activation energy appears to be the dominant parameter for categorization detonation stability, being below some critical value for stable mixtures and above it for unstable ones.

In Eckett (2001) it was found that the effective activation energy $\Theta = E_a/RT_{VN}$ (T_{VN} is the temperature at the von Neumann state) together with the CJ Mach number M_{CJ} can be used for classification of detonation stability.

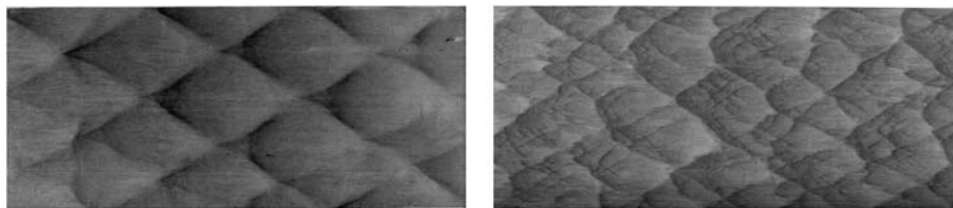


FIGURE 3.32

Representative picture of cellular structure for regular (left) cellular structure in $2H_2 - O_2 - 17Ar$ and irregular (right) structure in $C_3H_8 - 5O_2 - 9N_2$. Image height is 152 mm (Austin, 2003).

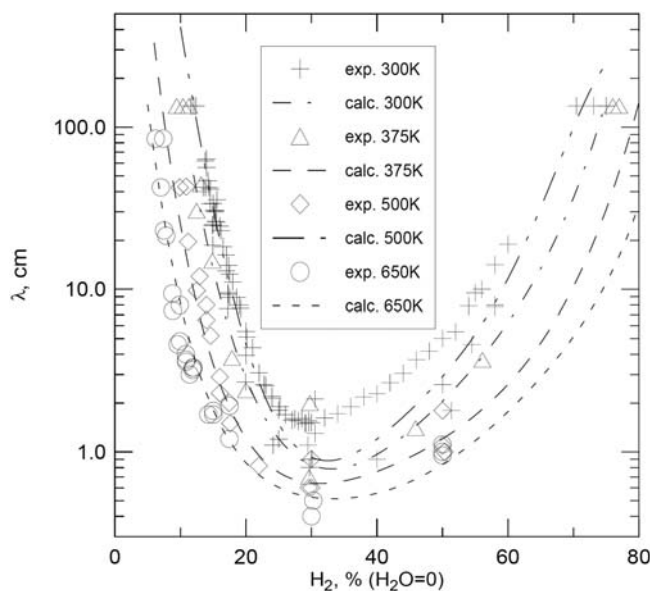


FIGURE 3.33

Dependence of detonations cell sizes on mixture composition for different initial mixture temperature.

In Fig. 3.34 several examples of detonation with different stability regimes are shown. Weakly unstable detonations can be observed in mixtures with parameters $\Theta = \sim 5$ and $M_{CJ} = \sim 4.6$; it is well-known that hydrogen – oxygen mixtures with high degree dilution with noble gases create very regular cellular structure. The mixtures with $\Theta = \sim 7$ are moderately unstable and with $\Theta = \sim 11$ are highly unstable.

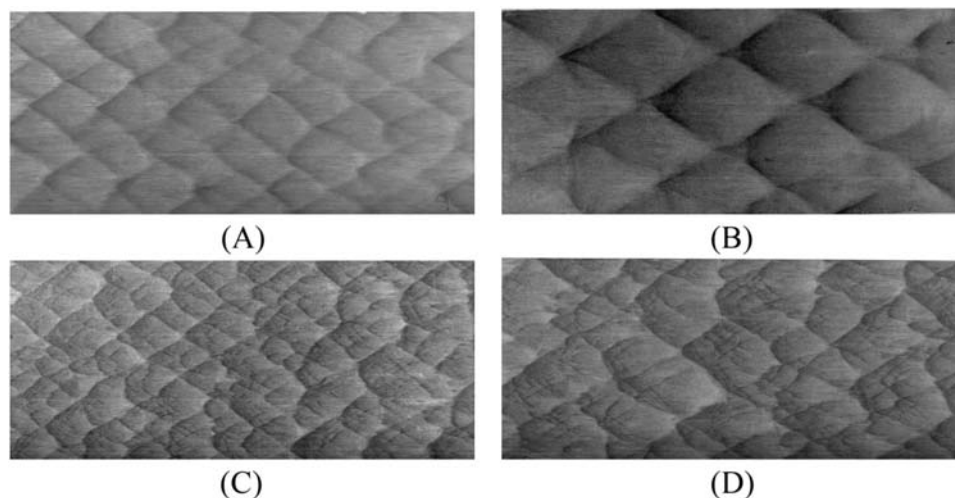
3.7.2 Strategies in the detonation prevention

A number of factors exist, which prevent a comprehensive survey of the current methods of the detonation control. For the onset of detonation at least presence of the burnable mixture and a source of ignition is required. Thus the active prevention of such conditions' realization (see Sections 3.1, 3.2, and 3.3) could be a starting point in the accidental risk reduction.

However, if the burnable mixture exists and it was ignited, then the following phases in the general strategy could be distinguished:

- extinguishing or inhibition of the initial flame;
- prevention of the initial flame acceleration;
- destroying of the flame-shock complex for accelerating flame and for stable propagating detonation.

Often in practice, chemical reactions are inhibited by the chemically active additives. For the industrial conditions it is difficult to predict when and where in the setup a flame can appear; therefore the

**FIGURE 3.34**

Examples of weakly unstable detonation in $2\text{H}_2 - \text{O}_2 - 12\text{Ar}$ and $2\text{H}_2 - \text{O}_2 - 17\text{Ar}$ (upper row), and of highly unstable detonation in $\text{H}_2 - \text{N}_2\text{O} - 1.33\text{N}_2$ and $\text{C}_3\text{H}_8 - 5\text{O}_2 - 9\text{N}_2$ (lower row). Detonation. Image height is about 152 mm (Austin, 2003).

flame inhibitor should be injected directly into the mixture where ignition had occurred. The use of fire suppression systems based on the injection of water mist could be also highly effective for flame suppression. Another safety system is flame arrestor, which is usually an obstacle in the flame path made of wire mesh, sheet metal plate with punched holes, metal spiral bands, etc.

Alternatively, for the prevention of accelerating or fast flames' development, methods allowing pressure reduction in the burning area are widely used, and bursting membrane is one of the widely used technique. Note that risk of uncontrolled vented explosion (see Section 3.5.4) still often remains.

For the destroying of emerging or finally formed chemical reaction–shock complex, the appliances similar to flame arresters or pipes with largely expanded part are used. The principle of operation for such devices can be understood taking into account conditions for the strong flame acceleration (see Section 3.6.3) and for the establishment of stable detonation (see Section 3.6.4).

References

- Abdel-Gayed, R. G., Bradley, D., & Lung, F. K.-K. (1989). Combustion regimes and the straining of turbulent premixed flames. *Combustion and Flame*, 76, 213–218.
- Abramovich, G. N. (1963). *The theory of turbulent jets*. Cambridge(MA): Massachusetts Institute of Tech. Press.
- Alekseev, V. I., Kuznetsov, M. S., Yankin, Y. G., & Dorofeev, S. B. (2001). Experimental study of flame acceleration and DDT under conditions of transverse venting. *Journal of Loss Prevention in the Processes Industries*, 14, 591–596.

- Allelein, H. J., et al. (2007). *International standard problem ISP-47 on containment thermal hydraulics—Final report*. Paris: Organisation for Economic Co-Operation and Development.
- Astbury, G., & Hawksworth, S. (2007). Spontaneous ignition of hydrogen leaks: A review of postulated mechanisms. *International Journal of Hydrogen Energy*, 32, 2178–2185, 9.
- Austin, J. (2003). *The role of instability in gaseous detonation*. Pasadena, CA: California Institute of Technology.
- Baraldi, D., et al. (2009). Numerical analysis of release, dispersion and combustion of liquid hydrogen in a mock-up hydrogen refuelling station. *Journal of Loss Prevention in the Process Industries*, 22, 303–315.
- Barley, C. D., & Gawlik, K. (2009). Buoyancy-driven ventilation of hydrogen from buildings: Laboratory test and model validation. *International Journal of Hydrogen Energy*, 34, 5592–5603.
- Batchelor, G. K. (1954). Heat convection and buoyancy effects in fluids. *Quarterly Journal of the Royal Meteorological Society*, 80, 339–358.
- Batchelor, G. K. (1959). Small-scale variation of convected quantities like temperature in turbulent fluid. Part 1. General discussion and the case of small conductivity. *Journal of Fluid Mechanics*, 5, 113–133, 1.
- Batchelor, G. K. (1993). *The theory of homogeneous turbulence*. Cambridge: Cambridge University Press.
- Bauwens, C. R. L. & Dorofeev, S. B. (2019). Quantifying the potential consequences of a detonation in a hydrogen jet release. *Adelaide, International Conference on Hydrogen Safety*.
- Behrendt, F., & Warnatz, J. (1985). The dependence of flame propagation in H_2 O_2 N_2 mixtures on temperature, pressure, and initial composition. *International Journal of Hydrogen Energy*, 10, 749–755.
- Bénard, P., et al. (2007). High pressure jets in the presence of a surface. San Sebastian (Spain), International Conference on Hydrogen Safety, p. 10.
- Benteboula, S., Bengaouer, A. & Cariteau, B. (2009). Comparison of two simplified models predictions with experimental measurements for gas release within an enclosure. Ajaccio, International Conference on Hydrogen Safety.
- Birch, A. D., Brown, D. R., Dodson, M. G., & Swaffield, F. (1984). The structure and concentration decay in high pressure jets of natural gas. *Combustion Science and Technology*, 36, 249–261.
- Birch, A. D., Hughes, D. J., & Swaffield, F. (1987). Velocity decay of high pressure jets. *Combustion Science and Technology*, 52, 161–171.
- Borghi, R. (1988). Turbulent combustion modelling. *Progress in Energy and Combustion Science*, 14, 245–292.
- Bradley, D. (1992). How fast can we burn? *Symposium (International) on Combustion*, 24, 247–262.
- Bradley, D., Lau, A. K. C., Lawes, M., & Smith, F. T. (1992). Flame stretch rate as a determinant of turbulent burning velocity. *Philosophical Transactions of the Royal Society of London. Series A: Physical and Engineering Sciences*, 338, 359–387.
- Brearley, D. & Tolson, P. (1995). *The frictional ignition properties of (18/8) stainless steel*. Buxton: Health Safety Laboratory.
- Breitung, W., et al. (2000). *Flame acceleration and deflagration-to-detonation transition in nuclear safety*. State-of-the art report by a group of experts, Issy-les-Moulineaux, France: NEA/CSNI/R (2000)7.
- Bricard, P., & Friedel, L. (1998). Two-phase jet dispersion. *Journal of Hazardous Materials*, 59, 287–310.
- Buckel, J. W., & Chandra, S. (1996). Hot wire ignition of hydrogen-oxygen mixtures. *International Journal of Hydrogen Energy*, 21, 39–44.
- Bull, D. C., & Elsworth, J. E. (1981). *Initiation of unconfined gas detonations in hydrocarbon-air mixtures by a sympathetic mechanism*. United States: American Institute of Aeronautics and Astronautics.
- Burke, S. P., & Schumann, T. E. W. (1928). Diffusion flames. *Industrial & Engineering Chemistry*, 20, 998–1004.
- Cariteau, B., & Tkatschenko, I. (2012). Experimental study of the concentration build-up regimes in an enclosure without ventilation. *International Journal of Hydrogen Energy*, 37, 17400–17408.

- Cariteau, B., et al. (2011a). Experimental results on the dispersion of buoyant gas in a full scale garage from a complex source. *International Journal of Hydrogen Energy*, 36, 2489–2496.
- Cariteau, B., Brinster, J., & Tkatschenko, I. (2011b). Experiments on the distribution of concentration due to buoyant gas low flow rate release in an enclosure. *International Journal of Hydrogen Energy*, 36, 2505–2512.
- Catlin, C. A. (1991). Scale effects on the external combustion caused by venting of a confined explosion. *Combustion and Flame*, 83, 399–411.
- CENELEC. (2019). Explosive atmospheres. Electrostatic hazards, guidance. Brussels: European Committee for Electrotechnical Standardization.
- Chang, S. L., & Rhee, K. T. (1983). Adiabatic flame temperature estimates of lean fuel/air mixture. *Combustion Science and Technology*, 35, 203–206.
- Chapman, D. L. (1899). On the rate of explosion in gases. *Physics of Fluids*, 47, 90–104.
- Chen, C. J., & Rodi, W. (1980). *Vertical turbulent buoyant jets: a review of experimental data*. Oxford and New York: Pergamon Press.
- Chirivella, J. E., & Witcofsky, R. D. (1986). Experimental results from fast 1500-gallon LH₂ spills. *AIChE Symposium Series: Cryogenic Properties, Processes and Applications*, 82, 120–141.
- Cho, P., & Law, C. K. (1986). Catalytic ignition of fuel/oxygen/nitrogen mixtures over platinum. *Combustion and Flame*, 66, 159–170.
- Ciccarelli, G., & Dorofeev, S. (2008). Flame acceleration and transition to detonation in ducts. *Progress in Energy and Combustion Science*, 34, 499–550.
- Ciccarelli, G., et al. (1994). Detonation cell size measurements and predictions in hydrogen-air-steam mixtures at elevated temperatures. *Combustion and Flame*, 99, 212–220.
- Clavin, P. (1994). Premixed combustion and gas dynamics. *Annual Review of Fluid Mechanics*, 26, 321–352.
- Cleaver, R. P., Marshal, M. R., & Linden, P. F. (1994). The build-up of concentration within a single enclosed volume following a release of natural gas. *Journal of Hazardous Materials*, 36, 209–226.
- Cooper, M. G., Fairweather, M., & Tite, J. P. (1986). On the mechanisms of pressure generation in vented explosions. *Combustion and Flame*, 65, 1–14.
- Coward, H. F., & Jones, G. W. (1952). *Limits of flammability of gases and vapors*. Washington: US Government Printing Office.
- Dahoe, A. E. (2005). Laminar burning velocities of hydrogen–air mixtures from closed vessel gas explosions. *Journal of Loss Prevention in the Process Industries*, 18, 152–166.
- Deimling, L., et al. (2011). Visualisation of jet fires from hydrogen release. *International Journal of Hydrogen Energy*, 36, 2360–2366.
- Damköhler, G. (1947). The effect of turbulence on the flame velocity in gas mixtures. NACA TM No. 1112. Translated from *Zeitschrift fuer Elektrochemie und angewandte Physikalische Chemie*, 46(11), (Nov. 1940), 601–626.
- Dienhart, B. (1995). Ausbreitung und Verdampfung von flüssigem Wasserstoff auf Wasser und festem Untergrund, Juelich: Research Center Juelich.
- Dimotakis, P. E. (2005). Turbulent mixing. *Annual Review of Fluid Mechanics*, 37, 329–356.
- DIN_EN_1839. (2004). *Bestimmung der Explosionsgrenzen von Gasen und Dämpfen* (German Edition). Berlin: European Commission.
- Doering, W. (1943). On detonation processes in gases. *Annals of Physics*, 43, 421–436.
- Dorofeev, S. B. (1995). *Blast effects of confined and unconfined explosions*. Pasadena, Proceedings of the 20th International Symposium on Shock Waves, 1995.
- Dorofeev, S. B. (2007). Evaluation of safety distances related to unconfined hydrogen explosions. *International Journal of Hydrogen Energy*, 32, 2118–2124.
- Dorofeev, S. B., et al. (2000). Effect of scale on the onset of detonations. *Shock Waves*, 10, 137–149.
- Dorofeev, S. B., et al. (2001). Evaluation of limits for effective flame acceleration in hydrogen mixtures. *Journal of Loss Prevention in the Processes Industries*, 14, 583–589.

- Dorofeev, S. B., Sidorov, V. P., Kuznetsov, M. S., Matsukov, I. D., & Alekseev, V. I. (1999). Effect of scale on the onset of detonations. *Proceedings of the 17th International Colloquium on the Dynamics of Explosion and Reactive System, Heidelberg*.
- Dowdy, D. R., Smith, D. B., Taylor, S. C., & Williams, A. (1991). The use of expanding spherical flames to determine burning velocities and stretch effects in hydrogen/air mixtures. *Symposium (International) on Combustion*, 23, 325–332.
- Dupre, G., Knystautas, R., & Lee, J. H. S. (1985). Near-limit propagation of detonations in tubes. *Progress in Aeronautics and Astronautics*, 106, 244–259.
- Eckart, C. (1948). An analysis of the stirring and mixing processes in incompressible fluids. *Journal of Marine Research*, 7, 265–275.
- Eckett, C. A. (2001). *Numerical and analytical studies of the dynamics of gaseous detonations*. Los Angeles, CA: California Institute of Technology.
- Edwards, D. H., & Morgan, J. M. (1977). Instabilities in detonation waves near the limits of propagation. *Journal of Physics D: Applied Physics*, 10, 2377–2387.
- Edwards, D. H., Williams, G. T., & Breeze, J. C. (1959). Pressure and velocity measurements on detonation waves in hydrogen-oxygen mixtures. *Journal of Fluid Mechanics*, 6, 497–517.
- Ekoto, I. W., et al. (2014). *Liquid hydrogen release and behavior modeling: State-of-the-art knowledge gaps and research needs for refueling infrastructure safety*. Albuquerque: Sandia National Laboratories.
- Ewan, B., & Moodie, K. (1986). Structure and velocity measurements in underexpanded jets. *Combustion Science and Technology*, 45, 275–288.
- Fauske, H. K., & Epstein, M. (1988). Source term considerations in connection with chemical accidents and vapour cloud modelling. *Journal of Loss Prevention in the Processes Industries*, 1, 75–83.
- Flamm, L. & Mache, H. (1917). Combustion of an explosive gas mixture within a closed vessel. Wien: Ber. Akad., Wiss, pp. 9–44.
- Franquet, E., Perrier, V., Gibout, S., & Bruel, P. (2015). Free underexpanded jets in a quiescent medium: A review. *Progress in Aerospace Sciences*, 77, 25–53.
- Friedrich, A., Breitung, W., Stern, G., Veser, A., et al. (2012). Ignition and heat radiation of cryogenic hydrogen jets. *International Journal of Hydrogen Energy*, 37(22), 17589–17598. Available from <https://doi.org/10.1016/j.ijhydene.2012.07.070>.
- Fukutani, S., Kunioshi, N., & Jinno, H. (1991). Flame structure of an axisymmetric hydrogen-air diffusion flame. *Symposium (International) on Combustion*, 23, 567–573.
- Gao, Y., Ng, H. D., & Lee, J. H. S. (2015). Experimental characterization of galloping detonations in unstable mixtures. *Combustion and Flame*, 162, 2405–2413.
- Gardner & Hulme, G. (1995). A survey of current predictive methods for explosion hazard assessments in the UK offshore industry. Sudbury (UK): Health and Safety Executive, Sheffield (United Kingdom).
- Gavrikov, A. I., Efimenko, A. A., & Dorofeev, S. B. (2000). A model for detonation cell size prediction from chemical kinetics. *Combustion and Flame*, 120, 19–33.
- Gebhart, B., Jaluria, Y., Mahajan, R. L., & Sammakia, B. (1988). *Buoyancy-induced flows and transport*. New York: Hemisphere.
- Gelfand, B. E. (1996). Droplet breakup phenomena in flows with velocity lag. *Progress in Energy and Combustion Science*, 22, 201–265.
- Gilbert, R. B., & Strehlow, R. A. (1966). Theory of detonation initiation behind reflected shock waves. *AIAA Journal*, 4, 1777–1783.
- Glassman, I., Yetter, R. A., & Glumac, N. G. (2014). *Combustion: Fifth Edition*. Netherlands: Elsevier.
- Goodwin, D. G., Speth, R. L., Moffat, H. K., & Weber, B. W., 2021. Cantera: An object-oriented software toolkit for chemical kinetics, thermodynamics, and transport processes. [Online] Available at: <<http://www.cantera.org>>.

- Grune, J., et al. (2013). Experimental investigation of hydrogen–air deflagrations and detonations in semi-confined flat layers. *Journal of Loss Prevention in the Process Industries*, 26, 317–323.
- Grune, J., et al. (2017). Detonation wave propagation in semi-confined layers of hydrogen–air and hydrogen–oxygen mixtures. *International Journal of Hydrogen Energy*, 42, 7589–7599.
- Grune, J., Sempert, K., Kuznetsov, M., & Breitung, W. (2011). Experimental study of ignited unsteady hydrogen jets into air. *International Journal of Hydrogen Energy*, 36, 2497–2504.
- Guirao, C. M., Knystautas, R., & Lee, J. H. (1989). *A summary of hydrogen-air detonation experiments*. Albuquerque: McGill University, Sandia National Laboratories.
- Gülnder, Ö. L. (1991). Turbulent premixed combustion modelling using fractal geometry. *Symposium (International) on Combustion*, 23, 835–842.
- Gupta, S., Brinster, J., Studer, E., & Tkatschenko, I. (2009). Hydrogen related risks within a private garage: Concentration measurements in a realistic full scale experimental facility. *International Journal of Hydrogen Energy*, 34, 5902–5911.
- Hanna, S. R., & Strimaitis, D. (1989). *Workbook on test cases for vapour cloud source dispersion models*. New York: American Institute of Chemical Engineers.
- Harris, R. J. (1983). *The investigation and control of gas explosions in buildings and heating plant*. London; New York: E. & F.N. Spon in association with the British Gas Corp.
- Harrison, A. J., & Eyre, J. A. (1987). “External explosions” as a result of explosion venting. *Combustion Science and Technology*, 52, 91–106.
- Hawksworth, S. et al. (2004). Mechanical ignition hazards in potentially explosive atmospheres—EC project MECHEX. 3.
- Hawksworth, S., et al. (2005). Ignition of explosive atmosphere by mechanical equipment. *Hazards XVIII. Process Safety-sharing Best Practice*, 1, 337–347.
- Hawthorne, W. R., Weddell, D. S. & Hottel, H. C. (1949). Third Symposium on Combustion and Flame and Explosion Phenomena. The Williams and Wilkins Co., Baltimore, Maryland, pp. 266–288.
- Higgins, A. & Bruckner, A. (1996). *Experimental investigation of detonation initiation by hypervelocity blunt projectiles*. In: 34th Aerospace Sciences Meeting and Exhibit. Reno(NV): 34th Aerospace Sciences Meeting and Exhibit.
- Hooker, P., Willoughby, D. B. & Royle, M. (2011). *Experimental releases of liquid hydrogen*. 4th International Conference on Hydrogen Safety, San Francisco, CA, USA.
- Hottel, H. C. & Hawthorne, W. R. (1949). Diffusion in laminar flame jets. Third Symposium (International) on Combustion, pp. 254–266.
- Houf, W. & Schefer, R. (2005). Predicting radiative heat fluxes and flammability envelopes from unintended releases of hydrogen. 16th Annual Hydrogen Conference and Hydrogen Expo. Washington, D.C., USA.
- Houf, W. G., & Winters, W. S. (2013). Simulation of high-pressure liquid hydrogen releases. *International Journal of Hydrogen Energy*, 38, 8092–8099.
- Hu, E., Huang, Z., He, J., & Miao, H. (2009). Experimental and numerical study on laminar burning velocities and flame instabilities of hydrogen–air mixtures at elevated pressures and temperatures. *International Journal of Hydrogen Energy*, 34, 8741–8755.
- Ichard, M., Hansen, O. R., Middha, P., & Willoughby, D. (2012). CFD computations of liquid hydrogen releases. *International Journal of Hydrogen Energy*, 37, 17380–17389.
- Iijima, T., & Takeno, T. (1986). Effects of temperature and pressure on burning velocity. *Combustion and Flame*, 65, 35–43.
- ISO. (2015). Basic considerations for the safety of hydrogen systems. ISO/TC 197 Hydrogen technologies. Québec, Canada.
- Jasper, A. W., et al. (2014). Predictive a priori pressure-dependent kinetics. *Science (New York, N.Y.)*, 346, 1212–1215.

- Jordan, T., Jallais, S. & Bernard, L. (2019). Status of the pre-normative research project PRESLHY for the safe use of liquid hydrogen. Adelaide, 8th International Conference on Hydrogen Safety ICHS 2019.
- Jouguet, E. (1906). On the propagation of chemical reactions in gases. *Journal de Mathématiques Pures et Appliquées, 1*, 347–425.
- Kaneshige, M., & Shepherd, J. E. (1997). *Detonation database*. Pasadena: Graduate Aeronautical Laboratories, California Institute of Technology.
- Kang, Y., et al. (2016). On predicting the length, width, and volume of the jet diffusion flame. *Applied Thermal Engineering, 94*, 799–812.
- Kapila, A. K., Schwendeman, D. W., Quirk, J. J., & Hawa, T. (2002). Mechanisms of detonation formation due to a temperature gradient. *Combustion Theory and Modelling, 6*, 553–594.
- Kee, R. J., Rupley, F. M. & Miller, J. A. (1989). *Chemkin-II: A Fortran chemical kinetics package for the analysis of gas-phase chemical kinetics*. United States: <<https://www.osti.gov/servlets/purl/5681118>>.
- Khokhlov, A. M., Oran, E. S., & Wheeler, J. C. (1997). A theory of deflagration-to-detonation transition in unconfined flames. *Combustion and Flame, 108*, 503–517.
- Kim, M., Do, K., Choi, B., & Han, Y. (2012). High-order perturbation solutions to a LH2 spreading model with continuous spill. *International Journal of Hydrogen Energy, 37*, 17409–17414.
- Kneebone, A. & Prew, L. R. (1974). Shipboard jettison test of LNG onto the sea. *Proceedings of the Fourth International Conference on LNG, Algiers*.
- Knystautas, R., Lee, J. H. S., Moen, I. O., & Wagner, H. G. G. (1979). *Direct initiation of spherical detonation by a hot turbulent gas jet* (pp. 1235–1245). Pittsburgh: The Combustion Institute.
- Kogarko, S. M. & Zeldovich, Y. B. (1948). Detonation of gaseous mixture. *Academiia Nauk, SSSR Doklady, 63*, pp. 553–556.
- Kolmogorov, A. N. (1941). The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers. *Cr Acad. Sci. URSS, 30*, pp. 301–305.
- Konnov, A. A. (2019). Yet another kinetic mechanism for hydrogen combustion. *Combustion and Flame, 203*, 14–22.
- Koroll, G. W., Kumar, R. K., & Bowles, E. M. (1993). Burning velocities of hydrogen-air mixtures. *Combustion and Flame, 94*, 330–340.
- Kotchourko, A., Lelyakin, A., Yanez, J. & Halmer, G. (2017). COM3D: Turbulent Combustion Code. User Guide. [Online] Available at: <<https://com3d-code.com>>.
- Kotchourko, A., Baraldi, D., Bénard, P., Eisenreich, N., Jordan, T., Keller, J., Kessler, A., Lachance, J., Molkov, V., Steen, M., Tchouvelev, A., Wen, J. (2014a). State of The Art and Research Priorities in Hydrogen Safety. European Commission Joint Research Centre Institute for Energy and Transport. Joint Research Centre, Westerduinweg 3, PBox 2, 1755 ZG, Petten, The Netherlands, JRC84686 EUR 26344 EN. ISBN 978-92-79-34719-1 (PDF), ISSN 1831-9424 (online), <https://doi.org/10.2790/99638>, <<https://publications.jrc.ec.europa.eu/repository/>>.
- Kotchourko, N., et al. (2014b). Concentration measurements in a round hydrogen jet using Background Oriented Schlieren (BOS) technique. *International Journal of Hydrogen Energy, 39*, 6201–6209.
- Kumar, R. K. (1985). Flammability limits of hydrogen-oxygen-diluent mixtures. *Journal of Fire Sciences, 3*, 245–262.
- Kuo, K. K. (2005). *Principles of combustion* (3rd ed.). New York: John Wiley & Sons.
- Kuznetsov, M., & Grune, J. (2019). Experiments on combustion regimes for hydrogen/air mixtures in a thin layer geometry. *International Journal of Hydrogen Energy, 44*, 8727–8742.
- Kuznetsov, M., et al. (2002). DDT in methane-air mixtures. *Shock Waves, 12*, 215–220.
- Kuznetsov, M., Alekseev, V., Matsukov, I., & Dorofeev, S. (2005). DDT in a smooth tube filled with a hydrogen–oxygen mixture. *Shock Waves, 14*, 205–215.
- Kuznetsov, M. S., et al. (1997). Experimental and numerical studies on transmission of gaseous detonation to a less sensitive mixture. *Shock Waves, 7*, 297–304.

- Kuznetsov, M. S., Alekseev, V. I., & Dorofeev, S. B. (2000). Comparison of critical conditions for DDT in regular and irregular cellular detonation systems. *Shock Waves*, 10, 217–224.
- Landau, L. D., & Lifshitz, E. M. (1987). *Fluid mechanics: Volume 6 (Course of Theoretical Physics)* (2nd ed.). Kronberg: Butterworth-Heinemann.
- Law, C. K. (2006). *Combustion physics*. New York: Cambridge University Press.
- Le Cong, T. & Dagaut, P. (2008). Effect of water vapor on the kinetics of combustion of hydrogen and natural gas: Experimental and detailed modeling study. <https://doi.org/10.1115/GT2008-50272>, ASME, pp. 319–328.
- Lee, J. H., Lee, B. H. K., & Knystautas, R. (1966). Direct initiation of cylindrical gaseous detonations. *Physics of Fluids*, 9, 221–222.
- Lee, J. H. S., & Higgins, A. J. (1999). Comments on criteria for direct initiation of detonation. *Philosophical Transactions: Mathematical, Physical and Engineering Sciences*, 357, 3503–3521.
- Lee, J. H. S., & Moen, I. O. (1980). The mechanism of transition from deflagration to detonation in vapor cloud explosions. *Progress in Energy Combustion Science*, 6, 359–389.
- Lee, J. H. S. (1977). Initiation of gaseous detonation. *Annual Review of Physical Chemistry*, 28, 75–104.
- Lee, J. H. S. (1985). On the transition from deflagration to detonation. *Progress in Astronautics and Aeronautics*, 106, 3–18.
- Lees, F. P. (1996). *Loss Prevention in the Process Industry* (2nd ed.). London: Butterworth.
- Lemmon, E. W., Huber, M. L. & McLinden, M. O. (2013). NIST Standard Reference Database 23: Reference Fluid Thermodynamic and Transport Properties-REFPROP, Version 9.1. Washington(USA): NIST Standard Reference Database 12 (Version 5.0). Software package by NIST, US Department of Commerce.
- Lewis, B., & Von Elbe, G. (1934). On the theory of flame propagation. *The Journal of Chemical Physics*, 2, 537–546.
- Lewis, B., & von Elbe, G. (1987). *Combustion, flames and explosions of gases* (3rd ed.). San Diego, CA: Academic Press.
- Li, J., Zhao, Z., Kazakov, A., & Dryer, F. (2004). An updated comprehensive kinetic model of hydrogen combustion. *International Journal of Chemical Kinetics*, 36, 566–575. Available from <https://doi.org/10.1002/kin.20026>.
- Libby, P. A., & Williams, F. A. (1994). *Turbulent reacting flows*. Oxford: Academic press.
- Liñán, A., & Williams, F. A. (1993). *Fundamental aspects of combustion. Oxford Engineering Science Series, No. 34*. New York, US: Oxford University Press.
- Lind, C. D. & Whitson, J. 1974. *Explosion hazards associated with spills of large quantities of hazardous materials*. Phase II. Final report, Washington: Dept. Transp., U.S. Coast Guard Final Rep., ADA 047585.
- Linden, P. F. (1999). The fluid mechanics of natural ventilation. *Annual Review of Fluid Mechanics*, 31, 201–238.
- Liu, D. D. S., & MacFarlane, R. (1983). Laminar burning velocities of hydrogen-air and hydrogen-air steam flames. *Combustion and Flame*, 49, 59–71.
- Lovachev, L. A., et al. (1973). Flammability limits: An invited review. *Combustion and Flame*, 20, 259–289.
- Lowesmith, B. J., Hankinson, G., Spataru, C., & Stobbart, M. (2009). Gas build-up in a domestic property following releases of methane/hydrogen mixtures. *International Journal of Hydrogen Energy*, 34, 5932–5939.
- Lowesmith, B. J., Hankinson, G., Spataru, C. & Stobbart, M. (2007). Gas build-up in a domestic property following release of methane/hydrogen mixtures. San, 2nd International Conference on Hydrogen Safety.
- Lutz, A. E., Kee, R. J. & Miller, J. A. (1988). SENKIN: A Fortran program for predicting homogeneous gas phase chemical kinetics with sensitivity analysis, Oak: OSTI-DOE.
- Maas, U., & Warnatz, J. (1988). Ignition processes in hydrogen-oxygen mixtures. *Combustion and Flame*, 74(1), 53–69. Available from [https://doi.org/10.1016/0010-2180\(88\)90086-7](https://doi.org/10.1016/0010-2180(88)90086-7).
- Maeda, S., Minami, S., Okamoto, D., & Obara, T. (2016). Visualization of deflagration-to-detonation transitions in a channel with repeated obstacles using a hydrogen-oxygen mixture. *Shock Waves*, 26, 573–586.
- Makarov, D., & Molkov, V. (2013). Plane hydrogen jets. *International Journal of Hydrogen Energy*, 38, 8068–8083.

- Marangon, A., & Carcassi, M. N. (2014). Hydrogen–methane mixtures: Dispersion and stratification studies. *International Journal of Hydrogen Energy*, 39, 6160–6168.
- Markstein, G. H. (1964). *Nonsteady flame propagation*. New York: Pergamon Press.
- Marshall, N., Liquide, A., & Scientifique, D. (1976). *Encyclopédie des gaz*. Amsterdam: Elsevier.
- Matalon, M. (2007). Intrinsic flame instabilities in premixed and nonpremixed combustion. *Annual Review of Fluid Mechanics*, 39, 163–191.
- Merilo, E. G., Groethe, M. A., Colton, J. D., & Chiba, S. (2011). Experimental study of hydrogen release accidents in a vehicle garage. *International Journal of Hydrogen Energy*, 36, 2436–2444.
- Meshkov, E. E. (1969). Instability of the interface of two gases accelerated by a shock wave. *Fluid Dynamics*, 4, 101–104.
- Meyer, J. W., & Oppenheim, A. K. (1971). On the shock-induced ignition of explosive gases. *Symposium (International) on Combustion*, 13, 1153–1164.
- Molkov, V. (2012). Fundamentals of Hydrogen Safety Engineering, Free download eBook, bookboon.com, <<https://bookboon.com/en/fundamentals-of-hydrogen-safety-engineering-i-ebook>>.
- Molkov, V. & Saffers, J.-B. (2011). The correlation for non-premixed hydrogen jet flame length in still air. University of Canterbury, New Zealand, International Association for Fire Safety Science, pp. 933–943.
- Molkov, V., & Saffers, J.-B. (2013). Hydrogen jet flames. *International Journal of Hydrogen Energy*, 38, 8141–8158.
- Molkov, V., Makarov, D., & Puttock, J. (2006). The nature and large eddy simulation of coherent deflagrations in a vented enclosure-atmosphere system. *Journal of Loss Prevention in the Process Industries*, 19, 121–129.
- Molkov, V. V., et al. (2004). Vented gaseous deflagrations with inertial vent covers: State-of-the-art and progress. *Process Safety Progress*, 23, 29–36.
- Molkov, V. V., Makarov, D. V. & Prost, C. (2005). On numerical simulation of liquefied and gaseous H₂ releases at large scales. Pisa, 1st International Conference on Hydrogen Safety.
- Morton, B. R., Taylor, G. I., & Turner, J. S. (1956). Turbulent gravitational convection from maintained and instantaneous sources. *Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences*, 234, 1–23.
- Mueller, M. A., Kim, T. J., Yetter, R. A., & Dryer, F. L. (1999). Flow reactor studies and kinetic modeling of the H₂/O₂ reaction. *International Journal of Chemical Kinetics*, 31(2), 113–125. Available from [https://doi.org/10.1002/\(SICI\)1097-4601\(1999\)31:2<113::AID-KIN5>3.0.CO;2-0](https://doi.org/10.1002/(SICI)1097-4601(1999)31:2<113::AID-KIN5>3.0.CO;2-0).
- Neumann, J. (1942). In A. J. Taub (Ed.), *Theory of detonation waves*. New York: Macmillan, John von Neumann, Collected Works.
- Oppenheim, A. K., Manson, N., & Wagner, H. G. (1963). Recent progress in detonation research. *AIAA Journal*, 1(10), 2243.
- Oran, E. S., & Gamezo, V. N. (2007). Origins of the deflagration-to-detonation transition in gas-phase combustion. *Combustion and Flame*, 148, 4–47.
- Oriani, R. A., Hirth, J. P., & Śmiałowski, M. (1985). *Hydrogen degradation of ferrous alloys*. Park Ridge, NJ: Noyes Publications.
- Pasquill, F., & Smith, F. B. (1983). *Atmospheric diffusion* (3rd ed.). New York: John Wiley & Sons, Ltd.
- Peters, N. (1988). *Laminar flamelet concepts in turbulent combustion* (pp. 1231–1250). Washington: Combustion Institute—Elsevier.
- Peters, N. (1991). Length scales in laminar and turbulent flames. *Progress in Astronautics and Aeronautics*, 135, 155–182.
- Peters, N. (1999). The turbulent burning velocity for large-scale and small-scale turbulence. *Journal of Fluid Mechanics*, 384, 107–132.
- Peterson, P. F. (1994). Scaling and analysis of mixing in large stratified volumes. *International Journal of Heat and Mass Transfer*, 37, 97–106.

- Poinsot, T., Veynante, D., & Candel, S. (1991). Diagrams of premixed turbulent combustion based on direct simulation. *Symposium (International) on Combustion*, 23, 613–619.
- Powell, F. (1984). Ignition of gases and vapours by hot surfaces and particles—A review. Lucerne, 9. International Symposium on the Prevention of Occupational Accidents and Diseases in the Chemical Industry.
- Proust, C., Carleton, F. B., Bothe, H. & Hawksorth, S. (2000). Prenormative research on the use of optics in potentially explosive atmospheres: Propex. Brussels: OPOCE, Directorate-General for Research and European Commission.
- Randerson, D. (1984). Atmospheric science and power production, USDOE Technical Information Center, Oak Ridge, United States, month = {7}, number = {DOE/TIC-27601 ON: DE84005}, <https://doi.org/10.2172/6503687>, <https://www.osti.gov/servlets/purl/6503687>.
- Rayleigh. (1882). Investigation of the character of the equilibrium of an incompressible heavy fluid of variable density. *Proceedings of the London Mathematical Society*, 1–14, 170–177.
- Richtmyer, R. D. (1960). Taylor instability in shock acceleration of compressible fluids. *Communications on Pure and Applied Mathematics*, 13, 297–319.
- Roper, F. G. (1977). The prediction of laminar jet diffusion flame sizes: Part I. Theoretical model. *Combustion and Flame*, 29, 219–226.
- Roper, F. G., Smith, C., & Cunningham, A. C. (1977). The prediction of laminar jet diffusion flame sizes: Part II. Experimental verification. *Combustion and Flame*, 29, 227–234.
- Royle, M., & Willoughby, D. (2011). The safety of the future hydrogen economy. *Process Safety and Environmental Protection*, 89, 452–462.
- Scase, M., Dalziel, S., & Hunt, J. (2006). Time-dependent plumes and jets with decreasing source strengths. *Journal of Fluid Mechanics*, 563, 443–461.
- Schatzmann, M. (1979). An integral model of plume rise. *Atmospheric Environment (1967)*, 1, Volume 13, 721–731.
- Schefer, R. W., et al. (2006). Characterization of leaks from compressed hydrogen dispensing systems and related components. *International Journal of Hydrogen Energy*, 31, 1247–1260.
- Shebeko, Y. N., et al. (1995). Burning velocities and flammability limits of gaseous mixtures at elevated temperatures and pressures. *Combustion and Flame*, 102, 427–437.
- Shentsov, V., Molkov, V. & Kuznetsov, M. (2015). The pressure peaking phenomenon: validation for unignited releases in laboratory-scale enclosure. Yokohama, 6th International Conference on Hydrogen Safety.
- Sivashinsky, G. I. (1983). Instabilities, pattern formation, and turbulence in flames. *Annual Review of Fluid Mechanics*, 15, 179–199.
- Solberg, D. M., Pappas, J. A. & Skramstad, E. (1980). Experimental investigations on flame acceleration and pressure rise phenomena in large scale vented gas explosions. Basle, Swiss Society of Chemical Industries, p. 16.
- Song, W., Du, J., Xu, Y., & Long, B. (1997). A study of hydrogen permeation in aluminum alloy treated by various oxidation processes. *Journal of Nuclear Materials*, 246, 139–143.
- Statharas, J. C., et al. (2000). Analysis of data from spilling experiments performed with liquid hydrogen. *Journal of Hazardous Materials*, 77, 57–75.
- Strehlow, R. A. (1969). Transverse waves in detonations: II. Structure and spacing in H₂-O₂, C₂H₂-O₂, C₂H₄-O₂ and CH₄-O₂ systems. *AIAA Journal*, 7, 492–496.
- Strehlow, R. A. (1970). Multi-dimensional wave structure. *Progress in Aeronautics and Astronautics*, 15, 345–357.
- Strehlow, R. A., Maurer, R. E., & Rajan, S. (1969). Transverse waves in detonations: I. Spacings in the hydrogen-oxygen system. *AIAA Journal*, 7, 323–328.
- Studer, E., et al. (2012). Interaction of a light gas stratified layer with an air jet coming from below: Large scale experiments and scaling issues. *Nuclear Engineering and Design*, 253, 406–412.

- Swain, M., Filoso, P., Grilliot, E., & Swain, M. (2003). Hydrogen leakage into simple geometric enclosures. *International Journal of Hydrogen Energy*, 2, 28, 229–248.
- Swift, I., & Epstein, M. (1987). Performance of low pressure explosion vents. *Plant/Operations Progress*, 6, 98–105.
- Szabó, T., et al. (2012). Parameterization of laminar burning velocity dependence on pressure and temperature in hydrogen/air/steam mixtures. *Combustion Science and Technology*, 184, 1427–1444.
- Takahashi, F., Mizomoto, M., & Ikai, S. (1983). Alternative energy sources III. *Nuclear Energy/Synthetic Fuels*, 5, 447–457.
- Tanford, C. (1947). Theory of burning velocity. I. Temperature and free radical concentrations near the flame front, relative importance of heat conduction and diffusion. *The Journal of Chemical Physics*, 15, 433–439.
- Tanford, C., & Pease, R. N. (1947). Theory of burning velocity. II. The square root law for burning velocity. *The Journal of Chemical Physics*, 15, 861–865.
- Taylor, G. I. (1950). The instability of liquid surfaces when accelerated in a direction perpendicular to their planes. I. *Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences*, 201, 192–196.
- Teodorczyk, A. (1992). Calculation of thermodynamic parameters of combustion products behind overdriven detonation wave. *Biuletyn Informacyjny ITC Politechniki Warszawskiej*, 76, 21–45.
- Teodorczyk, A., & Benoan, F. (1996). Interaction of detonation with inert gas zone. *Shock Waves*, 6, 211–223.
- Teodorczyk, A., & Thomas, G. (2003). Flame and detonation arresters—European Normative Standard for Testing. Pardubice, Department of Theory & Technology of Explosives, University of Pardubice, pp. 411–419.
- Teodorczyk, A., Lee, J. H. S., & Knystautas, R. (1991). The structure of fast turbulent flames in very rough, obstacle-filled channels. *Symposium (International) on Combustion*, 23, 735–741.
- Thiele, M., et al. (2002). Spark ignited hydrogen/air mixtures: Two dimensional detailed modeling and laser based diagnostics. *Combustion and Flame*, 1, 128, 74–87.
- Thomas, G. O., Sutton, P., & Edwards, D. H. (1991). The behavior of detonation waves at concentration gradients. *Combustion and Flame*, 84, 312–322.
- Tieszen, S. R., et al. (1986). Detonation cell size measurements in hydrogen-air-steam mixtures. *Progress in Astronautics Aeronautics*, 106, 205–219.
- Tse, S. D., Zhu, D. L., & Law, C. K. (2000). Morphology and burning rates of expanding spherical flames in H_2/O_2 /inert mixtures up to 60 atmospheres. *Proceedings of the Combustion Institute*, 28, 1793–1800.
- Venetsanos, A. G., et al. (2009). An inter-comparison exercise on the capabilities of CFD models to predict the short and long term distribution and mixing of hydrogen in a garage. *International Journal of Hydrogen Energy*, 34, 5912–5923.
- Verfondern, K., & Dienhart, B. (1997). Experimental and theoretical investigation of liquid hydrogen pool spreading and vaporization. *International Journal of Hydrogen Energy*, 22, 649–660.
- Verfondern, K., & Dienhart, B. (2007). Pool spreading and vaporization of liquid hydrogen. *International Journal of Hydrogen Energy*, 32, 2106–2117.
- Verhelst, S., & Sierens, R. (2007). A quasi-dimensional model for the power cycle of a hydrogen-fuelled ICE. *International Journal of Hydrogen Energy*, 32, 3545–3554.
- Veser, A., et al. (2011). The structure and flame propagation regimes in turbulent hydrogen jets. *International Journal of Hydrogen Energy*, 36, 2351–2359.
- Voitsekhovskii, B. V., Mitrofanov, V. V., & Topchiyan, M. E. (1969). Structure of the detonation front in gases (survey). *Combustion, Explosion and Shock Waves*, 5, 267–273.
- von Pidoll, U., Brzostek, E., & Fröchtenigt, H.-R. (2004). Determining the incendivity of electrostatic discharges without explosive gas atmospheres. *IEEE Transactions on Industry Applications*, 40, 1467–1475.

- Winters, W. S., & Houf, W. G. (2011). Simulation of small-scale releases from liquid hydrogen storage systems. *International Journal of Hydrogen Energy*, 36, 3913–3921.
- Witcofski, R. D., & Chirivella, J. E. (1984). Experimental and analytical analyses of the mechanisms governing the dispersion of flammable clouds formed by liquid hydrogen spills. *International Journal of Hydrogen Energy*, 9, 425–435.
- Worster, M. G., & Huppert, H. E. (1983). Time-dependent density profiles in a filling box. *Journal of Fluid Mechanics*, 132, 457–466.
- Xiukui, S., Jian, X., & Yiyi, L. (1989). Hydrogen permeation behaviour in austenitic stainless steels. *Materials Science and Engineering: A*, 114, 179–187.
- Yamanishi, Y., Tanabe, T., & Imoto, S. (1983). Hydrogen permeation and diffusion through pure Fe, pure Ni and Fe–Ni alloys. *Transactions of the Japan Institute of Metals*, 24, 49–58.
- Yáñez, J., Kotchourko A., Kuznetsov M., Lelyakin A., Jordan T. (2011). Modeling of the flame acceleration in flat layer for hydrogen-air mixtures. San Francisco, USA, 4th International Conference on Hydrogen Safety.
- Zeldovich, I. A. B., Barenblatt, G. I., Librovich, V. B., & Makhviladze, G. M. (1985). *Mathematical theory of combustion and explosions*. New York, NY: Consultants Bureau.
- Zel'dovich, I. B. (1940). On the theory of the propagation of detonation in gaseous systems. *Zh. Eksp. Teor. Fiz.*, 10, 542–568.
- Zel'dovich, I. B., Kogarko, S. M., & Simonov, N. N. (1956). An experimental investigation of spherical detonation of gases. *Soviet Physics. Technical Physics*, 1, 1689–1713.
- Zeldovich, Y. B. (1951). Theory of flame propagation—Technical Memorandum 1282. Washington, DC, United States: National Advisory Committee for Aeronautics.
- Zeldovich, Y. B., Librovitch, V. B., Makhviladze, G. M., & Sivashinsky, G. I. (1970). On the development of detonation in a non-uniformly preheated gas. *Acta Astronautica*, 15, 313–321.
- Zhang, X., & Ghoniem, A. F. (1993). A computational model for the rise and dispersion of wind-blown, buoyancy-driven plumes I. Neutrally stratified atmosphere. *Atmospheric Environment*, 27, 2295–2311.

Accident consequences

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4.1 Accident initiators

The development of the hydrogen economy introduces the use of hydrogen in a range of different physical conditions, containment types, and operating environments, all of which present their own individual hazard possibilities. At one extreme, these include a liquid at 20 K, with risks to most materials coming into contact, and at the other extreme, the generation of heat at over 2000 K. Other storage conditions involve high pressures of the order of 70 MPa, introducing the possibility of direct blast effects, through shocks or fragments, or overpressures following ignition of turbulent combustible mixtures.

Therefore essential to the consideration of accident consequences is the estimation of hazards and hazard levels, for example, overpressures, thermal radiation, and the estimation of the damage level or the vulnerability of the receiving objects. The chapter is therefore organized into two broad categories, that is, sources of harm and impacts of harm, and in the latter, consideration is given to both infrastructure and humans.

Open access incident reporting has become a key learning facility for the chemical industry over 60 years and more recently, specific hydrogen incident reporting has followed this good example with the availability of two significant reporting databases, “Hydrogen Tools” supported by the US Department of Energy and “HIAD,” supported by the European Commission {see HIAD [*Hydrogen Incident and Accident Database* (HIAD 2.0)], [HIAD Hydrogen Incidents and Accident Database, 2020](#)}} and H2Tools (Hydrogen Safety Panel, [Hydrogen Tools, 2020](#)).

A random selection of events reported in the HIAD and Hydrogen Tools databases is presented in [Table 4.1](#).

Reviewing the many incidents reported in the databases, it is clear that initiating events may or may not be directly due to the behavior of hydrogen, but a hydrogen hazard may arise through other indirect

Table 4.1 Small selection of H₂-related incidents from the databases.

Nature of incident	Date	Outcome	Cause
H ₂ release from compressor	May 1971, UK	Jet fire, explosion	Coupling failure
Electrolyzer leakage	September 2005, France	Explosion	Corrosion/fatigue/material failure
Release from hydrocracking reactor	January 1987, Asia	Explosion	Human error
Hydrogenation reactor leakage	April 1987, EU	Fire, explosion	Defective agitator
Hydro-finer heater rupture	January 1974, EU	Fire	Human error
Overtaken H ₂ road tanker	September 1976, UK	Fire	Human error
H ₂ leakage from fuel-feeding system for a space rocket engine	May 1991, Japan	Explosion	Repair management error
H ₂ leak from a cryogenic hydrogen skid	February 1987, United States	Unignited release	Leak from packing in a globe valve
Rupture disk operation on LH ₂ tank	January 1974, United States	BLEVE	Equipment failure, human error
H ₂ charging system	August 1986, United States	Near miss	Globe valve failure due to H ₂ embrittlement

causes such as human error or system equipment failure. Good design choices have evolved through the sharing of incident information and dedication to the study of suitable materials for hydrogen usage (e.g., [San Marchi & Somerday, 2012](#)) and certain types of initiating cause are therefore quite rare, such as Boiling Liquid Expanding Vapor Explosion (BLEVE) type failures and H_2 embrittlement failures.

Although these initiating events are widely varying in their nature, the outcomes occupy a more limited domain and are dominated by fire and blast overpressures. The following sections provide information on these and others along with supporting background information.

4.2 Overpressure generation

4.2.1 Physical explosions

Physical explosions will normally produce a pressure wave or shock wave, that is, the sudden violent expansion of a fluid not connected with a chemical reaction. The strongest man-made physical explosion is surely the nuclear explosion of an atomic bomb. The strongest explosion resulted in blast wave overpressures of 170 kPa in the central zone (14-km radius for the example of a 20 Mt explosion in 5.5-km height) and gradually decaying to the outside with still 20 kPa at 60-km distance.

With the development of hydrogen as an energy carrier in the transport sector, the possibilities for small scale, high pressure explosion events become a realistic scenario, given the typical maximum storage pressures of around 70 MPa being considered for current designs.

Another source of such a physical explosion is a bursting pressure vessel, which may result from a fire-induced BLEVE. The higher the liquid density, the more destructive is the BLEVE. If the liquid is flammable, a fireball will follow. Several pressure spikes are created during a BLEVE event: from the flashing liquid, from the expanding vapor phase, and, if applicable, from the combustion. Missiles and projectiles may also result from a physical explosion and may cause injuries, fatalities, or damage at considerable distances depending on the explosion energy liberated. Projectile hazard increases with average liquid temperature and associated vapor pressure.

Another example of a physical explosion is the so-called rapid phase transition (RPT), a thermal vapor explosion resulting from the spontaneous phase change of a fluid getting in contact with a much hotter or colder liquid, for example, a cryogen spilled onto water. Prerequisites of such an explosive boiling are, a temperature of the “hot” fluid above the boiling point of the “cold” fluid and, a certain mixing of both fluids allowing a close and sufficiently long contact. Although the energy release is small compared with a chemical explosion, fragmentation and phase change of the “cold” fluid (vapor evolution) can occur at such a high rate that shock waves may be formed. For liquified natural gas (LNG) onto a water surface, overpressures with damaging potential of up to 5 kPa were observed. RPT explosions with different materials (molten metal plus water) in the metal and chemical industries were even the cause for people killed by flying melt or the blast wave. Natural examples of RPT were the catastrophic explosions of the island volcanoes Krakatoa, Indonesia, in 1883 and Surtsey, Iceland, in 1963. Although much of the study of RPT events has been with LNG, liquid hydrogen presents the same range of phase change overpressure and damage possibilities.

For gaseous storage, hydrogen can be stored in four types of pressure vessels as presented in [Fig. 4.1](#), reproduced from [Dolci \(2016\)](#). The pressure vessels are generally cylinders, but they can also be polygons or toroid shaped. Metallic pressure vessels are known as Type I. Type II pressure vessels

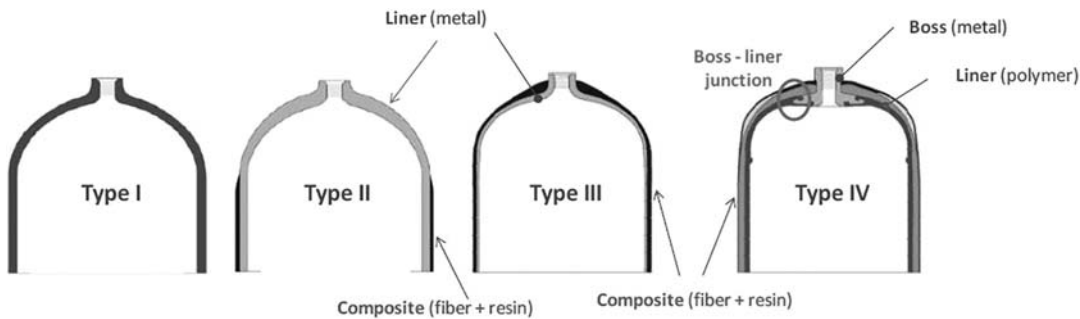


FIGURE 4.1

Present vessel storage types for gaseous hydrogen.

Table 4.2 Capability limits applying to hydrogen storage vessels.	
	Technology capabilities
Type I	Pressure limited to 50 MPa
Type II	Pressure not limited
Type III	For $P < 45$ MPa
Type IV	For $P < 100$ MPa

consist in a thick metallic liner hoop wrapped on the cylindrical part with a fiber resin composite. The fully composite materials-based pressure vessels (designated by COPV) are made of a plastic or metallic liner wrapped with carbon fibers embedded in a polymer matrix (filament winding). When the liner contributes to the mechanical resistance (more than 5%), the COPV is of Type III (mostly metal liner). Otherwise, the COPV is of Type IV (mainly polymer liner or seldom extremely thin metal liner).

The operational range of these vessels is shown in Table 4.2.

The behavior of these vessels under failure conditions, such as fire impingement, has been and continue to be the subject of safety studies around the world and the following examples described by Kim, Volodymyr, Makarov, and Molkov (2015) provide data on the effects of a relatively small vessel failing at high pressure.

Fire exposure tests of Types III and IV hydrogen fuel tanks without pressure relief devices were conducted by Weyandt (2005, 2006). The stand-alone Type IV tank had a length of 0.84 m, diameter 0.41 m, 72.4 L volume, 35 MPa storage pressure, and was installed 0.20 m above ground. The tank ruptured catastrophically after 6 min, 27 s of exposure to a propane bonfire. Experimental measurements demonstrated the blast pressures of 300 kPa, 83 kPa, and 41 kPa at 1.9 m, 4.2 m, and 6.5 m, respectively, and the maximum diameter of the fireball about 7.7 m. Tank fragment projectiles were found at distances of 34 – 82 m. In the test of the Type III tank, which was installed under a car, the blast wave overpressure ranged from 140 kPa at 1.2 m to 12 kPa at 15 m and the maximum fireball diameter was 24 m. The projectiles were discovered at distances up to 107 m. Such experimental results indicate that tank rupture may have devastating consequences, causing damage to life and property at substantial distances from the tank location.

Present computer codes now have the capability to model the pressure waves from such ruptures with good accuracy as demonstrated by the calculated results from Kim et al. (2015) using the experimental test results from the above Weyandt work.

Examples of the outputs from this modeling exercise are shown in Figs. 4.2 and 4.3.

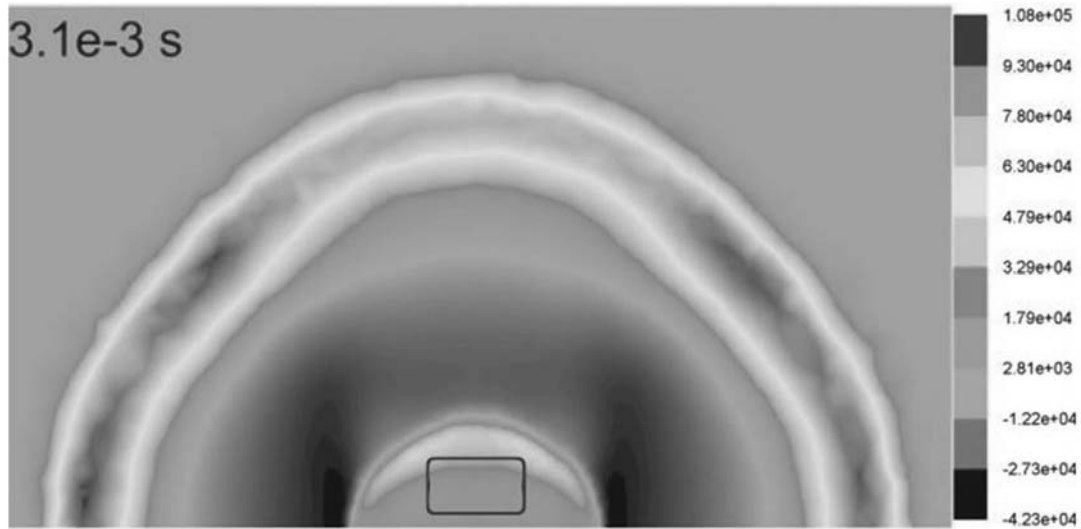


FIGURE 4.2

Computed blast wave amplitude on hydrogen vessel rupture at 3.1 msec. Numerical values are Pa. Tank volume = 72 L at 35 MPa pressure.

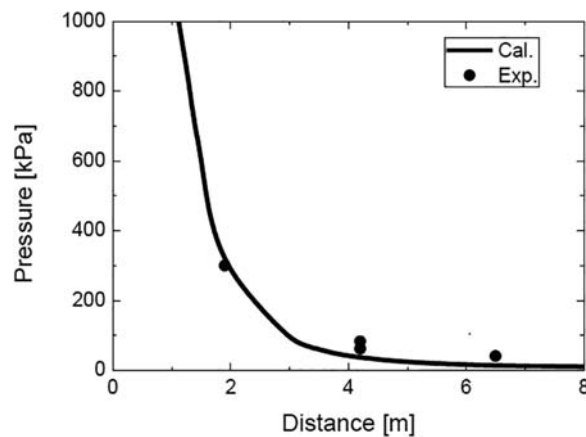


FIGURE 4.3

Computed blast leading edge pressure (Cal) compared with measured values (Exp) versus distance from the vessel.

4.2.2 Boiling liquid expanding vapor explosion

Concerning the catastrophic vessel failure and release of liquefied content giving rise to a BLEVE, while the possibilities of this event are less likely for hydrogen storage than for hydrocarbons, due to the double skin nature of the storage tanks, the physical phenomenon can be described in the same way, and the outcomes therefore are similar and cannot be ignored when considering accident scenarios.

The explosion energy in case of a BLEVE has been assessed by the difference between final and initial state of the bursting vessel assuming isentropic expansion. This plus a certain portion of the bursting pressure energy can potentially contribute to the blast wave generation (Canadian Society for Chemical Engineering, 2004). Without further ignition of the cloud being considered in this scenario, this would be classed as a cold BLEVE.

Models for BLEVE blast prediction are described in the Centre for Chemical Process Safety (CCPS) guidelines (AIChE CCPS, 1994) and the so-called “Yellow Book” of Netherlands Organisation for Applied Scientific Research (TNO) (CPR14E:2005:YellowBook). These references also describe models to predict the magnitude and duration of the fireball, which often follows a BLEVE as described later in this section.

A development from the guidelines has been a numerical method to calculate the blast effects originating from an exploding vessel of liquefied gas (van den Berg, van der Voort, Weerheijm, & Versloot, 2004). Adequate blast calculation requires full knowledge of the blast source characteristics, that is, the release and subsequent vaporization rate of the flashing liquid. As the conditions that allow explosive vaporization are not entirely clear and the vaporization rate of a flashing liquid is unknown, safe assumptions have often been adopted as the starting point in the modeling. The blast effects from a BLEVE are numerically computed by imposing the vapor pressure of a flashing liquid as boundary condition for the gas dynamics of expansion. The modeling shows that the rupture of a pressure vessel containing a liquefied gas in free space only develops a blast of significant strength if the vessel near-instantaneously disintegrates.

All of the experimental data relate to studies on hydrocarbons, ammonia, and carbon dioxide and these have provided some insights into the main source of blast overpressure to be expected. The work of Birk (Birk, Davison, & Cunningham, 2007) has shown that “cold BLEVE” events using a tank of isopentane results in blast pressure in the near field (20 m) of around 10–20 kPa and that this originated from the pressurized vapor volume of the tank. For unconfined tank ruptures, there was little evidence that the subsequent flashing of the liquid contents could happen quickly enough to generate a blast wave effect. For confined ruptures, however, it was considered that the liquid phase change could contribute to the overall blast effect and given the confined nature of many storage situations, this may need to be considered.

It is noted that the evolution of this blast pressure with distance can be modeled using established pressure decay models as discussed below using energy scaled distance.

Van den Berg (2008) has made similar assessments but has taken a more conservative approach by assuming that the liquid can flash explosively to an extent governed by its degree of superheat and evaporation requirements. The conditions for this outcome have been assumed to be that the liquid temperature is above its homogeneous nucleation temperature, in which case evaporation is homogeneous throughout the liquid following depressurization. This behavior should also be true for liquid hydrogen.

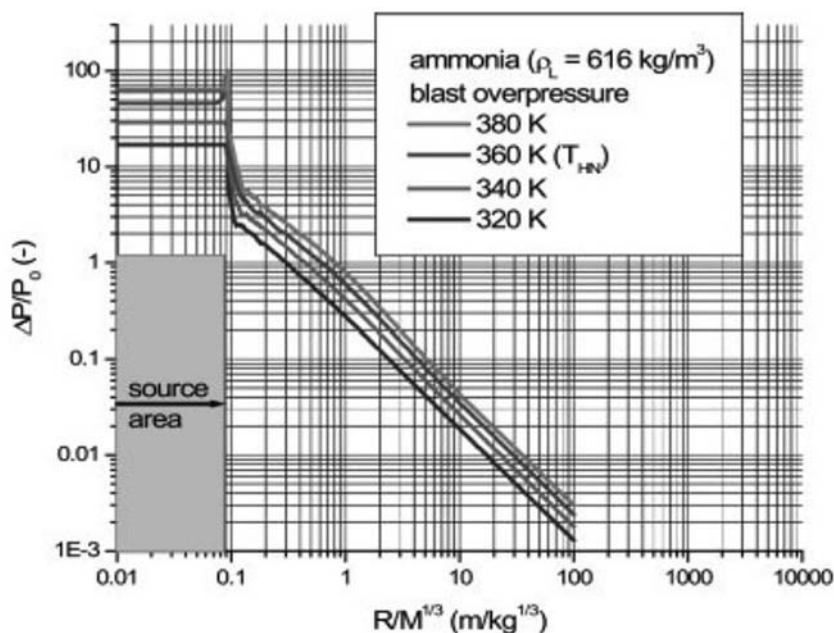


FIGURE 4.4

Blast pressure produced by BLEVE event based on flash evaporation only.

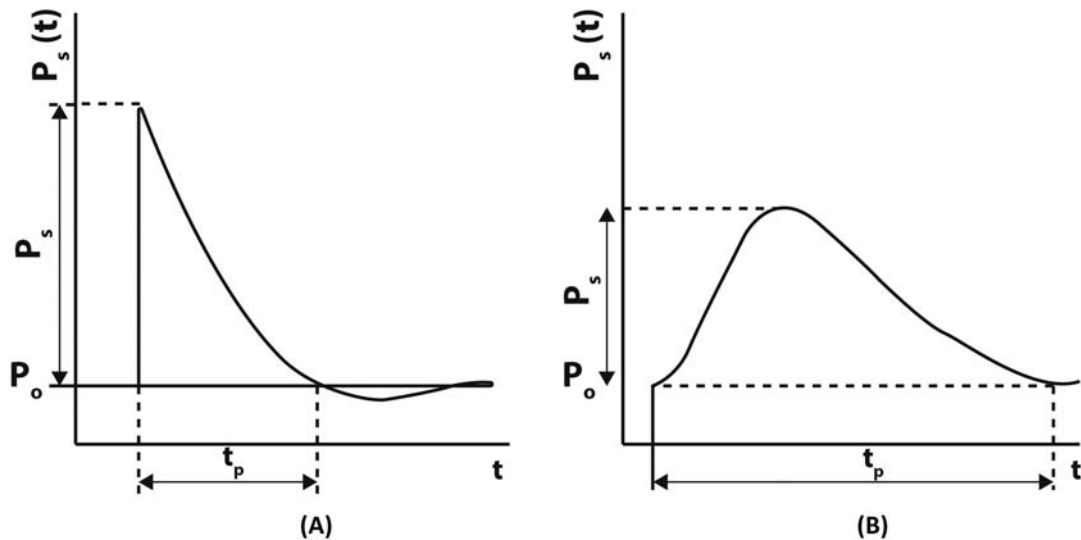
From Van den Berg A.C. (2008). Blast charts for explosive evaporation of superheated liquids. Process Safety Progress, 27(3), 219–224.

The study performs a numerical analysis of the development of this blast wave with distance and presents the results as normalized pressure changes with the distance parameter $R/M^{1/3}$; M is the evaporated liquid mass. An example of the kind of output produced by this kind of analysis is shown in Fig. 4.4 for the case of stored liquid ammonia. Analyses of this kind should also be possible for the hydrogen case.

4.2.3 Chemical explosions

In chemical explosions, which are usually exothermic oxidation reactions, a great portion of the combustion energy is carried by the developing blast wave, which is uniformly distributed in all directions. Depending on the various types of combustion processes (slow deflagration or fast turbulent flame or detonation), the pressure history will be different. It is characterized by the peak overpressure and the pressure increase rate and decay rate. This effect is strongest at ground level (hemispherical) explosions where due to reflection the respective yield ratio can be twice as high as for a spherical explosion.

The rate at which the energy release occurs will influence the shape of the pressure–time profile and also the peak pressure, due to the local power density of the combustion and its influence

**FIGURE 4.5**

Characteristic shape of pressure – time function for a detonation shock wave (left) and a deflagration pressure wave (right) (P_o , initial pressure; P_s , peak side-on overpressure; t_p , duration of positive phase).

on the local expanding pressure wave. Pressure waves originating within a particular combustible region will propagate away from the source and, due to the heating of the transport medium as the wave passes through it, the pressure waves behind the leading edge will tend to travel faster and “catch up” with the leading edge resulting in a sharpening of the pressure wave into more of a shock wave event.

In the limit of fast combustion, the combustion is coupled to a high intensity shock wave and both travel at supersonic speeds. Hydrogen combustion is more susceptible to such detonation phenomena than most other combustible gases and presents a special hazard for this reason, as the shock pressures that result can easily exceed the design limits of containment equipment. Fig. 4.5 shows the difference in pressure – time profiles for the two contrasting combustion events described above.

4.2.3.1 Confined combustion

In a deflagration with flame speeds of 1–10 m/s, the volume expansion of the gas acts like a piston displacing the unburnt gas. The deflagration pressure wave in a confined space is characterized by a slow increase of pressure and fluid velocity in the region preceding the flame front. The pressure in the vessel is independent of the location and mainly determined by the fraction of burnt gas. The static pressure loading in slow deflagration processes is described by the “adiabatic, isochoric, complete combustion” (AICC) pressure representing an upper bound in a confined space. A mitigation of the AICC pressure is given by incomplete combustion, venting, radiation/conduction heat losses, or the addition of diluents. Therefore the maximum static pressure will be generally lower than the AICC pressure. On the other hand, initial turbulence increases the rate of combustion and thus can

increase peak pressures due to faster heat generation compared to the heat loss processes. The peak pressure in a closed vessel for most hydrocarbon mixtures is in the order of 0.8 MPa, sufficient for many buildings to exceed their failure limits. For a hydrocarbon – oxygen mixture, it is even 1.6 MPa. A hydrogen – air mixture, initially at normal temperature and pressure (NTP), will reach a pressure of 0.815 MPa and its volume will increase by a factor of 6.89 (Baker, Cox, Westine, Kulesz, & Strehlow, 1983).

The pressure build-up depends on the flame propagation and also the degree of confinement. Particularly hazardous configurations are those, which are heavily confined like tubes, pipes, or channels, where—if long enough—even in insensitive methane – air mixtures, high flame speeds and pressures can be reached. Venting can reduce the pressure. Inside a spherical vessel, the pressure rise following the ignition of a flammable mixture is proportional to the cube of the burning velocity. For stoichiometric mixtures flames speed will reach 600–700 m/s, which will result in a pressure rise of few bar in the leading shock. Comparing explosion tests in tubes and in spherical vessels, it was observed that pressures are generally lower in a spherical propagation of the gas mixture (unconfined) than in a planar propagation. The pressure behind the flame front is decaying away from the flame, as wave energy dissipates.

4.2.3.2 Fast deflagration

In the intermediate stage of a fast deflagration with the flame front still traveling at subsonic speed, a preceding shock wave is developing in the still unburnt mixture. Although the peak overpressure is normally lower than for a detonation event, the pressure decay takes place over a longer period of time. This means that the impulse, that is, the integral of pressure over time, which is a measure for the load upon a structure, may be similar in both cases. The peak overpressure increases with increasing flame speed and transient pressures can be locally higher than the AICC pressure.

4.2.3.3 Detonation

In contrast, the detonation is a combustion mode with the flame traveling at supersonic speeds in the order of 2000 m/s. The flame front proceeds by shock wave compression of the unburnt gas. It is characterized by a distinct pressure spike and a subsequent almost exponential decrease. The shock wave, which is at the same time coupled to the flame front, is followed by the reaction zone, in which a pressure discontinuity is observed where the pressure even drops to values lower than atmospheric pressure (“molecular collapse”) due to the much denser oxidation product (water) upon hydrogen combustion. The essential parameters are peak overpressure and positive/negative phase of the specific impulse depending on the liberated explosion energy. The combustion process is completed without an expansion of the gas cloud. Peak overpressures in the near field are typically in the range of 1.5–2 MPa. The pressure wave gradually decays and eventually turns into an acoustic wave.

In geometries that allow the transition from deflagration to detonation, pressures near the location where detonation takes place may be much higher than the Chapman – Jouguet (CJ) pressure of a stabilized (and idealized) detonation wave, which is due to a precompression effect by the propagating shock wave (van Wingerden, 1999). In pipes with no obstacles, the transition distance increases with increasing diameter (e.g., 8 m for propane – air mixture in a 50-mm diameter pipe) (Moen, 1993).

In confined spaces, peak pressures can range between “normal” deflagration peak pressure and very high pressures following deflagration to detonation transition (DDT). The worst case is considered to be the DDT on a reflected shock wave produced by a fast flame with an estimated peak pressure to be by a factor of 10 higher than the detonation pressure. The transfer of a detonation wave into adjacent mixtures is possible and has been observed for planar clouds, whereas in spherical clouds, fast deflagrations are more likely to occur.

It should be noted that an explosion in a vessel that is connected by a small opening to another vessel creates a peak overpressure and a pressure increase rate much higher than in a single vessel explosion, a phenomenon known as “pressure piling.” A pressure of more than 3.5 MPa was measured in a two-chamber geometry for a stoichiometric hydrocarbon – air mixture, where 0.8 MPa were expected for the explosion in a single vessel. Unlike the length of the interconnecting tube, its diameter is pertinent for the peak overpressure.

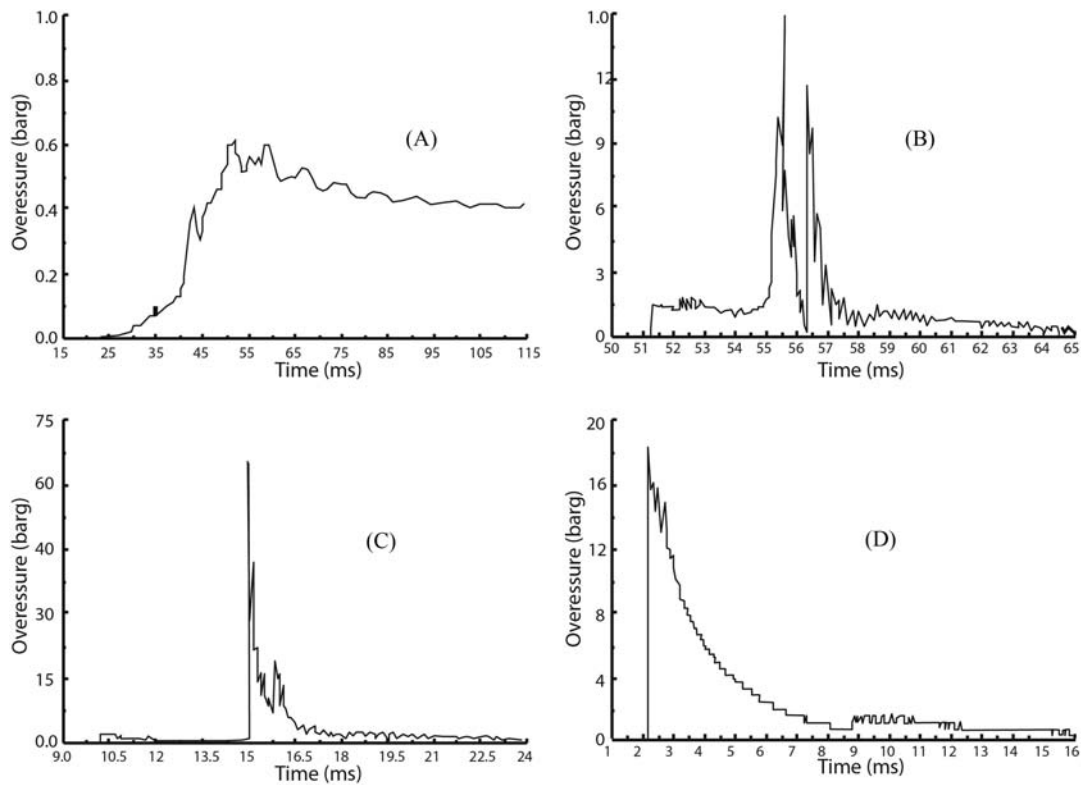
A good example of the progression of pressure profiles resulting from a deflagration to detonation is provided by the combustion tube experiments of Teodorczyk (Teodorczyk & Thomas, 2003). The phases during an accelerating explosion event in a tube, initiated by a low energy ignition source, may be categorized in general terms as follows:

1. Slow deflagration (SDEF).
2. Fast deflagration (FDEF).
3. Overdriven or transition to detonation (DDT).
4. Steady CJ detonation (CJDET).

Examples of pressure histories corresponding to each of these phases are shown in Fig. 4.6A – D, obtained during flame acceleration tests. Fig. 4.6A shows a slow deflagration, which exhibits a gradual pressure rise with a maximum over pressure below 1 bar. Fig. 4.6B is a pressure record from a fast deflagration test where a precursor shock has formed. Fig. 4.6C shows the pressure histories typical of an overdriven detonation immediately following a transition to detonation event. The peak pressures are several times greater than the pressures observed for a steady detonation wave, shown in Fig. 4.6D.

4.2.3.4 Vapor cloud explosion

The possible outcome of a vapor cloud explosion (VCE) is an event described as a deflagration (either slow or fast) or in the limit, a detonation. These differ dramatically in peak overpressure, in the duration of the impulse (time-integrated pressure), in the steepness of the wave front, and in the decrease of overpressure with propagation distance. Secondary blast wave parameters are the peak reflected pressure, peak dynamic (blast wind) pressure, shock front velocity, and blast wave length. The burning rate of the combustible mixture will determine the pressure outcome of the event and for hydrogen this burning rate will be initially determined by the stoichiometry of the fuel/air mixture and the turbulence level within the combustible mixture into which the flame is passing. The expanding gas cloud will induce forward motion in the mixture ahead of the flame front and when obstacles are present, then this will produce turbulence ahead of the flame. This in turn will increase flame speed on its arrival due to the increase in heat and mass transfer area available, and the resulting acceleration will intensify the process. Flame speeds in hydrogen can increase by two orders of magnitude by this mechanism and this is reflected in the rate of local pressure rise and consequent peak pressure. Such

**FIGURE 4.6**

Progression of pressure profiles for flame propagation in a tube under different test conditions showing different outcomes.

pressure may transmit without significant reduction if the environment has a degree of confinement and a high pressure arising at one part of a system may arrive elsewhere with the same amplitude.

For nonpremixed hydrogen – air mixtures, combustion of an unconfined VCE typically liberates only a fraction of its thermal energy content, in many cases less than 10% (Lind, 1975). This is the result of an energetic dilution process during the release, which is often from high pressure, and resulting in a large fraction of the released hydrogen being driven below its lower flammable limit (LFL). Depending on the combustion mode (deflagration/detonation), the explosion is connected with a more or less destructive pressure shock wave.

Inhomogeneities can result in local detonations decaying to deflagrations. When the shock wave leaves the cloud, it turns into an expanding decaying wave. In the long-distance range, the pressure wave for both deflagration and detonation exhibit about the same shape decaying with $1/r$. Local explosions like from jet flames result in locally high pressures and can also result in high flame speed in less confined areas and may even trigger a detonation wave.

4.2.3.5 Real gas cloud behavior

In reality, a gas cloud shows the typically expected features of a nonpremixed, inhomogeneous concentration distribution, air entrainment at the boundaries, and stratification if evolving from a pool of liquefied gas. Furthermore in case of an explosion, a real gas cloud is not an “ideal” explosion source due to a larger-than-infinitesimal volume and a lower energy density and energy deposition rate, thus leading to nonideal blast waves. Deviations from the ideal situation are able to either enhance or attenuate the pressure build-up. Nonstoichiometry as well as ignition at the cloud edge will certainly have a damping effect on the pressure build-up. The maximum blast impulse, which becomes larger with increasing shock duration, is not near the explosion center, but about 13–15 charge radii. A near-ground flat long-stretched cloud of heavy gases or vaporized cryogenics may experience multipoint ignition connected with a sequence of pressure peaks, and more turbulence-generating terrain roughness or obstacles in the flow path, both effects of which lead to an enhancement of the pressure build-up.

Unlike a heavy gas cloud, which would be of a pancake form, a hydrogen vapor cloud would soon cover an area, which is larger than that of a hemispherical cloud with the same explosive inventory. Only in case of just vaporized LH_2 after a large-scale spill, the cold gas cloud would travel and stretch near ground, until sufficient air has entrained from the outside to make the gas positively buoyant and develop soon to a vertically stretched cloud shape.

The flame spreading in a nonspherical cloud is spherical until it reaches the cloud edge at some point; then it continues in the direction, where still gas can be found. The pressure is decreasing immediately behind the flame front because of the upward expansion of the combustion products.

4.2.3.6 Documented release incidents

There have been differing views regarding the likelihood of VCEs with hydrogen due to the assumption that ignition will occur very close to the moment of release and therefore avoid the build-up of a flammable premixed cloud with delayed ignition, which is necessary for the generation overpressure. The evidence, however, does not support this view as the abstract from the report by [Thomas, Eastwood, and Goodrich \(2015\)](#) confirms.

[Zalosh and Short \(1978\)](#) reviewed over 400 accidents involving hydrogen from a variety of sources covering the time period from 1965 to 1977. The referenced report provides details only for selected accidents. Zalosh and Short concluded that the data “indicate that hydrogen explosions have been a more serious problem than other types of hydrogen accidents in terms of the number of incidents, casualties, and reported property damage.” Their analysis showed that slightly more than half the hydrogen incidents were explosions, and that explosions accounted for three-quarters of the injuries and fatalities. Three-quarters of the incidents reviewed involved hydrogen gas, with most of the remaining incidents involving liquid. Hence, the analysis of Zalosh and Short would indicate that a significant fraction of the incidents involved explosions of hydrogen gas, and that explosions caused a disproportionally large fraction of the casualties.

[Table 4.3](#) summarizes some of the VCE incidents referenced in the Thomas report.

Apart from those releases with ignitions in unconfined, unobstructed areas, where the overpressures generated are small, the predictions of such pressures are difficult and originate from the complexities of the fluid flow behavior around the system geometry.

Table 4.3 Highlights of some of the more well-known incidents involving VCEs over the last 50 years with varying degrees of pressure generation.

Incident	Date	Congestion	Confinement	Quantity involved		Outcome
				Gas mass (kg)	Gas volume (m ³)	
Polysar Sarnia	1984	Mixed (low-medium)	Partial	30 kg	1050	Detonation
Jackass Flats	1964	None	None	90 (10% of release)	—	Deflagration
Silver Eagle Refinery	2009	Mixed (low-medium)	None	43	—	Deflagration
Muskingum River	2007	Very low	Partial	—	—	Deflagration
H ₂ Incidents database	2013	Varies	Varies	—	—	—
NASA H ₂ Incidents	1974	Varies	Varies	—	—	Some detonation
Hanau, Germany	1991	Mixed (low-medium)	None	400	4500	Deflagration
Sodegaura, Japan	1992	Mixed (low-medium)	None	—	—	Deflagration

Computational fluid dynamics (CFD) codes have much improved in recent years and are likely to provide the best predictive tool for the future in analyzing complex plant layouts.

4.2.3.7 Experimental work

Apart from the experience obtained by observations and lessons learned from explosion accidents, numerous experiments have been performed worldwide to investigate the transient behavior of overpressures following the explosive combustion of fuel – air mixtures. Tests were conducted under various conditions such as confined, partially confined, or unconfined, larger-scale or smaller-scale geometry, fuel type and constitution with the main goal of development of or comparison with simulation approaches. The most dangerous configurations were found to be, as expected, those with a major obstruction, even for less sensitive fuel gases such as methane. For DDT cases, traveling distance for the flame must be sufficiently long, which would be around 3 m for a stoichiometric hydrogen – air mixture. Only a few of those test series are mentioned in the following showing the broad range of activities.

In order to demonstrate the differences that fuel reactivity and turbulence level make to the overpressures, which can result from a gas explosion, Table 4.4 is presented for recent work (Hawksworth et al., 2016) on the overpressure developed within a 600-mm diameter, 12-m long duct, which is fed with hot air traveling at 21 m/s toward an open outlet and with different fuels. The fuel/air ratio is controlled by the rate of fuel addition and this is well-mixed before arriving at an ignition point 2 m downstream. Two geometries are shown, one with no obstacles and the other with 15 rows of vertical tubing positioned halfway along the duct. The table shows the fuel injection rates and the combustion energy injection rates, which these correspond to. With no obstacles,

Table 4.4 Comparison of peak pressure developed with the ignition of hydrogen/air and methane/air, and the effect of turbulence generation.

Fuel injected	Fuel mass flow rate (kg/s)	Energy injection rate (MJ/s)	Duct obstacles	Peak pressure developed (kPa)
Methane	0.142	7.1	None	21.6
Hydrogen	0.0297	3.56	None	13.0
Methane	0.105	5.25	15 rows	65
Hydrogen	0.030	3.60	15 rows	173

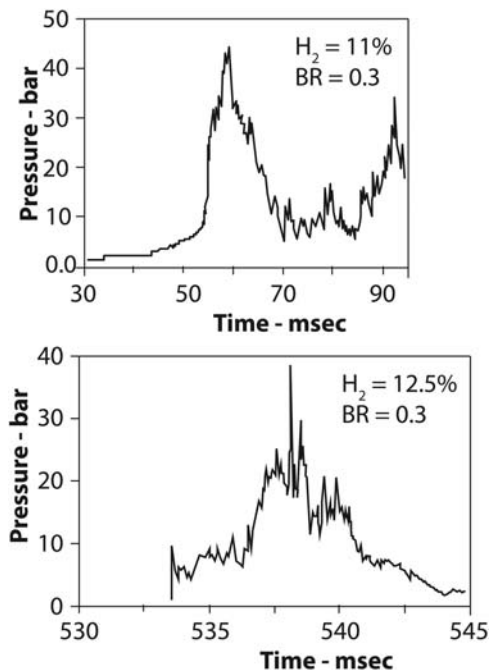
a modest 0.1–0.2 bar overpressure is developed. When obstacles are present, then even for a reduced energy injection rate for methane, the peak pressure is higher than the unobstructed case, showing the effect of turbulence on flame speed for this species. For the hydrogen case, where the energy injection rates are almost identical, the effect is even more dramatic, showing that the turbulence generation has increased the peak pressure by a factor of 13, due to its effect on the flame speed within and downstream of the obstructions.

A further example of the sensitivity of explosion overpressures to geometry and hydrogen concentration is shown in 7. Large-scale experiments have been conducted by the Russian Kurchatov Institute using premixed hydrogen – air mixtures. The RUT facility, with a confined volume of 480 m³, was employed for a series of tests ranging from slow deflagration to detonation. Hydrogen concentrations varied between 10% and 14% by volume. During slow deflagration (no obstacles present), the overpressures measured increased with H₂ concentration, from around 0.1 MPa to 0.17–0.23 MPa. Insertion of obstacles [blockage ratio (BR) of 30% and 60%] resulted in accelerated flames creating overpressures of 1.1–1.6 MPa for gas mixtures with 14% H₂ concentration. There was even the observation of a detonation at an H₂ concentration as low as 12.5% (Breitung & Kotchourko, 1996) (Fig. 4.7).

The Russian RUT facility, using a tube of 66 mm diameter and a maximum length of 3 m, was used for systematic studies with gas mixtures of hydrogen, oxygen, and nitrogen. Peak overpressures and impulses were investigated as a function of the location of DDT, which influences the degree of precombustion. Peak pressures observed were well above the CJ pressures for detonation of the undisturbed mixture (Breitung & Kotchourko, 1996).

Large-scale testing on DDT in hydrogen – air mixtures was conducted in the FLAME facility (“Flame Acceleration Measurements and Experiments”), a 30.5-m long, 2.44-m high, and 1.83-m wide rectangular channel with a closed ignition end and an open far end and with venting/obstruction possibilities between fully closed and 50% open. Tests were carried out with variation of H₂ concentrations (12–30 vol%), obstacle BR (0%–33%), and top venting. The observed lower detonability limits were at 15 vol% of H₂ with obstacles and 25 vol% of H₂ without obstacles. DDT also occurred at a 24.8 vol% H₂ concentration with a 13% open top. Top venting was found to have two counteracting effects: increase of burning rate due to turbulence dominant at low venting rates, and flame speed reduction due to loss of gas for large venting rates (Yang, Musicki, & Nimmual, 1991).

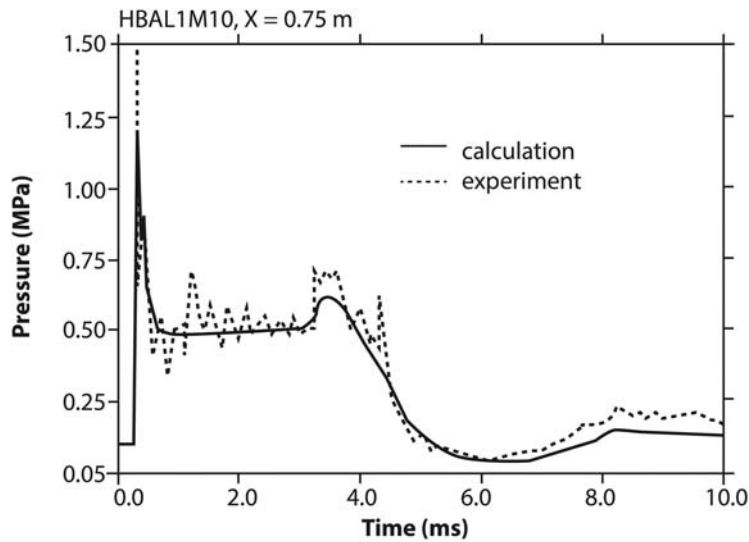
An explosion tube of 2.5 m diameter and 10 m length with one open end was used in Norway to study peak overpressures of ignited stoichiometric propane – air mixtures. The tests have shown

**FIGURE 4.7**

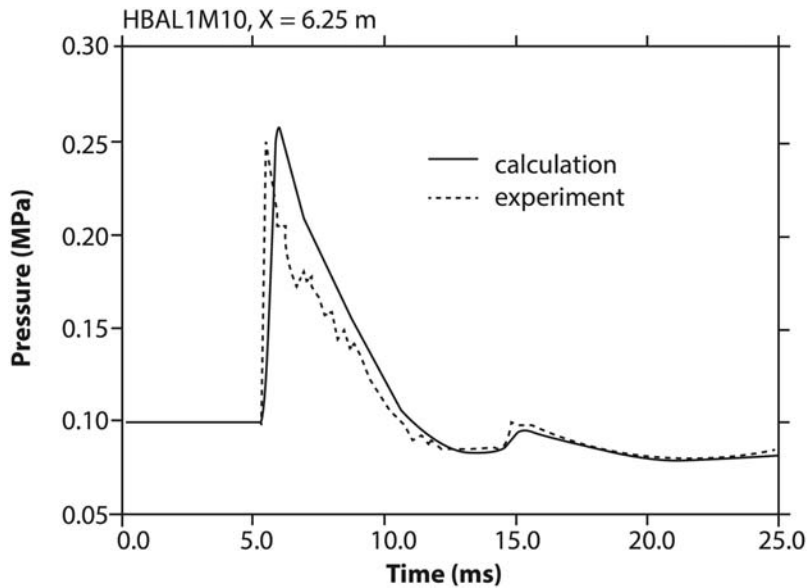
Explosion overpressures from RUT tests showing effects of blockage ratios and hydrogen premixed concentrations, showing slow deflagration (top left), fast deflagration (top right), detonation (bottom).

the significant influence of the BR inside the tube on the flame speed and pressure increase, respectively, which can come close to the detonation range (GexCon, 2006). Smaller-scale detonation test tube experiments have been conducted at the Karlsruhe Institute of Technology, the Technical University of Munich, the DLR in Stuttgart or the High-temperature Combustion Facility, HTCF, at Brookhaven National Laboratory employing different types of obstruction and differently diluted hydrogen – air mixtures to study flame acceleration and various DDT mechanisms.

Within the nuclear power plant safety program and the PNP gas cloud program, the German Fraunhofer Institute for Chemical Technology (FH-ICT) conducted various series of tests using mixtures of propane, ethylene, methane, and hydrogen with air to investigate detonation and DDT in spherical, hemispherical, and tube geometries. Unconfined hemispherically shaped H_2 – air mixtures at volumes between 7.5 m^3 and 2100 m^3 were ignited measuring a maximum overpressure of $6.3\text{ m}^3\text{kPa}$, which corresponds to a flame velocity of 84 m/s (Pfortner & Schneider, 1983; Schneider & Pfortner, 1978; Pfortner, 1983, 1985). Balloon tests were conducted with hemispherically shaped H_2 – air mixtures with a volume of 50 m^3 and concentrations of $20\text{ vol}\%$ and $29.6\text{ vol}\%$. Ignition occurred at the center on the ground by means of an explosive to trigger detonation. Pressures were measured at various positions inside and outside the balloon (Figs. 4.8 and 4.9). Visually measured flame speeds agreed well with the theoretical values (Breitung & Redlinger, 1995).

**FIGURE 4.8**

Measured and calculated pressure transient inside the balloon in an FH-ICT hemispherical balloon test with H_2 – air detonation.

**FIGURE 4.9**

Measured and calculated pressure transient outside the balloon in an FH-ICT hemispherical balloon test with H_2 – air detonation.

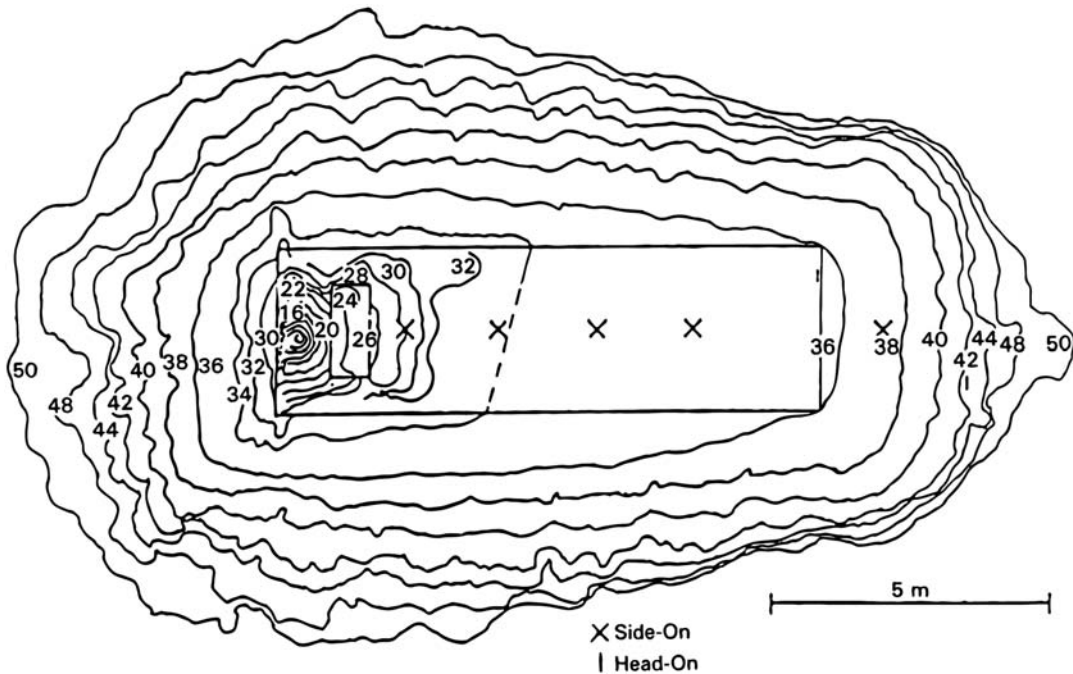


FIGURE 4.10

Measured flame front profiles in an FH-ICT lane test with parallel walls (37% H_2 – air mixture), fan generated turbulence, and DDT near the wall (contour 34).

The influence of partial confinement on the combustion behavior of H_2 – air mixtures was examined in further ICT tests employing a $10 \times 3 \times 3 \text{ m}^3$ lane with parallel walls (Pfortner, 1983; Pfortner & Schneider, 1983; Schneider, 2005), which also have resulted in one case in a transition from deflagration to detonation (Fig. 4.10).

The extensive experimental research programs on gas explosions within the EU projects MERGE (Mercx, 1994) and EMERGE (Mercx, 1997) have shown that overpressures are mainly determined by fuel type, geometric scale as well as the arrangement and number of obstacles, which are passed by the propagating flame.

Other unconfined explosion tests are known of the BASF company in Germany. In 1943 and 1948, explosion accidents occurred at BASF resulting from the bursting of liquid gas vessels, subsequent flash evaporation, and mixing of the fuel with ambient air and eventually ignition of the cloud. The cause of the bursting was a heating of the overfilled vessels by the radiation of the sun, that is, there was not enough vapor buffering inside of the tanks. The experimental simulation and modeling of these events have been performed in the 1970s by BASF and Fraunhofer ICT by use of differently sized vessels with volumes between 0.2 vol% and 1.2 vol% m^3 corresponding to a mass of up to 452 kg of liquid propylene. Pressures observed were in the range of 0.5–1.5 vol% kPa for the smaller and 4–7 kPa for the larger vessels (Maurer, Schneider, Hess, & Leuckel, 1975; Maurer, Hess, Giesbrecht, & Leuckel, 1977; Giesbrecht, 1980, 1981).

With respect to other physical explosions, tests were conducted in the 1970s and 1980s with the spillage of LNG into a pond of water (e.g., Coyote series, Burro series, Maplin Sands series) to measure among other parameters the strength of RPT pressure waves. After releasing 40 m^3 of LNG onto water, observed RPT overpressures were as high as 5 kPa (Koopman et al., 1982).

Gaz de France initiated an RPT research program in 1981 in Lorient with large-scale tests using LNG. The spillage of amounts between 1 m^3 and 9 m^3 onto water has shown that the occurrence and strength of RPT were strongly related to the volume of the mixing zone. Maximum explosion pressure recorded was equivalent to 4.15 kg of trinitrotoluene (TNT). Research activities also included fundamental studies of the phenomena and computer code development. Due to the larger temperature difference, consequences of LH_2 spills onto water may be more severe.

4.2.3.8 Modeling work

The propagation of a pressure wave in a compressible medium can be described by the Rankine – Hugoniot equations (or “jump conditions”) based on the conservation equations for mass, momentum, and energy. From this relationship, it can be derived that the density ratio of air, if assumed to be an ideal gas, behind and in front of the shock front is limited to about 6. For air as a real gas, however, allowing dissociation or ionization at high temperatures, this ratio can be significantly higher.

A first model description of the shock wave parameters for an explosion in air was given in 1870 by Rankine with shock front velocity, maximum dynamic pressure as functions of the peak static overpressure, speed of sound in air, and the atmospheric conditions. Later modeling efforts have used theoretical or empirical approaches to find agreement with experimental data (Pandey, Kumar, Paul, & Trikha, 2006). Fig. 4.11 shows a comparison of measured explosion pressures with different models as a function

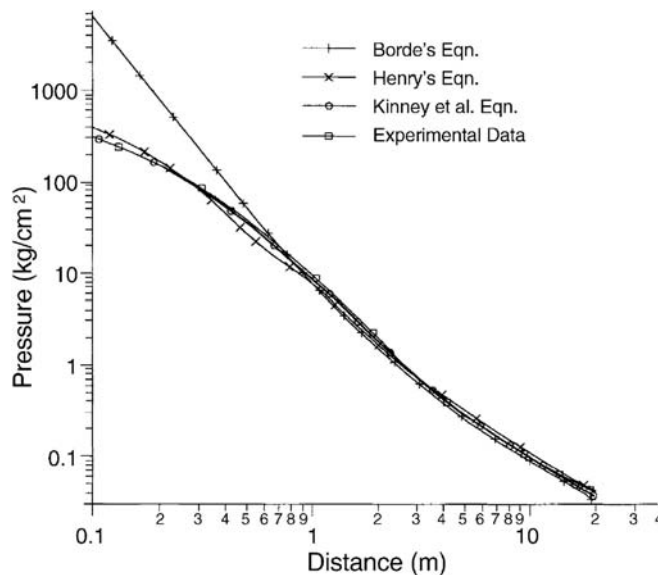


FIGURE 4.11

Blast pressures versus distance for a detonation of 1 kg of TNT. *TNT*, trinitrotoluene.

of distance from the source. It was generally stated that agreement between theory and experiment is less good in the near-field compared to the medium and far-field because of the more complex flow pattern.

Accurate empirical and theoretical models exist for detonation waves. According to the CJ theory of 1899 and 1905, detonation represents a linear discontinuity, transforming the reactants completely to products at an infinite reaction rate. Detonation velocity and pressure can be calculated from equilibrium chemistry as a function of the gas mixture only and the heat of reaction. Respective data for hydrogen at NTP in an unsupported detonation are:

- CJ velocity: 1968 m/s;
- CJ detonation pressure: 1.58 MPa.

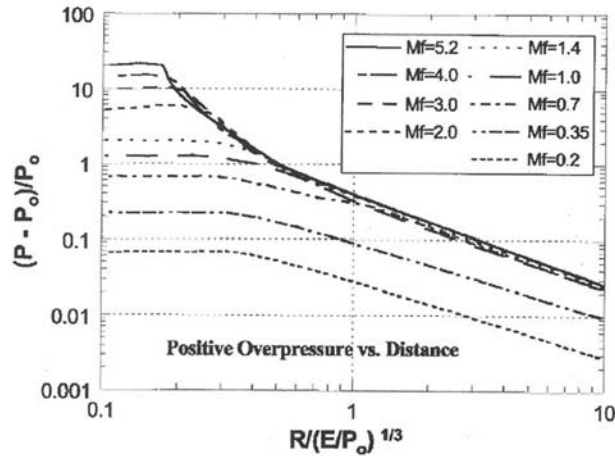
The CJ theory predicts the thermodynamic state immediately behind the detonation wave, but cannot describe the structure of the wave. Processes inside the detonation front are extremely complex involving multidimensional shock interactions in an intensive turbulent reacting medium. Still, the simple one-dimensional CJ model prediction of velocity and overpressure is quite close to what is observed, within a few per cent for velocity and 10%–15% difference for the pressure measurements (Tieszen, 1993). CJ (and AICC) pressures of a fixed gas mixture increase linearly with the initial pressure at constant initial temperature, and are inversely proportional to the initial temperature at constant initial pressure. However, the CJ theory is not capable of determining the dynamic detonation parameters such as detonability limits, initial energy, or critical tube diameter. No theory exists so far that provides estimates of these parameters. CJ parameters of a gas or gas mixtures can be calculated with the code STANJAN developed at Stanford University.

In the ZND (Zel'dovich-von Neumann-Doering) theory, the detonation wave is described as a two-dimensional dome-shaped shock wave, where at its front both temperature and pressure rise. It is followed by a reaction zone whose thickness is determined by the reaction rate. Here the detonable substance reacts at high pressure and temperature until everything is transformed into product gases. The chemical reaction causes a rapid fall in pressure (“von Neumann spike”). The reaction zone remains unchanged (steady) when moving through the substance. A variable ranging between 0 and 1 describes the state and the progress of the chemical reaction. Detonation velocities and pressures are less than for a plane shock front.

A very simple way of modeling blast effects is the TNT equivalent method derived from the decay of shock waves from high-explosive or nuclear explosions in the atmosphere. It is an estimation of the mass of TNT per unit mass of fuel, whose detonation would result in the same blast wave at the same distance. One kg of TNT translates into energy of 4520 kJ meaning that 1 Nm³ of hydrogen gas corresponds to 2.22 kg of TNT. The weakness of the TNT equivalent model, if applied to a VCE, is to ignore the pressure-time characteristic differences between a gas cloud and a detonative TNT explosion. It is deemed to overestimate near-field and underestimate far-field effects. Furthermore the model does not consider the influence of turbulence and confinement. The TNT model considers only the total amount of fuel involved and particularly does not take into account the yield factor in a VCE, which is generally only a small fraction, in many cases < 1%.

The most common form of blast scaling law is that according to Hopkinson and Cranz (Baker et al., 1983):

$$Z = \frac{R}{E^{1/3}} \quad \text{OR} \quad Z = \frac{R}{W^{1/3}} \quad (4.1)$$

**FIGURE 4.12**

Baker – Strehlow – Tang curves of overpressure versus distance for various flame Mach numbers (M_f).

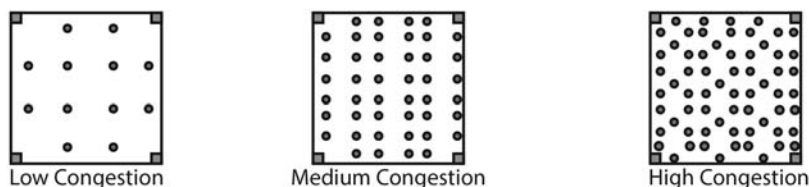
Where Z is the scaled distance, R is the distance from the center of explosion, E is the total heat of combustion, W is the total weight of a standard explosive (such as TNT). The law has been validated against numerous experiments over a large range of explosion energies and can be applied to predict blast wave properties of large-scale explosions based on the data of small-scale experiments (assuming same explosive, same geometry). There is, however, a lack of experimental data at very small scaled distances; the above relation appears to become inappropriate for $Z < 0.16 \text{ m/kg}^{1/3}$.

A wide variety of explosion experiments has been evaluated to derive blast charts. Commonly known and accepted are the Baker – Strehlow blast curves for VCEs in the open atmosphere or the TNO blast waves for hemispherical explosions.

An improvement toward a more realistic modeling was made with a new set of blast curves, called the Baker – Strehlow – Tang curves (Fig. 4.12), by considering a more precise blast pressure decay behavior and by a clearer indication of the flame speed choices to be made for use with the curves, that is, acknowledging that VCEs give rise to a range of deflagration types. These curves plot the normalized blast pressure against the scaled distance. The result is a considerable reduction at long distances. The curves were validated in all combustion regimes (Tang & Baker, 1999).

The value of the use of such graphs is their convenience over CFD type approaches and more recent work has added some clarification by defining congestion types more clearly and providing guidelines on flame speeds based on congestion and fuel type. Fig. 4.13 (Pierorazio, Thomas, Baker, & Ketchum, 2005) shows the additional guidelines available.

An alternative methodology for overpressure estimation arising from gaseous or flashing liquid releases is the TNO multienergy method based on the multienergy concept, which asserts that, for gas deflagration, overpressure and blast develop only under appropriate boundary conditions, that is, only where the flammable mixture is partially confined and/or obstructed (AIChE CCPS, 1994; Mercx, van den Berg, Hayhurst, Robertson, & Moran, 2000; Mercx, Weerheijm, & Verhagen, 1991). This assumption can be made provided that transition to detonation or DDT does not take

**FIGURE 4.13**

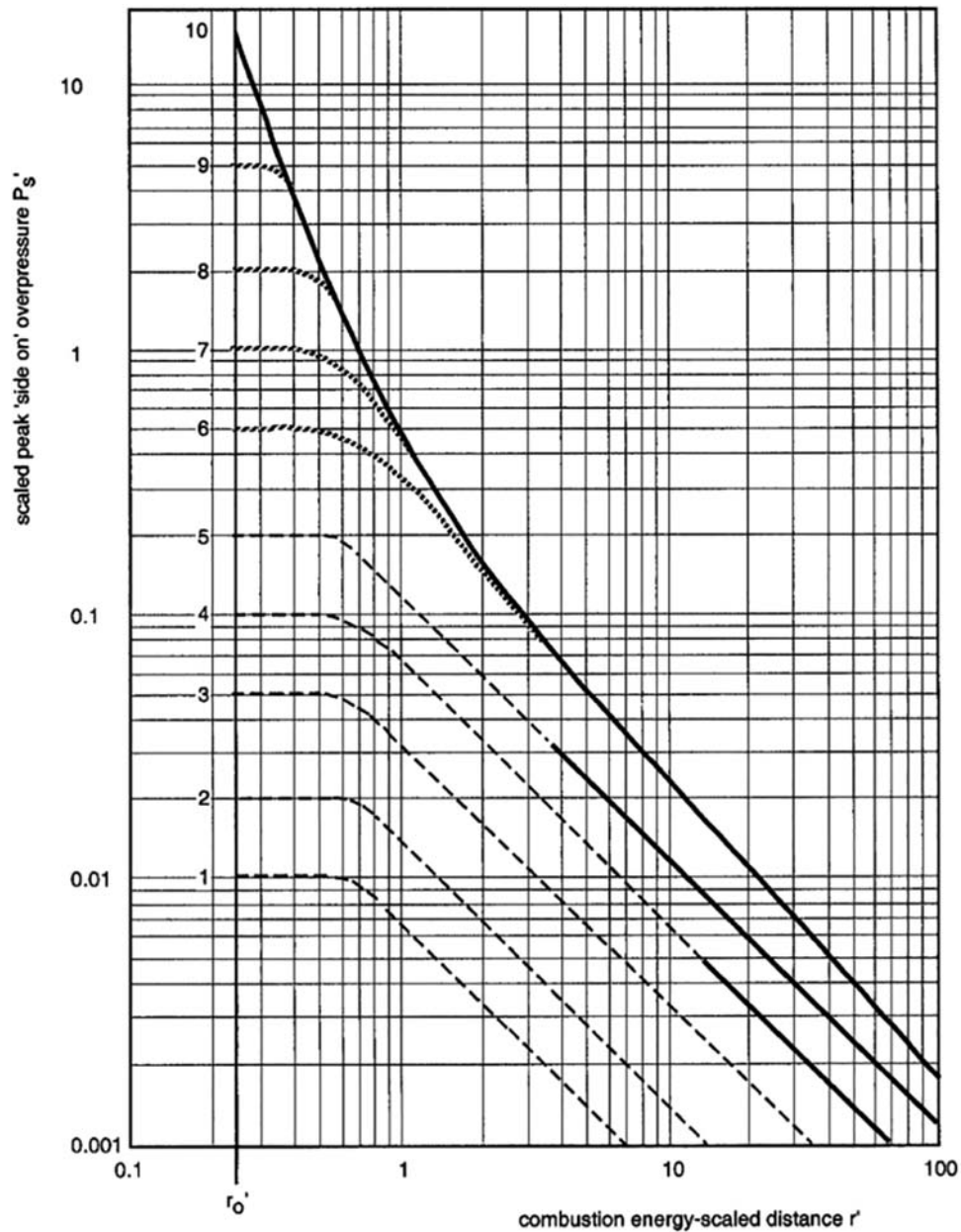
Choices of confinement types for blast model and associated table for flame Mach number choice.

place, which for hydrogen needs more careful consideration than for hydrocarbons. Based on the multienergy concept, a VCE is modeled as a series of hemispherical model charges. Each model charge is characterized by a charge size and a charge strength. The charge size is related to the heat of combustion present in the source, while the charge strength is related to the explosion overpressure. Based on these characterizations, scaled blast parameters (peak overpressure, positive phase duration) as a function of scaled distance have been calculated with the TNO FCT (“Flux-corrected Transport”) Euler solver “BLAST” (Fig. 4.14) (Mercx et al., 2000).

The strength of the blast wave is expressed as a number between 1 and 10 representing categories of “insignificant” to “detonative.” Calculation results suggest that damaging explosions can occur only, when flame acceleration takes place within a plant structure (Mercx et al., 1991, 2000). The charge strength can be determined either by numerical calculation (CFD) or by using the experimentally based GAME correlation, a relation between the overpressure and details of obstacle configurations. The charge size is influenced by the Critical Separation Distance (CSD), the distance between two obstructed regions above which a vapor cloud explosion can be modeled as two separate sources of blast. Below this CSD, the charge can be modeled as a single blast source. Guidance on the CSD has been obtained in experimental research projects such as RIGOS. It has been found that the CSD between a “donor” and an “acceptor” increases with the explosion overpressure up to a maximum of half the donor dimension. This CSD concept is central to the correct use of the TNO curves and detailed guidance on the determination of the “donor” and “receiver” dimensions can be found in the report prepared by Van den Berg and Mos (2005).

The Karlsruhe Institute of Technology has developed the calculation models DET1D and DET3D to determine the characteristic detonation parameters within the reaction zone and outside in the unburnt mixture. These models have been mainly applied to assess the load on a nuclear containment upon confined combustion of homogeneous mixtures of H_2 , O_2 , N_2 , H_2O . Code validation was made against the Russian RUT experiments and FH-ICT balloon tests (see Figs. 4.8 and 4.9). Parameter calculations of a 3D detonation have shown that the 3D structure is not important for the pressure load and that a relatively coarse grid provides sufficient accuracy (Breitung & Redlinger, 1995).

The state-of-the-art approach to modeling of combustion phenomena are 3D CFD codes, which need to be adapted to the spatial and temporal lengths characteristic for chemical reactions. Typical time steps for hydrodynamic flow calculations are in the order of 10^{-6} s and for the $H_2 - O_2$ reaction mechanism 10^{-10} s (Breitung & Redlinger, 1995). Many approaches are based on the assumption of incompressible flows and are restricted to slow deflagrations or to pure detonations. For fast deflagrations with Mach numbers > 0.3 , incompressibility can no longer be assumed, as pressure waves are not negligible.

**FIGURE 4.14**

Blast overpressure versus scaled distance for different explosion strengths according to the multienergy method.

Several methods of turbulence modeling have been developed. In the “Reynolds Averaged Navier Stokes” (RANS) equations, turbulence is modeled by means of a turbulence viscosity, which can be calculated in a k - ϵ two-equations system. In contrast, the “Direct Numerical Simulation” employs the direct and complete solution of the conservation equations taking into consideration all turbulent structures. Somewhere in between is the method “Large Eddy Simulation” (LES) where only the large turbulence structures are being resolved by the calculation grid, whereas small turbulences are filtered away and covered by a turbulence viscosity model approach.

Today’s modeling and simulation show good agreement with a variety of fast combustion phenomena. It has the potential to provide accurate data for realistic scenarios depending on how accurately the respective submodels are working. For example, the above-mentioned LES method for turbulent flows with high Reynolds numbers was successfully applied to the FH-ICT deflagration test with stoichiometric H_2 – air mixture (Molkov, Makarov, & Schneider, 2005). The application of computer models, however, should be limited to cases or ranges, for which the codes were validated. Examples of CFD codes applied to combustion are AIXCO (RWTH, Germany), AUTOREAGAS (TNO & ANSYS Century Dynamics, The Netherlands), CFX (Harwell, the United Kingdom), FLACS (CMI, Norway), FLUENT (Fluent, the United States), COM3D (KIT, Germany), PHOENICS (CHAM, the United Kingdom).

4.3 Heat radiation generation

There exists an extensive published literature on flame radiation from hydrocarbon flames and pool fires, see, for example, De Ris (1979); Mudan (1984); Tien and Lee (1982); Viskanta and Menguc (1987). However, there is a limited number of studies on hydrogen flame radiation, particularly on large scale.

Thermal radiation is a primary mode of heat transfer and is the dominant mechanism of heat transfer in large fires involving hydrocarbons and soot particulates, and which produce intermediate unstable radicals (e.g., O, H, OH, N) as well as stable nonluminous gaseous combustion products (CO_2 , CO , H_2O , NO_x).

The contribution to the radiative transfer in flames can be regarded as being due to luminous and nonluminous radiation. Nonluminous flame radiation originates from transitions in the molecular energy levels and the emission of photons, mainly associated with rotational and vibrational transitions in the infrared (IR) and microwave part of the spectrum and due to the main combustion products CO_2 , H_2O , CO . The radiation from these transitions gives rise to discrete but closely spaced emission lines.

In practical engineering systems, where pressure and geometric scales are large, pressure broadening of spectral lines cause them to overlap with each other. In luminous flames a continuum of radiation in the visible and IR is also emitted by the unburnt carbon particulates called soot that contribute greatly to the luminosity of the flames. The actual quantity and distribution of combustion products and/or soot produced in fires depend on the type and configuration of fuel and local supply of oxygen. In contrast to hydrocarbon fuels, the hydrogen burns more cleanly in air, producing a nonluminous, almost invisible, pale blue flame believed to be due to molecular or atomic hydrogen.

4.3.1 Combustion characteristics of hydrogen

In order to understand thermal radiation hazards from hydrogen flames, it is important to understand the relative differences in the physical properties and combustion characteristics of hydrogen and hydrocarbon flames. Table 4.5 provides comparison of the physical properties of hydrogen with hydrocarbon methane.

Hydrogen has a much wider range of flammability in air (4% – 75% by volume) than methane (5% – 17% by volume), propane, or gasoline, and the minimum ignition energy (for a stoichiometric mixture) is about an order of magnitude lower (1/16th that of methane). In many accidental situations the LFL is more important. The LFL for hydrogen is similar to that of methane, about twice that of propane, and four times that of gasoline. In addition, the minimum ignition energy for hydrogen at the LFL is also similar to that of methane.

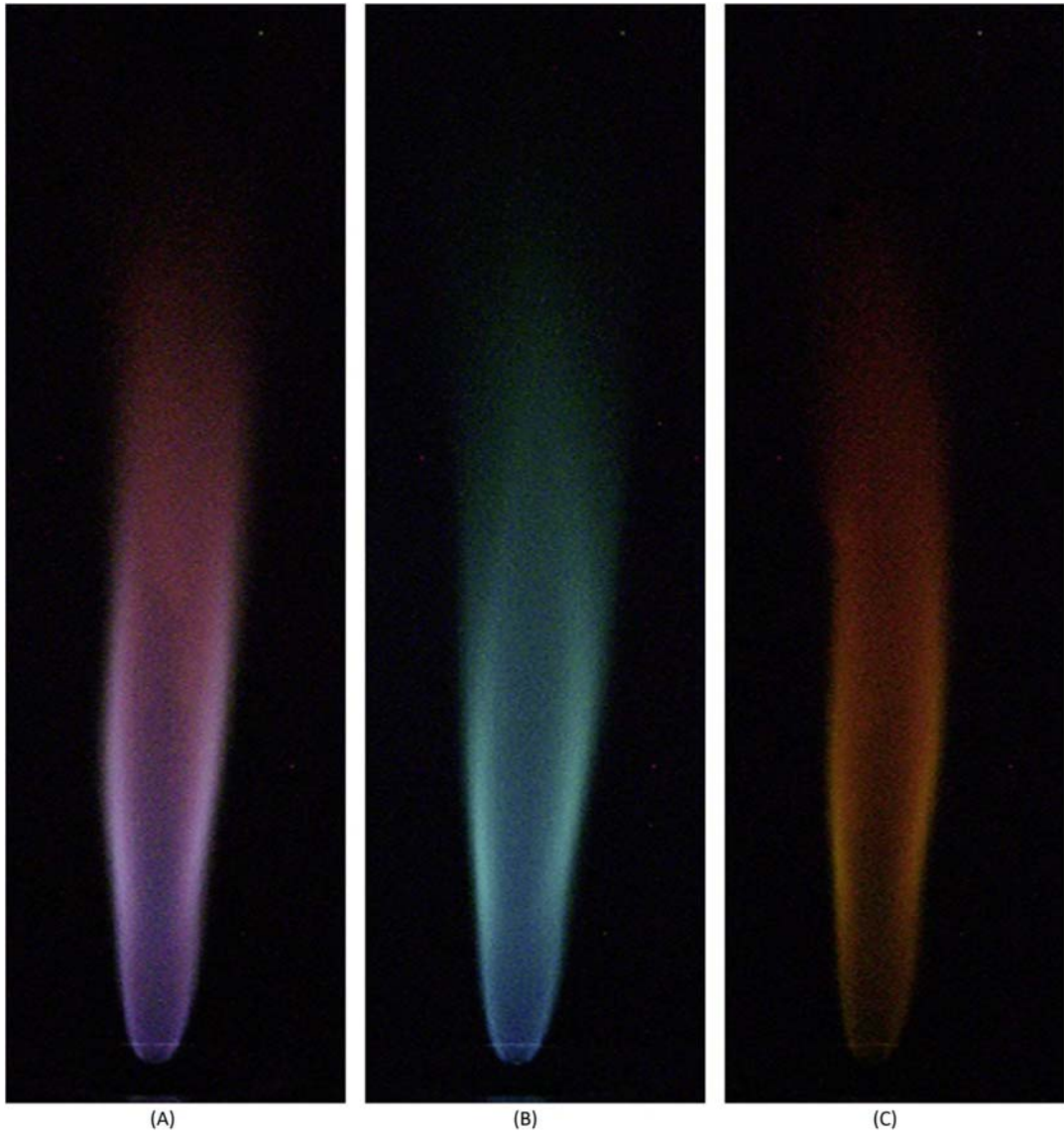
Hydrogen – air mixtures can burn as a jet flame at a fixed point, with combustion taking place along the edges of the jet where it mixes with sufficient air. Alternatively, in a stationary combustible mixture in the open with no confinement, a flammable hydrogen mixture will undergo slow deflagration during which a flame propagates through the mixture. Deflagration refers to a flame that relies on heat and mass-transfer mechanisms to combust and move into areas of unburned fuel. If the flame speed is accelerated, perhaps due to extreme initial turbulence or turbulence induced by obstacles or confinement, the result is a fast flame event and a resulting explosion. In the extreme case the flame speed becomes supersonic and results in detonation. Once initiated, detonation is self-sustaining (no further turbulence or confinement is required) as long as the combusting mixture is within the detonable range.

The heat of combustion of hydrogen per unit weight is higher than any other material, but hydrogen has a relatively low heat of combustion per unit volume. Thus the combustion of a given volume of hydrogen will release less energy than the same volume of either natural gas or gasoline.

In contrast to other hydrocarbon fuels, a hydrogen flame radiates significantly less IR radiation (heat) and virtually no visible radiation (light). The pale blue emission referred to above is almost invisible in artificial light or daylight. Figs. 4.14 and 4.15 show measured emission in H_2 – air flames across the spectral range from ultraviolet (UV) to near IR (Fig. 4.16).

Table 4.5 Physical and combustion properties of hydrogen and methane.

	Hydrogen	Methane
Auto-ignition temperature	520°C	630°C
Heat of combustion (lower heating value)	119.9 MJ/kg	50.1 MJ/kg
(upper heating value)	141.9 MJ/kg	55.6 MJ/kg
Lower flammable limit (in air)	4.0 vol%	5.3 vol%
Upper flammable limit (in air)	75.0 vol%	15 vol%
Stoichiometric mixture (in air)	29.5 vol%	9.5 vol%
Density (@ 20°C, 100 kPa)	0.08988 kg/m ³	0.71 kg/m ³
Diffusivity (@ 20°C, 100 kPa)	0.61 cm ² /s	0.16 cm ² /s
Viscosity (@ 20°C, 100 kPa)	8.814 μPa-s	11.023 μPa-s
Flame temperature (in air)	2045°C	1875°C
Minimum ignition energy (in air)	0.017 mJ	0.274 mJ

**FIGURE 4.15**

Luminosity photograph from laminar diffusion H₂-jet flame. (A) Unfiltered. (B) Short wavelength pass filter with 550 nm cutoff. (C) Long wavelength pass filter with cutoff at 530 nm.

From Schefer R.W., Kulatilaka W.D., Patterson B.D., Settersten T.B. (2009). *Visible emission of hydrogen flames*. Combustion and Flame, 156, 1234–1241.

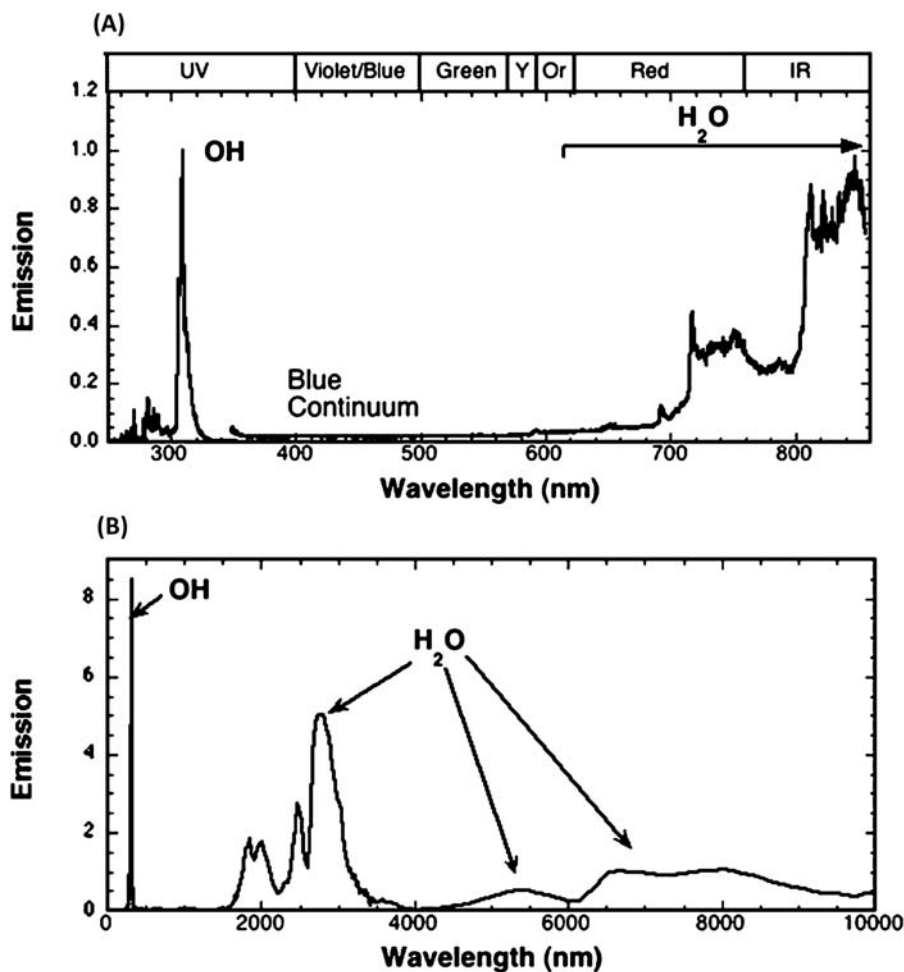


FIGURE 4.16

Emission spectra in typical hydrogen-air flames. (A) Premixed H_2 -air jet flame showing measured spectral peaks from UV to near IR—note range from 350 nm to 850 nm scaled by factor 6.5 relative to OH region. (B) Calculated emission peaks for diffusion H_2 -air flame.

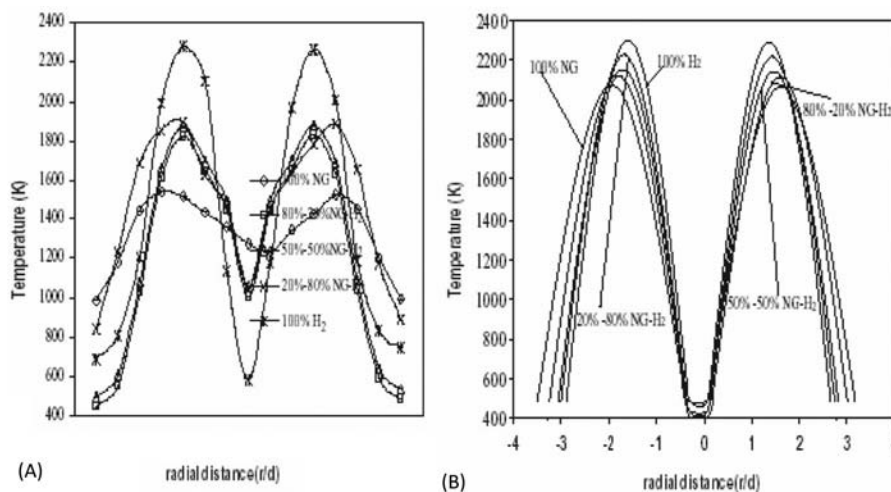
From Schefer R.W., Kulatilaka W.D., Patterson B.D., Settersten T.B. (2009). Visible emission of hydrogen flames. Combustion and Flame, 156, 1234–1241.

The combustion chemistry of a hydrogen flame gives rise to H, O, and N atoms, and OH radicals, which emit specific peaks at specific wave lengths. Table 4.6 shows the emission lines associated with the atomic species involved.

As Table 4.6 shows, a hydrogen flame also emits some limited amount of radiation in the UV region around 180–315 nm, mainly due to OH radical and which is exploited by UV detectors to

Table 4.6 Atomic emission lines observed in a hydrogen-air flame (Itoh et al., 2001).

Element	Wavelength (nm)
H	656.3
N	742.4, 744.3, 746.9, 818.5, 818.8, 821.6, 822.3, 824.2
O	777.2, 777.4, 777.5, 844.6

**FIGURE 4.17**

Radial temperature distribution at the near-burner region. (A) Spectrally derived from OH measurements. (B) Computed using detailed chemical kinetics.

detect hydrogen flame. The OH radical is one of the important intermediates in the combustion reaction mechanism, which is a reliable indicator of the flame zone, flow structure, and flame temperature near stoichiometry.

The laser-induced fluorescence technique has been used (Choudhuri & Gollahalli, 2004) to measure OH radical concentration in the UV emission region near 315 nm. Here, the flame temperature was also determined by measuring the rotational energy distribution of the OH radical at the excitation of this emission (1,0) band ($A^2 \Sigma \leftarrow X^2 \Pi$) system. Maximum statistical uncertainties in measurements were estimated to be 20–30 K in the temperature range. Fig. 4.17 from this work gives an indication of the flame temperatures within a diffusion flame burner using H_2/CH_4 mixture and including pure hydrogen, both from measurements and calculated from chemical kinetics.

It can be seen from the figure that the peak temperature of the 100% hydrogen-air flame is 2320 K and that of the 100% natural gas-air flame is 1600 K. The lower peak temperature of the natural gas flame is attributed to the higher heat losses both in the form of banded nonluminous radiation from CO_2 and continuous luminous radiation from solid soot particulates,

which are absent in the H_2 -air flame. Both flames have heat losses from banded nonluminous radiation from H_2O . The computed stoichiometric temperature contours using the detailed chemical kinetics show a similar trend for the different mixture conditions. Compared to experimental results, predicted temperatures are higher and steeper in shape for all mixture conditions.

The lower radiation from a hydrogen flame makes the flame itself hotter than a hydrocarbon flame, and objects engulfed by a hydrogen flame tend to heat faster. However, the lower radiation of heat from the flame means that less heat is radiated to objects or people outside the flame. The consequence of the almost invisible hydrogen flame is that the human physical perception of the heat from a hydrogen fire does not occur until the person is in closer proximity with the flame than is the case with a hydrocarbon flame.

4.3.2 Experimental studies of radiation

The theory of thermal radiation from flames is very complex, and an exact solution, even for reasonably simple situations, is generally impossible. The combustion products and soot, acting as participating media, add further complexity to the situation. Therefore a wide range of calculation methods and mathematical models with varying levels of complexity and accuracy have been developed. The methods differ in complexity and accuracy and therefore, for the purposes of safety analyses, the use of empirical correlations, which are based on sound experimental data, is likely to be most useful for safety engineers.

Although the radiation from hydrogen flames is less than for hydrocarbon ones, the high mass flow rates associate with a number of accident scenarios, for example, jet fires, imply high rates of energy release through combustion and therefore the radiation load from these needs to be evaluated.

Much of the investigation of hydrogen jet fires has centered on the main hazard evaluation, which is flame radiation. Fig. 4.18 identifies the first of the main parameters in this description, that is, flame length and width.

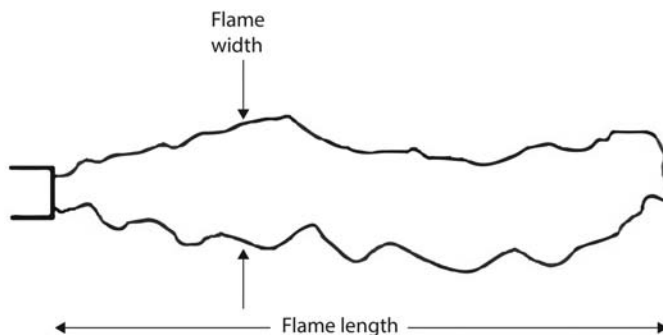


FIGURE 4.18

Simple parameters representing a burning jet.

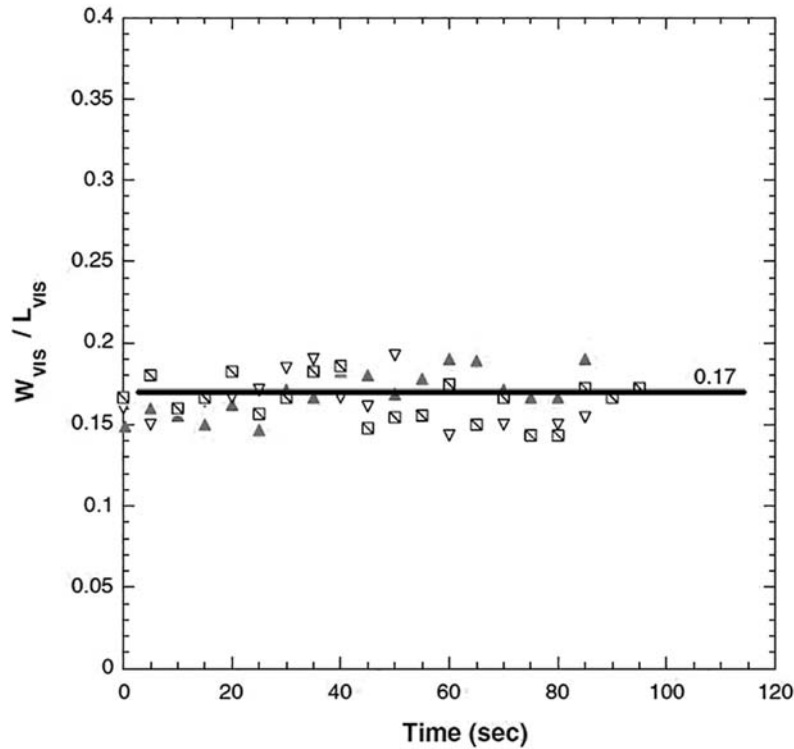


FIGURE 4.19

Flame width/length ratio during discharge of a high pressure hydrogen jet.

Even these simple measures are not completely trivial to determine due to (1) the fluctuating nature of the flame arising from turbulence; and (2) the difference in the radiation emission bands associated with different aspects of the flame evolution.

The relationship between flame width W and flame length L_{vis} has been described in Schefer (Schefer, Houf, Bourne, & Colton, 2006) in simple terms of $W = 0.17 L_{vis}$ for a range of flame fuels and this has been confirmed in the reported results shown in Fig. 4.19. The width W refers to the widest part of the visible flame and this simple relation is used by most authors.

A number of additional jet and flame parameters are used in the estimation of received radiation flux in the vicinity of a flame. These are detailed below.

The fraction of chemical combustion energy emitted as flame radiation (X_{rad}) is defined as

$$X_{rad} = \frac{S_{rad}}{m_{fuel} DH_c} \quad (4.2)$$

where S_{rad} is the total radiated power, m_{fuel} is the fuel flow rate, and ΔH_c is the heat of combustion of hydrogen.

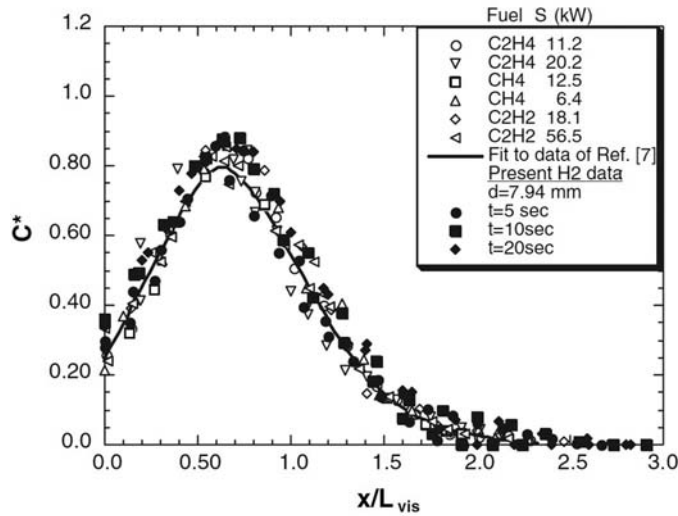


FIGURE 4.20

Measured value of function C^* for a range of nozzle releases and fuels as a function of normalized axial distance (Schefer et al., 2006; Sivathanu & Gore, 1993).

For turbulent jet flames, a nondimensionalized radiated power parameter C^* has been used (Turns & Myhr, 1991) to relate the radiative heat flux $q_{\text{rad}}(x, r)$, at an axial position x and radial position r , with the total radiated power S_{rad} by means of:

$$q_{\text{rad}}(x, r) = \frac{C^* S_{\text{rad}}}{4\pi r^2} = \frac{C^* X_{\text{rad}} m_{\text{fuel}} DH_c}{4\pi r^2} \quad (4.3)$$

It has been found that C^* is only a function of the nondimensionalized axial position x/L_{vis} for a range of combustion species, nozzle dimensions, and mass flows and this is shown in the data of Fig. 4.20, which also includes the hydrogen data of Schefer et al. (2006).

The value of the behavior of C^* , as shown in Fig. 4.20, is that a measure of $q(x, r)$ at a particular station provides a means of calculating S_{rad} and hence allows the estimation of $q(x, r)$ at any other position around the flame. However, with an independent knowledge of the radiated fraction X_{rad} , the S_{rad} value can be calculated independently of a $q(x, r)$ measurement and then used in the above equation.

The collapse of data in Fig. 4.20 is within a 15% range for the region of x/L_{vis} in the range 0.5–0.7, which indicates the best region for measuring heat flux and hence S_{rad} values. At this axial position the C^* value is in the range 0.8–0.85.

The estimation of X_{rad} for use in the above estimation of $q_{\text{rad}}(x, r)$ arises from a separate observation on turbulent jet flames due to Turns & Myhr (1991). These authors defined a timescale τ_f

$$\tau_f = \frac{\rho_f W_{\text{vis}}^2 L_{\text{f}} f_s}{3\rho_j d_j^2 u_j} \quad (4.4)$$

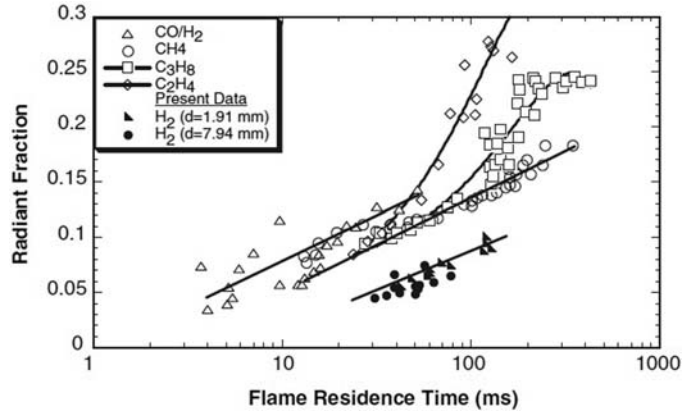


FIGURE 4.21

Data showing the relation between flame “residence time” parameter and radiant fraction X_{rad} from Turns and Myhr (1991) and including hydrogen data from Schefer et al. (2006).

where ρ_f and ρ_j are the flame and jet exit densities, respectively. W_{vis} is the width of the visible flame, d_j is the nozzle diameter, u_j is the gas exit velocity, and f_s is the stoichiometric mass fraction of fuel. L_f is equal to the visible flame length. The flame density is calculated from $\rho_f = p_\infty W_{\text{mix}}/R T_{\text{ad}}$ with p_∞ the ambient pressure, W_{mix} the mean molecular weight of the stoichiometric mixture of hydrogen combustion in air, R is the gas constant, and T_{ad} is the adiabatic flame temperature.

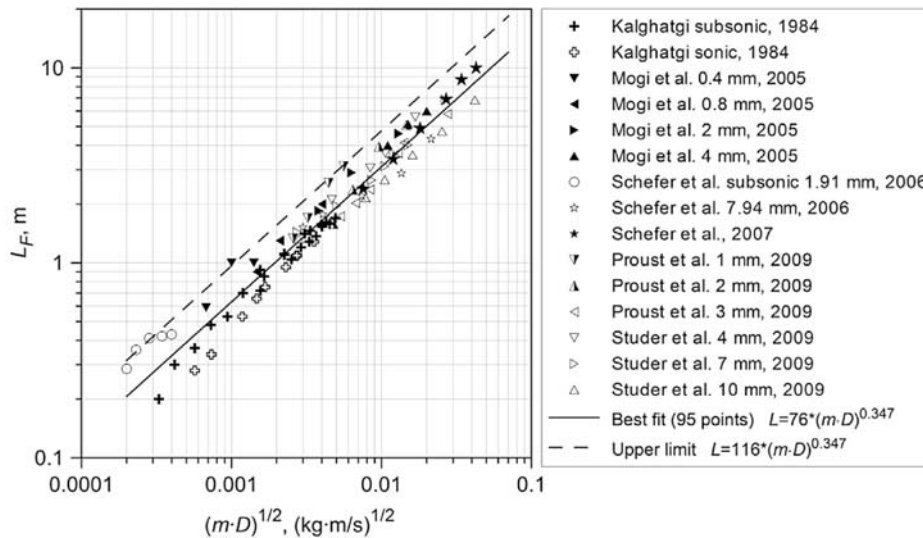
The time parameter is a measure of the fuel mass contained within the flame volume (calculated as a cone) divided by the rate of fuel mass flow from the nozzle. The usefulness of this relation is that the authors observed that the radiated fraction X_{rad} , for a particular fuel, was approximately a linear function of this “residence time.” Fig. 4.21 shows the supporting data for this conclusion and including additional hydrogen test data from Schefer et al. (2006).

A necessary parameter to enable use of equation for the parameter τ_f is the flame length, and this has been the subject of several studies over the years, which have included additional definitions involving Froude number. The most robust determination of flame length has derived from the review work of Molkov and Saffers (2013), where a nondimensional group approach has been used to rationalize the data involving different nozzles, fuels, and mass flow rates. Fig. 4.22 shows that a good collapse of the data from a wide range of published material can be obtained using only the mass flow rate and the actual nozzle diameter.

The flame length then takes the form:

$$L_f = 76 (mD)^{0.347} \quad (4.5)$$

where m is in kg/s and D is in m.

**FIGURE 4.22**

Review of jet flame data showing flame length as a function of only mass flow rate m , and nozzle diameter D .

From Molkov V., Saffers J.-B. (2013). *Hydrogen jet flames*. International Journal of Hydrogen Energy, 38(19), 8141–8158.

The correlations and graphs above show that a methodology becomes available to estimate the radiation flux at any point in the vicinity of a flame for the purposes of estimating hazard. The method then takes the form:

1. Define the nozzle diameter d_j .
2. Use a suitable model to estimate the choked flow mass flow rate, for example, Abel-Noble real gas equations and nozzle exit velocity.
3. Calculate flame length from Eq. (4.5).
4. Estimate flame width from $W = 0.17 L_f$.
5. Estimate flame residence time from Eq. (4.4).
6. Determine the flame radiation fraction X_{rad} from Fig. 4.21 for relevant gas.
7. Calculate S_{rad} from the X_{rad} value and heat of combustion.
8. Use Fig. 4.20 to determine an appropriate C^* value for the chosen x/L_f position.
9. Use Eq. (4.3) with the relevant r value to determine the received radiation flux at chosen coordinate position.

It is noted that the maximum $q(x,r)$ value occurs at an axial position, x/L_f , corresponding to the maximum of C^* and this is close to $x/L_f = 0.7$.

4.3.3 Liquid hydrogen releases

For large-scale liquid hydrocarbon storage, it is normal to consider pressurized liquid releases resulting from vessel failure under the liquid vapor pressure. These can give rise to pool fires and BLEVEs in the event of fire engulfment of the storage vessel.

**FIGURE 4.23**

Liquid hydrogen pool fire following 60 L/min release over 5 min duration.

Due to the cryogenic nature of liquid hydrogen storage, accidental liquid releases will give rise to liquid pool fires and this has been demonstrated experimentally in the report by [Hall \(2014\)](#) involving multiple tests involving LH_2 with associated ignition. These tests released LH_2 at a rate of 60 L/min with durations up to 5 min, and radiometers monitored the thermal radiation at a distance from the release. [Fig. 4.23](#) shows a photograph of one of these tests.

It has already been noted that hydrogen flames radiate less than flames from the combustion of hydrocarbon gases and the radiative fraction is influenced by any contaminants present within the burning cloud. However, for the test shown the depth of flame reveals the emissions present in the yellow part of the spectrum.

The radiative fraction was estimated for the steady burning periods of the LH_2 release experiments, that is when the initial cloud had burned out and the hydrogen was being consumed as it was being released and evaporated.

It is common to approximate the radiative fraction of a flame based on radiometer readings taken at a significant distance from the flame such that an inverse square law can be reasonably applied. However, in this case the flame was elongated along the line of the radiometers and was generally close to the ground. For this reason, a semicylindrical radiating heat source was assumed for the purposes of estimating the radiative fraction and the total radiated heat was estimated using

$$S_{\text{rad}} = (1 + \alpha) \frac{\pi d L q_r}{2} \quad (4.6)$$

where: S_{rad} = heat radiated (kW); d = distance to radiometer (m); L = Length of flame (m); q_r = heat flux at radiometer (kW/m^2); α = reflection coefficient of concrete surface below the flame.

Data from two of the ignited releases were analyzed in this way, assuming a reflection coefficient for the concrete of 0.55 (Markvart & Castalzer, 2003), giving an estimate of the radiative fraction as 0.054. This estimate compares reasonably well with previously reported values for gaseous and LH₂ releases (Friedrich et al., 2012; Studer et al., 2009) and with the values measured using high pressure jet releases referred to in the discussion above.

The potential for BLEVE behavior of LH₂ storage has been considered in some quarters as not a viable event and this is primarily due to the fact that liquid hydrogen tanks are double walled, and a direct impinging heat load on the inside tank is less probable. This heat load is necessary to “boil” the liquid on the inner surface of the inner tank creating a weakening of the metal and an associated over pressure, which may lead to rupture of the tank and a subsequent vapor explosion. Such storage vessels are designed for relatively low pressure and low temperature operation, and a small number of studies (Ustolin, Song, & Paltrinieri, 2019; Verfonderen, 2008) have examined a possible BLEVE scenario and the outcome of tank rupture.

Ustolin examined the failure of a 1-m³ hydrogen storage tank with a burst pressure of 32 bar corresponding to a liquid temperature of 43 K, which is above the critical temperature of LH₂ ($T_{\text{critical}} = 29.7$ K). In this case, the results were used to estimate various blast radii.

Some empirical relationships exist in the literature for LH₂ to determine the diameter of a fireball D , the height the fireball achieves H , and the duration of the fireball t based on the mass of fuel involved. For LH₂ BLEVE, Hankinson and Lowesmith (2013) used the equations originally developed for LNG with the equivalent mass of methane, based on equal net energy (120 MJ/kg for H₂ and 50 MJ/kg for CH₄, where 1 kg H₂ is equivalent to 2.4 kg CH₄).

The dimensional study concluded the following for the fireball dimensions and duration:

- Diameter D (m) = $6.14 M^{1/3}$;
- Height H (m) = $0.41 M^{1/3}$;
- Duration t (sec) = $0.75 M^{1/3}$.

The power dependencies implied by these have yet to be confirmed experimentally but allow a provisional estimate of the scale and duration of releases, which in turn allow a determination of the power/energy output from the BLEVE event when combined with the radiated fraction figures from other hydrogen fire phenomena. These can in turn be used to calculate dose values, based either on a point source model if distances justify this, or using a model based on view factors in a similar way for hydrocarbon BLEVEs.

4.4 Infrastructure impact

4.4.1 Heat radiation

4.4.1.1 Interaction with objects

Even in the absence of people, the effects of radiation on infrastructure and hardware need to be considered in installation planning, and recourse can be made to the behavior of particular materials to radiation loads, with the future possibility of developing probit-type functions describing degrees of impact or failure. As the aim of most thermal studies on emissive sources is to estimate the received radiation flux at a particular distance from the source, it is convenient to

Table 4.7 Thermal radiation flux levels for damage to a number of structural materials.

Thermal radiation intensity (kW/m ²)	
4	Glass breakage (30-min exposure).
12.5–15	Piloted ignition of wood, melting of plastics (30-min exposure).
18–20	Cable insulation degrades (30-min exposure).
10–20	Ignition of fuel oil (120 s or 40 s, respectively).
25–32	Unignited ignition of wood, steel deformation (30-min exposure).
35–37.5	Process equipment and structural damage (e.g., storage tanks) (30-min exposure).
100	Steel structure collapse (30-min exposure).

From LaChance J.L. (2009). Uniform risk acceptance and harm criteria—IEA HIA Task 19 Perspective. 3rd International Conference on Hydrogen Safety, Ajaccio, France. 16–18 September. These radiation intensity levels are broadly in line with the UK Building Regulation guidance criterion of a global 12.6 kW/m², which accurately represents the piloted ignition threshold of soft wood for continuous radiant exposures of 10–20 min.

express the response of structural objects to received fluxes in combination with exposure times, and with the associated outcome also identified, for example, melting, breakage, ignition. It is considered a justified assumption to use the material response data obtained from one radiative source in judging its response to the same radiation intensity from a different source. Therefore the estimates of radiated power from the various hydrogen sources of interest can be used to judge the response of structural materials, which have been studied using different sources for example, electrical.

The International Energy Agency (IEA) has an ongoing program for hydrogen technology adoption through its Hydrogen Implementing Agreement, within which are a number of dedicated tasks aimed at facilitating the use of hydrogen by improving knowledge in certain areas. Task 19 is one of these aimed at developing predictive methods and generating data relating to infrastructure.

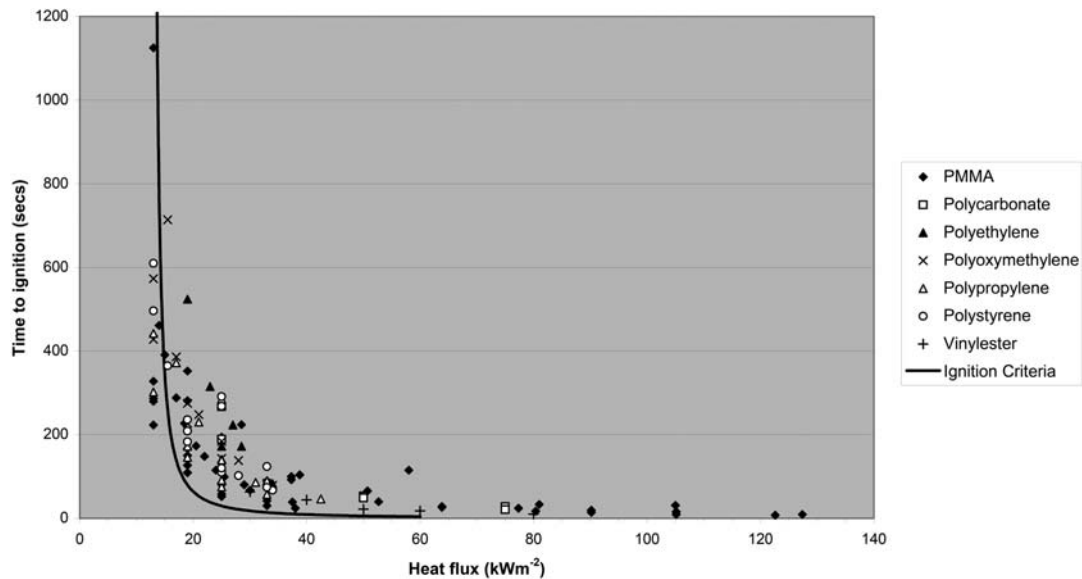
Although the relevant probit functions do not yet exist for structures, [LaChance \(2009\)](#) has summarized the flux levels associated with damage to a number of structural materials as shown in [Table 4.7](#).

The work by [Hare and Burrell \(2006\)](#) has provided a more substantial database relating to the ignition of a wide range of materials in industrial and domestic use. The report covers materials including asphalt, carpet, nine types of wood, clothing, epoxy, fiber board insulation, paint, phenolic resin, and six polymer types. Due to the variability of ignition behavior for particular materials, much of the data are presented as a plot of time to ignition against received heat flux, which reveals this variability graphically. An example is shown in [Fig. 4.24](#).

4.4.1.2 Methods of determination of objects response

The threshold for ignition in gases and liquids is more easily determined than for solids, thus involving the observation of sustainable combustion for gases or simply a surface temperature sufficient to maintain an ignitable vapor for liquids.

Solids cannot be characterized as easily because of low thermal conductivity and no mass transfer. The point of ignition can be measured as a surface temperature, a mass flux (pyrolysis), or a

**FIGURE 4.24**

Ignition times for a range of plastics as a function of received radiation.

From Hare J., Burrell, G. (2006). Review of HSE building ignition criteria. Health & Safety Laboratory Report HSL/2006/33.

heat flux (radiation). The experimentally measured, lowest thermal intensity capable of ignition is the minimum heat flux and the lowest thermal intensity theoretically possible is the critical heat flux.

Measurements on solids have been carried out by a variety of methods. The most common is the cone calorimeter, which exposes a small sample of test material, between 5 cm² and 200 cm², to heat from a cone shaped electric radiator either in the vertical or horizontal orientation. Other sources of radiation include gas flame furnaces, carbon arcs, and tungsten lamps. Of these the radiant flame furnace best characterizes the wavelength profile of a flame. The standards ASTM E 1354–92, ASTM E 1474–92, ANSI/NFPA 264 A, and ISO 5600 are all modified versions of the cone calorimeter test.

The principle of the test procedure is shown in Fig. 4.25, which is a schematic of the device first developed by National Institute of Standards and Technology. The key elements are the heated cone, which can deliver radiation flux at a calibrated level to the sample placed below the cone, and a means of identifying the onset of ignition or pyrolysis, achieved by means of a load cell holding the sample. Downstream gas analysis from the exhaust is normally available to examine the decomposition or combustion products.

Changes in mechanical properties of materials can readily be determined by the normal laboratory yield stress tools adapted to enable the application of controlled thermal flux to the samples.

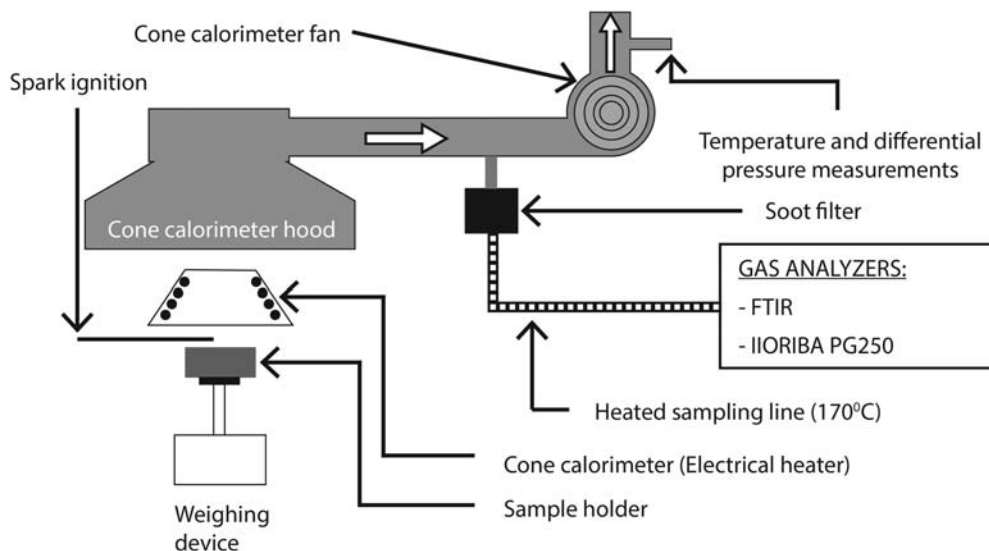


FIGURE 4.25

Principle of the cone calorimeter.

4.5 Blast effects

4.5.1 Debris and missiles

No fully validated model exists in terms of predicting projectile hazards from bursting pressure vessels. However, this difficult physical problem may be divided in two parts:

- fragmentation process evaluation;
- projectiles trajectories.

The first part is probably the most complex one, and only numerical simulation seems to allow for prediction using appropriate material and fracture models as, for example, shown in [Gurson \(1977\)](#); [Erdogan, Delale, and Owczarek \(1977\)](#); [Mott \(1943\)](#). The second part couples the initial fragmentation process and following ballistic considerations. Solutions as suggested in [Baker et al. \(1983\)](#); [Baum \(1999\)](#); [UFIP \(2002\)](#) are the most widely applied and easy-to-handle models. The software PROJEX using a method developed by INERIS may also be used and seems to give better results ([INERIS, 2004](#)). A global model based on statistical considerations is also being conceived in [Hauptmanns \(2001\)](#).

Most modeling approaches deal with the effects of blast and fire. In many hydrogen explosion scenarios, however, the throw of missiles or debris is also important and in some cases is dominant. For hydrogen gas explosions this is typically the case for scenarios where some degree of confinement or enclosure is present. When in combination the combustion process changes from a deflagration to a detonation the throw of missiles or debris can be devastating. Examples are gas

explosions inside industrial equipment, inside a garage or car park, or in a nuclear plant. These scenarios may lead to a major hazard of debris or missiles. Other types of explosions involving hydrogen are BLEVEs and physical explosions. In these scenarios, it is typically a vessel rupture leading to the throw of missiles. Any risk assessment methodology for hydrogen should contain models for the throw of debris or missiles.

4.5.1.1 The initial conditions

In general, the initial conditions for throw are defined by the distributions of the missile or debris mass, launch velocity, and launch direction. These conditions are determined by the failure process and the subsequent acceleration by the expanding gases, or reaction products. This is accompanied by pressure relief as a result of the increasing vent area between the accelerating items.

The initial conditions for throw of missiles from vessels, as a result of physical explosions, BLEVEs, and internal gas explosions can be determined with PGS02 ([Committee for the Prevention of Disasters, 2005](#)) (CPR14E, 2005:YellowBook), referred to as the “Yellow Book,” and the CCPS Guideline ([AIChE CCPS, 1994](#)). In these references, the models of Baum (BLEVEs), Baker (physical explosions), and Gelfand (physical explosions and internal gas explosions) are presented. For BLEVEs, the model of Baum predicts the initial velocity of missiles based on the amount of liberated energy. The model of Baker for physical explosions relates the initial velocity of missiles to a scaled overpressure, the vessel geometry, and the number of fragments. With the model of Gelfand for internal gas explosions, the initial velocity is based on the scaled overpressure and a scaled energy release rate.

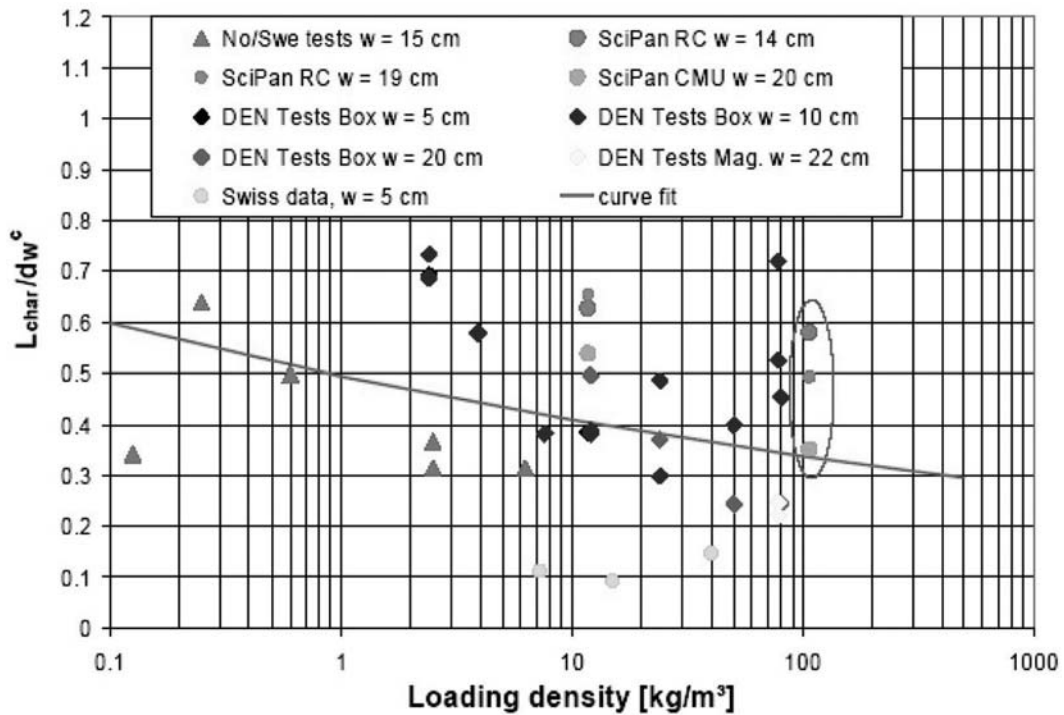
The break-up process of buildings constructed from reinforced concrete or brick during an internal gas explosion is a more complicated phenomenon. In the case of a (weak) deflagration, the combustion continues during the break-up process and debris launch. The coupling between pressure build-up, venting, and break-up determines the part of the structure that will participate in the throw, and the distributions of debris launch velocity (DLV) and launch direction.

When a hydrogen detonation takes place inside a building the combustion process is completed before the structure starts to break up. As a result, the structure is loaded with a quasi-static load of typically 0.8 MPa overpressure. This by far exceeds the structural strength of any typical industrial building. As a result, the structure will largely break up and accelerate. Hydrogen detonations in buildings can be compared with bare quantities of high explosives in ammunition magazines. A similar quasi-static load is obtained when a bare charge resulting in a loading density of 0.25 kg/m^3 of TNT is detonated. In [Fig. 4.26](#), the result after the detonation of 2 kg of TNT in an 8-m^3 Kasun building (Norwegian “small quantities” ammunition storage building) is displayed ([Langberg, Christensen, & Skudal, 2004](#)).

For detonations of bare explosives in ammunition magazines, relations for the distributions of debris mass, launch velocity, and launch direction have been derived within the Klotz Group ([Van Doormaal & Weerheijm, 2006b](#); [Van Doormaal, van der Voort, Verolme, & Weerheijm, 2006a](#)). Those relations are based on debris pick-up data from a collection of large scale trials, and depend on loading density and wall thickness. The cumulative debris mass distribution was found to decay exponentially with debris mass. The characteristic length of debris decreases slowly with loading density. [Fig. 4.27](#) shows this characteristic length for a large collection of trial data. Note that the effective loading density for hydrogen detonations of 0.25 kg/m^3 is situated at the low loading density regime of the trial database.

**FIGURE 4.26**

Kasun storage building with an internal volume of 8 m^3 (left), result after detonation of 2 kg TNT (right). *TNT*, trinitrotoluene.

**FIGURE 4.27**

Characteristic length versus loading density for the Klotz Group trial database.

The initial debris velocity decreases with debris mass around a typical velocity, the DLV. This relation has been based on backward calculations. The launch angle distribution was found to be a rather sharp Gaussian distribution centered around a direction close to the wall normal directions. This directionality is illustrated in Fig. 4.28, where a frame is shown during the debris launch from

**FIGURE 4.28**

Development of the debris throw during a Kasun trial with 50 kg TNT illustrating the strong directionality of throw. *TNT*, trinitrotoluene.

a Kasun building after detonation of 50 kg TNT. Note that this loading density exceeds the energy content of a hydrogen detonation by far.

The addition of the new Sci Pan 3 trial to the database has been reported ([van der Voort, Van Doormaal, Verolme, & Weerheijm, 2006](#)). The relations are currently being implemented in a software code, the KG-Engineering Tool.

4.5.1.2 The throw of debris and missiles

The initial distributions are the required input for throw models. Throw models determine a collection of impact locations in the field. Together with the impact velocity and impact angle the consequences for the infrastructure and for human beings can be determined. Most existing throw models make use of a Monte Carlo technique. Predefined initial distribution of debris, missile mass, launch velocity, and launch direction are sampled to obtain the initial conditions for a number of trajectory calculations. To obtain a proper representation of the debris or missile density in the field, the required number of Monte Carlo simulations needs to be very large. A global model based on statistical considerations is also being conceived in [Hauptmanns \(2001\)](#).

TNO has developed a universal throw model ([van der Voort et al., 2006](#)), the basis of which is the source function theorem, an underlying mathematical relation between the debris or missile density and the initial distributions. This model reduces the required number of trajectory calculations dramatically. Trajectory calculations have to be carried out with caution as the selected shape, orientation, and lift and drag coefficients of debris and missiles have a significant influence on throw distances.

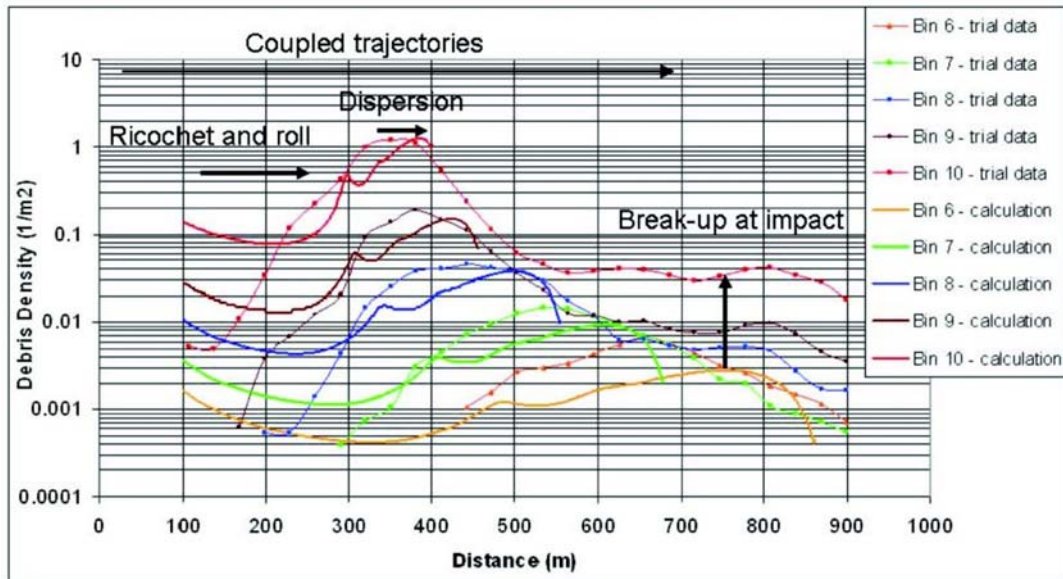


FIGURE 4.29

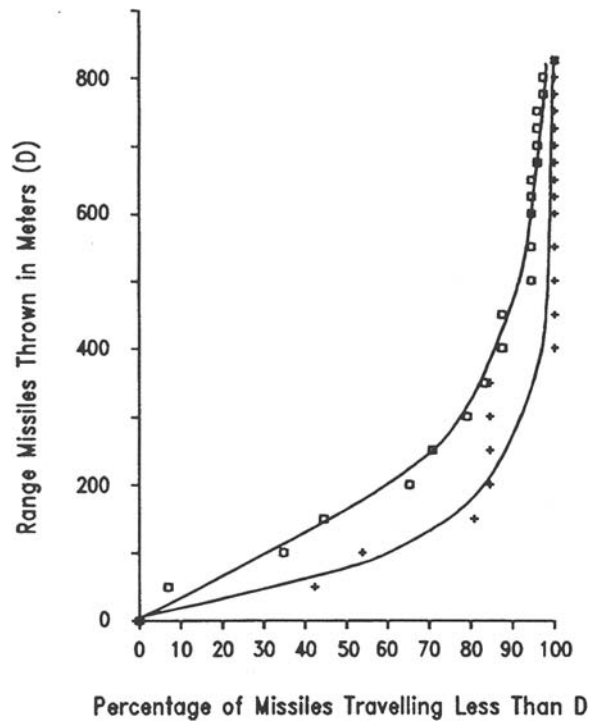
Debris density calculated with the universal throw model together with experimental result. Sci pan 3 trial.

In Fig. 4.29 an application to the Sci Pan 3 trial is shown together with experimental results (van der Voort et al., 2006). Compared to hydrogen detonations, this trial is situated at the other end of the loading density regime ($> 100 \text{ kg/m}^3$), but illustrates the validity of the model in general. The calculations are in reasonable agreement with the experimental data. Differences can be recognized and understood as the model does not take into account

- ricochet and roll (transport of missiles and debris after the first trajectory);
- coupled trajectories;
- break-up at impact.

In reality, these phenomena play a role, but are, on the other hand, not always important in the field of risk assessment. Solutions as suggested in Baker et al. (1983); Baum (1999); UFIP (2002) are widely applied and easy-to-handle models. The software PROJEX using a method developed by INERIS may also be used and seems to give better results (INERIS, 2004).

Experimental data for the projectile hazards from bursting pressure vessels is displayed in Fig. 4.30. Data available are either from actual plant accidents on a large scale or from small-scale testing. The analysis shows among other findings that 80% of all vessel ruptures resulting from fires lead to missiles, that nonfire ruptures have an increased number of fragments, that spherical vessels produce more fragments than cylindrical vessels, or that the end tubs of vessels travel further than other types of fragments. There is also the observed tendency of missiles to export fire (Leslie & Birk, 1991). More recently (Manning & Lawrence, 2017) the capabilities of CFD have been coupled with fragment launch velocity estimation and applied

**FIGURE 4.30**

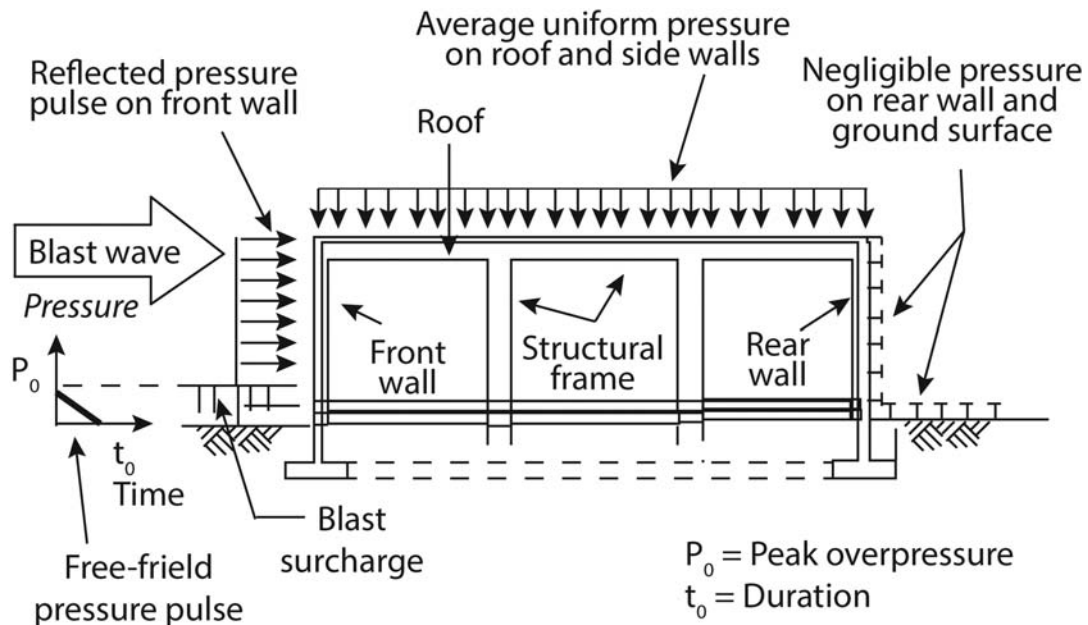
Projectiles traveling after catastrophic pressure vessel failure + : experimental data; □ : case studies.

to pressure vessel rupture, where improvements over the Baker original methodology have been demonstrated.

4.5.2 Building structural response

The effects from an explosion, which have an impact on structures, are pressure changes (blast wave) and air movement (“explosion wind”) as well as thermal radiation and flying missiles. Only a third of the chemical explosion energy is involved in the generation of the detonation blast wave; the other two-thirds are released much slower during the subsequent mixing and burning of the detonation products with the air (Federal Emergency Management Agency FEMA, 2003). In general, structural responses are highly dynamic, highly inelastic, and highly interactive. The mechanical effect of a blast wave is determined by the overpressure and the duration of the positive phase.

The blast parameters are dependent on the distance between structure and blast center. At close distances, the target is exposed to a high-intensity pressure load over a localized region; at greater distances, the load is reduced, but covers a larger surface area. The diagram in Fig. 4.31 describes the pressure load on a rectangular structure.

**FIGURE 4.31**

Simplified blast loading diagram on a rectangular building.

The dynamic interaction of a blast wave with the structure depends on the pressure-time history, that is, rise time and duration of positive phase and peak pressure or the impulse (which is the time integral of the pressure).

4.5.2.1 The initial diffraction loading

Diffraction loading is given by forces resulting from direct and reflected pressures during the initial phase. Reflection of the pressure wave at the front side amplifies the incident peak pressure with a normally reflected wave to represent an upper limit. Also density and temperature of the reflected wave are increased compared to the incident wave. The flow around the obstacle determines the further pressure development at the front and at the back side.

The net horizontal loading is that on the front minus that on the back face. The reflection coefficient, that is, the ratio between reflected and incident overpressure, is dependent on the blast wave type (pressure or shock wave), its intensity, and on the incident angle. For a pressure wave, this coefficient can have a value up to about 3 depending on the incident angle (the more usual case will have an oblique incidence). For a shock wave, it can be in the range of 2 – 8 and even higher for explosives (Fig. 4.32). Dynamic loads of fast transient pressures are imposed, if the combustion energy is inhomogeneously distributed, and are specific to the structure geometry.

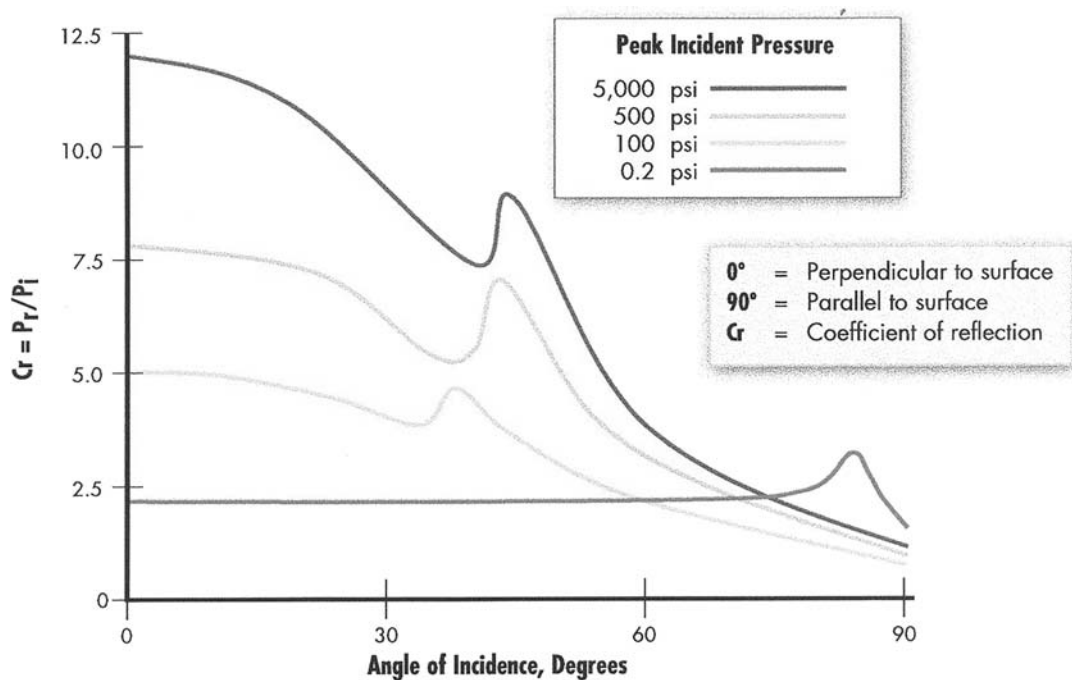


FIGURE 4.32

Reflected pressure coefficient as a function of the incidence angle.

4.5.2.2 The drag loading

After the diffraction phase is completed, the structure is subjected to a stagnation pressure. The distance of the incident wave when interacting with the structure causes major pressure differences developing from the edges of the structure. Resulting from this rarefaction wave, pressures decrease. During this so-called drag phase, strong transient winds (explosion wind) with flow velocities of several 100 m/s are effective. Drag forces will particularly have an impact on smaller structures such as pipe work. Load duration during vapor cloud explosions may be long enough to be comparable with the time required for the dynamic response time of the structure. In case of large structures, the rarefaction from the edges is insignificant.

4.5.2.3 Confined areas

In a confined or partially obstructed area, an explosion will create a structure loading, where two phases can be distinguished, the reflected blast loading followed by the gas loading phase. When the pressure wave hits the (rigid) wall, gases are brought to a rest and the wave is reflected. At normal incidence, the reflected shock wave further compresses the burnt gases increasing the pressure by about a factor of 2.3. For the case of a detonation in a hydrogen mixture, Figs. 4.33 and 4.34 show the effects of normal reflection on the resulting pressure and temperature in the wall vicinity.

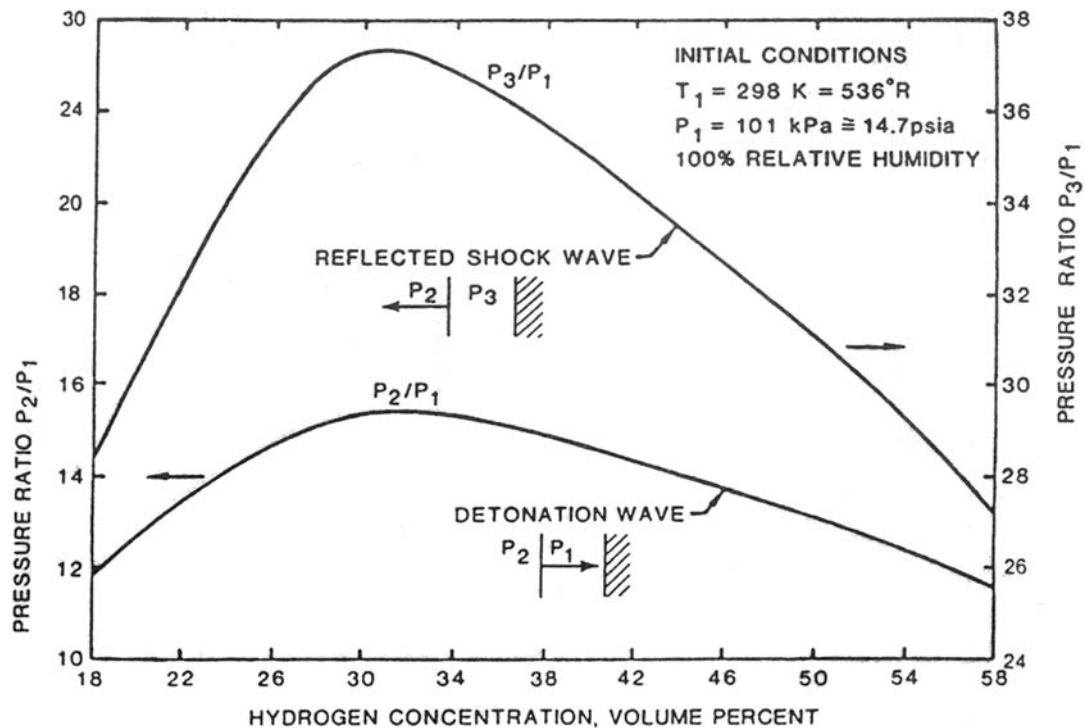


FIGURE 4.33

Theoretical detonation pressure and normally reflected detonation pressure for hydrogen – air mixtures (IAEA, 1990).

The mixture of reflected pressure waves and deflected air flows are the result of reverberation of the initial high-pressure, short-duration reflected wave with the amplitude decaying with each reflection until eventually pressure levels out at gas pressure loading. The latter phase lasts longer, the less venting is available. The more complex the structure, the more difficult is the prediction of the critical conditions for mechanical failure for a given load history.

For a vented volume or a volume with weak surfaces, the venting formula (NFPA68, 2002; prEN14994, 2004) may be a tool in order to predict the pressure applied. In case of a detonation, the loading may be divided in two parts (USACE, 1990), a shock pressure and a constant load.

4.5.3 Assessment of object responses

Forces acting on a structure will lead to a deformation to an extent, which depends on the material properties and structure composition. For a static or quasi-static load, that is, a constant or slowly changing load as from a simple deflagration, it will be in equilibrium with the internal forces resulting in a deformation of the structure. For a dynamic load, that is, a fast load transient, however, a dynamic contribution from inertia forces will add to the equilibrium, which can show positive or

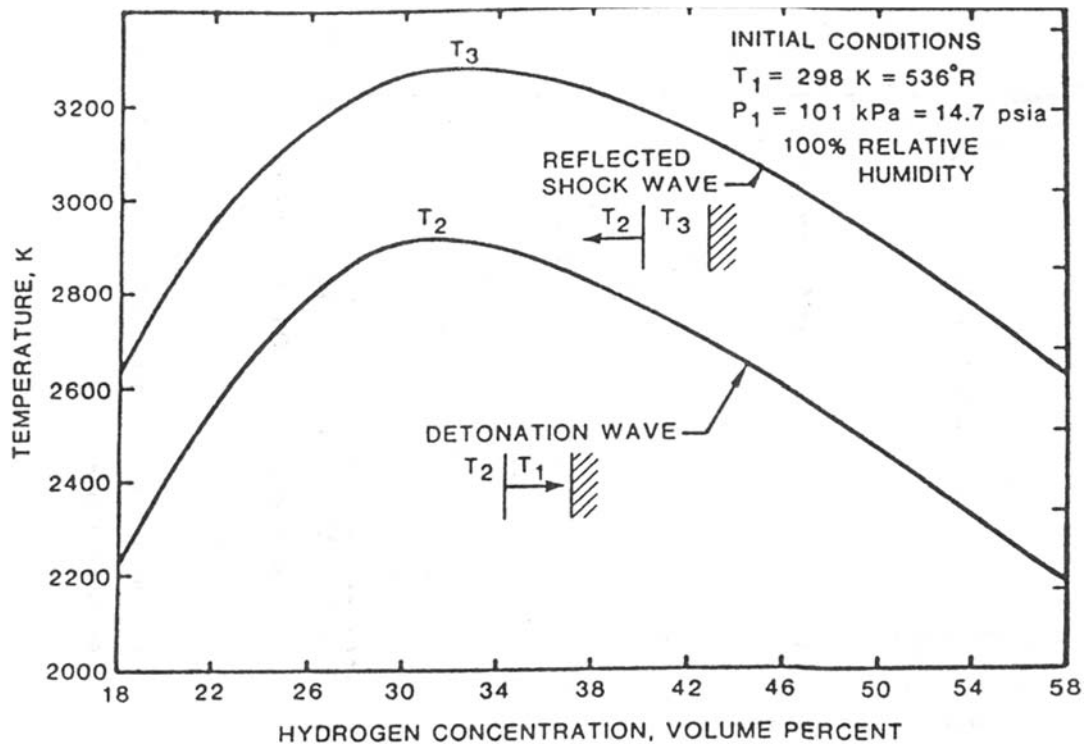


FIGURE 4.34

Theoretical detonation temperature and normally reflected detonation temperature for hydrogen – air mixtures.

negative acceleration, that is, mass and stiffness of the structure will play a major role. The load from a gas explosion is considered a dynamic load due to its short overpressure duration, which is typically in the range of 100–200 Ms. Detonations tend to excite the high natural frequencies of a building, whereas deflagrations are more effective for the lower frequencies. It appears to be technically more difficult to design a building against both explosion modes rather than only one.

Methods used in order to determine structural response and resistance belong to three main categories:

- Empirical methods that are widely used in risk assessment and that are mainly based on pressure peak values or P-I diagrams.
- Analytical methods.
- Numerical methods that can be handled with the help of a CFD code coupled with a finite element method.

4.5.3.1 Empirical methods—methods based on pressure peak values

An empirical and very global approach of determining the strength of structures is to relate overpressures to the degree of observed damage. This very useful and easy handling method is widely

used. The relationship between pressure and damage, which is derived from TNT explosions, cannot satisfactorily be transferred to vapor cloud explosions. The pressure decay from a TNT explosion is much faster than from a vapor cloud explosion. The high impulse and the suction effect due to the below-atmospheric pressure phase will certainly result in a different damage pattern. Thus damage criteria such as those derived by Schardin (Fig. 4.35) from TNT explosions are not directly applicable (Giesbrecht, 1988). Many pressure criteria were defined in the past related to various structures and specific components, however, varying over a large uncertainty range. A rough classification is given in Table 4.8.

A useful pressure peak for domino effects for risk assessment may also be defined at a value of 20 kPa (MEDD, 2004).

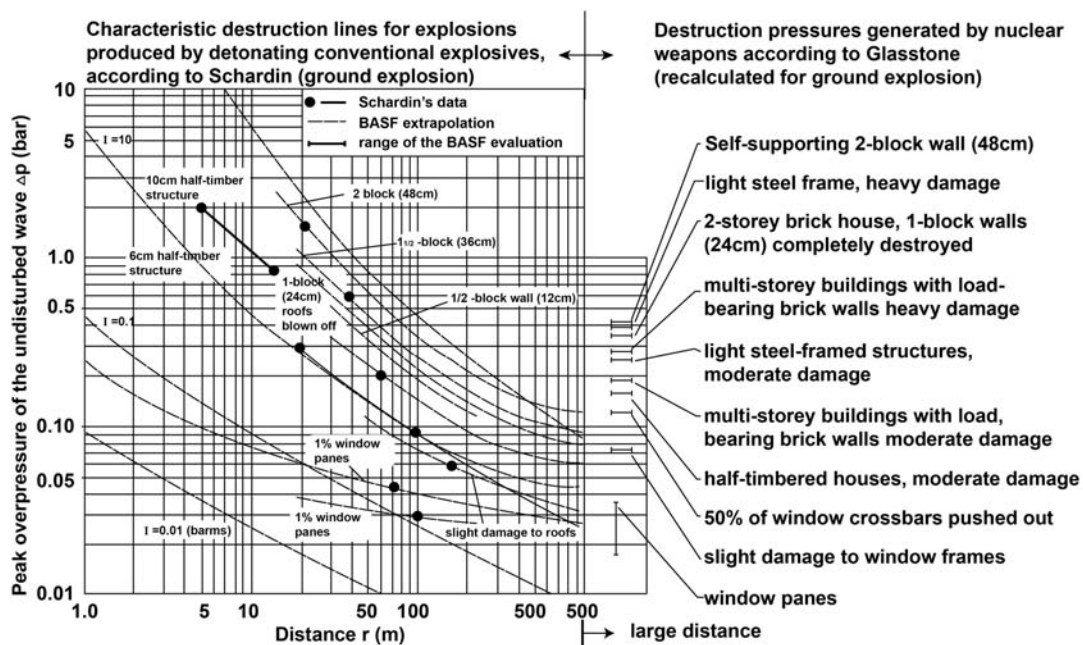


FIGURE 4.35

Schardin destruction curves for detonation waves.

Table 4.8 Damage classification (CPR16E, 2005: GreenBook) (Committee for the Prevention of Disasters, 1992).

Zone	Damage level	Overpressure of incident blast wave (kPa)
A	Total destruction	> 83
B	Heavy damage	> 35
C	Moderate damage	> 17
D	Minor damage	> 3.5

4.5.3.2 Methods based on P-I diagrams

Another global approach may couple pressure peak with impulse. The impulse, that is, the pressure negative and positive integrated over time, is a measure for the explosion energy (Fig. 4.36), which also varies in time and space over the exposed structure surface. Damage to the structure resulting from a blast wave may be subdivided into direct effects and what is named progressive collapse, a kind of secondary failure following the change of the load pattern on a structure due to the direct effects. Features of a P-I diagram are the asymptotes in P and I direction and the monotonic relation between P and I, which suggests a subdivision into three regimes: impulse-controlled, peak load-controlled, and an intermediate dynamic stage (Li & Meng, 2002). P-I diagrams are being widely used in damage assessments not only for structural damage, but also for predicting blast-induced human injuries. They are providing useful information on the vulnerability of targets.

Damage levels can also be visualized in pressure-impulse (PI) diagrams, where different regions can be defined by iso-lines. An example is given in Fig. 4.37 showing the experimental results for the observed damage in percentage, after different types of houses were exposed to a certain explosion (pressure/impulse) load (CPR16E, 2005:GreenBook). Important for the damage effect of a short-term load (=shock wave) is only the impulse, whereas it is the maximum overpressure for that of the longer-term load (=pressure wave). The solid lines in the figure indicate the lower

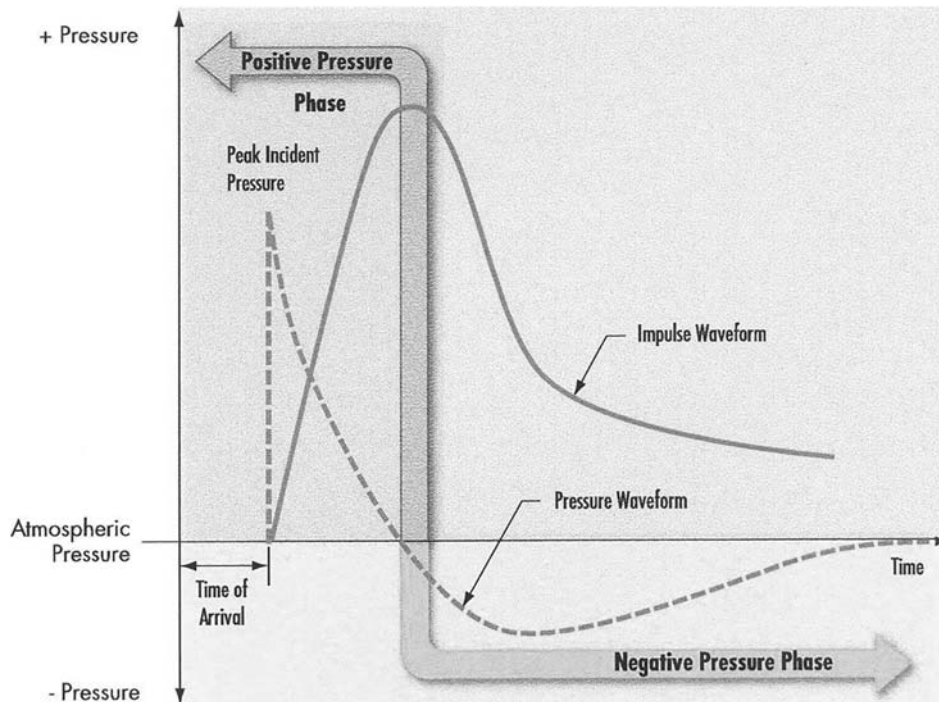
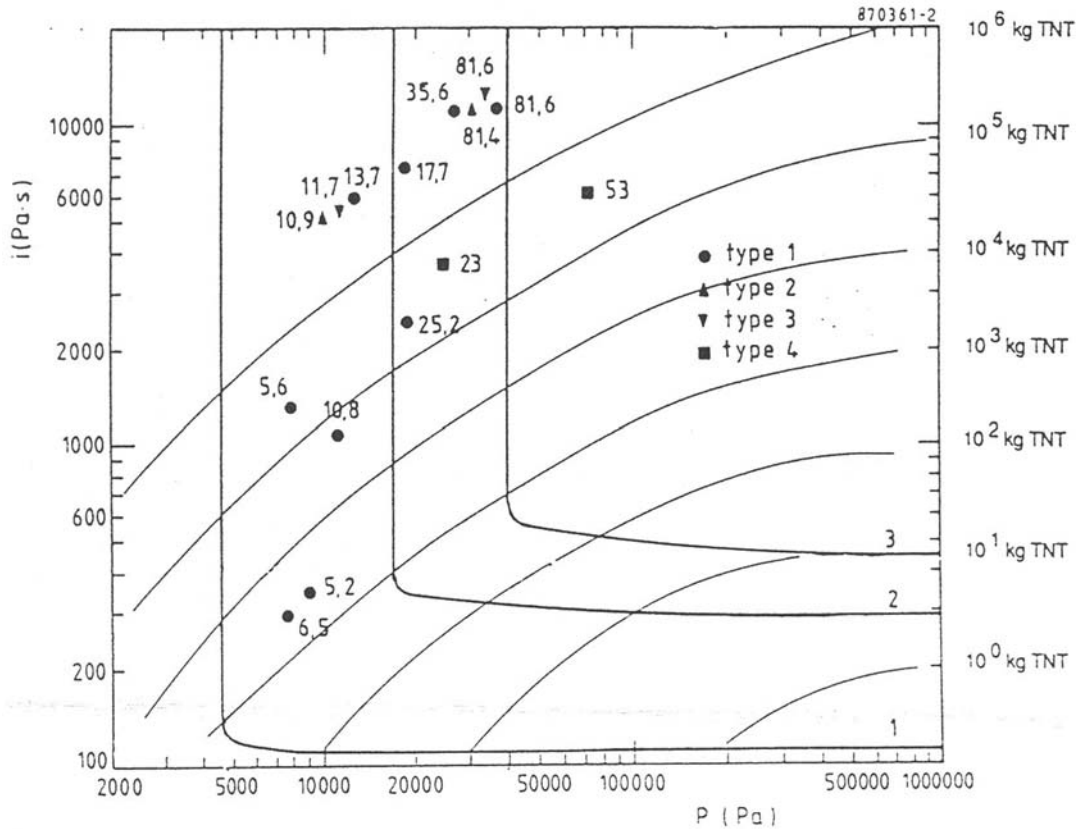


FIGURE 4.36

Transient behavior of impulse.

**FIGURE 4.37**

Pressure-impulse diagram with experimental damage values for different types of houses.

boundaries for light damages, for severe damages, and for collapsing structures of the houses investigated.

The estimation of the level of damage is usually done taking either fixed-limits methods or the PROBIT method. In a fixed-limits method, the hazard level is compared to fixed limits like IDLH (Immediately Dangerous to Life or Health) values or pressure thresholds, as given in the literature. It is a simple method, but may lead to wrong conclusions in transient hazards. In such cases, the PROBIT method is the more appropriate one. First a hazard load, L , is estimated, which could be the overpressure in an explosion or the integrated thermal flux at a certain location. Then the PROBIT or probability unit, y , is given by

$$y = k_1 + k_2 \cdot \ln(L) \quad (4.7)$$

Eq. (4.7) can be related to a certain probability of death, injury, or damage. k_1 and k_2 are empirical parameters, which are specific to, for example, toxic chemicals or fire and explosion effects and

can also be taken from the literature (Canadian Society for Chemical Engineering, 2004). For the assessment of the probability to obtain a certain level of damage, so-called probit functions have been introduced and suitable damage criteria have been defined (CPR16E, 2005:GreenBook).

Computer simulation techniques have been developed by SRI International, United States, for analyzing hydrogen explosions and the subsequent response of structures and humans. The thermodynamic code TIGER is used to calculate explosion pressures and expansion characteristics, which are input to the DYNA3D model to determine the impact on structures. A simplification of the blast load damage analysis has led to the development of the PI methodology, which is based on the observation that for a broad range of structures, the final damage state depends on peak pressure (P) and total impulse intensity (I) only. This methodology allows for a quick and easy evaluation of accident scenarios by employing a continuously growing library of PI load and damage curves that are based on either experimental data or computer simulations (Sanai, 1996).

4.5.3.3 Analytical methods

A structure can be schematically represented by a system of masses coupled with springs or dampers. For the static case, if linear-elastic or nonlinear-elastic forces are acting, displacements of the masses become zero again, when the load disappears. In case of plastic or elasto-plastic behavior, displacement is zero or very small, until the maximum load is reached. Under a static load, the structure will then fail; under dynamic load, it may retain a residual displacement. In general, structures must be designed to react elastically under typical loads like wind. Plastic displacement must be limited to abnormal load conditions. The maximum displacement depends on load duration, t_D , and the natural frequency of the structure, T . For low t_D/T ratios, the displacement is smaller than for static loads. For large t_D/T ratios, the displacement can be larger than understatic load conditions. Other important parameters are the static strength and the ductility. Load schemes are distinguished between a step function for a long-duration pressure wave and an impulse load for a short-impact shock wave.

Detonations tend to excite the high natural frequencies of a building, whereas deflagrations are more effective for the lower frequencies. It appears to be technically more difficult to design a building against both explosion modes rather than only one. However, for hydrogen explosions experience proves that detonations or strong deflagrations must be preferred. Several analytical procedures may be distinguished. They have to be coupled with structural engineering calculations. Still these methods are likely to lead to very conservative structural designs, and are rather suitable for hand calculations.

A comparison between detonations of explosives and blast waves resulting from nuclear weapon explosions, characterized by quasi-static pressure due to a longer impulse time, shows that, assuming the same damage, the detonation pressure or the pressure resistance of an object is much higher than the resistance against a blast wave from nuclear tests (Pfortner, 1975). The pressure resistance behavior of a building under detonative dynamic and quasi-static loading derived from numerous detonative explosion studies can be summarized in an empirical equation for the quasi-static reference overpressure of the building p_{st} :

$$p_{st} = 0.15 \cdot p_r^{2/3} \quad (4.8)$$

where p_r is the perpendicularly reflected overpressure or the pressure resistance of the building subjected to a detonation. If the TNT equivalent, as derived from the damage of some of the severe

accidents, is interpreted as the incident pressure wave in the order of 70 kPa, resulting from a deflagration, the respective quasi-static pressure would be much smaller, with $p_{st} = 22$ kPa.

4.5.3.4 Dynamic load factor

The analytical procedure is usually simplified by introducing a so-called dynamic load factor (DLF), which is defined as the ratio of maximum dynamic displacement over static displacement. It transforms a dynamic peak load into a static load with the same effect on the structure. The DLF is dependent on the dynamic load time and the natural frequencies of the structure. For long explosion times and in case of an idealized triangle-shaped shock wave load, the DLF approaches its boundary limit of 2 (MEDD, 1994).

4.5.3.5 Single degree-of-freedom model

In a simple method, a static working load is assumed to simulate the effects of blast loads onto structures. This conservative approach, however, cannot predict structure performance and seems to be not optimal for transient blast loads. An improvement of modeling is given with the quasi-static methods, which specify a triangular pressure pulse and determine the dynamic structural capacity as a function of the material strengths of the structure. The structural response can be found by using the charts. Single degree-of-freedom models analyze critical components in terms of their structural resistance and predict the response of the structure, which then determines the damage level (USACE, 1990).

4.5.3.6 Numerical methods

More information is obtained from multi-degree-of-freedom models, sophisticated CFD models with the possibility to consider flame propagation, and pressure profiles also from local explosions. Dynamic finite-element analyses with tools such as ADINA, LS-DYNA, AUTODYN, ABAQUS Explicit may provide the best judgment on whether or not a structure is able to withstand a blast wave.

However, one must be very careful when using software that is still under development and which is still not fully validated to industrial applications. Numerical methods are also usually very sensitive to parameters and calculation grid changes.

4.6 Physiological impact

4.6.1 Damage by low temperature releases

Skin contact with liquid hydrogen or cold hydrogen gas may cause severe cold burns, comparable with those caused by boiling water. Unprotected skin may freeze onto surfaces cooled by the liquid, causing severe damage on removal. Prolonged skin exposure to cold hydrogen may result in frostbite. A symptom is local pain, which usually gives warning of freezing but sometimes no pain is felt or it is short-lived. Frozen tissues are painless and appear waxy, with a pale whitish or yellowish color. Thawing of the frozen tissue can cause intense pain. Shock may also occur. The eyes are particularly susceptible—even small splashes of liquid hydrogen, or short exposures to cold vapor or gas, may cause instant freezing of eye tissues and permanent damage.

Transient exposure to very cold gas produces discomfort in breathing and can provoke an attack of asthma in susceptible people. Prolonged inhalation of cold vapor or gas may cause serious lung damage. Prolonged exposure of the entire body to cold can result in hypothermia.

4.6.2 Asphyxiation by hydrogen

Hydrogen is not poisonous, but as with any gas (except oxygen) a risk of asphyxiation exists mainly in confined areas as a result of oxygen depletion. Normal air contains around 20.8% of oxygen, by volume. Besides the dilution by mixing with other gases, oxygen may be consumed in combustion of hydrogen or other burning gases and may be depleted via condensation on very cold surfaces like liquid hydrogen pools. Thus diluting the oxygen volumetric content below 19.5% will cause effects on human beings. Alarm levels are generally set at 19% oxygen. This is less than 2% below normal levels, so it is important that sensors are stable in order to avoid false alarms. The different stages of asphyxiation at ground level are related to the remaining oxygen concentration as shown in Table 4.9.

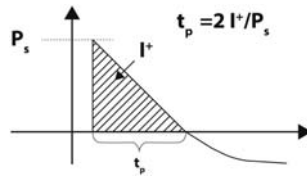
4.6.3 Human harm from blast

4.6.3.1 Direct blast effects

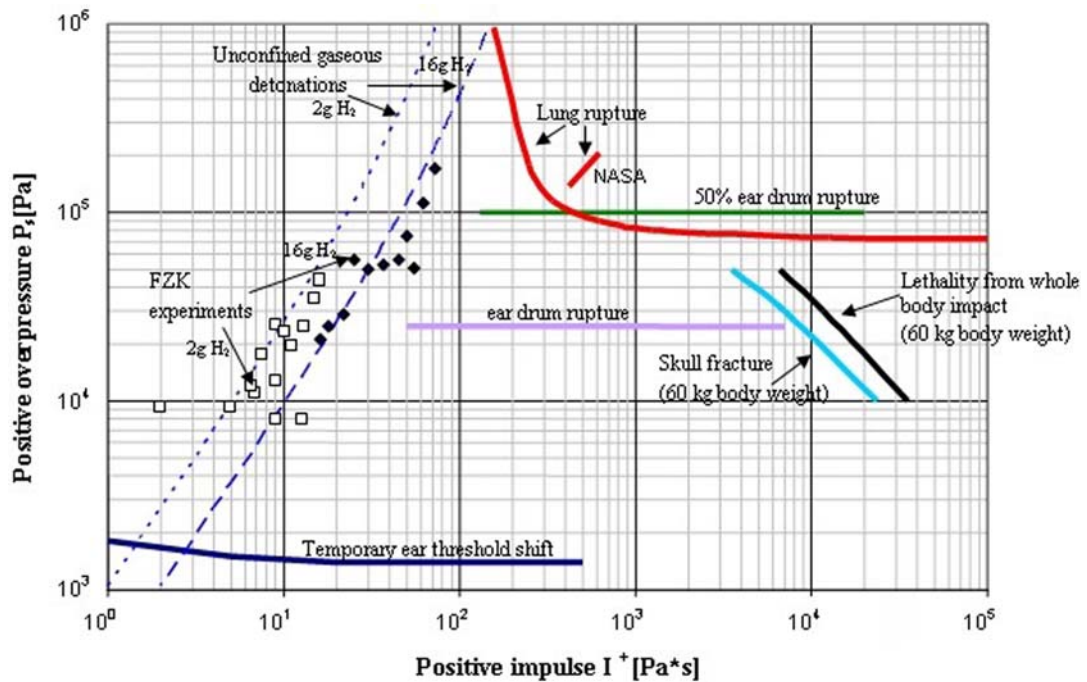
These direct effects, also called primary effects, are caused by the dynamic pressure waves. Although the human body is able to adapt to slow pressure changes (diving, high altitudes, etc.), the dynamic changes in a blast front may cause severe damage. Those organs, where large differences in densities are encountered, like the lungs or the inner ear, are particularly vulnerable. Ear damage does not lead to death, but due to the ear's high sensitivity it is often used as an indicator for an exposure. Lung damage depends on a combination of peak overpressure P_s and on the pulse duration t_p (Fig. 4.38).

Table 4.9 Effect of O₂ concentration from NASA (NASA, 1997), and with DNV data (DNV Technica, 2001).

O ₂ (vol%)	Effects and symptoms
19–21	No discernible symptoms can be detected by the individual.
15–19	Reduction of physical and intellectual performance without the sufferer being aware. Early symptoms in persons with heart, lung, or circulatory problems may be induced.
12–15	Deeper respiration, faster pulse, poor coordination.
10–12	Headaches, giddiness, poor judgment, slightly blue lips. Risk of death below 11 vol%, tolerance time 30 min.
8–10	Nausea, vomiting, unconsciousness, ashen face, fainting within a few minutes without prior warning, mental failure, tolerance time 5 min.
6–8	Fainting occurs after approximately 3 min. Death in 8 min; 50% death and 50% recovery with treatment in 6 min, 100% recovery with treatment in 4 – 5 min.
3–6	Coma in 40 sec, respiration ceases, death or permanent brain damage, even if rescued.
0–3	Death within 45 sec.

**FIGURE 4.38**

Simplified relationship between positive impulse and time of the overpressure peak.

**FIGURE 4.39**

Threshold data from Baker, NASA, and gaseous detonation data.

Similar as for the structural effects, threshold limits for ear and lung damage are displayed in PI diagrams (Fig. 4.39).

A more recent analysis of fatality levels associated with blast waves has been carried out by [van der Voort et al. \(2016\)](#) updating the longstanding data of Bowen from the 1960s. The results from this are in Fig. 4.40 for three fatality probabilities.

Not only for the general mixing and detonation initiation but also for the physiological effects the spatial confinement is a very important factor. Generally one has to expect an increased immediate and late mortality in closed space blast scenarios compared to open-air explosions

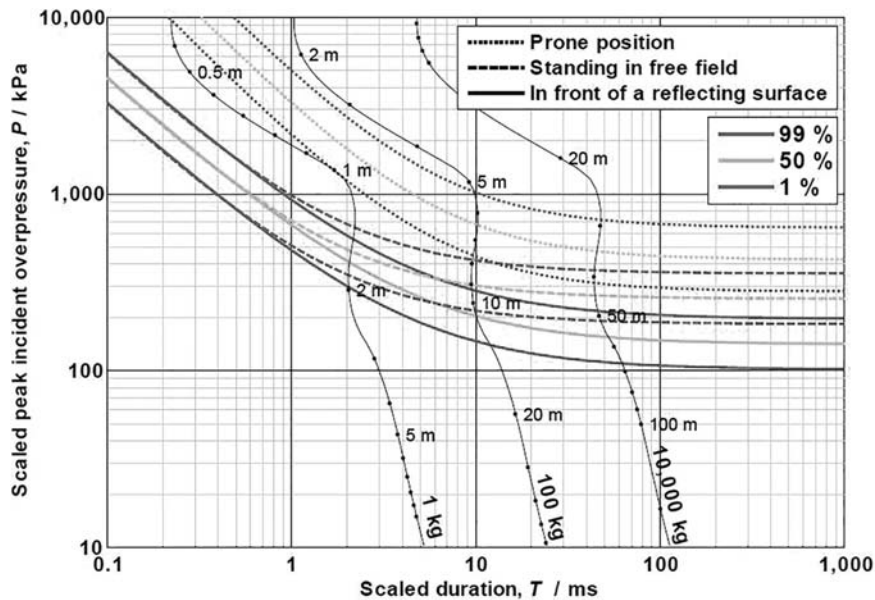


FIGURE 4.40

Peak overpressure versus duration for fatality levels (1%, 50%, and 90%). Load curves for a hemispherical surface burst 1 kg, 100 kg, and 10,000 kg TNT are included. *TNT*, trinitrotoluene.

(Kaiser, 2000; Kluger, 2003). However, usually only little mortality is due to these primary effects compared to the more severe combination of indirect blast effects, like missiles, body translation, and associated impact and heat effects.

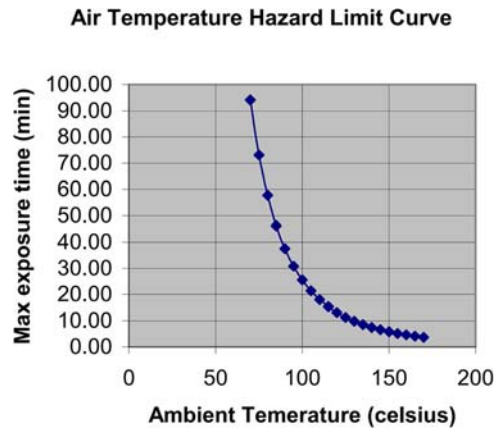
4.6.3.2 Indirect blast effects

Indirect blast effects include secondary effects, these are generated by missiles (e.g., accelerated parts of the pressure vessel, parts of the building, glass) and tertiary effects linked to the body translation. Especially the impact, the deceleration when hitting a wall or any other a solid structure, can cause skull fractures with traumatic consequences, even death. With a simplified model the body displacement caused by the blast may be calculated and a lethality threshold may be attributed to the resulting velocities, see Baker et al. (1983). See Fig. 4.38. with the skull fracture as the representative indirect blast effect.

4.6.4 Thermal effects from fires

Thermal effects from hydrogen fires involve two main phenomena, which could harm human beings: elevated air temperature and heat radiation. Depending on the scenario only one or both in a suitable combination has to be considered.

Air temperature effects are represented in graphical form in CPR16E (CPR16E: 2005 GreenBook) as shown in Fig. 4.41.

**FIGURE 4.41**

Time to incapacitation as a function of the air temperature.

Table 4.10 Elevated air temperature effects.

Temperature (°C)	Effects and symptoms
127	Difficult breathing.
140	Tolerance time 5 min.
149	Mouth breathing difficult, T limit for escape.
160	Rapid, unbearable pain with dry skin.
182	Irreversible injury in 30 s.
203	Respiratory systems tolerance time less than 4 min with wet skin.

Below 70°C, no severe effect has to be expected. Between 70°C and 150°C the time to incapacitation may be 94 min and 6 min, respectively. The figure shows a plot of the empirical dependency. The data can also be represented as in terms of varying degrees of harm for differing air temperatures.

For open fires involving hydrogen, the air temperature is likely to be low except close to the flame boundary, and above 150°C, radiation effects become the dominant factor (Table 4.10).

Heat radiation may cause pain, first, second, and third degree burns as well as fatal burns. The different burn types are characterized by the depth they reach in the skin. Similar as for the pressure effects a combination of the peak load, here radiation intensity, and the characteristic duration are the most important factors. Above 1.6 kW/m², negative effects on human beings have to be expected. Table 4.11 relates exposure times and pain thresholds.

The human response to thermal radiation is represented by a dose equation D, of the following form

$$D = I^{4/3} t \left((kW/m^2)^{4/3} s \right).$$

Table 4.11 Threshold of pain (Kaiser, 2000).

Exposure time (s)	Radiation intensity (kW/m ²) lightly clothed
60	1.7
40	2.3
30	2.9
16	4.7
9	6.9
6	9.5
4	11.7
2	19.9

Table 4.12 Maximum radiation exposure time (DNV Technica, 2001).

Exposure time leading to death	Radiation intensity (kW/m ²) lightly clothed	Radiation intensity (kW/m ²) protectively clothed
2 min ... 10 min	2	4
1 min ... 2 min	4	8
0.5 min ... 1 min	10	13
< 0.5 min	16	25

This may be used in connection with several available probit functions for Y representing the degree of resulting harm, for example, $Y = -10.7 + 1.99 \ln D$ from Lees (1994).

The following based on the TNO probit function relates the exposure time and the radiation intensity for 100% fatality (Table 4.12).

Using thermal dose units and for a specific received flux level (kW/m²) the percentage chance of a particular harm may be plotted, which gives an indication of tolerable exposure times as shown in models from Jagger (2004) (Fig. 4.42).

Graphs of this kind can be used in conjunction with radiation prediction models for the various types of hydrogen radiation sources discussed above. In this way, safe distances from potential combustion events can be established. An example is shown below from the work of Houf and Schefer (2007), relating to flux levels set out by the International Fire Code Council Inc (2002) (Fig. 4.43).

Besides the IR content hydrogen combustion produces UV radiation capable of sunburn-like effects. Hydrogen fires are difficult to see at daylight and due to the optical properties, the heat of smaller flames is felt late.

4.6.5 Personal protective equipment

Using the appropriate protective equipment can reduce the possible consequences of the above-described hazards. The concerned personnel should be protected against exposure to cryogenic temperatures, high temperatures, thermal radiation from a hydrogen flame, and oxygen-deficient atmospheres. Procedures that are established for operations involving hydrogen should describe the

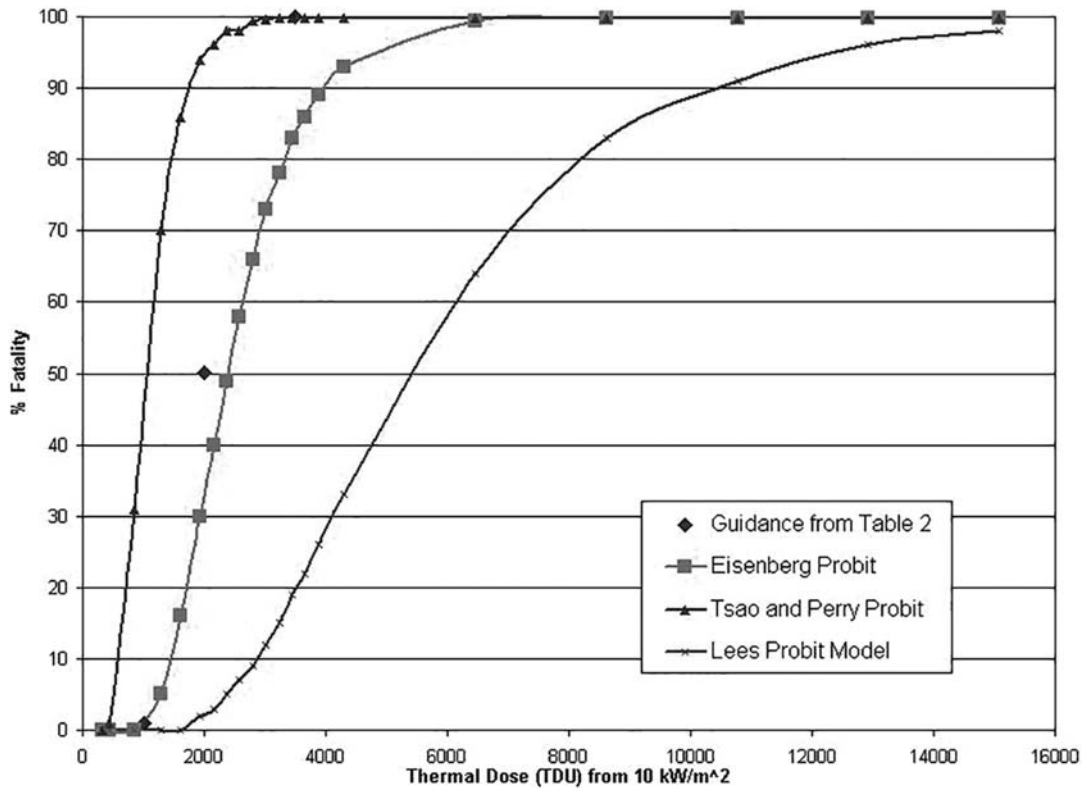


FIGURE 4.42

Fatality probabilities from thermal radiation at 10 kW/m^2 based on a range of probit models.

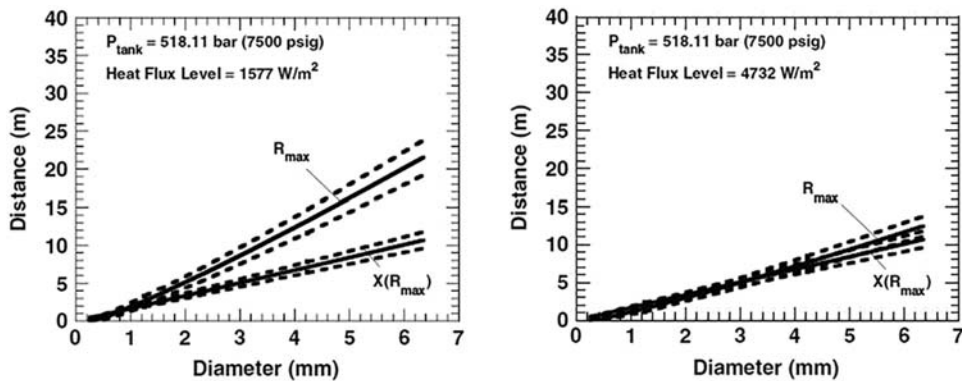


FIGURE 4.43

Radial distance from a jet fire source as a function of release nozzle diameter for two levels of received flux. $R_{\max}$ is radial distance, $X(R_{\max})$ is axial distance from jet exit.

personal protective equipment (PPE) that is needed for the operations to be performed. Some general guidelines for PPE that should be considered beneficial in working with hydrogen are summarized below. These guidelines do not address PPE that should be considered when involved in other activities such as working on electrical circuits or performing a cleaning or decontamination operation. Some specific recommendations for PPE are as follows:

- Eye protection or better complete face shield should be worn when liquid hydrogen is handled.
- Properly insulated gloves should be worn when handling anything that comes in contact with liquid hydrogen or cold gaseous hydrogen. The gloves should fit loosely, remove easily, and not have large cuffs.
- Full-length trousers, preferably without cuffs, should be worn with the legs kept on the outside of boots or work shoes.
- Closed-toe shoes should be worn (open or porous shoes should not be worn).
- Clothing made of ordinary cotton or flame-retardant cotton should be worn. Avoid wearing clothing made of nylon or other synthetics, silk or wool because these materials can produce static electricity charges that can ignite flammable mixtures. Synthetics can melt and stick to the flesh, causing greater burn damage. Any clothing sprayed or splashed with hydrogen should be removed until they are completely free of hydrogen.
- Self-contained breathing equipment should be worn when working in a confined space that may have an oxygen-deficient atmosphere.
- Portable hydrogen and fire-detection equipment should be used to warn of hydrogen leaks and fires.
- Alternatively it is often recommended to wave with a broom in front of oneself or to pluck some grass and throw it in the direction of the intended movement. If the broom or the grass comes in contact with the barely visible flame the smoke indicates the flame position.
- Personnel should ground themselves before touching or using a tool on a hydrogen system.
- The use of spark-proof tools is often recommended; however, the energy required for ignition of a flammable hydrogen/air mixture is so small that even spark-proof tools can cause an ignition. Consequently, all tools should be used with caution to prevent slipping, glancing blows or dropping, all of which can cause sparks.
- Water sprays and wet clothes may reduce the thermal effects induced by hydrogen flames considerably.

The immediate treatment of persons who come in contact with liquid hydrogen or have been exposed to the very cold gases is to loosen any clothing that may restrict blood circulation and seek immediate hospital attention for all but the most superficial injuries. Do not apply direct heat to the affected parts, but if possible place in lukewarm water. Sterile dry dressings should be used to protect damaged tissues from infection or further injury, but they should not be allowed to restrict the blood circulation. Alcohol and cigarettes should not be given.

4.7 Environmental impact

One major claim for the introduction of hydrogen-based technologies is the promising environmental benefits of such technologies. Hydrogen is considered a clean fuel capable of avoiding the greenhouse effect caused by carbon dioxide releases using fossil fuels. A supporting argument is as follows: hydrogen may be produced by water electrolysis and under usage it is oxidized back to

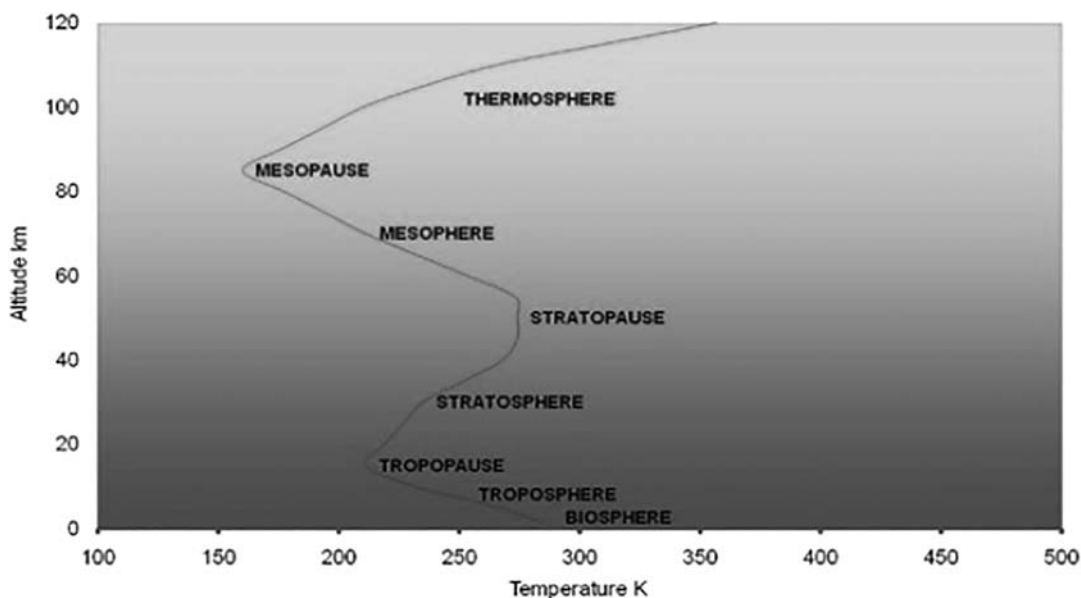
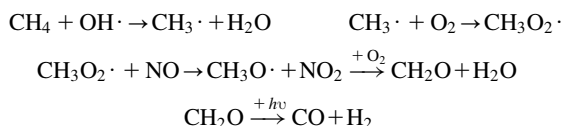


FIGURE 4.44

Definition of the atmosphere based on the average temperature profile.

water without producing any pollutants. However, this is to a certain extent an idealistic view as the major amount of hydrogen is currently produced based on fossil fuels, though in the future hydrogen to a much higher degree may be produced using sustainable energy sources as, for example, wind or water power. The environmentally friendly production of hydrogen depends therefore strongly on the technology applied. The usage of fossil fuels like natural gas conversion to produce hydrogen may result in an increase of greenhouse gas emissions and other pollutants compared with the present situation. More fossil fuels are needed to produce enough “hydrogen energy” for the customers. A solution to avoid such pollution could be technologies of carbon sequestration and other cleaning measures to capture the pollutants to be considered for the specific fuel used, which also would favor central, large-scale hydrogen production and distribution facilities (Fig. 4.44).

Hydrogen is a permanent gas and any diffusive, targeted, or accidental release will emit predominantly into the atmosphere. Presently, the major sources for hydrogen emissions are biomass burning, for example, forest fires, emissions from the traffic using - internal combustion engines. It is also produced by photochemical atmospheric reactions of formaldehyde. It is resulting, for example, from the methane oxidation cycle in the atmosphere as shown in the following reaction equations where one photolysis path for formaldehyde (CH_2O) is generating hydrogen.



The hydrogen content of the atmosphere is compared to other atmospheric gases and is found to be on the trace gas level. The atmosphere is divided into different spheres as shown in the figure based on an average temperature profile, which is very important concerning the gas exchange between these spheres (Table 4.13).

The hydrogen concentration profile with altitude is found constant at a level of about 0.5 ppm in the troposphere and stratosphere as shown in Table 4.14. Only in the very high atmosphere, is there an increase of hydrogen.

Hydrogen is the only gas capable of escaping into space, but it is found that nearly 100% of the hydrogen is degraded by the photochemical atmospheric processes or deposited back to the biosphere. The atmospheric degradation accounts for about 25%, while the dry deposition is about 73% (Schultz, Markert, & Pilegaard, 2004a; Schultz, Markert, & Pilegaard, 2004b). The current hydrogen releases are estimated as follows: 40 million t/yr from methane, VOC releases and their photochemical degradation, 16 million t/yr from biomass burning, 15 million t/yr from traffic and industries, and 6 million t/yr from soils and ocean. The sinks for hydrogen are: 19 million t/yr degraded by reaction with OH free radicals, and 56 million t/yr soil up-take. The numbers are uncertain and are estimated to be within ± 10 million t/yr for traffic and industry

Table 4.13 Average relative composition of the troposphere at mid-latitudes (Heicklen, 1976).

Gas	Concentration (ppm)
Nitrogen, N ₂	780840
Oxygen, O ₂	209460
Argon, Ar	9340
Carbon dioxide, CO ₂	325
Sum of noble gases (He, Ne, Kr, Xe)	24.6
Methane, CH ₄	1.4
Hydrogen, H ₂	0.5
Nitrous oxide, N ₂ O	0.25
Carbon monoxide, CO	0.08
Ozone, O ₃	0.025
Nitroxides, NO + NO ₂	0.006

Table 4.14 Average relative hydrogen concentration in different altitudes (Heicklen, 1976).

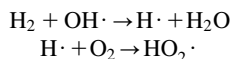
Altitude (km)	Hydrogen (ppm)
0	0.5
20	0.5
40	0.5
60	0.5
80	4.3
100	1.0

emissions and ± 15 million t/yr for soil deposition. Therefore the biological processes to degrade hydrogen are very important for its overall atmospheric balance. These processes are not well-understood presently as, for example, the capacity for hydrogen degradation is unknown. The biosphere could be acting as a large buffer keeping the atmospheric hydrogen concentration constant even though the releases are increased due to the activities from a hydrogen economy.

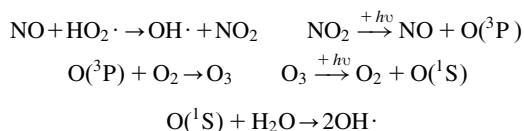
Some concerns on possible adverse environmental effects caused by large increased atmospheric hydrogen concentrations have been described in the literature (Logan, 1981; Schultz, 2003; Tromp, 2003; Prather, 2003; Schultz et al., 2004a,b; Wennberg et al., 1998). Based on gaseous hydrocarbon experience, it is estimated that a full hydrogen economy may allow up to 10% of generated hydrogen to leak into the atmosphere. In the following, the present knowledge of the possible effects of this scenario is discussed.

Hydrogen is not a greenhouse gas, as it does not absorb electromagnetic radiation within the IR spectrum. Therefore higher atmospheric hydrogen concentrations will not directly contribute to the climate forcing. The consumption of hydrogen will result in water regardless of whether it is used in fuel cells or burned under controlled or uncontrolled conditions, like under accidental fires or explosions. The water will be released into the troposphere with its normally huge content of water. Therefore the additional water will not raise considerably the overall content of water vapor in the troposphere. An adverse effect is identified when hydrogen is atmospherically degraded in the lower stratosphere, where the persistent ice (from the water) may have a cooling effect that again influences the temperature-dependent ozone depletion mechanism.

This has a direct effect on the above-mentioned possible cooling effect for the lower stratosphere, as has been predicted through model calculations. It has been shown (Tromp, 2003) that a fourfold increase of atmospheric hydrogen will have a notable effect on the ozone depletion, with a sevenfold increase in H_2 leakage producing an 8% reduction in ozone concentrations. For atmospheric photochemical degradation of hydrogen, the initiating and rate determining step is by an OH free radical reaction with hydrogen.



For atmospheric photochemical degradation of carbon monoxide or other hydrocarbons, the initiating and rate determining step is by an OH free radical reaction with the CO or the hydrocarbon leading to stable products and other free radicals. The latter reacts with other atmospheric components, for example, NO to give ozone and to regenerate OH free radicals in chemical and photolytic reaction steps. By that mechanism the atmosphere contains a stable OH concentration of about 10^6 molecules per cm^3 that is sufficient to remove pollutants.



Other reviews of the effect of hydrogen emissions (Dick, 2018) conclude that the knock-on impact on global warming is negligible, assuming renewable hydrogen is used and that, although most studies agree on a decrease in ozone concentrations, the effects are more likely to be around 1% rather than the 8% previously suggested.

The emissions of hydrocarbons and nitrogen oxides lead to the generation of ozone. While this is a very important protection against short wave UV radiation in the stratosphere, ozone is to be considered a pollutant in the troposphere. It is part of the effects that lead to “forest death” by acid rain as ozone destroys the protecting wax layer on leaves, and it has adverse health effects to people. In summertime, dangerous ozone levels can build up leading to restrictions on traffic, for example, in Germany, in order to mitigate high ozone levels. Therefore the establishment of the hydrogen economy will improve the air quality in the cities as the hydrocarbon and nitrogen oxide emissions are reduced. On the other hand nitrogen oxide is an important intermediate for the generation of the constant background concentration of OH free radicals.

It is clear that less environmental pollution using hydrogen is only to be achieved using sustainable primary energy sources for hydrogen production. Using fossil fuels without carbon dioxide sequestration will result in total in even more pollutant emissions. Therefore the overall environmental benefits need to be assessed using live cycle assessments for the larger-scale production systems.

The main conclusion is that environmental impacts of hydrogen are connected with a scenario involving significant increases of the emissions into the atmosphere. Most of the hydrogen is degraded in the biosphere. More precise knowledge on the mechanisms and capacities are needed to make better model calculations on the possible increase of hydrogen due to the new technologies based on hydrogen and fuel cells. At the moment, a precaution may be to reduce the emissions as much as possible, which is not only related to the environment, but also to safety, as hydrogen fires and gas phase explosions are very prominent concerns for hydrogen applications.

References

- AICHe CCPS. (1994). *Guidelines for evaluating the characteristics of vapor cloud explosions, flash fires, and BLEVEs*. New York: Center for Chemical Process Safety, American Institute of Chemical Engineers.
- Baker, W. E., Cox, P. A., Westine, P. S., Kulesz, J. J., & Strehlow, R. A. (1983). *Explosion hazards and evaluation*. New York: Elsevier Scientific Publishing Company.
- Baum, M. R. (1999). Failure of a horizontal pressure vessel containing a high temperature liquid: the velocity of end-cap and rocket missiles. *Journal of Loss Prevention in the Processes Industries*, 12, 137–145.
- Birk, A. M., Davison, C., & Cunningham, M. (2007). Blast overpressures from medium scale BLEVE tests. *Journal of Loss Prevention in the Process Industries*, 20, 194–206.
- Breitung, W., & Kotchourko, A. (1996). Numerische Simulation von turbulenten Wasserstoff-Verbrennungen bei schweren Kernreaktorunfällen. *Nachrichten-Forschungszentrum Karlsruhe*, 28(2–3), 175–191, Research Center Karlsruhe.
- Breitung, W., & Redlinger, R. (1995). Containment pressure loads from hydrogen combustion in unmitigated severe accidents. *Nuclear Technology*, 111, 395–419.
- Canadian Society for Chemical Engineering, 2004. Risk assessment—Recommended practices for municipalities and industry. Ottawa, Canada.
- Burjor, A. (1993). *Hydrogen problems related to reactor accidents*. Denmark, Roskilde: Riso National Laboratory.
- Choudhuri, A. R., & Gollahalli, S. R. (2004). Intermediate radical concentrations in hydrogen-natural gas blended fuel jet flames. *International Journal of Hydrogen Energy*, 29, 1291–1302.

- Committee for the Prevention of Disasters. (1992). Methods for the determination of possible damage to people and objects resulting from releases of hazardous materials, CPR16E. The Dutch Ministry of the Interior and Kingdom Relations, The Hague, The Netherlands.
- Committee for the Prevention of Disasters. (2005). Methods for the calculation of physical effects due to releases of hazardous materials (liquids and gases), CPR14E. The Dutch Ministry of the Interior and Kingdom Relations, The Hague, The Netherlands.
- De Ris J. (1979). Fire radiation—A review. In *Proceedings of the Seventeenth Symposium (International) on Combustion*. Pittsburgh. The Combustion Institute, page 1003.
- Dick R. G. (2018). Hydrogen for heating: Atmospheric impacts—A literature review, BEIS Research Paper Number 21.
- DNV Technica. (2001). *Human resistance against thermal effects, explosion effects, toxic effects, and obscuration of vision*. Scandpower A/S, Det Norske Veritas, Oslo, Norway.
- Dolci F. (2016). Research Priority Workshop on Hydrogen Safety, 26–27 September 2016 Petten, The Netherlands.
- Erdogan, F., Delale, F., & Owczarek, J. A. (1977). Crack propagation and arrest in pressurized containers. *Journal of Pressure Vessel Technology, Transactions of the ASME*, 99, 90–99.
- Federal Emergency Management Agency (FEMA). (2003). *Reference manual to mitigate potential terrorist attacks against buildings*. Washington D.C: McGraw-Hill.
- Friedrich, A., Breitung, W., Stern, G., Vesper, A., Kuznetsov, M., Fast, G., ... Rottenecker, M. (2012). Ignition and heat radiation of cryogenic hydrogen jets. *International Journal of Hydrogen Energy*, 37(22), 17589–17598.
- GexCon. (2006). Gas explosion handbook. Online: <<http://www.gexcon.com>>.
- Giesbrecht, H. (1980). Analyse der potentiellen Explosionswirkung von kurzzeitig in die Atmosphäre freigesetzten Brenngasmengen, Teil I: Gesetze der Wolkenausbreitung und -deflagration aus Berstversuchen und Metallbehältern. *Chemie Ingenieur Technik*, 52, 114–122.
- Giesbrecht, H. (1981). Analyse der potentiellen Explosionswirkung von kurzzeitig in die Atmosphäre freigesetzten Brenngasmengen, Teil II: Gesetze der Wolkenausbreitung und -deflagration aus Berstversuchen und Metallbehältern. *Chemie Ingenieur Technik*, 53, 1–10.
- Giesbrecht, H. (1988). Evaluation of vapour cloud explosions by damage analysis. *Journal of Hazardous Materials*, 17, 247–257.
- Gurson, A. L. (1977). Continuum theory of ductile rupture by void nucleation and growth. *Journal of Engineering Materials and Technology*, 123, 203–209.
- Hall J. (2014). *Ignited releases of liquid hydrogen*. Health & Safety Laboratory Research Report RR987.
- Hankinson G., Lowesmith B. J. (2013). *Integrated Design for Demonstration of Efficient Liquefaction of Hydrogen (IDEALHY)*. IDEALHY Deliverable Report D3.11.
- Hare J., Burrell, G. (2006). *Review of HSE Building Ignition Criteria*. Health & Safety Laboratory Report HSL/2006/33.
- Hauptmanns, U. (2001). A procedure for analyzing the flight of missiles from explosions of cylindrical vessels. *Journal of Loss Prevention in the Process Industries*, 14, 395–402.
- Hawthorth, S., Moodie, K., Gummer, J., Ewan, B., Michels, H., Linstedt, P., Winstanley, P. & Pekalski, A. (2016). Safety of combined cycle gas turbine and gas engine systems operating on hydrogen rich syngas and biogas. 8th International Seminar on Fire & Explosion Hazards, Hefei, China, 25-28 April 2016.
- Heicklen, J. (1976). *Atmospheric chemistry*. London: Academic Press.
- HIAD (Hydrogen Incidents and Accident Database), JRC, European Commission <<https://odin.jrc.ec.europa.eu/giada/>> Accessed May 2020.
- Houf, W., & Schefer, R. W. (2007). Predicting radiative heat fluxes and flammability envelopes from unintended releases of hydrogen. *International Journal of Hydrogen Energy*, 32, 136–151.

- Hydrogen Tools, Pacific Northwest National Laboratory and U.S. Department of Energy's Office of Energy Efficiency <<https://h2tools.org/>> Accessed 05.20.
- INERIS. (2004). Formalisation du savoir et des outils dans le domaine des risques accidentels (DRA-35). Version Projet, Les {\\e}clatements de reservoirs, L'Institut National de l'Environnement Industriel et des Risques, Verneuil-en-Halatte, France.
- International Fire Code Council Inc, 2002. ISBN # 1-892395-60-6 (soft) ISBN # 1-892395-59-2 (loose-leaf) ISBN # 1-892395-81-9 (e-document).
- Itoh, S., Shinoda, M., Kitagawa, K., Arai, N., Lee, Y.-I., Zhao, D., ... Yamashita, H. (2001). Spatially resolved elemental analysis of a hydrogen-air diffusion flame by laser-induced plasma spectroscopy (LIPS). *Microchemical Journal*, 70, 143–152.
- Jagger S. (2004). *Human vulnerability to thermal radiation offshore*. Health & Safety Laboratory Report HSL/2004/04.
- Kaiser W. (2000). Ermittlung und Berechnung von Störfallablaufszenarien nach Maßgabe der 3. Störfallverwaltungsvorschrift. Bd. 1, Publisher, Berlin Umweltbundesamt 2000.
- Kim W. K., Volodymyr S., Makarov D., & Molkov V. (2015). *High pressure hydrogen tank rupture: Blast wave and fireball*. 6th Intentional Conference on Hydrogen Safety, Yokohama, Japan.
- Kluger, Y. (2003). Bomb explosions in acts of terrorism. *The Israel Medical Association Journal*, 5, 235–240.
- Koopman, R. P., Cederwall, R. T., Ermak, D. L., Goldwlr, H. C., Jr., Hogan, W. J., McClure, J. W., ... Shinn, J. H. (1982). Analysis of BURRO series 40-m3 LNG spill experiments. *Journal of Hazardous Materials*, 6, 43–83.
- LaChance J. L. (2009). *Uniform risk acceptance and harm criteria—IEA HIA Task 19 Perspective*. 3rd International Conference on Hydrogen Safety, Ajaccio, France, 16–18 September.
- Langberg H., Christensen S. O., & Skudal S. (2004). *Test program with small concrete Kasun houses*. Rapport 24/2004, Forsvarsbygg, FoU.
- Lees, F. P. (1994). *Loss prevention in the process industries*. Butterworth-Heinmann.
- Leslie, I. R. M., & Birk, A. M. (1991). State of the art review of pressure liquefied gas container failure modes and associated projectile hazards. *Journal of Hazardous Materials*, 28, 329–365.
- Li, Q. M., & Meng, H. (2002). Pressure-impulse diagram for blast loads based on dimensional analysis and single-degree-of-freedom model. *Journal of Engineering Mechanics*, 128, 87–92.
- Lind, C. D. (1975). What causes unconfined vapour cloud explosions? *Loss Prevention*, 9, 101–105.
- Logan, J. A. (1981). Tropospheric chemistry: A global perspective. *Journal of Geophysical Research*, 86, 7210–7254.
- Manning, T. A., & Lawrence, S. L. (2017). Fragment acceleration modelling for pressurized tank burst. *Journal of Spacecraft and Rockets*, 54(3), 755–768.
- Markvart, T., & Castalzer, L. (2003). *Practical handbook of photovoltaics: Fundamentals and applications*. Elsevier, ISBN 1-85617-390-9.
- Maurer B., Hess K., Giesbrecht H., & Leuckel W. (1977). Modelling of vapour cloud dispersion and deflagration after bursting to tanks filled with liquefied gas. *2nd Int. Loss Prevention Symp.* Heidelberg.
- Maurer B., Schneider H., Hess K., & Leuckel W. (1975). Modellversuche zur Flash-Entspannung, atmosphärischen Vermischung und Deflagration von Flüssiggasen nach deren Freisetzung bei Behälterzerknall. Int. Sem. on Extreme Load Conditions and Limit Analysis Procedures for Structural Reactor Safeguards and Containment Structures, Berlin, 8 – 11 September.
- MEDD. (1994). *Guide de sécurité pour la conception de salles de contrôle résistant à l'explosion dans les raffineries de pétrole et sur les sites pétrochimiques*. Paris, France: Ministère de l'Ecologie et du Développement Durable.
- MEDD. (2004). Guide technique relatif aux valeurs de référence de seuils d'effets des phénomènes accidentels des installations classées. Version Octobre 2004.

- Mercx W. P. M. (1997). *Extended modelling and experimental research into gas explosions*. Final report of the EMERGE project CEC contract EV5VCT930274.
- Mercx W. P. M. (1994). *Modelling and experimental research into gas explosions*. Overall final report of the project MERGE. CEC Contract STEP-CT-0111 (SSMA), European Commission, Directorate General XII, Brussels, Belgium.
- Mercx, W. P. M., van den Berg, A. C., Hayhurst, C. J., Robertson, N. J., & Moran, K. C. (2000). Developments in vapour cloud explosion blast modeling. *Journal of Hazardous Materials*, 71, 301–319.
- Mercx W. P. M., Weerheijm J. & Verhagen T. L. A. (1991). Some considerations on the damage criteria and safety distances for industrial explosions. HAZARDS XI—New directions in process safety, UMIST, Manchester, UK, 16–18 April.
- Moen, I. O. (1993). Transition to detonation in fuel air explosive clouds. *Journal of Hazardous Materials*, 33, 159–192.
- Molkov, V., & Saffers, J.-B. (2013). Hydrogen jet flames. *International Journal of Hydrogen Energy*, 38(19), 8141–8158.
- Molkov V. V., Makarov D. V., & Schneider H. (2005). Hydrogen-air deflagrations in open atmosphere: Large eddy simulation analysis of experimental data. *First International Conference on Hydrogen Safety*. Pisa, Italy.
- Mott N. F. (1943). Fragmentation of H.E. shells, a theoretical formula for the distribution of weight of fragments. A.O.R.G. Memorandum, Army Operational Research Group, Richmond, United Kingdom.
- Mudan, K. S. (1984). Thermal radiation hazards from hydrocarbon pool fires. *Progress in Energy and Combustion Science*, 10, 59–80.
- NASA. (1997). *Safety standard for hydrogen and hydrogen systems. Guidelines for hydrogen system design, materials selection, operations, storage, and transportation*. Technical report NSS 1740.16, Office of Safety and Mission Assurance.
- NFPA 68, (2002). Venting of deflagrations, 2002 Edition. National Fire Protection Association, Quincy, MA, United States of America.
- Pandey, A. K., Kumar, R., Paul, D. K., & Trikha, D. N. (2006). Non-linear response of reinforced concrete containment structure under blast loading. *Nuclear Engineering Design*, 236, 993–1002.
- Pfortner H. (1975). Gas cloud explosions and resulting blast effects. Paper S3/1 at the 3rd Int. Conf. on Structural Mechanics in Reactor Technology (SMIRT), 8–11 September, Berlin.
- Pfortner H. (1983). Flame acceleration and pressure build up in free and partially confined hydrogen-air clouds. ICDEERS, 9th Colloquium on the Dynamics of Explosions and Reactive Systems.
- Pfortner, H. (1985). The effects of gas explosions in free and partially confined fuel-air mixtures. *Propellants, Explosives, Pyrotechnics*, 10, 151–155.
- Pfortner H., & Schneider H. (1983). Ballonversuche zur Untersuchung der Deflagration von Wasserstoff Luft Gemischen (Abschlussbericht). PNP-Sicherheitssofortprogramm: Prozessgasfreisetzung—Explosion in der Gasfabrik und Auswirkungen von Druckwellen auf das Containment. ICT Internal Report, Fraunhofer Institut für Chemische Technologie.
- Pierorazio, A. J., Thomas, J. K., Baker, Q. A., & Ketchum, D. E. (2005). An update to the Baker – Strehlow – Tang vapour cloud explosion prediction methodology flame speed table. *Process Safety Progress*, 24(1), 59–65.
- Prather, M. J. (2003). An environmental experiment with H₂O. *Science (New York, N.Y.)*, 302, 581–582.
- prEN 14994. (2004). Gas explosion venting protective systems. Draft European Standard. Prepared by Technical Committee CEN/TC 305 under a mandate given to CEN by the European Commission and the European Free Trade Association.
- San Marchi C., Somerday B. P. (2012). *Technical reference for hydrogen compatibility of materials*. Sandia Report, SAND2012–7321.

- Schefer, R., Houf, W., Bourne, B., & Colton, J. (2006). Spatial and radiative properties of an open-flame hydrogen plume. *International Journal of Hydrogen Energy*, 31, 1332–1340.
- Schefer, R. W., Kulatilaka, W. D., Patterson, B. D., & Settersten, T. B. (2009). Visible emission of hydrogen flames. *Combustion and Flame*, 156, 1234–1241.
- Schneider H. (2005). Large scale experiments: Deflagration and deflagration to detonation within a partial confinement similar to a lane. *First International Conference on Hydrogen Safety*. Pisa, Italy.
- Schneider, H., & Pfortner, H. (1978). *Flammen- und Druckwellenausbreitung bei der Deflagration von Wasserstoff Luft Gemischen*. PNP-Sicherheitssofortprogramm: Prozessgasfreisetzung - Explosion in der Gasfabrik und Auswirkungen von Druckwellen auf das Containment. Karlsruhe, Germany: Presentation Jahrestagung des Fraunhofer Instituts für Treib- und Explosivstoffe.
- Schultz, M. G. (2003). Air pollution and climate-forcing impacts of a global hydrogen economy. *Science (New York, N.Y.)*, 302, 624–627.
- Schultz M. G., Markert F., & Pilegaard K. (2004a). Ozone and climate impacts of a global hydrogen economy. Poster at the Quadrennial Ozone Symposium, Kos, Greece.
- Schultz M. G., Markert F., & Pilegaard K. (2004b). *Hydrogen and the environment*. Technical report, Riso National Laboratory.
- Sivathanu, Y. R., & Gore, J. P. (1993). Total radiative heat loss in jet flames from single point radiative flux measurements. *Combustion and Flame*, 94, 265–270.
- Studer, E., Jamois, D., Jallais, S., Leroy, G., Hebrard, J., & Blanchetière, V. (2009). Properties of large-scale methane/hydrogen jet fires. *International Journal of Hydrogen Energy*, 34(23), 9611–9619.
- Tang, M. J., & Baker, Q. A. (1999). A new set of blast curves from vapor cloud explosion. *Process Safety Progress*, 18, 235–240.
- Teodorczyk A., & Thomas G. (2003). Flame and detonation arresters—European Normative Standard for Testing. *Proceedings of the VI Seminar in New Trends in Research of Energetic Materials*. Pardubice, pp. 411–419.
- Thomas, K. T., Eastwood, C., & Goodrich, M. (2015). Are unconfined hydrogen vapour cloud explosions credible? *Process Safety Progress*, 34(1), 36–43.
- Tien, C. L., & Lee, S. C. (1982). Flame radiation. *Progress in Energy and Combustion Science*, 8, 41–59.
- Tieszen, S. R. (1993). Effect of initial conditions on combustion generated loads. *Nuclear Engineering and Design*, 140, 81–94.
- Tromp, T. K. (2003). Potential environmental impact of a hydrogen economy on the stratosphere. *Science (New York, N.Y.)*, 300, 1740–1742.
- Turns, S. R., & Myhr, F. H. (1991). Oxides of nitrogen emissions from turbulent jet flames: Part 1—Fuel effects and flame radiation. *Combustion and Flame*, 87, 319–335.
- UFIP. (2002). Guide méthodologique UFIP pour la réalisation des études de dangers en raffineries, Stockages et depôts de produits liquides et liquéfiés. L'Université de Formation Inter-professionnelle, Saint Laurent du Var, France.
- USACE. (1990). *Structures to resist accidental explosions*. Report TM5–1300, US Army Corps of Engineers, Hyattsville, USA.
- Ustolin, F., Song, G., & Paltrinieri, N. (2019). The influence of H₂ safety research on relevant risk assessment. *Chemical Engineering Transactions*, 74, 1393–1398.
- Van den Berg, A. C. (2008). Blast charts for explosive evaporation of superheated liquids. *Process Safety Progress*, 27(3), 219–224.
- Van den Berg A. C., & Mos A. L. (2005). *Research to improve guidance on separation distance for the multi-energy method (RIGOS)*. Health & Safety Executive Research Report 369.
- Van den Berg, A. C., van der Voort, M. M., Weerheijm, J., & Versloot, N. H. A. (2004). Expansion-controlled evaporation: A safe approach to BLEVE blast. *Journal of Loss Prevention in the Process Industries*, 17, 397–405.

- van der Voort, M. M., Holm, K. B., Kummer, P. O., Teland, J. A., van Doormaal, J. C. A. M., & Dijkers, H. P. A. (2016). A new standard for predicting lung injury inflicted by Friedlander blast waves. *Journal of Loss Prevention in the Process Industries*, 40, 396–405.
- van der Voort M. M., Van Doormaal J. C. A. M., Verolme E. K., & Weerheijm J. (2006). Analysis of the Sci Pan 3 debris throw data using the Klotz Group approach. *32nd Explosives Safety Seminar*. Philadelphia, Department of Defense Explosives Safety Board.
- Van Doormaal J. C. A. M. & Weerheijm J. (2006b). J Klotz Group engineering tool for debris launch prediction. *32nd Explosives Safety Seminar*. Philadelphia, Department of Defense Explosives Safety Board.
- Van Doormaal J. C. A. M., van der Voort M. M., Verolme E. K., & Weerheijm J. (2006a). *Design of KG-engineering tool for debris throw prediction*. Report TNO-DV2 2005 C112, TNO.
- van Wingerden C. J. M. (1999). *Detonations in pipes and in the open*. CMR internal report, Christian Michelsen Research.
- Verfondern, K. (2008). Safety considerations on liquid hydrogen. Forschungszentrum Jülich GmbH.
- Viskanta, R., & Menguc, M. P. (1987). Radiation transfer in combustion systems. *Progress in Energy and Combustion Science*, 13, 97–160.
- Wennberg, P. O., Hanisco, T. F., Jaegle, L., Jacob, D. J., Hints, E. J., Lanzendorf, E. J., . . . Bui, T. P. (1998). Hydrogen radicals, nitrogen radicals, and the production of O₃ in the upper troposphere. *Science (New York, N.Y.)*, 279, 49–53.
- Weyandt, N. (2005). *Analysis of induced catastrophic failure of a 5000 psig Type IV hydrogen cylinder*. Southwest Research Institute Report 01.0639.01.001.
- Weyandt, N. (2006). *Catastrophic failure of a 5,000-psig hydrogen cylinder installed on a typical SUV*. Southwest Research Institute Report 01.06939.01.005.
- Yang J. W., Musicki Z., & Nimnual S. (1991). *Hydrogen combustion, control, and value-impact analysis for PWR dry containments*. Technical Report NUREG/CR-5662, U.S. Nuclear Regulatory Commission.
- Zalosh R. G., & Short T. P. (1978). *Compilation and analysis of hydrogen accident reports, COO-4442-4, FMRC J.I. 4A7N0.RG, RC78-T-54*. Factory Mutual Research Corporation for the U.S. Department of Energy, October 1978.

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Risk assessment

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5.1 Introduction

Managing health and safety requires consideration about what might cause harm to people or damage to property and environment and taking reasonable steps to prevent such harm. This is risk assessment. As hopefully we all recognize, it is something we do in everyday lives at low level without giving it too much thought. However, for workplace and other higher hazard activities a formal risk assessment is required and is something you are required by law to carry out.

Depending on the industry, nature, hazards, and scale of the activities the approach to assessing potential risks and developing your risk assessment will be different. Different approaches to risk assessment and safety depend on the types of activities you are considering: for example, major hazards industries with hydrogen and/ or other hazardous materials above defined and aggregated quantity limits, where a full safety case is required considering the risks on and off site. For smaller scale activities involving just a few people and only a few easy to manage risks, the risk assessment should be more straight forward but still needs careful consideration.

A risk assessment is not about creating huge amounts of paperwork, but rather about identifying sensible measures to control the risks associated with your activities. As already noted, you are probably already taking steps to protect yourself and your employees, but a risk assessment will help you decide whether you have covered all you need to.

Hydrogen safety engineering practices in the production, storage, distribution, and use of hydrogen are essential to ensure the safety of operations. The aim of this chapter is to allow operators and developers of equipment and facilities at all levels and stages of development to have an understanding of what is involved to achieve an appropriate and effective risk assessment. It is not intended to give all the answers and there will be additional work needed to complete the risk assessment process. We deliberately focus on the risk assessment process to compliment the rest of this book. The level of understanding and complexity in the examples increases, including a complex quantitative risk assessment (QRA) example in Section 7. While many of the examples will be challenging for those early on their journey to understand risk assessment of hydrogen, it hopefully serves to illustrate the complexity and need for expert support, while at the same time allowing users to appreciate the need for such approaches and to strengthen the knowledge needed to engage with such processes.

When handling hydrogen, the importance of safety must never be forgotten as a number of hazards can lead to serious accidents. Also, it is important to remember that as a growing and relatively new technology, its widespread acceptance will be critically impacted by unnecessary incidents and accidents.

Why we need risk assessment

We obviously apply risk assessment to reduce risks, or even better, to avoid incidents and accidents. This approach involves potential hazards, failure mechanisms, and related incidents associated with any process or system being identified, analyzed, reported [and recorded in relevant knowledge databases HIAD ([Hydrogen Incident and Accident Database \(HIAD 2.0\)](#), 2016) and H2Tools ([Hydrogen Safety Panel](#), 2019), handbooks, papers, etc.], and eliminated or mitigated as part of sound safety planning and comprehensive safety engineering. All relevant objects or aspects that may be adversely affected by a failure should be considered, including low-frequency high consequences events. Therefore the general protection objective is to exclude or at least minimize potential hazards and associated risks to prevent impacts on the following:

- *People.* Hazards that pose a risk of injury or loss of life to people must be identified and eliminated or mitigated. A complete safety assessment considers not only those personnel who are directly involved in the work, but also others who are at risk due to these hazards.
- *Property.* Damage to or loss of equipment or facilities must be prevented or minimized. Damage to equipment can be both the cause of incidents and the result of incidents. An equipment failure can result in collateral damage to nearby equipment and property, which can then trigger additional damage.
- *Equipment.* Failures can even lead to additional hazards and risks, for example, through the domino effect. Effective safety planning, monitoring, and reporting considers and minimizes serious risk of equipment and property damage.
- *Environment.* Damage to the environment must be prevented. Any aspect of a natural or the built environment, which can be harmed due to a hydrogen system or infrastructure failure, should be identified and analyzed. A qualification of the failure modes resulting in environmental damage must be considered.

5.2 Safety regulations and standards in different countries

It is important to understand that this chapter will provide an overview of risk assessment but does not provide all the information you need to risk assess the activities you are looking to undertake, as each risk assessment is unique and tailored to a specific application. However, it will provide a backbone of guidance to allow you to then:

- Understand the health and safety issues associated with your activities.
- Appreciate the technical and legal requirements (in your country of operation) that you will need to meet.
- Understand methodologies to identify and assess risks of your operations.
- Then develop a suitable risk assessment to control your activities.

Hopefully you already have good familiarity with the safe use of hydrogen, from activities in more traditional industry, with how health and safety is regulated in your country or region, and

with the relevant authorities who do this and may be able to help and guide you to do this safely. If not, and even if you do, then we advise you do all of the following:

- You work with, employ, or partner with others with relevant knowledge and experience of handling hydrogen to advise and assist you and have a clear understanding of the regulations in your region/country.
- We advise you to contact your safety regulator and get a good understanding from them of their requirements and with sufficient lead time. If you do not know who your safety regulator is, or how to contact them as a first point of contact to investigation we direct you to the Safe and Health at Work area of the United Nations International Labour Organization ([United Nations International Labour, 2021](#)) website. In addition, if your work activities are focused in the following areas we suggest you consult the following institutions as starting point to understand if your region is a participant and make contact with the relevant authorities in your region and to get an understanding of activities involving hydrogen at this level.
 - For maritime safety consult the International Maritime Organization (IMO) ([International Maritime Organisation, 2021](#)).
 - For road transport applications consult the United Nations Forum for Harmonization of Vehicle Standards ([United Nations Forum for Harmonisation of Vehicle Standards, 2021](#)).
 - For aviation, consult the United Nations body called the International Civil Aviation Organization ([The International Civil Aviation Authority, no date](#)). Individual national regulators then take these regulations implementing and enforcing them in their own country. Within Europe much of the safety regulations are set by a European Commission body called the European Aviation Safety Agency. This means there is a common set of requirements across Europe on areas like pilot licensing and aircraft type approvals. National regulators, such as the UK CAA, then use those requirements to regulate civil aviation in their country.
- Contact the International Association for Hydrogen Safety ([International Association for Hydrogen Safety “HySafe”, 2021](#)) via its website and we will be able to help you in a number of ways, including directing to contacts, sources of guidance and information, and training on hydrogen safety.

This list aims to cover most applications where hydrogen already has a strong footprint, or where it is currently developing, and hopefully allows you to make contact with the appropriate local contacts and begin to build what you need to proceed with your application safely.

When you have done so, they should also be able to give an understanding of the international and national regulations and standards relating to your activities. For more information and guidance see [Chapter 7](#), Legal Requirements, Technical Regulations, Codes, and Standards for Hydrogen Safety, of this book.

More or less all countries have their national standards body, like Polski Komitet Normalizacyjny, Deutsches Institut für Normung, British Standards Institute, Association Française de Normalisation, etc. The general globalization and the progressing economic integration of the European Union (EU) member states have the effect that national standards align with international ones, which are developed in collaboration of international experts and are then adopted nationally.

Generally, while the application of a standard is not mandatory (unless a regulation refers to that standard), you should have a good understanding of those appropriate to your activities and apply them, or base your approach on them if appropriate, as they usually represent best practice.

5.3 Safety management principles

The safety management principles will underpin how you assess risk in the different activities of your operation. There are two different safety management principles generally used:

1. Consequence-based safety management will claim that the worst conceivable events at an installation processing hazardous material should not have consequences outside certain boundaries, and will thus design safety systems to assure this.
2. Risk-based safety management maintains that the residual risk should be analyzed both with respect to the probabilities and the nature of hazard, and hence give information for further risk mitigation. This implies that very unlikely events might, but not necessarily, be tolerated.

5.4 Risk-based safety management

Risk-based safety management is the philosophy in line with central regulations and legislation in western countries increasingly requiring risk assessment and documentation.

The risk-based safety management (often called risk management) principle is that some risks, specified through risk acceptance criteria, should be removed or reduced to meet safety requirements (residual risk level objectives). However, both prescriptive requirements (detailed standards) and goal-oriented requirements (risk-based decisions) have a role to play in design of processes and installations.

Beyond implementing risk reducing measures in order to meet a tolerable risk level, such measures should also be implemented if further risk reduction can be obtained at costs lower than the benefits obtained. This is often referred to as the ALARP principle (As Low As Reasonable Practicable—see section later 5.4.1). Risk reduction can then either be consequence reducing measures or accident probability reducing measures. Probability reducing measures should be preferred.

The principle is illustrated in the Fig. 5.1.

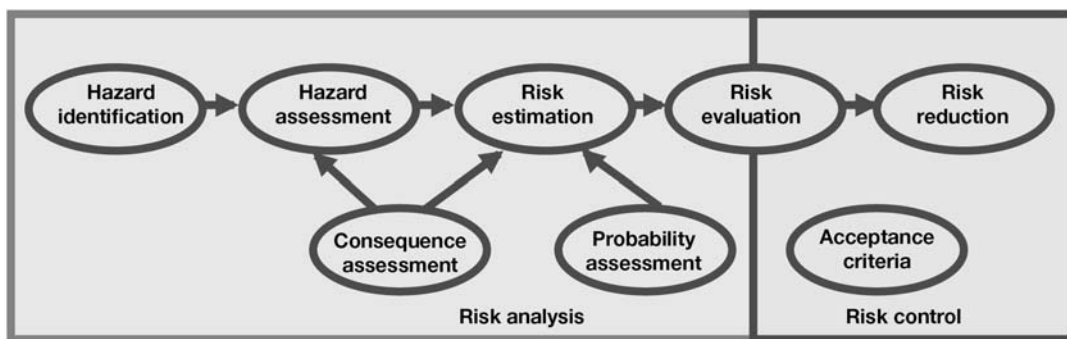


FIGURE 5.1

Risk-based safety management.

5.5 Assessing and profiling the risks

The range of risks that can affect an organization includes and goes beyond health and safety risks to include quality, environmental, and asset damage, with issues in one area clearly impacting others. Health and Safety Executive (HSE) document *Managing for Health and Safety* (Health and Safety Executive, 2013) highlights these interrelationships and their management through a process called the Plan, Do, Check, Act cycle, which is illustrated in Fig. 5.2.

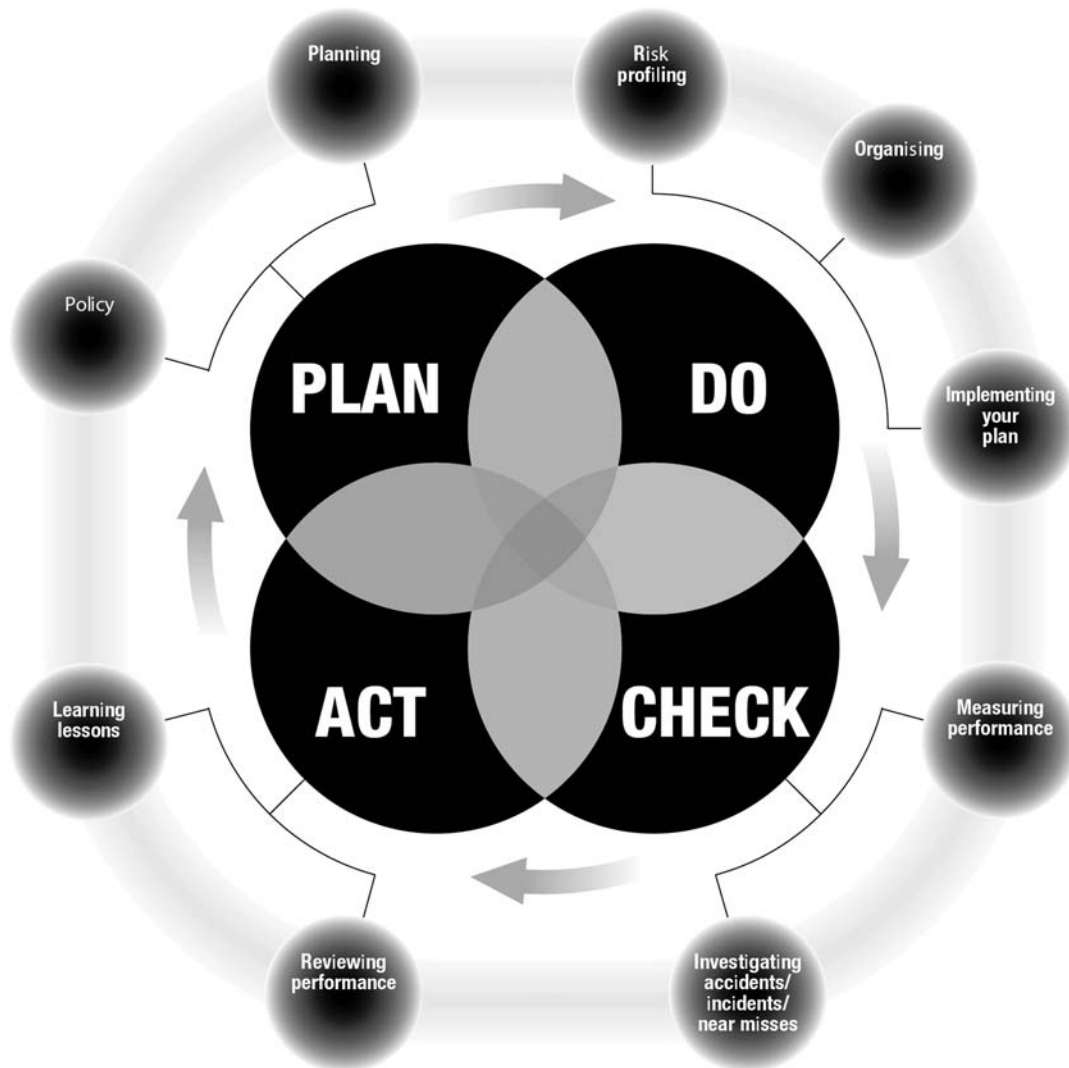


FIGURE 5.2

Plan, Do, Check, Act cycle.

From Health and Safety Executive. (2013). Managing for health and safety HSG65, Part 1, pp. 1–62. ISBN 978 0 7176 6604 1.

The document also discusses the importance of establishing a risk profile to understand the nature and levels of threats faced by an organization. It examines the likelihood of adverse effects occurring, the level of disruption and costs associated with each type of risk, and the effectiveness of the control measures in place.

Although they may not use these precise terms, many organizations will most likely have built a risk profile that covers:

- The nature and level of the risks faced by the organization;
- The likelihood of adverse effects occurring and the level of disruption;
- Costs associated with each type of risk;
- Effectiveness of the controls in place to manage those risks.

5.5.1 Identifying risks

Clearly, in some organizations risks will be tangible and immediate safety issues, for example, machine guarding, whereas in others the risks may be less obvious, for example, health-related and it could be a long time before the illness becomes apparent. Issues such as degrading plant integrity are nearly always important as these can lead to later emerging risks in many businesses. Similar risks may originate from things that happen very infrequently but with catastrophic effects (high-hazard, low-frequency events, such as an oil refinery explosion) to things that happen much more frequently but with lesser consequences (low-hazard, high-frequency events). In the developing world of hydrogen applications, both are important, but particularly the high-hazard, low-frequency incidents as these could destroy the business, cause immediate death and injury, and so would be high-priority in a risk profile.

In terms of who should do the assessment, HSE document HSG65 ([Health and Safety Executive, 2013](#)) advises that it should be completed by someone with a knowledge of the activity, process, or material that is being assessed. Operators and their safety representatives are a valuable source of information. If an adviser or consultant assists with the risk assessment, it is essential that managers and operators should still be involved.

It is also important to consider all your activities, taking account of possible harm to

- employees;
- contractors;
- members of the public;
- those using products and services;
- anyone else affected by the activity, such as neighbors.

Remember to think of how a risk could affect different groups, such as young or inexperienced workers, pregnant workers, workers with a disability, migrant workers, or aging workers. Also consider your supply chain—if that is not properly managed, the actions of others in those networks can impact on your health and safety risks.

Identifying and understanding hazards early in any risk assessment process is essential and bringing key parties from the list above into the process is essential. To do this safety expertise should be consulted early to ensure that safety engineering design features, practices, and

procedures are consistently applied as part of the project implementation. Work scope could include but not limited to:

- Reviewing planning documents ensuring that safety issues are addressed and budgeted to close the most important knowledge gaps and technological bottlenecks.
- Reviewing designs with the intent of approving and/or assisting with the approval of a project. Inspecting and performing regular audits of the installation to ensure safe execution of the project.
- Investigating lessons learned and reporting for accidents, incidents, and near-misses for applying the knowledge gained in corresponding updates of the safety plan.
- Developing further safety-related management issues.
- Formulating knowledge gaps and technological bottlenecks identified within a project scope to be addressed in similar projects in the future.

5.5.2 What the law says on assessing risks

The law states ([The Management of Health and Safety at Work Regulations 1999 No. 3242, 1999](#)) (UK case so please check for your country) that a risk assessment must be “suitable and sufficient,” that is, it should show that

- a proper check was made;
- you asked who might be affected;
- you dealt with all the obvious significant risks, taking into account the number of people who could be involved;
- the precautions are reasonable, and the remaining risk is low;
- you involved your workers/operators or their representatives in the process.

The level of detail in a risk assessment should be proportionate to the risk and appropriate to the nature of the work. Insignificant risks can usually be ignored, as can risks arising from routine activities associated with life in general, unless the work activity compounds or significantly alters those risks. Your risk assessment should only include what you could reasonably be expected to know—you are not expected to anticipate unforeseeable risks, although for new technologies and processes as you learn more about the risks you should update your risk assessments. To find out more on the UK approach see HSE’s risk assessment website ([HSE, 2021](#)).

For other countries contact the national or regional regulators for guidance.

5.5.3 Assessing the level of risk

The level of risk arising from the work activity should determine how sophisticated the risk assessment needs to be.

5.5.3.1 *Small businesses*

For small businesses, with few or simple risks, a suitable and sufficient risk assessment can be a very straightforward process based on informed judgment and using appropriate guidance. It is unlikely that any business working with hydrogen will fall into this category, based on the potential consequences of any incident involving hydrogen, or any other pressurized flammable gas.

5.5.3.2 *Medium-sized businesses or those with greater risks*

In these cases, the risk assessment will need to be more sophisticated. You may need specialist advice for some areas of the assessment, for example:

- Risks requiring specialist knowledge, for example, a particularly complex process or technique.
- Risks needing specialist analytical techniques, for example, being able to measure air quality and to assess its impact.

For the case of any business using hydrogen it is essential that you provide an adequate and demonstrable understanding and experience (i.e., competence) of working with hydrogen at a level appropriate to the activities you are undertaking.

5.5.3.3 *Large and high-hazard sites*

These sites will require the most developed and sophisticated risk assessments. For manufacturing sites using or storing bulk hydrogen (or other associated hazardous substances), large-scale mineral extraction, or nuclear plant, the risk assessment will be a significant part of the legally required safety case or report and may incorporate such techniques as quantified risk assessment (see [Section 5.4.5](#) of this chapter). Other statutory requirements, for example, the Control of Major Accident Hazards Regulations and nuclear installations licensing arrangements, include more specific and detailed arrangements for risk assessment.

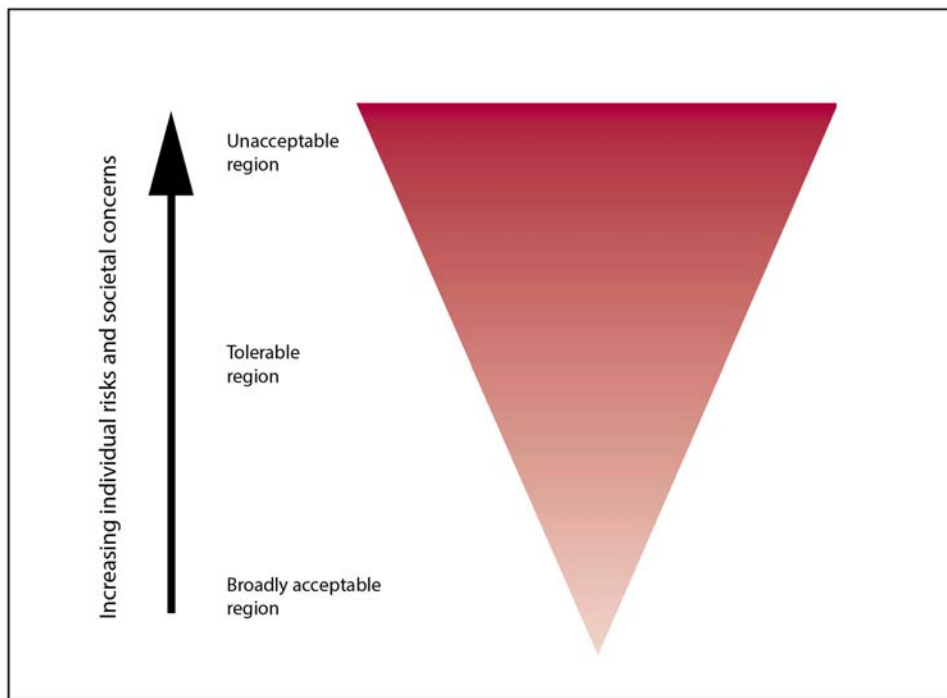
Safety planning, monitoring, and reporting for those concerned with all activities and especially new potentially hazardous applications with new fuels such as hydrogen, are essential and need to be based on a good understanding of hydrogen's hazards and risks. Safety compliance needs to be achieved with the regulations prevailing under all circumstances, complying with relevant international or national standards and existing company safety policies, codes, and procedures. Where new technologies or safety approaches are proposed, which do not rely on existing codes and standards, they must be evidenced to be safe to all parties including the safety regulator.

The ultimate aims are to assist in identifying minimum safety requirements, hazards, and associated risks, and in generating safety plans and risk assessments that will assist in achieving inherently safer conduct of all work related to the development and operation of hydrogen systems and infrastructure. Revisiting safety plans and risk assessments periodically is part of an overall effort to pay continuous and priority attention to the associated safety aspects and to account for all modifications of the considered system and its operations.

5.5.4 The ALARP principle—As Low As Reasonable Practicable

It is a duty holders' responsibility to demonstrate that the risk has been reduced to "as low as reasonably practicable (ALARP)" in those situations where the work activity is unusual (i.e., good practice is not yet established) or where there is a risk of a disaster (e.g., petrochemical and nuclear installations).

Below a certain risk level where the hazard is in the unacceptable area and risk cannot be tolerated whatsoever, normally defined by risk acceptance criteria, the ALARP principle should

**FIGURE 5.3**

The ALARP principle.

be applied in order to optimize risk reduction. In such cases cost–benefit analyses should be used in order to evaluate different risk reducing options. Risk reducing measures should be implemented if the cost is not disproportionate relative to the benefits of risk reduction.

The ALARP principle is illustrated in [Fig. 5.3](#).

5.5.5 Quantitative risk assessment

QRA is a formalized mathematical method used in order to evaluate numerical individual, environmental, employee, and public risk level values for comparison with regulatory risk criteria. As satisfactory demonstration of acceptable risk levels is often a requirement for approval of major hazard plant construction plans, the use of QRA is important for such cases in order to indicate and take actions for risk minimization. However, even if QRA uses scientific methods and verifiable data, it does have uncertainties, limitations, and application boundaries and therefore its use has to consider very carefully those aspects. That is why QRA is used as a complementary tool in decision-making processes, in areas where its strengths recommend it.

5.5.6 Risk assessment methodologies

Risk assessment methodologies are in principle applicable to any object or activity. Risk assessment methodologies are frequently used for applications of flammable gases, though there are not many examples of specific hydrogen studies so far.

Risk assessment studies should preferably be undertaken by multidisciplinary teams, although the effort should be in proportion to the risk being assessed. There are several risk assessment methods available, and one should select the method most applicable to the object analyzed and the purpose of the assessment.

5.5.7 Risk assessment process

The risk assessment process is an iterative process, as shown in Fig. 5.4. The risk is assessed, as well as the effect of risk reduction measures, until the risk inflicted by the system assessed (with implemented risk reduction measures) is evaluated as tolerable.

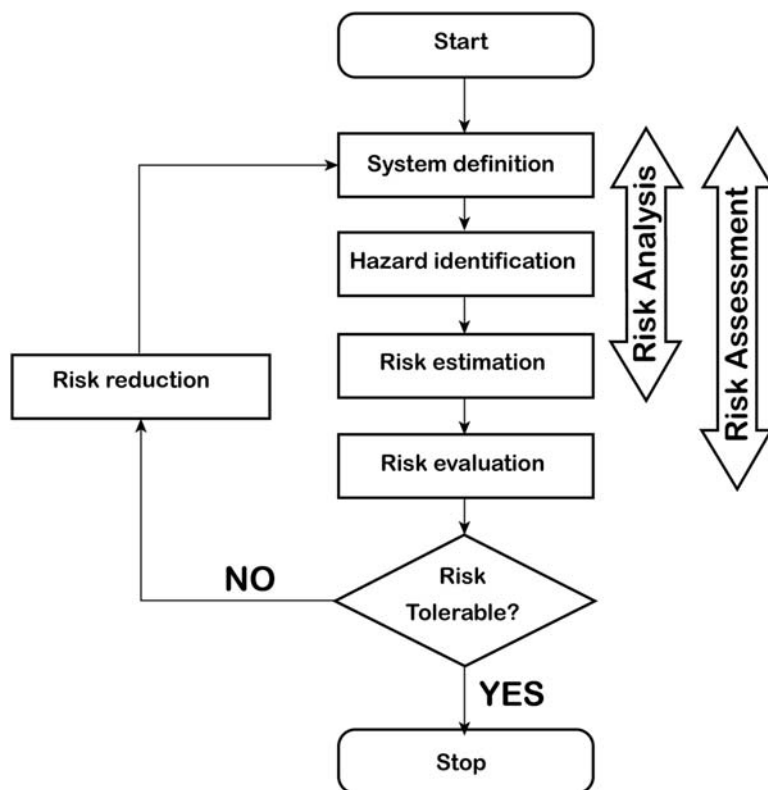


FIGURE 5.4

Risk assessment process.

But even when the assessed risk is evaluated as tolerable, the risk assessment process is not finished. The society's safety objectives and even an enterprise's safety objectives are more ambitious than maintaining the risk at a fixed level: risk assessment and risk reduction is also an iterative process through time. Indeed, change in state of the art enabling further risk reduction will eventually lower the tolerable risk level. Besides, new knowledge about the hazards evaluated may also render risk assessments or expensive risk reducing measures.

5.6 Practical risk assessment

The fundamentals of risk assessment, so essentially how to assess the risks, follow the steps laid out below from "Risk assessment—Brief guide to controlling risks in the workplace," HSE document INDG163 ([Health and Safety Executive, 2014](#)). Take note that the approach presented is that for a fairly small and simple process with less severe consequences, to illustrate general principles.

5.6.1 Identifying the hazards

Accurately identifying the potential hazards is the first and one of the key aspects of the risk assessment process. How you do this depends on a number of things including the scale and complexity of the facility.

The purpose of the hazard identification is to identify all hazards of relevance. Each hazard should be described in terms of accident(s) it may lead to. In order to identify the hazards that may arise, a systematic review should be made of technical as well as operational conditions, which may influence the risk. Historical records and experience from previous risk analysis do provide a useful input to the hazard identification process. Examples of this type of methodology are checklists, hazard indices, and review of historical occurrences. The hazard identification should not only consider the initial events, but also include the chain of events causing local and remote impairment, loss or damage.

5.6.2 Hazard identification methodologies

Hazard identification of a particular system, facility, or activity may yield a very large number of potential accidental events and it may not always be feasible to subject each one to detailed quantitative analysis. In practice, hazard identification is a screening process where events with low or trivial risks are dropped from further consideration. However, the justification for the events not studied in detail should be given. Quantification is then concentrated on the events that will give rise to higher levels of risk.

Fundamental methods such as Hazard and Operability (HAZOP) studies, fault trees (FTs), event tree (ET) logic diagrams, and Failure Mode and Effect Analysis (FMEA) are tools that can be used to identify the hazards and assess the criticality of possible outcomes. These methods also have the advantage of being sufficiently general for use on hydrogen facilities without specific adaptation.

The HAZOP technique consists of the application of a formal systematic detailed examination of the process and engineering intention of new or existing facilities to assess the hazard potential of operation outside the design intention or malfunction of individual items of equipment and their consequential effects on the facility as a whole. It is a qualitative technique for identification of

safety vulnerabilities or hazards based on the combination of guide words and parameters of the system, process, or product under consideration. The technique is to divide the process into natural subsections and use a set of guide words to identify possible deviations with hazardous potential. The technique is well-suitable for hydrogen applications, especially for the more complex systems. It answers the question how design intent or operating conditions may fail to be achieved at each step of the design process or technique. The guide words must always be appropriately selected to the process that is analyzed and additional guide words can be used (ISO, 2019). Fig. 5.5 shows a very small excerpt from the results of a HAZOP conducted for the storage unit in a hydrogen filling station. The excerpt focusses on the communication line between compressor and storage vessel, labeled “Node P1” and deviations of the hydrogen flowrate parameter.

The FMEA is a qualitative technique for systematically analyzing each possible failure mode within a system, and identifying the resulting effect on that system, the mission, and people. FMEA is highly suitable for reliability assessment and can be used for in-depth study of a critical part of a system. The FMEA may be extended with a criticality analysis; a quantitative procedure that ranks failure modes according to their probability and consequences (i.e., the resulting effect of the failure mode on system, mission, or personnel) and is then named a Failure Mode and Effects Criticality Analysis (FMECA).

HYDROGEN REFUELLING STATION						
Node P1: Communication line between compressor unit and storage vessels at high, medium, and low pressure.						
Process: Filling of storage vessels at high, medium, and low pressure, respectively.						
Deviation	Parameter	Causes	Consequences	Protections	Comments	TOP
No	Hydrogen flowrate	Compressor Failure	Pressure vessel filling interruption.	PLC closes storage unit solenoid valves	Filling interruption.	
		Isolation valve Ivb1 in compression line closed.	Compression line pressure increase.	PLC closes storage unit solenoid valves. PRD1 vent actuates. PS2 switches off the compressor.	Unwanted compression line pressure increase if PRD1 vent fails.	
Greater than		PS2 switches off the compressor at pressure higher than the storage set point.	Storage vessel overpressurization.	Pressure gauge & pressure transducer reveal overpressure. PLC switches off the compressor & closes the storage unit solenoid valves.	If pressure is too high, storage vent actuates.	
		Compressor gas high flow & PR pressure regulator fails.	Pressure increase & possible storage vessel overpressurization.	PS2 switches off the compressor. Pressure gauge & pressure transducer reveal overpressure.	If pressure is too high, storage vent actuates.	
Less than		PS2 switches off the compressor at pressure lower than the storage set point	No complete vessel filling.	PLC closes the storage unit solenoid valves.	No complete vessel filling.	
		PR pressure regulator fails.	Delay in storage vessel filling.	No.	-	
		Hydrogen leak.	Loss of hydrogen in environment .	PLC, activated by hydrogen sensors, switches off the compressor & closes the storage unit solenoid valves.	If adverse conditions happen, ignition is possible.	TOP 1

FIGURE 5.5

Example of a HAZOP result for one “node” of a hydrogen refueling station.

From Casamirra, M., et al. (2007). *Risk analysis of the storage unit in a hydrogen refuelling station*. In International Conference on Hydrogen Safety, pp. 1–10. Available at: <http://conference.ing.unipi.it/ichs2007/fileadmin/user_upload/CD/PAPERS/11SEPT/4.1.294.pdf>.

The FMEA and FMECA were originally developed by the National Aeronautics and Space Administration as a means of assuring that hardware built for space applications had the desired reliability characteristics. In the offshore industry, FMEA and FMECA have been increasingly utilized during the past years. FMECA was also used in the European Integrated Hydrogen Project (EIHP2) for development of guidelines for inspection and maintenance of hydrogen applications.

The initial step of an FMEA is a functional description of the system and the division of the system into subsystems and items. Each item is given an identification code. For each item the purpose/function of the item is then described, and possible failure modes are listed and analyzed with respect to causes and possible consequences. Means for detection of failure modes and mitigation/repair are also analyzed.

For each potential failure the effects, probabilities and—in case—likelihood of early detection have to be quantified along respective scales, and then recorded in a specific FMEA worksheet. An example for such a worksheet with a relatively high quantitative character is shown in Figs. 5.6 and 5.7.

Some practical considerations that need to be covered in relation to hazards are described below and are in part taken from HSE document “A brief guide to controlling risks in the workplace INDG163 (rev4),” and while not exhaustive are helpful in carrying out the simplest hazard identification process.

- Is this a well-established, well-understood process or activity? If so then achieving this can draw on past experience. However, for new processes that may not have been addressed before, which many innovative hydrogen projects will be and so extra care is required to ensure that potential hazards are identified. It is therefore essential that the activities, processes, or substances used as part of delivering a task are well-understood so that any that could cause injury are highlighted for attention. Conversely, for established processes that operatives may do every day, it is easy to become complacent and forget key safety aspects.

SYSTEM: _____
SUBSYSTEM: _____
REFERENCE DRAWING: _____

FAILURE MODE AND EFFECTS ANALYSIS (FMEA)
FAULT CODE # _____

DATE: _____
SHEET: _____
PREPARED BY: _____

Subsystem/ Module & Function	Potential Failure Mode	Potential Local Effect(s) of Failure	Potential End Effect(s) of Failure	SEV	Ct	Potential Cause(s) of Failure	OCC	Current Controls/Fault Detection	DET	RPN	Recommended Action(s)	Area/Individual Responsible & Completion Date(s)	Actions Taken	SEV	OCC	DET	RPN
Subsystem name and function	"How can this subsystem fail to perform its function?" "What will an operator see?"	The local effect of the subsystem or end user.	Downtime # of hours at the system level. 2. Safety 3. Environ- mental 4. Scrap loss			"How can this failure occur?" Describe in terms of something that can be corrected or controlled. 2. Refer to specific errors or malfunctions		"What mechanisms are in place that could detect, prevent, or minimize the impact of this cause?"			"How can we change the design to eliminate the problem?" "How can we detect (fault isolate) this cause for this failure?" "How should we test to ensure the failure has been eliminated?" "What PM procedures should we recommend?"	"What is going to take responsibility?" "When will it be done?"	"What was done to correct the problem?" Examples: Engineering Change, Software revision, no recommended action at this time due to obsolescence, etc.				

Critical Failure symbol (Cf)
Used to identify critical failures
that must be addressed (i.e.,
whenever Safety is an issue).

Severity Ranking (1-10)
(see Severity Table)

Occurrence Ranking (1-10)
(see Occurrence Table)

Detection Ranking (1-10)
(see Detection Table)

Risk Priority Number
 $RPN = Severity \times Occurrence \times Detection$

How did the "Action Taken"
change the RPN?

FIGURE 5.6

Example of FMEA worksheet (SEMATECH, 1992).

No.	Failure Mode	Cause	Effects	Controls	F	C	Recommendation
Date: July 8, 2003 Process: 2. Electrolysis Study Section: 2A. Water Electrolysis, Hydrogen Generation Design Intent: Generate 1.25 kg/h of >99.99% hydrogen at 150 psig by electrolysis of water							
1.	KOH leak	Mechanical failure or human error	Employee injury	Operator wears rubber gloves and face shield Safety shower and eyewash station in area Polycarbonate shield around cells	L	M	
2.	Exposed electrical circuit 240 volts DC, 600 amps	Human error	Employee injury	Polycarbonate shield around cells limits access	L	H	
3.	Rectifier startup	Human error Turn unit on before turning breaker on and then contact rectifier	Employee injury		L	H	Provide mechanism to prevent turning power on to unit until rectifier breaker is on
4.	Demineralizer failure	Media is saturated or deactivated	Sludge buildup in bottom of cell, no safety hazard	Conductivity detector and high conductivity shutdown at one micro-Siemens	–	–	
5.	KOH pump failure	Mechanical failure	Low water level in cell and high temperature damages cell, no safety hazard	Low level shutdown High temperature shutdown	–	–	
6.	Cell control failure	Instrument failure leading to stack failure	Cell damage with loss of fluid and potential exposure	Low level shutdown High temperature shutdown	L	M	Provide caustic containment sump and drain

FIGURE 5.7

FMEA for hydrogen fueling options.

From California Energy Commission. (2004). Failure modes and effects analysis for hydrogen fueling options (*California Energy Commission, 2004*).

- Check manufacturers' instructions or data sheets for chemicals and equipment as they can be very helpful in explaining the hazards and putting them in their true perspective.
- Look back at your accident and ill-health records—these often help to identify the less obvious hazards.
- Take account of nonroutine operations (e.g., maintenance, cleaning operations, or changes in production cycles).
- Remember to think about long-term hazards to health (e.g., high levels of noise or exposure to harmful substances).

A good source of information is the again the HSE website ([HSE, 2021](#)). HSE publishes practical guidance on hazards and how to control them.

Note also that it is important to think about all safety aspects, not just those associated with hydrogen and its properties. For example, working at height, working with chemicals, machinery, and asbestos. Depending on the type of work you do, there may be other risks that are relevant to your business.

5.6.3 Who might be harmed?

Think how employees (or others who may be present, such as contractors or visitors) might be harmed. Ask everyone involved in the process what they think the hazards are, as they may notice things that are not obvious to you and may have some good ideas on how to control the risks.

For each hazard you need to be clear about who might be harmed and what might be damaged—it will help you identify the best way of controlling the risk. That does not mean listing everything and everyone by name, but rather identifying groups of people (e.g., people working in the storeroom or passers-by) and safety critical subsystems.

Remember:

- Some workers may have particular requirements, for example, new and young workers, migrant workers, new or expectant mothers, people with disabilities, temporary workers, contractors, homeworkers, and lone workers (<https://www.hse.gov.uk/toolbox/workers>).
- Think about people who might not be in the workplace all the time, such as visitors, contractors, and maintenance workers.
- Take members of the public into account if they could be harmed by your work activities.
- If you share a workplace with another business, consider how your work affects others and how their work affects you and your workers. Talk to each other and make sure controls are in place. Ask your workers if there is anyone you may have missed.

5.6.4 Evaluate the risks

Having identified the hazards, you then have to decide how likely it is that harm will occur, that is, the level of risk and what to do about it. Risk is a part of everyday life and you are not expected to eliminate all risks. What you must do is make sure you know about the main risks and the things you need to do to manage them responsibly.

Generally, you need to do everything “reasonably practicable” to protect people from harm. This means balancing the level of risk against the measures needed to control the real risk in terms of money, time, or trouble. However, you do not need to take action if it would be grossly disproportionate to the level of risk.

Your risk assessment should only include what you could reasonably be expected to know—you are not expected to anticipate unforeseeable risks.

5.6.5 Risk assessment methodologies

The risk assessment tasks will depend on the purpose of the risk assessment. The risk assessment will normally involve a comparison of a calculated risk level with criteria for acceptable risk level. The acceptable or tolerable risk level would be based on the enterprise’s own safety standards and/or risk criteria established by the authorities. The risk assessment may also include comparison of alternative designs or activity plans.

If the risk is not controlled (acceptance criteria are not met) or the objective is to reduce the risk further to a level ALARP, options of risk reducing measures should be addressed and their desirable effect should be estimated. This should indeed be a multidisciplinary exercise, preferably involving people responsible for (future) operation of the object evaluated. The process of the risk assessment includes thus a reevaluation of the risks and of risk reduction measures based on cost–benefit analysis.

If the risk is controlled and the acceptance criteria are met, the chosen concept including the assumptions might be acceptable, but not the optimum from a cost–benefit point of view. In order to optimize the design, sensitivity calculations may be carried out.

Risk analysis methodologies are often grouped into three categories: qualitative, deterministic, and probabilistic. A qualitative analysis will normally characterize hazards with respect to likelihood and severity of consequences without quantification. A deterministic analysis will quantify the consequences of the most severe event possible, while the probabilistic analysis will quantify the probability and consequences of different scenarios developing from the possible initial events. The probabilistic analysis, also called quantitative risk analysis, is further described in [Section 5.8](#).

Look at what you are already doing, and the control measures you already have in place. Ask the following questions:

- Can I get rid of the hazard altogether?
- If not, how can I control the risks so that harm is unlikely?
- Some practical steps you could take include:
 - trying a less risky option;
 - preventing access to the hazards;
 - organizing your work to reduce exposure to the hazard.

Improving health and safety does not necessarily cost a lot. For instance, placing a mirror on a blind corner to help prevent vehicle accidents is a low-cost precaution, considering the risks. Failure to take simple precautions can cost you a lot more if an accident does happen.

Involve your workers, so you can be sure that what you propose to do will work in practice and will not introduce any new hazards [“Risk assessment: template and example” ([HSE, 2021](#))].

If you control a number of similar workplaces containing similar activities, you can produce a model risk assessment reflecting the common hazards and risks associated with these activities.

You may also come across model assessments developed by trade associations, employers’ bodies, or other organizations concerned with a particular activity. You may decide to apply these model assessments at each workplace, but you can only do so if you

- satisfy yourself that the model assessment is appropriate to your type of work;
- adapt the model to the detail of your own work situations, including any extension necessary to cover hazards and risks not referred to in the model.

5.6.6 Record your significant findings

Make a record of your significant findings—the hazards, how people might be harmed by them, and what you have in place to control the risks. Any record produced should be simple and focused on controls. In some cases, according to HSE, if you have fewer than five employees, you do not have to write anything down. But it is useful to do this so you can review it at a later date, for example, if something changes. If you have five or more employees, you are required by law to write it down. However, for the case of any hydrogen system or application we recommend that the risk assessment must be written down.

Any paperwork you produce should help you to communicate and manage the risks in your business. For most people this does not need to be a big exercise—just note down the main points about the significant risks and what you concluded.

An easy way to record your findings is to use a risk assessment template, an example of which is shown in [Fig. 5.8](#).

Risk assessment template

Company name:

Assessment carried out by:

Date of next review:

Date assessment was carried out:

What are the hazards?	Who might be harmed and how?	What are you already doing to control the risks?	What further action do you need to take to control the risks?	Who needs to carry out the action?	When is the action needed by?	Done

More information on managing risk : www.hse.gov.uk/simple-health-safety/risk/

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FIGURE 5.8

Example of risk assessment: template and example (HSE, 2021).

This approach is hopefully familiar to you, and if not please seek assistance. For simple examples of how this approach has been used for fairly everyday examples please also see the HSE website (HSE, 2021).

From this template it is hopefully clear that a risk assessment must be suitable and sufficient and have robust thinking recorded to support the information in each box, that is, it should show that:

- a proper check was made;
- you asked who might be affected;
- you dealt with all the obvious significant hazards, taking into account the number of people who could be involved;
- the precautions are reasonable, and the remaining risk is low;
- you involved your employees or their representatives in the process.

Where the nature of your work changes fairly frequently or the workplace changes and develops (e.g., a construction site), or where your workers move from site to site, your risk assessment may have to concentrate more on a broad range of risks that can be anticipated.

Take a look at the selection of example risk assessments on risk section of HSE's website (HSE, 2021). They show you what a completed risk assessment might look like for your type of workplace. You can use these as a guide when doing your own.

The site also has online risk assessment tools, to help employers complete and print off their own records.

If your risk assessment identifies a number of hazards, you need to put them in order of importance and address the most serious risks first.

Identify long-term solutions for the risks with the biggest consequences, as well as those risks most likely to cause accidents or ill health. You should also establish whether there are improvements that can be implemented quickly, even temporarily, until more reliable controls can be put in place.

Remember, the greater the hazard the more robust and reliable the measures to control the risk of an injury occurring will need to be.

5.6.7 Regularly review your risk assessment

Few workplaces stay the same. Sooner or later, you will bring in new equipment, substances, and procedures, which could lead to new hazards. Therefore it makes sense to review what you are doing on an ongoing basis, look at your risk assessment again, and ask yourself:

- Have there been any significant changes?
- Are there improvements you still need to make?
- Have your workers spotted a problem?
- Have you learnt anything from accidents or near-misses?
- Make sure your risk assessment stays up-to-date.

In the previous section the aim was to identify potential hazards and risks bringing together expertise of safety and risk assessment, with that of your application or plant, be it planned or already built. This input should draw together an understanding of the nature and scale of your activities and the risks and hazards it may produce.

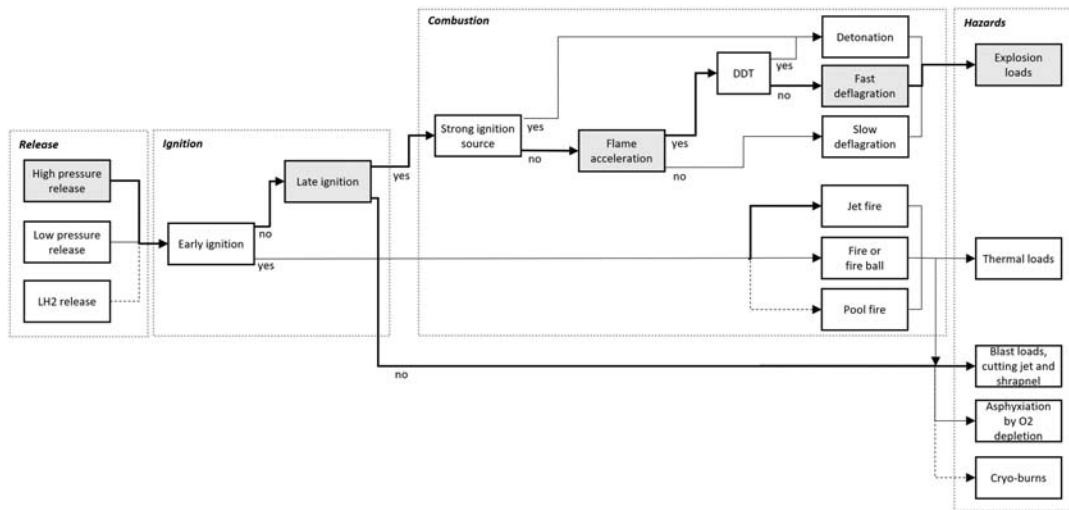
In this section the aim is to raise your awareness to thinking about these hazards in terms of the physical phenomena they could produce and the wider impact this could have. The key here is to understand the nature and size of any hazard produced (drawing on the other chapters of this book, namely, Chapter 3, Phenomena Relevant to Accidents and Chapter 4, Accident Consequences) and understanding the potential downstream impact/escalation and domino effect.

5.7 Hydrogen hazards

Many hydrogen accidents are initiated by a release of hydrogen from a container or pipe, and this situation is considered as an example in [Fig. 5.9](#) taken from [Hydrogen Safety Panel \(2019\)](#). However, incidents can occur in other ways including entry of air, or another oxidant, into the hydrogen system as observed in the fatal Gangneung, South Korea, electrolyzer accident in 2019 ([Choi, Cho, & K., 2020](#)) or, conversely, entry of hydrogen into an oxidant system as occurred in 1975 in the Ilford, UK accident ([HM Factory Inspectorate, 1976](#)). As these accidents illustrate all situations that could lead to flammable gas mixtures being produced must be investigated.

5.7.1 Thinking about the critical steps in an incident

The potential consequences of a hydrogen release are detailed in [Fig. 5.9](#). Ignition and flame acceleration are safety critical as they decide how severe the consequences might be. The vertical upward direction in the figure implies escalation. Therefore the hazards associated with a “no ignition” accident (e.g., oxygen depletion) are considered less critical than those associated with a delayed ignition and fast combustion regimes, which imply far reaching explosive loads. The most likely escalating sequence observed in the majority of severe accidents, which are in fact gas explosion events, are marked with gray boxes in [Fig. 5.9](#).

**FIGURE 5.9**

Potential consequences and hazards associated with hydrogen release.

Assessing the stage in this diagram we aim to give an overview of the steps required to prevent each stage, which is presented as commentary of the precautions required, the details of the relevant standards etc. identified in [Chapter 7](#), Legal Requirements, Technical Regulations, Codes, and Standards for Hydrogen Safety, of this book.

Release. Containment of hydrogen is clearly critical. In different types of applications hydrogen may be stored in different ways (as a compressed gas in steel or composite pressure vessels) or as a liquid hydrogen in a cryostat. To avoid accidental release clearly the materials need to be suitable, which includes containers, valves, actuators, pipes, hoses, seals, and gaskets (refer [Chapter 2](#): Hydrogen Technologies). Failure to address even a minor detail, such as unsuitable metallic component in an actuator or valve, could result in the release of hydrogen.

No ignition. Here the obvious hazards, we should expect, include asphyxiation or cold burns from cryogenic releases. For both, means of providing ventilation or separation should be considered.

Early ignition. Early ignition could occur as of the close proximity of unprotected ignition source close to the release, electrostatic discharges, or spontaneous self-ignition of high-pressure releases. Such events should be considered for the design of vent stacks and similar. Early ignition generally avoids deflagrations or explosions and results in fire or jet fire. The potential for such fires (pool, jet, or fireball) clearly need to be considered and separation from people and vulnerable equipment need to be planned into a design.

Late ignition. Here the ignition is more likely to be as a result of unprotected or malfunctioning protected ignition source. In this situation we could expect flame acceleration to occur leading to a deflagration and some over pressure. Depending on the nature of the ignition source (*strong ignition source*) and potential for *flame acceleration* and transition to *detonation* the explosion pressure could be very severe.

5.7.2 Safety principles to reduce or eliminate hazards

The previous section looked at hazards from the point of view of the engineering steps involved in an escalating incident. Here, we present Fig. 5.10, which looks at safety principles that should be considered when identifying safety vulnerabilities or hazards in a risk assessment procedure. Any expert in charge of a safety review shall have those principles in mind. However, they do not replace neither legal requirements, comprehensive hydrogen safety engineering, nor detailed risk assessment eventually required by regulations, codes, and standards. They allow stakeholders to take safety into account at different stages of the design, implementation, and operation of a process using hydrogen in a very basic manner.

5.7.3 Hazard estimation from modeling

A key part of producing a risk assessment for a complex hydrogen facility is to be able to estimate the size and nature of particular hazards, to understand their impact, and the effort required to

Number	Safety Principle	Explosion Protection Tier
1	Make a proper selection of materials for hydrogen technologies.	1 st Tier
2	Limit hydrogen inventories, especially indoors, to what is technologically necessary.	
3	Avoid/limit the formation of a flammable mixture, e.g. by using ventilation or reducing release size.	
4	Carry out ATEX zoning analysis. Avoid ignition sources using proper materials or devices in different ATEX zones, remove electrical systems where appropriate or provide electrical grounding, etc.	2 nd Tier
5	Avoid congestion. Reduce turbulence promoting flow obstacles in respective ATEX zones.	3 rd Tier
6	Avoid confinement. Place storage in the open if possible or use proper size openings in the enclosure.	
7	Combine hydrogen leak or fire detection and counter-measures.	
8	Provide efficient passive barriers in case of active barriers deactivation for whatever reason to limit adverse effects of explosion .	Organisational Safety Principles
9	Train and educate staff in hydrogen safety to establish a new hydrogen safety culture .	
10	Report near misses and incidents to databases and include lessons learned in your documentation.	

FIGURE 5.10

Safety principles.

Taken from Hydrogen Safety Panel. (2019). Safety planning for hydrogen and fuel cell projects, p. 52. Available at: <https://fch.europa.eu/sites/default/files/Safety_Planning_for_Hydrogen_and_Fuel_Cell_Projects_Release1p31_20190705.pdf>.

address them. Sources for such information include standards, the scientific literature, and commercial models. As much of this information is addressed in the other chapters of this book, this information is not reproduced here. For more information please see ([International Association for Hydrogen Safety “HySafe”, 2021](#)).

5.8 Learning from accident information

It is strongly acknowledged across all stakeholders that learning from accidents, incidents, and other safety events is of utmost importance to all operators of hazards plant. Learning from these past failures is essential to prevent them happening again in the future and to put in place mitigation to address their consequences, an important part of the work toward achieving effective risk control. We should all learn from events, failures, and errors committed in the past, and not only within our own industry or company, but also look beyond and draw lessons from elsewhere.

The use of incident and failure rate databases as a management tool has shown that it provides an opportunity for an organization or company to check its performance, learn from its mistakes, and improve its management systems and risk control. Comprehensive knowledge of events having the potential for inducing hazardous situations or loss of production will also contribute to the corporate learning and memory. On a company or plant level, the lessons to be learned from consulting such databases are both qualitative as well as quantitative. They can range from the identification of component/system failures, accident scenarios, or initiating events (IE) not being predicted in advance to quantitative statistical calculations and estimations to be used in reliability, maintainability, or risk studies. They also strengthen the culture of personnel working in risky technology industries by making them aware of the factors (technical, organizational, human, etc.) and the dynamics that lead to accidents or failures, and critically using this information in risk assessments. Safety authorities are utilizing accident databases as an operative and a management tool in several ways, such as following up of the overall safety level within the area of the authority's interest, resource allocation concentrated and prioritized on the most accident-prone areas, and in accident prevention. Accident information taken from a database may also support surveillance visits, in conducting accident investigations, and in the work of developing of rules and regulations.

A problem at present in the area of novel and developing hydrogen applications is that frequency data for failures are still extremely limited and so these impacts on our ability to carry out QRA models. Over time it is expected that these data will become more available as the number of applications, and unfortunately probably also incidents grow. As a hydrogen practitioner, it is important that you contribute to this effort, both by learning from these data in developing risk assessments, but also adding to the data any learnings or incidents.

5.8.1 Recent progress—hydrogen accident databases

5.8.1.1 *Hydrogen Incident and Accident Database*

The Hydrogen Incident and Accident Database (HIAD) ([Melideo, Moretto, & Wen, 2019](#)) was originally designed to be a collaborative and communicative web-based information platform holding high quality information of accidents and incidents related to hydrogen technologies. The main

goal was to become not only a standard industrial accident database but also an open communication platform suitable for safety lessons learned and risk communication as well as a potential data source for risk assessment. It was set up to improve the understanding of hydrogen unintended events, to identify measures and strategies, to avoid incidents/accidents, and to reduce the consequence if an accident occurs. The database, which is operated by the European Commission's Joint Research Centre (JRC), currently contains data on close to 600 safety-related events. These data are related not only to real incidents and accidents but also to hazardous situations.

The concept of the hydrogen accident database was generated during the project HySafe, an EC cofunded Network of Excellence of the Sixth Framework Program. HIAD was built by JRC and populated by many HySafe partners. After the end of the project, the database has been maintained and populated by JRC with publicly available events.

The original idea was to provide a tool also for QRA, able to conduct simple analyses of the events. Unfortunately, that goal could not be reached because of a lack of required statistics: it was not possible to establish a link with potential event providers coming from private sector, not willing to share information considered confidential.

Some important background on HIAD is that since June 2016 JRC have been developing a new version of the database (i.e., HIAD 2.0—see Fig. 5.11); the structure of the database and the web interface have been redefined and simplified, resulting in a streamlined user interface compared to the previous version of HIAD. The new version is mainly focused to facilitate the sharing of lessons learned and other relevant information related to hydrogen technology; the database will be public, and the events will be anonymized. The database will contribute to improve the safety

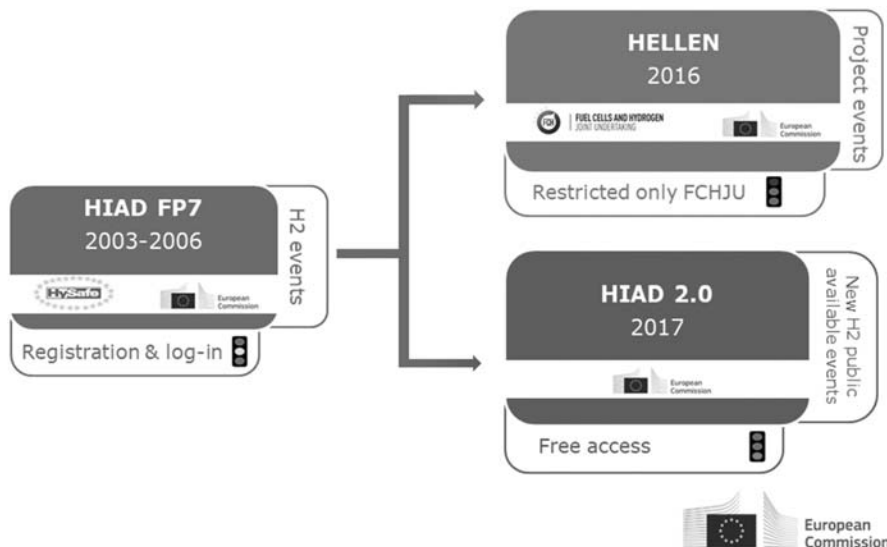


FIGURE 5.11

New databases and their relationship with HIAD.

awareness, fostering the users to benefit from the experiences of others as well as to share information from their own experiences.

The upgrade work on HIAD was started by the JRC with close collaboration of the EC's Fuel Cells and Hydrogen Joint Undertaking (FCH JU). The development of a new version of the database in order to specifically collect incidents from FCH JU projects (namely HELLEN) began in 2016. The new database has a significantly simplified structure: based on an in-depth analysis of the data quality collected in the previous years, entry fields were redefined and reduced, resulting in a more streamlined user interface compared to the older HIAD version. In addition, a template for data collection was developed; it includes explanations for each entry field and guides the reporting activities. The access to the HELLEN database is limited only to staff of the FCH JU and to the JRC.

The FCH JU launched the European Hydrogen Safety Panel (EHSP) [Fuel Cells and Hydrogen 2 Joint Undertaking (FCH 2 JU), 2017] initiative in 2017, the mission of which is to assist the FCH JU both at program and at project level in assuring that hydrogen safety is adequately managed, and to promote and disseminate hydrogen safety culture within and outside of the FCH JU program. It is composed of a multidisciplinary pool of experts. A key part of the activities of the EHSP is the analysis of safety data and events contained in HIAD 2.0. In close collaboration with JRC, the EHSP members have systematically reviewed more than 250 events and the lessons learned stemmed from this assessment are being released. Consult the EHSP section of FCH JU website to keep up-to-date with the important findings of this activity.

5.8.1.2 H2Tools

The H2Tools Portal (Hydrogen Tools, 2021) was developed by the Pacific Northwest National Laboratory through support from the US Department of Energy's Office of Energy Efficiency and Renewable Energy (EERE). EERE accelerates development and facilitates deployment of energy efficiency and renewable energy technologies and market-based solutions that strengthen US energy security, environmental quality, and economic vitality. The goal of the portal is to support implementation of the practices and procedures that will ensure safety in the handling and use of hydrogen in a variety of fuel cell applications.

The portal brings together and enhances the utility of a variety of tools and web-based content on the safety aspects of hydrogen and fuel cell technologies to help inform those tasked with designing, approving, or using systems and facilities, as well as those responding to incidents. A key part of the portal is the section on lessons learnt containing 200+ incidents, which are searchable against key hydrogen safety criteria.

5.9 Modeling as a tool for QRA

QRA is a formal and systematic approach to estimating the likelihood and consequences of hazardous events, and expressing the results quantitatively as risk to people, the environment, or your business. It also assesses the robustness and validity of quantitative results, by identifying critical assumptions and risk driving elements. The technique is widely used in a range of industries, and for a comprehensive guide to QRA offshore, as an established industry, see reference Spouge (1999).

This section has the goal to present a general description on QRA based on available methodologies, which use

- ET or other equivalent approaches;
- approaches available for handling large databases; as well as
- transfer of information from the design documents to the safety/reliability models.

The QRA approaches are illustrated with results from available case studies. QRA is considered as part of an integrated evaluation of a given installation (deterministic and probabilistic) designed to give a complementary set of insights. The main strong feature of QRA results resides in the fact that it gives a set of insights based on risk ranking and evaluation.

5.9.1 Key tenets of a QRA methodology

A QRA type of analysis is a methodical process of modeling the installations based on barrier approach and considering all the challenges potentially leading to undesirable effects to public, environment, or workers.

5.9.1.1 Features of QRA

While there are different approaches adopted for QRA, there are a number of specific features:

- Definition of risk sources.
- Scenarios and mitigation systems (like ventilation ones).
- Implementation of results from deterministic (e.g., thermohydraulic) analyses.
- Use of extensive diverse databases.
- Definition of the list of postulated IE. The IE frequency analysis estimates how likely it is for the IE to occur. The frequencies are usually obtained from analysis of previous operating history of the installation. If no data are available a set of FT are built to derive expected frequencies for IE.
- Definition of scenarios and failures in detail, leading to the development of ET. The definition of the consequences of each scenario is part of this task. Consequence modeling evaluates the resulting effects if the accidents occur, and their impact on personnel, installation, environment, and population, based on the adopted set of acceptable final states of the installation, given a set of challenges to it. Definition of end states—states in which we expect to have potential hazardous situation is also part of building ET.
- Definition of a set of systems operating as barriers for various challenges to the installation, for which FT are built. The action includes system definition, defining the system boundaries, part of the installation or the activity whose risks are to be analyzed, and detailed system failure mechanisms.
- Integration of the FT into the ET is performed to derive the full set of paths leading to various potential dangers from risk perspective end states.
- Risk quantification and assessment of acceptability is finally performed for the risk metrics adopted, which usually is related to
 - individual risk—the risk experienced by an individual person.

- group (or societal) risk—the risk experienced by the whole group of people exposed to the hazard. Up to this point, the process has been purely technical, and is known as risk analysis.
- Writing of the QRA report based on the steps performed. QRA reporting includes also documentation on actions performed for all steps during the process.

5.9.2 Strengths and limits of QRA

The strengths of QRA for hydrogen application are as follows:

- The approach builds on prior experience usually outside that of designers, operators, and regulators.
- The statistical approach means that its use for novel applications can have added value drawing on the numerical evidence base of mature applications.

QRA is one of the few techniques able to provide guidance to designers and operators on how best to minimize the risks of accidents due to the numerical weightings and provides an understanding of where to focus risk prevention. For novel applications such prioritization is a particular strength.

QRA combines experience with structured judgments to help anticipate accidents before they occur, which is why it is most effective when applied to major accidents that are difficult to address subjectively, because they lie outside the experience of most designers, operators, and regulators. This captures the low frequency/high impact events where chances of such accidents occurring are low, but their consequences can be catastrophic, involving the potential for massive loss of life, damage to the environment, financial loss, and on occasions leading to the failure of the company or major changes to the entire industry. Thus there is a moral and practical incentive to use the best available methods to minimize these risks. With new and novel applications of hydrogen this is especially relevant as a large hydrogen accident could halt the uptake of the technologies across many sectors.

QRA is used for comparative ranking of risks. For a hydrogen system this is relevant because it means that prioritization of the risks is possible, helping designers, installers, and operators to focus on the areas that have the most significant risk. This feature of QRA provides an understanding of compound risk, for example, when the presence of hydrogen amplifies an existing risk factor.

QRA is applied to activities where there is operating experience to provide a statistical base for the analysis (e.g., semisubmersible drilling rigs). However, safety in these areas can be managed reasonably well on the basis of accident experience. It can also be applied in relatively novel applications [e.g., early concrete platforms, floating production systems, tension leg platforms, extension of using from one area (nuclear) to another (nonnuclear)] with little operating experience, which is where its greatest value is found, especially where standard technology is applied in novel environment.

The limitations of QRA are the following:

- A lack of agreed approaches and poor circulation of data, resulting in wide variations in study quality.
- QRA has a higher impact and significance from the point of view of risk ranking, than from the point of view of absolute risk values.

As QRA is a relatively new technique, there is a lack of agreed approaches and poor circulation of data, resulting in wide variations in study quality. Also, because it is quantitative, QRA appears to be objective, but in reality, it is very judgmental, using in a very intensive manner subjective type of probabilities, dependent on expert judgment. Expert judgments may be explicit in areas where data are unavailable, but there are also many implicit judgments in the analysis and application of data that are available, and these are often unrecognized. Therefore the evaluation of the sensitivity and uncertainties of the results is actually a decisive factor of the analysis. QRA thus has rather a higher impact and significance from the point of view of risk ranking, than from the point of view of absolute risk values.

5.9.3 Examples—Application barriers, fault tree, event trees, and bow-tie diagrams

The bow-tie diagram is a powerful technique to illustrate and understand the risks that could lead to a potential event and is widely used in a number of industries including aviation. The approach is a combination of a FT on left-hand side and an event tree on the right-hand side, the two connected by what is usually called the “top event.” The primary causes identified in the FT (left-most events) are referred to as “initiating events” (threats). The event tree starts from the top event and leads to “consequences,” which usually are events we want to reduce risks of, either in probability or in severity. Fig. 5.12 illustrates how the bow-tie is organized, constructed, and used.

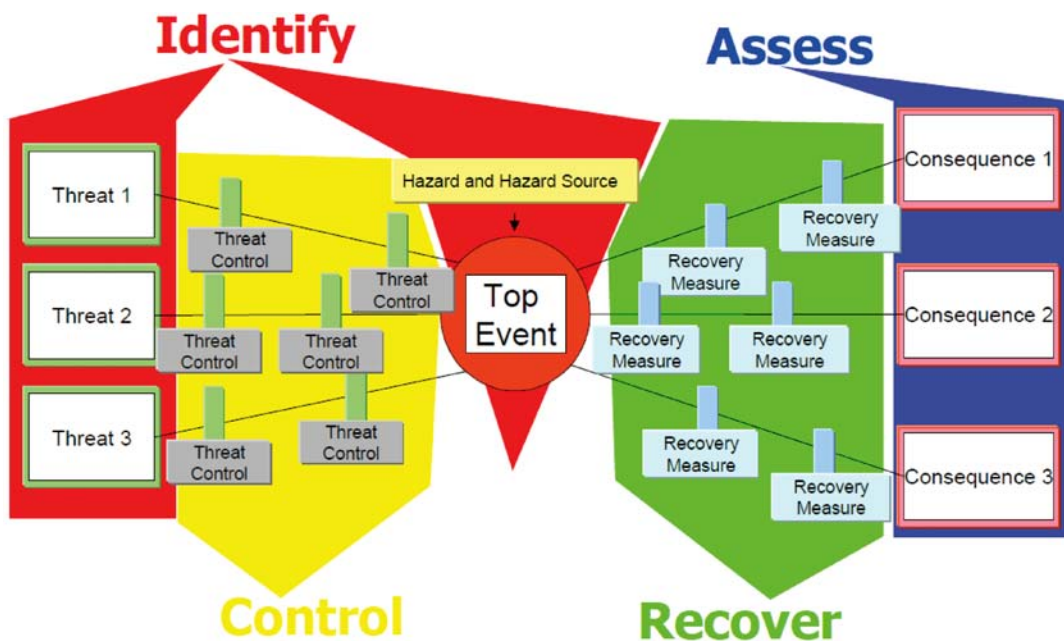


FIGURE 5.12

Bow-tie diagram (Risktec Solutions Ltd, 2007).

This powerful technique provides a detailed analysis for risk assessment, but requires a high level of expertise. As this example illustrates, using this or other similar technique, a full analysis can become quite complex. It is therefore essential that you take advice and get help to carry out the analyses unless you are very experienced in its use and in hazards generally, and even then, we would always recommend a peer review of the analysis and results by a competent party.

An example of the application of bow-tie analysis is presented to illustrate the technique and its application, which is taken from [Saw et al. \(2016\)](#) and describes the risk assessment of hydrogen composite storage applications when subjected to fire. In the paper they identify three consequences of composite cylinders containing hydrogen when exposed to a fire.

1. No effect/leak through the body of the cylinder: when the hydrogen storage is exposed to a sufficiently aggressive fire impact, the fusion of the liner can occur ([Bustamante Valencia et al., 2016](#)). This phenomenon can create a leak where the storage is supposed to remain airtight, or avoid a burst by allowing a pressure drop through that leak. A leak through the body of the cylinder would feed the surrounding fire without causing more significant effects.
2. Leak through Temperature Pressure Relief Device (TPRD): this means the hydrogen storage leaks through the well-functioning TPRD as a result of a thermal impact.
3. Burst: this means the hydrogen storage bursts as a result of a thermal impact and a failure to open, of the TPRD.

This example is quite complex and required high level of expertise and effort to carry out. Here we just present an overview of the paper, selecting key snippets to hopefully illustrate principles, but also demonstrate the complexity and need for a multidisciplinary team effort. Full details are obviously available in the paper.

First, an overview of the four-step bow-tie method is applied as detailed below taken from [Bustamante Valencia et al. \(2016\)](#).

First step. Identify the top events. The quantification is concerned with assessing the frequency of a dangerous consequence (leak or burst) of the hydrogen storage when exposed to a thermal impact. The top events must then be given a thermal impact followed by a storage failure. In this regard, the IE associated are the fires producing this level of impact on the hydrogen storage.

Second step. List, and position on the FT, the safety barriers selected to be involved in reducing the probability of occurrence of the top event. These barriers will prevent one or more IE from having an impact on the storage. This means we do not analyze when the barriers behave properly, as nothing will happen if they do (i.e., we assume a safety barrier is efficient and has been correctly selected). We only assess the cases when they fail to respond on demand, by taking into account their Probability of Failure on Demand during the quantification.

Third step. List, and position on the event tree, all the relevant risk mitigation measures; this includes measures reducing the probability of one or more consequences, as well as measures reducing the intensity of the consequences (e.g., a firewall). In this work, we do not take into account the latter, as the consequence/impact of a hydrogen leak or burst is out of the scope of the FireComp project ([FireCOMP, 2016](#)). We simply assess the frequency of occurrence of any leak or burst, whatever the leak flow rate or volume.

Fourth step. Assess the consequences using all the information displayed on the bow-tie diagram.

5.9.3.1 Review of the preliminary risk analysis

The first step was to conduct a review of the preliminary risk analysis performed within the FireComp project (FireCOMP, 2016), to give the elements required for the first step of the construction of bow-ties. The scope of the study needed to be defined first (application and environment), as bow-ties will vary depending on the use of the storage. These were paired with the list of IE that can have a thermal impact on the storage, and then regrouped.

5.9.3.2 Applications involving high-pressure hydrogen storage

Hydrogen cylinders are not usually used as a single element but integrated in bundles or other type of frames. Ten combinations of an application and an environment needed to be studied, which was presented in Table 1 of Saw et al. (2016). For clarity, each combination of an application and an environment was then referred to as an application. For example, an AIR LIQUIDE (AL) bundle used in transportation is then an application; the same bundle used in a customer facility is another application.

5.9.3.3 List of initiating events

The authors then identified 13 IEs (I1 – I13) that may have a thermal impact on the hydrogen storage in at least one of its applications. Note that malicious acts of arson are out of the scope of these kinds of assessments.

A preliminary risk analysis was carried out by AIR LIQUIDE and HEXAGON to check whether the IE were physically possible or not, for each application. Following review by FireComp Partners (FireCOMP, 2016), some events were discounted for some applications because they were considered too improbable. Table 2, reproduced from Saw et al. (2016) in Fig. 5.13, gives the updated list of events which was selected for the QRA in terms of if that the scenario is physically possible, or not, which the authors interpreted as the scenario is physically impossible or has been considered too improbable in the preliminary risk analysis held by AIR LIQUIDE (AL) and HEXAGON (HEX) prior to the study. From this a list of the seven top events were then selected, which are detailed in Fig. 5.14, reproducing Table 4 from Saw et al. (2016).

In the paper, Saw et al. then focus on applications involving forklift trucks, and describe the methodology employed to quantitatively assess the risk of composite storage systems exposed to fire.

5.10 Human factors

5.10.1 Human factors and safety

In the context of technology and safety, the term “human factors” refers to factors that involve humans and that have an impact on safety. The involvement of humans in a technological setting comprises humans as planners and operators (i.e., professional engagement with technology) and as users (typically nonprofessional users, clients, or customers).

In any technological application, the contribution of humans is to some degree paradoxical: humans contribute essentially to maintaining safety not only in the design phase but also in the operational phase by controlling processes; but, at the same time, humans also make errors and thereby create dangerous situations and sometimes accidents. The various estimates of how often human error is the primary causal factor in industrial and transport vary somewhat, but typically

N°	Initiating event, Ii	Is the initiating event selected for this application?									
		AL Tube trailers type III – on the road	AL Tube trailers type III – in a plant	HEX type IV trailers – on the road	HEX type IV trailers – in a plant	AL bundles – in transportation	AL bundles – in customer facility	HEX bundles – in transportation	HEX bundles – in customer facility	Refuelling station (H ₂ source)	Forklift
Road accident involving a trailer or a car or forklift during transport											
I1	Fire of the tyres	Yes	No	Yes	No	Yes	No	Yes	No	Yes	Yes
I2	Oil or fuel fire (the source of fuel is the one of the truck)	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No
I3	Fire propagation from another car or another truck on the other side of the truck, or at a fuel station	Yes	No	Yes	Yes	Yes	No	Yes	Yes	No	No
I4	Fire after Roll-over	Yes	No	Yes	No	Yes	No	Yes	No	No	Yes
I5	Compartment fire (engine or interior parts)	No	No	Yes	Yes	No	No	Yes	No	No	No
I6	Battery fire (hybrid and fuel cell vehicles)	No	No	Yes	Yes	No	No	Yes	No	No	No
Hydrogen (H ₂) flame impact											
I7	From TPRD of another bundle or cylinder (normal opening or not) of hydrogen storage	No	No	No	No	No	Yes	Yes	Yes	No	No
I8	From (HP) fittings, valves or piping connections	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fire on a plant (customer plant or filling plant)											
I9	Pallets fire	No	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes
I10	Electric fire	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
I11	Building (combustible walls) fire	No	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes
I12	Fire while filling H ₂ / tow-away	No	No	Yes	Yes	No	Yes	Yes	Yes	No	Yes
Forest fire											
I13	Forest fire	No	No	No	Yes	No	Yes	No	Yes	No	No

FIGURE 5.13

Initiation events and applications.

Table 2 from Saw, J. L. et al. (2016). *The EU Firecomp project and risk assessment of hydrogen composite storage applications using bow-tie analysis*. Institution of Chemical Engineers Symposium Series, January 2016, p. 161.

N°	Top event	N°	Initiating event
T1	45 kW/m ² partial	1	Fire of the tyres
		10	Electric fire
		13	Forest fire
T2	45 kW/m ² engulfment	5	Compartment fire (engine or interior parts)
		6	Battery fire (hybrid and fuel cell vehicles)
		9	Pallets fire
T3	125 kW/m ² partial	2	Oil or fuel fire (the source of fuel is the one of the truck)
		3	Fire propagation from another car or another truck on the other side of the truck, or at a fuel station
T4	125 kW/m ² engulfment	11	Building (combustible walls) fire
T5	280 kW/m ² local	7	From TPRD of another bundle or cylinder (normal opening or not) or hydrogen storage
T6	280 kW/m ² partial	8	From (HP) fittings, valves or piping connections
		12	Fire while filling H ₂ / tow-away
T7	280 kW/m ² engulfment	4	Fire after Roll-over

FIGURE 5.14

List of top events.

Table 4 from Saw, J. L. et al. (2016) *The EU Firecomp project and risk assessment of hydrogen composite storage applications using bow-tie analysis*. Institution of Chemical Engineers Symposium Series, January 2016 (161) described by level of thermal input and engulfment.

range between 50% and 90%. Organizational factors may be involved in active human errors and will typically be categorized as poor/lacking procedures, training, man-power planning etc. or sometimes more global shortcomings such as poor safety culture.

5.10.2 Human errors and organizational failures

The modern view of human error among safety specialists is that while human error is unavoidable, the circumstances that prompt human errors are to a large extent controllable. This view of human error has in large part been shaped by Rasmussen (1983) and Reason (1990), each of whom has promoted the “systems view” of human error. Rasmussen advocated on a general level the idea that human error is human–system mismatch. His the so-called SRK framework (skill, rule, knowledge) has been used widely for analyzing human error and was subsequently further developed by Reason.

Rasmussen’s three levels of performance essentially correspond to decreasing levels of familiarity or experience with the environment or task.

1. *Skill-based performance*: At this level people carry out routine, highly practiced tasks in what can be characterized as a largely automatic fashion. Except for occasional checking, very little conscious effort is required (e.g., writing on a keyboard, changing gears, or doing any of the countless daily well-rehearsed tasks at home or at work).
2. *Rule-based performance*: At this level we execute a well-known routine procedure, but have to take into account some change in a situation and modify our preprogrammed behavior, typically a situation with which we are familiar or have been trained to deal with, we engage in rule-based behavior (e.g., filling up petrol in one’s car at the usual petrol station).
3. *Knowledge-based performance*: This level of performance is required and used when we face a novel situation and have no applicable rules. It may be a form of problem-solving employing

analytical reasoning and stored knowledge. In a technological setting this is the type of behavior.

Rasmussen's skill-rule-knowledge framework for classifying performance relates essentially to a cognitive classification of how familiar we are with tasks and, hence, at which level of conscious effort and attention we are devoting to our tasks. These distinctions between different levels of performance are important and useful, because they allow us

- to model complex performance (typical in technological work settings) by reference to the shortcuts that people have learned to use to save time and effort, and
- to analyze errors according to different levels of performance.

Reason adapted Rasmussen's performance-based model, tying the classification of errors to cognitive processing. An often-used definition of human error is the following taken from the domain of medicine but based on Reason's work focusing originally on industrial safety:

An error is defined as the failure of a planned action to be completed as intended or the use of a wrong plan to achieve an aim.

An error is thus either a failure in planning (wrong intention) or a failure to carry out one's plan. The latter Reason classified as lapses and slips, depending on whether it is memory failure (lapse) or a response failure (a slip).

Fig. 5.15 shows Reason's categories and how they expand upon the SRK framework.

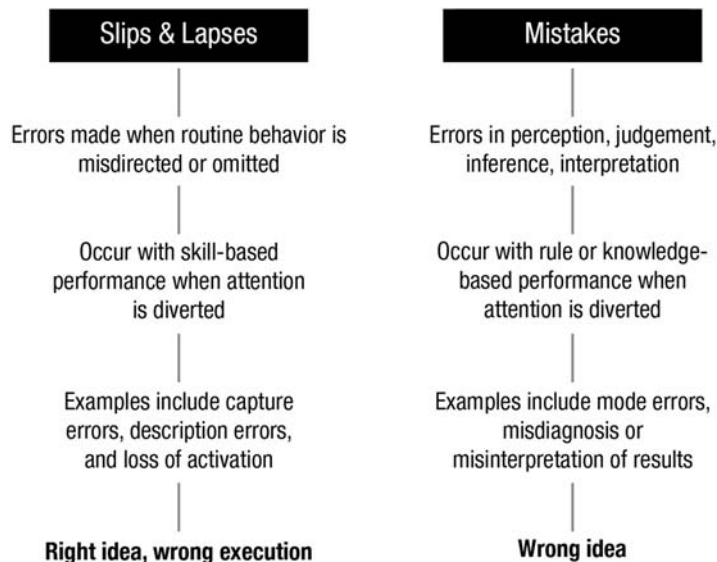


FIGURE 5.15

Categories and how they expand upon the SRK framework (Reason, 1990).

5.10.3 Human reliability assessment

In his well-known textbook on human reliability assessment (HRA) methods, Kirwan (1994) emphasizes that one of the primary goals of human reliability analysis is to provide a means of properly assessing the risks attributable to human error. To achieve this aim three overall phases must be carried out: (A) Identifying what errors can occur. (B) Deciding how likely the errors are to occur. (C) Enhancing human reliability by reducing this error likelihood (human error reduction).

Thus HRA is used to identify, model, predict, and when possible, to reduce human errors in operations and will typically include normal production operations, maintenance, testing, and emergency conditions. It is well-known that maintenance and testing are phases especially vulnerable to human error that can seriously influence system safety. For instance, through insertion of an incorrect component, miscalibration, or failure to align the system back to its operational configuration.

A number of methods and techniques are available, with which a structured analysis of human reliability of a specific industrial setting is undertaken. A common preliminary phase for conducting an HRA is, when the scope of the problem and the exercise to be undertaken has been defined, to perform a risk assessment. This may be carried out with techniques described in the section above on hazard identification methodologies, for example, fault and ET. Next, it is customary to define the interaction between humans and the system in terms of a task analysis—for example, hierarchical or time-line task analysis. The next step involves the identification of possible errors, possibly by using group-based techniques such as HAZOP, as described above. Following the identification of errors, it is necessary to estimate the likelihood of their occurrence. To arrive at a quantification of human error probabilities either human error databases can be used or expert judgment or a mixture of both. In the table in Fig. 5.15 we show an illustration of human error quantification from a commonly used HRA method. Finally, risk reduction options must be reviewed and possibly a prioritization, selection, and a plan for implementation of risk reduction (Figs. 5.16 and 5.17).

➤ Scope
➤ Task analysis
➤ Risk/hazard identification
➤ Modelling
➤ Human error identification
➤ Human error quantification
➤ Impact assessment
➤ Risk control options
➤ Risk reduction plan
➤ Safety assessment reporting

FIGURE 5.16

Generic classifications (HEART).
After Williams, (2015). Heart—A proposed method for achieving high reliability in process operation by means of human factors engineering technology, *Safety and Reliability*, 35(3), 5–25.

Generic task	Proposed nominal human unreliability (5 th -9 th percentile bounds)
a) Totally unfamiliar, performed at speed with no real idea of likely Consequences	0.55 (0.35-0.97)
b) Shift or restore system to a new or original state on a single attempt without supervision or procedures	0.26 (0.14-0.42)
c) Complex task requiring high level of comprehension and skill	0.16 (0.12-0.28)
d) Fairly simple task performed rapidly or given scant attention	0.09 (0.06-0.13)
e) Routine, highly-practised, rapid task involving relatively low level of skill	0.02 (0.007-0.045)
f) Restore or shift a system to original or new state following procedures, with some checking	0.003 (0.0008-0.007)
g) Completely familiar, well designed, highly-practised, routine task occurring several times per hour, performed to highest possible standards by highly-motivated, highly-trained and experienced person, totally aware of implications of failure, with time to correct potential error, but without the benefit of significant job aids	0.0004 (0.00008-0.009)
h) Respond correctly to system command even when there is an augmented or automated system providing accurate interpretation of system stage	0.00002 (0.000006-0.0009)

FIGURE 5.17

Proposed reliabilities for generic tasks.

5.10.4 Safety culture

It has become widely accepted that an organization's safety culture can have an impact on safety performance. Two installations may be entirely alike in terms of production, ownership, workforce, procedures and yet differ in terms of safety performance and measurable safety climate. When seeking to assess and control the impact of human factors on the level of risk of a plant or other installation, it is therefore essential to include organizational factors as well.

The concept of safety culture was introduced in the aftermath after the nuclear power plant accident in Chernobyl in 1986, when the concept was invoked to explain a corporate attitude and approach that tolerated gross violations and risk-taking behavior. A large number of studies have since developed models and measures of safety culture. One of the most widely cited definition of safety culture was offered by the Advisory Committee on the Safety of Nuclear Installations in the United Kingdom (HSE, 1993), who defined safety culture as follows:

The safety culture of an organization is the product of individual and group values, attitudes, perceptions, competencies and patterns of behavior that determine the commitment to, and the style

and proficiency of, an organization's health and safety management. Organizations with a positive safety culture are characterized by communications founded on mutual trust, by shared perceptions of the importance of safety and by confidence in the efficacy of preventive measures.

The first-generation models characterized positive safety culture for a given organization as founded on mutual trust, shared belief in the importance of safety, and shared belief that preventive measures make a difference. In subsequent models (from mid-90s and later) there is additionally an emphasis on organizational learning: that is, willingness and ability to learn from experience (errors, incidents, and accidents).

There is no "standard model" of safety culture, but there is widespread agreement that safety culture includes factors relating to

- visible top management and shop floor management commitment to safety;
- work force ownership and participation in safety solutions;
- open communication;
- mutual trust between management and employees;
- willingness and ability to learn from experience (an organization with a memory);
- work force involvement in company and motivation.

5.10.4.1 Measuring safety culture/climate

A distinction is often made between culture and climate: culture is slow to change and involves mostly tacit (unspoken, hard, or impossible to articulate) beliefs and norms, whereas climate is shaped by context and more explicit. Therefore empirical studies, and especially surveys and interviews, are usually said to uncover, at best, safety climate, whereas safety culture is only characterized indirectly.

When considering empirical approaches to safety climate, decisions must be made about the following three dimensions:

1. What should be measured (what are the factors to be measured?).
2. How should measures be made (what are the methods and techniques to perform the measurement?).
3. Where should measures be made (in which part of the organization should sampling and appraisal be made?).

As described above, the issue of what should be measured is addressed in somewhat different but not necessarily incompatible ways by different analysts. There is general but not precise agreement about the factors that are involved in safety climate culture.

Similarly, different methods and techniques are available for assessing safety climate—interviews, field observations, participation, surveys. But within this range of methods, surveys (written or web-based questionnaires, oral interview surveys) yield a uniform output that may be quantified more or less directly.

Finally, assessment of safety climate in an organization may be targeted at both, the management level (top and middle management including supervisors) and the shop floor level, and at different operational units and even office staff and planners.

Taking the range of choices into account, the most typical kind of safety climate, respectively safety culture assessment method is that of questionnaire-based surveys of staff perceptions and attitudes. A questionnaire-based survey will typically be targeted at the operational staff, including possibly group leaders, and, of course, will use some form of safety climate questionnaire—a safety climate assessment tool.

There are a large number of such tools (questionnaires) available, some of these being domain-specific (maritime, oil production platforms, aviation, process industry etc.). Questionnaires require the respondent to answer specific questions in terms of the selection of one among a fixed set of reply options—typically a selection from a Likert-type ranking scale: “Strongly Agree, Agree, Neither agree nor disagree, Disagree, or Strongly disagree.” Question items will form groups that correspond to underlying factors (say, perception of top management commitment to safety). For validated survey tools, factors will have been established through possibly pilot surveys and subsequent statistical analysis (for instance, factor analysis). Finally, analysts will be able to establish a benchmark when survey tools have been applied to a range of installations or workplaces within comparable parameters. For instance, the EU project ARAMIS, adapting a construction and production plant questionnaire, has used this to collect data from five European Seveso-type plants. On the ARAMIS approach (Duijm et al., 2004), a single global safety climate index has been established for use in integrated assessment of safety management—instead of a range of different indices, each corresponding to a single safety climate factor.

References

- Bustamante Valencia, L., Valencia, V., Blanc-Vannet, P., Domergue, D., Heudier, L., Jamois, D., et al. (2016). Thermal history resulting in the failure of lightweight fully-wrapped composite pressure vessel for hydrogen in a fire experimental facility. *Fire Technology*, 52(2), 421–442. Available from <https://doi.org/10.1007/s10694-015-0513-y>.
- California Energy Commission. (2004). *Failure modes and effects analysis for hydrogen fueling options*.
- Casamirra, M., Castiglia, F., Corchia, L., Giardina, M., Lombardo, C., Messina, G. (2007). Risk analysis of the storage unit in a hydrogen refuelling station. In *International Conference on Hydrogen Safety*, pp. 1–10. Available at: <http://conference.ing.unipi.it/ichs2007/fileadmin/user_upload/CD/PAPERS/11SEPT/4.1.294.pdf>
- Choi, J. O., Cho, W. C., K, S. K. (2020). Safety issues of electrolysis systems in coupled operation with renewable energy sources. In *HySafe Research Priority Workshop 2020*.
- FireCOMP. (2016). Available at: <https://cordis.europa.eu/project/id/325329/reporting>
- Fuel cells and Hydrogen 2 Joint Undertaking (FCH 2 JU). (2017). *European Hydrogen Safety Panel*. Available at: <<https://www.fch.europa.eu/page/european-hydrogen-safety-panel>>
- Health and Safety Executive. (2013). *Managing for health and safety HSG65*, Part 1, pp. 1–62. ISBN: 978 0 7176 6604 1.
- Health and Safety Executive. (2014). Risk assessment: A brief guide to controlling risks in the workplace. *Toxicologic Pathology*, p. 5. Available at: <<http://www.hse.gov.uk/pubns/indg163.htm>>
- HM Factory Inspectorate. (1976). *The explosion at Laporte Industries Ltd.*, Ilford, April 5, 1975, p. 24.
- HSE. (1993). *ACSNI study group on human factors: Third Report— Organising for safety*, ISBN-13 978-0717608652, Publisher, HSE Books, 1 Dec. 1993. HSE.
- HSE. (2021). *Managing risks and risk assessment at work*. Available at: <<https://www.hse.gov.uk/simple-health-safety/risk/index.htm>>

- Hydrogen Incident and Accident Database (HIAD 2.0). (2016). Available at: <<https://odin.jrc.ec.europa.eu/giada>>
- Hydrogen Safety Panel. (2019). *Safety planning for hydrogen and fuel cell projects*, p. 52. Available at: <https://fch.europa.eu/sites/default/files/Safety_Planning_for_Hydrogen_and_Fuel_Cell_Projects_Release1p31_20190705.pdf>
- Hydrogen Tools. (2021). Available at: <<https://h2tools.org/>>
- International Maritime Organisation (IMO). (2021). Available at: <<https://www.imo.org/en>>
- International Association for Hydrogen Safety “HySafe”. (2021). Available at: <<https://hysafe.info/>>
- ISO. (2019). IEC 31010:2019 Risk management—Risk assessment techniques. Available at: <<https://www.iso.org/standard/72140.html>>
- Kirwan, B. (1994). *A guide to practical human reliability assessment*. CRC Press.
- Melideo, D., Moretto, P., & Wen, J. (2019). HIAD 2.0—Hydrogen incidents and accidents database. In *Australia International Conference on Hydrogen Safety ICHS 209* (pp. 1–12). September 21–23, 2019, Adelaide.
- Rasmussen, J. (1983). Skills, rules, and knowledge; signals, signs, and symbols, and other distinctions in human performance models. *IEEE Transactions on Systems, Man, and Cybernetics*, 13(3), 257–266.
- Reason, J. (1990). *Human error*. Cambridge University Press.
- Risktec Solutions Ltd. (2007). Practical HSE risk management—An introduction to the bow-tie method. In *International Conference for Achieving Health & Safety Best Practice in Construction*, Dubai, UAE.
- Saw, J. L., et al. (2016). The EU firecomp project and risk assessment of hydrogen composite storage applications using bow-tie analysis. *Institution of Chemical Engineers Symposium Series*, January 2016 (p. 161).
- SEMATECH. (1992). *Failure mode and effects analysis (FMEA): A guide for continuous improvement for the semiconductor equipment industry SEMATECH technology transfer #92020963B-ENG*. SEMATECH.
- Spouge, J. (1999). *A guide to quantitative risk assessment for offshore installations*. ISBN I 870553 365.
- The International Civil Aviation Authority. (no date). Available at: <<https://www.icao.int/Pages/default.aspx>>
- The Management of Health and Safety at Work Regulations 1999 No. 3242. (1999). Available at: <<https://www.legislation.gov.uk/uksi/1999/3242/contents/made>>
- United Nations Forum for Harmonisation of Vehicle Standards. (2021). Available at: <http://www.unece.org/trans/main/wp29/meeting_docs_wp29.html>
- United Nations International Labour Organisation. (2021). Available at: <<https://www.ilo.org>>
- Williams, J. C. (2015). Heart—A proposed method for achieving high reliability in process operation by means of human factors engineering technology. *Safety and Reliability*, 35(3), 5–25. Available from <https://doi.org/10.1080/09617353.2015.11691046>.

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Safety measures and safety barrier functions

6

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6.1 Introduction to system safety

The utilization of hydrogen energy needs the build-up of an infrastructure that supports the production, storage, transport, and use of hydrogen. As a fuel for electrical power production, hydrogen may be produced from many different sources, for example, wind energy, photovoltaic, biomass, or fossil fuels. Hydrogen, as an intermediate, for example, together with carbon dioxide, may be used to synthesize other sustainable liquid fuels for some possible applications, such as, for aircraft. An indication of the complexity of such applications is exemplified in Fig. 6.1.

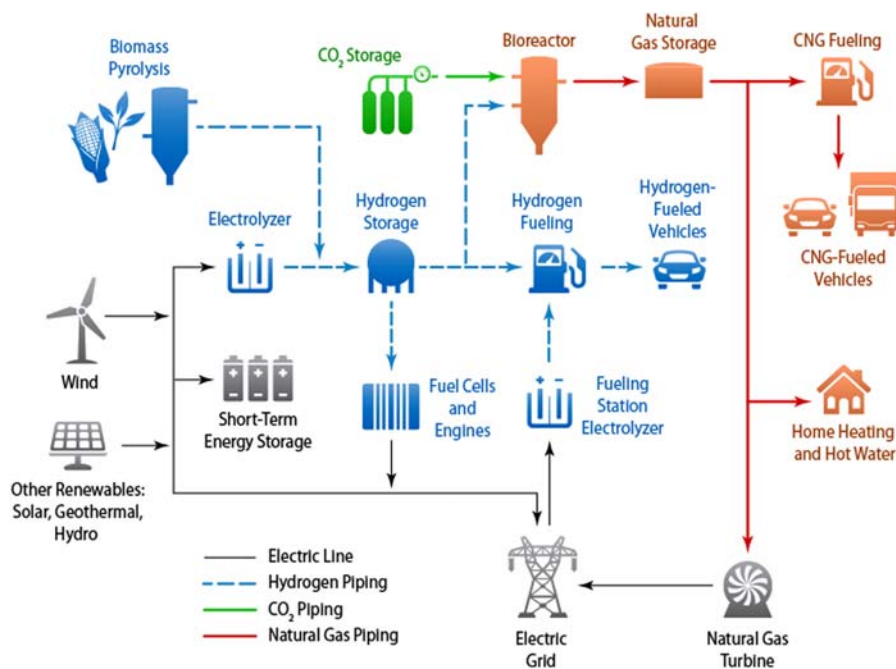


FIGURE 6.1

Hydrogen infrastructure indicating the many production pathways for hydrogen (NREL, 2020).

Illustration courtesy of the National Renewable Energy Laboratory (NREL).

This chapter will present and discuss the various methods and technical solutions to introduce safe hydrogen-based technologies. Such methods are implemented to make the supply chain with its multiple components and hydrogen storage conditions safe and acceptable for society. The application of hydrogen energy and the supporting infrastructure with all its substructures and parts need to be at least as safe as the similar present up-to-date technologies. A sociotechnological systems approach is taken here, as systems safety concerns technology and the interaction with humans utilizing various components. It also concerns compliance with the present infrastructure, such as, the design of tunnels and car parks (Dagdougui et al., 2018; Markert & Hansen, 2018).

Each of the infrastructures needs to be compatible with relevant hydrogen applications which are mentioned in Chapter 2 of this book, Hydrogen Technologies. In a broad view, the hydrogen application can be categorized into applications concerning production, storage and transport, and hydrogen (DS/ISO/TR 15916, 2015). A range of production methods are available, for example, reformation of natural gas and electrolysis. Furthermore, hydrogen produced as a by-product is available in large amounts in chemical industries. Each of the processes may have particular risks, and therefore there is a need to consider specific safety measures. See also Chapter 4 “Accidental Consequences” and Chapter 5 “Risk Assessment.”

Production is also connected to hydrogen storage and (bulk) transport. The hydrogen may be stored and transported under pressurized gaseous or cryogenic liquid conditions. Furthermore, technologies are available that stores hydrogen in the form of metal hydrides (Buchner, 1984; Jurczyk et al., 2008; Tanaka et al., 2009) or utilizing certain chemicals as the reversible conversion of hydrogen to ammonia (Duijm, Paulsen, & Markert, 2005; Strickland, 1984), or using the reversible dehydration of methyl-cyclohexane to toluene (Taube et al., 1983; Taube et al., 1985). Hereunder, it may be argued that pressurized transport is most cost-efficient for shorter and medium distances while liquid hydrogen (LH₂) transport is favorable for long distances (Markert et al., 2007; Simbeck & Chang, 2002). The safety measures have thus, on the one hand, to deal with very high-pressure storages and transport applications that may cause mechanical and thermal explosions as well as jet fires. On the other hand, the safety measures for cryogenic storage and transport must consider the very low temperatures and the LH₂ boil-off that has to be removed continuously.

The critical facts on the physical and chemical nature of the various forms of hydrogen lead to certain phenomena that must be considered for safety reasons. These are described in more detail in this book’s different chapters but are summarized here briefly. The system safety requirements need to consider the very high pressure and cryogenic temperatures, the wide flammability range, low ignition energy, high buoyancy, and pressure peaking effects of unignited or ignited pressure releases. Therefore as stated in the standard on “Basic considerations for the safety of hydrogen systems,” ISO/TR 15916:2015 (DS/ISO/TR 15916, 2015), the following safety categories for gaseous and LH₂ primary hazards need to be considered to find appropriate safety measures or safety barriers:

- **Flammability:** The flammable range is over a large concentration interval. Hydrogen has very low ignition energy at stoichiometric concentrations. Ignition may result in thermal and pressure effects that need to be considered; some mixtures with other gases have easy ignitability as well, hydrogen has a high deflagration to detonation (DDT) probability.
- The small size of the molecule causes a low viscosity and high diffusion rate of hydrogen in air. Hydrogen clouds have high buoyancy, which is an advantage in open areas but might be the opposite in enclosed areas where the gas will be quickly mixed and accumulated.

- Interaction with materials (embrittlement of certain metals).
- Asphyxiation hazard if oxygen is replaced.
- Hazards associated with the storage conditions:
 - Very high pressure or very low temperatures compared with the energy carriers we are used to at least in domestic applications.
 - Others for other storage methods, like metal hydride storages.

These phenomena are of safety concern in any hydrogen application. The above phenomena as flammability and asphyxiation are also known for other fuel applications. For hydrogen, the hazards related to its flammability are distinct from other fuels and need careful considerations, while other hazards, for example, asphyxiation, are similar to other fuels and gases.

6.2 Preventive and mitigative safety barrier functions

In the following sections, several preventive and mitigative technologies, methodologies, and procedures are described, which support the safe and acceptable performance of hydrogen installations and applications in a socio-technological context. These technologies can be categorized into preventive and mitigative safety barrier functions. These functions increase the safety level for single subcomponents of a hydrogen system. However, only the combined function of all the implemented safety barrier functions will provide the best prevention to achieve a societal acceptance of the overall safety risks of a particular application. The safety functions will decrease the number of critical events, such as a hydrogen release, and mitigate the potential harm resulting from such critical events. Hereunder, it is essential that the barriers are independent of each other so that a failure in one barrier is not causing another barrier's failure.

The fundamental hypothesis within system safety is that there are no zero risks, but the risk shall be reduced to an acceptable level. Due to the vast number of technological, organizational, and human factors influencing any system, improbable critical events may happen that initiates an event chain with an accidental outcome.

A holistic approach has to be taken to ensure the system safety of hydrogen technologies that will be part of our daily lives. The uncontrolled flow of energy (UFOE) model (Khan, Rathnayaka, & Ahmed, 2015; Rasmussen & Grønberg, 1997a,b) idealizes any accident, as seen in Fig. 6.2. Within this model, any system is in a safe state when the primary process equipment confines the hazardous source or hydrogen. If this confinement is compromised, the hazardous source may expose people, constructions, or the environment—the vulnerable objects. The additional safety barrier functions are implemented to prevent partly (e.g., minimize the number of) accident events: the Loss of Confinement (LoC) and partly to help to mitigate the harm: the UFOE that is a result of LoC events. The UFOE is here thought of as the term that describes all possible hazardous events from hydrogen jet fires over explosions to degrees of burns on people and pressure waves impacting people and constructions.

A vital philosophy is an “Inherently Safer” approach (Kletz, 2003). A reduction in safety risks may be reached by utilizing technology to prevent or mitigate potential accidents. On the other hand, some of the recommended safety precautions, for example, given in the respective ISO/TR standards, are not purely technological but are part of procedures that need to be established and

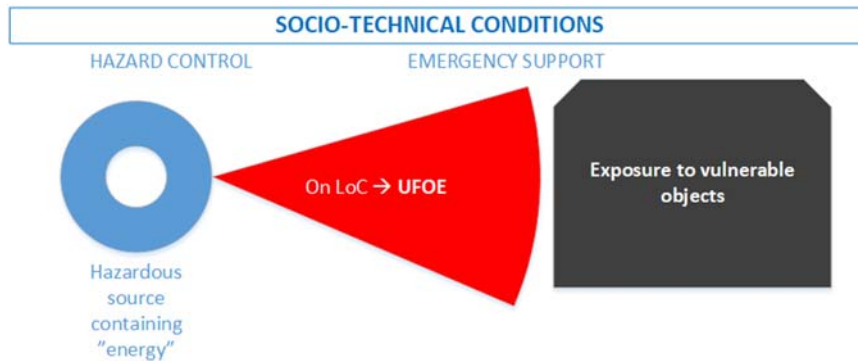


FIGURE 6.2

UFOE model as simple accident model (Khan et al., 2015; Rasmussen & Grønberg, 1997a,b).

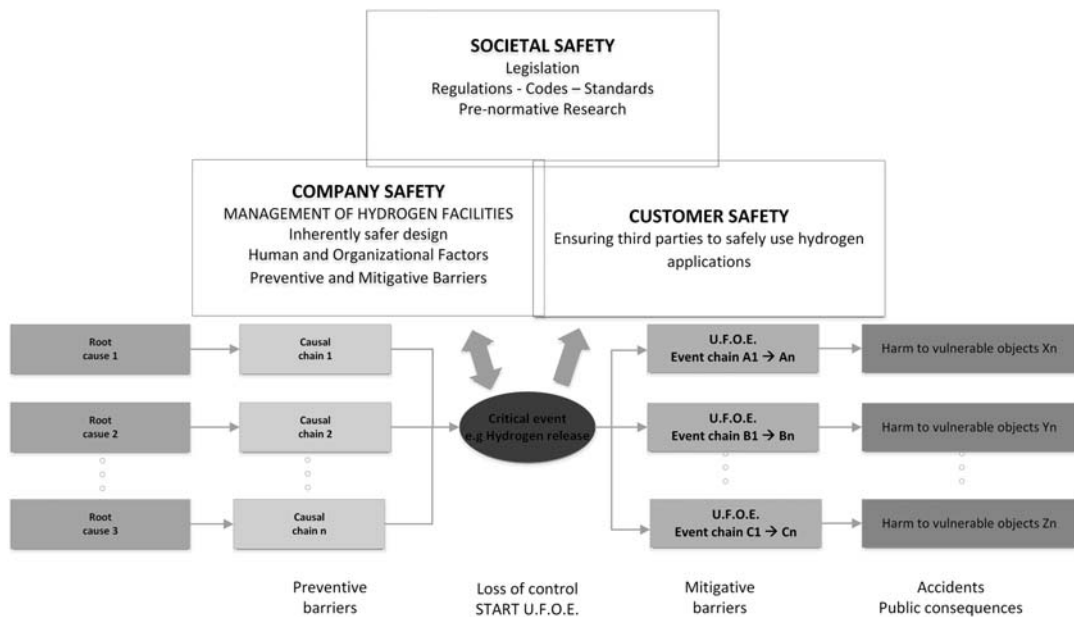


FIGURE 6.3

Sociotechnological and customer safety influences inspired by Leveson (2004) and Rasmussen and Svedung (2000).

maintained by the organizations responsible for the safety in the society. Rasmussen and Svedung (Leveson, 2004; Rasmussen & Svedung, 2000) have widened the scope within the socio-technological context. They argue that any critical event has to be seen as a complex interaction of technology and society as a whole, as shown in Fig. 6.3. It is seen that a critical event may occur

Table 6.1 Classification of safety barriers.	
Implemented barrier type	
Passive	Physical means, such as, dike, fire wall. Human/operational.
Active	Technical, such as, safety instrumented systems according to IEC 61511. Other technological safety systems. Facilities to reduce external risks

both caused by the primary technology, the failure of safety functions, and the human impact on the system. This has to be seen as an interaction in the wider context of the safety policies of a company, the guidelines, standard and regulations within the respective branch of industries, as well as the national and international laws.

The term “safety barrier function” needs some clarification. Safety barriers can be understood differently, as Sklet (2006) is pointing out in his literature review on understanding and definitions of the term “safety barrier”. He recommends the following understanding to classify barrier types or systems as illustrated in Table 6.1. The classification makes the primary distinction into “passive” and “active” barrier types. Often active barriers are found to be a combination of technical means and human/operational elements. Passive barriers may be considered to function continuously over time, while active barriers are activated before use. Both types may be installed permanently or just temporarily in case of an emergency.

The physical, passive barriers (e.g., containment, fences, and firewalls) are usually functioning continuously as they do not need to be activated. They may also be temporary, e.g., a temporary obstruction fencing a working area during an activity. The passive, human operational barriers (e.g., safety distances in accordance with Haddon’s principle separation in time and space) may be functioning continuously, or be implemented as part of high-risk activities.

Active, human/operational barriers may be in a continuous mode or activated on demand. Often, these barriers are an integrated part of a work process (e.g., self-control of work and third-party control of work) in order to reveal potential failures, e.g., introduced by humans.

Safety barriers may also be classified in several other ways. The classification illustrated in Fig. 5 may not always be best suitable for the purpose of the classification. Then, some other lines of classification described in Section 3 can be used (cited from Sklet, 2006).

It should be recognized that technical safety barriers are installed for safety reasons and are not essential to run the primary process. In general, technical safety barriers are usually designed to be independent of the process control measures and with a certain redundancy, dependent on the risk connected with the technical process. The barriers need to be designed for the various stages of the hydrogen supply chain, that is for the hydrogen production facilities, bulk transport by tankers, ships, trains, pipelines, and small, medium and large scale pressurized or cryogenic storages, as well as other storage types.

It is, as mentioned above, distinguished between passive and active safety measures (barriers). These have different requirements for maintenance during their lifetime. Therefore to reduce safety risks, it is of great importance to introduce and maintain proper maintenance of the hydrogen applications and particularly of the safety barriers. This issue is exemplified in Fig. 6.4, where a primary process failure is thought of being detected and repaired in due time. Any failure of the essential process equipment will stop the process.

In contrast, a safety barrier malfunction may not be detected in due time as such a failure is not influencing the primary hydrogen process or service and may only be detected in case the barrier is needed in the situation of a critical event. This thought implies that such failures do not directly develop into an accident, and the system may work for many years, but the risk will be higher as not all safety measures are in place. Therefore the safety management strategies for safety barriers need operational planning and well-organized test, control, and maintenance procedures. Thus appropriate inspections and maintenance of barriers are essential. Hereunder, the organizational and human factors must be recognized appropriately as part of any safety barrier system. Only with

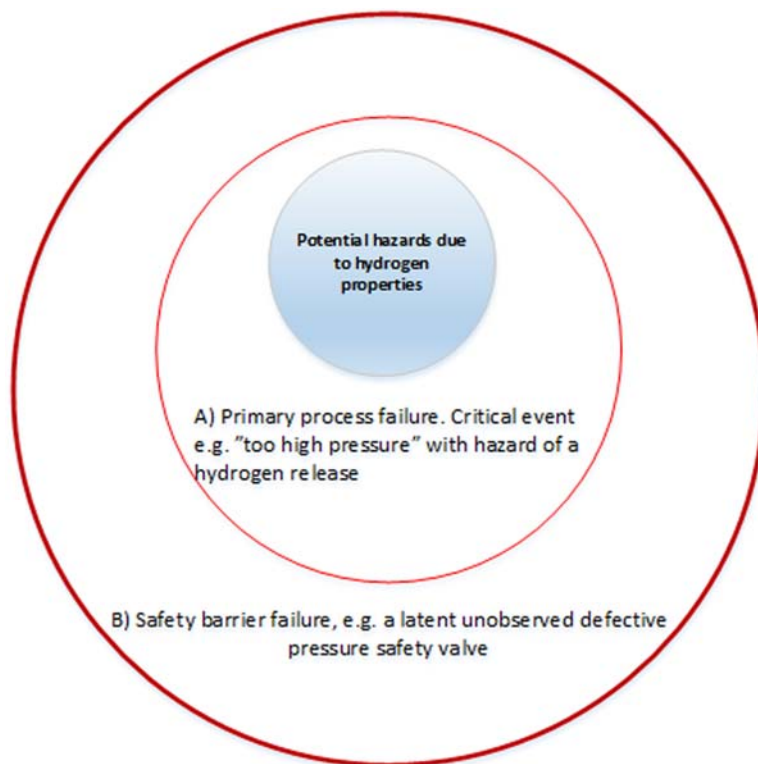


FIGURE 6.4

Control versus safety barrier.

fully functioning safety barriers or lines of defense, the overall risk level predicted by the risk assessment is valid (Duijm & Markert, 2009; Duijm, 2009).

The necessity of safety measures and implementation of safety barriers is commonly the result of a qualitative or quantitative risk assessment (QRA). Such risk assessments apply the knowledge and prediction tools to evaluate the consequences and potential hazards resulting from accidents described in the former chapters. Part of this is to assess the safety barriers' effect to help reach the acceptable risk level to allow for the application within the specific context.

One part of the risk assessment procedures is to gain knowledge on the acceptability of the implied risks and understand the best safety management strategies to ensure a high safety level. The other part is assessing the reliability of hydrogen components, including any safety installations that minimize the occurrence of hazardous events. Both are describing the risk of any system.

In order to categorize the components and subsystems to improve safety management, a technical process may be regarded as the components that are essential to perform the process, which are the primary controls discussed above. This approach is in line with the Management Oversight and Risk Tree -MORT-concept (Trost & Nertney, 1985) to distinguish between controls and safety barriers (Trost & Nertney, 1985). Control barriers are related to control of the designed (wanted) energy flows, while safety barriers are associated with the control of unwanted energy flows (see Fig. 6.4). For instance, the primary process could be the tube/pipe necessary to direct a material flow from place A to place B. It could also be the tank that can keep the amount of hydrogen stored as fuel in an hydrogen fuel cell (HFC) vehicle. The strength and other mechanical parameters of the tube and tank are determined by the required temperature and pressure of the process to be efficient and reliable. Real-world conditions may deviate from the optimal conditions, and therefore safety barriers are added to the primary system to prevent or limit the possible hazardous outcome of such deviations. It could be simply that the pipe or tank is made more robust to cope with, for example, higher pressure due to inadequate pressure regulations. It could also introduce additional safety barriers where each of the barriers can prevent or mitigate events due to deviations, such as deviating temperatures and pressures, including mechanical failures due to aging processes or incorrect use of components.

6.2.1 System safety—what is safe enough?

The use of safety barriers will provide additional safety to the hydrogen technologies. The question is how many safety barriers are necessary to reach a safety level that is widely accepted in society. This leads to the question of what is safe enough?

The answer to this would often be a risk assessment compared to acceptance criteria or applying the newest standards in the respective field of hydrogen application. The former will predict a risk level for a technological system, for example, hydrogen fueling station and compare it against acceptance criteria in the form of the individual risk or the societal risk (Ale, 2005; Macintyre et al., 2007; Johansen & Rausand, 2014).

There are different concepts like "As Low As Reasonably Achievable" or "As Low As Reasonably Practicable" (Franks, 1999; Mohammadfam & Zarei, 2015; Singh et al., 2010; Zhiyong, Xiangmin, & Jianxin, 2010). The UK Health and Safety Executive (Health & Safety Executive (HSE), 2021b) provides explanations and guidelines on these concepts.

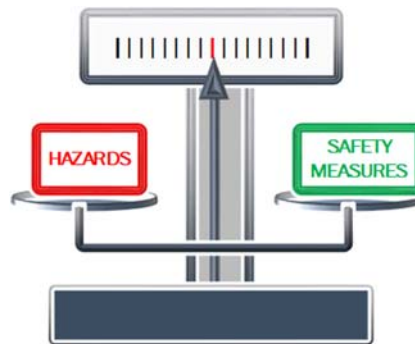


FIGURE 6.5

Cost-benefit considerations using the ALARP approach, the probability of occurrence and severity of hazards should be in balance with the safety measures designed to eliminate the hazard (added safety devices, include procedures and staff training).

One approach is that the safety measures should balance the costs of the potential hazards (Health & Safety Executive, 2021a). Nevertheless, the safety measures' costs may exceed the cost for the hazards in cases in which people could be potential harmed. The HSE uses the term “grossly disproportionate” to evaluate such (Fig. 6.5).

6.3 Inherently safe systems

The forthcoming chapter will address system safety measures and safety barriers for hydrogen systems; hereunder, it will present the concept of building inherently safer systems as preventive and mitigation measures. The inherently safer systems concept has also been applied to hydrogen systems, as, for example, reported by Landucci, Tugnoli, and Cozzani (2010). Such principles and inherently safer design (ISD) approaches are also considered in the ISO/TR 15916:2015 (see Chapter 7: Legal Requirements and Standards (Regulations, Codes, and Standards) for Hydrogen Safety for a comprehensive overview of related standards). The methodology developed during the past decades.

In 1973 Haddon published his basic principles to avoid hazards (Haddon Jr., 1995; n. reprinted), as shown in Table 6.2. The energy damage process is understood as *The phenomena of concern are those involved when energy is transferred in such ways and amounts, and at such rates, that inanimate or animate structures are damaged [...]. The harmful interactions with people and property of hurricanes, earthquakes, projectiles, moving vehicles, ionizing radiation, lightning, conflagrations, and the cuts and bruises of daily life illustrate this class.*

These basic principles are generic for any application and activity in society to reduce human injuries and fatalities as well as economic losses. Therefore these also could be used in the form of guidewords to plan new hydrogen installations and infrastructures. The energy damage processes concerning incidents and accidents with hydrogen technologies are, for example, jet flames from sonic releases, explosions, missiles/projectiles triggered by an explosion.

Table 6.2 Haddon's 10 basic principles (Haddon Jr., 1995).

1	Prevent marshaling (build-up) of energy.	Avoid the marshaling of the form of energy. For hydrogen, it may be for the safe storage and design of facilities and technologies to prevent the accumulation of released gas that on ignition may lead to severe consequences.
2	Reduce the amount of energy marshaled.	For any application, consider reducing the amount of hydrogen to be produced, stored, and transported close to populated/residential areas, for example, large-scale production at remote plants and truck transportation or on-demand small scale production (electrolyzers) at fueling stations and no truck transport.
3	Prevent the release of this energy.	Follow the safety standards for hydrogen applications. Prevent the accidental release scenarios and establish proper means to avoid an accumulation of diffusive releases. Refer to Seveso and Atex directives.
4	Modify the rate of the spatial distribution of release of energy from its source.	For example, design the diameter for the temperature – pressure release device (TPRD)—safety valves not to get pressure peaking phenomena and secondary effects.
5	Separate the energy in space or in time from people and vulnerable objects.	For example, establish safety distances to people and vulnerable equipment. Establish hydrogen detectors to trigger appropriate alarms to enable timely evacuation.
6	Establish separation by the interposition of a material “barrier.”	Establish appropriate barriers such as protective walls against blasts, missiles, and thermal impacts. As well as personal protection means.
7	Modify the contact surface between source and humans/susceptible structures.	Any hydrogen facility should be designed appropriately to avoid harm, for example, caused by cryogenic temperatures, momentum jet releases.
8	Strengthen the susceptible structures (people and property) to reduce losses to these that may result from damage by energy transfer.	To reduce losses (people and property) by increasing the demand, for example, fire and blast resistance. Increase people's knowledge to understand the behavior of hydrogen in applications better and understand the hazards.
9	Mitigate or stop the damage process when it is occurring.	Different means possible, safeguards by detectors, alarms, emergency response etc.
10	Measures to allow (fast) recovery to the preevent status or at least to some stabilized but structurally modified status.	Back to work: delivery of hydrogen; safe use in infrastructures (tunnels, car parks), one facility down can be replaced by a neighboring one—robust networks—business continuation.

Focusing on inherently safer hydrogen technologies also implies including the compatibility of these technologies within the general socio-technological system. In other words, the infrastructures and societies around us. Hereunder, about the safety of public applications of hydrogen production

bulk transport and refueling. Hydrogen cars and other hydrogen vehicles have to operate safely within the traffic infrastructures. Thus they need to be parked in ordinary garages, underground car parks, parking houses. They also need to have a sufficient safety level not to increase the risk on the road and rail systems, for example, in case of a traffic accident within a tunnel. The safety of these technologies needs to be at least as safe as the standard technologies around us.

More comprehensive safer design strategies were developed and promoted for the process industries, for example, Kletz and coworkers (Kletz & Amyotte, 2010; Kletz, 1985, 1988). Fig. 6.6 shows the generic risk management strategies for hydrogen applications. The priority, if possible, is on the inherent safety strategies. These are successively followed by the implementation of passive safety barriers, active barriers, and procedural barriers.

(Gao, Abdul Raman, & Hizaddin, 2020) reviewed presently used inherently safer systems approaches.

Another review by Park et al. (2020) evaluates inherent safety assessment tools (ISAT) that may be used from the conceptual design stage. The authors categorize the approaches as: (a) H-ISATH Hazard-based inherent safety assessment tools, (b) R-ISAT Risk-based ISAT, and (c) CO-ISAT Cost optimal ISAT. They conclude that implementing safety measures during the early stages of process design is done with fewer efforts than improving system safety at later stages of the design process. The inherent safer design approach is to be combined with other risk assessment methods.

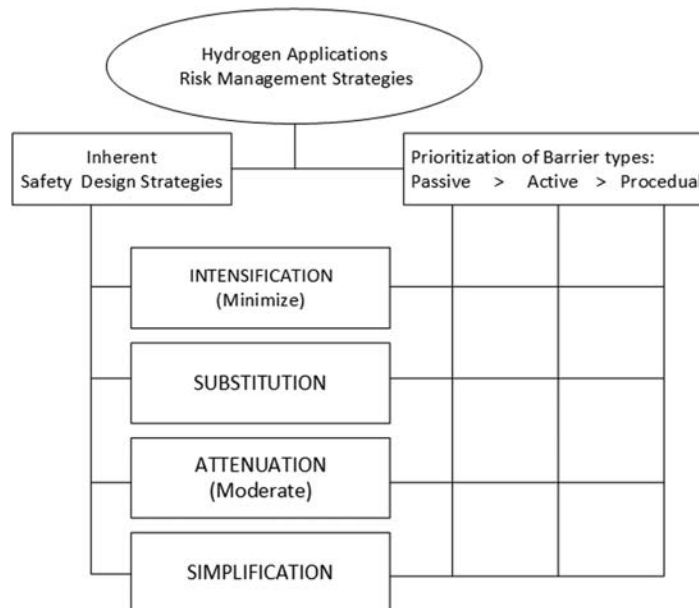


FIGURE 6.6

Inherently safer system approach for hydrogen (Moore & Fuller, 2017; Rigas & Amyotte, 2012).

The ISD (Kletz & Amyotte, 2010) is guided by four guidewords explained by Park et al. (2020) and illustrated in Fig. 6.6.

- Intensification: Use smaller quantities of hazardous materials by reducing the size of relevant equipment. Also called “minimize.”
- Substitution: Use of less hazardous materials, reactions, and processes.
- Attenuation: Use more moderate process conditions through dilution, refrigeration, and process alternatives, but the total amount of chemicals might not be reduced. Also called “moderate.”
- Simplification: Eliminate unnecessary complexity to decrease the likelihood of errors.

Many different measures achieve the inherently safer approach for hydrogen. The most basic safer techniques are already made during the design phase. Proper design will, for example, effectuate built-in simplifications, the optimized size of the installations, and the process equipment. This design includes the best choice of the most appropriate hydrogen production facilities. The resulting optimized design may be improved by proper safety barrier functions, as shown in Table 6.1. These are of different basic types such as (1) passive safety barriers, (2) active safety barriers, and (3) procedural barriers, preferably prioritized in that respective order. Hereunder, inspection and maintenance strategies for the three types of barriers have to be implemented lasting the whole lifetime of the installation.

Park et al. (2020) present a seven-step method to incorporate ISD into a risk assessment during the design phase.

- | | |
|-------------|---|
| Step 1 | Identify hazards—including the causes and effects of (ideally) all hazards. |
| Steps 2 – 3 | Inherently safer design (ISD). <ul style="list-style-type: none"> a Avoid hazards (first order ISD approach). <i>This avoidance may be achieved by an alternative process design.</i> b Reduce hazard (second order approach). <i>This reduction may be achieved by either reducing the consequence of risk or reducing the likelihood of risk.</i> <ul style="list-style-type: none"> • The inherently safer approaches target the former achievement: intensify, substitute, or attenuate process or plant. • The last is mainly addressed through process simplification or plant simplification. |
| Step 4 | Avoid domino effects by segregation of the facilities. <i>This avoidance may be achieved by introducing “distance” or using sections of the plants themselves as barriers to segregate/protect people and emergency systems from hazard effects.</i> |
| Steps 5 – 6 | Layers of protection (Franks, 1999; Gowland, 2006; Wei, Rogers, & Mannan, 2008). <ul style="list-style-type: none"> a Passive safeguards (first add-on safety), <i>that is, use safeguards that do not need initiation.</i> b Active safeguards (second add-on safety), <i>that is, use active systems depending on timely hazard detection and initiation.</i> c Procedural safeguards. <i>Operator and maintenance procedures should be the last resort.</i> |
| Step 7 | Residual risk reduction. <i>Iterate through ISD, domino effects and layers of protection until risks are tolerable per objective in Step 1.</i> |

In that sense, the ISD Steps 1 – 3 are seen similar to a normal QRA where the consequences and the likelihood are assessed for each hazard. The domino effects are easily included and a ranked order for layers of protection is established, understanding that implementation of passive systems preferred to active systems. At the same time, procedural safety means are the last resort that should be considered.

6.3.1 Avoiding risks/resilient systems

A modern system safety philosophy is found in the framework of system resilience. According to Afgan and Veziroglu (2012), resilience can be understood as *the capacity of a complex system to be recovered after a sudden change* from a wanted stable to an unwanted stage and back to the stable stage. It is, in other words, explained by some authors as the ability of a system to bouncing back after an accident. It is closely related to systemic risk management. By that, Afgan and Veziroglu define Loss of Resilience, which is an analog to the term Loss of Confinement in the UFOE model described in Fig. 6.2. Thus the resilience of a system may be improved by preventing possible failures and mitigating the failure's consequences. The latter will directly minimize the hazards caused by accidents. The latter will clearly contribute to shortening the recovery time after the accident.

Any technology includes hazards and may result in accidents. There are no zero risks, but technical systems need to be designed and operated so that the residual risk has a minimal likelihood. For hydrogen technology, the technological system may be seen as an energy system, and one of the most critical parameters is the security of supply. The overall risk may be calculated by different means as, for example, described by Hughes, de Jong, and Wang (2016). Therefore the resilience of a hydrogen system is a high-level goal to achieve a sustainable and robust energy system securing a functioning energy supply within a sociotechnological context.

6.4 Organizational and human factors

6.4.1 Design and construction

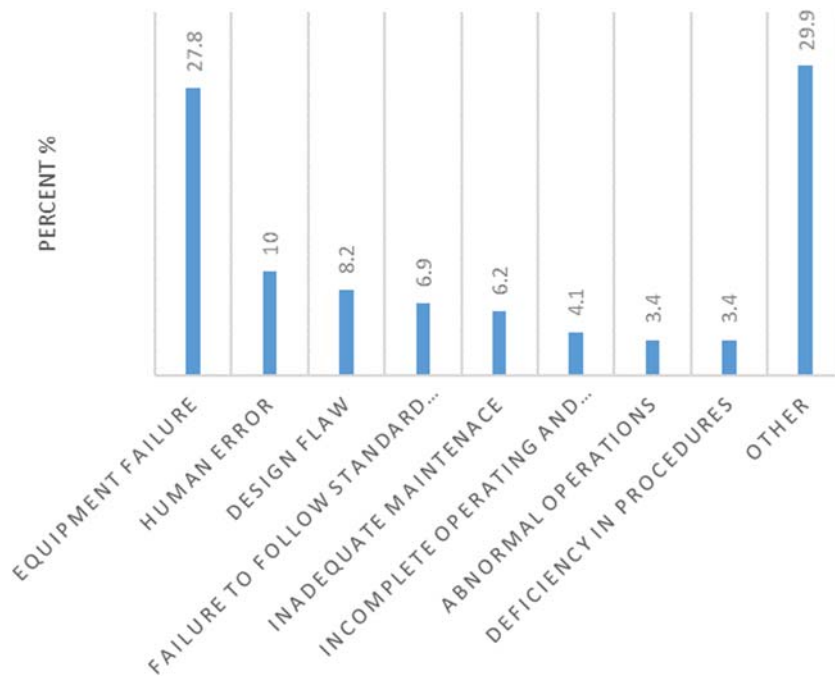
A significant impact on the safety risks of hydrogen installations and systems is given by the interaction of humans with the technological system. This impact is exemplified by the statistics shown in Fig. 6.7. Though the failure of the equipment is the leading cause of hydrogen releases, human error is a not neglectable cause. Human error is also hidden as a cause in most of the other causes seen in Fig. 6.7 (Rigas & Amyotte, 2012). It may be expected that with the increasing maturity of the hydrogen equipment, the accident cause “equipment failure” will decrease in the future.

It regards the design of an installation that should be made using the ISD principles described above. Hereunder, the installation's operation, the man – machine interface needs to be designed in an unambiguous way to avoid mistakes in the process of the hydrogen systems.

Human error is seen as a cause of failure that initiates incidents and accidents in an older view. Presently, the theory is that it is a combination of human and organizational factors (HOF) causing operational errors. In the old view, technical systems are basically safe, and the main threat to safety comes from the inherent unreliability of people. Therefore safety risks can only be reduced by protecting, for example, the hydrogen systems from unreliable persons by selection, procedures, automation, training, and discipline (DOI Standard, 2009).

Another view sees human error more broadly. It is not the cause but the symptom of failure. By that, HOF are seen together as the cause of an accident. Human error is a symptom of problems inside the system. Thus the following can be said (DOI Standard, 2009; Rigas & Amyotte, 2012):

- Safety is not inherent in systems. The systems themselves are contradictions between multiple goals that people must pursue simultaneously. People have to create safety.

**FIGURE 6.7**

Failure causing hydrogen accidents recorded in the H₂ incidents database (Rigas & Amyotte, 2012).

- Human error is systematically connected to features of people, tools, tasks, and operating environment. Progress on safety comes from understanding and influencing these connections.

The HOF factors include several organizational aspects such as the company culture to cope with safety and reliability, the workload and time constraints doing the respective job, the motivation of the staff, the quality of training, and the quality of procedures. Hereunder, the adequateness of supervision and the appropriate staff size must be considered (Gambetti, Casalli, & Chisari, 2012). This idea is included in a resilient approach to process safety, for example, by Jain et al. (2018).

Necessary for the overall safety level is the quality of maintenance procedures of the hydrogen systems. Practical approaches that assess the technical maintenance of systems based on HOF factors is the Barrier and Operational Risk Analysis approach (Aven, Sklet, & Vinnem, 2006) or the safety barrier assessment involving human factors (Duijm, 2015; Markert & Nielsen, 1992; Markert, Duijm, & Thommesen, 2013). These methods link the reliability of a technical installation with the quality of maintenance procedures.

6.4.2 Safety procedures and training

For hydrogen systems accident prevention, several essential basic safety procedures should be implemented to reduce the occurrence of hazards. These procedures are applicable for handling

hydrogen during large- and small-scale production, storage, and transportation. These procedures are developed on the knowledge of the physical – chemical properties of hydrogen. Part of the safety procedure's success is seen in the proper training of professionals and sufficient public knowledge of hydrogen fuel behavior.

Safety procedures may be any combination of technically means and behavioral routines all stakeholders shall follow handling hydrogen systems. Such procedures will be described in RCS on these systems. They should be expected on the company level and supported by the regulators and branch associations and on the legislative level. *The following of safety procedures and training of professional personnel involved in the handling of hydrogen systems are probably the most important of prevention measures to reduce the occurrence of and potential consequences of incidents or accidents. Such procedures and training exist [...]. Similar procedures are, therefore, to be developed for new applications of hydrogen, such as transport and energy conversion for professionals who come into contact with hydrogen. For the public, specific education courses are needed to address the specific properties of hydrogen, compared to other more familiar fuels such as natural gas or gasoline.* (Cited from IA HySafe, 2006).

6.5 Relevant regulations and standards

The following subsections describe some of the more central regulations, standards, and guidelines relevant to hydrogen safety. The ATEX directives are European regulations aimed at reducing fire and explosion risk related to flammable atmospheres. The standards and guidelines are focusing on hydrogen-specific requirements. A comprehensive overview on the relevant RCS is given in Chapter 7, Legal Requirements and Standards (Regulations, Codes, and Standards) for Hydrogen Safety of this book.

6.5.1 Generic measures

The above-described approaches to avoid hazards and to have ISD concepts for hydrogen applications are also part of the ISO/TR 15916:2015 (p. 30). The standard understands an ISD as an essential safety characteristic that is built-in. This safety characteristic is specified as being, for example, inherently safety features as fail-safe design, passive, and automatic safety operations. It also embraces caution and warning devices to warn and alert on deviating process conditions, equipment malfunctions, or failures. Such devices need to detect such states as timely as possible to allow the personnel (e.g., operators, technical staff, and emergency personnel) for a reasonable response time. This timely detection will enable them to cope with the situation and possibly get the operation back to safe conditions, initiate a shutdown, and start an evacuation. In order to have the best response, knowledge on key technical parameters and system layout needs to be readily available, for example, time of fire resistance of safety-critical components.

A fail-safe design as part of an ISD includes the reliability rating of equipment and components and contains measures of redundant safety features. These could be built in as, for example, pressure relief devices as TPRDs in hydrogen gas tanks to avoid overpressure that may lead to vessel (incl. joint, connections) rupture, doubling safety-critical components and systems. Also,

Table 6.3 Safety recommendations made in ISO/TR 15916.
Minimize the quantity of hydrogen that is stored and involved in an operation.
Minimize pipeline diameter and operational pressure to satisfy technological requirements to mass flow rate where applicable.
Isolate hydrogen from oxidizers, hazardous materials, dangerous equipment.
Identify and, if possible, separate or eliminate potential ignition sources.
Separate people and facilities from the potential effects of unignited releases, fire, deflagration, or detonation originating from the failure of hydrogen equipment or storage systems.
Elevate hydrogen systems and vent them above other facilities.
Prevent hydrogen/oxidizer mixtures from accumulating in confined spaces (under the eaves of roofs, in equipment shacks or cabinets, or within equipment covers or cowlings).
Minimize personal exposure by limiting the number of people exposed, the time that the person is exposed.
Use of alarms and warning devices (including hydrogen and fire detectors) and area control around a hydrogen system.
Use of personal protective equipment.
Practice good housekeeping such as keeping access and evacuation routes clear and keeping weeds and other debris away from hydrogen systems.
Observe safe operational requirements, such as working in pairs when operating in a hazardous situation.

monitoring the hydrogen systems and the remote control/operation strategies is to be addressed hereunder. The maintainability of such systems is of great importance to avoid potential accidents. Also, cleaning procedures and exchange of built-in filters are to be considered. Instruments and controls should be positioned so that their measurements are not negatively interfered with and the controls are easily activated automatically or manually.

Hydrogen is used under very high pressures or very low temperatures. By this, the choice of appropriate construction material is an essential step in an ISD approach. Therefore the choice of the materials should consider the basic design for temperature and pressure, possible embrittlement, and hydrogen attack behavior.

Safe use of hydrogen and hydrogen systems is supported by any design and operation that reduces the severity of incidents and accidents to a minimum. The general safety recommendations of the ISO/TR 15916 listed in Table 6.3 are aligned with the Haddon principles: (cited from ISO/TR 15916).

6.5.2 The ATEX directives

To prevent explosions under normal working conditions, two European directives for controlling explosive atmospheres, ATEX, had been developed back in 1995. The directive has two parts: (1) Directive 99/92/EC giving the minimum requirements to provide health and safety protection of workers potentially at risk from explosive atmospheres. (2) Directive 2014/34/EU concerns the equipment and protective systems intended for application in the potential atmospheres defined in the former directive. Respective standards detail the requirements EN 60079-10-1:2015 Classification of areas—Explosive gas atmospheres; EN 60079-10-2:2015 Classification of areas—Explosive dust atmospheres and several standards to regulate equipment requirements as, for

example, EN IEC 60079-0:2018 Explosive atmospheres—part 0: Equipment—General requirements. In addition, different standards to detail the use of detectors are found as EN 60079-29-2:2015 Gas detectors—Selection, installations, use, and maintenance of detectors for flammable gases and oxygen.

A guideline document “Guidelines to Directive 2014/34 EU /ATEX)” has been released with the latest update 2nd edition December 17, 2017 (Bridges et al., 2009; European Commission, 2014).

Some brief basics are explained in the following sections to better understand the ignition and flammable behavior of hydrogen. Hydrogen gas is flammable in a vast concentration interval (4–75 vol.%) in the air with very low ignition energy at stoichiometric conditions, which are 29 vol.% of hydrogen in air. Decreasing or increasing the hydrogen fraction leads to higher ignition energies necessary to start the composition. If the concentrations are lower than 4 vol.% or larger than 75 vol.%, ignition is no longer possible at ambient conditions. The additional requirements for the ignition of hydrogen are illustrated by the fire triangle shown in Fig. 6.8. For any fuel, the lower and upper flammability limits embrace the combustible concentrations in air. An oxidizer needs to be present for ignition, which is the air’s oxygen in most accidental cases. The third necessary factor is the availability of an ignition source. In case only one of the three factors are not present, combustion will not occur. By that, prevention strategies are designed to avoid flammable concentrations, avoid ignition sources, and avoid the presence of oxidizers. The latter strategy is the basis for inerting processes where equipment is flushed with an inert (not combustible) gas such as nitrogen or argon. The former two strategies are the core of the EU’s ATEX regulation to avoid flammable atmospheres under normal operating conditions. This avoidance of flammable atmospheres can be achieved by the design of the technical system but can be supported by the installation of ventilation means. A zone concept is applied when a flammable atmosphere cannot be avoided based on the duration and frequency of occurrence. Hereunder, the duration is very dependent on the ventilation conditions. In a second step, the zones with frequent and prolonged

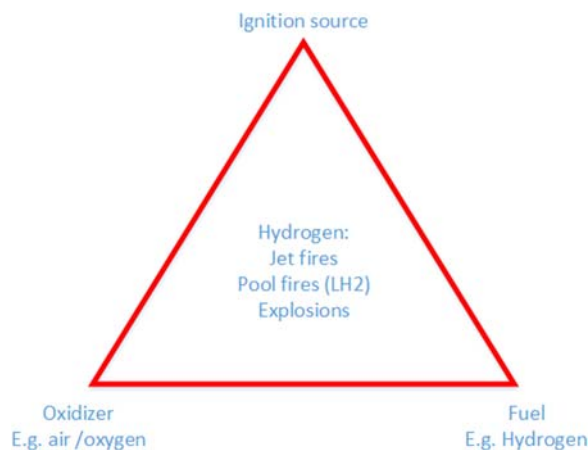


FIGURE 6.8

The fire triangle.

explosive atmospheres have imposed the restriction to avoid ignition sources, for example, explosion-proof electrical equipment, spark-free tools for maintenance.

The ATEX “workplace” directive sets requirements for classifying hazardous places for flammable gases and dust. Hazardous places are classified in terms of zones based on the frequency and duration of the occurrence of an explosive atmosphere.

Zone 0: A place in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapor, or mist is present continuously or for long periods or frequently.

Zone 1: A place in which an explosive atmosphere consisting of a mixture of air or flammable substances in the form of gas, vapor, or mist is likely to occur in normal operation occasionally.

Zone 2: A place in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapor, or mist is not likely to occur in normal operation but, if it does occur, will persist for a short period only.

There are requirements to ex-approved electrical and mechanical equipment within the hazardous zones (ex-zones). In addition, other types of measures for control and removal of ignition sources, like proper grounding and bonding, avoid sources of static electricity, forbidden smoking, shutdown of equipment, and others can be implemented, see also later chapters.

6.5.3 ISO/TR 15916:2015

ISO/TR 15916:2015 provides guidelines for the use of gaseous and LH₂. In addition, it provides guidelines for hydrogen storage in either of these or other forms (hydrides). It identifies the basic safety concerns, hazards, and risks and describes the properties of hydrogen relevant to safety. Detailed safety requirements associated with specific hydrogen applications are treated in other international standards.

6.5.4 NFPA 2 “Hydrogen Technologies,” 2020 edition

The National Fire Protection Association code (NFPA) 2, Hydrogen Technologies Code, (NFPA, 2020) covers the requirements applicable to the generation, installation, storage, piping, use, and handling of hydrogen. The 2020 edition includes several updated chapters to reflect the latest in fire and life safety. NFPA also includes separate chapters for hydrogen applications, such as gaseous and LH₂ fueling stations, vehicle fueling facilities, HFC power systems, hydrogen generation systems, combustion applications, laboratory operation, parking, and repair garages. The code also gives a comprehensive list of applicable standards relevant for hydrogen applications (NFPA, 2020).

6.5.5 EIGA guidelines relevant for hydrogen safety barriers

European Industrial Gases Association (EIGA) has several guidelines and best practice documents for the safe design and operation of hydrogen systems. Some of these documents are listed below. The publications also include guidelines for safety barriers related to safety distances, layout, ventilation, venting systems, purging with inert gas, a.o.

Doc. 121/14 Hydrogen pipeline systems:

This document is intended to guide the design and operation of hydrogen transmission and distribution systems.

The document summarizes the current industrial practices and is not intended to be a mandatory standard or code. It is based upon the experience and practices of members of companies from Regional Gas Associations (EIGA, 2014).

Doc. 06/19 Safety in storage, handling, and distribution of liquid hydrogen:

The document is intended to guide companies directly associated with installing LH₂ storages at the user's premises and distributing LH₂ by road, rail, and sea (EIGA, 2019).

Doc. 211/17 Hydrogen vent systems for customer applications:

The publication provides recommendations and a methodology for the safe design of venting systems. The design of hydrogen vent stacks aims at preventing the following three main hazards:

- deflagration due to the delayed ignition of the explosive hydrogen/air mixture coming out of the vent stack;
- thermal radiation generated by the jet fire (the flame coming out of the vent stack); and
- DDT transition inside the stack due to the acceleration of the combustion of the air/hydrogen mixture that can be present inside the venting system, in the presence of an ignition source (relevant if air can enter the venting system) (EIGA, 2017a).

Doc. 15/06 Gaseous hydrogen stations

This document is prepared for the guidance/best practices of designers and operators of gaseous hydrogen stations. Its application will achieve the primary objective of improving the safety of gaseous hydrogen station operation (EIGA, 2006).

Doc. 171/12 Storage of hydrogen in systems located underground

This document addresses the safety issues that are specific to the storage of hydrogen in systems located underground.

Although hydrogen storage and related equipment should be located above ground in a well-ventilated area, this form of installation may be needed when it is beneficial to keep above-ground areas free of equipment, such as the case for hydrogen vehicles refueling stations.

The document covers the requirements specific to installing a hydrogen storage system in an underground space with top or side access. Hereafter called a vault, LH₂ storage and ancillary systems and compressed hydrogen storage are composed of a single container or multiple cylinders or tubes.

The document is intended for use by all personnel involved in the design, installation, and operation of hydrogen storage systems that may need to be located underground.

6.5.6 DNV hydrogen guideline documents

Det Norske Veritas (DNV) have also developed a handbook for the safe design and operation of hydrogen fueled ships—"Handbook for Hydrogen-fuelled Vessels" (MarHySafe JDP, 2021).

This is the first edition of the "Handbook for Hydrogen-fuelled Vessels" and suggests a road map toward safe hydrogen operations using fuel cells. It details how to navigate the complex requirements for design and construction, and it covers the most important aspects of hydrogen operations, such as safety and risk mitigation, engineering details for hydrogen systems, and

implementation phases for maritime applications, based on the current regulatory alternative design process framework. This publication is the result of the first phase of the DNV-led joint industry project MarHySafe. The project is ongoing, and the handbook will be continually updated to reflect the latest industry expertise on hydrogen as ship fuel.

6.6 Specific safety barriers in practice for hydrogen applications

Technical safety barriers/functions are systems and devices that will prevent or limit the consequences of an accidental leak. Fig. 6.9 illustrates an example of such barriers in a bow tie diagram. The following chapters focus on describing the barriers/safety systems, what considerations should be taken related to hydrogen facilities, and hydrogen properties. The intention is not to present requirements but to describe the principles and how they are linked.

Not all barriers are possible to include in all applications/areas. For example, at industrial plants, most of them will be implemented. On the other hand, it may not be the case in small-scale unmanned applications like hydrogen fueling stations, domestic heating systems based on hydrogen combustion (like the H21 project), cars, and areas like parking garages and tunnels. For example, barriers requiring human interventions will be less effective because of the time aspect. It is of utmost importance for such applications to design according to inherent safety principles and secure

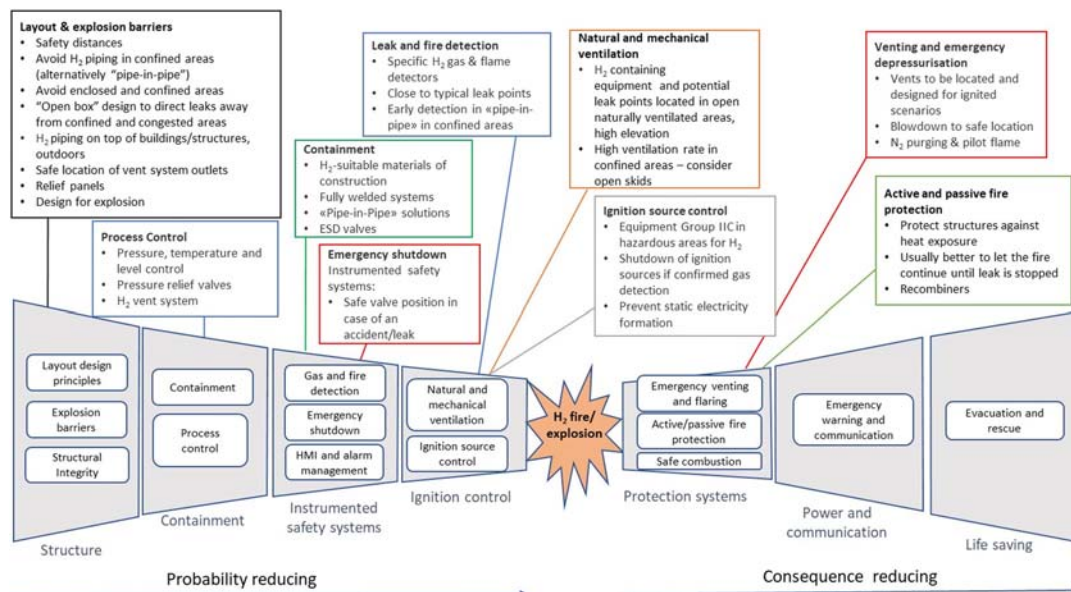


FIGURE 6.9

Safety in design—for example, hydrogen risk-reducing measures bow tie for hydrogen prevention and mitigation measures (Samdal, Grainger, Hamborg, Nilsen, & Sommersel, 2021).

adequate reliability of instrumented safety functions that automatically are designed to mitigate hazardous waste deviations and accidents.

The following chapters cover mainly technical barriers. Emergency procedures and evacuation are not discussed. This chapter's "containment" function does not include material considerations but is more linked to safe layout and to avoid leak points in enclosed areas. However, designing hydrogen systems for very high pressures or very low temperatures it is crucial to consider material phenomena, as for example, hydrogen embrittlement.

6.6.1 Detection of hydrogen releases

Gas detection systems are important elements in a barrier system for the detection and control of leaks. Gas/leak detection represents the first line of defense in the case of a hydrogen release. Ideally, actions can be taken to stop the hydrogen release before a flame or explosion.

Detection by sensors or other means is only one important part of a safety barrier system. There are additional components that have to be in place as well. On the detection side, the sensor needs to be reliable and to trigger in case of an event a signal to activate an alarm (control room, site alarm, siren, alarm to rescue service, automatic shutdown, and potentially venting/blowdown etc.). The signal/alarm has to be diagnosed and verified/classified by an operator judgment of an expert or automatically by a computer system (software; Bayesian network, machine learning concepts). Proper actions have to be taken on a valid alarm, such as, facility shutdown, emergency blowdown, and evacuation initiation procedures (see Fig. 6.10).

Gas detectors and voting systems for confirmed gas detection can be integrated with other control and safety systems. Thus gas detection has an interface to several other safety barriers, such as layout, emergency ventilation, closing emergency shutoff valves, emergency venting or flaring, firefighting systems, emergency procedures, and other actions that might be initiated in case of confirmed gas detection.

There is a range of commercially available sensors. These are based on principles as described in Table 6.4.

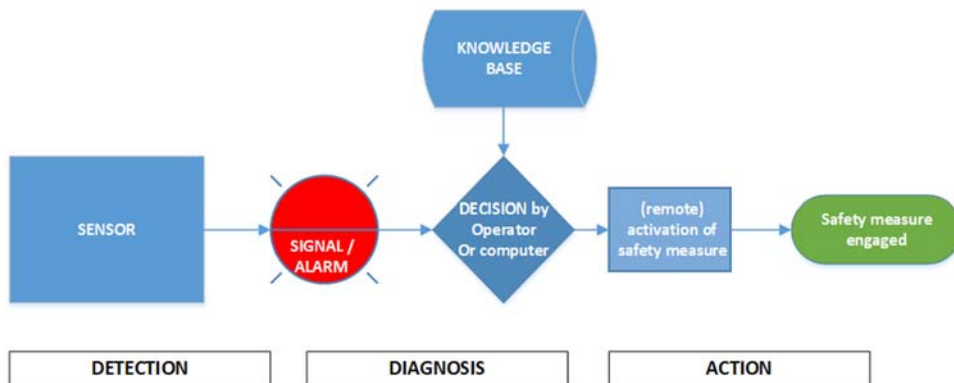


FIGURE 6.10

A complete detection—diagnose action safety barrier.

Table 6.4 Commercially hydrogen sensors (Buttner et al., 2011).

Sensor class	Types and detection modes
Electrochemical	Amperometric and potentiometric sensors are the two main electrochemical sensor platforms.
Metal oxide	Metal oxide sensors are fabricated with a wide bandgap semiconductor material, such as tin oxide or other metal oxide materials as the active element of the sensor.
Pellistor	Heated ceramics embedded with platinum or other noble metal catalysts provide a catalytic surface for hydrogen in air combustion.
Thermal conductivity	The detection sensor measures a temperature-induced change due to the exposure to hydrogen.
Optical devices	Optical sensors cannot detect hydrogen directly, as the gas is nondetecting in the UV-vis and the infrared spectrum. However, indirectly, the devices measure optical property changes of film materials upon hydrogen exposure, for example, palladium films.
Palladium film/palladium alloy films	Such sensors use the unique, highly selective permeability of hydrogen into palladium that changes the electrical conduction with hydrogen concentration.

Two of the most commonly applied technologies for combustible gas detection are infrared and catalytic beads. Only the latter technology is appropriate for hydrogen. The former infrared gas detector responds to gases that absorb IR radiation (e.g., methane and propane) but not hydrogen (Anderson, 2011).

6.6.1.1 Catalytic point detection

The detector contains a catalyst (for hydrogen palladium and/or platinum) to facilitate hydrogen combustion with oxygen in the air. This detector applies a chemical reaction near the catalyst surface, not an open flame. A temperature reading senses the heat of combustion, and the concentration can be calculated.

6.6.1.2 Electrochemical detection

The released gas flows through a membrane to get in contact with an electrolyte, where the reaction produces a current.

6.6.1.3 Semiconduction oxide

The gas reacts with chemisorbed oxygen in a semiconductor material and changes the electrical resistance of that material.

6.6.1.4 Thermal conductivity

This instrument is based on the heat of conduction from a heat source into the surrounding environment. The heat conduction rate is dependent on the thermal conductivity properties of the gas.

6.6.1.5 Mass spectrometer

The released gas is ionized and accelerated through an electric field along a curved path. The curvature amount depends on the particle mass and is used to separate the particles by mass.

Mass spectrometers, electrical detection, semiconducting oxide, thermal conductivity detectors are also point-detector instruments with the possibility of reading concentration.

6.6.1.6 Acoustic detection

The detection technology is based on ultrasound measurement (typically above 25 kHz) noise generated by the leakage itself. Rotating and vibrating machinery and artificial noise sources will produce noise with most of its energy spectrum in the audible frequency range (20 – 20,000 Hz). A leak will produce noise with a frequency energy spectrum like white or pink noise, with energy in frequency bands up to 80 or 100 kHz. These noise properties are used related to the detection of gas leaks. A high-quality microphone is used along with an electrical circuit with a frequency high-pass filter. If a certain noise level in the ultrasonic area is detected, the monitoring unit assumes a gas leak. The noise level and detection delay can be adjusted.

Acoustic detectors can detect a leak, but not exactly where or how large the leak is. The detection rate depends on the background noise; the lower the noise, the longer the detection radius.

6.6.1.7 Glow plugs

The glow plug is not a gas detection instrument per se but is used to ignite flammable gas volumes, which can then be detected by flame/heat detection instruments.

6.6.1.8 Bubble testing

Bubble testing is one of the simplest leak detection methods; however, it is not a continuous monitoring system. It needs to be applied directly to the leak source, can only be used with inert gases at low pressure, and is limited to temperatures above freezing. Bubble solutions can detect very small leaks but do not measure concentrations.

6.6.1.9 Gas leakage detection by system monitoring

Process variables (pressure, temperature, flow, level) can be measured, and abnormalities can then be used to indicate that there might be a leak.

Liquid H₂ leaks can be detected through loss of vacuum and by observing the formation of frost, the formation of solid air, or a decrease in outer wall temperature on vacuum-jacketed equipment.

A future option for hydrogen detection is Raman spectroscopy. Other more advanced methods are the Schlieren and Shadowgraph Imaging, Fiber Optic Sensing, and Ultrasonic Leak Detection. These methods are currently under development or may be commercially available but without sufficient testing for hydrogen leaks. Nevertheless, most of them may be applicable for hydrogen wide-area monitoring (Buttner et al., 2011, 2020; Hübert, Boon-Brett, & Buttner, 2018).

6.6.1.10 Detection system guidelines

Infrared and catalytic point detectors are quite similar in their ways of working. They can measure gas concentration at the detector's location, and a leak can be detected at a predefined concentration level, for example, 20% lower flammability limit (LFL). However, confirmed detection depends on the leak exposing the detector, and therefore several detectors will be necessary to cover the area possibly exposed to flammable gas.

A combination of point and line detectors are often used in the industry. Line detection is based on the IR detection principle and therefore cannot be used for hydrogen. Acoustic detectors might be a solution to cover larger areas, especially in open areas where it can be difficult to detect a leak, especially for a buoyant gas like hydrogen.

Hydrogen detectors should be located:

- close to potential leak points (valves, flanges, process equipment, fuel dispenser);
- at locations where there is a natural flow path or detection might indicate a hazard, like air intake, ventilation outlet, vent stack outlet systems;
- at a high elevation level, as especially gaseous hydrogen is buoyant and will rise. For LH_2 , being less buoyant, the location of gas detection at low points is also relevant.

When designing new systems, it is advantageous to also consider the location of equipment with a potential to leak, close to each other, and then locate detectors in this area. A systematic investigation of indoor hydrogen releases has been made and reported ([Tchouvelev et al., 2020](#)).

Voting systems are often used, for example, two of three detectors need to indicate detected gas for an emergency shutdown to be initiated. The intention with the voting system is to avoid a system where a simple error in an instrument might lead to an unnecessary shutdown.

6.6.2 Fire detection

The fire detection system aims to monitor for the presence of fire to alert and allow control actions to be initiated manually or automatically to minimize the likelihood of fire escalation and the probability of people being exposed and to minimize the probability of a fire condition to develop and escalate.

Fire detection systems might have interfaces with the other safety systems/functions such as natural and mechanical ventilation, ESD (emergency shutdown system), emergency depressurization, flare/safe burning system, active fire protection, layout and explosion barriers, emergency power and lighting, alarm, and communications systems.

A flame of pure hydrogen is almost invisible in daylight. The emissivity outside of the flame also decreases very quickly. Therefore for a person, it may be difficult to know that he/she is approaching a flame until he/she is inside the flame. Here the heat radiation will be very large. A hydrogen fire will cause changes in the surroundings that can be used to detect the fire. The changes are called the signature of the fire. Although the nonluminous hydrogen flame makes visual detection difficult, there is a strong heat effect and turbulence of the surrounding atmosphere.

A critical aspect of a hydrogen leak is that it may ignite, and the heat may expose equipment that contains hydrogen. This heat exposure could weaken the material and subsequently rupture, thus resulting in more severe accident scenarios as jet fires or explosions.

There should be means to detect hydrogen flames in all areas in which leaks, spills, or accumulation of hydrogen may occur.

Thermal and optical sensors can be used to detect burning hydrogen. Many thermal detectors are needed to cover large areas or volumes. They should be located at or near the site of a potential fire. In general, a hydrogen fire does not generate smoke, so smoke detectors are less effective as fire detection.

Optical sensors for detecting hydrogen flames can operate in the ultraviolet or infrared spectral region. Closed-circuit infrared and ultraviolet remote-viewing systems equipped with appropriate filters have been used successfully.

Linear heat detection systems, consisting of heat-sensitive polymer insulation and inner conductors, respond when a rated activation temperature is reached.

Thermal (heat) detectors can respond to either a fixed temperature or when a specified temperature change rate occurs ([Pacific Northwest National Laboratory, 2021](#)), NFPA 2.

Flame detectors should be installed in certain applications (e.g., near hydrogen dispensers in hydrogen fueling stations). Detectors should provide a rapid and reliable indication of the existence, location, and size of a hydrogen flame.

More details about different types of fire detectors can be found in “Sourcebook of Hydrogen Applications, app. 3” ([Bain et al., 1998](#)). Requirements related to fire detection (detection system, locations) for hydrogen applications are given in NFPA 2, NFPA72. Some background in formations may be found in the following publication: ([Hosch, 2019](#)).

6.6.3 Layout considerations and explosion barriers

The layout and arrangement should reduce the probability and the consequences of accidents through location, separation and orientation of areas, equipment, and functions.

Explosion barriers shall reduce the consequences of the pressure loads from an ignited gas cloud to prevent the accident from spreading to other areas, process systems, storage vessels, and/or critical equipment.

The best way to reduce the explosion probability of hydrogen systems is to utilize the buoyancy and diffusion properties. This utilization means that hydrogen processing systems should be installed in open areas at a preferably higher elevation level than the surroundings. The prevailing wind directions should be considered when utilizing natural ventilation. This consideration will ensure that leaks will be directed away from safety-critical systems, confined areas, or other areas’ systems that could be compromised by a fire or explosion or where ignition will lead to higher pressures.

Walls can be installed to prevent a hydrogen leak from entering obstructed areas. However, consideration must be taken to not lead to additional explosion hazards in the obstructed areas.

Hydrogen pipelines should not pass in obstructed or enclosed areas. If this cannot be avoided, welded connections should be used and/or a “pipe-in-pipe” solution can be considered in these areas. In the latter solution, any leak will be collected in the outer pipe and directed to a safe location where it cannot accumulate or lead to hazardous conditions outside the outlet. Gas detection can be located in or close to the outlet of the “pipe-in-pipe” system to detect any leak on the pipeline.

The decision of safe distances to prevent escalation and safe distance to third party can be quantified by a risk analysis and based on acceptance criteria. Guidance is given, for example, in ISO 15916, and requirements to safety distances are given in NFPA 2 and the other hydrogen applications referred to in the NFPA standards listed in NFPA 2. Another guideline for determining safety and separation distances is the EIGA document Doc. 75/21 ([EIGA, 2017b](#)).

6.6.4 Process control systems

Process control systems include means for protection, incorporating mechanical protection devices, and safety instrumented functions to ensure that the process conditions do not exceed specified process safety limits. The aim is to control any abnormal process operating conditions to prevent and/or minimize possible accidental releases.

Typical actions for protection, for example, Process Shut Down (PSD) functions and Pressure Safety Valves (PSV), include:

- stop hydrogen flow,
- shutdown processing and utility equipment,
- isolate leaks,
- shutdown heat input to the process,
- pressure relief.

The process control system includes instrumented-based process shutdown, overpressure protection, valves with process safety functions, PSV, tank overfilling protection, alarms, and automatic actions. The need for and reliability of these systems are often identified in Hazop and Safety Integrity Level and Layer of Protection Analyses.

The extent of a PSD situation will depend on the type of abnormality. It may vary from shutdown of process sections with minimum effect on the application to a total process shutdown.

The process control systems have interfaces with containment, emergency shutdown, emergency depressurization and vent systems, layout and explosion barriers, and operational barriers such as human – machine interface and alarm management.

The process safety functions shall provide reliable and fast detection of process upsets and execution of the actions that are considered necessary to control the situation and avoid escalation and will initiate actions that shall bring the system back to normal conditions.

The process control system intends to act before ESD and return the system to safe conditions before an emergency shutdown is needed. In industrial plants, the ESD and PSD systems are usually independent. For ESD, please see the next section.

6.6.5 Emergency shutdown

The purpose of the ESD is to prevent the escalation of abnormal conditions into a major hazardous event and limit the extent and duration of any such events. ESD have interfaces to leak and fire detection, ignition source control, emergency depressurization and flaring/venting systems, and safety and automation systems. In addition, ESD initiations might activate (direct or indirect) other safety systems/functions such as emergency ventilation, emergency power and lighting, alarm and communication system for use in emergency situations.

6.6.6 Natural and mechanical ventilation

The basic principle of ventilation is to use fresh air volume flows to dilute and remove flammable gas pockets of hydrogen caused by any minor or accidental release and avoid accumulation in the enclosed areas.

6.6.6.1 *Natural ventilation*

The effect of good and effective ventilation is to dilute the released gas to a nonflammable concentration (below 4 vol.% of hydrogen). Good natural ventilation can be achieved in combination with a good layout. For hydrogen, this will be achieved if the hydrogen-containing/processing equipment is located in open and as much as possible uncongested areas at a high elevation level. Then the buoyancy and diffusivity of hydrogen will lead to effective dilution of a leak to a nonflammable concentration. Circulation zones and locations where hydrogen can be accumulated should be avoided. The potential for accumulation could be verified by dispersion calculations or by measurements.

The opposite (negative) effect, with accumulation and good mixing of hydrogen and air, will be achieved in congested and enclosed areas. Such a situation should be avoided.

6.6.6.2 *Mechanical ventilation*

In enclosed areas, natural ventilation will, in general, not be effective enough, and there will be requirements (regulations, standards) related to mechanical ventilation. To be an effective means to remove/avoid flammable gas concentrations/clouds, mechanical ventilation systems need a proper design using the right ventilation technique and ventilation capacity.

For proper mechanical ventilation design, the properties of hydrogen should be utilized, with an inlet at a low level of the room and an outlet at a high level. It must be secured that the outlet leads to a safe location, not leading to hazardous conditions outside the enclosure. The ventilation system must be approved for the appropriate hazardous area classification, see respective Chapters 6.5.2 and 6.6.8.

Emergency ventilation can also be a measure related to mechanical ventilation, for example, increasing the ventilation rate in case of confirmed gas detection or other signals indicating a leak in the enclosed area. Also, for mechanical ventilation circulation, zones and locations where hydrogen can be accumulated should be avoided.

6.6.6.3 *Ventilation considered in the ATEX directive 99/92/EC*

Ventilation is considered in the ATEX directive 99/92/EC (European Union, 2000), and the availability and quality of the ventilation is part of the evaluation for area classification. ATEX is applicable for normal operating conditions to avoid explosions.

The extent of the hazardous areas/zones defined by ATEX may be reduced based on good and available/reliable ventilation. It is distinguished between natural and forced ventilation. The former depends on the actual weather conditions, while the latter can be specifically designed and constantly be provided on a reliable power supply condition.

The ventilation can be achieved by forced/mechanical ventilation or by passive natural means. The effectiveness of natural ventilation is strongly dependent on meteorological conditions such as wind speed, pressure differences, and temperature differences. The effectiveness of forced ventilation is more predictable depending on the installed fan and its reliability.

Ventilation can thus reduce the need for hazardous areas and thus the need for ex-certified equipment, also reflected in codes and standards (e.g., EN 60079-10). However, this should be used with care as the leaks considered in the methodology used to determine hazardous areas do not consider large leaks. Therefore for larger accidental leaks, prevention of ignition sources, for example, by using ex-certified electrical and mechanical equipment in and outside the actual hazardous areas, should be considered (see also Section 6.6.8).

6.6.7 Draining and spill containment systems

The purpose of the drain and spill containment system is to provide measures for containment and proper disposal of flammable or environmentally hazardous liquid spills, including liquefied gases. The intention is to prevent further spreading of the spill, and spilled flammable liquid could be led to safe disposal. It is very important to secure that the contained spill does not lead to an additional hazard, for example, fire exposure of storage vessels (if ignited) or accumulation (prolonged duration of flammable gas in congested and confined areas).

Drain/collection systems have no mission/purpose for gaseous hydrogen systems and are not recommended (NFPA 2's Chapter 6.15.1), as no liquid is formed during a spill. For LH₂ systems, when it comes to spill control, drainage and secondary containment are NOT recommended, for example, NFPA 2's Chapter 6.15.2 states "*Diking shall not be used to contain an LH2 spill*". However, NFPA 2's Chapter 6.15.2.1 recommends further "*diking or berms shall be permitted to direct the spill away from exposure*". It is not recommended to locate liquid H₂ containers in diked areas with other hazardous materials (NFPS 2's Chapter 8.2.2.3.7).

6.6.8 Ignition source control

The purpose of ignition source control systems is to prevent ignition and subsequent risk of a fire or an explosion by making the potential ignition sources harmless or reducing the likelihood of occurrence of effective ignition sources.

Ignition source control automatic functions might be realized through other systems such as the ESD system. Ignition source control interfaces might be incorporated within the main safety barriers (systems or functions): such as natural and mechanical ventilation, leak detection, ESD, fire detection, emergency power and lighting, alarm and communication system for use in emergency situations, layout design principles, and explosion barriers.

The autoignition temperature of hydrogen is higher than for most other fuels; however, the minimum ignition energy is significantly lower for hydrogen than for other flammable gases, meaning that ignition might occur without any clear ignition source. Pressurized hydrogen can, if released, self-ignite due to shock effects (Astbury & Hawksworth, 2007).

In addition to ex-classified equipment approved for hydrogen (gas group IIC, T) in hazardous areas, additional requirements might be recommended to reduce the ignition probability of a release. These are:

- proper grounding and bonding of metallic equipment;
- avoid materials that might lead to a build-up of static electricity;
- use of antistatic materials on floors and walls, antistatic window glass;
- avoid open fire, hot work such as welding, air intakes of combustion engines, or other combustion processes;
- use of spark-proof tools according to classification defined in the ATEX directive 94/9/EC and according to recognized standards;
- further procedural requirements could be to forbid open flames, smoking, cell phones, and similar electric devices. For example, the use of working clothes and footwear that cannot generate static electricity (use of antistatic clothing and shoes, conductive footwear), procedures

to secure bonding and earthen of refueling equipment including the refueling hose at a hydrogen fueling station.

These are requirements that will be implemented at industrial plants. In addition, they are implemented to a certain degree in machinery rooms, workshops, repair garages, inside the boundary limits of hydrogen fueling stations, and other applications where it is accepted in general that safety precautions are implemented and adhered to. In industrial applications, these requirements are often extended outside the hazardous zones as the extent of the zones, in general, is decided by small leaks or releases and not major releases of flammable gas.

The more operational mitigation measures might be more challenging to implement in case of the public use of hydrogen (cars, at ferries, in tunnels, in domestic houses). For these applications, inherent safety in design should be more focused so that accidental releases are prevented or that any release, if ignited, will not lead to significant hazardous consequences.

There are a series of requirements to ignition source control in NFPA 2 related to, for example, safety distances from ignition sources like open flames and welding to various operations, applications and bulk volumes of hydrogen, related to avoidance of static electricity formation.

NFPA 2 and ISO 15916 set requirements to minimum distances from ignition sources like open flames and welding to bulk volumes of LH_2 to avoid ignition sources when unloading, refueling etc. (DS/ISO/TR 15916, 2015; NFPA, 2020).

6.6.9 Inerting

Ignition and subsequent combustion and explosion can only occur on condition of three main parameters.

1. The hydrogen (fuel) is to be present with its flammability limits.
2. The oxygen in the air is present or any other oxidizer needed for the combustion.
3. An appropriate ignition source to start the combustion or explosion.

Inerting replaces the oxygen in the air by diluting it with an inert gas such as nitrogen or argon, or other noble gas. Inerting has some disadvantages, such as the risk of suffocation, which may necessitate evacuating people before activation. However, it can be a relevant safety barrier for applications where the risk of detonations with the potential of large damages cannot be out ruled, such as liquid storage compartments on transport ships or ships used for public transport (ferries).

Inert systems can also be used in pipe-in-pipe systems, like in a vent stack or piping passing through enclosed or obstructed areas. Here the annulus between the inner pipe (transporting the hydrogen or vented hydrogen) and the outer pipe can be filled with inert gas or fluid. This procedure can be combined with gas detection located at the top of the vent stack or detection in the annulus of a transport pipe.

6.6.10 Blowdown and venting/flaring (safe burning)

The purpose of emergency depressurization is to reduce the pressure and inventory in a segment (e.g., storage vessel) if a fire exposes the segment. A reduction in pressure implies reduced material stress and, hence, reduced risk of rupture due to heating caused by fire. It will also reduce the leak

rate and leak duration from a leaking segment (and, hence, also reduce the associated fire in case the leak is ignited).

The purpose of a flare or cold vent system is to provide safe disposal (to a safe location) of releases from pressure relief, depressurization systems, and other vents. Blowdown systems are common systems in industrial plants, and they can be activated automatically, initiated by specific deviations from normal operation or confirmed gas or fire detection. It can also be initiated from control rooms or manually. Automatic activation requires a systematic analysis to identify the signals that shall lead to blowdown. A thorough analysis must also be carried out to verify that the gas can be released or burnt safely. For flares, normally, a pilot flame is in place to secure ignition.

Blowdown and venting/flaring systems might have interfaces with the other safety systems/functions such as leak and fire detection, ESD, passive and active fire protection, layout design principles, and explosion barriers.

These systems will be more common in industrial applications as it might be more difficult to achieve safe location in domestic areas, garages, tunnels etc. However, hydrogen vehicles will often be protected with pressure relief devices triggered by temperature, pressure, or a combination (Sunderland, 2008).

Vent stacks are common technology at, for example, hydrogen fueling stations. During the operation of hydrogen energy stations for transportation and stationary power applications, it might be necessary to vent stored hydrogen to a safe location safely.

In general, it is recommended to direct the vents upward to utilize the buoyancy of hydrogen. This upward venting must be combined with measures to prevent clogging due to dust, water, ice, and bird droppings. An example of a vent stack is shown in Fig. 6.11. Here the direction of the vent termination (illustrated by the red arrow) points upward and will direct any ventilated hydrogen away from customers and the surroundings where people and vehicles could be located.

Other solutions are implemented where the outlet piping direction is more horizontal, as illustrated in the report “Zero Emission Bay Area (ZEBA) Fuel Cell Bus Demonstration Results” (Eudy & Post, 2014, p. 14).

Flaring/ignition of the vented gas might not be that common in nonindustrial applications.



FIGURE 6.11

Illustration of vent arrangement (see arrow) at a hydrogen fueling station at Heroya Industrial Park, Norway.

Courtesy Sandra H. Nilsen August, 2020.

Backflow of air into a vent system needs to be avoided as ignition might lead to an explosion in the vent line. This risk may be reduced by purging the vent line with a small inert gas flow.

Requirements to venting systems for hydrogen applications can, for example, be found in NFPA 2, covering design, location, safety distances.

6.6.11 Passive fire protection

The role of passive fire protection (PFP) is to ensure that relevant structures, piping, and equipment components have adequate fire resistance regarding load-bearing properties, integrity, and insulation properties during a dimensioning fire and contribute to reducing the consequences in general. The fire protection should prevent unacceptable escalation or spread of fires by providing in-situ protection until the affected part of the installation can be evacuated. In combination with other fire protection measures, the ESD and emergency depressurization systems can ensure controlling emergencies.

Main areas should be separated by safety distances or firewalls such that unacceptable escalation/spreading of fire and explosion from one main area to surrounding main areas are avoided.

Areas with important safety functions and areas with a high fire risk should be separated from the surroundings by adequate fire divisions. When designing PFP systems, the following aspects must be taken into consideration:

- Avoiding spreading of the accident to other areas, for example, storage areas or auxiliary systems.
- Avoiding injuries involving danger to personnel or third party, control systems, emergency systems, or during the evacuation from buildings.

Pressurized vessels, process equipment, and piping should have adequate fire resistance to prevent unacceptable escalation of a dimensioning fire scenario. Such protection includes pipe and vessel supports. Adequate protection of process equipment and piping in a fire situation can be obtained by either one or a combination of the following measures: emergency depressurization, wall thicknesses of equipment, adequate construction material, PFP, and active fire protection.

Protection of safety-critical equipment:

Requirements for fire protection and the use of PFP must be considered in relation to safety criticality and the fire risk in the area. This applies to the protection of equipment such as:

- Safety-critical cables, junction boxes, and trays (to equipment which is to operate after ignited leaks, for example, ESD system).
- Fire detection of adjacent areas.
- Fire Water pumps (when relevant).
- Safety-critical pipe supports (as active fire protection, flare/vent lines).
- Safety-critical ducting (needed during a fire).
- Emergency shutdown system (ESD) and electronic data processing (EDP) (e.g., emergency depressurization) valves and activation system.
- Flare-/vent lines.

Materials used in areas that may be exposed to a fire should be noncombustible. Suppose it is justified from a safety point of view to use materials that do not meet the requirements of

noncombustibility. In that case, such materials should have limited flame-spread properties, low smoke development, and heat generation.

6.6.12 Active fire protection

The main purpose of active firefighting systems is to provide means for firefighting in addition to cooling equipment and structures. Firefighting systems may be used to reduce explosion pressure. However, this effect is uncertain when it comes to hydrogen.

Active firefighting also has interfaces with other safety systems/functions such as the containment function, natural and mechanical ventilation, leak and fire detection, ESD, PFP, layout design principles, explosion barriers, and structural integrity.

The design and capacity of firefighting systems should be based on an analysis of the credible fire scenarios and the potentially hazardous consequences. Aspects to be taken into consideration are (not extensive list) as follows:

- exposure of human beings;
- escalation, for example, could the fire expose storage vessels with a subsequent risk of rupture; and
- loss of escape ways and means for evacuation and rescue.

The best way of handling a hydrogen fire is to let it burn under control until the hydrogen flow can be stopped. A hazardous combustible mixture starts forming immediately if the hydrogen fire is extinguished and the hydrogen flow is not stopped. Extinguishment may create a larger hazard due to an increased potential for an explosion if gas from a release is subsequently re-ignited. This risk applies especially in congested areas and where hydrogen can accumulate in case of a leak. For a hydrogen fire, the most important would be to stop the hydrogen flow, shutdown the process, and activate the fail-safe condition of the installation.

Water should never be sprayed to enter a liquid-hydrogen vent system that gets sufficiently cold to freeze that water due to the risk of clogging the vent system.

Water systems can be used to cool storage systems that might burst if they are exposed to a fire. For applications where hydrogen represents the major fire and explosion hazard, it might be safer to use PFP to protect these systems.

Firewater systems, including hoses, sprinklers, deluge systems etc., could be activated to cool hydrogen equipment exposed to a fire. Remotely controlled water-spray equipment is preferred to hoses, cool the equipment, and reduce the fire spread. If it is necessary to use hoses, personnel using them should stay behind protective structures. This applies both to hydrogen fires and other fires.

Only trained professional firefighters or trained emergency personnel should fight hydrogen fires.

The deluge or water spray systems design should be based on recognized international standards, for example, NFPA 13 and 15.

Water deluge, which gas detection systems can activate, can be effective at reducing worst-case explosion pressures. However, the performance of water deluge will depend greatly on parameters like scale, confinement, congestion and how water is applied, and more work might be needed to validate the effect related to hydrogen releases (Gexcon US, 2018).

6.6.12.1 Recombiner

A recombiner is a device that promotes the recombination of hydrogen with oxygen—usually available as a constituent of air—forming water. As such, a recombiner may avoid, remove, or at least slow down the formation of flammable mixtures caused by an accidental leak of hydrogen in an enclosed area (HySafe, 2006; OECD, 2015).

Recombiners can be both active and passive devices. Active recombiner uses heat to initiate the conversion. Passive recombiner uses the effect that hydrogen and oxygen react already at low temperatures and even beyond conventional concentration limits in an exothermal reaction in the presence of catalysts such as platinum or palladium. Appropriate measures (e.g., system design) have to be taken to prevent the system temperatures from exceeding the self-ignition temperature of hydrogen. This might cause an unintended ignition due to the exothermal reaction at elevated hydrogen concentrations. Without appropriate measures, the use of recombiner is limited to mixtures below the ignition limit (Ledjeff, 1987; Baggemann et al., 2017; Reinecke & Poss, 2012; Reinecke et al., 2010; Reinecke, Tragsdorf, & Gierling, 2004).

At an industrial scale, hydrogen is used in designated areas where the risk of formation of flammable mixtures can be reduced by design or where venting can easily be applied. As a consequence, only a few specific application fields exist where recombiners are used. The main application of recombiners have been in nuclear plants and related to batteries.

In nuclear reactors, recombiners are used to remove hydrogen that is produced in service (boiling water reactors, active recombiners) or possibly released during a severe accident (light water reactors, passive autocatalytic recombiners).

Active thermal recombiners are used with gas capacities of 100 m³/h. The inlet gas is heated up to 700°C and above, initiating the recombining reaction. The outlet gas is cooled before leaving the device. Catalytic recombiners usually need a minimum concentration of about 0.5 vol.% for a start-up. As passive recombiners are self-feeding devices, the conversion rate depends on the self-generated throughput that depends on the catalyst temperature. In known systems, typical flow velocities are between 0.5 m/s and 1.0 m/s. To keep the system active over long periods, some devices propose to be kept in a sealed environment to prevent the catalyst from being spoiled (poisoned) by the atmosphere.

Detailed information on the long-term research (qualifying tests, experimental studies) that has been performed in the nuclear field with extensive bibliographical references are given in Liang et al. (2014).

With the increasing domestic use of hydrogen, for example, in an increasing number of mobile applications, an added need for specific recombiner systems may be utilized, such as in repair garages, vehicle storage compartments, tunnels, domestic housing combustion applications.

6.6.13 Explosion protection

6.6.13.1 Enclosed areas

A hydrogen leak inside an enclosed area, if not stopped or otherwise controlled, is a scenario that implies a high probability of explosion or even a transition to a detonation. Therefore it is critical to detect leaks in enclosed areas as early as possible and implement mitigation measures that will prevent the accumulation of hydrogen or reduce the consequences of ignition as far as possible.

Gas detection should be coupled to alarm systems, and confirmed gas detection should automatically activate Point 3 (below). Points 5 – 8 should be considered depending on the results of risk assessment.

1. *Reduce the number of leak points inside the enclosure.* Typical leak points such as valves flanges should be located on the outside of enclosures. If such leak points have to be located indoors, the volume should be as small as possible, and pipe-in-pipe systems should be considered.
2. *Minimize the orifice of the inflow of hydrogen to the enclosure.* This will reduce the leak probability and limit the maximum release rate in case of a leak inside the enclosure.
3. *Emergency ventilation* applied to enclosures can be initiated automatically in case of a confirmed detection of leaking hydrogen. For example, the initiation may be based on gas detection related to a predefined fraction of LFL (e.g., 20% LFL) or by other signals indicating that a leak of hydrogen has occurred or may occur (see also Section 6.6.6 on ventilation). The time until the emergency ventilation reaches full capacity should be in the order of seconds. This is especially important for high-pressure hydrogen systems as high pressure will lead to large release rates. Improved ventilation could also be achieved by automatic opening of roofs or other areas at a high elevation in case of confirmed gas detection.
4. *Shutdown and safe relief of the gas inventory to a safe location.* In case of confirmed gas or fire detection, the process should automatically shutdown (ESD, see Section 6.6.4). Valves should go to a fail-safe position, and if the hydrogen inside the process equipment cannot be contained safely, it should be relieved to a safe location through a vent valve or in case that fails through a PSV. The system should also be isolated from inflow or backflow by isolation valves (ESD). A critical aspect of ESD valves is the response time, which should be as quick as possible.
5. *Explosion relief areas.* Explosion relief areas could be installed when hydrogen processing equipment is located in buildings/enclosures. Explosion relief areas could be explosion panels that are light panels that blow out at small overpressures. The aim is to relieve the explosion before dangerous overpressures are developed. The areas of the explosion relief panel should be as large as possible. Their properties (effect) should be specified and documented. It should be secured that explosion relief panels do not lead to flying debris around the installation. Note that explosion relief panels will have no reducing effect on the explosion pressure in case of a detonation. Explosion venting/relief applies to equipment and for buildings. The venting guideline is found in NFPA-68 (NFPA, 2018). In the Biannual Report of Hydrogen Safety (HySafe, 2006), some examples are described. For the venting of hydrogen from enclosures such as garages, the pressure peak phenomena is of importance (Makarov et al., 2018).
6. *Design enclosure against explosion.* The enclosure's construction can be designed to withstand a very high explosion blast so that the energy of the explosion is absorbed in the construction. Such a design will prevent a blast wave, and flying debris in the surroundings will be prevented. This type of enclosure will usually require thick and heavy concrete constructions and may not be a practical solution for a unit above ground.

6.6.13.2 Risk for DDT that can propagate into open areas

In general, a flammable gas cloud of hydrogen has a higher probability of DDT than most other flammable gases. It should be noted that in case a strong explosion is initiated inside a partly

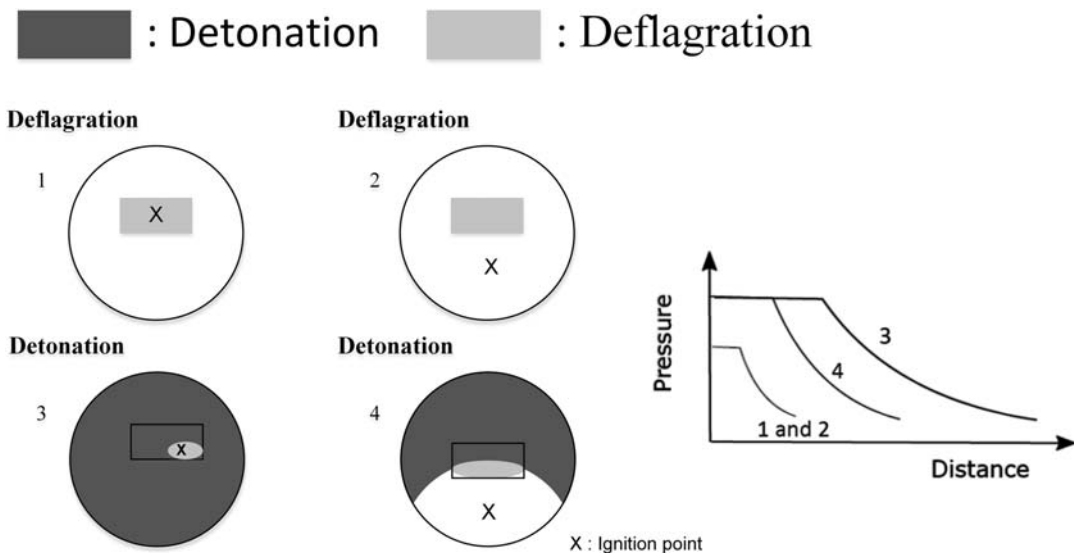
enclosed/obstructed area, there is a risk that it might propagate into open areas if flammable gas is collected here. If the whole volume of flammable gas, also in the unobstructed area, contributes to the combustion, this might lead to very high pressure at longer distances than if only the gas in the enclosed volume contributed.

The principle is illustrated in Fig. 6.12.

Reprinted from Oran, E. S., Chamberlain, G. and Pekalski, A. (2020) Mechanisms and occurrence of detonations in vapor cloud explosions. *Progress in Energy and Combustion Science*, 77, p. 100804. <https://doi.org/10.1016/j.pecs.2019.100804>.

The circle in Fig. 6.12 illustrates the risk of detonation dependent on the ignition point (Oran et al., 2020). Fig. 6.12 represents a flammable gas cloud and the rectangle an enclosed/obstructed area. The red area is the cloud that detonates, and the yellow the cloud that deflagrates

- Illustration 1 represents ignition in the enclosed area. Then the result will be a deflagration of the limited volume in the obstructed area, with a limited pressure that decreases quickly by distance from the obstructed area center.
- Illustration 2 in the figure represents ignition in the open area. The result will probably be a flash fire propagating toward the obstructed area, and the pressure will increase to a deflagration in the obstructed area. The pressure distance curve will be as for Illustration 1.
- Illustration 3 represents a situation in the enclosed area when a DDT occurs. That might be caused by a strong ignition source and/or a very high degree of obstruction/congestion increasing the turbulence in the combustion. The detonation can then propagate to the whole



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FIGURE 6.12

Illustrations of the risk of detonation dependent on ignition point.

flammable volume, causing very high pressure and much higher pressure at longer distances (curve number 3).

- Illustration 4 represents ignition in the open area, with a resulting flashfire propagated toward the obstructed area, where pressure builds up and a DDT occurs in the obstructed area. The detonation then continues at the other side of the enclosed area, also leading to very high pressure at a longer distance than in Illustrations 1 and 2, but shorter than in Illustration 3 as less gas detonates in Illustration 4.

For gaseous hydrogen systems, the detonation propagation risk into an unconfined gas cloud might be less relevant as hydrogen in an open and unconfined area will dilute more effectively compared to nonflammable situations. Furthermore, the flammable volumes will be limited. However, LH₂ systems are characterized by very low temperatures, a tendency to less buoyant gas clouds, and hydrogen gas concentrations may have a prolonged duration in open areas. Thus in the design of LH₂ systems, the effect mentioned above needs to be considered. Therefore additional measures are needed to avoid strong ignition sources, obstructed areas, and secure good natural ventilation.

6.7 Safety examples for hydrogen technologies

In the following section, several examples are given with an indication of installed hydrogen safety barriers.

6.7.1 Hydrogen fueling stations

A lot of focus has been taken on the safety of fueling stations during recent years, (see, e.g., Brennan et al., 2011; Ham et al., 2011; Marangon et al., 2007; Markert et al., 2017). Fig. 6.13 shows the hydrogen fueling station at Heroya Industrial Park, Norway. This hydrogen fueling station was opened in 2007 and was one of Norway's very first hydrogen fueling stations. Hydrogen was produced by water electrolysis or piped in from a nearby industrial facility. The station was closed after a few years of operation, but there are now plans to open it again after several new initiatives in Norway to develop renewable energy and fuels. Fig. 6.14 shows vent stacks, fencing, and safety signs.

The station is constructed to be fully automated. In case of confirmed gas detection, loss of ventilation in enclosed areas, or process deviations outside normal operation limits, the station will close and go to a safe position, and fueling will automatically stop. Several measures in design are also implemented to avoid any formation of static electricity.

6.7.2 The Deep Purple project

TechnipFMC's Deep Purple project (TechnipFMC, 2020) develops solutions for offshore production of hydrogen and intermittency management of wind farms by storing the hydrogen subsea. The project is illustrated in Fig. 6.15.

**FIGURE 6.13**

Hydrogen fueling station at Heroya Industrial Park, Norway.

Cortesi Sandra H. Nilsen August, 2020.

**FIGURE 6.14**

Hydrogen fueling station at Heroya Industrial Park, Norway, showing vent stacks, fencing, and safety signs. Cortesi Sandra H. Nilsen, August 2020.

The Deep Purple concept was founded at the Research Council of Norway's Idea Lab in June, 2016. Partners were TechnipFMC, Ocean Hyway Cluster, Energy Valley, and Sintef. TechnipFMC took ownership of the project and led the development process since then, including collaboration with the founders.

The Deep Purple project has been developed into different concepts. The main scope is a complete solution for stabilizing offshore wind production by using subsea storage of hydrogen. This will enable energy storage from offshore wind and a stable zero-emission energy supply to offshore

**FIGURE 6.15**

The Deep Purple offshore concept. Energy system management, electrolyzers, and fuel cells are located in the floating yellow device in the middle of the image. The hydrogen storage tanks are located on the seabed below the floater. Picture reprinted by permission of TechnipFMC.

installations and island communities. Offshore large-scale hydrogen production from wind farms for pipeline export is also part of the concept. The concept also allows international shipping to bunker hydrogen at open sea for export or vessel propulsion. Another scope is an integrated system for safe subsea hydrogen storage and distribution at ports around the world.

Intermediate storage of hydrogen at the sea bottom is an innovative solution and safe. Any leak from the storage tanks will not represent fire or explosion risk and is far from the customers. The authors have not been in contact with the project to get more details about the safety systems.

6.7.3 Hydrogen ferry in Norway

The ferry connections Bodø-Moskenes (FuelCellsWorks, 2020) is Norway's longest and runs over very demanding sea areas. Electric ferries will not work well enough on this route. Therefore the government has decided that the next tender will presuppose the use of hydrogen technology.

The safety requirement is that a hydrogen ferry must be as safe as a regular ferry. The shipping company that is to operate the connection must obtain approvals from, among others, the Norwegian Maritime Directorate and the Directorate for Civil Protection and Emergency Planning.

From 2021, the ferry route Hjelmeland-Nesvik-Skipavik in Rogaland, Norway, will be served by the world's first hydrogen-electric ferry. In addition to batteries, more than half of the operation will be fuel cells with LH₂ as fuel (Fenstad, 2020). Due to the safety risk, DNV GL, the Norwegian Defence Research Establishment (FFI) and the Norwegian Public Roads Administration recently launched the first large-scale tests at DNV GL's test facilities in Spadea Dam in the United Kingdom. The tests included conditions representing a jetty's geometries, representing the ship side or other obstacles that cause the gas to condense. If the concentration of the gas becomes too high, it can detonate in case of ignition. The tests were recorded using an HD camera. The tests involved pouring three tonnes of LH₂ from a tanker at the test facility and then igniting the formed hydrogen cloud.

6.8 Conclusion and current development

Conventional carbon-based energy companies have increased their focus on low carbon technologies, including hydrogen technologies. An increasing number of companies are setting strategic directions to demonstrate the needed commitment to long-term value creation that supports the Paris Agreement. This includes acceleration of decarbonization by establishing a well-functioning market for carbon capture, transport, and storage, and developing competitive hydrogen-based value chains and solutions.

This chapter discussed the general safety principles from a technical and operational perspective. Hereunder, it should be noted that the recommended safety strategies taken by industrial companies might be different from those that are best for third-party applications.

A system safety approach unfolds ISD aspects and safety barrier function methodologies to prevent hydrogen accidents or mitigate the accidental consequences. It is described how they should be applied to design and operate installations in practice where hydrogen is used as fuel. The safety barriers are suggested based on the properties of hydrogen that influence safety and prevent accidents or accidental situations.

Solutions to improving hydrogen safety are not necessarily only of technical nature, but the important roles of organizational and human factors are highlighted as well. Hereunder, such factors are important for a well-conducted design and construction of facilities. In the operational phase, safety procedures and training are important and the maintenance of such procedures themselves.

The specific properties of hydrogen must be taken into account to ensure the safe design and operation of hydrogen systems (see Chapter 1: Hydrogen Fundamentals). These properties differ substantially from those of natural gas and other conventional oil and gas products. Development projects need to consider and mitigate the increased possibility of high explosion pressures or detonation if hydrogen releases accumulate in enclosed or congested areas. On the other hand, hydrogen's buoyant properties can be exploited by locating potential leak points in the open to avoid gas accumulation, thereby reducing the explosion risk.

Some specific passive and active safety barriers are about these properties described. Fires and explosions need to be avoided and should not create additional events. Hereunder, the fast and reliable detection of an event is a central need and important to signal for further additional safety procedures such as emergency shutdown, start for inerting, and many other active safety barriers.

The respective knowledge on best practices and safety barriers is the result of pre-normative research activities. Many projects have been conducted in the past, but new aspects and phenomena have to be investigated as hydrogen technologies develop. Recently, projects, such as HyTunnel-CS and PRESLHY, investigate the safety of hydrogen vehicles in tunnels and similar confined spaces and the behavior of LH₂, respectively. The Fuel Cells & Hydrogen Joint Undertaking programme funds both projects. The Norwegian SH2IFT project is another example of such research. Hydrogen technologies are still under fast development, and it is important to maintain continuous research in the field. Another aspect is incident monitoring (e.g., HIAD database) that provides essential knowledge to validate and improve the safety standards.

All these different activities together support the safe introduction of pressurized and LH₂ applications in various areas comprising road, train, and maritime transport, including heavy-duty transport, mining industries and supply chain activities to supply hydrogen-refueling stations.

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References

- Afgan, N., & Veziroglu, A. (2012). Sustainable resilience of hydrogen energy system. *International Journal of Hydrogen Energy*, 37(7), 5461–5467. Available from <https://doi.org/10.1016/j.ijhydene.2011.04.201>.
- Ale, B. J. M. (2005). Tolerable or acceptable: A comparison of risk regulation in the United Kingdom and in the Netherlands. *Risk Analysis*, 25(2), 231–241.
- Anderson, C. (2011). Detecting hydrogen gas and flames, hazardex. Available at: <<http://www.hazardexonthenet.net/article/43288/Detecting-Hydrogen-Gas-and-Flames.aspx>> Accessed 13.12.20.
- Astbury, G. R., & Hawksorth, S. J. (2007). Spontaneous ignition of hydrogen leaks: A review of postulated mechanisms. *International Journal of Hydrogen Energy*, 32(13), 2178–2185. Available from <http://www.sciencedirect.com/science/article/B6V3F-4NWW6H2-1/2/0b4cb8c9c274c471ec82521f7cbd626b>.
- Aven, T., Sklet, S., & Vinnem, J. E. (2006). Barrier and operational risk analysis of hydrocarbon releases (BORA-Release): Part I. Method description. *Journal of Hazardous Materials*, 137(2), 681–691. Available from <http://www.sciencedirect.com/science/article/B6TGF-4JW15MV-3/2/4847b455e6e65442ba8599c897087a31>.
- Baggemann, J., et al. (2017). Numerical study on the influence of different boundary conditions on the efficiency of hydrogen recombiners inside a car garage. *International Journal of Hydrogen Energy*, 42(11), 7608–7616. Available from <https://doi.org/10.1016/j.ijhydene.2016.04.084>.
- Bain, A., Barclay, J. A., Bose, T. K., Edeskuty, F. J., Fairlie, M. J., Hansel, J. G., ... Swain, M. R. (1998). *Sourcebook for hydrogen applications*. Hydrogen Research Institute and National Renewable Energy Laboratory.
- Brennan, S., et al. (2011). Hydrogen and fuel cell stationary applications: Key findings of modelling and experimental work in the HYPER project. *International Journal of Hydrogen Energy*, 36(3), 2711–2720. Available from <https://doi.org/10.1016/j.ijhydene.2010.04.127>.
- Bridges, J. et al. (2009). Risk assessment of products of nanotechnologies. European Commission.
- Buchner, H. (1984). Hydrogen use—transportation fuel. *International Journal of Hydrogen Energy*, 9(6), 501–514. Available from <http://www.sciencedirect.com/science/article/B6V3F-498M9GN-18T/2/7a6967c63b9b1752a683e4c3720d73b6>.
- Buttner, W., et al. (2020). 2020 HySafe research priorities virtual workshop series progress in safety research—Phenomenon and risk control updates from RPW 2018 sensors and monitoring what is a hydrogen sensor. *HySafe—Priority Research Workshop*.
- Buttner, W. J., et al. (2011). An overview of hydrogen safety sensors and requirements. *International Journal of Hydrogen Energy*, 36(3), 2462–2470. Available from <https://doi.org/10.1016/j.ijhydene.2010.04.176>.
- Dagdougui, H. et al. (2018) Chapter 7—Hydrogen logistics: Safety and risks issues. In H. Dagdougui et al. (Eds.) *Hydrogen infrastructure for energy applications*. Academic Press, pp. 127–148. <https://doi.org/10.1016/B978-0-12-812036-1.00007-X>.
- DOI Standard. (2009). *Concepts and principles, human performance improvement handbook*. Washington D.C.: U.S. Department of Energy. Available from https://www.standards.doe.gov/standards-documents/1000/1028-BHdbk-2009-v1/@_images/file.

- DS/ISO/TR 15916 (2015). *Basic considerations for the safety of hydrogen systems*.
- Duijm, N. J. (2015). Recommendations on the use and design of risk matrices. *Safety Science*, 76, 21–31. Available from <https://doi.org/10.1016/j.ssci.2015.02.014>.
- Duijm, N. J., Paulsen, J. L., & Markert, F. (2005). Safety aspects of ammonia as a fuel in fuel-cell-powered vehicles. *Advances in Safety and Reliability—Proceedings of the European Safety and Reliability Conference, ESREL 2005*.
- Duijm, N. J. (2009). Safety-barrier diagrams as a safety management tool. *Reliability Engineering & System Safety*, 94(2), 332–341. Available from <https://doi.org/10.1016/j.res.2008.03.031>.
- Duijm, N. J., & Markert, F. (2009). Safety-barrier diagrams as a tool for modelling safety of hydrogen applications. *International Journal of Hydrogen Energy*, 34(14), 5862–5868. Available from <https://doi.org/10.1016/j.ijhydene.2009.02.002>.
- EIGA. (2006). *Gaseous hydrogen stations IGC Doc 15/0*, 6. Available at: Available from https://www.eiga.eu/nc/publications/eiga-documents/doc-1506-gaseous-hydrogen-stations/?sword_list%5B%5D=hydrogen.
- EIGA (2014). Hydrogen pipeline systems IGC Doc 121/14.
- EIGA (2017a). *Hydrogen vent systems for customer applications*. Available at: https://www.eiga.eu/nc/publications/eiga-documents/doc-21117-hydrogen-vent-systems-for-customer-applications/?sword_list%5B%5D=hydrogen.
- EIGA (2017b). *Methodology for determination of safety and separation distances*. Available at: <https://www.eiga.eu/index.php?eID=dumpFile&t=f&f=4237&token=4e8ab1b66b99ae4cf70e5e9e431da909b216594f>.
- EIGA (2019). *Safety in storage, handling and distribution safety in storage, handling and distribution*. Available at: https://www.eiga.eu/nc/publications/eiga-documents/doc-0619-safety-in-storage-handling-and-distribution-of-liquid-hydrogen/?sword_list%5B%5D=hydrogen.
- Eudy, L., & Post, M. (2014). Zero Emission Bay Area (ZEBA) fuel cell bus demonstration results: Third report. *Golden*. Available from <https://doi.org/10.2172/1134120>, CO (United States).
- European Commission (2014). *Atex guidelines—European Commission*. Available at: <http://ec.europa.eu/DocsRoom/documents/41403/attachments/1/translations/> Accessed 30.11.20.
- European Union (2000). *Directive 1999/92/EC of the European Parliament and of the Council of 16 December 1999 on minimum requirements for improving the safety and health protection of workers potentially at risk from explosive atmospheres*, Official Journal L 023. Available at: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:31999L0092>.
- Fenstad, A. (2020). *Tenner på tre tonn flytende hydrogen for å sikkerhetsteste hydrogenferge, avis artikel*. Available at: <https://www.tu.no/artikler/tenner-pa-tre-tonn-flytende-hydrogen-for-a-sikkerhetsteste-hydrogenferge/483113?key=NbBOatD6>.
- Franks, A. (1999). Lines of defence/layers of protection analysis in the COMAH Context. *Amey VECTRA limited for the Health and Safety Executive*. (HSE).
- FuelCellsWorks (2020). *Norway's Lofoten Islands to have hydrogen ferries by 2024*. Available at: <https://fuelcellworks.com/news/norways-lofoten-islands-to-have-hydrogen-ferries-by-2024/>.
- Gambetti, F., Casalli, A., & Chisari, V. (2012). The human factor in process safety management. *Chemical Engineering Transactions*, 26, 279–284. Available from <https://doi.org/10.3303/CET1226047>.
- Gexcon US (2018). *Explosion and mitigation*. Available at: <https://www.gexcon.com/us/products-services/Explosion-and-Mitigation/35/en> Accessed: 01.07.21.
- Gao, X, Abdul Raman, A., & Hizaddin, H. (2020). Systematic review on the implementation methodologies of inherent safety in chemical process. *Journal of Loss Prevention in the Process Industries*, 65, 104092. Available from <https://doi.org/10.1016/j.jlp.2020.104092>.
- Gowland, R. (2006). The accidental risk assessment methodology for industries (ARAMIS) / layer of protection analysis (LOPA) methodology: A step forward towards convergent practices in risk assessment? *Journal of Hazardous Materials*, 130(3), 307–310.

- Haddon, W., Jr. (1995). Energy damage and the 10 countermeasure strategies. 1973. *Injury Prevention*, 1(1), 40–44. Available from <http://www.scopus.com/inward/record.url?eid=2-s2.0-0029258070&partnerID=40&md5=769e4b4cd82561eaf926aa39e8a4157e>.
- Ham, K., et al. (2011). Benchmark exercise on risk assessment methods applied to a virtual hydrogen refueling station. *International Journal of Hydrogen Energy*, 36(3), 2666–2677. Available from <https://doi.org/10.1016/j.ijhydene.2010.04.118>.
- Health and Safety Executive (2021a). *Cost Benefit Analysis (CBA) checklist*. Available at: <<https://www.hse.gov.uk/managing/theory/alarpccheck.htm>> Accessed 25.06.21.
- Health and Safety Executive (2021b). *Risk management: Expert guidance*. Available at: <<https://www.hse.gov.uk/managing/theory/index.htm>> Accessed 25.06.21.
- Hosch, M. J. (2019). Hydrogen and fire safety. *International Fire Protection*. Available from <https://ifpmag.mdmublishing.com/hydrogen-and-fire-safety-detecting-the-most-flammable-element-on-earth/>.
- Hübert, T., Boon-Brett, L. & Buttner, W. (2018). Sensors for safety and process control in hydrogen technologies. CRC Press. <https://doi.org/10.1201/b19141>.
- Hughes, L., de Jong, M., & Wang, X. Q. (2016). A generic method for analyzing the risks to energy systems. *Applied Energy*, 180, 895–908. Available from <https://doi.org/10.1016/j.apenergy.2016.07.133>.
- HySafe, I. A. (2006). Chapter V: Hydrogen safety barriers and safety measures. *Biennial Report on Hydrogen Safety*. Available from http://hysafe.net/download/1200/BRHS_Chap5_V1p2.pdf. (Accessed 13.12.20).
- Jain, P., et al. (2018). Process Resilience Analysis Framework (PRAF): A systems approach for improved risk and safety management. *Journal of Loss Prevention in the Process Industries*, 53, 61–73. Available from <https://doi.org/10.1016/j.jlp.2017.08.006>.
- Johansen, I. L., & Rausand, M. (2014). Foundations and choice of risk metrics. *Safety Science*, 62(0), 386–399. Available from <https://doi.org/10.1016/j.ssci.2013.09.011>.
- Jurczyk, M., et al. (2008). Nanoscale Mg-based materials for hydrogen storage'. *International Journal of Hydrogen Energy*, 33(1), 374–380. Available from <http://www.sciencedirect.com/science/article/B6V3F4PHSF79-4/2/b91e63b7e104752416bdeafa0242bfff>.
- Khan, F., Rathnayaka, S., & Ahmed, S. (2015). Methods and models in process safety and risk management: Past, present and future. *Process Safety and Environmental Protection*, 98, 116–147. Available from <https://doi.org/10.1016/j.psep.2015.07.005>.
- Kletz, T. A. (1985). *What went wrong? Case histories of process plant disasters*. Houston, TX: Gulf Pub. Co.
- Kletz, T. A. (1988). *Learning from accidents in industry*. London, GB: Butterworths.
- Kletz, T. A. (2003). Inherently safer design—Its scope and future. *Process Safety and Environmental Protection*, 81(6), 401–405. Available from <https://doi.org/10.1205/095758203770866566>.
- Kletz, T. A., & Amyotte, P. (2010). *Process plants*. CRC Press. Available from <https://doi.org/10.1201/9781439804568>.
- Landucci, G., Tugnoli, A., & Cozzani, V. (2010). Safety assessment of envisaged systems for automotive hydrogen supply and utilization. *International Journal of Hydrogen Energy*, 35(3), 1493–1505. Available from <https://doi.org/10.1016/j.ijhydene.2009.11.097>.
- Ledjeff, K. (1987). Elimination of hydrogen or oxygen from explosive mixtures by catalytic techniques. *International Journal of Hydrogen Energy*, 12(5), 361–367. Available from [https://doi.org/10.1016/0360-3199\(87\)90063-2](https://doi.org/10.1016/0360-3199(87)90063-2).
- Leveson, N. (2004). A new accident model for engineering safer systems. *Safety Science*, 42(4), 237–270. Available from [https://doi.org/10.1016/S0925-7535\(03\)00047-X](https://doi.org/10.1016/S0925-7535(03)00047-X).
- Liang, Z. et al. (2014). *Status report on hydrogen management and related computer codes* (June).
- Macintyre, I., et al. (2007). Canadian hydrogen safety program. *International Journal of Hydrogen Energy*, 32(13), 2134–2143. Available from <https://doi.org/10.1016/j.ijhydene.2007.04.017>.

- Makarov, D., et al. (2018). Pressure peaking phenomenon: Model validation against unignited release and jet fire experiments. *International Journal of Hydrogen Energy*, 43(19), 9454–9469. Available from <https://doi.org/10.1016/j.ijhydene.2018.03.162>.
- Marangon, A., et al. (2007). Safety distances: Definition and values. *International Journal of Hydrogen Energy*, 32(13), 2192–2197. Available from <https://doi.org/10.1016/j.ijhydene.2007.04.007>.
- MarHySafe JDP. (2021). *Handbook for hydrogen-fuelled vessels*. (1st ed.) Edited by DNV. Available from <https://www.dnv.com/maritime/publications/handbook-for-hydrogen-fuelled-vessels-download.html>.
- Markert, F., et al. (2007). Safety aspects of future infrastructure scenarios with hydrogen refuelling stations. *International Journal of Hydrogen Energy*, 32(13), 2227–2234. Available from <https://doi.org/10.1016/j.ijhydene.2007.04.011>.
- Markert, F., et al. (2017). Risk and sustainability analysis of complex hydrogen infrastructures. *International Journal of Hydrogen Energy*, 42(11), 7698–7706. Available from <https://doi.org/10.1016/j.ijhydene.2016.06.058>.
- Markert, F., Duijm, N. J., & Thommesen, J. (2013). Modelling of safety barriers including human and organisational factors to improve process safety. *Chemical Engineering Transactions*, 31(2004), 283–288. Available from <https://doi.org/10.3303/CET1331048>.
- Markert, F., & Hansen, Z. N. L. (2018). Chapter 16—Risk analysis of complex hydrogen supply chains. In C. Azzaro-Pantel (Ed.), *Hydrogen supply chains* (pp. 521–552). Academic Press. Available from <https://doi.org/10.1016/B978-0-12-811197-0.00016-6>.
- Markert, F., & Nielsen, O. J. (1992). The reactions of OH radicals with chloroalkanes in the temperature range 295–360 K. *Chemical Physics Letters*, 194(1–2). Available from [https://doi.org/10.1016/0009-2614\(92\)85753-W](https://doi.org/10.1016/0009-2614(92)85753-W).
- Mohammadfam, I., & Zarei, E. (2015). Safety risk modeling and major accidents analysis of hydrogen and natural gas releases: A comprehensive risk analysis framework. *International Journal of Hydrogen Energy*, 40(39), 13653–13663. Available from <https://doi.org/10.1016/j.ijhydene.2015.07.117>.
- Moore, D. A. and Fuller, B. A. (2017). *Latest development in inherently safer technology*. Available at: <<https://engineering.purdue.edu/P2SAC/presentations/documents/AcuTech>> Latest development in IST—P2SAC Spring 2017.pdf. Accessed 30.11.20.
- NFPA (2018). *NFPA 68 guide for venting of deflagrations, NFPA68*. National Fire Protection Association. Available at: <<https://www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards/detail?code=68>> Accessed 02.03.20.
- NFPA. (2020). *NFPA 2 Hydrogen technologies*. Available from <https://www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards/detail?code=2>.
- NREL. (2020). *Renewable electrolysis*. Available from <https://www.nrel.gov/hydrogen/renewable-electrolysis.html>.
- OECD (2015). Status report on hydrogen management and related computer codes.
- Oran, E. S., Chamberlain, G., & Pekalski, A. (2020). Mechanisms and occurrence of detonations in vapor cloud explosions. *Progress in Energy and Combustion Science*, 77, 100804. Available from <https://doi.org/10.1016/j.pecs.2019.100804>.
- Pacific Northwest National Laboratory (2021). Hydrogen tools—Flame-detection, best practice. Available at: <<https://h2tools.org/bestpractices/flame-detection>>.
- Park, S., et al. (2020). Incorporating inherent safety during the conceptual process design stage: A literature review. *Journal of Loss Prevention in the Process Industries*, 104040. Available from <https://doi.org/10.1016/j.jlp.2019.104040>, Elsevier.
- Rasmussen, B., & Grønberg, C. D. (1997a). *Accident knowledge and emergency management*. Roskilde, Denmark: Risø National Laboratory.
- Rasmussen, B., & Grønberg, C. D. (1997b). Accidents and risk control. *Journal of Loss Prevention in the Process Industries*, 10(5), 325–332. Available from [https://doi.org/10.1016/S0950-4230\(97\)00022-3](https://doi.org/10.1016/S0950-4230(97)00022-3).

- Rasmussen, J., & Svedung, I. (2000). *Proactive risk management in a dynamic society*. Swedish Rescue Services Agency.
- Reinecke, E. A., et al. (2010). Open issues in the applicability of recombiner experiments and modelling to reactor simulations. *Progress in Nuclear Energy*, 52(1), 136–147. Available from <https://doi.org/10.1016/j.pnucene.2009.09.010>.
- Reinecke, E. A., & Poss, G. (2012). Operational behaviour of passive auto-catalytic hydrogen recombiners. *Nuclear Engineering International*, 57(697), 18–21.
- Reinecke, E. A., Tragsdorf, I. M., & Gierling, K. (2004). Studies on innovative hydrogen recombiners as safety devices in the containments of light water reactors. *Nuclear Engineering and Design*, 230(1–3), 49–59. Available from <http://www.sciencedirect.com/science/article/B6V4D-4BP9MD5-1/2/787c24eeb9e863492edfa5929573c014>.
- Rigas, F., & Amyotte, P. (2012). *Hydrogen safety*. CRC Press. Available from <https://doi.org/10.1201/b12267.f>.
- Simbeck, D. R. & Chang, E. (2002) Hydrogen supply: Cost estimate for hydrogen pathways—Scoping analysis. Colorado. Available at: <https://p2infohouse.org/ref/38/37695.pdf>.
- Singh, B. et al. (2010). 20 years on lessons learned from Piper Alpha. The evolution of concurrent and inherently safe design. In Papers presented at the 2009 international symposium of the Mary Kay O'Connor Process Safety Center, 23(6), pp. 936–953. <https://doi.org/10.1016/j.jlp.2010.07.011>.
- Samdal, U., Grainger, D., Hamborg, E., Nilsen, S. H., & Sommersel, O. K. (2021). Hydrogen safety strategies and risk management in Equinor. *Proceedings International Conference Hydrogen Safety Edinburgh*.
- Sklet, S. (2006). Safety barriers: Definition, classification, and performance. *Journal of Loss Prevention in the Process Industries*, 19(5), 494–506. Available from <http://www.sciencedirect.com/science/article/B6TGH-4J32JCN-1/2/20055bcd00a71c60b84bf45deac0d266>.
- Strickland, G. (1984). Hydrogen derived from ammonia: Small-scale costs. *International Journal of Hydrogen Energy*, 9(9), 759–766. Available from <http://www.sciencedirect.com/science/article/B6V3F-498W1R2-4N/2/5d6a5656731de5704ef7be20bb2b3eb4>.
- Sunderland, P. B. (2008). Pressure relief devices for hydrogen vehicles. *3rd European Summer School in Hydrogen Safety, Belfast, UK*, 1–29. Available from <http://www.hysafe.org/science/eAcademy/docs/3rdesshs/presentations/ESSHS2008SunderlandP2.pdf>.
- Tanaka, H., et al. (2009). Hazard assessment of complex hydrides as hydrogen storage materials. *International Journal of Hydrogen Energy*, 34(7), 3210–3218. Available from <http://www.sciencedirect.com/science/article/B6V3F-4VP5XBF-4/2/8083834d569c2444331b1b44cb2e4332>.
- Taube, M., et al. (1985). A prototype truck powered by hydrogen from organic liquid hydrides. *International Journal of Hydrogen Energy*, 10(9), 595–599. Available from <http://www.sciencedirect.com/science/article/B6V3F-498M864-V9/2/dc089a701baa3f0dd2d5ed4c6b374510>.
- Taube, M., et al. (1983). A system of hydrogen-powered vehicles with liquid organic hydrides. *International Journal of Hydrogen Energy*, 8(3), 213–225. Available from [https://doi.org/10.1016/0360-3199\(83\)90067-8](https://doi.org/10.1016/0360-3199(83)90067-8).
- Tchouvelev, A. V., et al. (2020). Development of risk mitigation guidance for sensor placement inside mechanically ventilated enclosures—Phase 1. *International Journal of Hydrogen Energy*. Available from <https://doi.org/10.1016/j.ijhydene.2020.09.108>.
- TechnipFMC. (2020). *Deep Purple project collaboration, web article*. Available from <https://www.gceocan.no/news/posts/2020/july/deep-purple-project-collaboration/>.
- Trost, W. A., & Nertney, R. J. (1985). *Barrier analysis*, United States. Available from <https://www.osti.gov/biblio/6061362>.
- Wei, C., Rogers, W., & Mannan, M. (2008). Layer of protection analysis for reactive chemical risk assessment. *Journal of Hazardous Materials*, 159(1), 19–24.
- Zhiyong, L., Xiangmin, P. & Jianxin, M. (2010). Quantitative risk assessment on a gaseous hydrogen refueling station in Shanghai. In *ISMF-09, international symposium on multiphase flow, heat mass transfer and energy conversion*, 35(13), pp. 6822–6829. <https://doi.org/10.1016/j.ijhydene.2010.04.031>.

Legal requirements, technical regulations, codes, and standards for hydrogen safety

7

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7.1 Scope

This chapter presents the legal and standardization framework [regulations, codes, and standards (RCS)] available for fuel cells and hydrogen (FCH) technologies. As in the previous chapters, the focus here is on hydrogen safety, that is, hazards associated in particular with its flammability and the needs to pressurize or to liquefy it for practical purposes. Materials compatibility is a third important hydrogen-specific aspect. This legal and standardization framework has been developed over the past two decades, and is based on the scientific understanding of hydrogen behavior described in the other chapters of this book. Other aspects of safety, related for example to electrical and electromagnetic (EMC) hazards and functional safety, are not treated here unless specific to hydrogen.

Moreover, this chapter focuses in particularly on public safety, that is, on the RCS framework needed to guarantee the required risk control of the fuel cell technologies when broadly deployed among the public in mobility, transport, and power applications. It is obviously not the first publication dedicated to this topic: for example, chapter 13 of [Rigas and Amyotte \(2018\)](#) covers a scope comparable to the matter presented here. On a more limited and specific topics other publications can be mentioned: international RCS collaborations have been summarized by [Day \(2010\)](#) and more recently by [Linssen and Hake \(2016\)](#); finally, a description of RCS for hydrogen refueling stations (HFS) can be found in [Wurster \(2016\)](#). Despite the availability of these and other previous works, this chapter provides new insights: on one side, an updated and complete picture of the international and national RCS landscape related to hydrogen safety, and on the other, an in-depth description of technical aspects for the most important FCH applications.

The text is structured in this way: it starts with a summary of the international regulations and standards related to hydrogen safety, followed by a sketch of the RCS frame in selected macro-regions: Europe, North America, and Asia ([Section 7.2](#)). [Section 7.3](#) introduces the RCS literature dedicated to general hydrogen safety aspects such as hydrogen detection, liquid hydrogen (LH₂) handling, and hydrogen compatibility of materials. [Section 7.4](#) presents safety standards and regulations designed for specific FCH applications. Finally, the last [Section 7.5](#) summarizes gaps and action priorities.

7.1.1 What does regulations, codes and standard mean?

The acronym RCS, formed by the terms Regulations, Codes and Standards, is commonly used to indicate generically a regulatory and/or standardization frame. While regulations specify legally binding requirements, standards contain technical provisions and requirements of voluntary adoption. Regulations are usually issued by governments or local authorities or international intergovernmental organizations, while standards development is industry-led. The regulations presented in this book are mainly of a technical character, specifying for example requirements on the safe design, type-approval, and the operation of hydrogen systems. The European Union (EU) also makes use of directives. While regulations are immediately enforced overall in the EU, directives require transposition and adoption into the law of each EU Member States. Codes find in general a place between regulations and standards, and their specific roles and functions depend on the country and region of adoption. Codes can for example be industry standards adopted by (local) authorities: in these cases have a status close to regulations.

Standards can be issued by a national or international standardization body. In the EU, harmonized European standards are standards issued by the European standardized bodies such as CEN and CENELEC, upon request of the European Commission (EC). A similar hierarchical structure can be found in other countries as well, and in addition there are also industrial codes and standards and guidelines issued by other bodies, such as the American Society of Mechanical Engineers (ASME) in the United States and the European Industrial Gases Association (EIGA) in Europe. For an extended excursus on the meaning of codes and their relationship to standards, regulations, and best practices, see Tchouvelev, De Oliveira, and Neves (2019).

Unless otherwise specified, the term RCS is used here to indicate generically a set of legal and voluntary requirements and guidelines. The international RCS framework is the focus of this chapter, which enters in some national details only in absence of a global agreement on specific topics.

7.1.2 Elements of regulations, codes and standard terminology

Although a coherent and standardized terminology frame for the hydrogen and fuel cells technologies is already in place, discrepancies still exist, due to the complexity of the technologies and the variety of their possible applications. Specific standards for FCH technologies have often grown “organically” from already existing standards dedicated to similar technologies. Hydrogen is considered in some context a dangerous good, in another a chemical feedstock, and in another again a fuel. Moreover, some of the definitions rely on local legislative approaches. The goal of this section is to clarify how this document utilizes few specific terms, for which confusion still exist in the specialized literature.

A field historically characterized by lack of harmonized terminology is the one of *compressed hydrogen storage*. Its main component is called a *container*, a *cylinder*, a *tube*, or a *tank* depending on the RCS document and the application considered. This term multiplicity, however, does not affect their safety. More important is the set of terms used to define various pressure levels, an area that may cause some safety-related misunderstanding. This document uses the definitions of the United Nation Regulation UN R134 on the approval of motor vehicles with regard to the safety-related performance of hydrogen-fueled vehicles (UN R134, 2015). *Nominal working pressure* (NWP) is the settled pressure or rated pressure in a fully fueled container at a uniform temperature of 15°C. The *maximum allowable working pressure* (MAWP) means the highest pressure to which a pressure container or storage system is permitted to operate under normal operating conditions. Finally, the burst pressure is the pressure at which the tanks break open or fly into pieces due to the force of internal pressure. The set of pressures defined for an on-board system does not correspond exactly to the set adopted for pressure level used at the fueling station. The Annex E of the International Organization for Standardization (ISO) 19880-1 *Gaseous hydrogen—Fuelling stations—Part 1: General requirements* provides a comparison and explication of the two sets.

For the purpose of simplicity, this chapter uses *explosion* as a general term indicating *deflagrations and detonations*. For a definition of these phenomena for the specific case of hydrogen combustion, the ISO/TR 15916 on *Basic considerations for the safety of hydrogen systems* can be consulted (the 2015 edition is being reviewed now). Because of the polyvalent meaning of the term explosion in the common language, its scientific definition is characterized by a certain degree of

disagreement. For a discussion on this topic from the point of view of hydrogen technologists, see Keller, Gresho, Harris, & Tchouvelev (2014).

Type-approval or certificate of conformity is granted to a product that meets a minimum set of regulatory, technical, and safety requirements. Generally, type-approval is required before a product is allowed to be sold in a particular country, so the requirements for a given product will vary around the world. Certifications are cognate expressions often used in the specialized literature.

Finally, there is concept of *safety distance*, as *distance to acceptable risk level between a hazard source and a target (human, equipment or environment)*. This is the definition taken and simplified from ISO 19880-1 *Gaseous hydrogen—Fuelling stations—Part 1: General requirements*. The term “safety distance” is equivalent to “setback distance” or “separation distance,” used in other documents, and is inherently “risk-informed.” In other words, it is based not only on physical determination of the risk, but also on a probabilistic approach requiring adoption of quantitative risk assessment (QRA) techniques. The ISO 19880-1 defines also a *hazard distance*, based on *the determination of a physical effect causing harm to people, equipment or environment*, thus without the probabilistic dimension.

7.1.3 How this chapter refers to regulations, codes and standard documents

These documents provide the full name of the international and national standards the first time they are mentioned, and only with their number in the following text. Legally binding documents are on the contrary referenced in the reference section.

It is not the aim of this document to provide an exhaustive list of all existing national and international standards regarding hydrogen technologies. Previews are available at the websites of their publishing organizations. Also, regularly updated lists of hydrogen-related standards are available online. The website managed by the US Fuel Cell and Hydrogen Energy Association is probably the most complete repository (<http://www.fuelcellstandards.com>, last accessed 02 March 2021). The European Fuel Cells and Hydrogen Observatory maintains a structured list of international and European standards (<https://fchobservatory.eu/observatory/Policy-and-RCS/Standards>, last accessed 02 March 2021).

7.2 The regulations, codes and standard international frame

7.2.1 United Nations Regulations

The United Nations Economic Commission for Europe (UNECE) is one of the five regional commissions of the United Nations. Its major goal is to promote pan-European economic integration. All United Nations Member States may participate in the work of UNECE, as well as international professional organizations and other nongovernmental organizations. At the moment there are 56 UN Member States and more than 70 organizations involved. In the context of the RCS, it is important to know that the UNECE sets out technical regulations and conventions to facilitate international cooperation. Therefore the UN Regulations developed by UNECE represent the global level of RCS integration. The agreement on a UN Regulation does not imply automatic implementation. Each country has first to adopt it, a process that can require several years.

Transport of hydrogen as dangerous good is regulated by the ADR, short for *European Agreement concerning the International Carriage of Dangerous Goods by Road* (UN-ECE, 2020). Its main goal is to guarantee and harmonize safe carriage of dangerous goods and to prevent pollution to the environment. Part 2 classifies dangerous goods: compressed hydrogen and LH₂ belong to the class 2.1 of flammable gases with respectively UN1049 and UN1066 label. Hydrogen – methane blends have number UN2034. According to the ADR, a flammable gas is a gas ignitable when in a mixture of 13% or less by volume with air, or has a flammable range with air of at least 12 percentage points regardless of the lower flammable limit. Part 4 of the ADR prescribes minimum requirements for the packing, which for compressed hydrogen translates, among others, on requirements on the design, operation, and testing of compressed tanks to guarantee their integrity.

UN has issued similar provisions also for transport of dangerous goods by other transport modes, such as the *Agreement on the International Carriage of Dangerous Goods by Inland Waterways* (ADN) (UN, 2009).

Specifically on hydrogen technologies applied to mobility, the World Forum for Harmonization of Vehicle regulations (WP.29) of the UNECE has adopted in 2013 the so-called GTR13, the *Global Technical Regulation on Hydrogen and fuel cells* (UN-ECE, 2013). The GTR13 has been the result of the work of subgroup on hydrogen safety in the period 2007–11. The GTR13 proposal has further developed into the UN Regulation R134 on *Uniform provisions concerning the approval of motor vehicles and their components with regard to the safety-related performance of hydrogen-fueled vehicles*. This document has become the worldwide reference for national approach to type-approval of hydrogen-fueled vehicles. It is summarized and explained in Section 7.4.3.

The International Maritime Organization (IMO) is the United Nations agency responsible for the safety and security of shipping and the prevention of marine and atmospheric pollution by ships. It has the role to develop a globally implemented regulatory framework for the shipping industry. To this purpose, the IMO issues codes that are equivalent to the UN Regulations and equally legally binding. For example, dangerous goods transportation by open sea falls under the responsibility of the IMO, which issued the *Code on International Maritime Dangerous Goods*, in brief the IMDG Code (IMO, 2020), with the same goals as the above mentioned ADR. Recently, in relation with the international efforts toward abatement of pollution and CO₂ emission, ship builders and maritime engine manufacturers started considering alternative to diesel and heavy oil toward zero-emission fuel options. To assist development of lower-emission technologies for maritime propulsion, the IMO has issued in 2015 a new mandatory code for ships fueled by low-flashpoint fuels: the *Code on International Code of Safety for Ship Using Gases or Other Low-flashpoint Fuels*, in brief the IGF Code (IMO, 2017). Further details on the regulation of merchant shipping are given in Section 7.4.4.

7.2.2 International standardization bodies

The ISO is an independent, nongovernmental organization with the goal to develop international industrial and commercial standards. Its members represent their national standards organizations. The ISO has a so-called General Consultative Status that the United Nations give to nongovernmental organizations. Similarly, the International Electro-technical Commission (IEC) publishes international standards for all electrical, electronic technologies.

Both the ISO and the IEC play an important role in the development of safety standards for the FCH technologies. There are two Technical Committees (TC) fully devoted to hydrogen and fuel cell technologies:

1. The *ISO/TC 197 – Hydrogen technologies* develop standards for the production, storage, transport, measurement, and use of hydrogen.
2. The *IEC/TC 105 – Fuel cell technologies* develop standards for any type of fuel cells and their applications for power and heat, transport, electrolysis (as reverse fuel cells), flow batteries.

ISO/TC 197 covers safety aspects directly related to the hazards of hydrogen, both gaseous and liquid, while IEC/TC 105 focusses more (but not exclusively) on electrical safety.

There are additional ISO TC of relevance for hydrogen safety, which are not specifically dedicated to hydrogen but deal with applications and products for flammable gases, whose standards have to be applied in absence of a hydrogen-specific document:

1. *ISO/TC 58—Gas cylinders*, for the standardization of gas cylinders and other pressure receptacles, their fittings and requirements relating to their manufacture and use.
2. *ISO/TC 220 – Cryogenic vessels*, for the standardization of insulated vessels for the storage and the transport of refrigerated liquefied gases of class 2 of the UNECE ADR.
3. *ISO TC 22/SC 37 – Electrically propelled vehicles*, for standards on safety of the whole vehicle.

The standardization framework for hydrogen and fuel cells applications is rather broad and complex, because hydrogen as an energy vector can be used in many different energy sectors. Several additional TC publish general standards of relevance for the hydrogen technologies. Fig. 7.1 gives a graphic idea of this complexity. In the following part of this chapter the most important standards will be described in relation to individual safety aspects (Weidner, Honselaar, & Ortiz Cebolla, 2016).

7.2.3 Regulations, codes, and standards in the United States

RCS for hydrogen buildings, structures, and other components are based upon model code organizations: International Code Council and National Fire Protection Agency (NFPA). They create fire and building code documents, such as the International Fire Code (IFC), International Building Code, and NFPA 1, which is the overarching Fire Code (NFPA, 2021a). States and cities in the United States typically adopt codes from one of these organizations and amend them based upon their regional needs. For example, California amends the IFC to create its own California Fire Code, which sets the requirements for everything in the state, from fueling stations to firework displays. The state fire codes are often written into state law and are legally binding.

Specific use codes, such as NFPA 2 *Hydrogen Technologies* Code, are referenced by both model codes organizations for most hydrogen requirements (NFPA, 2021b). The NFPA 2 provides high level, hydrogen-specific requirements for buildings, outdoor storage, and several other topics, such as dispensing, electrolyzers, repair garages etc. Most fueling stations in the United States base their design on the requirements in NFPA 2. NFPA 2 and NFPA 55 *Compressed Gases and Cryogenic Fluids* Code (NFPA, 2020) complement each other. The former focuses on gaseous and LH₂, while the latter addresses all other gases and cryogenic fluids.

NFPA 2 (and many of the fire codes) reference component standards, which form the third tier of RCS. These component standards are spread among several standard organizations, such as CSA Group (formerly known as Canadian Standards Association), Society of Automotive

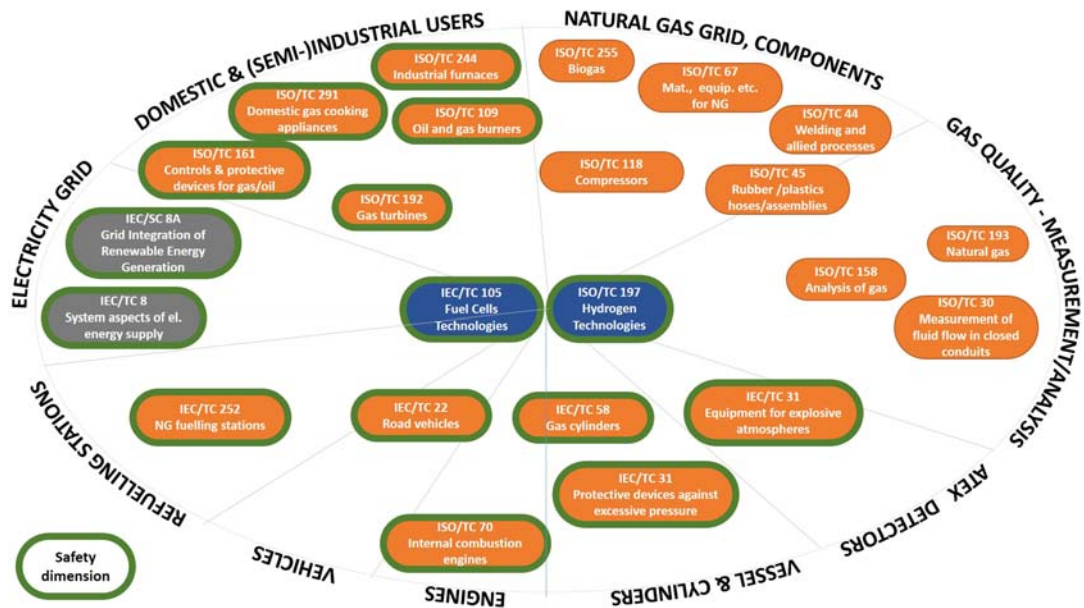


FIGURE 7.1

Map of the international ISO and IEC standardization technical committees related directly and indirectly with FCH applications. Those activities directly related to hydrogen safety are identified by a thick border. *FCH*, fuel cells and hydrogen.

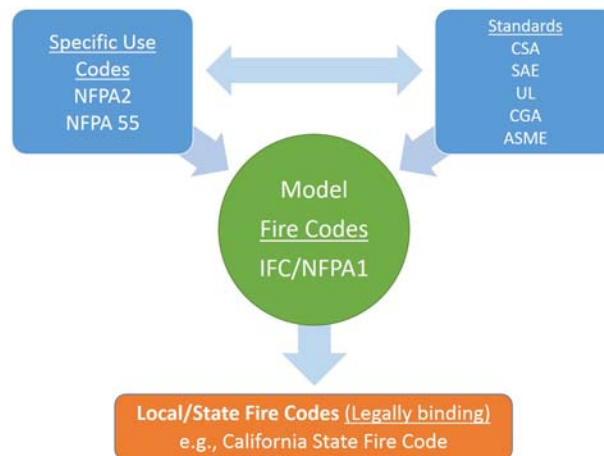
Picture redrawn, updated, and simplified from Weidner, E., Honselaar, M. and Ortiz Cebolla, R. (2016). CEN-CENELEC Sector Forum Energy Management / Working Group Hydrogen Final Report. <https://doi.org/10.2790/66386>.

Engineers (SAE), UL (formerly known as Underwriters Laboratories), Compressed Gas Association (CGA), and ASME. There are approximately 50–60 component level standards that encompass hydrogen dispensers, vent stacks, fueling protocol, and storage vessels. Both SAE and CGA have committees dedicated to hydrogen technologies, and several of their standards are described later on when dealing with specific applications in Sections 7.3 and 7.3.5. Fig. 7.2 shows in a graphic form the US RCS system.

Vehicle regulations and transport of hazardous materials in the United States are managed by the US Department of Transportation who is in the process of adopting UNECE Global Technical Regulation 13 (UN-ECE, 2013). A recent overview of the regulatory frame in the United States for all the sectors of potential interest for the deployment of hydrogen technologies is available in BairdEhrhart, Glover, & Lafleur, (2021).

7.2.4 Regulations, codes, and standards in Canada

Similar to the US NFPA 2, the Canadian hydrogen installation code is published by the Bureau de Normalisation du Québec as National Standard of Canada (BNQ, 2007). It dates back to 2008, but a new version is expected in 2021. It defines for example the installation requirements of gaseous

**FIGURE 7.2**

The structure of the US regulations, codes, and standards framework.

HFS and provides guidelines for the installation of fuel cells and internal combustion engines for emergency or back-up power to commercial buildings and residential homes. It applies to all liquid and gaseous installations, but not to industrial uses as process gas in refineries or feedstock for chemicals. Also hydrogen production above 21 kg/h are excluded.

The Standards Council of Canada is the Canadian standardization body. It has been pioneering in the production of CSA standards on hydrogen technologies that are applied as well in the United States. For example, the CSA standards belonging to the HGV 4.x series dedicated to gaseous HFS and their components have been published in the 2012 – 16 period, while their international equivalent are now being finalized. CSA standards are often published in combination with the American National Standards Institute (ANSI). For example:

1. ANSI/CSA HGV 4.1-2012 *Standard for hydrogen dispensing systems*.
2. ANSI/CSA HGV 4.2-2012 *Standard for hoses for compressed hydrogen fuel stations, dispensers, and vehicle fuel systems*.

7.2.5 Regulations, codes, and standards in Europe

7.2.5.1 EU Regulations of relevance for hydrogen and fuel cell technology

Differently than in the US case, the EU does not have a law specifically dedicated to the safe design, testing, construction, and deployment of FCH technologies and systems. These have to respect different directives and regulations, in first instance those governing hazards related to potential explosive atmospheres and the transport and storage of flammable gases by means of pressurized devices. These legislative documents have generated a cluster of harmonized European standards, which further specify design aspects, technical solutions, and testing and certification

methods. Overviews of the European regulatory framework covering to hydrogen technologies are already available. See, for example, [Floristean and Brahy \(2019\)](#) for the legal and administrative processes and specific to the hydrogen safety dimension ([European Hydrogen Safety Panel, 2019](#)). Nevertheless, a summary of the European RCS framework is given here, because it is needed for understanding the RCS rules for specific hydrogen applications presented later.

A set of EU Directives aims at preventing damage by explosions in case of unintentional releases of flammable gases. It mainly consists of two different Directives called ATEX (from the French terms for *ATmosphères EXplosives*):

1. *ATEX Directive 2014/34/EU*, called ATEX-products ([EU, 2014a](#)), covers equipment and protective systems intended for use in potentially explosive atmospheres.
2. *ATEX Directive EC/1999/92*, called ATEX-workplace ([EU, 1999](#)), concerns the protection of workers potentially at risk from explosive atmospheres.

These Directives define in detail legal responsibilities, the conformity assessments, and the related traceability and certification system. For the scope of this chapter, it is enough to mention that they prescribe the general obligation to provide protection to workers potentially at risk from ATEX. To provide this protection, technical and/or organizational measures have to be taken, appropriate to the nature of the operation with the following basic principles ranked in order of priority:

1. Prevention of the formation of explosive atmospheres or, where the nature of the activity does not allow that, avoidance of the ignition of them.
2. Mitigation of the detrimental effects of an explosion to ensure the health and safety of workers.

The suppression of equipment ignition sources and of the electrostatic sources is among the prevention measures to be taken. Regarding the mitigation measures, the ATEX considers explosion suppression and the isolating systems as flame arresters and barriers. A risk assessment is mandatory, one fundamental element of it being the classification of the hazardous places based on the frequency and the occurrence of the explosive atmosphere. This classification (also called the “zoning”) relies on the definition of “zones.” These are the most common zones:

1. Zone 0: a place in which an ATEX is present continuously or for long periods or frequently.
2. Zone 1: a place in which an ATEX is likely to occur in normal operation occasionally.
3. Zone 2: a place in which an ATEX is not likely to occur in normal operation but, if it does occur, will persist for a short period only.

While the US NPFA 2 *Hydrogen Technologies Code* is a prescriptive code and provides specific provisions for practices specific for hydrogen systems, the ATEX Directives are an overarching piece of legislation, defining only essential health and safety requirements and should be used in combination with European harmonized standards (see next section). Their application confers a presumption of conformity with the mandatory requirements of the Directives.

Other EU Directives, in addition to the ATEX Directives, must be considered when designing, installing, and operating (hydrogen) high-pressure systems:

1. The *Pressure Equipment Directive* (PED) concerning the safe design, manufacture, and conformity assessment of stationary pressure equipment with a maximum allowable pressure greater than 0.5 bar ([EU, 2014b](#)).

2. The *Transportable Pressure Equipment Directive* (TPED) is an EU Directive that applies to transportable pressure equipment used for the transport of dangerous goods by road, rail, and inland waterway (EU, 2010b). Gas cylinders, road tankers, battery vehicles, container vessels, and vessels for cryogenically cooled gases belong to the equipment falling under the TPED. This EU Directive is aligned to the UN Regulation for the transport of dangerous goods ADR mentioned in Section 7.2.1.

The PED and the TPED are not specifically on hydrogen. A complete description of its structure lies outside the scope of this chapter. Here it is enough to mention that they make use of a classification of pressurized equipment according to ascending level of hazard, and is based, among other criteria, on pressure and volume of the equipment. As in the case of the ATEX, their implementation required the development of related European harmonized standards setting requirements on specific type of pressurized equipment, such as boilers and burners, gas pipeline components and vessels (including cryogenic liquids), their materials and the related hazard prevention and mitigation tools. While specific standards are dedicated to selected gases, none of them focuses on hydrogen technologies. As example for many, here only the standard EN 15001-1 *Gas Infrastructure* will be mentioned, on covering requirements for the design, construction, and operation of gas installation pipework. A list with more than 100 harmonized standards related to the PED is regularly updated by the EC. Fig. 7.3 summarizes the hierarchical dependences for the Directives mentioned so far.

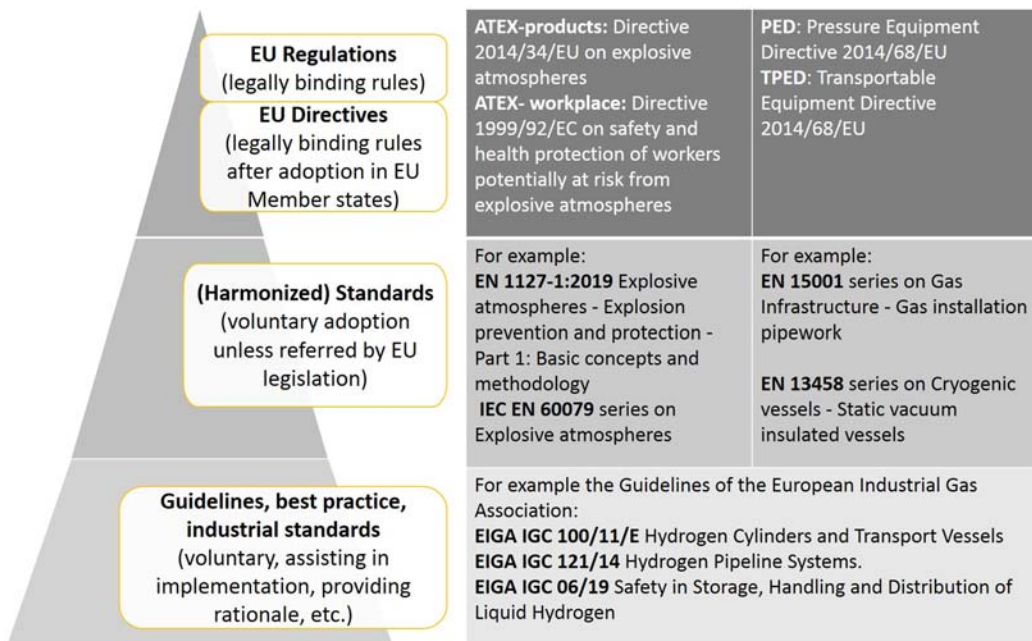


FIGURE 7.3

Hierarchical structure of the EU regulations, codes, and standards (RCS) framework. The directives shown are given as examples and are only a part of the RCS systems.

EU legislation, which will have to be applied to up-scaled hydrogen production and storage, is the Directive for the *prevention and control of accidents involving dangerous chemicals*. The so-called Seveso-III-Directive (2012/18/EU) aims at the prevention of major accidents involving dangerous substances and at limiting the consequences of such accidents on human health and the environment. Depending on the amount of dangerous substances present, establishments are categorized in lower and upper tier, the latter are subject to more stringent requirements.

The Directive set obligations for operators and the Member State authorities: for example, operators are required notification of all concerned establishments, a major accident prevention policy, internal emergency plans for upper tier establishments, safety reports and communications of all accidents. The Member States are required to develop external emergency plans (including ensuring that the operator takes all the measure deemed necessary), make publicly available all information, and finally to conduct regular inspections. For hydrogen the thresholds are 5 tons for lower-tier requirements and 50 tons for upper-tier requirements. Because of the relatively small amount of hydrogen stored on site, the Seveso III Directive was not applicable to HFS. The situation may change soon, in view of the planned massive deployment of the hydrogen technologies necessary for the reduction of green house gases (GHGs) emission from sectors such as the heavy duty transport and heavy industry. Especially, the CO₂ emissions from these sectors are difficult to abate by direct electrification. An example of this development are the European target for increasing green electrolyzers capacity to 20 GW in 2030 (European Commission, 2020).

Specifically on FCH applications to mobility, the EU has developed a regulation on European type-approval of fuel cell electric vehicle (FCEV), consisting of:

1. EC Regulation 79/2009 on type-approval of hydrogen-powered motor vehicles (EU, 2009b).
2. EU Regulation 406/2010 implementing Regulation EC 79/2009 (EU, 2010a). Annexes III and IV contain the requirements for hydrogen components and systems designed to use, respectively, liquid and compressed hydrogen. These provisions cover also components installation on hydrogen-powered vehicles.

This regulatory set was developed in the years 2007–10 to allow type-approval of FCEV in the EU and facilitate the deployment of this technology in absence of a global technical regulation (GTR). After the publication of the UNECE GTR13 and the UN R134 (see Section 7.4.3.1 on type-approval of FCEV), the latter was adopted in the EU (EU, 2019b) and the two type-approval processes have been made equivalent in the EU. Nevertheless, there are differences between the two type-approval processes: for example the fact that the EU legislation set comprehensive requirements on materials choice. The EU 79/2009 will be withdrawn in 2022, leaving only the UN R134, and the European approach to material selection will be the topic of a dedicated legislation (EU, 2019a).

At the level of EU Member States, there is a national and sometimes regional legislation aiming at the practical implementation of the overarching regulatory frame. For hydrogen systems, these “local” legal and administrative processes are listed on the online database of the European project HyLaw (<https://www.hylaw.eu/database>, accessed 20 March 2021). This project has also performed an assessment in terms of time required for a successful approval, and published a series of policy reports for 18 Member States and for the EU as a whole (Floristean et al., 2019; Floristean, Brahy, & Kraus, 2018).

7.2.5.2 EU standardization frame of relevance for hydrogen

The European Committee for Standardization (CEN) and that for Electrotechnical Standardization (CENELEC) are European standardization bodies. The core members represent the National Standardization Bodies of the EU Member States, the EFTA Countries, Turkey, Serbia, and the Republic of North Macedonia. Each EU Member State also has its own standardization body and a set of (historical) national standards. However, following the publication of a European Standard, each NBS has to withdraw any conflicting national standard.

They are equivalent, at European level, to the international ISO and the IEC, with which they have dedicated agreements: the so-called Vienna Agreement between ISO and CEN and the Frankfurt Agreement between IEC and CENELEC. These agreements aim at preventing duplication of effort and reducing time when preparing standards. For example, CEN and CENELEC can easily adopt ISO and IEC standards. The vice versa is also possible and new standard projects may be jointly planned.

The historical place for the development of European hydrogen-related standards is the CEN TCs on Cryogenic vessels (CEN/TC 268) and in particular its Working Group 5 on specific hydrogen technologies applications.

The Working Group Hydrogen of the CEN-CENELEC Sector Forum Energy Management (SFEM) was established in 2015 with a foresight function on priorities and future needs. The SFEM hydrogen performs gap analysis of the existing standardization frame for the whole FCH technology chain and the still existing barriers. It has performed barriers and gaps analyses and published a roadmap on future per-normative and standardization needs (Weidner, Honselaar, Klassek, & De Jong, 2019).

In 2018 the CEN/CENELEC Joint Technical Committee JTC 6 on hydrogen in energy systems was formed, including hydrogen safety (CEN/CLC/JTC 6/WG 3). The rationale behind its establishment was the absence of a single TC at CEN and CENELEC covering the wide range of topics identified by the SFEM WG hydrogen. Nevertheless, topics already covered by the scope of the CEN/TC 268 will not be transferred to CEN-CLC/JTC 6. Both TCs have a liaison status A with the ISO/TC 197.

Upon request of the EC, CEN and CENELEC provide so-called harmonized standards, published in the *Official Journal of the EU*. These standards can be used to demonstrate compliance with the related EU legislation, although they remain voluntary (EU, 2012). The EU Regulations and Directives mentioned in the previous section are completed by a family of European harmonized standards with detailed technical requirements. An example of harmonized European standards is the family of the EN 60079, dedicated to explosive atmospheres and related equipment protections by various safety measures, including detection. They are transposed from their parents belonging to the IEC 60079 series and facilitate the implementation of the ATEX Directives. Another example of this relationship is the EU Directive on the *Deployment of Alternative Fuels Infrastructure* (AFID) (EU, 2014c). Hydrogen is one of the alternative fuels considered by the AFID, together with electricity and natural gas (NG). The implementation of the Directive required the development of standards enabling the interoperability between infrastructure and vehicles throughout the EU. To enable interoperability of hydrogen infrastructure throughout the EU territory, the AFID required the publications of European standards on gas quality (EN 17124), connection devices (EN 17268), and fueling points, including filling protocols (EN ISO 17127). Fig. 7.3 shows examples of the hierarchical dependence of document in the EU RCS framework.

7.2.6 Regulations, codes, and standards in Japan

In Japan, the legislation regulating hydrogen technologies is mainly issued by the Ministry of Economy, Trade and Industry (METI) and by the related New Energy & Industrial Technology Development Organization. The two major national regulations, called Acts, under which hydrogen technologies such as storage, transportation, and fueling fall, are High-pressure Gas Safety Act and the Fire Service Act, where hydrogen is treated as an industrial gas (Nagashima, 2018). This explains why provisions adopted for HFS were originally the same as those intended for large-scale chemical plants. Also the use of LH_2 is regulated by the same high pressure Act.

Another national regulation, concerning safety requirements applicable to new hydrogen and fuel cells vehicles, which is part of the Road Vehicles Act, was developed in the 2002–05 period by the Ministry of Land, Infrastructure, Transport and Tourism (MLIT) as part of the Road Vehicles Act. It was equivalent to the EU regulation on type-approval of FCEV, but it was covering also motorcycles (MLIT, 2004). As in the European case, this national legislation will be superseded by adoption in Japan of the UN Regulation R134 (see Section 7.2.1 on the UN Global Regulations and Section 7.4.3.1 on type-approval of FCEV).

The standardization process in Japan consists in two overarching elements:

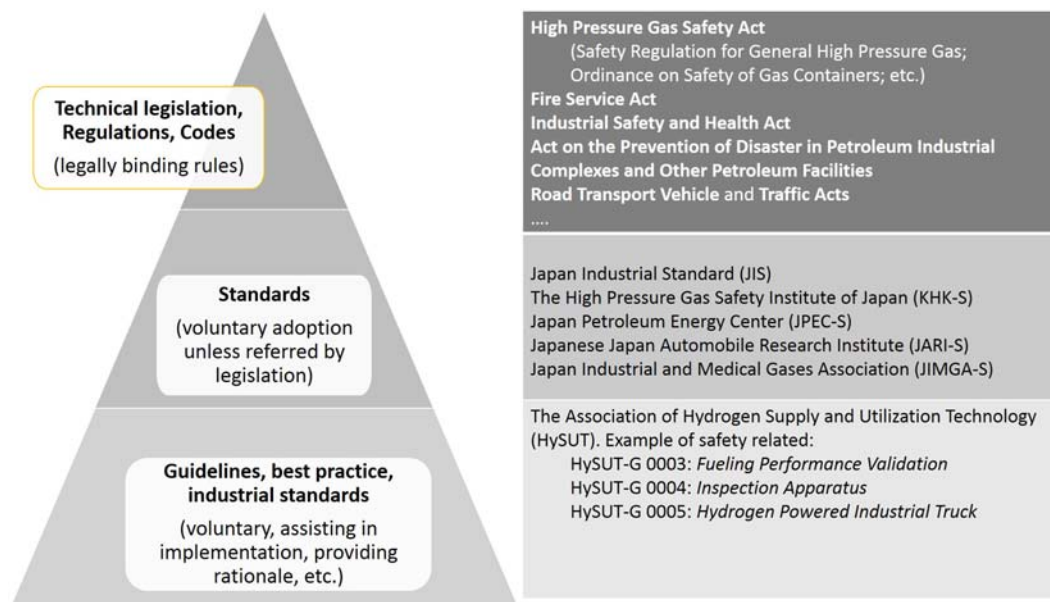
1. The Japanese Industrial Standards Committee (JISC) coordinates the standardization process. JISC is a body attached to METI and is in charge of surveying and deliberating on industrial standards.
2. The Japanese Standards Association (JSA) is the Japanese industrial standard development organization. JSA publishes the JIS standards and promotes standardization and management system in Japan.

For example, the series JIS C 88xx is dedicated to proton exchange membranes (PEM) fuel cells, such as the JIS C 8831 on *Safety Evaluation Test for Stationary Polymer*. Some other JIS standards related to fuel cells technologies have been aligned to the international framework, such as the series JIS C 62282 on stationary Fuel Cell Power Systems, transposed from the IEC series with the same name. There are, however, additional institutions that can publish Japanese standards. These are summarized in a nonexhaustive list in Fig. 7.4.

Among the early hydrogen-specific standards, which has enabled the first deployment of FCEV, the following two can be mentioned, both published for the first time by the Japanese Automobile Research Institute (JARI) 2004 and adopted as part of the mentioned Acts, so that they became compulsory:

1. JARI S 001 *Technical Standard Containers of Compressed Hydrogen Vehicle Fuel Devices*.
2. JARI S 002 *Technical Standard for Components for Compressed-Hydrogen Vehicle Fuel Device (valve and PRD)*.

These documents were based on the equivalent standards for compressed NG, and in their first edition were limited to 35 MPa NWP for the container. Regarding the Japanese Petroleum Energy Centre, it has developed the JPEC-S 003 standard for hydrogen fueling protocol. Its approach is similar to the SAE J2601 (see Section 7.4.3.3 on interoperability aspects), but its scope from the start was including buses refilling up to a total storage of 25 kg, something that SAE achieved only recently.

**FIGURE 7.4**

The hierarchical structure of the Japanese regulations, codes, and standards framework (the list of documents is given as an example and is not exhaustive).

In 2012–14, revisions were made to the High-pressure Gas Safety Act and the Fire Service Act to cover requirements for 70 MPa hydrogen stations, and related aspects on materials selection, fueling stations operation, road traffic, and hydrogen transportation rules (Yoshida, 2015). In 2014, METI was the initiator of a Strategic Road Map for Hydrogen and Fuel Cells covering a time span up to 2040 and beyond. Along with broadening the range of hydrogen applications and increasing of the hydrogen demands, the roadmap was foreseeing a strengthening of the delivery infrastructure and reduction of the related costs, including the renovation of the standardization framework, while always guaranteeing the same level of safety. The roadmap has been updated regularly, with the last published version in 2019 in view of “hydrogen” Olympic Games in 2020 (METI, 2019). One of the drivers for updating the RCS framework has been recently the objective aligning national regulations to the international standards, as enabler for international trade of hydrogen. The 2019 update of the hydrogen roadmap contains an action dedicated to a stronger national coordination and harmonization of RCS, including a comparison with the US and EU RCS frameworks. In line with the ambition to create a global “hydrogen society,” Japan has indeed recently stepped up its effort in international standardization organizations ISO and IEC, but also UNECE GTR13, with a specific focus on safety (Arias, 2019).

One example of this regular RCS improvement is that fact that originally, stations filling at 70 MPa were not allowed to trespass this pressure value. Since 2016, they are permitted to operate up to 82 MPa. This measure has increased efficiency of the filling process, by reaching a full filling of the FCEV tanks. Another example of more recent development is the standard JPEC-TD 0004

issued by the Japan petroleum Energy Center, which sets the conditions under which the self-service fueling is allowed at a hydrogen station (Nagashima, 2018).

Other new or updated RCS documents reduced the redundant safety margins on safety distances at the HFS and for hydrogen transport. The Japanese RCS framework is at the forefront in terms of new hydrogen applications. Among recent Japanese developments, three specific achievements should be mentioned:

1. A Japanese technical regulation for hydrogen and fuel cell motorcycles was achieved in 2017 (Yamada & Mashiba, 2019), and then translated in the new UN Regulation R146 (UN R146, 2019). The mentioned UNECE GTR13 and the related UN R134 could not be used, because it was limited to four wheelers. In comparison to these, a motorcycle has a lighter weight, smaller body, and a smaller storage system. These characteristics required a different approach to safety design and the adaptation of the approval tests. Among the specific features there are a fixed downwards release direction of the pressure release device (PRD), specific provisions for the protection of the on-board tank, and alternative crash tests with other acceleration values.
2. Japan is working at international regulations for the waterborne transportation of LH₂. As an enabler for a global trade in hydrogen, a Japanese consortium has achieved in 2020 the first waterborne certification of a LH₂ tanker (IMO, 2016; Nishifuji, 2019). Hydrogen is not listed among the products foreseen by the IGC code of the IMO. Therefore for the attainment of the LH₂ carrier certification, requirements had to be defined according to an alternative process described by the IGC code, by means of a so-called tripartite agreement among administrations and port authorities. This action is the first step toward an update or extension of the of IMO IGC code to hydrogen (Nishifuji, 2019).

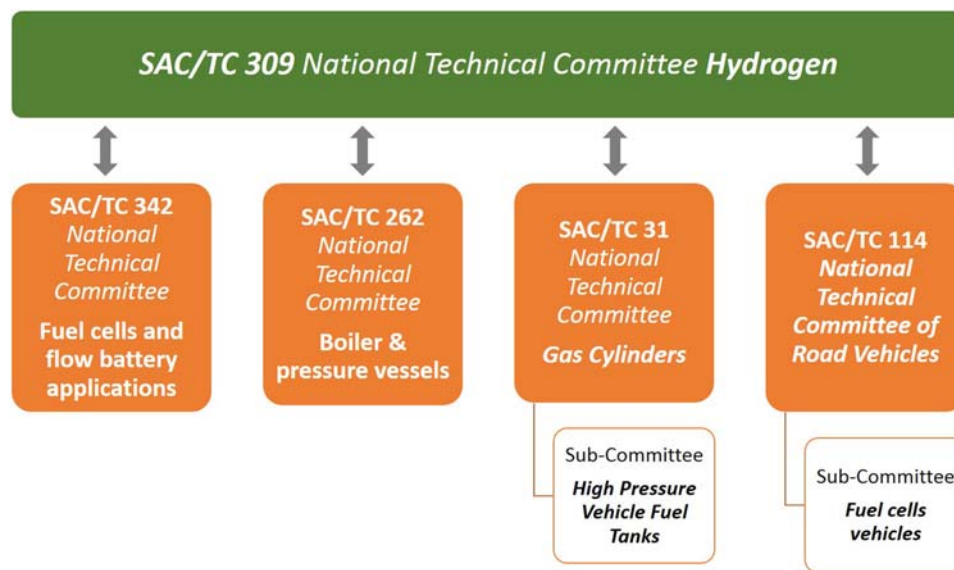
In the frame of the High Pressure Gas Safety Act, in 2020 METI has issued guidelines related to drones mounting fuel cells and compressed hydrogen storage (METI, 2020). The aim is to avoid risks from the fall of these drones. Among the envisaged measures are requirements on specific skills for operator, safe securing of the storage tanks, and preventing their corrosions.

7.2.7 Regulations, codes, and standards in the People's Republic of China

The Chinese standardization body relevant for FCH technologies is the Standardization Administration of the People's Republic of China (SAC), which represents China in the ISO and the IEC. China makes use of two types of standards: industry-driven standards with a voluntary character and mandatory standards equivalent to national law. These are mainly related to health, safety, and security dimensions. The national standards starting with GB/T are recommended, and several of the SAC standards related to fuel cells systems belong to this category.

Inside the SAC, more than one National Technical Committee deals with FCH topics. The most directly relevant for hydrogen safety is the SAC/TC 309, on hydrogen production, storage, transportation and fueling, applications, testing, and safety. Other committees focused on fuel cells and hydrogen technologies have a safety dimension as well, as shown in Fig. 7.5. Other committees on gases, work safety, and coatings also issue standards related to hydrogen components.

Due to the complexity of the fuel cells hydrogen technology chain, the SAC/TC 309 has established a national hydrogen framework to coordinate the related standardization effort (Yang, Wang,

**FIGURE 7.5**

Major National Technical Committees of the Administration of China contributing to hydrogen technologies standards.

Zhang, & Zhang, 2019), with a component dedicated explicitly to safety. In the period 2008–20, these committees have published a considerable number of hydrogen-related standards, covering all the hydrogen supply chains and several end uses. A complete overview of Chinese standards till 2019 is available in Yang, Wang, Zhang, & Zhang (2019).

Despite this high number of standards, it is common agreement (Zhao et al., 2020) that important gaps are still existing, hindering a deployment of a complete hydrogen supply chain. For example, in 2020, China does not yet have a standardized approval process for HFS, the maximal pressure for the road transport of gaseous hydrogen is 20 MPa. Another example is the partial adoption by China of the Regulation UN R134 (UN R134, 2015), excluding compressed hydrogen tanks of type 4 and NWP of 70 MPa (see Section 7.4.3.1 “Type approval of FCEV”). This is explained by an earlier strategic focus on hydrogen for public transport. Also in other cases, the Chinese standardization framework for FCH is not fully harmonized with the international framework. The SAC standards cover operative ranges (NWP, temperatures), which are still sometime more limited than those accepted elsewhere, hindering a full deployment of the technology. Another bottleneck is the classification of hydrogen as hazardous good, which hinders permission on on-site hydrogen production at the HFS.

These gaps are expected to be addressed soon. In the frame of the 13th Five Year Plan for Technological Innovation in Transportation (2017), China has announced a Strategic Roadmap for FCH, which contains also an RCS plan to enable the development of a fleet with 70 MPa on-board storage (Verheul, 2019). This implies a further alignment to the UN GTR13 and R134, with HFS and refueling protocols inspired by (but not simply copied from) ISO 1998-1, and SAE J2601 and

J2601/2 (Clague, 2019). The implementation of the roadmap is already visible in some of the standards mentioned above, valid for 70 MPa, and in the 2018 document T/CECA-G0018 on fueling specification, which reflects the SAE J2601.

7.3 General safety aspects

Comprehensive guidelines on hydrogen safety topics are publicly available. They play an important role in providing rationale and coherent introductions to legal requirements and standards provisions, which are typically of difficult access and not easy consultation. One of the first documents of this type were the *NASA Guidelines for Hydrogen System Design, Materials Selection, Operations, Storage, and Transportation* back in 1997. It is now canceled and superseded by the *Guide to the Safety of Hydrogen and Hydrogen Systems*, published by the American Institute of Aeronautics & Astronautics (ANSI/AIAA G-095A-2017). More recently, the US National Renewable Energy Laboratory (NREL) has published in 2015 a Safety Guide (Rivkin, Burgess, & Buttner, 2015) containing an introduction to general safety issues, the US RCS framework, and the permitting process. In Europe, the European Hydrogen Safety Panel has recently published guidelines on safety planning (European Hydrogen Safety Panel, 2019). This document is tailored to the research projects of the European Fuel Cells and Hydrogen Joint Undertaking (FCHJU), nevertheless containing an introduction to European RCS framework and safety planning applicable to generic type of projects.

The Technical Report ISO/TR 15916 *Basic considerations for the safety of hydrogen systems* is not a standard, but provides guidelines for the use of hydrogen in all its form (gas, liquid, and as hydrides), identifying basic safety concerns, hazards, and risks. It describes the properties of hydrogen relevant to safety and presents an overview of prevention and mitigation measures for the operations of equipment and design of facilities, such as ventilation and barricades. The edition of 2015 is under revision now.

7.3.1 Hydrogen-materials compatibility

The Technical Report ISO/TR 15916 just mentioned provides a list of selected materials and their suitability for hydrogen applications. For compressed hydrogen, the major criterion to be considered for the correct choice of metals and metallic alloys is their susceptibility to hydrogen embrittlement. For example, aluminum and copper alloys can always be used, as well as austenitic steel with nickel concentration above 7%. On the contrary, other types of steels and cast irons are suited for compressed hydrogen only under specific conditions, and often too brittle for cryogenic services. The final effect of hydrogen embrittlement on the in-operando performance of metallic components depends on the complex interaction of many operative environmental and mechanical parameters, design and manufacturing parameters, materials microstructure, and hydrogen purity. It is for this reason that regulations and standards for the design of hydrogen products can only issue general rules. To this respect, the ISO/TR 15916 advises material classes with negligible interaction with hydrogen, design measures tending to minimize local fatigue, and manufacturing measures tending to avoid cold plastic deformations and reducing hardness. The material classes considered in the ISO/TR 15916 and the method used for their classification in terms of severity of hydrogen

embrittlement are based on a knowledge basis of several years ago. In particular, there is a need to update the list of materials susceptible to hydrogen provided by ISO/TR 15916, and to this purpose the ISO/TC 197 has started a working group with this only purpose.

A decisive factor for the selection of materials is also the availability of data describing their performance in presence of hydrogen. An essential source for these data is the *Technical Database for Hydrogen Compatibility of Materials* produced and regularly updated by the US national research laboratory SANDIA summarized in the report (San Marchi & Somerday, 2012). It is important to remark that this document does not provide specific recommendations for material selections. It just provides experimental data, which can be used for the correct design of a specific product. Other RCS documents require the experimental measurement of the performances of specific materials to be used and their correct evaluations for design purposes. The EIGA has produced Codes of Practice on materials testing and selection for transportable (EIGA, 2011) and stationary (EIGA, 2006) high-pressure cylinders and for hydrogen pipelines (EIGA, 2014). Within their different scopes, these documents provide recommendations to avoid or limit hydrogen embrittlement by classifying the materials for hydrogen service in different classes based on compositional and strength ranges. A list of additional reference documents published by ANSI and ASME can be found in San Marchi, Somerday, and Nibur (2014). These guidelines do not contain design requirements but only data on materials properties.

The family of the standards ISO 11114 *Gas cylinders—Compatibility of cylinder and valve materials with gas content* provides recommendations for the safe combination of gas cylinders and their contents. It applies to seamed and welded metallic and composite cylinders, and to compressed, liquefied, and dissolved gas and gas blends. Its Part 1 focuses on metallic, Part 2 on nonmetallic materials. Part 3 is of critical importance for hydrogen application because it provides test requirements for the assessment of the materials properties. There is a certain disagreement on the extension of the use of the ISO 11114 to mobility and transport applications. While the UNECE GTR13 (UN-ECE, 2013) and the UN R134 (United Nations, 2015b) regulations do not contain any requirements on material compatibility, the European Regulation for type-approval of FCEV (EU, 2010a) prescribes hydrogen compatibility to be proved according to ISO 11114-4:2017 *Test methods for selecting steels resistant to hydrogen embrittlement*. This standard was developed for stationary applications and does not consider materials behavior under cyclic stresses. A dedicated task of the Informal Working Group for the Phase 2 of the GTR13 is working on a possible global consensus on this topic.

In the United States, the ASME Code for Pressure Piping and the Boiler and Pressure Vessel Code have published sections specifically dedicated to hydrogen: the ASME B31.12-2019 on *Hydrogen Piping and Pipelines*, and the Article KD-10 of the ASME BPVC VIII.3-2019 on *Special requirements for vessels in high pressure gaseous hydrogen transport and storage service*. The latter contains requirements based on fracture mechanics. Finally, the recent ANSI/CSA CHMC1-2014 *Test methods for evaluating material compatibility in compressed hydrogen applications—Metals* assists designers by providing a method for a performance comparison, but it does not exclude the need to comply with additional standards existing at the level of the full components.

Regarding polymers, in hydrogen applications they consist of thermoplastics and elastomers predominantly used for liners (in high-pressure storage cylinders and tanks) and sealing (e.g., O-rings), the ISO/TR 15916 reports a good compatibility with gaseous hydrogen. An international standard for hydrogen compatibility with polymer materials is not yet available. In the United States and Canada, uniform testing methods for assessing their compatibility are provided by the second part of the already mentioned ANSI/CSA CHMC2-2019 *Test methods for evaluating material*

compatibility in compressed hydrogen applications—Polymers. Among the properties assessed, there is the hydrogen permeability and the effect of static and cycling exposure to hydrogen on the stability, wear, and aging of the materials.

Regarding materials selection for LH₂ applications, a major effect to consider is the fragility of materials of both metals and plastics at cryogenic temperatures, related to a drop in fracture toughness and ductility as explained in [Chapter 1](#), Phenomena relevant to accidents, of this book. Related general RCS documents are mentioned in [Section 7.3.3](#) on liquid hydrogen.

7.3.2 Hydrogen detection

Hydrogen detection is an essential component of the safety design of hydrogen systems. Hydrogen sensors play a particular critical role as part of the risk mitigation strategy for installations in confined and semiconfined spaces, but they are also found in fueling stations and can be used on vehicles.

A comprehensive description of the function principles and the state of the art of the hydrogen detection technology can be found here ([Hübert, Boon-Brett, & Buttner, William, 2018](#)) and in [Chapter 6](#), Safety measures and safety barrier functions, including all aspects of safety, from design to maintenance to performance testing.

The standard IEC 60079-29 *Explosive atmospheres* standard is dedicated to detection of flammable gases and consists of four parts:

1. IEC 60079-29-1:2016 *Gas detectors—Performance requirements of detectors for flammable gases* specifies general requirements for their construction and performance, including testing methods. The scope is broad, from portable and transportable to fixed equipment for the detection and measurement of flammable gas or vapor concentrations with air, intended for use in explosive atmospheres.
2. IEC 60079-29-2:2015 *Gas detectors—Selection, installation, use and maintenance of detectors for flammable gases and oxygen.* Here the scope is extended to oxygen detectors and the focus is on industrial and commercial installations. An appendix gives an overview of the various detection technologies, their major area of usage, and their shortcomings.
3. IEC 60079-29-3:2014 *Gas detectors—Guidance on functional safety of fixed gas detection systems.* It is similar to the previous one, but the focus is on functional safety for fixed equipment and it also applies to the detection of toxic gases. It contains as well a metrology dimension.
4. IEC 60079-29-4:2009 *Gas detectors—Performance requirements of open path detectors for flammable gases* specifies performance requirements of specific detection equipment able to measure spectral absorption.

On top of the mentioned standards for all flammable gases and vapor, there is also a specific standard dedicated to hydrogen detectors, the ISO 26142:2010 *Hydrogen detection apparatus—stationary applications*. It defines the performance requirements and test methods of hydrogen detection apparatus designed to measure and monitor hydrogen concentrations in stationary applications.

The use of hydrogen sensors is not mandatory for mobility applications. FCEV manufacturers are free to decide if hydrogen detection is part of their safety design. In addition, in case of adoption of a detection system in the vehicle, the place and the function of the sensor can vary considerably (dry vs humid environment, vibration, temperature cycling, etc.). It is therefore rather challenging to

define performance requirements ensuring detection availability for safety strategies for all possible environmental and working conditions. Recommendations are nevertheless available in the SAE Report on *Characterization of On-board Vehicular Hydrogen Sensors*. This report leaves to the sensors manufacturers and the system integrators (who can also be the car manufacturers) the freedom and responsibility to specify the functions and the related performance tests.

The standards mentioned above allow the assessment of the performance and safety of individual type of sensors. However, the correct installation of sensors to guarantee their optimal role in the overall safety design depends on individual applications, which determines the environmental and operative conditions under which the sensors are expected to work. Humidity, temperature, vibration, poisoning gas species, air pressure variations are only few of the many variables affecting the detection function. Moreover the correct location for their installation in space may be not trivial in case of complex spatial geometry. There is therefore a specific need for methods enabling design-based placing of sensors. The methods must rely on a risk assessment approach, possibly assisted by computational fluid-dynamics modeling tools to simulate effects of selected release scenarios.

7.3.3 Liquid hydrogen

Like in the case of compressed hydrogen technology, for LH₂ there are two specific RCS areas: one aims at guaranteeing safe confinement (tanks, material suitability, ancillary system such as safety valves, piping) and the other at mitigation of accidental releases.

Belonging to the former is the rather old ISO 13985:2006 *Liquid hydrogen—Land vehicle fuel tanks*, which provides requirements for the construction and the testing of refillable on-board tanks. LH₂ tanks consist of an inner and an outer tank, the space in between guaranteeing the thermal insulation needed to minimize heat transfer. Due to the unavoidable boil-off of the cryogenic liquid, the standard prescribes the presence of two PRD operating at two different overpressure values, related to a maximal permissible fault pressure and an impermissible fault range. Another requirement regards the need to avoid overfilling, by means of liquid level detection and overfill protection gage. This standard is accompanied by the ISO 13984:1999 *Liquid hydrogen—Land vehicle fueling system interface*, which offers a standardized interface for fueling. These two standards are based on knowledge of two decades ago, and are deemed not aligned to the recent progress achieved in LH₂ technologies. Moreover, despite the progress in knowledge, LH₂ storage on-board of vehicles is still not a completely mature technology. ISO has decided to withdraw them, to avoid the risk that new regulations refer to obsolete documents. Considering the almost not existing deployment of on-board LH₂ storage, these two standards have not been scarcely used. They must be updated or replaced by new documents. For example, the ISO 13984 does not consider nonmetallic materials and breakaway coupling. A spotted weakness of the ISO 13985 is its outdated boil-off management strategies.

The ISO 13985 refers to a series of ISO standards dedicated to vessels for transport of cryogenic gases, whereby it must be noted that the most common cryogenic gases have a liquefaction temperature higher than that of hydrogen:

1. ISO 21010 *Cryogenic vessels—Gas/material compatibility*, it provides general guidance for chemical compatibility with gases, not mechanical properties, for which the following have to be used.

2. ISO 21028-1:2016 *Cryogenic vessels—Toughness requirements for materials at cryogenic temperature—Part 1: Temperatures below -80°C* , which, however, does not apply to unalloyed steels and cast materials.
3. ISO 21009:2015 *Cryogenic vessels—Static vacuum-insulated vessels*; it consists of two parts dedicated, respectively, to design, fabrication, inspection and tests, and to operational requirements.
4. ISO 21029-1:2018 *Cryogenic vessels—Transportable vacuum insulated vessels of not more than 1000 litres volume—Part 1: Design, fabrication, inspection and tests*.
5. ISO 20421-1:2019 *Cryogenic vessels—Large transportable vacuum-insulated vessels—Part 1: Design, fabrication, inspection and testing*.
6. ISO 20421-2:2017 *Cryogenic vessels—Large transportable vacuum-insulated vessels—Part 2: Operational requirements*.

These are completed by additional standards dedicated to the related valves such as the three parts of the ISO 21013 *Pressure-relief accessories for cryogenic service* and the ISO 21011 *Valves for cryogenic service*.

In comparison with the compressed hydrogen supply chain, the LH_2 technologies present a much lower degree of deployment beyond industrial uses. LH_2 is handled, stored, and used in bulk quantities in industrial areas, but there are few examples of LH_2 stored in public areas such as fueling stations. This explains why the set of RCS documents specifically dedicated to LH_2 is rather limited, as it is the knowledge base on LH_2 behavior. The physical phenomena related to liquid or cryogenic spilling are rather complex and difficult to describe with the present state of knowledge. Experiments are very expensive, and require measurements that are very challenging with the available detection technologies. This explains why, originally, regulations aiming at preventing and mitigating liquid and cryogenic accidental releases may show a strong prescriptive character and are based on very large, not necessarily science-informed safety margins.

Directly related to safety of LH_2 are the requirements in the US NFPA 2 (NFPA, 2021b) Code, which provide separation distances and methods for their reduction (e.g., fire rating of building or sprinklers systems). The NFPA 55 prescribes equivalent safety distances (NFPA, 2020). Specifically on venting and vent stacks, provisions can be found in the CGA G-5.5 *Hydrogen Vent Systems*, which is referenced by the NFPA. In Europe, although not a standard, the Code of Practice on the safety in storage, handling and distribution of liquid hydrogen of the EIGA represents a coherent and comprehensive approach to the matter and it is largely consulted. It has been recently updated adopting a performance-base approach (EIGA, 2019).

In general, in comparison with the compressed hydrogen requirements, the safety distances for liquid are larger. This approach corresponds to the intuitive assumption that room temperature hydrogen diffuses quicker vertically, while the cryogenic gas tends to spread also in the horizontal direction due to its higher density. The storage capacities covered range from 0.150 m^3 to approximately 284 m^3 . Despite this broad range entailing three orders of magnitude, the maximum separation distances differ only by a factor of three. Moreover, some separation distances seem independent from the storage quantities. The values of safety distances prescribed by these documents differ considerably. For example:

1. The EIGA guidelines prescribe 8 m distance from other flammable gas storages, independently from the dimension of the LH_2 storage.

2. The NPFA2:2016 Code requires 15 – 23 m for, respectively, small (150 – 13250 l) and large (57000 – 284000 l) LH₂ storages.

However, the effect of the LH₂ stored amount does not play a role anymore for the distance from places of public assembly: the NPFA2 prescribes 23 m, while for EIGA it is 20 m. All these safety distances are generated by means of hydrogen dispersion models, but they are clearly based on different assumptions, approaches, and classifications. For specific applications using LH₂, additional binding and not-binding approaches and guidelines are available at national level, for example in the Netherlands (PSG35, 2015), in the United Kingdom (HSE, 2006), and in the United States (Harris, Dedrick, Lafleur, & San, 2014).

Despite these works, there is the need for more accurate descriptions of LH₂ release behavior, applicable to higher mass quantities. This is the reason why recent pre-normative research projects aim at providing new and accurate data and a more science-informed methodology to safety requirements for LH₂. An example is the European project PRESLHY (2018–21, <https://preslhy.eu/>, accessed 20 March 2021), addressing the gaps by developing reference experimental data for validated appropriate models and engineering correlations (Venetsanos, 2018; Cirrone, Makarov, & Molkov, 2019). Similar efforts are ongoing at several US National Research Centers, for example at NREL on hydrogen plume behavior from LH₂ storage venting (Buttner et al., 2017), and at SANDIA on characterization of cryogenic releases (Hecht & Panda, 2019), as described in the chapter dedicated to LH₂ in this book. These efforts will support the development of a new generation of international standards for the safe handling of LH₂. An additional future challenge comes from the increase in hydrogen quantities expected to be generated, stored, and transported in a full-scale global hydrogen economy. The present RCS frame is not well-equipped to regulate safe handling of tons of hydrogen. For example, the NPFA2 has a maximum of 75,000 US gallons of LH₂, corresponding to approximately 20 t of hydrogen. Industrial hydrogen is already stored in system with more than 100 t. To enable storage outside specialized industrial areas, it would be necessary to adopt performance-based safety approach offered by the Chapter 5 of the NPFA 2 [see for example the analysis performed by Rivkin, Burgess, & Buttner (2017)]. In the EU, storage of hydrogen above 5 t will fall under the Seveso III directive (European Union, 2012).

7.3.4 Emergency responders

The deployment of hydrogen technologies requires the development of specific emergency procedures, including the strategy of intervention of emergency responders. When arriving at the accident site, before starting firefighting and rescue actions, they must urgently decide on an intervention strategy based on the scarce data available to them. In the case of a traffic accident, for example, the identification of the type of fuel vehicles involved is a critical step. Different fuels require different intervention and rescue strategies. Therefore a correct and well-visible labeling plays an important role. The series of the ISO 17840-2:2019 *Road vehicles—Information for first and second responders* addresses this need, and has been recently updated with new pictograms to be applied to components for vehicles propelled by alternative fuels, including hydrogen. These are to be used in combination of so-called emergency response guides:

1. ISO 17840-2:2019 *Road vehicles—Information for first and second responders—Part 2: Rescue sheet for buses, coaches and heavy commercial vehicles*.

2. ISO 17840-3:2019 *Road vehicles—Information for first and second responders—Part 3: Emergency response guide template*.
3. ISO 17840-4:2018 *Road vehicles—Information for first and second responders—Part 4: Propulsion energy identification*.

Specifically to hydrogen vehicles, The SAE J2990-1-2016 *Gaseous Hydrogen and Fuel Cell Vehicle First and Second Responder Recommended Practice* contains recommendations on how to disable the hydrogen and high voltage system, and on-scene and postaccident inspection procedures. To inform first and second responders, the European project HyResponse (2013–16, <http://www.hyresponse.eu/>, accessed 20 March 2021) has developed an “Emergency Response Guidelines,” which include nomograms for the evaluation of safety distances (Bertau, Li, Tretsiakova-McNally, Molkov, & Makarov, 2016). Similarly, in the United States the Pacific Northwest National Laboratory offers a series of e-learning modules for first responders, which can be considered guidelines (https://energyenvironment.pnnl.gov/projects/project_description.asp?id=382, accessed 20 March 2021).

7.3.5 Hydrogen systems in confined and semiconfined spaces

As explained in depth in Chapter 4, Accidental consequences and Chapter 5, Risk assessment of this book, the risk associated with the release of hydrogen in confined spaces requires a specific approach, due to the possibility of phenomena such as detonation and deflagration. Considerable attention has been dedicated to publish guidelines and standards dedicated to risk prevention and mitigation in applications requiring installation of hydrogen systems in confined spaces. These may be a building for auxiliary power supply, or an indoor hydrogen dispenser for fuel cells forklift in warehouses. Another typical example is the parking of FCEV in public or private garages. In addition, the mobility and transport applications treated later may have to deal with, for example, a hydrogen and fuel cell systems underdeck for ship propulsion.

For North America, specific provisions can be found in the NPFA 2 (NFPA, 2021b), which also refers to the NPFA 55 (NFPA, 2020). The latter introduces the concept of control areas, defined in term of safety measures available (sprinklers, gas cabinets, ventilation). Depending on the type, NFPA 2 establishes maximal quantities of compressed or LH₂ allowed. For quantities above the maximal allowed, special provisions regarding construction, ventilation, separation, are also contained in NPFA 55. Generic provisions for detection, explosion control, and fire alarm and protection systems are also given in Chapter 6 of the NPFA 2, and further specifically for compressed and LH₂, respectively, in Chapter 7 and Chapter 8. Dedicated chapters are also available for the installation of fuel cell power and hydrogen production systems, of fueling facilities and for parking garage.

In Europe, the two ATEX Directives explained in Section 7.2.5 (EU, 1999, 2014a) have to be applied possibly in combination with European harmonized standards. These are for example the EN 60079 translated from the international IEC series 60079 on *Explosive atmospheres*:

1. IEC 60079-0:2015 *Explosive atmospheres—Part 0: Equipment—General requirements* it is the “parent” standard containing general requirements.
2. IEC 60079-10-1:2015 *Explosive atmospheres—Part 10–1: Classification of areas—Explosive gas atmospheres* is dedicated to the classification of areas of the explosive atmospheres.

These standards and the related others of the IEC 60079 series are intended to be applied where an ignition hazard may occur due to the presence of flammable gas. Design aspects have typically to consider use of intrinsically safe equipment and minimize the number of connections. These documents are not specific to hydrogen, and are not prescriptive on minimal design requirements, such as dimension of natural ventilation openings or characteristics of a forced ventilation system. A risk assessment is required, aiming at identifying all hazards, and estimating and eliminating the related risks. This can be achieved by means of an inherently safe design, passive and active controls, and minimization of the remaining risks, by means of mitigation measures, labeling, training, etc.

In all these cases, the safety approach includes always ventilation strategies, aiming at ensuring the potential hydrogen concentration remains under a specific level able to prevent ignition and explosion in all cases. Ventilation can be of two types: natural or forced/mechanical ventilation.

Gas detection is considered as a possible mitigation measure, often in combination with automatic action as depressurization of hydrogen gas supplies and/or shut-off of electrical appliances. For the specific standards related to the gas detection see [Section 7.3.2](#).

The standards mentioned above belonging to the IEC 60079 series aim at preventing flammable gas releases and at avoiding or limiting the consequences of explosions. They are widely adopted but give less attention to (hydrogen) jet fires and to the evaluation of the distances at which persons and environment can be safe.

In addition to harmonized or international standards, not-binding guidelines are also available, aiming at providing engineers, manufacturers, technicians, and authorities with practical information and assisting in making the correct choices. An example of these is the UK *Installation permitting guidance for hydrogen and fuel cell stationary applications* ([HSE, 2006](#)). It was produced by modifying the final report of the European project HYPER (2006–09). This document does not have a legal status in the United Kingdom, but it refers to national regulations and standards. It covers relatively small to stationary systems with a maximum of 10 kW_{el} and total power outputs of approximately 50 kW of combined heat and electrical power. Typical applications are small back-up power supplies or combined heat-power units for domestic use.

In the EU, there are also legally binding national or regional documents, based on the EU Directives, which specify in more practical details the procedures to be followed.

Recently, some prenormative research projects have issued additional recommendations for further standardization regarding more specific hydrogen-related applications. Among others, these two projects funded by the European FCHJU deserve mentioning:

1. FCHJU Project HyIndoor on safe indoor use of FCH systems (2012–15) has issued recommendations on ventilations (vents position and geometry), on limiting overpressure by means of venting devices, and on mitigation measures such as flow restrictions, sensors, jet-fires ([Fuster et al., 2017](#); [Jallais et al., 2014](#)).
2. FCHJU Project SUSANA (2013–16), in support to the required safety analysis, has developed a CFD model evaluation protocol and guidelines, including validation datasets aiming at an assessment of the CFD results used in the risk assessment and safety design of fuel and hydrogen applications ([Baraldi et al., 2017](#); [Tolias et al., 2019](#)).

A particularly harsh challenge is represented by cases such as parking hydrogen vehicles in a private or public garage or hydrogen vehicles in tunnels. In these cases, the safety design measures

adopted for transport applications has to be assessed in infrastructures which have been designed according to more generic safety rules. A standardized approach to permitting legislation is therefore difficult and not yet available. The first difficulty is met when defining credible initiating accidental scenarios. Moreover, the assessment of the consequences is further complicated by many factors, such as the dimension and the position of ventilation opening, the type and strength of the building structure, the possibility of domino effects.

National authorities, or more in general authorities having jurisdiction, have started reconsidering general approaches to safety in public mobility infrastructures such as garages, underpasses, and tunnels from the specific point of view of hydrogen vehicles. US National laboratory SANDIA has recently mapped the hazards in tunnels related to hydrogen vehicles and the presently available knowledge database. The European Project HYTUNNEL-CS (2019–22, <https://hytunnel.net/>, accessed 20 March 2021) is working on safety requirements for FCEV in tunnels and underground parking, aiming not only at data and models for risk assessment and prevention, but also at management of emergencies. The project has a high RCS dimension ([Hy-Tunnel-CS, 2020](#)) and will contribute to a future standard in this area.

7.3.6 Risk assessments

The majority of the standards and the legislative documents presented in this chapter prescribes a risk assessment as a necessary requirement for the certification/approval of the whole hydrogen systems. The risk assessment approaches and techniques are described in [Chapter 5](#), Risk assessment of this book. Here is a simple list of the generic standards that can be used. For qualitative risk analyses, the following documents can be used:

1. IEC 60812:2018 *Failure modes and effects analysis (FMEA and FMECA)*.
2. IEC 61025:2006 *Fault tree analysis*.
3. SAE J1739-2006 *Potential Failure Mode and Effects Analysis in Design (Design FMEA) and Potential Failure Mode and Effects Analysis in Manufacturing and Assembly Processes (Process FMEA) Reference Manual*.

For quantitative risk analyses, the following may be useful:

1. IEC 31010:2019 *Risk management—Risk assessment techniques*.
2. ISO 12100:2010 *Risk Assessment*.

For more information on risk assessment methods, tools and techniques see [Chapter 5](#), Risk assessment of this book.

7.4 Regulations, codes, and standards for specific hydrogen applications

This section presents the safety RCS framework regarding selected elements of the FCH supply and value chains. For each of them, the applicable regulations will be handled first, followed by an assessment of the available international standards, which are shown schematically in [Fig. 7.6](#).

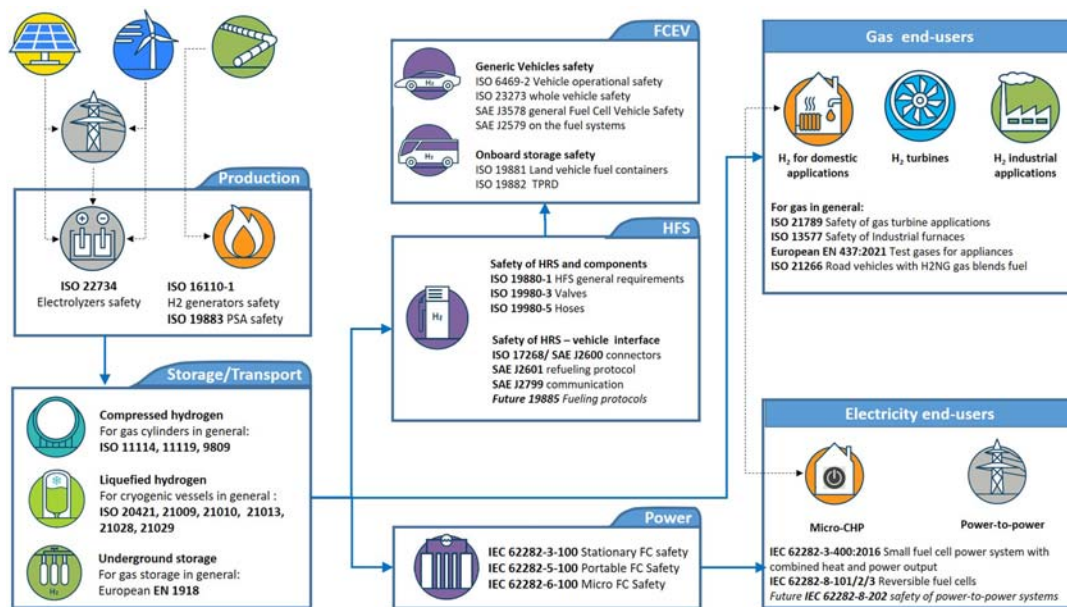


FIGURE 7.6

Simplified FCH value chains and the related international standards on hydrogen safety. Whenever a hydrogen-specific standard is absent, more generic gas standards are given. *FCH*, fuel cells and hydrogen.

7.4.1 Hydrogen production

For hydrogen electrolyzers, the reference safety document is the standard ISO 22734:2019 *Hydrogen generators using water electrolysis—Industrial, commercial, and residential applications*. In the ISO terminology, electrolyzers are "... hydrogen gas generation appliances, herein referred to as hydrogen generators, using electrochemical reactions to electrolyze water to produce hydrogen" (copied from the Scope of the ISO 22734). This standard covers alkaline electrolysis and solid polymeric materials such as PEM and anion exchange membranes (AEM). Solid-oxide electrolyzers and fuel cells used in reverse mode are excluded. It provides safety requirements and related testing methods for all the components and materials. A quantitative or at least semiquantitative risk assessment of the overall system is also needed.

The coupling of electrolyzers to renewable energy sources is a key element for the development of energy systems with low GHG emissions. However, classic electrolyzers have been developed to operate at a rated power value and were not designed for the higher variability characterized by renewable power sources. The mentioned coupling may generate additional safety issues. For example, the gas permeability of the separation membrane of certain electrolyzers increases when operating at low power. This so-called cross-over phenomenon will cause an increase of hydrogen in the oxygen stream and of oxygen in the hydrogen stream. The ISO 22734 is now under revision, aiming at a guaranteeing safer operation in combination with variable power sources. The standard

will be completed by a technical report ISO TR 22734-2 on *Testing guidance for performing electricity grid service*. Part of the required preparatory work has been delivered by the European project QUALYGRID (2017–20, <https://www.qualygrids.eu/>, accessed 20 March 2021), which has developed a set of qualifying tests of electrolyzers for grid services.

Regarding safety of hydrogen production from fossil fuels, such as steam methane reformers, the ISO 16110-1:2007 *Hydrogen generators using fuel processing technologies—Part 1: Safety* can be used. By using the ISO terminology this standard covers “hydrogen generators that convert an input fuel to a hydrogen-rich stream of composition and conditions suitable for the type of device using the hydrogen” (copied from the Scope of the ISO 16110-1). The input fuel can be NG and NG rich mixtures, and also fuels derived from oil refineries, alcohols, etc. The applications considered are indoor and outdoor commercial, industrial, light industrial, and residential uses. The standard sets an upper limit to the capacity of 400 m³/h of hydrogen at reference conditions. It tackles all significant hazards, from electrical and EMC hazards to materials compatibility and high-pressure aspects, including prevention of explosions, and defines the tests to be used for their assessment. It covers the system and its components (valves, pipes, rotating equipment, etc.). It is completed by the Technical Specifications ISO/TS 19883:2017 *Safety of pressure swing adsorption systems for hydrogen separation and purification*, which applies to any type of commercial or industrial purification streams using this hydrogen purification process.

7.4.2 Stationary applications

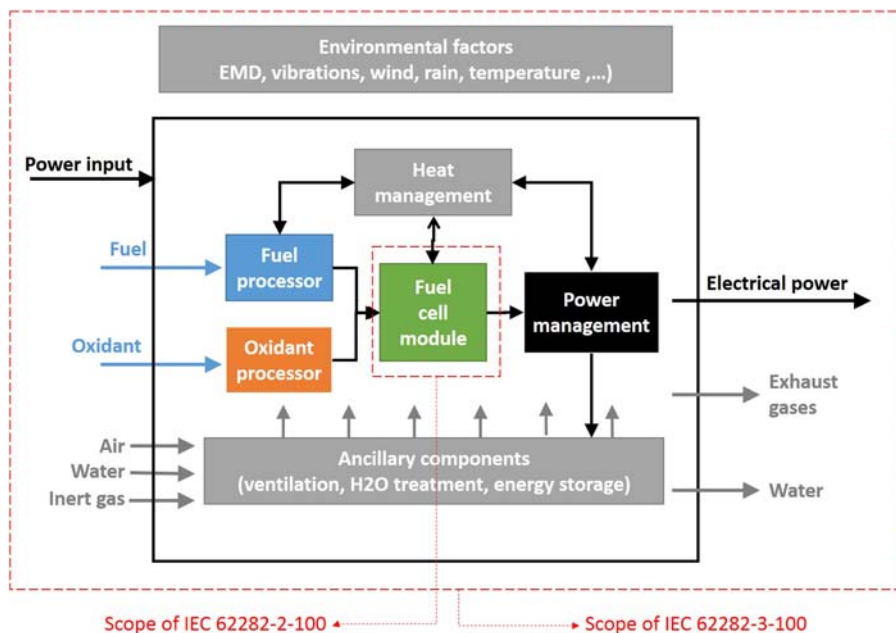
7.4.2.1 Stationary fuel cell systems

A schematic definition of a typical stationary fuel cell power system is given in Fig. 7.7. It is the definition adopted by the IEC 62282-3-100:2019 *Fuel cell technologies—Part 3—100: Stationary fuel cell power systems—Safety*, dedicated to the safety of stationary fuel cells. This standard applies to “stationary packaged, self-contained fuel cell power systems which generate electricity through electrochemical reactions” (text copied from the Scope of the IEC 62282-3-100:2019). The scope covers indoor and outdoor commercial, industrial, and residential use in nonhazardous areas, for both stand-alone power systems or connected to the electrical grid.

The scope of this standard covers all possible combinations, from low temperature power units to high temperature fuel cells integrated with methane reformed or any other type of fuel processing system, and with the possibility as well of energy storage and heat recovery. Many different input fuels are covered, from pure hydrogen to NG, biogases, synthetic gases, and also alcohols.

The standard requires that the manufacturer performs a risk assessment, aiming at the identification of all hazards, and the estimation and elimination of the related risks. The range of safety requirements is broad:

1. General requirements on pressure equipment and piping, flue gas venting systems, gas-conveying parts.
2. Prevention and protection against fire or explosion hazards (not only those generated by the fuel cell power systems, but also by burners in case a reformer is present).
3. Electrical safety and EMC compatibility.

**FIGURE 7.7**

Typical stationary fuel cell power system, modified from the sketch contained in IEC 62282-3-100 and IEC 62282-2-100.

Typical tests are leakage tests (both pneumatic and hydraulic), strength tests (normal operation and under electrical overload), operation of burners, emissions, and venting, including so-called wind and rain tests.

While the scope of the IEC 62282-3-100 is the full power unit, for type approval of fuel cell modules the following standard can be used: IEC 62282-2-100:2020 *Fuel cell technologies—Part 2—100: Fuel cell modules—Safety* (Fig. 7.7). Also in this case, road vehicles applications are explicitly excluded.

The series of standards IEC 62282-8-101 to 103 is dedicated to reversible fuel cells, which can be used for power generation (fuel cell mode) and for hydrogen production (electrolysis or reverse mode). These standards are not specifically dedicated to safety, but prescribe performance tests for, respectively, solid oxide, PEM, and alkaline single cells and stacks. This type of fuel cells can be used in power-to-power systems, able to provide services to the electrical grid. They consist of an electrolyzer converting system power into hydrogen, a hydrogen storage system, and a fuel cell converting the chemical energy stored in hydrogen back to the power grid. It is also possible to replace the two electrochemical systems by one reversible fuel cells. Safety design and certification of the complete power-to-power system is very complex and time consuming, because calling upon several RCS documents and requirements belong to the areas of electrolyzers, fuel cells, and hydrogen storage. Therefore IEC/TC 105 is working at the standard IEC 62282-8-201 on *Test procedures for the performance of power-to-power systems*, and is already considering a future IEC 62282-8-202 on *Power-to-power systems—Safety*.

7.4.2.2 Stationary hydrogen storage

As explained in [Chapter 2](#), Hydrogen technologies, hydrogen storage is an essential component of many applications, from stationary FCH production up to transport infrastructure.

The ISO/TC 197 has worked on a standard dedicated to the safe hydrogen storage in cylinders and tubes. However, the draft ISO 19884 *Gaseous hydrogen—Cylinders and tubes for stationary storage* did not pass the balloting and was deleted. It would have applied to stationary storage with maximal volume of 10,000 m³ and a MAWP up to 110 MPa, covering metallic and composites tanks with the only exclusion of welded liners. It was drafted considering the specific operative conditions expected for hydrogen storage, including shallow pressure cycles. The ISO/TC 197 working group WG15 has restarted the drafting.

An important aspect of the safety of compressed gas storage during operation is related to periodic inspection requirements, which allow for the identification of damaged tanks and their possible requalification. In most of the cases, a visual inspection is the standard operation required. More advanced nondestructive assessment methods have been developed for many years, but not all have achieved the maturity required to be referred to in a standard. One exception is the modular acoustic emission technique described in the Technical Specifications ISO/TS 19016:2019 *Gas cylinders—Cylinders and tubes of composite construction—Modal acoustic emission (MAE) testing for periodic inspection and testing*. This document applies to the periodic inspection of all types of tanks and can be used in combination of all the above-mentioned standards, including the on-board storage ones (see [Section 7.4.3.1](#) Type approval of FCEV).

To the authors' knowledge, an international standard for LH₂ stationary storage does not exist, but the general ISO standards for cryogenic vessels mentioned in [Section 7.3.3](#) could be used.

Regarding bulk hydrogen underground storage, in Europe there is the standard EN 1918:2016 *Gas infrastructure—Underground gas storage*, consisting in five parts providing functional recommendations for storage in aquifers, salt caverns, and rock caverns. The fifth part contains recommendations for surface facilities. In all cases, safety management is mandatory, and specifically to minimize risk of blowout, surface and subsurface safety valves and a gas monitoring system are required. Other provisions concern the determination of maximal storage pressure and the confinement of the geological storage. The EN 1918 does not contain specific requirements for hydrogen.

7.4.3 Mobility and road transport

This section covers RCS for private FCEV mobility, which covers light duty vehicles (LDV) for private mobility and fuel cells buses, as well as fuel cells of heavy duty vehicles (HDV). RCS for fueling infrastructure is also handled here, including with interoperability aspects such as the required standardized interface between vehicle and dispenser.

7.4.3.1 Type approval of FCEV

As mentioned in [Section 7.2.1](#), the UN R134 ([UN R134, 2015](#)) has become the reference document for a world-wide type-approval of fuel cell vehicles. The regulation applies to hydrogen-fueled vehicles of category M and N2, which incorporate compressed hydrogen storage system (CHSS). The category M consists of vehicles with at least four wheels used for the carriage of passengers. It includes not only passengers' cars but also buses. Vehicles of category N2 are commercial trucks

with a mass comprised between 3.5 tons and 12 tons. For the L categories, motorcycles, there is the UN R146, briefly described at the [Section 7.2.6](#) on RCS in Japan.

As stated in the introduction of the GTR13, “The goals of this GTR are to develop and establish a GTR for hydrogen-fueled vehicles that attains or exceeds the equivalent levels of safety of those for conventional gasoline fueled vehicles and is performance-based and does not restrict future technologies” [text copied from page 5 of the final text of the GTR13 ([UN-ECE, 2013](#))].

The scope of the UNECE GTR13 and the UN R134 is the safety-related performance of these vehicles, with an obvious focus on their hydrogen storage system and its individual components. The major difference in scope between the two documents is related to LH₂ storage. While the GTR13 presents optional requirements for liquefied hydrogen storage system, the R134 deals exclusively with CHSS. Another difference regards the safety of the electric power train, which are not covered by the UN 134. Both documents do not afford harmonization of crash tests and of material compatibility aspects. From now on, in this text, if not differently stated, reference is made to the regulation R134: its part 1 and 2 are dedicated to specifications of, respectively, the CHSS and its components, while the part 3 gives specifications to all the other hydrogen-containing components of the vehicle.

The on-board hydrogen storage system consists at least of a high-pressure storage container and closure devices with the following safety functions ([Fig. 7.8](#)):

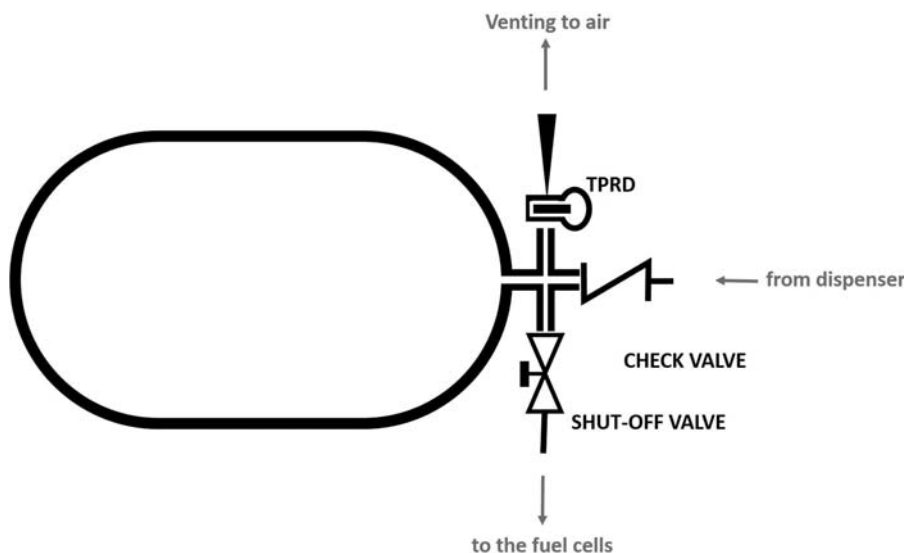
1. Prevention of the back flow from the container to the filling device: a nonreturn valve, also called check valve, is used for this function.
2. Prevention of the flow from the container to the fuel cells: an automatic shut-off valve is used for this function.
3. Allow rapid depressurization of the container in case of fire, to which goal a Temperature-controlled Pressure Release Device (TPRD) is installed.

The concept of the type-approval of the UN R134 relies on the quantitative assessment of the initial strength indicators, followed by the applications of accelerated stress tensors and the final assessment of the residual strength of the storage system. Therefore the storage system has to meet requirements for baseline metrics (initial burst pressure and cycle life), durability, on-road system performance (pneumatic sequential tests), performance in fire, and durability of primary closures.

The durability performance verification aims to ensure confinement in case of external events, which shorten the life of the storage system: it includes mechanical (drop test), chemical, and extreme temperature exposures. The related tests are typically sequential pneumatic tests. The on-road performance verification aims to ensure confinement under conditions accurately representing the full life of the system. It includes worse case operations and its pressure cycling tests and leak/permeation tests are performed with hydrogen as testing media. At the end of both assessments a residual proof pressure test and a residual strength burst test are performed (hydraulic in both cases). The minimum acceptable residual burst pressure is set at 225% of the NWP for carbon fibers composites.

Similar verification tests are required for the CHSS component, and specifically for the TPRD, which plays a critical role in ensuring safe hydrogen release in case of fire, before the failure of the confining function of the CHSS.

Which is the degree of implementation of the UN R134 by the UN contracting parties? A snapshot from 2017 indicates the following:

**FIGURE 7.8**

Scheme of a typical on-board hydrogen storage system.

1. In the EU the UN R134 has been made equivalent to the EU regulations EC79/2009 and EC406/2010, described in [Section 7.2.5 \(EU, 2009b, 2010a\)](#). Therefore in the EU type approval can be obtained via EU regulation or R134 regulation. They are not the same but their safety levels are equivalent. In view of the alignment of the EU solely to the UN R134, this aspect will have to be properly handled to avoid misunderstanding on the correct materials choice. As already mentioned in [Section 7.2.5](#), the EU regulations EC79/2009 will be withdrawn in 2022.
2. In Japan, adoption of the UN R134 falls in the frame of the Free Trade Agreement with EU. An agreement is in place in principle, but the finale date of adoption is not known.
3. South Korea has first adopted the EU Directive EC79; now the UN R134 is also accepted, with a couple of deviations: localized fire/bonfire test is not adopted and a sled test for fuel system is added as a requirement.
4. Canada will adopt high voltage requirements based on recent National Highway Traffic Safety Administration (NHTSA) final rule to ensure harmonization.
5. People's Republic of China: GTR13 (Phase 1) is now a mandatory standard but for the fire test the Chinese document has a different test. Another deviation regarding Type 4 cylinders, on-board pressure only up to 35 MPa.
6. In the United States: the NHTSA is performing a verification of the GTR13, to get details needed for self-certification.

From the start, the drafting of the GTR13 has been structured in two phases. The present GTR13 text is the result of the completion of the first phase. The second phase started in 2017, with the following goals:

1. Revision of the scope to address additional vehicle classes.
2. Investigation of a potential harmonization of crash test specifications.
3. Requirements for material compatibility and hydrogen embrittlement.
4. Requirements for the fueling receptacle.
5. Evaluation of performance-based test for long-term stress rupture proposed in Phase 1.
6. Consideration of research results reported after completion of Phase 1.
7. Consideration of a reduction of the minimum burst pressure requirement.

The work of the GTR13 Phase 2 will probably terminate in 2022. Its degree of advancement can be followed at UNECE GTR-2 website (<https://wiki.unece.org/display/trans/HFCV-GTR13-Phase+2+session>, accessed 20 March 2021). Examples of present focus are roll-over tests for buses, a rationale for the number of cycles in the storage system tests for heavy-duty applications, an improved fire resistance test based on heat flow. Another interesting new item is dedicated to conformable tanks approval.

There are several standards and guidelines dedicated to FCEV safety and specifically on its high-pressure components. Some of them have assisted the development of the UNECE GTR13 and the R134, for example, the SAE J2578 *Recommended Practice for General Fuel Cell Vehicle Safety*, which proposes safety criteria and testing methodologies for the design of FCEVs. It includes safety provisions for operation and maintenance, all the electrical safety aspects, and post-crash criteria. These guidelines for the whole vehicles are complemented by the standard SAE J2579 *Fuel Systems in Fuel Cell and Other Hydrogen Vehicles*. It defines safety requirements for the design, construction, operation, and maintenance of hydrogen fuel storage and handling systems of on-road vehicles. It provides a well performance-based requirements for verification of design prototype and production, and in this sense can be considered a precursor of the UNECE GTR13.

The ISO 23273:2013 *Fuel cell road vehicles—Safety specifications—Protection against hydrogen hazards for vehicles fuelled with compressed hydrogen* is dedicated to the hydrogen safety of fuel cells road vehicles. The focus is on hydrogen-related hazards and the protection of persons and environment inside and outside the vehicle. It is limited to the vehicle design and operation (manufacturing, maintenance, and repair are excluded), and considers only compressed hydrogen on-board storage systems. It does not provide a testing matrix, but only basic requirements.

The more recent ISO 19881:2018 *Gaseous hydrogen—Land vehicle fuel containers* regards on-board CHSS. It sets requirements for LDV, HDV, and industrial powered trucks (forklifts and other material handling vehicles). All tanks are covered, from Type 1 to Type 4, and the four NWP of 25 MPa, 35 MPa, 50 MPa, and 70 MPa. The range of containers is broader than the definition used in the GTR13. The ISO 19881 covers the following three classes:

1. Category A is for containers intended for LDV and HDV regardless of the potential for further qualification according to the GTR13.
2. Category B is for type 4 containers intended to be further qualified in accordance with the GTR13, for FCEV with a gross vehicle weight less than 4536 kg.
3. Category C is for containers to be used in hydrogen powered industrial trucks.

The ISO 19881 is complemented by the standard ISO 19882: 2018 *Gaseous hydrogen—Thermally activated pressure relief devices for compressed hydrogen vehicle fuel containers*, dedicated to the performance testing verification of TPRDs. The Technical Committee ISO/TC 197 is

also working at a future ISO 19887—*Gaseous hydrogen—Fuel system components for hydrogen fueled vehicles*. It will cover all the FCEV components not covered in other standards dealing with FCEV components. When the EU regulations EC79/2009 will be withdrawn, the future ISO 19887 standard will be available to cover in Europe the components not covered by the UN R134.

7.4.3.2 Hydrogen fueling stations

Differently to the vehicle case, covered by an international type-approval regulation, permission for the construction and operation of fueling stations is issued by local authorities and regulated by national or regional laws. In general, HFS have to respect the same general legal framework required for all hydrogen stationary installations (see [Section 7.4.2](#)). However, given the complexity of an HFS, consisting of high-pressure components such as compressors, pipelines, and compressed or liquid storage, of hydrogen supply or on-site production, refueling dispensers, flexibles, vehicles on site and probably also general public, the existence of dedicated legal provisions or permitting guidelines would help enormously developers, operators, and license providers.

Several pioneering countries with an advanced hydrogen technologies deployment have already developed detailed recommendations for the safety design and the permitting process specific for HFS. One example is California, for which the US DOE National Research Centre NREL has published in 2012 an RCS template for HFS ([Rivkin et al., 2012](#)), completed in 2016 by a guide to permitting hydrogen dispenser with the purpose to assist all players (project developers, permitting officials, code enforcement officials) in developing permit applications ([Rivkin, Buttner, & Burgess, 2016](#)). Among other topics, this guide identifies applicable codes and standards relevant to the permitting requirements.

In Europe the approach varies from country to country. Germany relies on the requirements of the guideline VdTÜV *Compressed gases 514, Requirements for hydrogen fueling stations* ([VdTÜV, 2010](#)). Italy was one of the first countries to issue legal requirements in 2006, but only for refueling at NWP of 35 MPa, which was the state of the art at that time. This law has been updated later in 2018 to be able to accommodate the technological progress that allowed for refueling at 70 MPa ([Repubblica Italiana, 2018](#)). Guidelines for the permitting of HFS exist also in the Netherlands as of 2010 and have been recently updated in the publication dedicated to the PGS 35 on *Hydrogen: Installations for delivery of hydrogen to road vehicles* ([PSG35, 2015](#)).

A comparison of different approaches adopted for the safety design of HFS in various countries, performed in 2014 ([Pique, Weinberger, De-Dianous, & Debray, 2017](#)), has shown that many documents share several basic elements, such as technical and human barriers and preventive, mitigating, and protective measures, but differ in other elements. The practice shows results can significantly differ depending on the method and the assumptions chosen. In particular the choice of the initiating release scenarios (e.g., the choice of the dimension of the openings causing a hydrogen leak) and the assumption on the failure frequencies of individual components play a critical role. That difference in assumptions lead to diverging risk assessment conclusions is confirmed by a comparison published in 2018 of the QRA results performed by three different companies on the same HFS installation ([Honselaar, Pasaoglu, & Martens, 2018](#)).

The recently published ISO 19880-1:2020 *Gaseous hydrogen—Fuelling stations—Part 1: General requirements* offers a common approach to safe design of HFS. This standard requires measures to reduce to acceptable level the risk of harm from fires and explosions. A risk assessment is therefore mandatory, which has to consider mitigation measures to reduce the probability of formation of flammable mixtures (such as shut-off systems, PRD) and their ignition (for example

the measures of the standard IEC 60079-10-1) and consequences (such as over-pressure protection for building, detection, emergency gas release, fire barriers). These elements are not individually mandatory, but their consideration as risk mitigation elements in the risk assessment is recommended. The ISO 19880-1 prescribes the incorporation of safety distances in the design of an HFS. The concept of safety distance of an installation is related to its QRA and the related acceptable level of risk. For a further understanding of the concept, see [Chapter 5](#), Risk assessment and [Chapter 6](#), Safety measures and safety barrier functions, of this book. The standard does not prescribe a methodology for their calculations: safety distances or the methodology for their calculations are usually imposed by national or local authority, but they can as well be part of a generic design proposed to the permitting authority. Nevertheless, the ISO 19880-1 provides examples for the calculation of safety distances by means of specific methodology, and offers as well guidelines for the performance of the overall risk assessment of the factory and site acceptance tests.

While many of the provisions in this standard do not differ in principle from requirements for stationary installations, an important specific aspect of HFS safety is the interface with the vehicles and the need to guarantee safe operation during fueling, including the need to avoid damage to the vehicle on-board storage system caused by the malfunctioning of the dispenser. The standard therefore contains informative sections dedicated to the refueling protocols and specifically to the standard SAE J2601 (see next section) and to the existing onboard over-pressure protection mechanisms (see previous section).

The general requirements contained in the ISO 19880-1 are completed by additional standards dedicated to the design, testing, and safety performance of individual components of an HFS: compressors, dispensers, fittings, hoses, valves. While some are still under development, these are the safety-related standards already published:

1. ISO 19880-3:2018 *Part 3: Valves*.
2. ISO 19880-5:2019 *Part 5: Dispenser hoses and hose assemblies*.

7.4.3.3 Interoperability aspects (connection, communication, filling protocol)

To guarantee the possibility for an FCEV to fuel at a refueling station, a standardized interface (vehicle-dispenser coupling) must be available. As already explained in the previous section, it is also critical to ensure the so-called interoperability, that is, the adoption of the same interface throughout different industrial standards and the broadest geographic coverage. Failure to achieve so would require the use of adapters, which can cause additional safety issues, or a major showstopper for the deployment of the technology.

In the case of a hydrogen vehicle, the interoperability aspects extend beyond the simple mechanical coupling between the hose of the dispenser and the tank of a car. It comprises as well the adoption of a standardized filling protocol, data exchange protocol tool, and the related systems, which have the role to ensure safety operation of the refueling process. This is indeed one of the most delicate items, because the coupling of safety design of cars and of refueling infrastructure may create additional safety requirements.

Regarding mechanical coupling, the standard SAE J2600-2015, *Compressed Hydrogen Surface Vehicle Fueling Connection Devices* (from now on called J2600), proposes a standardized connection device generally adopted by FCEV manufacturers. A connection device, or connector, consists of a nozzle (the end of the hose at the dispenser side) and in a receptacle (fueling input at the

vehicles side). The standard prescribes the geometric dimensions of the connector and provides the tests required for the certification of these devices (for example leak tests and durability under thermal and pressure cycles), and their acceptance criteria. The SAE J2600 defines hydrogen in the pressure classes of 11, 25, 35, 50, and 70 MPa. The connector has to work between -40°C and $+85^{\circ}\text{C}$. An important safety aspect relates to the gas trapped between nozzle and receptacle at the end of the filling, and its safe venting. The standard foresees three types of nozzles, two of which remain pressurized and adopt different methods to vent the remaining gas, and one to be used in combination of hoses, which can depressurize at dispenser shutdown.

There is also an ISO standard for connectors, the ISO 17268:2020 *Gaseous hydrogen land vehicle refuelling connection devices*, which is equivalent to the SAE J2600. Some tests are slightly different in the two standards, and in addition, the ISO foresees a high flow version of the connector for 35 MPa, dedicated to commercial vehicle applications. The new version of 2020 contains a new specific freezing test, to verify that the connection does not freeze together for more than 30 s after fueling. This test is comparable to the Pre-cooled Hydrogen Exposure Test of the SAE J2600 (10 s).

The possibility for the dispenser to be automatically informed on the type of tank and its state of charge provides an improved control of the fueling process in terms of safety refueling speed. The standard SAE J2799-2019 *Hydrogen Surface Vehicle to Station Communications Hardware and Software* provides design and testing requirement for a communication protocol for the exchange of data between the vehicle and the fueling station. Its scope is limited to CHSS. The hardware is based on infrared system, and the set of data to be shared consists of tank volume, receptacle type (related to the NWP), fueling commands (start, stop, abort), and the gas pressure and temperature. The standard has been developed for LDV, but it may also be adopted for busses and forklifts. It may also be used in conjunction with the connecting device standard mentioned above and the SAE fueling protocol mentioned just below, but communication is not a requisite for the fueling, noncommunication fueling being still a possibility.

The refueling of an FCEV has to respect strict operative safety boundaries, as well as to achieve two additional and independent objectives: maximization of the autonomy driving range and the minimization of the time spent for refueling. The safety requirements consist in limiting the refueling parameters to an operative window, which avoids overpressurization (gas pressure above 125% of the NWP) and respect the temperature limits of -40°C and $+85^{\circ}\text{C}$. Inside this safe operative window, the refueling must be performed in the quickest way to achieve the higher state of charge of the tank.

To this purpose, the SAE has developed fueling protocols for FCEV with CHSS, which are widely adopted by commercial HFS. The protocols are described in the standard SAE J2601_202005 *Fueling Protocols for Light Duty Gaseous Hydrogen Surface Vehicles*, applicable to LDV with compressed on-board storage at NWP of 35 and 70 MPa and for four nominal temperatures of the gas delivered by the dispenser: -40°C , -30°C , -20°C , and ambient temperature. Fig. 7.9 shows the fueling window for a tank with generic NWP. Filling operations bringing the final pressure – temperature state of the gas in the tank beyond the operative limit indicated in the figure are unsafe and must be avoided. To this purpose the J2601 provides tabulated values prescribing the filling parameters (average flow rate and target pressure) based on specific value of the initial conditions (initial pressure and volume of the tank) and boundary condition (the environmental temperature). Background information and validation can be found in [Schneider et al. \(2014\)](#). The SAE J2601 tables are widely implemented in present HFS, but are considered affected by a strong safety redundancy, because based on a very simplified modeling of the fueling process.

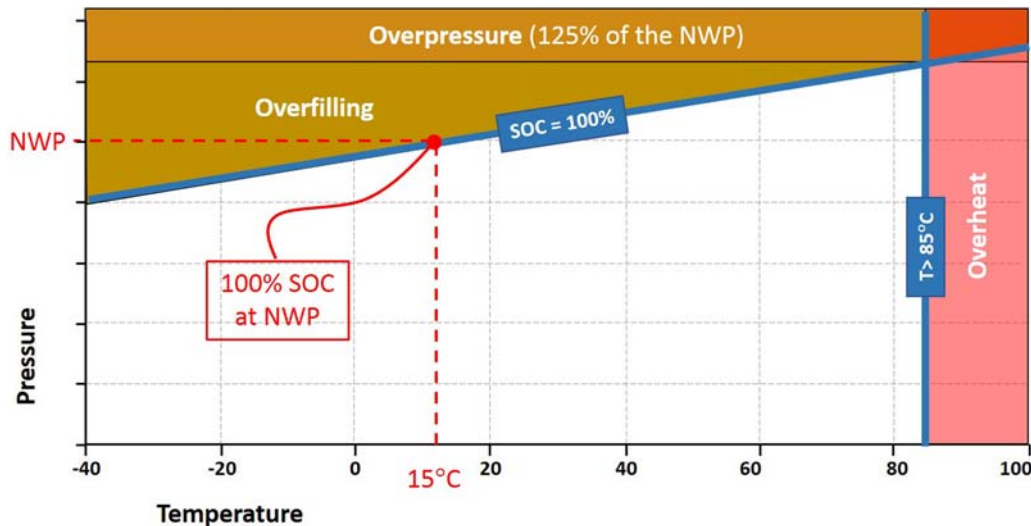


FIGURE 7.9

Boundary conditions and fueling window for a generic onboard hydrogen storage system. The pressure values depend on the NWP of the tank.

Adapted and generalized from the SAE J2601 standard.

7.4.3.4 Heavy-duty transport

Following the Paris agreements (United Nations, 2015a) there is the need to find near-zero GHG emission solutions also for sectors whose GHG emissions are more difficult to abate, such as energy-intensive industry and heavy-duty transport. While for the industrial sector an RCS frame is already in place (hydrogen as an industrial feedstock is used since many years), this is not the case for freight transport on road, rail, and water.

Regarding heavy-duty road vehicles, some of the standards already in place for mobility and its infrastructure can already be used for this more demanding sector. However, the need for higher power and energy and longer traveling autonomy generate additional design and safety challenges, especially for on-board storage, but also for new storage or production requirement at the HFS. The available RCS frame for FCEV for heavy-duty applications (long haul busses, international goods transport) is now restricted to compressed onboard storage at 35 MPa normal working pressure. Dedicated standards will be required in case of the adoption of an NWP of 70 MPa for the on-board storage, or of cryogenic storage solutions.

As a consequence of the high mass flow involved in the refueling process of HDV, specific filling protocol provisions are required, together with dedicated specification for a high flow/high-pressure connecting device. Regarding this latter, the Working Group 5 of the ISO/TC 197 *Gaseous hydrogen land vehicle refuelling connection devices* is working on a new version of the mentioned ISO 17268 with new hardware design for high mass flow fueling at 70 MPa.

Regarding the high flow refueling protocol, the SAE has already issued the technical report SAE J2601/2_201409 *Fueling Protocol for Gaseous Hydrogen Powered Heavy Duty Vehicles* on refueling of HDV (typically transit busses and any commercial vehicles with more than 10 kg hydrogen storage on-board). However, this document does not provide detailed tables of protocol algorithm aiming at how to maximize the state of charge of the tank, but it simply sets safety limits and performance requirements for a safe fueling of HDV, including performance verification. Fig. 7.9, shown for the LDV standard, is applicable also to this document, which defines three fueling options:

1. *High flow* option, with an upper mass flow limit of ≤ 120 g/s. It requires a correspondent high flow connection at 35 MPa as defined by the ISO 17268.
2. *Normal fueling* option, with an upper mass flow limit of 60 g/s. It occurs also at variable speed, but it may use of both the SAE J2600 and the ISO 17268.
3. *Slow fueling* option C, also called time fill, with an upper limit of ≤ 30 g/s.

To address the lack of a specific protocol for HDV, The Working Group 24 of the ISO/TC 197 has started the drafting of the future standard ISO 19885 for *Gaseous hydrogen fuelling protocols for hydrogen-fuelled vehicles*. Differently from the SAE JS2601 series, it will not provide specific protocols for high flow fueling, but will also address protocol design (including communication) and the process required for their verification and implementation.

7.4.4 Nonroad transport

7.4.4.1 Industrial trucks

A particular place in this context is taken by forklift and their refueling in warehouse, because it is a case of a transport application in (semi)confined space. The safety of the forklift itself is covered by IEC 62282-4-101:2014 *Fuel cell technologies—Part 4—101: Fuel cell power systems for propulsion other than road vehicles and auxiliary power units (APU) —Safety of electrically powered industrial trucks*. It is dedicated to the safety of electrically powered industrial trucks including material-handling equipment for outdoor and indoor use. The output voltage is limited to 150 V. It does not only consider hydrogen but also direct methanol fuel cells, and covers safety related both to hydrogen and electricity hazards.

The related fueling system is partially covered by ISO 19880 series dedicated to HFS and its components such as dispensers, valves, hoses, and compressors. However, the ISO 19880-1 states specifically that additional requirements may be expected for indoor dispensing.

Finally, for fueling of the forklifts, the SAE J2601/3-2013 *Fueling Protocol for Gaseous Hydrogen Powered Industrial Trucks* is available. This document is structured in the same way as the J1601/2 (see previous section), but it is applicable to pressures of 25 and 35 MPa. It describes fueling methods, with fixed or variable flow-limiting devices, but it does not provide algorithms or tables.

7.4.4.2 Rails

Despite recent development on hydrogen trains, guidelines facilitating the permitting process are still missing. Operative permissions have to be obtained in each case by means of an individual-type approval procedure. Although the absence of an RCS frame for hydrogen trains is not seen by

major stakeholders as a bottleneck now, the situation may change due to a rapid market uptake or a policy intervention. Obtaining operative license is a time-consuming approach and may hinder further deployment, as identified by a recent European report dedicated to this application (Ruf, Zorn, De Neve, Andrae, & Erofeeva, 2019).

7.4.4.3 Waterborne sector

The waterborne sector comprises a broad range of applications, from international freight transport to mobility (e.g., ferryboats), fishery, and port operations. In term of RCS framework, a major divide can be set between maritime transport (i.e., on sea) and inland waterways transport. More precisely, countries have sovereignty on inland bays, estuaries, lakes, and navigable water up to a specific distance from its shore. In international waters, the International Convention for the Safety of Life at Sea (SOLAS) is the reference international treaty regulating internationally merchant ships. It establishes safety standards for their construction and operation, including on-board components. SOLAS has been developed in the frame of IMO already presented in Section 7.2.1. It is responsibility of the flag state of the ship to guarantee compliance to the SOLAS Convention and some states have embedded the SOLAS requirements into their national law. In addition, the ship owners need a classification certificate issued by a registered body called a *classification society* to obtain an insurance. This certificate states that the construction and the maintenance of a ship is done according to relevant standards. The classification societies maintain standards to be used for certification. The international RCS framework is mainly provided by the IMO (see Section 7.2.1), which provides ship designers, owners, and certifying societies with specific codes, aiming at minimization of risks to the ship and its crew. Of specific interest for the safety of alternative fuels is the *International Code of Safety for Ship Using Gases or Other Low-flashpoint Fuels* (IMO, 2017). This Code contains internationally mandatory criteria for ships using gas or low-flashpoint liquids as fuel. Its major goal is to achieve a level of safety and reliability equivalent to the one achieved with incumbent technologies. Therefore it is not only about machinery for propulsion but also on auxiliary functions, throughout the design, construction, and operation phases. Despite the generic titles, it has been specifically developed to enable the deployment of NG for propulsion. Therefore certification of hydrogen systems still requires an alternative design process. The Maritime Safety Committee of the IMO is working at a broadening of the IGF scope to hydrogen as maritime fuel, building on the experience delivered by ongoing demonstration projects. This new document will, however, cover only the fuel cells installations, but not the hydrogen storage, which will still have to follow an alternative design process. Further details can be found in Tronstad, Høgmoen Åstrand, & Haugom, (2017), which provides also a list of selected guidelines and national rules. Classification societies have also started developing own rules for alternative power generation, both for propulsion and auxiliary power systems. For example, the company Det Norske Veritas (DNV) has expanded its *Rules for Classification of Ships* with a Part 6 dedicated to additional class notations (DNV-GL, 2020). Chapter 2 is on *Propulsion, power generation and auxiliary systems*, and covers battery storage, gas storage, and fuel cells. It provides requirements for the control of hydrogen-related risks, for example on piping design features, hydrogen detection, and fire detection and protection systems. For example, one option is a secondary enclosure for piping, containing inert gas: as alternative, the whole piping system must be welded, and ventilation must be accordingly designed. The main scope of this document is, however, still limited on compressed or liquid NG storage coupled to the fuel cell system via a reformer for the on-board production of

hydrogen. The on-board storage of hydrogen is not yet covered by any RCS document, and this causes considerable difficulties to prototype designers. To be able to place the storage system below deck, the designer has not only to respect all the requirements typical of confined hydrogen systems, including hazardous area classification, but also the generic requirements for energy system on board of ships, which were not specifically developed for hydrogen. Also for a placement above deck, the result of the risk assessment can cause such limitations on the space available that the business case for the specific ship disappears.

The development of solution for hydrogen-driven ships requires also the development of a dedicated infrastructure at ports, whose various components are covered by the RCS elements mentioned in part of this chapter dedicated to stationary installation. However, hydrogen bunkering, and specifically liquid bunkering, including the required interface and the fueling process, is not yet covered by any specific applications and will have to be “translated” from the RCS frame available for liquid NG. For example, the ISO 20519:2017 *Ships and marine technology—Specification for bunkering of liquefied natural gas fuelled vessels* and the ISO 16904:2016 *Petroleum and natural gas industries—Design and testing of LNG marine transfer arms for conventional onshore terminals*. In the EU, the already mentioned Seveso III Directive shall be applied in case of bunkering more than 5 t of hydrogen.

The regulator for inland navigation in Europe is the *Commission centrale pour la navigation du Rhin* (CCNR). The major piece of legislation for the certification of vessels is the EU Directive 1629 on *Technical requirements for inland waterway vessels*. For detailed technical provisions see [EU \(2016a\)](#). This directive refers to the standard ES-TRIN issued by the *European Committee for drawing up Standards in the field of Inland Navigation* (CESNI). Its 2021 edition contains a chapter dedicated to fuels with a flash point below 55°C, but it is all about liquid NG. CESNI is drafting requirements for the use of fuel cells on inland navigation vessels, planned for 2021. Nevertheless, also in the case of inland waterways, prototype ships using hydrogen as fuels with on-board hydrogen storage will have to follow an ad-hoc alternative route for certification.

7.4.4.4 Aviation

The same role covered by IMO in the waterborne sector is covered by the International Civil Aviation Organization (ICAO) for civil aviation. The ICAO is the United Nations agency responsible for setting global rules regulating the international air traffic. Its technical body is the Air Navigation Commission, which formulates international codes and standards for the ICAO. In the United States, the Federal Aviation Administration is the governmental body regulating civil aviation, including airports. In the EU the European Union Aviation Safety Agency is the certifying body for civil aviation ([EU, 2018b](#)), issuing also technical regulations and standards.

The almost complete lack of a hydrogen-specific RCS frame in aviation mirrors the huge technology challenges for the development of this technology. The SAE, together with the European Organisation for Civil Aviation Equipment EUROCAE, has issued recommendations for the installation of fuel cells, derived from the lesson learned in other sectors:

1. SAE AIR6464 (2013) Aircraft Fuel Cell safety Guidelines EUROCAE/SAE, WG80/AE-7AFC.
2. SAE AS6858 (2017) *Installation of Fuel Cell Systems in Large Civil Aircraft*.
3. SAE AIR7765 (2019) *Considerations for Hydrogen Fuel Cells in Airborne Applications*.

These documents are not standards, but guidelines based on best practices for the safe integration of fuel cells in aviation. All three documents consider on PEM fuel cells: the AIR6464 focuses on the use of fuel cells for auxiliary power supplies. The AS6858 goes further and provides requirements for the integration of the full system consisting of the fuel cells and the hydrogen storage, independently of their use on board. The AIR7765 has a more generic character, aiming at transferring the experienced gained on FCH in other sectors to the aviation case. All three documents refer for generic equipment to Part 25 of the *Certification Specification for Large Aeroplanes* (CS-25) of the European Union Aviation Safety Agency (EASA, 2007), which defines conditions for airworthiness and acceptable means of compliance.

The use of fuel cells to directly provide power to propellers, however, is not covered by any specific standards or guidelines. Recent projects had to follow an alternative certification procedure based on a specific risk assessment. An even bigger RCS gap regards to the possible use of hydrogen as a fuel in aviation gas turbines. These gaps can be filled only by means of expensive pre-normative research, a need that has been recognized by research and innovation funding agencies and considered in their recent programs.

7.4.5 Hydrogen—natural gas blends

Hydrogen injection to the gas grid goes under the name of hydrogen – NG admixture or blending (H_2NG). It aims at a reduction of the carbon content of the gas network with a consequent reduction of CO_2 emissions at the point of use. This theme is similar but should not be confused with the concept of pure hydrogen in pipelines, which has the only purpose to transporting hydrogen from its production sites to the consumption sites (see Section 7.4.6). The two themes share nevertheless the general requirement to guarantee and control gas confinement along pipelines and for all its components.

The list of regulation and standards for the safety of NG grids is long and all of them will have to be carefully assessed because some may limit considerably the maximal amount of hydrogen in NG. Other limitations can be found for industrial end uses such as turbines, or for domestic appliance such as burners. For historical reasons, every country has set its own limits for hydrogen in their NG network. The maximum value of allowed concentrations in volumetric varies from 0% in Japan and 0.1% in the United Kingdom to 6% in France up to 10% in Germany (Dolci, Jordan, Keller, & Moretto, 2019). An increase of these limits will require a review of the ATEX compliance of all affected components. It may also require a new approach to safety, for example related to detection strategies. In general, a safe deployment of H_2NG admixture faces two different type of challenges, tackled individually in the next two sections:

1. One related to an increased concentration of hydrogen in the already existing distribution and transportation gas pipeline network.
2. Another one related to the multiplicity of end uses, from domestic to industrial and transport applications, which will have all to be tested and re-certified for higher hydrogen concentration.

7.4.5.1 Injection of hydrogen into gas transport and distribution pipelines

Regarding the integrity and safety of gas pipelines, there is a general agreement on the fact that hydrogen concentrations up to 20% in volume can be reached relatively easily without critical

modifications of the existing technologies. However, the advantage achieved with this concentration in terms of CO₂ reduction is limited. Therefore the present prenormative research and standardization efforts aims at hydrogen contents up to 60%.

An early review study centered on US scenarios for both transmission and distribution pipelines is the NREL study *Blending Hydrogen into Natural Gas Pipeline Networks: A Review of Key Issues* of 2014 (Melaina, Antonia, & Penev, 2013). The study confirms the relative easiness to achieve 20% volumetric concentration of hydrogen. In the United States the legislation regulating gas pipelines, including their safety, is provided the Code of Federal Regulations 49 CFR Part 192 *Transportation of natural and other gas by pipeline: minimum federal safety standards*. Subpart J is dedicated to test requirements, Subparts O and P to, respectively, transmission and distribution integrity management. A related US standard is ASME B31.8S-2018 *Managing System Integrity of Gas Pipelines*. It is applicable to pipelines made of any type of materials, and to the entire system including compressor and delivery stations, valves, and connections.

In Europe, the C is only responsible for harmonization aspects of the gas network, such as its interoperability across member states, common responsibilities of operators, and users and common congestion management procedures (EU, 2009a). For a list of the standards affected by hydrogen injection in the European gas networks, see the already mentioned CEN SFEM report (Weidner, Honselaar, Klassek, & De Jong, 2019). As a follow-up of the gap analysis and roadmap provided by this report, CEN/TC 234 on Gas Infrastructure has decided to work at a roadmap to review the existing standardization frame on NG and to allow for a broader range H₂NG admixture compositions. A series of proposals for the drafting of new standards on injection and blending facilities for renewable gases such as hydrogen and bio-methane is in balloting at the time being. As example of the existing standards, which will be affected by higher content of hydrogen, the following related to the integrity of transmission and distribution pipelines can be mentioned:

1. Requirements for an SMS for transmission lines: EN 16348:2013 Gas infrastructure SMS for gas transmission infrastructure and Pipeline Integrity Management System (PIMS) for gas transmission pipelines—Functional requirements.
2. Requirements for a SMS for distribution lines: EN 15399:2018 Gas infrastructure SMS for Gas Networks with maximum operating pressure up to and including 16 bar.
3. Standards series EN 12007-1 to 5 on Gas infrastructure—Pipelines for maximum operating pressure up to and including 16 bar.

7.4.5.2 End uses of H₂NG gas blends

The use of H₂NG blends requires a reassessment and possibly an adaptation of the present safety provisions for all the NG end uses. An important NG application is power production. Gas turbines able to function with hydrogen-rich blend exist already; however their use has been so far been limited to refineries and selected other chemical processing plants where “waste” hydrogen could be used for power generation instead of being released to the chimney. This explains the absence of a dedicated standardization frame for hydrogen as turbine gases. The basic reference standard in this case is the ISO 21789:2009 *Gas turbine applications—Safety*, valid for all type of open cycle turbines. It identifies all associated hazards and specifies the appropriate preventative measures and processes for their prevention or effects minimization. A risk assessment has to assess the effect of all hydrogen releases inside the turbine enclosure, which contains the gas supply components and the auxiliary systems.

Variation in NG composition will also affect the industrial heat sector, for example boilers and furnaces. The safety of furnaces is the focus of the series of standards ISO 13577 on *Industrial furnaces and associated processing equipment—Safety*. The ISO 13577-2:2014, for example, is dedicated to the safety of the combustion and fuel handling systems, while the ISO 13577-3:2016 provides safety requirements for the generation and use of reactive atmosphere gases.

In the EU, domestic and industrial end uses such as cooking, heating, and hot water production are regulated by the EU Regulation 2016/426 on *Appliances burning gaseous fuels* (EU, 2016b). This is completed by the list of harmonized standards that may be used for their testing and certification (EU, 2018a). One example out of this list is standard EN 437:2021 *Test gases—Test pressures—Appliance categories*, which specifies the test gases and the test pressures for each category of appliances. The present version considers also the so-called “town gas,” with a high content of hydrogen and also a high content of nitrogen. Although some companies have already achieved certification of new hydrogen boilers by means of alternative processes, existing appliances were not designed for hydrogen admixture and its use will conflict with the liability by manufacturers. There are approximately 200 million of appliances in the EU, and for their recertification, new test gases have to be defined and product standards have to be updated regarding the type of tests, the testing conditions, and their acceptance requirements.

The extension of this standardization framework to higher hydrogen content in the gas requires also the execution of prenormative research to guarantee the availability of data. Accurate risk assessment methods need to be informed with detailed knowledge of the physical and chemical behavior of H₂NG releases (ignition probability and concentration limits, flame properties) and their consequences. Several of these gaps will be overcome by ongoing projects, such as:

1. The UK project HyDeploy (first phase 2018–21, <https://hydeploy.co.uk/>), studying the performance and the safety of appliances fed with up to 20% volume of hydrogen with NG (Jeffrey, Spriggs, McEwen, Hooker, & Manson-Whitton, 2019).
2. The European project THYGA (Testing Hydrogen admixture for Gas Applications, 2020–22, <https://thyga-project.eu/>), performing experimental studies on the effect of hydrogen concentrations up to 60% in volume on residential and commercial gas appliances (De Wit, 2020).

Finally, in case of H₂NG as fuel in compressed NG vehicles, the following RCS documents are applicable, some of which set a very strict maximal hydrogen concentration for the on-board storage:

1. The UN R110 (UN R110, 2014) sets limit of 2% in volume for hydrogen, which becomes much lower under specific values of humidity in the gas (this is valid for metallic on-board storage tank only).
2. The ISO 11439:2013 *Gas cylinders—High pressure cylinders for the on-board storage of natural gas as a fuel for automotive vehicles* has the same approach to hydrogen contents as the UN R110 for the type approval or prototyping of on-board storage of NG. The maximum of 2% in volume of hydrogen in NG for tanks made of steel with high tensile strength reduces to 0.1% in case of so-called wet gases, with a water vapor concentration above 32 mg/m³, and this limit is valid for all types of tanks.
3. The ISO 21266:2018 *Road vehicles—Compressed gaseous hydrogen (CGH₂) and hydrogen/natural gas blends fuel systems*; it consists of two parts, on safety requirements and the second to test methods.

7.4.6 Hydrogen delivery

7.4.6.1 Hydrogen transport by trailers

The already mentioned ADR (United Nations, 2019) is reference regulation for hydrogen road transportation and distribution in Europe. Nowadays, delivery of hydrogen, for example to refueling stations, occurs mainly by means of trailers carrying compressed hydrogen cylinders. In the EU, also the TPED Directive (EU, 2010b) has to be applied as well, as described in Section 7.2.5. According to article 2 of the ADR, international transport of accepted dangerous goods shall be authorized, provided compliance with the requirements regarding their packaging and labeling (Annex A) and the construction, equipment, and operation of the vehicle carrying them (Annex B). The ADR refers to standards containing technical requirements for its implementation, for example two belonging to the ISO 11119 family on *Gas cylinders—Refillable composite gas cylinders and tubes—Design, construction and testing*:

1. ISO 11119-2:2020 *Part 2: Fully wrapped fiber reinforced composite gas cylinders and tubes up to 450 l with load-sharing metal liners*. It is applicable to Type 3 for compressed and liquefied gas and a lifetime of 15 years.
2. ISO 11119-3:2013 *Part 3: Fully wrapped fibre reinforced composite gas cylinders and tubes up to 450 l with nonload-sharing metallic or nonmetallic liners*. It is similar to the previous one, but applicable to cylinders of Type 4.

For bigger capacities, the ADR refers to the following two:

1. ISO 11515:2012 *Gas cylinders—Refillable composite reinforced tubes of water capacity between 450 and 3000 L—Design, construction and testing*. It is applicable to all types of tanks and a pressure not higher than 160 MPa.
2. ISO/TS 17519:2019 *Gas cylinders—Refillable permanently mounted composite tubes for transportation*. Applies only to Types 3 and 4 and not to liquefied gases. The volume range is between 450 and 1000 L and the pressure is limited to 100 MPa. There is also a limitation to the (pressure $\times$ volume) value, which must be less than 300000 MPa $\cdot$ L.

In Europe, the EN 12245:2009 *Transportable gas cylinders—Fully wrapped composite cylinders* applies to cylinders of Types 3 and 4 with a volume up to 3000 l.

Annexes A and B of the ADR are regularly updated. However, a continuous synchronization between last versions of the standards and their referencing in the last version of the ADR is not guaranteed. It can occur that the ADR still refers to previous versions, which have been withdrawn by the ISO. This is the case, for example, of the ISO 11119-2: the most recent version of the ADR refers to the version of 2013, which has been withdrawn by ISO and replaced with its version 2020. Consequently, the new versions of the ISO standards do not guarantee compliance with the ADR in Europe. This mismatch between the GTR and the referred standards is caused by their different pace of development, but it has usually a short duration.

A more recent work at CEN produced a standard dedicated to permanently mounted hydrogen cylinders, for example, in a trailer. It is applicable only to carbon-fibers composites and to pressure equal or higher than 30 MPa: EN 17339:2020 *Transportable gas cylinders—Fully wrapped carbon composite cylinders and tubes for hydrogen*. It is applicable with a limit of 100 MPa and a capacity of 3 m³, and the product of pressure $\times$ capacity cannot exceed 100 MPa $\cdot$ m³. Additional to the

mentioned standards, the Technical Committee ISO/TC 58 has developed a whole series of standards for the design, manufacturing, inspection, safety testing of all type of compressed cylinders for various applications. For example, the series of technical reports ISO/TR 13086 consists of guidelines for the design of composite cylinders, from thermal resistance to structural mechanics topics such as burst ratio, stress rupture, cycle fatigue. Standards for cryogenic gas transportation are developed by the Technical Committee ISO/TC 220. Finally, other aspects of gaseous and LH₂ delivery, such as the connection interfaces with the receiving part (for example the connection between the trailer and the stationary storage at the HFS) have not yet dedicated standards.

Also in other parts of the world, the RCS frame for transport of hydrogen is still complete and designers have to rely on the general frame for dangerous gases and liquid. An example of certification process for a trailer carrying composite cylinders in Japan is described in [Azuma et al. \(2014\)](#).

7.4.6.2 Hydrogen transport by pipelines

For bulk quantities and long distances, other hydrogen delivery technologies result to be more economically advantageous than road transport. All the already mentioned general safety and material compatibility standards are of application ([Section 7.3](#)). Pipelines for the transportation of pure hydrogen exist already to a small scale both in Europe and in the United States (see [Section 2.3.5 in Chapter 2](#), Hydrogen technologies, of this book). These have been designed respecting hydrogen-specific requirements. On top of these, already existing NG pipelines can be repurposed to transport hydrogen. A typical safety concern in this case is the compatibility of an ample range of steel components with hydrogen, particularly in view crack growth under cyclic stress condition and in the welding areas. Other challenges are the compatibility and aging behavior of compressors and valves, and the permeability of joints and nonmetallic materials. The suitability of already existing gas pipelines for hydrogen operation has to be assessed case by case.

Specifically on transport of pure hydrogen in pipelines is the US code ASME B31.12 (2019) *Hydrogen Piping and Pipelines*, which is part of the ASME B31 Code for Pressure Piping. Part 12 considers both piping for distribution of gaseous and LH₂ in industrial environment, as well as gaseous hydrogen pipelines, covering design and operation requirements, material choice, testing, and inspection. Additional US standards are:

1. CGA G-5.4 *Standard for Hydrogen Piping Systems at User Locations* contains specifications for piping systems for gaseous or LH₂ (2019).
2. CGA G-5.6 *Standard on Hydrogen Pipeline Systems* (2005, reaffirmed 2013).

In Europe, the standard family of EN 13480 *Metallic industrial piping* (consisting of nine parts) contains safety requirements for metallic industrial piping systems for any fluids and without restriction on their internal pressure (see [Chapter 2: Hydrogen technologies, Section 2.6.6](#)). The EIGA *Guidelines for Hydrogen Pipeline Systems* ([EIGA, 2014](#)) is not a standard, nevertheless it is a essential consultation document, which has been harmonized with the CGA G-56.

This section is about pure hydrogen in a gas pipeline. It complements the matter presented in [Section 7.4.5](#), which contains a general discussion on the integrity of existing gas pipelines with increasing content of hydrogen. It must be noted, however, that the two sections differ in terms of overall goal and applications. This section is on transport of hydrogen from one place to another. [Section 7.4.5](#) is about reducing the CO₂ emission of natural gas by enriching it with hydrogen.

7.4.6.3 Hydrogen transport by ships

Regarding waterborne transportation, general requirements for Europe are covered by the UN Directive for the international transport of dangerous goods by inland waterways (ADN), issued by the Economic Commission for Europe (UNECE). As in the case of the ADN, this UN document sets requirements for the design, testing, and safety assessment of packaging for transportation of various dangerous goods, including flammable compressed or liquefied gases. Similar to the ADR, the ADN refers as well to international standards for the related technical specifications, and are regularly updated. In view of the expected global trade in hydrogen, ocean tankers are seen as one of the options for efficient transport means of LH₂ on long distances. The IMO covers this area, with codes harmonized to the ADN. Of interest are the *International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk* (ICG) and the *International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk* (IBF Code).

As reported in [Section 7.2.6](#), the first prototype waterborne carrier was certificated in 2020 for the transport of LH₂ from Australia to Japan (see [Section 7.2.6](#)). This case is very interesting because certification has been achieved by a combination of different RCS documents with a remarkable international dimension ([Nishifuji, 2019](#)). In conjunction with the alternative certification procedure, Japan and Australia proposed a provisional safety standard adopted by the IMO as Interim Recommendations for Carriage of Liquefied Hydrogen Bulk ([IMO, 2016](#)).

7.5 Conclusions, gaps, bottlenecks, and future needs

This chapter provides an overview of technical regulations and standards aiming at guaranteeing minimal acceptable safety of hydrogen systems, products, and applications. It focuses specifically on hydrogen-related safety, and does not consider in detail hazards common to other technologies, such as the electricity, EMC, and general gas-related hazards.

Especially on standards, this chapter aims at informing on still existing gaps. With one caveat: it is not easy to assess quality and the fitness-for-use of the standards presented. The impact of a standard depends on its broad adoption, and this relies on several additional factors, such as the maturity of the technology or the complexity of the testing requirements. The test campaign for type-approval of FCH vehicles, for example, is very lengthy and costly, and the availability of certified laboratories is scarce. Quickly developing technologies require frequent update of the standards. An example is the ISO standard on LH₂ on-board storage, based on knowledge of the technology dating back several years. The ISO has taken very recently the decision to withdraw them, to avoid that new regulations refer to them (see [Section 7.3.3](#)).

Originally, standards aiming at guaranteeing safe use of hydrogen technologies were not written specifically considering the specific characteristics of hydrogen, but “derived” from general standards dedicated to flammable gases. In the last decade, however, a remarkable effort in basic, applied, and in particular prenormative research has provided a broader basis of reliable data and a better understanding of hydrogen behavior under the specific conditions related to individual applications. It has been then possible to define science-informed or science-based hydrogen safety standards ([San Marchi et al., 2017](#)). This could be achieved, thanks to a better-coordinated national and international research strategy, and the return of experience from field tests ([Jordan et al., 2011](#)).

An additional component of this success has been the strengthening of the relationship between research funding bodies and the work at the standards development organizations.

The present standards are the result of multiannual incremental improvements based on technology progress and feedback received from the field-testing and the operation of the hydrogen applications and products. However, the degree of deployment of the FCH technologies is still rather limited. New national strategies, aiming at accelerating the uptake of renewable energies and achieving the targets agreed in the Paris agreement (United Nations, 2015a), will drive the introduction of hydrogen technologies in sectors not considered so far [see, for example, the European Hydrogen Strategy (European Commission, 2020)]. This will require an important expansion of the present RCS framework for safety. In addition, the needs for producing, transporting, storing, and handling much bigger quantities of hydrogen than done so far will push technology developments, which will require its further adaptation and realignment of the present standards. Identification of gaps and ranking of priorities are intrinsic components of the standardization life cycle. These analyses are regularly performed by standardization development organizations, but are usually not publicly available. The CEN/CENELEC SFEM Working Group on hydrogen has published a gap analysis and provided a roadmap for the whole hydrogen technology chain, which is worth consulting for a rather complete overview of prenormative and standardization needs (Weidner, Honselaar, Klassek, & De Jong, 2019). Another example of expert analysis are the research priorities workshop on hydrogen safety organized by the International Association HySafe (Dolci, Jordan, & Keller, 2018; Keller, 2019).

Based on these documents and the overview of this chapter, the following topics require immediate attention:

1. Infrastructure for fueling heavy-duty transport: it must guarantee safety while minimizing fueling time and maximizing traveling autonomy, and will require new hardware, software, and vehicle-dispenser communication tools. The already ongoing efforts for road of trucks has to be extended to even higher hydrogen mass flow rates needed for trains and ships, including harbor bunkering.
2. Permitting/certification processes for road and nonroad HDV: this topic ranges from the already ongoing efforts for type-approval of HDV to the adoption of new codes for the approval of hydrogen propulsion systems in the maritime and aviation sectors.
3. A new set of standards for the transport and use of LH₂, including on-board storage.
4. Beyond this need of statistically reliable field data, there is also the need for improved methods and tools for QRA, for example by modeling releases and consequences by means of properly validated and certified computational fluid-dynamics tools.

This list is not exhaustive, but provides the general lines and the urgency for the standardization development for the next few years.

(All web-addresses were last accessed April 01, 2021).

References

- Arias, J. (2019). Hydrogen and fuel cells in Japan, EU – Japan Centre for Industrial Cooperation. <https://www.eu-japan.eu/sites/default/files/publications/docs/hydrogen_and_fuel_cells_in_japan.pdf> Accessed 01.04.2021.

- Azuma, M., Oimatsu, K., Oyama, S., Kamiya, S., Igashira, K., Takemura, T., & Takai, Y. (2014). Safety design of compressed hydrogen trailers with composite cylinders. *International Journal of Hydrogen Energy*, 39(35), 20420–20425. Available from <https://doi.org/10.1016/j.ijhydene.2014.050.147>, Elsevier.
- Baraldi, D., Melideo, D., Kotchourko, A., Ren, J., Yanez, J., Jedicke, O., ... Duclos, A. (2017). Development of a model evaluation protocol for CFD analysis of hydrogen safety issues the SUSANA project. *International Journal of Hydrogen Energy*, 42(11), 7633–7643. Available from <https://doi.org/10.1016/j.ijhydene.2016.050.212>, Elsevier Ltd.
- Baird, A. R., Ehrhart, B. D., Glover, A. M., Lafleur, C. B. (2021). Federal oversight of hydrogen systems, SANDIA report SAND2021-2955. Available at: <https://www.energy.sandia.gov/wp-content/uploads/2021/03/H2-Regulatory-Map-Report_SAND2021-2955.pdf>. Accessed 01.04.2021.
- Bertau, S., Li, Z., Tretsiakova-McNally, S., Molkov, V., Makarov, D. (2016). European Emergency Response Guide, Project HyRepsonse, D6.3. <file://net1.ccc.eu.int/jrc-services/PTT-Users/moretpi/Desktop/HyResponseEuropeanEmergencyResponseGuide.pdf>.
- BNQ (2007). *Canadian Hydrogen Installation Code Standard CAN/BNQ 1784-000*, Bureau National Bureau de Normalisation du Québec.
- Buttner, W. J., Ciotti, M., Hartmann, K., Schmidt, K., Wright, H., & Weidner, E. (2017). *Empirical profiling of cold hydrogen plumes formed from venting of LH₂ storage vessels*. NREL/CP-5400-68771. Elsevier Ltd. Available from <https://doi.org/10.1016/j.ijhydene.2018.100.231>.
- Cirrone, D. M. C., Makarov, D., & Molkov, V. (2019). Thermal radiation from cryogenic hydrogen jet fires. *International Journal of Hydrogen Energy*, 44(17), 8874–8885. Available from <https://doi.org/10.1016/j.ijhydene.2018.080.107>, Elsevier Ltd.
- Clague, R. (2019). China hydrogen and fuel cell strategy. In *Proceedings of the fifteenth international hydrogen and fuel cells conference*, 19.03. 2019, Brimingham, UK.
- Day, R. (2010). Advancing commercialization of hydrogen and fuel cells technologies through international cooperation of Regulations, Codes and Standards. In D. Stolten (Ed.), *Hydrogen and fuel cells: Fundamentals, technologies and applications* (p. 908). Wiley-VCH.
- De Wit, C. (2020). *Overview of current standardization & certification framework and the impact of H2NG mixtures*, FCH2JU Project THYGA Deliverable D4.1.
- DNV-GL. (2020). Rules for Classification of Ships Part 6 Additional class notations Chapter 5 Equipment and design features, DNV-GL. <<https://rules.dnv.com/servicedocuments/dnv/#!/industry/1/Maritime/1/Rules>> for classification: Ships (RU-SHIP). Accessed 01.04.2021.
- Dolci, F., Jordan, T., Keller, J., Moretto P. (2018). Research Priority Workshop on Hydrogen Safety: 26–27 September 2016 Petten, The Netherlands, JRC Report EUR 29146. <https://doi.org/10.2760/77730>.
- Dolci, F., Jordan, T., Keller, J., & Moretto, P. (2019). Incentives and legal barriers for power-to-hydrogen pathways: An international snapshot. *International Journal of Hydrogen Energy*, 4, 0–7. Available from <https://doi.org/10.1016/j.ijhydene.2019.030.045>.
- EASA (2007). Certification Specifications for Large Aeroplanes, CS-25 Amendment 3. Available at: http://www.easa.europa.eu/agency-measures/docs/certification-specifications/CS-25/CS-25_Amdt3_19.09.07_Consolidatedversion.pdf. Accessed 01.04.2021.
- EIGA (2006). Gaseous hydrogen stations IGC Doc 15/06.
- EIGA (2011). Hydrogen cylinders and transport vessels IGC Doc 100/11/E.
- EIGA (2014). Hydrogen pipeline systems IGC Doc 121/14.
- EIGA (2019). Safety in storage, handling and distribution of liquid hydrogen IGC DOC 06/19.
- EU (1999). Directive No. 1999/92/EC, of 16 December 1999, on minimum requirements for improving the safety and health protection of workers potentially at risk from explosive atmospheres, Official Journal of the European Communities L23, (L 23), pp. 23/57-23/64. Available at: <<http://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:31999L0092&from=en>>. Accessed 01.04.2021.

- EU (2009a) Directive 2009/73/EC concerning common rules for the internal market in natural gas and repealing Directive 2003/55/EC, Official Journal of the European Union. Available at: <<http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2009:211:0094:0136:en:PDF>>. Accessed 01.04.2021.
- EU (2009b). Regulation No 79/2009 of the European Parliament and of the Council of 14 January 2009 on type-approval of hydrogen-powered motor vehicles, and amending Directive 2007/46/EC, Official Journal of the European Union, pp. 32–46. Available at: <<https://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2010:122:0001:0107:EN:PDF>> Accessed 01.04.2021.
- EU (2010a). Commission Regulation (EU) No 406/2010 of 26 April 2010 Implementing regulation (EC) No 79/2009 of the European Parliament and of the Council on type-approval of hydrogen-powered motor vehicles. Official Journal of the European Union, pp. 1–107. Available at: <<https://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2010:122:0001:0107:EN:PDF>> Accessed 01.04.2021.
- EU (2010b). Directive 2010/35/EU of the European Parliament and of the Council of 16 June 2010 on transportable pressure equipment. Official Journal of The European Union, 165, pp. 1–17. Available at: <<http://www.conformance.co.uk/adirectives/doku.php?id=transportpressure>> Accessed 01.04.2021.
- EU (2012). Regulation (EU) No 1025/2012 of the European Parliament and of the Council of 25 October 2012 on European Standardisation, amending Council Directives 89/686/EEC and 93/15/EEC and Directives 94/9/EC, 94/25/EC, 95/16/EC, 97/23/EC, 98/34/EC, 2004/22/EC, 200, Official Journal of the European Union. Available at: <<http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2012:316:0012:0033:EN:PDF>> Accessed 01.04.2021.
- EU (2014a). Directive 2014/34/EU of the European Parliament and of the Council of 26 February 2014 on the harmonisation of the laws of the Member States relating to equipment and protective systems intended for use in potentially explosive atmospheres (recast), Official Journal of the European Union L96. Available at: <<http://scholar.google.com/scholar?hl=en&btnG=Search&q=intitle:No+Title#0>> Accessed 01.04.2021.
- EU (2014b). Directive 2014/68/EU of the European Parliament and of the Council of 15 May 2014 on the harmonisation of the laws of the Member States relating to the making available on the market of pressure equipment, Official Journal of the European Union. Available at: <<http://data.europa.eu/eli/dir/2014/68/oj>> Accessed 01.04.2021.
- EU (2014c). *Directive 2014/94/EU—Deployment of alternative fuels infrastructure*, Official Journal of the European Union, pp. 1–20.
- EU (2016a). Directive 2016/1629 laying down technical requirements for inland waterway vessels, amending Directive 2009/100/EC and repealing Directive 2006/87/EC, Official Journal of the European Union. Available at: <<https://eur-lex.europa.eu/legal-content/EN/TEXT/PDF/?uri=CELEX:32016L1629&from=EN>> Accessed 01.04.2021.
- EU (2016b). Regulation (EU) 2016/426 of the European Parliament and of the Council of 9 March 2016 on appliances burning gaseous fuels and repealing Directive 2009/142/EC, Official Journal of the European Union.
- EU (2018a). *Commission communication in the framework of the implementation of Directive 2009/142/EC of the European Parliament and of the Council relating to appliances burning gaseous fuels. Official Journal of the European Union, C, 119, 4–18.*
- EU (2018b). Regulation 2018/1139 on common rules in the field of civil aviation and establishing a European Union Aviation Safety Agency. *Official Journal of the European Union.*
- EU (2019a). Regulation (Eu) 2019/2144 of The European Parliament and of The Council of 27 November 2019 on type-approval requirements for motor vehicles and their trailers, and systems, components and separate technical units intended for such vehicles, as regards th, Official Journal of the European Union, (L325), 1–40.
- EU (2019b). Regulation No 134 of the Economic Commission for Europe of the United Nations (UN/ECE)—Uniform provisions concerning the approval of motor vehicles and their components with regard to the safety-related performance of hydrogen-fuelled vehicles (HFCV) [2]. *Official Journal of the European*

- Union, L129, 43–89. Available from [https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:42015X1222\(01\)&from=EN](https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:42015X1222(01)&from=EN), Accessed 01.04.2021.
- European Commission. (2020). A hydrogen strategy for a climate-neutral Europe, COM/2020/301.
- European Hydrogen Safety Panel. (2019). Safety planning for hydrogen and fuel cell projects, European Fuel Cells and Hydrogen Joint Undertaking. Available at: https://fch.europa.eu/sites/default/files/Safety_Planning_for_Hydrogen_and_Fuel_Cell_Projects_Release1p31_20190705.pdf Accessed 01.04.2021.
- European Union. (2012). Seveso III Directive: directive 2012/18/eu of the European Parliament and of the council of 4 July 2012 on the control of major-accident hazards involving dangerous substances, amending and subsequently repealing Council Directive 96/82/EC. *Official Journal of the European Union*, 197, 1–37. Available from <http://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32012L0018>, Accessed 01.04.2021.
- Floristean, A. & Brahy, N. (2019). EU regulations and directives which impact the deployment of FCH technologies, HyLaw. Available at: https://www.hylaw.eu/sites/default/files/2019-02/D4.4-EUregulationsanddirectiveswhichimpactthedevelopmentofFCHtechnologies_0.pdf. Accessed 01.04.2021.
- Floristean, A., Brahy, N., & Kraus, N. (2018). List of Legal Barriers, HyLaw. Available at: <https://www.hylaw.eu/sites/default/files/D4.2-Listoflegalbarriers.pdf> (hylaw.eu). Accessed 01.04.2021.
- Floristean A., Brahy, N., Kraus, N., Skiker, S., Damman, S., Hayter, D. & Nozharova, D. (2019). EU policy paper, HyLaw. Available at: <https://www.hylaw.eu/info-centre> Accessed 01.04.2021.
- Fuster, B., Houssin-Agobmonson, D., Jallais, S., Vyazmina, E., Dang-Nhu, G., Bernard-Michel, G., et al. (2017). Guidelines and recommendations for indoor use of fuel cells and hydrogen systems. *International Journal of Hydrogen Energy*, 42(11), 7600–7607. Available from <https://doi.org/10.1016/j.ijhydene.2016.050.266>.
- Harris, A. P., Dedrick, D. E., Lafleur, C., San Marchi, C. (2014). Safety, codes and standards for hydrogen installations: Hydrogen fueling system footprint metric development, Sandia Report SAND, 2014–3416.
- Hecht, E. S., & Panda, P. P. (2019). Mixing and warming of cryogenic hydrogen releases. *International Journal of Hydrogen Energy*, 44, 8960–8970. Available from <https://doi.org/10.1016/j.ijhydene.2018.070.058>.
- Honselaar, M., Pasaoglu, G., & Martens, A. (2018). Hydrogen refuelling stations in the Netherlands: An inter-comparison of quantitative risk assessments used for permitting. *International Journal of Hydrogen Energy*, 43(27), 12278–12294. Available from <https://doi.org/10.1016/j.ijhydene.2018.040.111>, Elsevier Ltd.
- HSE (2006). Installation permitting guidance for hydrogen and fuel cell stationary applications, RR715. Available at: <http://www.hse.gov.uk/research/rrpdf/rr715.pdf> Accessed 01.04.2021.
- Hy-Tunnel-CS (2020). *Deliverable 1.4 Report on critical analysis of RCS for tunnels and similar confined spaces*, Hy_Tunnel-CS.
- Hübert, T., Boon-Brett, L., & Buttner, William, J. (2018). *Sensors for safety and process control in hydrogen technologies*. CRC Press.
- IMO (2016). Resolution MSC.420(97) Interim recommendations for carriage of liquefied hydrogen in bulk. Maritime Safety Committee Resolutions. Available at: <https://www.imorules.com/GUID-7844CD8A-5166-4EE0-A714-A7A946F26E38.html> Accessed 01.04.2021.
- IMO (2017). International Code of Safety for Ship Using Gases or Other Low-flashpoint Fuels (IGF Code), International Maritime Organization. Available at: <https://www.imo.org/en/MediaCentre/PressBriefings/Pages/26-MSC-95-ENDS.aspx> Accessed 01.04.2021.
- IMO (2020). International Maritime Dangerous Goods (IMDG) Code, International Maritime Organization.
- Jallais, S., Houssin, D., Shentsov, E.V., Molkov, V., Makarov, D., Melideo, D., ... Palmisano V. (2014). Widely accepted on fuel cell indoor installation and use, HyIndoor. Available at: https://www.hyindoor.eu/wp-content/uploads/2014/06/150202-HyIndoor-Guidelines_D5.1_Final-version3b.pdf Accessed 01.04.2021.
- Jeffrey, K. E., Spriggs, C., McEwen, S., Hooker, P., & Manson-Whitton, C. (2019). Safety and regulatory challenges of using hydrogen/natural gas blends in the UK. In: *Proceedings of the international conference on hydrogen safety ICHS 2019*, 24–26 September, Adelaide (Australia), pp. 1–12.

- Jordan, T., Adam, P., Azkarante, I., Barthelemy, H., Bauwens, L., Bengaouer, A., Nikolay, Z., et al. (2011). Achievements of the EC network of excellence HySafe. *International Journal of Hydrogen Energy*, 36(3), 2656–2665. Available from <https://doi.org/10.1016/j.ijhydene.2010.050.087>.
- Keller, J. (2019). HySafe Research Priorities Workshop Findings. In: *Proceedings of the ninth international conference on hydrogen safety*, Adelaide, Australia, pp. 24–26. <<https://hysafe.info/ichs2019/conference-papers-and-presentations/>> Accessed 01.04.2021.
- Keller, J. O., Gresho, M., Harris, A., & Tchouevlev, A. V. (2014). What is an explosion? *International Journal of Hydrogen Energy*, 39(35), 20426–20433. Available from <https://doi.org/10.1016/j.ijhydene.2014.040.199>, Elsevier.
- Linssen, J., & Hake, J.-F. (2016). Hydrogen research, development, demonstration and market deployment activities. In D. Stolten, & B. Emonts (Eds.), *Hydrogen science and engineering: Materials, processes, systems, and technology* (p. 1220). Wiley-VCH.
- Melaina, M. W., Antonia, O., & Penev, M. (2013). Blending hydrogen into natural gas pipeline networks: A review of key issues.
- METI (2019). *The strategic road map for hydrogen and fuel cells—Industry-academia-government action plan to realize a “Hydrogen Society*. Japanese Ministry of Economy, Trade and Industry. https://www.meti.go.jp/english/press/2019/pdf/0312_002b.pdf. Accessed 01.04.2021.
- METI (2020). Guidelines for safe use of high pressure gas in drones driven by hydrogen fuel cells. Japanese Ministry of Economy, Trade and Industry. <https://www.meti.go.jp/english/press/2020/pdf/0410_001a.pdf>. Accessed 01.04.2021.
- MLIT (2004). *Safety regulation for hydrogen and fuel cells vehicles*. Japanese Ministry of Land, Infrastructure, Transport and Tourism. <https://unece.org/japanese-legislation-hfcv>. Accessed 01.04.2021.
- Nagashima, M. (2018). Japan’s hydrogen strategy and its economic and geopolitical implications, Études de l’Ifri. <https://www.ifri.org/sites/default/files/atoms/files/nagashima_japan_hydrogen_2018_pdf>. Accessed 01.04.2021.
- NFPA (2020). NFPA 55 Compressed Gases and Cryogenic Fluids Code, NFPA.
- NFPA (2021a). NFPA 1 Fire Code, NFPA.
- NFPA (2021b). NFPA 2 Hydrogen Technologies Code, NFPA.
- Nishifuji, K. (2019). Liquefied hydrogen carrier pilot project in Japan. In *Proceedings of the H2@Port workshop*, 10–12 September 2019.
- Pique, S., Weinberger, S. B., De-Dianous, V., & Debray, B. (2017). Comparative study of regulations, codes and standards and practices on hydrogen fuelling stations. *International Journal of Hydrogen Energy*, 42(11), 7429–7439. Available from <https://doi.org/10.1016/j.ijhydene.2016.020.158>, Elsevier Ltd.
- PSG35. (2015). Hydrogen: Installations for delivery of hydrogen to road vehicles. *Hazardous Substances Publication Series*, 35. Available from <http://www.gevaarlijkestoffen.be/stoffen/grenswaarden.htm>, Accessed 01.04.2021.
- Repubblica Italiana. (2018). DECRETO del 23 ottobre 2018 Regola tecnica di prevenzione incendi per la progettazione, costruzione ed esercizio degli impianti di distribuzione di idrogeno per autotrazione. *Gazzetta Ufficiale della Repubblica Italiana*, 257, 4–12.
- Rigas, F. & Amyotte, P. (2018). Hydrogen safety. CRC Press. <<https://doi.org/10.1201/b12267>>.
- Rivkin, C. et al. (2012). Regulations, codes, and standards (RCS) template for California hydrogen dispensing stations, NREL. <<http://www.osti.gov/bridge>>. Accessed 01.04.2021.
- Rivkin, C., Burgess, R., & Buttner, W. (2015). Hydrogen technologies safety guide, NREL Technical Report NREL/TP-5400–60948. Available at: <<https://www.nrel.gov/docs/fy15osti/60948.pdf>>. Accessed 01.04.2021.
- Rivkin, C., Burgess, R., & Buttner, W. (2017). Regulations, codes, and standards (RCS) for large-scale hydrogen systems. In *Proceedings of the seventh international conference on hydrogen safety (ICHS 2017)*, p. 8.
- Rivkin, C., Buttner, W., & Burgess, R. (2016). Guide to permitting hydrogen motor fuel dispensing facilities.

- Ruf, Y., Zorn, T., De Neve, P.A., Andrae, P., Erofeeva, S. & Garriso, F. (2019). Study on the use of fuel cells & hydrogen in the railway environment Report 3 on Overcoming technological and nontechnological barriers to widespread use of FCH in rail applications Recommendations on future R & I. Available at: <<https://www.fch.europa.eu/sites/default/files/Report3.pdf>>. Accessed 01.04.2021.
- San Marchi, C., Hecht, E. S. S., Ekoto, I. W. W., Groth, K. M. M., LaFleur, L., Somerday, B. P. P., ... James, C. W. W. (2017). Overview of the DOE hydrogen safety, codes and standards program, part 3: Advances in research and development to enhance the scientific basis for hydrogen regulations, codes and standards. *International Journal of Hydrogen Energy*, 42(11), 7263–7274. Available from <https://doi.org/10.1016/j.ijhydene.2016.070.014>, Elsevier Ltd.
- San Marchi, C. & Somerday, B.P. (2012). Technical reference for hydrogen compatibility of materials, Sandia Report SAND2012-7321. <https://doi.org/10.1016/j.juro.2008.040.178>.
- San Marchi, C., Somerday, B. P., & Nibur, K. A. (2014). Development of methods for evaluating hydrogen compatibility and suitability. *International Journal of Hydrogen Energy*, 39(35), 20434–20439. Available from <https://doi.org/10.1016/j.ijhydene.2014.030.234>, Elsevier.
- Schneider, J., C. W. W., G., Mathison, F. R., Veenstra, M. J., Shim, J., Immel, R., ... Potzel, P., et al. (2014). Validation and sensitivity studies for SAE J2601, the light duty vehicle hydrogen fueling standard. *SAE International Journal of Alternative Powertrains*, 3(2). Available from <https://doi.org/10.4271/2014-01-1990>, 2014-01–1990.
- Tchouvelev, A. V., De Oliveira, S. P., & Neves, N. P. (2019). Chapter 6 Regulatory framework, safety aspects, and social acceptance of hydrogen energy technologies, In *Science and engineering of hydrogen-based energy technologies*. Elsevier Inc. <https://doi.org/10.1016/B978-0-12-814251-6.00006-X>.
- Tolias, I. C., Giannissi, S. G., Venetsanos, A. G., Keenan, J., Shentsov, V., Makarov, D., ... Molkov, V., et al. (2019). Best practice guidelines in numerical simulations and CFD benchmarking for hydrogen safety applications. *International Journal of Hydrogen Energy*, 44(17), 9050–9062. Available from <https://doi.org/10.1016/j.ijhydene.2018.060.005>, Elsevier Ltd.
- Tronstad, T., Høgmoen Åstrand, H., G-P. Haugom, L. Langfeldt. (2017). EMSA Study on the use of fuel cells in shipping.
- UN (2009). European Agreement Concerning the International Carriage of Dangerous Goods by Inland Waterways (ADN), United Nations. Available at: <http://www.unece.org/trans/danger/publi/adn/adn2009/09files_e.html>. Accessed 01.04.2021.
- UN R134 (2015). Uniform provisions concerning the approval of motor vehicles and their components with regard to the safety-related performance of hydrogen-fuelled vehicles (HFCV), ECE/TRANS/WP.29/2014/78. Available at: <<https://unece.org/fileadmin/DAM/trans/main/wp29/wp29regs/2015/R134e.pdf>>. Accessed 01.04.2021.
- UN R146 (2019). Uniform provisions concerning the approval of motor vehicles and their components with regard to the safety-related performance of hydrogen-fuelled vehicles of categories L1, L2, L3, L4 and L5, E/ECE/TRANS/505/Rev.3/Add.145. Available at: <<https://unece.org/fileadmin/DAM/trans/main/wp29/wp29regs/2018/R146e.pdf>> Accessed 01.04.2021.
- UN-ECE (2013). GTR13 Global technical regulation on hydrogen and fuel cell vehicles, ECE/TRANS/180/Add.13.
- UN-ECE (2020). Agreement Concerning the International Carriage of Dangerous Goods by Road (ADR), Inland Transport Committee, ECE/TRANS/300. Available at: <<https://unece.org/transportdangerous-goods/adr-2021-files>>. Accessed 01.04.2021.
- United Nations (2015a). ‘Paris Agreement’. Available at: <https://treaties.un.org/doc/Treaties/2016/02/2016021506-03PM/Ch_XXVII-7-d.pdf>. Accessed 01.04.2021.
- United Nations (2015b). Regulation No. 134 Uniform provisions concerning the approval of motor vehicles and their components with regard to the safety-related performance of hydrogen- fuelled vehicles (HFCV), ECE/TRANS/WP.29/2014/78.

- VdTÜV (2010). Anforderungen an Wasserstofftankstellen, VdTÜV Merkblätter Druckgase 514.
- Venetsanos, A. G. (2018). Homogeneous non-equilibrium two-phase choked flow modeling. *International Journal of Hydrogen Energy*, 43(50), 22715–22726. Available from <https://doi.org/10.1016/j.ijhydene.2018.100.053>, Elsevier Ltd.
- Verheul, B. (2019). Overview of hydrogen and fuel cell developments in China, Holland Innovation Network in China. Available at: <<https://www.nederlandwereldwijd.nl/binaries/nederlandwereldwijd/documenten/publicaties/2019/03/01/waterstof-in-china/Holland+Innovation+Network+in+China+-+Hydrogen+developments.+January+2019.pdf>>. Accessed 01.04.2021.
- Weidner, E., Honselaar, M., Klassek, D., De Jong, F. & Gindroz, B. (2019). CEN—CENELEC Sector Forum Energy Management/Working Group Hydrogen: 2018 update report. Available from <https://doi.org/10.2760/449204>.
- Weidner, E., Honselaar, M., & Ortiz Cebolla, R. (2016). CEN-CENELEC Sector Forum Energy Management / Working Group Hydrogen Final Report. Available from <https://doi.org/10.2790/66386>.
- Wurster, R. (2016). Regulations, codes and standards for hydrogen refueling stations. In D. Stolten, & B. Emons (Eds.), *Hydrogen science and engineering: Materials, processes, systems, and technology* (p. 1220). Wiley-VCH.
- Yamada, E., & Mashiba, T. (2019). Development of technical regulations for fuel cell motorcycles in Japan—Hydrogen safety. *World Electric Vehicle Journal*, 10(48), 1–6.
- Yoshida, T. (2015). Update on regulation review for HRS construction and operations in Japan. In *Proceedings of the sixth international conference on hydrogen safety*, Yokohama, Japan, 19–21 October. pp. 1–33.
- Yang, Y., Wang, G., Zhang, L., & Zhang, S. (2019). *Comparison of hydrogen specification in national standards for China*, *E3S Web of Conferences*, 118, 0–5. Available from <https://doi.org/10.1051/e3sconf/201911803042>.
- Zhao, F., Mu, Z., Hao, H., He, X., Przesmitzki, S. V., Amer, A. A., et al. (2020). Hydrogen fuel cell vehicle development in China: An industry chain perspective. *Energy Technology*, 8. Available from <https://doi.org/10.1002/ente.202000179>.

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HYDROGEN SAFETY FOR ENERGY APPLICATIONS

Engineering Design, Risk Assessment, and Codes and Standards

The ultimate guide to safe and efficient handling and usage of hydrogen for energy and vehicular applications.

Hydrogen Safety for Energy Applications: Engineering Design, Risk Assessment, and Codes and Standards presents different aspects of the contemporary knowledge regarding hazard, risk, and safety connected with hydrogen systems. In the initial part of the book the hydrogen fundamentals and the main hydrogen technologies are described and discussed. It then explores scientific aspects of the possible sources and consequences of accidental events that can occur when hydrogen is used, including in its vehicular applications. Risk assessment, as well as the safety measures/safety barriers applicable in such situations, are considered.

This versatile compilation of the different aspects of hydrogen systems safety experiences provides profound understanding of the risk assessment and necessary mitigation measures which, along with policy issues examined, conclude the circle of knowledge on hydrogen safety. This is essential for developing safe hydrogen energy systems, making quantitative assessments of the parameters of accidental release, distribution, and combustion, and developing legal requirements, codes, and standards.

Key features

- Provides factual material, such as models, correlations, tables, nomograms, and formulas, which can be used to perform evaluations and propose mitigation measures to diminish risk probability in hydrogen energy systems
- Presents reference data and detailed descriptions and guidelines of the contemporary risk assessment methodologies
- Covers accident phenomena and consequences of accidents specific to hydrogen systems in a widely applicable way for the wide variety of hydrogen activities

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